LINE GRAPHS, ROOT SYSTEMS, AND ELLIPTIC GEOMETRY: STRUCTURED PROOFS

Abstract. We give structured proofs, in the style of Lamport, of every numbered result of Cameron, Goethals, Seidel and Shult, *Line graphs, root systems, and elliptic geometry* [CGSS], and of the results from other sources that the formalization in Lean needs. Each result carries the paper's number, the name of its Lean declaration and the results it uses.

Contents

1. Conventions 2
Structured proofs 2
Notation 2
2. Lines at 60° and 90° 2
3. Root systems 26
3.1. The classification behind Remark 3.11 53
3.2. Lines contained in a Root System 114
4. Graphs with least eigenvalue −2 130
4.1. Theorem 4.1 130
4.2. Theorems 4.2 to 4.9 136
4.3. Theorems 4.10 and 4.11, and Remark 4.12 151
4.4. Theorem 4.13 and Remark 4.14 156
4.5. The proof of Remark 4.14 159
5. Spectral characterization 179
5.1. Lemmas 5.1 to 5.7 183
5.2. Theorem 5.8 and its consequences 193
5.3. The Hall–Connor nonexistence theorem 207
6. Hadamard matrices 234
6.1. The Hadamard products of pairs of rows 235
6.2. Hypothesis 6.1 238
6.3. Theorem 6.2 239
References 248

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1. Conventions

The purpose of this section is not to give full definitions and background, but rather to establish notation; we defer to [CGSS] for the mathematics and to `cgss/notation.md` for the dictionary between the notation of the paper, ours and the Lean names.

Structured proofs. We follow the format of Lamport [L95]. A result is stated with numbered hypotheses under Assume and its conclusion under Prove. A Proof sketch may follow; it is an informal road map. The proof is a list of steps $\langle 1\rangle 1$, $\langle 1\rangle 2$, ..., ending in a Qed step, and each step at level k has its own proof. The assumption of the enclosing level-k step is cited as $\langle k\rangle$, its numbered parts as $\langle k\rangle{:}i$, and those of the result itself as $\langle 0\rangle{:}i$. Definitions are made by Let, witnesses are introduced by Pick, and case splits use Case, with the proof of the closing Qed explaining why the cases are exhaustive. We expand a step until each leaf is a single inference from named facts: definitions, earlier steps, earlier results, Mathlib lemmas, or numbered computations. A step that departs from the paper's argument is tagged **[new]**, and a step that fills a gap in it is tagged **[repair]**.

Each result records the name of its Lean declaration with `\lean{...}` and the results it uses with `\uses{...}`; these macros print nothing.

Notation. We will use functional notation for the entries of matrices and vectors; the uv-entry of a matrix M is written $M(u,v)$, and the entry of a vector $\mathbf{v}$ indexed by u is written $\mathbf{v}(u)$. Graphs are X, Y and Z. A set of lines at 60° and 90° is recorded by the set Φ of vectors of squared norm 2 on its lines, taken with both signs, so that $\Phi = -\Phi$. We keep the paper's names for the paper's own objects: Σ, Γ_a, Γ_b, Γ_c, Δ, and the sets K, L, M, N of the proof of Lemma 2.10. The paper uses three graph conventions; `cgss/notation.md` gives them three names, which this blueprint uses.

2. Lines at 60° and 90°

We follow Section 2 of [CGSS], with the vector encoding of Section 1. We use the graph names and Lean names of `cgss/notation.md`; in Lean, `Hyp24` is in the namespace `LineSystem`. In what follows, V is a finite-dimensional real inner product space. We write $(\mathbf{x},\mathbf{y})$ for its inner product, as the paper does, since angle brackets are taken by step numbers. A *line system* S in V is a finite set $\Phi \subseteq V$ such that

$$\Phi = -\Phi, \qquad (\mathbf{r},\mathbf{r}) = 2 \quad (\mathbf{r}\in\Phi), \qquad (\mathbf{r},\mathbf{s}) \in \{-2,-1,0,1,2\} \quad (\mathbf{r},\mathbf{s}\in\Phi).$$

We count lines as antipodal pairs $\{\mathbf{r},-\mathbf{r}\}$, so the number of lines is $|\Phi|/2$. We do not require Φ to span V until Theorem 2.13. We call S *indecomposable* if, for all $P,Q\subseteq V$ with $P\cup Q=\Phi$ and $(\mathbf{r},\mathbf{s})=0$ for all $\mathbf{r}\in P$, $\mathbf{s}\in Q$, we have $P=\varnothing$ or $Q=\varnothing$. **[new]** This intrinsic definition agrees with the paper's orthogonal-subspace definition when Φ spans V; without spanning, the paper's literal wording allows the partition into the whole system in its proper span and an empty part. We also require finiteness explicitly: in finite dimension this loses no line systems, since their normalized signed vectors are pairwise at distance at least 1 on the compact unit sphere. An isomorphism of line systems is a bijection of their vector sets preserving inner products.

Elementary identities. We will use bilinearity, symmetry, nonnegativity of squared norm, and the implication $(\mathbf{x},\mathbf{x})=0 \Rightarrow \mathbf{x}=0$. These are, respectively, Mathlib's `inner_add_left`, `inner_add_right`, `inner_sub_left`, `inner_sub_right`, `inner_neg_left`, `inner_neg_right`, `real_inner_smul_left`, `real_inner_smul_right`, `real_inner_comm`, `real_inner_self_nonneg`, and `inner_self_eq_zero`, all in `Mathlib/Analysis/InnerProductSpace/Basic.lean`.

Here `inner_self_eq_zero` is the equivalence $(\mathbf{x},\mathbf{x}) = 0 \iff \mathbf{x} = 0$; its `NormedAddCommGroup` hypothesis holds for our real inner product spaces. In particular, for vectors of squared norm 2,

$$(\mathbf{r},\mathbf{s}) = 2 \iff \mathbf{r} = \mathbf{s}, \qquad (\mathbf{r},\mathbf{s}) = -2 \iff \mathbf{r} = -\mathbf{s}. \tag{1}$$

Indeed, $(\mathbf{r}-\mathbf{s},\mathbf{r}-\mathbf{s}) = 4 - 2(\mathbf{r},\mathbf{s})$ and $(\mathbf{r}+\mathbf{s},\mathbf{r}+\mathbf{s}) = 4 + 2(\mathbf{r},\mathbf{s})$ by bilinearity; the forward implications follow from zero squared norm, and the reverse implications follow by substitution. Thus distinct antipodal pairs represent distinct lines with angles $60°$ or $90°$. We will also use the *finite-span lemma*: for a finite family $(\mathbf{r}_i)_{i\in I}$ in V, every member of its span is $\sum_{i\in I} \alpha(i)\mathbf{r}_i$ for some $\alpha : I \to \mathbb{R}$. This is `Submodule.mem_span_range_iff_exists_fun` in `Mathlib/LinearAlgebra/Finsupp/LinearCombination.lean`.

Definition 2.1. ASSUME: S is a line system in a finite-dimensional real inner product space V.
We call S *star-closed* if

$$\forall \mathbf{r},\mathbf{s} \in \Phi, \quad (\mathbf{r},\mathbf{s}) = -1 \implies \mathbf{r}+\mathbf{s} \in \Phi.$$

The third line of the star on $\mathbf{r}$ and $\mathbf{s}$ is the line of $\mathbf{r}+\mathbf{s}$: its squared norm is $2+2-2=2$, and its inner product with each summand is 1. The three vectors lie in the plane spanned by $\mathbf{r},\mathbf{s}$.

Definition 2.2. ASSUME: S is a line system in a finite-dimensional real inner product space V.
We call S *maximal* if, for every $\mathbf{v} \in \operatorname{span}(\Phi)$,

$$(\mathbf{v},\mathbf{v}) = 2 \ \text{ and } \ \big(\forall \mathbf{r} \in \Phi,\ (\mathbf{v},\mathbf{r}) \in \{-2,-1,0,1,2\}\big) \quad \implies \quad \mathbf{v} \in \Phi.$$

We include ± 2 to allow a line already in S; by (1) these values occur only on that same line. **[repair]** The paper takes the lines to span $\mathbb{R}^n$ (the first paragraph of Section 2) and asks for maximality in $\mathbb{R}^n$; since we do not require Φ to span V, we ask for maximality within $\operatorname{span}(\Phi)$. In the whole of V the condition would be too strong: $\mathbf{e}_1+\mathbf{e}_2$ is at $60°$ or $90°$ to every line of A_ℓ in $\mathbb{R}^{\ell+1}$, and Theorem 3.6 would fail (errata, Implicit hypothesis 7). Thus this is Definition 2.2 in the signed vector encoding.

Lemma 2.3. ASSUME: 1. S is a line system in a finite-dimensional real inner product space V.
2. S is maximal.
PROVE: S is star-closed.

PROOF SKETCH: We will orient the two lines so that their inner product is -1 and test their sum against every root. This is the angle argument in the paper, including coincident lines.

⟨1⟩1. ASSUME: $\mathbf{r},\mathbf{s} \in \Phi$ and $(\mathbf{r},\mathbf{s}) = -1$.
PROVE: $\mathbf{r}+\mathbf{s} \in \Phi$.
⟨2⟩1. $(\mathbf{r}+\mathbf{s},\mathbf{r}+\mathbf{s}) = 2$.
PROOF: By bilinearity and ⟨1⟩, the squared norm is $2-1-1+2=2$. □
⟨2⟩2. ASSUME: $\mathbf{t} \in \Phi$.
PROVE: $(\mathbf{r}+\mathbf{s},\mathbf{t}) \in \{-2,-1,0,1,2\}$.
⟨3⟩1. CASE: $\mathbf{t} \in \{\mathbf{r},-\mathbf{r},\mathbf{s},-\mathbf{s}\}$.
PROOF: The four values are $1,-1,1,-1$, respectively, by bilinearity and ⟨1⟩. □
⟨3⟩2. CASE: $\mathbf{t} \notin \{\mathbf{r},-\mathbf{r},\mathbf{s},-\mathbf{s}\}$.
⟨4⟩1. $(\mathbf{r},\mathbf{t}),(\mathbf{s},\mathbf{t}) \in \{-1,0,1\}$.
PROOF: The line-system condition and (1) exclude ± 2. □
⟨4⟩2. Q.E.D.
PROOF: The sum of two integers between -1 and 1 is an integer between -2 and 2; apply bilinearity to ⟨4⟩1. □
⟨3⟩3. Q.E.D.
PROOF: The cases ⟨3⟩1 and ⟨3⟩2 are exhaustive. □
⟨2⟩3. $\mathbf{r}+\mathbf{s} \in \operatorname{span}(\Phi)$.

PROOF: Both $\mathbf{r}$ and $\mathbf{s}$ lie in Φ by $\langle 1\rangle$, and a span is closed under addition. □

$\langle 2\rangle 4$. Q.E.D.

PROOF: Apply Definition 2.2 to $\langle 2\rangle 3$, $\langle 2\rangle 1$ and $\langle 2\rangle 2$. □

$\langle 1\rangle 2$. Q.E.D.

PROOF: Step $\langle 1\rangle 1$ is Definition 2.1. □

Hypothesis 2.4. ASSUME: 1. V is a finite-dimensional real inner product space and S is a line system in V.

2. S is indecomposable and star-closed.

3. $\mathbf{a}, \mathbf{b} \in \Phi$ and $(\mathbf{a}, \mathbf{b}) = -1$.

LET: $\mathbf{c} \triangleq -\mathbf{a} - \mathbf{b}$, and

$$\Gamma_a \triangleq \{\mathbf{x} \in \Phi : (\mathbf{x}, \mathbf{a}) = 0,\ (\mathbf{x}, \mathbf{b}) = 1,\ (\mathbf{x}, \mathbf{c}) = -1\},$$

$$\Gamma_b \triangleq \{\mathbf{x} \in \Phi : (\mathbf{x}, \mathbf{b}) = 0,\ (\mathbf{x}, \mathbf{c}) = 1,\ (\mathbf{x}, \mathbf{a}) = -1\},$$

$$\Gamma_c \triangleq \{\mathbf{x} \in \Phi : (\mathbf{x}, \mathbf{c}) = 0,\ (\mathbf{x}, \mathbf{a}) = 1,\ (\mathbf{x}, \mathbf{b}) = -1\},$$

$$\Delta \triangleq \{\mathbf{t} \in \Phi : (\mathbf{t}, \mathbf{a}) = (\mathbf{t}, \mathbf{b}) = (\mathbf{t}, \mathbf{c}) = 0\},$$

$$\Sigma \triangleq \{\mathbf{a}, \mathbf{b}, \mathbf{c}\} \cup \Gamma_a \cup \Gamma_b \cup \Gamma_c \cup \Delta.$$

We will write $X = \mathrm{Orth}(\Gamma_a)$ for the *orthogonality graph* on Γ_a: distinct vertices x, y are adjacent exactly when $(\mathbf{x}, \mathbf{y}) = 0$. We will write a vertex as x and its representing vector as $\mathbf{x}$. The represented graph of a vector family uses inner product 1; its *obtuse graph*, denoted Obt, uses inner product -1. We use the orthogonality convention throughout Section 2. The *empty graph* $\overline{K_m}$ has m vertices and no edges, with $m = 0$ allowed.

[repair] We require the pair $\mathbf{a}, \mathbf{b}$ explicitly (Implicit hypothesis 4 of `cgss/errata.md`). A one-line system has no such pair; we will treat it separately in Lemma 2.12 and Theorem 2.13. **[repair]** Also, $\Delta = -\Delta$ here (Error 6), so Σ need not have just one vector on each line. Lemma 2.5 asserts a signed cover.

Lemma (star identities). ASSUME: Hypothesis 2.4 holds in the finite-dimensional real inner product space V.

PROVE: $\mathbf{c} \in \Phi$ and $(\mathbf{a}, \mathbf{b}) = (\mathbf{b}, \mathbf{c}) = (\mathbf{c}, \mathbf{a}) = (\mathbf{c}, \mathbf{b}) = -1$.

PROOF SKETCH: **[new]** We will state the star identities separately so that later results can cite them without citing a step inside the proof of Lemma 2.5.

$\langle 1\rangle 1$. $\mathbf{c} \in \Phi$ and all three inner products between $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are -1.

$\langle 2\rangle 1$. $\mathbf{a} + \mathbf{b} \in \Phi$, so $\mathbf{c} \in \Phi$.

PROOF: Use star-closure, $(\mathbf{a}, \mathbf{b}) = -1$, and closure under negation. □

$\langle 2\rangle 2$. Q.E.D.

PROOF: By bilinearity, $(\mathbf{a}, \mathbf{c}) = -2+1 = -1$ and $(\mathbf{b}, \mathbf{c}) = 1-2 = -1$; the remaining products follow from symmetry and the hypothesis $(\mathbf{a}, \mathbf{b}) = -1$. □

$\langle 1\rangle 2$. Q.E.D.

PROOF: Step $\langle 1\rangle 1$ and symmetry give the stated identities. □

Lemma 2.5. ASSUME: Hypothesis 2.4 holds in the finite-dimensional real inner product space V.

PROVE: $\Phi = \Sigma \cup (-\Sigma)$; equivalently, S consists exactly of the lines of Σ.

PROOF SKETCH: We will exclude equal nonzero inner products with $\mathbf{a}, \mathbf{b}$, then choose the sign in each of the remaining seven cases.

$\langle 1\rangle 1$. ASSUME: $\mathbf{u} \in \Phi$, $\epsilon \in \{-1, 1\}$, and $(\mathbf{u}, \mathbf{a}) = (\mathbf{u}, \mathbf{b}) = \epsilon$.

PROVE: $\mathbf{u} = -\epsilon\mathbf{c}$.

⟨2⟩1. $(\mathbf{a}+\mathbf{b}-\epsilon\mathbf{u}, \mathbf{a}+\mathbf{b}-\epsilon\mathbf{u}) = 0$.
PROOF: Bilinearity gives $2 + 2\epsilon^2 - 2\epsilon(2\epsilon) = 0$, since $\epsilon^2 = 1$ and $(\mathbf{a}+\mathbf{b}, \mathbf{a}+\mathbf{b}) = 2$. □
⟨2⟩2. Q.E.D.
PROOF: Zero squared norm gives $\mathbf{a}+\mathbf{b} = \epsilon\mathbf{u}$; multiply by ϵ and use $\mathbf{c} = -\mathbf{a}-\mathbf{b}$ and $\epsilon^2 = 1$. □
⟨1⟩2. ASSUME: $\mathbf{v} \in \Phi$.
PROVE: $\mathbf{v} \in \Sigma$ or $-\mathbf{v} \in \Sigma$.
⟨2⟩1. CASE: $\mathbf{v} \in \{\pm\mathbf{a}, \pm\mathbf{b}, \pm\mathbf{c}\}$.
PROOF: Choose the displayed sign of a member of $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\} \subseteq \Sigma$. □
⟨2⟩2. CASE: $\mathbf{v} \notin \{\pm\mathbf{a}, \pm\mathbf{b}, \pm\mathbf{c}\}$.
⟨3⟩1. $((\mathbf{v}, \mathbf{a}), (\mathbf{v}, \mathbf{b}))$ is one of $(0,0), (0,1), (0,-1), (1,0), (-1,0), (1,-1), (-1,1)$.
PROOF: By (1), both entries are in $\{-1, 0, 1\}$. Step ⟨1⟩1 excludes the two pairs $(1,1)$ and $(-1,-1)$. □
⟨3⟩2. We choose representatives by the following table.
PROOF: In each row $(\mathbf{v}, \mathbf{c}) = -(\mathbf{v}, \mathbf{a}) - (\mathbf{v}, \mathbf{b})$ by bilinearity; substitution gives the defining three inner products of the indicated set.

$((\mathbf{v}, \mathbf{a}), (\mathbf{v}, \mathbf{b}))$	$(0,0)$	$(0,1)$	$(0,-1)$	$(1,0)$	$(-1,0)$	$(1,-1)$	$(-1,1)$
representative	$\mathbf{v}$	$\mathbf{v}$	$-\mathbf{v}$	$-\mathbf{v}$	$\mathbf{v}$	$\mathbf{v}$	$-\mathbf{v}$
set	Δ	Γ_a	Γ_a	Γ_b	Γ_b	Γ_c	Γ_c

Each representative belongs to Φ by closure under negation. □
⟨3⟩3. Q.E.D.
PROOF: Apply ⟨3⟩2 to the exhaustive list ⟨3⟩1. □
⟨2⟩3. Q.E.D.
PROOF: The cases ⟨2⟩1 and ⟨2⟩2 are exhaustive. □
⟨1⟩3. Q.E.D.
PROOF: Step ⟨1⟩2 gives $\Phi \subseteq \Sigma \cup (-\Sigma)$. Conversely, the star identities and the definitions give $\Sigma \subseteq \Phi$, and $\Phi = -\Phi$ gives $-\Sigma \subseteq \Phi$. □

Lemma 2.6. ASSUME: Hypothesis 2.4 holds in the finite-dimensional real inner product space V.
PROVE:
1. The maps $\varphi_c(\mathbf{x}) = \mathbf{x} + \mathbf{c}$, $\varphi_a(\mathbf{y}) = \mathbf{y} + \mathbf{a}$, and $\varphi_b(\mathbf{z}) = \mathbf{z} + \mathbf{b}$ give inner-product-preserving bijections $\Gamma_a \to \Gamma_b \to \Gamma_c \to \Gamma_a$.
2. $\Gamma_b = \mathbf{c} + \Gamma_a$ and $\Gamma_c = -\mathbf{b} + \Gamma_a$.
3. For each $D \in \{\Gamma_a, \Gamma_b, \Gamma_c\}$ and distinct $\mathbf{x}, \mathbf{y} \in D$, $(\mathbf{x}, \mathbf{y}) \in \{0, 1\}$. Diagonal inner products are 2.

PROOF SKETCH: We will prove the translation identity for all three cyclic triples at once. Their composite is the identity, so each map is bijective. A negative inner product would give opposite vectors in two incompatible parts of the partition.
⟨1⟩1. ASSUME: $(\mathbf{r}, \mathbf{s}, \mathbf{t})$ is one of $(\mathbf{a}, \mathbf{b}, \mathbf{c}), (\mathbf{b}, \mathbf{c}, \mathbf{a}), (\mathbf{c}, \mathbf{a}, \mathbf{b})$, and $\mathbf{x} \in \Gamma_r$.
PROVE: $\mathbf{x} + \mathbf{t} \in \Gamma_s$.
⟨2⟩1. $\mathbf{x} + \mathbf{t} \in \Phi$.
PROOF: In each of the three listed triples, $(\mathbf{x}, \mathbf{t}) = -1$ by the definition of Γ_r; apply star-closure and the star identities. □
⟨2⟩2. $(\mathbf{x}+\mathbf{t}, \mathbf{r}) = -1$, $(\mathbf{x}+\mathbf{t}, \mathbf{s}) = 0$, and $(\mathbf{x}+\mathbf{t}, \mathbf{t}) = 1$.
PROOF: By bilinearity, the star identities and the defining products, these values are respectively $0-1$, $1-1$, and $-1+2$. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2 are the definition of Γ_s. □
⟨1⟩2. The three translations are bijections and preserve inner products.
⟨2⟩1. Each translation has the next two translations as a two-sided inverse.

PROOF: By ⟨1⟩1 all composites have the stated domains, and every cyclic composite sends $\mathbf{x}$ to $\mathbf{x}+\mathbf{a}+\mathbf{b}+\mathbf{c}=\mathbf{x}$. □

⟨2⟩2. ASSUME: $(\mathbf{r},\mathbf{s},\mathbf{t})$ is a listed cyclic triple and $\mathbf{x},\mathbf{y}\in\Gamma_r$.
PROVE: $(\mathbf{x}+\mathbf{t},\mathbf{y}+\mathbf{t})=(\mathbf{x},\mathbf{y})$.

PROOF: By bilinearity the left side is $(\mathbf{x},\mathbf{y})-1-1+2=(\mathbf{x},\mathbf{y})$, using the definition of Γ_r. □

⟨2⟩3. Q.E.D.

PROOF: Combine the two-sided inverses in ⟨2⟩1 with ⟨2⟩2. □

⟨1⟩3. ASSUME: $(\mathbf{r},\mathbf{s},\mathbf{t})$ is a listed cyclic triple and $\mathbf{x},\mathbf{y}\in\Gamma_r$ are distinct.
PROVE: $(\mathbf{x},\mathbf{y})\in\{0,1\}$.

⟨2⟩1. $(\mathbf{x},\mathbf{y})\neq 2,-2$.

PROOF: The value 2 contradicts distinctness by (1). The value -2 gives $\mathbf{y}=-\mathbf{x}$, contradicting $(\mathbf{x},\mathbf{s})=(\mathbf{y},\mathbf{s})=1$. □

⟨2⟩2. $(\mathbf{x},\mathbf{y})\neq -1$.

⟨3⟩1. ASSUME: $(\mathbf{x},\mathbf{y})=-1$.
PROVE: $\mathbf{x}=-(\mathbf{y}+\mathbf{t})$.

PROOF: Step ⟨1⟩1 puts $\mathbf{y}+\mathbf{t}$ in Φ, and bilinearity gives $(\mathbf{x},\mathbf{y}+\mathbf{t})=-1-1=-2$; apply (1). □

⟨3⟩2. Q.E.D.

PROOF: The equality in ⟨3⟩1 gives $0=(\mathbf{x},\mathbf{r})=-(\mathbf{y}+\mathbf{t},\mathbf{r})=1$, by the definition of Γ_r and $(\mathbf{t},\mathbf{r})=-1$ from the star identities, a contradiction. □

⟨2⟩3. Q.E.D.

PROOF: The line-system condition, ⟨2⟩1, and ⟨2⟩2 leave 0 and 1. □

⟨1⟩4. Q.E.D.

PROOF: Step ⟨1⟩2 proves part 1. Its first map gives $\Gamma_b=\mathbf{c}+\Gamma_a$, and its first two maps give $\Gamma_c=(\mathbf{a}+\mathbf{c})+\Gamma_a=-\mathbf{b}+\Gamma_a$. Step ⟨1⟩3 and the norm condition prove part 3. □

Lemma 2.7. ASSUME: Hypothesis 2.4 holds in the finite-dimensional real inner product space V.

PROVE:

$$\Delta=\{\mathbf{x}-\mathbf{y}:\mathbf{x},\mathbf{y}\in\Gamma_a,\ (\mathbf{x},\mathbf{y})=1\}.$$

Put

$$\mathcal{R}=\{\mathbf{a},\mathbf{b},\mathbf{c}\}\cup\Gamma_a\cup(\mathbf{c}+\Gamma_a)\cup(-\mathbf{b}+\Gamma_a)\cup\{\mathbf{x}-\mathbf{y}:\mathbf{x},\mathbf{y}\in\Gamma_a,\ (\mathbf{x},\mathbf{y})=1\}.$$

Then $\Sigma=\mathcal{R}$ and $\Phi=\mathcal{R}\cup(-\mathcal{R})$. In particular, any two instances of Hypothesis 2.4 in V with the same $\mathbf{a},\mathbf{b}$ and Γ_a have the same signed root set Φ.

PROOF SKETCH: We will recover Γ_b,Γ_c by translation and Δ by differences. A root of Δ orthogonal to all of Γ_a would belong to an orthogonal part of Φ; we will give that partition explicitly.

⟨1⟩1. ASSUME: $\mathbf{x},\mathbf{y}\in\Gamma_a$ and $(\mathbf{x},\mathbf{y})=1$.
PROVE: $\mathbf{x}-\mathbf{y}\in\Delta$.

⟨2⟩1. $\mathbf{x}-\mathbf{y}\in\Phi$.

PROOF: Apply star-closure to $\mathbf{x},-\mathbf{y}\in\Phi$, whose inner product is -1. □

⟨2⟩2. Q.E.D.

PROOF: The products of $\mathbf{x}-\mathbf{y}$ with $\mathbf{a},\mathbf{b},\mathbf{c}$ are $0-0,1-1,-1-(-1)$, respectively, by bilinearity. With ⟨2⟩1, this defines Δ. □

⟨1⟩2. ASSUME: $\mathbf{t}\in\Delta$ and $(\mathbf{t},\mathbf{x})\neq 0$ for some $\mathbf{x}\in\Gamma_a$.
PROVE: $\mathbf{t}=\mathbf{u}-\mathbf{v}$ for some $\mathbf{u},\mathbf{v}\in\Gamma_a$ with $(\mathbf{u},\mathbf{v})=1$.

LET: Choose $\mathbf{x}\in\Gamma_a$ with $(\mathbf{t},\mathbf{x})\neq 0$.

⟨2⟩1. $(\mathbf{t},\mathbf{x})\in\{-1,1\}$.

PROOF: If the product were ± 2, (1) would give $\mathbf{t} = \pm\mathbf{x}$, contradicting $(\mathbf{t}, \mathbf{b}) = 0$ and $(\mathbf{x}, \mathbf{b}) = 1$. The line-system condition and the nonzero assumption leave ± 1. □

⟨2⟩2. CASE: $(\mathbf{t}, \mathbf{x}) = -1$.

⟨3⟩1. $\mathbf{x} + \mathbf{t} \in \Gamma_a$.

PROOF: Star-closure gives membership in Φ; adding $\mathbf{t}$, which is orthogonal to $\mathbf{a}, \mathbf{b}, \mathbf{c}$, preserves the three products defining Γ_a. □

⟨3⟩2. Q.E.D.

PROOF: Choose $\mathbf{u} = \mathbf{x} + \mathbf{t}$, $\mathbf{v} = \mathbf{x}$. Their difference is $\mathbf{t}$, and $(\mathbf{u}, \mathbf{v}) = 2 - 1 = 1$ by bilinearity and ⟨2⟩. □

⟨2⟩3. CASE: $(\mathbf{t}, \mathbf{x}) = 1$.

⟨3⟩1. $\mathbf{x} - \mathbf{t} \in \Gamma_a$.

PROOF: Star-closure applies to $\mathbf{x}, -\mathbf{t}$; subtracting $\mathbf{t}$ preserves the three products defining Γ_a. □

⟨3⟩2. Q.E.D.

PROOF: Choose $\mathbf{u} = \mathbf{x}$, $\mathbf{v} = \mathbf{x} - \mathbf{t}$. Their difference is $\mathbf{t}$, and $(\mathbf{u}, \mathbf{v}) = 2 - 1 = 1$ by bilinearity and ⟨2⟩. □

⟨2⟩4. Q.E.D.

PROOF: Step ⟨2⟩1 makes ⟨2⟩2 and ⟨2⟩3 exhaustive. □

⟨1⟩3. No root of Δ is orthogonal to all of Γ_a.

LET: $P \triangleq \{\mathbf{t} \in \Delta : \forall \mathbf{x} \in \Gamma_a,\ (\mathbf{t}, \mathbf{x}) = 0\}$ and $Q \triangleq \Phi \setminus P$.

⟨2⟩1. $P = -P$ and $\mathbf{a} \in Q$.

PROOF: The defining products of P are all zero, and thus invariant under negation. Also $\mathbf{a} \in \Phi$ and $(\mathbf{a}, \mathbf{a}) = 2$, so $\mathbf{a} \notin \Delta$. □

⟨2⟩2. ASSUME: $\mathbf{t} \in P$ and $\mathbf{u} \in \Sigma \setminus P$.

PROVE: $(\mathbf{t}, \mathbf{u}) = 0$.

⟨3⟩1. CASE: $\mathbf{u} \in \{\mathbf{a}, \mathbf{b}, \mathbf{c}\} \cup \Gamma_a \cup \Gamma_b \cup \Gamma_c$.

PROOF: The definition of P gives orthogonality to $\mathbf{a}, \mathbf{b}, \mathbf{c}, \Gamma_a$. Lemma 2.6 gives $\Gamma_b = \mathbf{c} + \Gamma_a$ and $\Gamma_c = -\mathbf{b} + \Gamma_a$; bilinearity gives zero on each of these translates. □

⟨3⟩2. CASE: $\mathbf{u} \in \Delta \setminus P$.

⟨4⟩1. $\mathbf{u} = \mathbf{x} - \mathbf{y}$ for some $\mathbf{x}, \mathbf{y} \in \Gamma_a$.

PROOF: Membership in $\Delta \setminus P$ gives a nonzero product with a member of Γ_a; apply ⟨1⟩2. □

⟨4⟩2. Q.E.D.

PROOF: By the definition of P, $(\mathbf{t}, \mathbf{x} - \mathbf{y}) = 0 - 0 = 0$. □

⟨3⟩3. Q.E.D.

PROOF: The definition of Σ makes ⟨3⟩1 and ⟨3⟩2 exhaustive. □

⟨2⟩3. $(\mathbf{t}, \mathbf{u}) = 0$ for all $\mathbf{t} \in P$ and $\mathbf{u} \in Q$.

PROOF: Lemma 2.5 gives $\mathbf{u} = \epsilon\mathbf{v}$ with $\epsilon \in \{-1, 1\}$ and $\mathbf{v} \in \Sigma$. By ⟨2⟩1, $\mathbf{v} \notin P$. Step ⟨2⟩2 and bilinearity give $(\mathbf{t}, \mathbf{u}) = \epsilon \cdot 0 = 0$. □

⟨2⟩4. Q.E.D.

PROOF: We have $P \cup Q = \Phi$. Indecomposability and ⟨2⟩3 imply $P = \varnothing$ or $Q = \varnothing$; ⟨2⟩1 excludes the second alternative. □

⟨1⟩4. Q.E.D.

PROOF: Steps ⟨1⟩1, ⟨1⟩2, and ⟨1⟩3 give both inclusions in the formula for Δ. Substitute this equality and Lemma 2.6 into the definition of Σ to obtain $\Sigma = \mathcal{R}$; Lemma 2.5 gives $\Phi = \mathcal{R} \cup (-\mathcal{R})$. Equal $\mathbf{a}, \mathbf{b}, \Gamma_a$ give equal $\mathbf{c} = -\mathbf{a} - \mathbf{b}$ and equal $\mathcal{R}$, so they give equal root sets. □

Lemma 2.8. ASSUME: 1. Hypothesis 2.4 holds in the finite-dimensional real inner product space V.
2. $\mathbf{x}, \mathbf{y} \in \Gamma_a$ and $(\mathbf{x}, \mathbf{y}) = 0$.
PROVE: 1. $\mathbf{z} \triangleq \mathbf{b} - \mathbf{c} - \mathbf{x} - \mathbf{y}$ lies in Γ_a.
2. Each $\mathbf{w} \in \Gamma_a \setminus \{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is orthogonal to exactly one of $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$.

PROOF SKETCH: The vectors $\mathbf{x} + \mathbf{c}$ and $\mathbf{y} - \mathbf{b}$ lie in Φ and have inner product -1, so star-closure puts their sum, which is $-\mathbf{z}$, in Φ. Computing inner products places $\mathbf{z}$ in Γ_a. For part 2, $(\mathbf{w}, \mathbf{x}) + (\mathbf{w}, \mathbf{y}) + (\mathbf{w}, \mathbf{z}) = 2$, and each term is 0 or 1.
⟨1⟩1. $\mathbf{x} + \mathbf{c} \in \Gamma_b$ and $\mathbf{y} - \mathbf{b} \in \Gamma_c$.
PROOF: By ⟨0⟩:2 and Lemma 2.6. □
⟨1⟩2. $(\mathbf{x} + \mathbf{c}, \mathbf{y} - \mathbf{b}) = -1$.
PROOF: We use $(\mathbf{b}, \mathbf{c}) = -1$ from the star identities.

$$\begin{aligned}(\mathbf{x} + \mathbf{c}, \mathbf{y} - \mathbf{b}) &= (\mathbf{x}, \mathbf{y}) - (\mathbf{x}, \mathbf{b}) + (\mathbf{c}, \mathbf{y}) - (\mathbf{c}, \mathbf{b}) && \text{[bilinearity]}\\ &= 0 - 1 - 1 + 1 && \text{[⟨0⟩:2, the definition of } \Gamma_a\text{, and } (\mathbf{b}, \mathbf{c}) = -1]\\ &= -1.\end{aligned}$$

□
⟨1⟩3. $\mathbf{z} \in \Phi$.
PROOF: By ⟨1⟩1, ⟨1⟩2 and star-closure, $(\mathbf{x} + \mathbf{c}) + (\mathbf{y} - \mathbf{b}) = -\mathbf{z}$ lies in Φ; and $\Phi = -\Phi$. □
⟨1⟩4. $\mathbf{z} \in \Gamma_a$.
PROOF: We have $(\mathbf{z}, \mathbf{a}) = 0$, $(\mathbf{z}, \mathbf{b}) = 1$ and $(\mathbf{z}, \mathbf{c}) = -1$, by bilinearity, ⟨0⟩:2 and the definition of Γ_a; together with ⟨1⟩3, this is the definition of Γ_a. □
⟨1⟩5. ASSUME: $\mathbf{w} \in \Gamma_a \setminus \{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$.
PROVE: $\mathbf{w}$ is orthogonal to exactly one of $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$.
⟨2⟩1. $(\mathbf{w}, \mathbf{x}) + (\mathbf{w}, \mathbf{y}) + (\mathbf{w}, \mathbf{z}) = 2$.
PROOF:

$$\begin{aligned}(\mathbf{w}, \mathbf{x}) + (\mathbf{w}, \mathbf{y}) + (\mathbf{w}, \mathbf{z}) &= (\mathbf{w}, \mathbf{b} - \mathbf{c}) && \text{[the definition of } \mathbf{z}]\\ &= 1 + 1 && \text{[⟨1⟩ and the definition of } \Gamma_a]\end{aligned}$$

□
⟨2⟩2. Each of $(\mathbf{w}, \mathbf{x})$, $(\mathbf{w}, \mathbf{y})$, $(\mathbf{w}, \mathbf{z})$ is 0 or 1.
PROOF: By ⟨1⟩, ⟨0⟩:2, ⟨1⟩4 and Lemma 2.6. □
⟨2⟩3. Q.E.D.
PROOF: By ⟨2⟩1 and ⟨2⟩2, three numbers in $\{0, 1\}$ have sum 2, so exactly one of them is 0. □
⟨1⟩6. Q.E.D.
PROOF: By ⟨1⟩4 and ⟨1⟩5. □

Definition (the triangle property). ASSUME: X is a simple graph on a set U.
We say that X has the *triangle property* if every edge xy has exactly one common neighbour z, and every vertex outside $\{x, y, z\}$ is adjacent to exactly one of x, y, z. For adjacent x, y we will write $f(x, y)$ for this unique common neighbour; the graph will be clear from the argument. **[new]** We name this predicate once so that we can use the graph argument without a vector representation.

Corollary of Lemma 2.8. ASSUME: Hypothesis 2.4 holds in the finite-dimensional real inner product space V, and X is its orthogonality graph on Γ_a.
PROVE: X has the triangle property.

PROOF SKETCH: **[new]** We will record the pairwise orthogonality calculation needed to pass from Lemma 2.8 to graph terminology; it does not change that lemma.
⟨1⟩1. ASSUME: x, y are adjacent vertices of X.

PROVE: $\mathbf{z} \triangleq \mathbf{b} - \mathbf{c} - \mathbf{x} - \mathbf{y}$ represents a distinct common neighbour z.

⟨2⟩1. $(\mathbf{x}, \mathbf{z}) = (\mathbf{y}, \mathbf{z}) = 0$.

PROOF: Bilinearity gives $(\mathbf{x}, \mathbf{z}) = 1 - (-1) - 2 - 0 = 0$ and $(\mathbf{y}, \mathbf{z}) = 1 - (-1) - 0 - 2 = 0$, using adjacency and the definition of Γ_a. □

⟨2⟩2. Q.E.D.

PROOF: Lemma 2.8 puts $\mathbf{z}$ in Γ_a. Step ⟨2⟩1 and squared norm 2 give $z \neq x, y$ and $z \sim x, y$. □

⟨1⟩2. ASSUME: w is a common neighbour of adjacent x, y.

PROVE: $\mathbf{w} = \mathbf{b} - \mathbf{c} - \mathbf{x} - \mathbf{y}$.

PROOF: If w differed from the vertex in ⟨1⟩1, it would lie outside its triangle and be adjacent to two of its vertices, contrary to Lemma 2.8. □

⟨1⟩3. Q.E.D.

PROOF: Steps ⟨1⟩1 and ⟨1⟩2 give the unique common neighbour; Lemma 2.8 gives the assertion about every outside vertex. □

Lemma 2.9. ASSUME: 1. Hypothesis 2.4 holds in the finite-dimensional real inner product space V.

2. X is its orthogonality graph and $x \in V(X)$ is adjacent to every other vertex.

PROVE: There is an integer $m \geq 0$ and a partition of $V(X) \setminus \{x\}$ into m pairs, each of which makes a triangle with x. These are all the edges of X; thus X is the friendship graph with m triangles.

PROOF SKETCH: The unique triangle on each edge through x pairs the other vertices. We will use the outside-vertex condition to exclude edges between different pairs.

⟨1⟩1. For $y \neq x$, let $f(x, y)$ be the unique common neighbour of x, y. Then $f(x, y) \neq y, x$ and $f(x, f(x, y)) = y$.

⟨2⟩1. $f(x, y)$ exists and differs from x, y.

PROOF: Apply Corollary of Lemma 2.8 to the edge xy. □

⟨2⟩2. Q.E.D.

PROOF: The vertex y is a common neighbour of $x, f(x, y)$, so uniqueness in Corollary of Lemma 2.8 gives $f(x, f(x, y)) = y$. □

⟨1⟩2. The sets $\{y, f(x, y)\}$ partition $V(X) \setminus \{x\}$ into pairs.

PROOF: Step ⟨1⟩1 gives a fixed-point-free involution: its two-element orbits cover the domain, and intersecting orbits coincide by applying the involution. □

⟨1⟩3. There is no edge between two different pairs.

PROOF: A vertex in a different pair lies outside $\{x, y, f(x, y)\}$ and is adjacent to x. Corollary of Lemma 2.8 excludes adjacency to either y or $f(x, y)$. □

⟨1⟩4. Q.E.D.

PROOF: The finite partition in ⟨1⟩2 has some size $m \geq 0$. Step ⟨1⟩1 supplies the triangle edges, and ⟨1⟩3 excludes all others. When $V(X) = \{x\}$ the partition is empty and $m = 0$. □

Lemma (triangle-property normal forms). ASSUME: 1. X is a finite simple graph with the triangle property.

2. X has an edge.

3. No vertex of X is adjacent to every other vertex.

PROVE: For every pair of distinct nonadjacent vertices u, v, put

$$K = \Gamma_1(u) \cap \Gamma_1(v), \quad L = \Gamma_1(u) \setminus \Gamma_1(v),$$
$$M = \Gamma_1(v) \setminus \Gamma_1(u), \quad N = V(X) \setminus \big(\{u, v\} \cup K \cup L \cup M\big),$$

and put $k = |K|$. Then the following hold.

1. $k \in \{2, 3, 5\}$, X is regular of degree $2k$, $|V(X)| = 6k - 3$, and **[repair]** X is strongly regular with parameters $(6k - 3, 2k, 1, k)$ (Error 12 of `cgss/errata.md`).

2. The sets $\{u\}, \{v\}, K, L, M, N$ partition $V(X)$, with $|L| = |M| = k$ and $|N| = 3k-5$. The labels $x' = f(u,x)$ and $x'' = f(v,x)$ give bijections $K \to L$ and $K \to M$.
3. K, L, M are independent. For $x, y \in K$, $x \sim y'$ and $x \sim y''$ hold exactly when $x = y$, and $x' \sim y''$ holds exactly when $x \neq y$.
4. For $w \in N$, put $A(w) = K \setminus \Gamma_1(w)$ and $d_K(w) = |K \cap \Gamma_1(w)|$. Then $|A(w)| \in \{0, 2\}$, $d_K(w) \in \{k-2, k\}$, and $|N \cap \Gamma_1(w)| = d_K(w)$. For $x \in K$, $w \sim x'$ and $w \sim x''$ hold exactly when $x \in A(w)$.
5. The graph on N has $k(k-2)$ edges, each with its third vertex in K. Moreover,
$$\sum_{w \in N} d_K(w) = k(2k-4), \qquad 2|\{w \in N : d_K(w) = k\}| = (k-2)(5-k).$$
6. If $k = 2$, N consists of one vertex w, with $A(w) = K$, and has no edges.
7. If $k = 3$, the graph on N is a claw with centre w_0, where $A(w_0) = \varnothing$. Its leaves are w_x for $x \in K$, with $A(w_x) = K \setminus \{x\}$.
8. If $k = 5$, the map $w \mapsto A(w)$ is a bijection from N to the two-element subsets of K, and adjacency in N is disjointness of these labels.

These rules and the definitions of K, L, M, N specify every adjacency.

PROOF SKETCH: **[new]** We separate the combinatorial argument from its application to line systems. We will keep the paper's partition and prove regularity for arbitrary pairs. The invariant d_K gives both the bound on k and the three normal forms. All counting identities involving subtraction are read in $\mathbb{Z}$.

⟨1⟩1. Every edge xy has a unique common neighbour $f(x,y)$. Every vertex has positive degree and has a distinct non-neighbour.

⟨2⟩1. The triangle on xy is $\{x, y, f(x,y)\}$ and every outside vertex sees exactly one vertex of it.

PROOF: Apply the triangle property in ⟨0⟩:1. □

⟨2⟩2. Every vertex has positive degree.

PROOF: A triangle exists since X has an edge. Its vertices have degree at least 2, and every other vertex has a neighbour in it by ⟨2⟩1. □

⟨2⟩3. Every vertex has a distinct non-neighbour.

PROOF: This is precisely ⟨0⟩:3, the absence of a universal vertex. □

⟨2⟩4. Q.E.D.

PROOF: Combine ⟨2⟩1, ⟨2⟩2, and ⟨2⟩3. □

⟨1⟩2. For every pair of distinct nonadjacent vertices u, v, define, excluding u, v,
$$K = \Gamma_1(u) \cap \Gamma_1(v), \quad L = \Gamma_1(u) \setminus \Gamma_1(v),$$
$$M = \Gamma_1(v) \setminus \Gamma_1(u), \quad N = V(X) \setminus \big(\{u, v\} \cup K \cup L \cup M\big).$$
These four sets are disjoint. If $k = |K|$, then $|L| = |M| = k > 0$ and $\deg(u) = \deg(v) = 2k$. They admit labels $L = \{x' : x \in K\}$ and $M = \{x'' : x \in K\}$ with the following complete edge rules: K, L, M are independent; $x \sim y'$ and $x \sim y''$ exactly when $x = y$; $x' \sim y''$ exactly when $x \neq y$. For $w \in N$, $w \sim x'$ and $w \sim x''$ hold exactly when $w \not\sim x$.

⟨2⟩1. For $p \in \{u, v\}$ and the other vertex q, the triangles through p partition $\Gamma_1(p)$ into pairs; q sees exactly one vertex in each pair.

⟨3⟩1. On $\Gamma_1(p)$, the map $y \mapsto f(p,y)$ is a fixed-point-free involution, and these pairs are all the edges within $\Gamma_1(p)$.

PROOF: The proof of ⟨1⟩1 gives existence and uniqueness of the third vertex. Thus $f(p, f(p,y)) = y$. If two neighbours of p are adjacent, uniqueness of the common neighbour on an edge through p makes them one of these pairs. □

⟨3⟩2. Q.E.D.

PROOF: Since $p \not\sim q$, the triangle property forces q to see exactly one of the two other vertices of each triangle through p. □

⟨2⟩2. For $x \in K$, put $x' = f(u,x)$ and $x'' = f(v,x)$. These define bijections $K \to L$ and $K \to M$, and give the stated rules inside $K \cup L$ and $K \cup M$.

⟨3⟩1. Each pair through u consists of one point of K and one of L; each pair through v consists of one point of K and one of M.

PROOF: Apply ⟨2⟩1 with $(p,q) = (u,v)$ and (v,u), respectively. □

⟨3⟩2. Q.E.D.

PROOF: Step ⟨3⟩1 gives the bijections. Since ⟨2⟩1 lists every edge in either neighbourhood, each part is independent and the only edges between its two parts are the labelled pairs. □

⟨2⟩3. $x' \sim y''$ if and only if $x \neq y$.

⟨3⟩1. For $x = y$, $x' \not\sim x''$.

PROOF: The vertex x'' sees x and does not see u. Apply the triangle property to $\{u, x, x'\}$ to exclude $x'' \sim x'$. □

⟨3⟩2. For $x \neq y$, $x' \sim y''$.

PROOF: The vertex x' sees neither v nor y, by its definition and ⟨2⟩2. Applied to $\{v, y, y''\}$, the triangle property forces $x' \sim y''$. □

⟨3⟩3. Q.E.D.

PROOF: The cases ⟨3⟩1 and ⟨3⟩2 are exhaustive. □

⟨2⟩4. For $w \in N$ and $x \in K$, the adjacencies to (x, x', x'') are either $(1,0,0)$ or $(0,1,1)$.

PROOF: The vertex w sees neither u nor v. It must see one vertex in each of $\{u, x, x'\}$ and $\{v, x, x''\}$ by the triangle property. □

⟨2⟩5. Q.E.D.

PROOF: The sets partition $V(X) \setminus \{u, v\}$ by their two adjacency indicators. Step ⟨2⟩2 gives the equal cardinalities and degrees; positivity follows from ⟨1⟩1. The remaining assertions are ⟨2⟩2, ⟨2⟩3, and ⟨2⟩4. □

⟨1⟩3. **[repair]** X is regular. We fill the paper's missing propagation from nonadjacent pairs to arbitrary pairs (Gap 3 of `cgss/errata.md`).

⟨2⟩1. Distinct nonadjacent vertices have equal degrees.

PROOF: This is the degree equality in ⟨1⟩2, for an arbitrary pair. □

⟨2⟩2. ASSUME: $p \sim q$.

PROVE: $\deg(p) = \deg(q)$.

LET: $r = f(p,q)$. Choose $s \neq p$ nonadjacent to p, using ⟨1⟩1.

⟨3⟩1. The vertex s is outside $\{p, q, r\}$ and sees exactly one of q, r.

PROOF: Both q, r see p, so $s \neq q, r$; apply the triangle property and $s \not\sim p$. □

⟨3⟩2. CASE: $s \sim r$.

PROOF: Step ⟨3⟩1 gives $s \not\sim q$. Thus $\deg(p) = \deg(s) = \deg(q)$ by ⟨2⟩1. □

⟨3⟩3. CASE: $s \sim q$.

⟨4⟩1. $\deg(p) = \deg(r)$.

PROOF: Step ⟨3⟩1 gives $s \not\sim r$; apply ⟨2⟩1 to p, s and s, r. □

LET: Choose $t \neq q$ nonadjacent to q, using ⟨1⟩1.

⟨4⟩2. The vertex t is outside $\{p, q, r\}$ and sees exactly one of p, r.

PROOF: Since p, r see q, the vertex t differs from both. The triangle property gives one neighbour in $\{p, q, r\}$; it cannot be q. □

⟨4⟩3. $\deg(q)$ equals $\deg(r)$ or $\deg(p)$.

PROOF: If $t \sim p$, then $t \not\sim r$, so $\deg(q) = \deg(t) = \deg(r)$. If $t \sim r$, then $t \not\sim p$, so $\deg(q) = \deg(t) = \deg(p)$. Both equalities use ⟨2⟩1; ⟨4⟩2 exhausts the cases. □

⟨4⟩4. Q.E.D.

PROOF: Use ⟨4⟩1 in either alternative of ⟨4⟩3. □

⟨3⟩4. Q.E.D.

PROOF: Step ⟨3⟩1 makes ⟨3⟩2 and ⟨3⟩3 exhaustive. □

⟨2⟩3. Q.E.D.

PROOF: Identical vertices have equal degrees; distinct vertices are covered by ⟨2⟩1 and ⟨2⟩2. □

LET: Fix arbitrary distinct nonadjacent u, v and retain the sets and labels of ⟨1⟩2. Put $k = |K|$. By ⟨1⟩3, every vertex has degree $2k$.

⟨1⟩4. $k \geq 2$.

⟨2⟩1. If $k = 1$, then X is a single triangle.

PROOF: By ⟨1⟩3, every vertex would have degree 2. A triangle exists by ⟨1⟩1; its vertices already have two neighbours inside it, whereas every outside vertex would have a neighbour in it. Thus there is no outside vertex. □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨1⟩2 gives $k > 0$; ⟨2⟩1 contradicts the chosen nonadjacent pair when $k = 1$. □

⟨1⟩5. $|N| = 3k - 5$, $|V(X)| = 6k - 3$, and X has strongly regular parameters $(6k - 3, 2k, 1, k)$. The graph on N has $k(k - 2)$ edges, and the third vertex of every such edge lies in K.

⟨2⟩1. There are $4k^2 - 6k$ edges between $K \cup L \cup M$ and N.

⟨3⟩1. A vertex in K has $2k - 4$ neighbours in N; a vertex in L or M has $k - 1$ neighbours in N.

PROOF: By ⟨1⟩2, $x \in K$ has the four neighbours u, v, x', x'' outside N. A vertex $x' \in L$ has u, x and the $k - 1$ vertices y'' with $y \neq x$ outside N; a vertex $x'' \in M$ has v, x and the $k - 1$ vertices y' with $y \neq x$. Subtract these counts from degree $2k$. □

⟨3⟩2. Q.E.D.

PROOF: Sum ⟨3⟩1 over the three parts, each of size k: $k(2k - 4) + k(k - 1) + k(k - 1) = 4k^2 - 6k$. □

⟨2⟩2. The subgraph on N has $k(k - 2)$ edges, each with its third vertex in K.

⟨3⟩1. If $w, z \in N$ are adjacent, then $f(w, z) \in K$.

⟨4⟩1. Both u and v lie outside $\{w, z, f(w, z)\}$.

PROOF: They differ from w, z by the definition of N. The vertex $f(w, z)$ is adjacent to w, whereas neither u nor v is; thus they also differ from $f(w, z)$. □

⟨4⟩2. Q.E.D.

PROOF: By ⟨4⟩1, the triangle property applies to u and v. Each sees neither w nor z, so each sees $f(w, z)$. This is membership in K. □

⟨3⟩2. Each $x \in K$ is the third vertex of exactly $k - 2$ edges in N.

PROOF: In the triangle pairing of the $2k$ neighbours of x, the pairs $\{u, x'\}$ and $\{v, x''\}$ use its four neighbours outside N. The remaining $2k - 4$ neighbours pair within N, giving $k - 2$ edges. □

⟨3⟩3. Q.E.D.

PROOF: Step ⟨3⟩1 covers every edge in N; uniqueness of its third vertex prevents double counting. Sum ⟨3⟩2 over $|K| = k$. □

⟨2⟩3. $|N| = 3k - 5$ and $|V(X)| = 6k - 3$.

PROOF: Counting the degree sum over N, using ⟨2⟩1 and ⟨2⟩2, gives

$$\begin{aligned} 2k|N| &= 2k(k-2) + (4k^2 - 6k) && \text{[internal edges count twice]} \\ &= 2k(3k-5) && \text{[ring arithmetic]}, \\ |N| &= 3k - 5 && [2k > 0], \\ |V(X)| &= 2 + 3k + (3k-5) = 6k - 3 && \text{[the partition]}. \end{aligned}$$

□

⟨2⟩4. Q.E.D.

PROOF: Step ⟨2⟩3 gives the order and ⟨1⟩3 the degree. Step ⟨1⟩1 gives one common neighbour on each edge. For any nonadjacent pair, ⟨1⟩2 gives half its degree as the number of common neighbours, namely k. These are the four strongly regular parameters. □

⟨1⟩6. **[repair]** For $w \in N$, put $A(w) = K \setminus \Gamma_1(w)$ and $d_K(w) = |K \cap \Gamma_1(w)|$. Then $|A(w)| \in \{0, 2\}$, $d_K(w) \in \{k-2, k\}$, and $|N \cap \Gamma_1(w)| = d_K(w)$. We replace the unsupported pigeonhole argument by this invariant (Gap 1 of `cgss/errata.md`).

⟨2⟩1. In the triangle pairing at w, its neighbours in K pair with its neighbours in N. Its neighbours in L pair with its neighbours in M.

⟨3⟩1. If $w \sim x \in K$, then $f(w, x) \in N$.

PROOF: Each of u, v is adjacent to x and not to w. The triangle property excludes adjacency to $f(w, x)$. Also $f(w, x) \neq u, v$, since both are nonadjacent to w. The definition of N applies. □

⟨3⟩2. If $w \sim z \in N$, then $f(w, z) \in K$.

PROOF: Apply the third-vertex assertion of ⟨1⟩5 to the edge wz. □

⟨3⟩3. The partner of a neighbour of w in $L \cup M$ lies in $L \cup M$.

LET: Let $y \in (L \cup M) \cap \Gamma_1(w)$ and put $z = f(w, y)$.

⟨4⟩1. $f(w, z) = y$.

PROOF: The vertex y is a common neighbour of w, z; uniqueness in ⟨1⟩1 gives the equality. □

⟨4⟩2. $z \notin \{u, v\}$.

PROOF: The vertex z is adjacent to w, whereas u, v are not. □

⟨4⟩3. $z \notin K$.

PROOF: If $z \in K$, ⟨3⟩1 gives $f(w, z) \in N$, contrary to ⟨4⟩1 and $y \in L \cup M$. □

⟨4⟩4. $z \notin N$.

PROOF: If $z \in N$, ⟨3⟩2 gives $f(w, z) \in K$, contrary to ⟨4⟩1 and $y \in L \cup M$. □

⟨4⟩5. Q.E.D.

PROOF: The partition in ⟨1⟩2 and ⟨4⟩2, ⟨4⟩3, ⟨4⟩4 leave only $L \cup M$. □

⟨3⟩4. Q.E.D.

PROOF: The involution $y \mapsto f(w, y)$ pairs the neighbours of w. Steps ⟨3⟩1 and ⟨3⟩2 pair all its K and N neighbours. Step ⟨3⟩3 keeps every remaining pair in $L \cup M$. Neither part has internal edges by ⟨1⟩2, so each such pair has one vertex in each part. □

⟨2⟩2. If $A(w) \neq \varnothing$, then $|A(w)| = 2$.

LET: Choose $x \in A(w)$.

⟨3⟩1. The common neighbours of w, x' are exactly $\{y'' : y \in A(w),\ y \neq x\}$.

PROOF: The signature in ⟨1⟩2 gives $w \sim x'$. By ⟨2⟩1 their unique common neighbour lies in M. For $y \in K$, we have $w \sim y''$ exactly when $y \in A(w)$, and $x' \sim y''$ exactly when $y \neq x$, by ⟨1⟩2. Thus the displayed set lists all, and only, common neighbours. □

⟨3⟩2. Q.E.D.

PROOF: An edge has exactly one common neighbour by ⟨1⟩1. The bijection $y \mapsto y''$ in ⟨1⟩2 and ⟨3⟩1 give $|A(w)| - 1 = 1$. □

⟨2⟩3. Q.E.D.

PROOF: Either $A(w)$ is empty or ⟨2⟩2 applies. Since $d_K(w) = k - |A(w)|$, its two possible values follow. The K, N pairing in ⟨2⟩1 gives the asserted equality of degrees. □

⟨1⟩7. $\sum_{w \in N} d_K(w) = k(2k-4)$.

PROOF: Each $x \in K$ has $2k-4$ neighbours in N: its four neighbours outside N are u, v, x', x'' by ⟨1⟩2, and its degree is $2k$. Count edges between K, N from the two ends. □

LET: $h \triangleq |\{w \in N : d_K(w) = k\}|$.

⟨1⟩8. $2h = (k-2)(5-k)$.

PROOF: By ⟨1⟩6 and ⟨1⟩5, the sum in ⟨1⟩7 is $(|N| - h)(k-2) + hk = (3k-5)(k-2) + 2h$. Equating the two expressions and subtracting gives the formula. □

⟨1⟩9. **[new]** $2 \leq k \leq 5$. We obtain the upper bound without a Gram matrix, so the result applies to an arbitrary graph with the triangle property.

⟨2⟩1. If $k > 5$, then $(k-2)(5-k) < 0$.

PROOF: We have $k - 2 > 0$ and $5 - k < 0$; multiply these inequalities. □

⟨2⟩2. Q.E.D.

PROOF: The cardinality h is nonnegative. Thus ⟨1⟩8 contradicts ⟨2⟩1 if $k > 5$. The lower bound is ⟨1⟩4. □

⟨1⟩10. **[repair]** We have $k \in \{2, 3, 5\}$ (Gap 1).

⟨2⟩1. If $h > 0$, then $k = 3$.

LET: Choose $w_0 \in N$ with $d_K(w_0) = k$.

⟨3⟩1. There is $z \in N$ adjacent to w_0.

PROOF: Steps ⟨1⟩6 and ⟨1⟩4 give at least two neighbours of w_0 in N. □

LET: Choose such a z and put $x = f(w_0, z)$.

⟨3⟩2. $K \cap \Gamma_1(z) = \{x\}$.

PROOF: Step ⟨1⟩5 puts x in K. If $y \in K \setminus \{x\}$, then $y \sim w_0$ and $y \not\sim x$, since K is independent. The triangle property for $\{w_0, z, x\}$ forces $y \not\sim z$. □

⟨3⟩3. Q.E.D.

PROOF: Step ⟨3⟩2 gives $d_K(z) = 1$. By ⟨1⟩6, $1 = k - 2$ or $1 = k$; ⟨1⟩4 excludes the second, so $k = 3$. □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨1⟩9 leaves $k = 2, 3, 4, 5$. At $k = 4$, ⟨1⟩8 gives $h = 1$, whereas ⟨2⟩1 gives $k = 3$. Thus $k = 4$ is impossible. □

⟨1⟩11. **[repair]** (Gap 2.) If $k = 2$, the labelled graph is uniquely determined: N consists of one vertex w, with $A(w) = K$, and has no edges.

⟨2⟩1. $|N| = 1$ and $d_K(w) = 0$.

PROOF: Step ⟨1⟩5 gives $|N| = 1$. A simple graph on one vertex has degree 0, so ⟨1⟩6 gives $d_K(w) = 0$. □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨2⟩1 gives $A(w) = K$. The signature in ⟨1⟩2 makes w adjacent to every vertex of $L \cup M$ and to none of $K \cup \{u, v\}$. The remaining edges are all listed in ⟨1⟩2. Thus every edge is determined by labels on the two elements of K. □

⟨1⟩12. **[repair]** (Gap 2.) If $k = 3$, the subgraph on N is a claw. Its centre sees all of K and its three leaves correspond bijectively to K by their unique neighbour there. These labels determine the whole graph.

⟨2⟩1. There is exactly one $w_0 \in N$ with $d_K(w_0) = 3$; the other three vertices have $d_K = 1$.

PROOF: Step ⟨1⟩8 gives $2h = (3-2)(5-3) = 2$. Thus $h = 1$; ⟨1⟩5 gives $|N| = 4$, and ⟨1⟩6 gives the values 1 or 3. □

⟨2⟩2. The vertex w_0 is adjacent to all three other vertices of N, and they are pairwise nonadjacent.

PROOF: By ⟨1⟩6, the degrees inside N are 3 for w_0 and 1 for the others. Its three edges use the entire degree of each of the other vertices. □

⟨2⟩3. For each $x \in K$ there is exactly one leaf w_x with $K \cap \Gamma_1(w_x) = \{x\}$.

PROOF: By ⟨1⟩2 and ⟨1⟩3, the vertex x has $2k - 4 = 2$ neighbours in N, of which one is w_0. Its other neighbour is a leaf. Conversely, each leaf has exactly one neighbour in K by ⟨2⟩1. Thus this incidence is a bijection. □

⟨2⟩4. Q.E.D.

PROOF: Steps ⟨2⟩1, ⟨2⟩2, and ⟨2⟩3 determine N and its incidences with K. The signature in ⟨1⟩2 determines its incidences with L, M; the same step already lists all other edges. □

⟨1⟩13. **[repair]** (Gap 2.) If $k = 5$, the map $w \mapsto A(w) = K \setminus \Gamma_1(w)$ is a bijection from N to the two-element subsets of K. Two vertices of N are adjacent exactly when their labels are disjoint. These labels determine the whole graph.

⟨2⟩1. $d_K(w) = 3$ and $|A(w)| = 2$ for every $w \in N$.

PROOF: Step ⟨1⟩5 gives $|N| = 10$ and ⟨1⟩7 gives $\sum_{w \in N} d_K(w) = 5(10 - 4) = 30$. Each summand is 3 or 5 by ⟨1⟩6; the minimum total is already 30, so every summand is 3. □

⟨2⟩2. For distinct $x, y \in K$, there is exactly one $w \in N$ with $A(w) = \{x, y\}$.

⟨3⟩1. x' and y'' are adjacent and their third vertex lies in N.

⟨4⟩1. They have no common neighbour in $\{u, v\} \cup K \cup L \cup M$.

PROOF: The vertex u does not see y'', and v does not see x'. Their respective neighbours in K are the distinct vertices x, y. The set L is independent and contains x'; the set M is independent and contains y''. These observations exclude each of the five parts. □

⟨4⟩2. Q.E.D.

PROOF: Step ⟨1⟩2 gives $x' \sim y''$. Their unique common neighbour exists by ⟨1⟩1; ⟨4⟩1 and the partition put it in N. □

⟨3⟩2. For $w = f(x', y'')$, $A(w) = \{x, y\}$.

PROOF: Since w sees x', y'', the signature in ⟨1⟩2 gives $x, y \in A(w)$. Step ⟨2⟩1 gives $|A(w)| = 2$, so these are all its elements. □

⟨3⟩3. Any $z \in N$ with $A(z) = \{x, y\}$ equals w.

PROOF: The signature makes z a common neighbour of x', y''; uniqueness of the third vertex gives $z = f(x', y'') = w$. □

⟨3⟩4. Q.E.D.

PROOF: Steps ⟨3⟩2 and ⟨3⟩3 give existence and uniqueness. □

⟨2⟩3. For $w, z \in N$, $w \sim z$ if and only if $A(w) \cap A(z) = \varnothing$.

⟨3⟩1. If $w \sim z$, then $A(w) \cap A(z) = \varnothing$.

PROOF: Their third vertex x is in K by ⟨1⟩5. Every $y \in K \setminus \{x\}$ is nonadjacent to x, so the triangle property makes it adjacent to exactly one of w, z; x is adjacent to both. Thus no element of K is missed by both. □

⟨3⟩2. For fixed w, there are exactly three $z \in N$ whose labels are disjoint from $A(w)$, and exactly three neighbours of w in N.

PROOF: The complement of $A(w)$ in K has three elements, and thus three two-element subsets. Step ⟨2⟩2 gives exactly one vertex for each. Steps ⟨1⟩6 and ⟨2⟩1 give degree 3 within N. □

⟨3⟩3. Q.E.D.

PROOF: Step ⟨3⟩1 gives inclusion of the neighbour set in the disjoint-label set; ⟨3⟩2 gives equal finite cardinalities, so the two sets coincide. □

⟨2⟩4. Q.E.D.

PROOF: Steps ⟨2⟩1 and ⟨2⟩2 give the bijection, and ⟨2⟩3 gives the graph on N, the two-subset model of the Petersen graph. The labels give all incidences with K, their complements give those with L, M by ⟨1⟩2, and there are no incidences with u, v. □

⟨1⟩14. Q.E.D.

PROOF: Steps ⟨1⟩2, ⟨1⟩3, and ⟨1⟩5 give the partition, edge rules, regularity and parameters. Steps ⟨1⟩6, ⟨1⟩7, and ⟨1⟩8 give the degree identities. Step ⟨1⟩10 leaves exactly the three cases proved in ⟨1⟩11, ⟨1⟩12, and ⟨1⟩13. The pair u, v was arbitrary. □

Lemma (isomorphism from the order). ASSUME: 1. X, Y are finite simple graphs with the triangle property, and each has an edge.

2. Neither graph has a vertex adjacent to every other vertex.
3. $|V(X)| = |V(Y)|$.

PROVE: There is a graph isomorphism $X \to Y$.

PROOF SKETCH: **[repair]** We will apply the normal-form lemma separately to the two graphs and match their labels. This completes the uniqueness argument of Gap 2.

⟨1⟩1. Choose distinct nonadjacent $u, v \in V(X)$ and $\widetilde{u}, \widetilde{v} \in V(Y)$.
PROOF: Each graph has a vertex since it has an edge. By ⟨0⟩:2, that vertex has a distinct non-neighbour, in each graph. □

⟨1⟩2. The normal-form lemma applies to X, u, v; write $k_X = |K|$. In particular $|V(X)| = 6k_X - 3$.
PROOF: Apply Lemma (triangle-property normal forms) to the hypotheses on X and the first pair in ⟨1⟩1. □

⟨1⟩3. The normal-form lemma applies to $Y, \widetilde{u}, \widetilde{v}$; write $k_Y = |\widetilde{K}|$. In particular $|V(Y)| = 6k_Y - 3$.
PROOF: Apply Lemma (triangle-property normal forms) to the hypotheses on Y and the second pair in ⟨1⟩1. □

⟨1⟩4. $k_X = k_Y$.
PROOF: By ⟨0⟩:3, ⟨1⟩2 and ⟨1⟩3, $6k_X - 3 = 6k_Y - 3$; cancellation gives the equality. □

LET: Write k for this common value, use tildes for the parts of Y, and choose a bijection $\beta : K \to \widetilde{K}$, using ⟨1⟩4.

⟨1⟩5. **[repair]** The bijection β extends to a graph isomorphism (Gap 2).
⟨2⟩1. Extend β by $u \mapsto \widetilde{u}$, $v \mapsto \widetilde{v}$, $x' \mapsto \beta(x)'$, $x'' \mapsto \beta(x)''$, and the following rule on N: send the single vertex to the single vertex when $k = 2$; send centre to centre and w_x to $\widetilde{w}_{\beta(x)}$ when $k = 3$; send label A to label $\beta(A)$ when $k = 5$. The resulting map ψ is a bijection.
⟨3⟩1. The prescribed maps on $\{u\}, \{v\}, K, L, M$ are bijections onto the corresponding parts of Y.
PROOF: The label maps from K to L, M are bijections in both ⟨1⟩2 and ⟨1⟩3; compose these with β. □
⟨3⟩2. When $k = 2$, the map on N is a bijection.
PROOF: Both sets are singletons by part 6 of the normal-form lemma. □
⟨3⟩3. When $k = 3$, the map on N is a bijection.
PROOF: Part 7 of the normal-form lemma gives one centre and a bijective leaf labelling by K in each graph; β matches the labels. □
⟨3⟩4. When $k = 5$, the map on N is a bijection.
PROOF: Part 8 of the normal-form lemma identifies both sets with two-element subsets, and $A \mapsto \beta(A)$ is a bijection on these subsets. □
⟨3⟩5. Q.E.D.
PROOF: The parts partition both vertex sets by the normal-form lemma. Step ⟨3⟩1 and the exhaustive cases ⟨3⟩2, ⟨3⟩3, ⟨3⟩4 give a bijection on their disjoint unions. □
⟨2⟩2. The map ψ preserves adjacency and nonadjacency.
⟨3⟩1. It preserves all pairs with neither endpoint in N.
⟨4⟩1. The pair u, v is a nonedge in each graph.
PROOF: This is the choice in ⟨1⟩1. □
⟨4⟩2. Every pair u, x with $x \in K$ is an edge in each graph.
PROOF: This is the definition of K. □
⟨4⟩3. Every pair u, x' with $x' \in L$ is an edge in each graph.
PROOF: This is the definition of L. □
⟨4⟩4. Every pair u, x'' with $x'' \in M$ is a nonedge in each graph.
PROOF: This is the definition of M. □
⟨4⟩5. Every pair v, x with $x \in K$ is an edge in each graph.

PROOF: This is the definition of K. □

⟨4⟩6. Every pair v, x' with $x' \in L$ is a nonedge in each graph.
PROOF: This is the definition of L. □

⟨4⟩7. Every pair v, x'' with $x'' \in M$ is an edge in each graph.
PROOF: This is the definition of M. □

⟨4⟩8. Pairs within any one part $D \in \{K, L, M\}$ are nonedges in each graph.
PROOF: Each such D is independent by part 3 of the normal-form lemma. □

⟨4⟩9. The relation $x \sim y'$ is preserved in both directions.
PROOF: In each graph it is $x = y$, by part 3 of the normal-form lemma; β preserves and reflects equality. □

⟨4⟩10. The relation $x \sim y''$ is preserved in both directions.
PROOF: In each graph it is $x = y$, by part 3 of the normal-form lemma; β preserves and reflects equality. □

⟨4⟩11. The relation $x' \sim y''$ is preserved in both directions.
PROOF: In each graph it is $x \neq y$, by part 3 of the normal-form lemma; β preserves and reflects inequality. □

⟨4⟩12. Q.E.D.
PROOF: The listed pair types exhaust the five parts $\{u\}, \{v\}, K, L, M$. Loops are nonedges because the graphs are simple. □

⟨3⟩2. It preserves all pairs with exactly one endpoint in N.

⟨4⟩1. Vertices of N are adjacent to neither u nor v in either graph.
PROOF: This is the definition of N. □

⟨4⟩2. For $k = 2$, $\beta(A(w)) = \widetilde{A}(\psi(w))$.
PROOF: Both missed sets are the whole of K or $\widetilde{K}$, respectively, by part 6 of the normal-form lemma. □

⟨4⟩3. For $k = 3$, $\beta(A(w)) = \widetilde{A}(\psi(w))$.

⟨5⟩1. The equality holds at the centre.
PROOF: Both missed sets are empty by part 7 of the normal-form lemma. □

⟨5⟩2. The equality holds at each leaf w_x.
PROOF: Its missed set is $K \setminus \{x\}$ and its image has missed set $\widetilde{K} \setminus \{\beta(x)\}$, by part 7 of the normal-form lemma. □

⟨5⟩3. Q.E.D.
PROOF: A claw consists of its centre and leaves; apply ⟨5⟩1 and ⟨5⟩2. □

⟨4⟩4. For $k = 5$, $\beta(A(w)) = \widetilde{A}(\psi(w))$.
PROOF: This is the definition of ψ on the two-subset labels in ⟨2⟩1. □

⟨4⟩5. Pairs between N and K are preserved in both directions.
PROOF: Adjacency is $x \notin A(w)$ by definition. Steps ⟨4⟩2, ⟨4⟩3, and ⟨4⟩4 preserve this condition in every possible case. □

⟨4⟩6. Pairs between N and L are preserved in both directions.
PROOF: The relation $w \sim x'$ is $x \in A(w)$ by part 4 of the normal-form lemma, preserved by the three missed-set equalities above. □

⟨4⟩7. Pairs between N and M are preserved in both directions.
PROOF: The relation $w \sim x''$ is $x \in A(w)$ by part 4 of the normal-form lemma, preserved by the same equalities. □

⟨4⟩8. Q.E.D.
PROOF: Steps ⟨4⟩1, ⟨4⟩5, ⟨4⟩6, and ⟨4⟩7 exhaust the other endpoint. □

⟨3⟩3. It preserves all pairs with both endpoints in N.

⟨4⟩1. When $k = 2$, there are no edges in either N.

PROOF: Both sets are singletons by part 6 of the normal-form lemma. □

⟨4⟩2. When $k = 3$, adjacency within N is preserved in both directions.

PROOF: Part 7 of the normal-form lemma says that the edges join the centre to a leaf, and ⟨2⟩1 preserves these roles. □

⟨4⟩3. When $k = 5$, adjacency within N is preserved in both directions.

PROOF: Part 8 of the normal-form lemma says that adjacency is disjointness of labels; $\beta(A) \cap \beta(B) = \beta(A \cap B)$ since β is injective. □

⟨4⟩4. Q.E.D.

PROOF: The normal-form lemma gives $k \in \{2, 3, 5\}$, so the three preceding cases are exhaustive. □

⟨3⟩4. Q.E.D.

PROOF: Steps ⟨3⟩1, ⟨3⟩2, and ⟨3⟩3 exhaust all pairs. □

⟨2⟩3. Q.E.D.

PROOF: The bijection ⟨2⟩1 and the equivalence of adjacency in ⟨2⟩2 define a graph isomorphism. □

⟨1⟩6. Q.E.D.

PROOF: Step ⟨1⟩5 gives the required isomorphism. □

Lemma 2.10. ASSUME: 1. Hypothesis 2.4 holds in the finite-dimensional real inner product space V, and X is its orthogonality graph on Γ_a.

2. X has an edge, is noncomplete, and no vertex is adjacent to every other vertex.

PROVE: $|\Gamma_a| \in \{9, 15, 27\}$, and the graph is unique up to isomorphism in each case. More explicitly, if another instance of Hypothesis 2.4 in a finite-dimensional real inner product space W satisfies the same graph hypotheses and has the same value of $|\Gamma_a|$, its orthogonality graph is isomorphic to X. Moreover, **[repair]** X is strongly regular with parameters $(6k - 3, 2k, 1, k)$ for some $k \in \{2, 3, 5\}$ (Error 12).

PROOF SKETCH: The two graph lemmas give the conclusion. We will also retain the paper's Gram-matrix proof of the upper bound on k, including computation C1. By Lemma 2.9, having a universal vertex is equivalent here to being a friendship graph. Thus our precise hypothesis matches the paper's exclusion of that family; we retain its noncompleteness hypothesis as well.

⟨1⟩1. For any distinct nonadjacent u, v and $k = |\Gamma_1(u) \cap \Gamma_1(v)|$, the normal-form conclusions hold. In particular $|\Gamma_a| \in \{9, 15, 27\}$ and X has the stated strongly regular parameters.

PROOF: Corollary of Lemma 2.8 supplies the triangle property. Apply Lemma (triangle-property normal forms) and substitute $k = 2, 3, 5$ in $6k - 3$. □

⟨1⟩2. Any second instance with the same graph hypotheses and order has an orthogonality graph isomorphic to X.

PROOF: Corollary of Lemma 2.8 gives the triangle property for each instance. Apply Lemma (isomorphism from the order) to the two graphs. □

⟨1⟩3. The paper's Gram-matrix argument gives a second proof of $k \leq 5$.

LET: Choose distinct nonadjacent u, v and use the partition and labels of Lemma (triangle-property normal forms). Write $\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{x}', \mathbf{x}''$ for their representing vectors and put $k = |K| \geq 2$, by ⟨1⟩1.

⟨2⟩1. The following are Gram matrices, first for the family

$$\mathbf{u}, \mathbf{v}, (\mathbf{x})_{x \in K}, (\mathbf{x}')_{x \in K}, (\mathbf{x}'')_{x \in K},$$

then for its five block sums

$$\mathbf{u}, \mathbf{v}, \sum_{x \in K} \mathbf{x}, \sum_{x \in K} \mathbf{x}', \sum_{x \in K} \mathbf{x}''.$$

$$B_0 = \begin{pmatrix} 2 & 1 & 0^T & 0^T & \mathbf{j}^T \\ 1 & 2 & 0^T & \mathbf{j}^T & 0^T \\ 0 & 0 & I+J & J-I & J-I \\ 0 & \mathbf{j} & J-I & I+J & I \\ \mathbf{j} & 0 & J-I & I & I+J \end{pmatrix},$$

$$B = \begin{pmatrix} 2 & 1 & 0 & 0 & k \\ 1 & 2 & 0 & k & 0 \\ 0 & 0 & k^2+k & k^2-k & k^2-k \\ 0 & k & k^2-k & k^2+k & k \\ k & 0 & k^2-k & k & k^2+k \end{pmatrix}.$$

Here I, J have size k and $\mathbf{j}$ has k entries all equal to 1. We number the displayed rows and columns from 1.

⟨3⟩1. For distinct vertices x, y, $(\mathbf{x}, \mathbf{y}) = 0$ on edges and 1 on nonedges.
PROOF: Use the definition of X and Lemma 2.6; diagonal products are 2. □

⟨3⟩2. The displayed blocks of B_0 follow from Lemma (triangle-property normal forms).
PROOF: Independent parts give $I+J$; matched edges give $J-I$; all edges except matched pairs give I. The products with $\mathbf{u}, \mathbf{v}$ follow from the definitions of K, L, M, using ⟨3⟩1. □

⟨3⟩3. The matrix B consists of the block sums of B_0.
PROOF: Bilinearity gives each sum-vector product as the corresponding double sum. The sums of the entries of $I+J, J-I, I$ are k^2+k, k^2-k, k, respectively; those of $\mathbf{j}$ and 0 are k and 0. □

⟨3⟩4. Q.E.D.
PROOF: Steps ⟨3⟩2 and ⟨3⟩3 prove the formulas. **[repair]** In particular $B(3,1) = 0$, correcting the printed 2 (Error 3 of `cgss/errata.md`): for every $x \in K$, the vector $\mathbf{x}$ is orthogonal to $\mathbf{u}$. □

⟨2⟩2. $2 \le k \le 5$.

⟨3⟩1. The matrix D obtained by deleting row and column 1 of B is positive semidefinite, so $\det D \ge 0$.

⟨4⟩1. D is the Gram matrix of $\mathbf{v}, \sum_{x\in K} \mathbf{x}, \sum_{x\in K} \mathbf{x}', \sum_{x\in K} \mathbf{x}''$.
PROOF: Delete the first vector from the block-sum family in ⟨2⟩1. □

⟨4⟩2. D is positive semidefinite.
PROOF: Apply `Matrix.posSemidef_gram` in `Mathlib/Analysis/InnerProductSpace/GramMatrix.lean` directly to the four-vector family in ⟨4⟩1. □

⟨4⟩3. Q.E.D.
PROOF: Apply `Matrix.PosSemidef.det_nonneg` in `Mathlib/Analysis/Matrix/PosDef.lean` to ⟨4⟩2. We index the real matrix by `Fin 4`, a finite type with decidable equality. □

⟨3⟩2. Computation C1: $\det D = 2k^5(5-k)$.
PROOF: Expanding along the first row $(2, 0, k, 0)$ gives

$$\begin{aligned} \det D &= 2\big(k^5(7-k)\big) + k\big(-4k^4\big) && \text{[the two } 3\times 3 \text{ minors]} \\ &= 2k^5(5-k) && \text{[ring arithmetic].} \end{aligned}$$

Each 3×3 minor is its six-term determinant expansion. Thus C1 is a polynomial identity, certified by `Matrix.det_succ_row_zero` in `Mathlib/LinearAlgebra/Matrix/Determinant/Basic.lean`, with D indexed by `Fin (3+1)`. For each minor use `Matrix.det_fin_three` in the same file, with index type `Fin 3`, then ring arithmetic. □

⟨3⟩3. Q.E.D.

PROOF: Steps ⟨3⟩1 and ⟨3⟩2 give $2k^5(5-k) \geq 0$. Since $k \geq 2$ by ⟨1⟩1, division by $2k^5 > 0$ gives $k \leq 5$. □

⟨2⟩3. Q.E.D.

PROOF: Step ⟨2⟩2 is the claimed second proof. □

⟨1⟩4. Q.E.D.

PROOF: Steps ⟨1⟩1 and ⟨1⟩2 give the assertions of the lemma; ⟨1⟩3 records the paper's bound as well. □

Remark 2.11. *Side claim, not formalized.* We quote [CGSS]:

> Similar methods may be used to prove the characterization of symplectic and orthogonal graphs over GF(2) by the triangle property, cf. [24, 21].

We attach no Lean declaration to this side claim.

Lemma 2.12. ASSUME: 1. S is a nonempty indecomposable star-closed line system in a finite-dimensional real inner product space V.

2. When a star is chosen, Γ_a and X have the meanings of Hypothesis 2.4.

PROVE: Either S has one line, or it admits a pair $\mathbf{a}, \mathbf{b}$ as in Hypothesis 2.4 and, for every such choice, its graph X is one of: the empty graph $\overline{K_m}$ on $m \geq 0$ vertices; the friendship graph with $m \geq 1$ triangles; the graphs of Lemma 2.10 with parameters $(9, 4, 1, 2)$, $(15, 6, 1, 3)$, or $(27, 10, 1, 5)$. For any two instances of Hypothesis 2.4 in finite-dimensional real inner product spaces V, W, every isomorphism of their graphs on Γ_a extends to an isomorphism of their line systems, induced by a surjective linear isometry of their spans carrying the ordered stars to each other. This holds for every graph type in the list.

[repair] Nonemptiness excludes the empty line system, as required by Implicit hypothesis 4 of `cgss/errata.md`.

PROOF SKETCH: **[repair]** We give all four strongly regular parameters in the exceptional cases to delimit the paper's uniqueness statement (Error 12). We will recover the signed root set from Lemmas 2.5 and 2.7. A graph isomorphism preserves the Gram matrix of the roots in $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\} \cup \Gamma_a$, so it extends linearly to the spans and then carries the recovered sets to each other.

⟨1⟩1. **[repair]** Either S consists of one antipodal pair or it admits Hypothesis 2.4. We include the one-line case omitted by the choice of a star in the paper (Implicit hypothesis 4).

LET: Choose $\mathbf{r} \in \Phi$, using nonemptiness.

⟨2⟩1. If every root outside $\{\mathbf{r}, -\mathbf{r}\}$ is orthogonal to $\mathbf{r}$, then $\Phi = \{\mathbf{r}, -\mathbf{r}\}$.

PROOF: Take $P = \{\mathbf{r}, -\mathbf{r}\}$ and $Q = \Phi \setminus P$. The hypothesis and bilinearity make these parts orthogonal. Indecomposability and $\mathbf{r} \in P$ force $Q = \varnothing$. □

⟨2⟩2. Otherwise there are $\mathbf{a}, \mathbf{b} \in \Phi$ with $(\mathbf{a}, \mathbf{b}) = -1$.

PROOF: Choose $\mathbf{s} \notin \{\mathbf{r}, -\mathbf{r}\}$ with $(\mathbf{r}, \mathbf{s}) \neq 0$. Equation (1) and the line-system condition give ± 1. Set $\mathbf{a} = \mathbf{r}$, and set $\mathbf{b} = \mathbf{s}$ if the value is -1, or $\mathbf{b} = -\mathbf{s}$ if it is 1. Closure under negation supplies both choices. □

⟨2⟩3. Q.E.D.

PROOF: The cases ⟨2⟩1 and ⟨2⟩2 are exhaustive. □

⟨1⟩2. In any instance of Hypothesis 2.4, put $T = \{\mathbf{a}, \mathbf{b}, \mathbf{c}\} \cup \Gamma_a$. Then $\Phi = \mathcal{R} \cup (-\mathcal{R})$ for the set $\mathcal{R}$ in Lemma 2.7, and $\operatorname{span}(T) = \operatorname{span}(\Phi)$.

⟨2⟩1. We have $\Phi = \mathcal{R} \cup (-\mathcal{R})$, where

$$\mathcal{R} = \{\mathbf{a}, \mathbf{b}, \mathbf{c}\} \cup \Gamma_a \cup (\mathbf{c} + \Gamma_a) \cup (-\mathbf{b} + \Gamma_a) \cup \{\mathbf{x} - \mathbf{y} : \mathbf{x}, \mathbf{y} \in \Gamma_a,\ (\mathbf{x}, \mathbf{y}) = 1\}.$$

PROOF: This is the recovery equality in the statement of Lemma 2.7, including both inclusions. □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨2⟩1 recovers the set using only T and its inner products. Every displayed vector lies in $\operatorname{span}(T)$. Since $T \subseteq \Phi$, the two spans are equal. □

LET: For the extension assertion, fix any two instances of Hypothesis 2.4 in finite-dimensional real inner product spaces V, W and a graph isomorphism $\psi : X \to Y$. Use tildes for the second root set and ordered star. For $x \in V(X)$, write $\mathbf{s}_x$ for the vector at $\psi(x) \in V(Y)$.

⟨1⟩3. **[new]** There is a surjective linear isometry between the two root spans, sending $\mathbf{a}, \mathbf{b}, \mathbf{c}$ to $\widetilde{\mathbf{a}}, \widetilde{\mathbf{b}}, \widetilde{\mathbf{c}}$ and $\mathbf{x}$ to $\mathbf{s}_x$ for every $x \in V(X)$. We make the metric meaning of "determined" explicit.

LET: Index two vector families $(\mathbf{r}_i), (\mathbf{s}_i)$ by the same finite disjoint union $I = \{1, 2, 3\} \sqcup V(X)$. The first three values are the ordered stars; the other values are $\mathbf{r}_x = \mathbf{x}$ and the vectors $\mathbf{s}_x$ just defined. Put $R(\alpha) = \sum_i \alpha(i)\mathbf{r}_i$ and $R'(\alpha) = \sum_i \alpha(i)\mathbf{s}_i$. These are linear maps from $I \to \mathbb{R}$ to V, W, respectively, by `Fintype.linearCombination` in `Mathlib/LinearAlgebra/Finsupp/LinearCombination.lean`.

⟨2⟩1. $(\mathbf{r}_i, \mathbf{r}_j) = (\mathbf{s}_i, \mathbf{s}_j)$ for all $i, j \in I$.

⟨3⟩1. Products within the first three positions are 2 on the diagonal and -1 off it; products with a vertex of X are $0, 1, -1$ in that order.

PROOF: These are the star identities and the defining products of Γ_a in each system. □

⟨3⟩2. Products within the graph positions are 2 on the diagonal, 0 on edges, and 1 on nonedges.

PROOF: Use Lemma 2.6 and the orthogonality convention; the bijection ψ preserves equality, adjacency, and nonadjacency. □

⟨3⟩3. Q.E.D.

PROOF: Steps ⟨3⟩1 and ⟨3⟩2, with symmetry, cover all pairs of positions. □

⟨2⟩2. $\ker R \subseteq \ker R'$: for every $\alpha : I \to \mathbb{R}$, $R(\alpha) = 0$ implies $R'(\alpha) = 0$.

⟨3⟩1. The two sums have the same squared norm.

PROOF: Expanding by bilinearity gives

$$\sum_{i,j} \alpha(i)\alpha(j)(\mathbf{r}_i, \mathbf{r}_j) = \sum_{i,j} \alpha(i)\alpha(j)(\mathbf{s}_i, \mathbf{s}_j)$$

by ⟨2⟩1. □

⟨3⟩2. Q.E.D.

PROOF: If the first sum is zero, ⟨3⟩1 makes the second sum have squared norm zero. The zero-norm implication in the elementary identities makes that sum zero. □

⟨2⟩3. There is a surjective linear map F of the two spans such that

$$F\left(\sum_i \alpha(i)\mathbf{r}_i\right) = \sum_i \alpha(i)\mathbf{s}_i.$$

⟨3⟩1. The ranges of R, R' are the spans of $(\mathbf{r}_i), (\mathbf{s}_i)$, respectively.

PROOF: Apply `Fintype.range_linearCombination` in `Mathlib/LinearAlgebra/Finsupp/LinearCombination.lean` to each finite family. □

LET: Let R'_0 be R' with codomain restricted to $\operatorname{range} R'$, using `LinearMap.rangeRestrict` in `Mathlib/Algebra/Module/Submodule/Range.lean`.

⟨3⟩2. $\ker R \subseteq \ker R'_0$.

PROOF: A vector in the subspace $\operatorname{range} R'$ is zero exactly when its underlying vector is zero. Thus ⟨2⟩2 gives the inclusion. □

⟨3⟩3. There is a linear map $\overline{R}' : (I \to \mathbb{R})/\ker R \to \operatorname{range} R'$ satisfying $\overline{R}'([\alpha]) = R'_0(\alpha)$.

PROOF: Apply `Submodule.liftQ` to R'_0 and ⟨3⟩2; `Submodule.liftQ_apply` gives the formula. Both declarations are in `Mathlib/LinearAlgebra/Quotient/Basic.lean`. □

⟨3⟩4. There is a linear map $F_0 : \operatorname{range} R \to \operatorname{range} R'$ with $F_0(R(\alpha)) = R'(\alpha)$ for all α.

LET: Let $E : (I \to \mathbb{R})/\ker R \to \operatorname{range} R$ be `LinearMap.quotKerEquivRange` for R, in `Mathlib/LinearAlgebra/Isomorphisms.lean`, and put $F_0 = \overline{R}' \circ E^{-1}$.

⟨4⟩1. F_0 is linear.

PROOF: It is a composite of the linear map in ⟨3⟩3 and the inverse of a linear equivalence. □

⟨4⟩2. $F_0(R(\alpha)) = R'(\alpha)$.

PROOF: The lemma `LinearMap.quotKerEquivRange_symm_apply_image` in `Mathlib/LinearAlgebra/Isomorphisms.lean` gives $E^{-1}(R(\alpha)) = [\alpha]$. Apply the formula in ⟨3⟩3. □

⟨4⟩3. Q.E.D.

PROOF: Steps ⟨4⟩1 and ⟨4⟩2 prove both assertions. □

⟨3⟩5. F_0 is surjective.

PROOF: Every member of $\operatorname{range} R'$ has the form $R'(\alpha)$; by ⟨3⟩4, $R(\alpha)$ is a preimage. □

⟨3⟩6. $F_0(\mathbf{r}_i) = \mathbf{s}_i$ for every i.

PROOF: In ⟨3⟩4, take the coefficient function which is 1 at i and zero elsewhere. The two finite sums are $\mathbf{r}_i, \mathbf{s}_i$. □

⟨3⟩7. Q.E.D.

PROOF: Identify the ranges with the spans by ⟨3⟩1. The map F_0 in ⟨3⟩4, with surjectivity from ⟨3⟩5, is the required F and has the values in ⟨3⟩6. □

LET: Choose a surjective linear map F with the formula in ⟨2⟩3.

⟨2⟩4. The map F preserves inner products and is injective.

⟨3⟩1. For vectors $\mathbf{x}, \mathbf{y}$ in the first span, $(F(\mathbf{x}), F(\mathbf{y})) = (\mathbf{x}, \mathbf{y})$.

PROOF: By the finite-span lemma, write $\mathbf{x}, \mathbf{y}$ as combinations with coefficients α, β. The formula in ⟨2⟩3 gives their images, and bilinearity gives

$$\sum_{i,j} \alpha(i)\beta(j)(\mathbf{r}_i, \mathbf{r}_j) = \sum_{i,j} \alpha(i)\beta(j)(\mathbf{s}_i, \mathbf{s}_j)$$

by ⟨2⟩1. □

⟨3⟩2. F is injective.

PROOF: If $F(\mathbf{x}) = F(\mathbf{y})$, linearity gives $F(\mathbf{x} - \mathbf{y}) = 0$. Step ⟨3⟩1 gives $(\mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y}) = 0$. The zero-norm identity then gives $\mathbf{x} = \mathbf{y}$. □

⟨3⟩3. Q.E.D.

PROOF: Combine ⟨3⟩1 and ⟨3⟩2. □

⟨2⟩5. Q.E.D.

PROOF: Steps ⟨2⟩3 and ⟨2⟩4 give a linear isometry onto the second span. Step ⟨1⟩2 identifies these spans with the root spans. Taking one coefficient equal to 1 and all others zero in ⟨2⟩3 gives its prescribed values on the two indexed families. This is the equal-Gram argument LA2 from the prerequisites, proved here for the needed families. □

LET: Choose an isometry F with the values and surjectivity in ⟨1⟩3.

⟨1⟩4. The isometry F in ⟨1⟩3 carries the first root set bijectively onto the second.

⟨2⟩1. $F(\Phi) \subseteq \widetilde{\Phi}$.

⟨3⟩1. $F(\mathbf{a}) = \widetilde{\mathbf{a}} \in \widetilde{\Phi}$.

PROOF: Use ⟨1⟩3 and the second instance of Hypothesis 2.4. □

⟨3⟩2. $F(\mathbf{b}) = \widetilde{\mathbf{b}} \in \widetilde{\Phi}$.

PROOF: Use ⟨1⟩3 and the second instance of Hypothesis 2.4. □

⟨3⟩3. $F(\mathbf{c}) = \widetilde{\mathbf{c}} \in \widetilde{\Phi}$.

PROOF: Use ⟨1⟩3 and the star identities in the second instance. □

⟨3⟩4. For $x \in V(X)$, $F(\mathbf{x}) = \mathbf{s}_x \in \widetilde{\Gamma_a}$.

PROOF: Step ⟨1⟩3 gives the value; $\mathbf{s}_x$ is the vector at $\psi(x)$, a vertex of the second graph. □

⟨3⟩5. For $\mathbf{x} \in \Gamma_a$, $F(\mathbf{c}+\mathbf{x}) = \widetilde{\mathbf{c}} + \mathbf{s}_x \in \widetilde{\Phi}$.
PROOF: Linearity and ⟨1⟩3 give the equality. Lemma 2.6 in the second instance, or the reverse inclusion of its recovery equality in Lemma 2.7, gives membership. □
⟨3⟩6. For $\mathbf{x} \in \Gamma_a$, $F(-\mathbf{b}+\mathbf{x}) = -\widetilde{\mathbf{b}} + \mathbf{s}_x \in \widetilde{\Phi}$.
PROOF: Linearity and ⟨1⟩3 give the equality. Apply the second translation in Lemma 2.6 in the target, equivalently the reverse inclusion of its recovery equality in Lemma 2.7. □
⟨3⟩7. If $\mathbf{x}, \mathbf{y} \in \Gamma_a$ and $(\mathbf{x}, \mathbf{y}) = 1$, then $F(\mathbf{x}-\mathbf{y}) = \mathbf{s}_x - \mathbf{s}_y \in \widetilde{\Phi}$.
⟨4⟩1. $(\mathbf{s}_x, \mathbf{s}_y) = 1$.
PROOF: The isometry in ⟨1⟩3 preserves the given product. □
⟨4⟩2. Q.E.D.
PROOF: Linearity gives the image formula. The reverse inclusion of the second recovery equality in Lemma 2.7 applies to ⟨4⟩1. □
⟨3⟩8. If a displayed expression $\mathbf{r}$ maps into $\widetilde{\Phi}$, so does $-\mathbf{r}$.
PROOF: Linearity gives $F(-\mathbf{r}) = -F(\mathbf{r})$ and $\widetilde{\Phi} = -\widetilde{\Phi}$. □
⟨3⟩9. Q.E.D.
PROOF: The forward inclusion of the first recovery equality in Lemma 2.7 expresses every root by one of these seven kinds and a sign. Steps ⟨3⟩1 through ⟨3⟩7 handle the expressions, and ⟨3⟩8 handles negation. □
⟨2⟩2. $\widetilde{\Phi} \subseteq F(\Phi)$.
⟨3⟩1. Each root of the second ordered star has its corresponding first-star root as a preimage in Φ.
PROOF: Apply the prescribed star values in ⟨1⟩3; Hypothesis 2.4 and the star identities put all three preimages in Φ. □
⟨3⟩2. Each $\mathbf{s} \in \widetilde{\Gamma_a}$ has a preimage $\mathbf{x} \in \Gamma_a$.
PROOF: Take the inverse image of its vertex under the bijection ψ and use ⟨1⟩3. □
⟨3⟩3. Each $\widetilde{\mathbf{c}} + \mathbf{s}$ with $\mathbf{s} \in \widetilde{\Gamma_a}$ has a preimage in Φ.
PROOF: Choose $\mathbf{x}$ by ⟨3⟩2. The reverse inclusion of the first recovery equality in Lemma 2.7 puts $\mathbf{c}+\mathbf{x}$ in Φ, and linearity gives its image. □
⟨3⟩4. Each $-\widetilde{\mathbf{b}} + \mathbf{s}$ with $\mathbf{s} \in \widetilde{\Gamma_a}$ has a preimage in Φ.
PROOF: Choose $\mathbf{x}$ by ⟨3⟩2. The reverse inclusion of the first recovery equality in Lemma 2.7 puts $-\mathbf{b}+\mathbf{x}$ in Φ, and linearity gives its image. □
⟨3⟩5. Each $\mathbf{s}-\mathbf{t}$ with $\mathbf{s}, \mathbf{t} \in \widetilde{\Gamma_a}$ and $(\mathbf{s}, \mathbf{t}) = 1$ has a preimage in Φ.
PROOF: Choose preimages $\mathbf{x}, \mathbf{y}$ by ⟨3⟩2. The isometry in ⟨1⟩3 gives $(\mathbf{x}, \mathbf{y}) = 1$. The reverse inclusion of the first recovery equality in Lemma 2.7 puts $\mathbf{x}-\mathbf{y}$ in Φ, and linearity gives its image. □
⟨3⟩6. The negative of a target with a preimage in Φ also has a preimage in Φ.
PROOF: Negate the preimage, using $\Phi = -\Phi$ and linearity. □
⟨3⟩7. Q.E.D.
PROOF: The forward inclusion of the second recovery equality in Lemma 2.7 lists exactly the expressions and signs covered by ⟨3⟩1 through ⟨3⟩6. □
⟨2⟩3. Q.E.D.
PROOF: Injectivity in ⟨1⟩3, ⟨2⟩1, and ⟨2⟩2 make the restriction of F a bijection preserving inner products, as required by the definition of a line-system isomorphism. □
⟨1⟩5. Every graph arising from a chosen star belongs to the stated list.
⟨2⟩1. CASE: X has no edges.
PROOF: Its finite vertex set has some cardinality $m \geq 0$, giving the first type. □
⟨2⟩2. CASE: X has an edge and has a universal vertex.

PROOF: Lemma 2.9 gives a friendship graph with m triangles. Since X has an edge, $m \neq 0$, so $m \geq 1$. □

⟨2⟩3. CASE: X has an edge and has no universal vertex.

PROOF: A complete graph with an edge has universal vertices, so this graph is noncomplete. Thus Lemma 2.10 applies; its statement gives the displayed strongly regular parameters. □

⟨2⟩4. Q.E.D.

PROOF: These three cases exhaust existence of an edge and of a universal vertex. □

⟨1⟩6. Q.E.D.

PROOF: Step ⟨1⟩1 separates the one-line case; ⟨1⟩5 lists every graph in the other case. Steps ⟨1⟩3 and ⟨1⟩4 give the claimed uniqueness and its realization by a linear isometry of spans. □

Inputs from Section 3. **[repair]** The line counts and dimensions in Theorem 2.13 require the models of Section 3 (Gap 4 of `cgss/errata.md`). We will use the following facts from Definitions 3.1 to 3.3, Lemma 3.4, and computations C2 and C4. We record them as deferred dependencies, not as conclusions of Section 2. The labels `def:3.1`, `def:3.2`, `def:3.3`, `lem:3.4`, `comp:C2` and `comp:C4` are those of Section 3, which proves these facts. Each model is a nonempty indecomposable star-closed line system in its real span. The model A_1 consists of one antipodal pair and has rank 1. Each remaining model has a specified ordered star for which the following data hold. Here rank means dimension of the real span, and δ means $|\Delta|/2$, the number of lines in Δ.

model	rank	number of lines	graph on Γ_a	δ
A_ℓ $(\ell \geq 2)$	ℓ	$\ell(\ell+1)/2$	$\overline{K_{\ell-2}}$	not used
D_ℓ $(\ell \geq 4)$	ℓ	$\ell(\ell-1)$	friendship, $\ell-3$ triangles	not used
E_6	6	36	$(9,4,1,2)$	6
E_7	7	63	$(15,6,1,3)$	15
E_8	8	120	$(27,10,1,5)$	36

In the exceptional rows the tuple gives strongly regular parameters; the graph is the orthogonality graph, so it has degree 4, 6, or 10, respectively. We take its uniqueness from Lemma 2.10, not from the model verification. C2 gives the line counts and C4 verifies star-closure, indecomposability and the marked-star data; spanning sets in the explicit models give the displayed ranks. These verifications must not use Theorem 2.13 or Theorem 3.5. For Definition 3.3 we use fixed representatives of a root and a star in E_8. The fixed-root construction of E_7 and the fixed-star construction of E_6 suffice here. Prerequisite LN5 is needed to compare a fixed-root construction with the phrase "any one of its lines" in Definition 3.3, but is not used in Theorem 2.13.

Theorem 2.13. ASSUME: 1. V is a finite-dimensional real inner product space of dimension ℓ.
2. S is a nonempty indecomposable star-closed line system in V, and $\operatorname{span}(\Phi) = V$.

PROVE: The possibilities for S, up to isomorphism, are the five cases below. For every case with more than one line, there exist $\mathbf{a}, \mathbf{b} \in \Phi$ with $(\mathbf{a}, \mathbf{b}) = -1$ for which the stated data on Γ_a and Δ hold. **[repair]** We exclude the empty system and separate the one-line case (Implicit hypothesis 4 of `cgss/errata.md`).

1. $\ell(\ell+1)/2$ lines, with the empty graph $\overline{K_{\ell-2}}$ on Γ_a when $\ell \geq 2$; for $\ell = 1$ there is one line and no chosen star;
2. $\ell(\ell-1)$ lines for $\ell \geq 4$, with the friendship graph of $\ell-3$ triangles on Γ_a;
3. 120 lines in dimension 8, with $|\Gamma_a| = 27$ and $|\Delta|/2 = 36$;
4. 63 lines in dimension 7, with $|\Gamma_a| = 15$ and $|\Delta|/2 = 15$;
5. 36 lines in dimension 6, with $|\Gamma_a| = 9$ and $|\Delta|/2 = 6$.

Each case is unique up to an orthogonal transformation when the ambient space is $\mathbb{R}^\ell$. The zero-triangle case of Lemma 2.9 has $\ell = 3$ and six lines; it is already in case 1 and also satisfies the formula $\ell(\ell-1) = 6$.

The Lean statement gives the data on Γ_a and Δ for some star, while in the paper they belong to the star fixed after Hypothesis 2.4, which is arbitrary. Every star gives the same data: `LineSystem.five_possibilities_forall_hyp24` is the theorem for every star $(\mathbf{a}, \mathbf{b})$ of Hypothesis 2.4 whose system spans V, with the graph on Γ_a and the counts of Γ_a and Δ taken at that star. It leaves out the one-line case, which has no star.

PROOF SKETCH: We will match the graph from Lemma 2.12 with a specified star in one of the models. Its graph isomorphism gives a linear isometry of spans, so it transfers both rank and number of lines. Thus the numerical conclusions come from Section 3.

⟨1⟩1. If S has one line, then $\ell = 1$ and a linear isometry of spans carries it onto A_1.

⟨2⟩1. The span of $\Phi = \{\mathbf{r}, -\mathbf{r}\}$ is one-dimensional.

PROOF: Squared norm 2 gives $\mathbf{r} \neq 0$. Thus the singleton $\{\mathbf{r}\}$ is a basis of this span, which equals V by ⟨0⟩:2. □

⟨2⟩2. For either root $\mathbf{s}$ of A_1, the rule $F(\alpha\mathbf{r}) = \alpha\mathbf{s}$ defines a surjective linear isometry of spans.

PROOF: Step ⟨2⟩1 gives a unique coefficient α for each vector. Addition and scaling of this coefficient make F linear and surjective. For any real α, β, both inner products are $2\alpha\beta$ by bilinearity. Thus F preserves inner products. □

⟨2⟩3. Q.E.D.

PROOF: The map in ⟨2⟩2 sends $\{\mathbf{r}, -\mathbf{r}\}$ bijectively onto $\{\mathbf{s}, -\mathbf{s}\}$. The line count is $1 = 1(1+1)/2$, and ⟨2⟩1 gives $\ell = 1$. □

⟨1⟩2. If S has more than one line, its chosen-star graph is isomorphic to the specified graph of one model in the input table.

LET: Choose a star using Lemma 2.12 and let X be its graph.

⟨2⟩1. CASE: $X = \overline{K_m}$.

PROOF: The specified star of A_{m+2} has the empty graph on $(m+2)-2 = m$ vertices by the Section 3 inputs. A bijection between the finite vertex sets preserves adjacency since both graphs have no edges. □

⟨2⟩2. CASE: X is the friendship graph with $m \geq 1$ triangles.

⟨3⟩1. The specified graph in D_{m+3} is the friendship graph with m triangles.

PROOF: Use $m + 3 \geq 4$ and the D_ℓ row of the Section 3 inputs. □

⟨3⟩2. Q.E.D.

PROOF: Send a chosen common vertex to the model's common vertex, choose a bijection of the m pairs of remaining vertices, and a bijection within each pair. Lemma 2.9 lists exactly the edges preserved by this map, so it is a graph isomorphism. □

⟨2⟩3. CASE: $|V(X)| \in \{9, 15, 27\}$ as in Lemma 2.10.

⟨3⟩1. Choose $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ according as the order is $9, 15, 27$. Its specified graph satisfies the hypotheses of Lemma 2.10.

PROOF: The respective orders and degrees in the input table are $(9,4), (15,6), (27,10)$. Each degree is positive and smaller than order minus 1, so each graph has an edge, is noncomplete, and has no universal vertex. A graph of Lemma 2.9 has a universal vertex, so that type is excluded. □

⟨3⟩2. Q.E.D.

PROOF: The chosen model is a star-closed indecomposable line system and has the same graph order as X. Apply the uniqueness assertion of Lemma 2.10. □

⟨2⟩4. Q.E.D.

PROOF: Lemma 2.12 makes ⟨2⟩1, ⟨2⟩2, and ⟨2⟩3 exhaustive. □

⟨1⟩3. For a model matched in ⟨1⟩2, ℓ equals its rank and S has its number of lines. The marked sets Δ also have the same number of lines.

⟨2⟩1. The graph isomorphism extends to a surjective linear isometry F between the spans, carrying the ordered stars to each other and mapping root sets bijectively.

PROOF: Apply the reconstruction and equal-Gram argument of Lemma 2.12. □

⟨2⟩2. The two spans have equal dimensions.
PROOF: A linear isometry is injective; together with surjectivity in ⟨2⟩1, it sends a basis bijectively to a basis. The first span is V by ⟨0⟩:2. □

⟨2⟩3. The root bijection induces a bijection of antipodal pairs.
PROOF: Linearity gives $F(-\mathbf{r}) = -F(\mathbf{r})$; injectivity prevents two different antipodal pairs from merging, and surjectivity on roots covers every target pair. □

⟨2⟩4. The root bijection sends Δ exactly to the model's marked Δ.
PROOF: The definition of Δ asks for three zero inner products with the ordered star. Both F and its inverse preserve these products and the ordered star by ⟨2⟩1. Thus membership is preserved in both directions. □

⟨2⟩5. Q.E.D.
PROOF: Step ⟨2⟩2 transfers rank, ⟨2⟩3 transfers line count, and ⟨2⟩4 with ⟨2⟩3 transfers the line count in Δ. □

⟨1⟩4. The transferred data give exactly the five numerical cases in the statement.

⟨2⟩1. The empty-graph case gives $\ell = m+2$ and $\ell(\ell+1)/2$ lines; the friendship case with $m \geq 1$ triangles gives $\ell = m+3$ and $\ell(\ell-1)$ lines.
PROOF: Substitute the ranks of $\mathrm{A}_{m+2}, \mathrm{D}_{m+3}$ from the input table into ⟨1⟩3, then use their line-count formulas from Lemma 3.4 and C2. □

⟨2⟩2. The orders $27, 15, 9$ give respectively the data
$$(\ell, |\Phi|/2, |\Delta|/2) = (8, 120, 36), (7, 63, 15), (6, 36, 6).$$
PROOF: Use the $\mathrm{E}_8, \mathrm{E}_7, \mathrm{E}_6$ rows of the Section 3 inputs in ⟨1⟩3. **[repair]** We correct Error 6: we write $|\Delta|/2$ because the signed set Δ has respectively $72, 30, 12$ vectors. The paper's $|\Delta| = 36, 15, 6$ counts lines. □

⟨2⟩3. **[new]** The zero-triangle case of Lemma 2.9 has six lines in dimension 3; we make this overlap explicit.
PROOF: Its graph is a single vertex, the empty graph $\overline{K_1}$. The first formula in ⟨2⟩1 gives $\ell = 3$ and $3 \cdot 4/2 = 6 = 3 \cdot 2$. □

⟨2⟩4. Q.E.D.
PROOF: Combine ⟨1⟩1, ⟨2⟩1, ⟨2⟩2, and ⟨2⟩3. □

⟨1⟩5. Each numerical case, with its stated graph type, is unique up to an orthogonal transformation in $\mathbb{R}^\ell$.

⟨2⟩1. Two systems in the same case are isomorphic to the same model.
PROOF: In the first family the model rank is ℓ, so the model is A_ℓ; in the second it is D_ℓ; the exceptional rows specify $\mathrm{E}_8, \mathrm{E}_7, \mathrm{E}_6$ individually. Use ⟨1⟩1 for $\ell = 1$ and ⟨1⟩2 otherwise. □

⟨2⟩2. Q.E.D.
PROOF: Compose the span isometry to that model with the inverse for the other system. Both root spans are the entire $\mathbb{R}^\ell$ by ⟨0⟩:2. Thus the composite is a surjective linear isometry of $\mathbb{R}^\ell$, an orthogonal transformation, sending one line system onto the other. □

⟨1⟩6. Q.E.D.
PROOF: Step ⟨1⟩4 gives the alternatives and ⟨1⟩5 gives uniqueness. The asserted dimensions and counts depend on the explicit Section 3 inputs recorded above; no classification result of Section 3 was used. □

3. Root systems

We will construct the models before using the classification. All dimensions in this section are dimensions of spans. We use both signs of every root, with squared norm 2, as in Section 2. The symbol ℓ denotes rank, and N denotes the number of lines. Coordinates in the text start at 1;

Lean's coordinates in `Fin` start at 0. An isometric inclusion means a linear isometry of spans carrying the source roots into the target roots. It is proper when that inclusion of root sets is proper.

[new] We distinguish a proved blueprint result from an existing Lean declaration. The tags `\lean` give the Lean names; every one of them names a declaration in Lean, and `dependency-diagram.md`, at the top of the repository, maps the numbered results to their declarations. In computational leaves we specify the integer lists, the Boolean test, and the soundness theorem. The script `blueprint/checks/sec3/check_sec3.py` recomputes these tests exactly; it is a sanity check, not a kernel proof of a new certificate. No classification of Root Systems from Remark 3.11 is used here.

Lemma (transport on spans). ASSUME: S, T are line systems in finite-dimensional real inner product spaces V, V', and $f : \Phi(S) \to \Phi(T)$ is a bijection preserving inner products.
PROVE: There is a unique surjective linear isometry $F : \operatorname{span}\Phi(S) \to \operatorname{span}\Phi(T)$ with $F(\mathbf{r}) = f(\mathbf{r})$. It preserves line count and rank, star-closure, indecomposability, and maximality within the span. Its inverse has the same properties. For $\mathbf{a}, \mathbf{b} \in \Phi(S)$ with $(\mathbf{a}, \mathbf{b}) = -1$, the map f carries the sets Γ_a and Δ of Hypothesis 2.4 for $\mathbf{a}, \mathbf{b}$ bijectively onto those for $f(\mathbf{a}), f(\mathbf{b})$, and preserves orthogonality in Γ_a.

PROOF SKETCH: **[new]** We state the transport fact separately so later results do not cite a construction hidden inside a proof. We will factor finite linear combinations through their kernels, and then transport the defining predicates.

⟨1⟩1. For any real coefficients $a(\mathbf{r})$, the two sums $\sum a(\mathbf{r})\mathbf{r}$ and $\sum a(\mathbf{r})f(\mathbf{r})$ have equal squared norms.
PROOF: Expand both squared norms by bilinearity from Section 2; each term agrees by the inner-product hypothesis. □

⟨1⟩2. The rule $F(\sum a(\mathbf{r})\mathbf{r}) = \sum a(\mathbf{r})f(\mathbf{r})$ is a linear map of spans.
⟨2⟩1. A zero source combination has zero target combination.
PROOF: Apply zero squared norm from Section 2 to ⟨1⟩1. □
⟨2⟩2. The target combination map factors through the quotient by the kernel of the source combination map.
PROOF: Apply `Submodule.liftQ` to the kernel inclusion in ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Identify the quotient with the source range by `LinearMap.quotKerEquivRange`, and the range with the span by `Fintype.range_linearCombination`. □

⟨1⟩3. The map F is an isometry and is onto the target span.
⟨2⟩1. It preserves the inner product of two combinations.
PROOF: Bilinearity from Section 2 expands that product into the preserved products of pairs of roots. □
⟨2⟩2. Each target combination has a source combination mapping to it.
PROOF: Reindex its coefficients by the inverse of f. □
⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1 and ⟨2⟩2. □

⟨1⟩4. The asserted invariants are preserved.
⟨2⟩1. Antipodal pairs correspond bijectively.
PROOF: Linearity gives $F(-\mathbf{r}) = -F(\mathbf{r})$; use the root bijection. □
⟨2⟩2. The ranks agree.
PROOF: A linear bijection carries a basis to a basis. □
⟨2⟩3. Star-closure is preserved in each direction.
PROOF: Transport the sum and inner product in Definition 2.1 through F or F^{-1}. □

⟨2⟩4. Indecomposability is preserved in each direction.
PROOF: Pull an orthogonal partition back through the corresponding root bijection in the indecomposability definition in Section 2. □
⟨2⟩5. Maximality within the span is preserved in each direction.
PROOF: Transport a candidate vector of Definition 2.2 through the surjective span isometry. □
⟨2⟩6. **[new]** The map f carries Γ_a and Δ for $\mathbf{a}, \mathbf{b}$ onto those for $f(\mathbf{a}), f(\mathbf{b})$, preserving orthogonality.
PROOF: These sets are defined in Hypothesis 2.4 by inner products with $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c} = -\mathbf{a}-\mathbf{b}$; f preserves inner products and $F(\mathbf{c}) = -f(\mathbf{a}) - f(\mathbf{b})$ by linearity. □
⟨2⟩7. Q.E.D.
PROOF: The six preceding steps are the six assertions. □
⟨1⟩5. Q.E.D.
PROOF: The finite-span lemma in Section 2 forces the formula for any linear extension; thus ⟨1⟩2 is unique. Combine ⟨1⟩3 and ⟨1⟩4. □

The quotient and combination declarations are in the Mathlib files `LinearAlgebra/Quotient/Basic.lean`, `LinearAlgebra/Isomorphisms.lean`, and `LinearAlgebra/Finsupp/LinearCombination.lean`, respectively.

Lemma (restriction to the span). ASSUME: S is a line system in a finite-dimensional real inner product space V, and $W = \operatorname{span}\Phi(S)$, with the inner product of V.
PROVE: The vectors of $\Phi(S)$, as vectors of W, form the root set of a line system $S|_W$ in W, and $\Phi(S|_W)$ spans W. The identity of $\Phi(S)$ is a bijection $\Phi(S|_W) \to \Phi(S)$ preserving inner products. Thus $S|_W$ and S have the same number of lines, the same rank, and the same sets Γ_a and Δ with the same orthogonalities, and each is nonempty, star-closed or indecomposable exactly when the other is. A surjective linear isometry from W that carries $\Phi(S|_W)$ onto a root set carries $\Phi(S)$ onto it.

PROOF SKETCH: **[new]** Theorem 2.13 assumes that the roots span the ambient space. We will state the passage to the span once, since Theorems 3.5 and 3.6 use it for S and for the models.
⟨1⟩1. $S|_W$ is a line system in W.
PROOF: The inner product of W is that of V on vectors of W, by `Submodule.coe_inner`; the conditions of the line-system definition in Section 2 are conditions on these inner products and on negation. □
⟨1⟩2. $\Phi(S|_W)$ spans W.
PROOF: This is `Submodule.span_span_coe_preimage` for the set $\Phi(S)$. □
⟨1⟩3. The identity of $\Phi(S)$ preserves inner products.
PROOF: Apply `Submodule.coe_inner` to two roots. □
⟨1⟩4. The two systems have the asserted common invariants.
PROOF: Apply Lemma (transport on spans) to the bijection of ⟨1⟩3; nonemptiness is preserved by any bijection. □
⟨1⟩5. The assertion on surjective linear isometries holds.
PROOF: The two root sets are the same vectors of W, and $W = \operatorname{span}\Phi(S)$ by definition. □
⟨1⟩6. Q.E.D.
PROOF: Steps ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4 and ⟨1⟩5 are the assertions. □

The two span declarations are in the Mathlib files `Analysis/InnerProductSpace/Subspace.lean` and `LinearAlgebra/Span/Defs.lean`.

Lemma (an equal-rank proper inclusion). ASSUME: 1. S and T are line systems in finite-dimensional real inner product spaces.
2. The spans $W_S = \operatorname{span}\Phi(S)$ and $W_T = \operatorname{span}\Phi(T)$ have the same dimension.

3. $G : W_S \to W_T$ is a linear isometry with $G(\Phi(S)) \subsetneq \Phi(T)$.

PROVE: S is not maximal within W_S.

PROOF SKETCH: **[new]** We will pull back a target root outside the image; it is a candidate of Definition 2.2 that is not a root of S.

⟨1⟩1. G is onto W_T.
PROOF: An isometry is injective, and the dimensions agree by ⟨0⟩:2; apply `LinearMap.injective_iff_surjective_of_finrank_eq_finrank`. □

⟨1⟩2. PICK $\mathbf{t} \in \Phi(T) \setminus G(\Phi(S))$ and $\mathbf{x} \in W_S$ with $G(\mathbf{x}) = \mathbf{t}$.
PROOF: The root $\mathbf{t}$ exists by ⟨0⟩:3, and $\mathbf{x}$ by ⟨1⟩1. □

⟨1⟩3. $(\mathbf{x}, \mathbf{x}) = 2$.
PROOF: G preserves inner products and $(\mathbf{t}, \mathbf{t}) = 2$. □

⟨1⟩4. For every $\mathbf{r} \in \Phi(S)$, $(\mathbf{x}, \mathbf{r}) \in \{-2, -1, 0, 1, 2\}$.
PROOF: $(\mathbf{x}, \mathbf{r}) = (\mathbf{t}, G(\mathbf{r}))$, and $\mathbf{t}, G(\mathbf{r}) \in \Phi(T)$ by ⟨0⟩:3; use the line-system definition in Section 2. □

⟨1⟩5. $\mathbf{x} \notin \Phi(S)$.
PROOF: Otherwise $\mathbf{t} = G(\mathbf{x})$ would lie in $G(\Phi(S))$, contrary to the choice of $\mathbf{t}$. □

⟨1⟩6. Q.E.D.
PROOF: Steps ⟨1⟩3, ⟨1⟩4 and ⟨1⟩5 give a vector of W_S that Definition 2.2 requires in $\Phi(S)$ for a maximal S. □

The rank declaration is in the Mathlib file `LinearAlgebra/FiniteDimensional/Lemmas.lean`.

Definition 3.1. ASSUME: ℓ is an integer, $\ell \geq 1$ for A_ℓ and $\ell \geq 4$ for D_ℓ; $(\mathbf{e}_i)$ is the standard orthonormal coordinate basis.

LET: The signed roots of the two systems are

$$\Phi(\mathrm{A}_\ell) = \{\mathbf{e}_i - \mathbf{e}_j : 1 \leq i, j \leq \ell + 1,\ i \neq j\},$$
$$\Phi(\mathrm{D}_\ell) = \{s\mathbf{e}_i + t\mathbf{e}_j : 1 \leq i < j \leq \ell,\ s, t \in \{1, -1\}\}.$$

These are line systems: each root has squared norm 2; taking negatives stays in the displayed set; pairwise products are integers of absolute value at most 2. We verify these assertions below with the other models. The inclusion $\Phi(\mathrm{A}_\ell) \subseteq \Phi(\mathrm{D}_{\ell+1})$ uses these coordinates. For $\ell < 4$, the Lean definition `Dn` still uses this coordinate formula. The classification only uses D_ℓ for $\ell \geq 4$.

Remark. The paper adds that A_3 is represented by the 6 diagonals of the cuboctahedron. This is a side claim, and we prove it in `LineGraphs/SideClaims/Cuboctahedron.lean`. The vertices of the cuboctahedron are the midpoints of the twelve edges of the cube with vertices $(\pm 1, \pm 1, \pm 1)$, and distinct edges have distinct midpoints. The midpoints are the twelve vectors with one coordinate 0 and two coordinates ± 1, which is $\Phi(\mathrm{D}_3)$ in the coordinates above. The vertex opposite to $\mathbf{v}$ is $-\mathbf{v}$, so each diagonal carries exactly two vertices, and the six diagonals are the lines of the line system D_3 (`LineSystem.cuboctahedron_eq_Dn`). This system is isomorphic to A_3 (`LineSystem.nonempty_iso_cuboctahedron_An`); in Lean the isomorphism is checked on the integer roots, by the kernel.

Definition 3.2. ASSUME: Coordinate spaces have their usual real inner products.

LET: The following three signed sets, with the indicated normalization, define the three models of E_8. The default is model (ii).

1. In $\mathbb{R}^9$, take $\Phi(\mathrm{A}_8)$ together with both signs of $\frac{1}{3}\sum_{i=1}^{9} \mathbf{e}_i - \sum_{i \in T} \mathbf{e}_i$, for $T \in \binom{[9]}{3}$.
2. In $\mathbb{R}^8$, take $\Phi(\mathrm{D}_8)$ together with $\frac{1}{2}\sum_{i=1}^{8} \epsilon(i)\mathbf{e}_i$, where $\epsilon(i) \in \{1, -1\}$ and $\prod_i \epsilon(i) = 1$.

3. In $\mathbb{R}^8$, take $\pm\sqrt{2}\mathbf{e}_i$ and $\frac{1}{\sqrt{2}}\sum_{i\in B}\epsilon(i)\mathbf{e}_i$ for all sign choices and all $B \in \mathcal{B}$, where the blocks of $\mathcal{B} = S(3,4,8)$ are

$$\begin{array}{lllllll} 1234 & 1256 & 1278 & 1357 & 1368 & 1458 & 1467 \\ 2358 & 2367 & 2457 & 2468 & 3456 & 3478 & 5678. \end{array}$$

[repair] This fixes the labelling required by Gap 5. We correct the second printed "(i)" to (ii) (Error 6), and remove the repeated sign condition (Error 13). **[new]** The factor $1/\sqrt{2}$ in (iii) changes the printed vectors of squared norm 4 to our uniform squared norm 2, without changing their lines.

Lemma (integer realizations). ASSUME: The coordinate sets are those of Definitions 3.1 and 3.2.

PROVE: They are line systems. Write $\mathbf{r} = c\mathbf{v}$ with integral $\mathbf{v}$ and $c^2q = 1$. Their presentations are as follows; each list of raw vectors has no repetitions.

model	q	c	raw vectors $\mathbf{v}$
A_ℓ	1	1	$\mathbf{e}_i - \mathbf{e}_j$ $(i \neq j)$
D_ℓ	1	1	$s\mathbf{e}_i + t\mathbf{e}_j$ $(i < j)$
E_8 (i)	9	1/3	$3(\mathbf{e}_i - \mathbf{e}_j)$, $\pm(\mathbf{1} - 3\mathbf{1}_T)$
E_8 (ii)	4	1/2	$2(s\mathbf{e}_i + t\mathbf{e}_j)$, ϵ $(\prod \epsilon = 1)$
E_8 (iii)	2	$1/\sqrt{2}$	$\pm 2\mathbf{e}_i$, $\epsilon\mathbf{1}_B$

Every raw squared norm is $2q$, and every raw product lies in $q\{-2,-1,0,1,2\}$. Each triple of points belongs to exactly one block of $\mathcal{B}$; distinct blocks intersect in 0 or 2 points.

PROOF SKETCH: **[new]** We make the integer normalization and injective indices explicit to match the Lean models. We will check coordinates before using the common integer-model transfer.

⟨1⟩1. The assertions hold for A_ℓ, D_ℓ.
  ⟨2⟩1. Every such vector has exactly two nonzero coordinates, each a sign.
    PROOF: Read the two displays in Definition 3.1. □
  ⟨2⟩2. Its squared norm is 2 and the products are integers between -2 and 2.
    PROOF: Sum the two coordinate products from ⟨2⟩1. □
  ⟨2⟩3. Supports and signs recover the displayed indices uniquely.
    PROOF: In A_ℓ the positive and negative positions give the ordered pair; in D_ℓ the sorted support gives the pair and its entries give the signs. □
  ⟨2⟩4. Q.E.D.
    PROOF: Negating the displayed signs stays in each list; use ⟨2⟩2 and ⟨2⟩3. □
⟨1⟩2. The assertions hold for model (i).
  ⟨2⟩1. For $t_T = \mathbf{1} - 3\mathbf{1}_T$, $t_T \cdot t_U = 9(|T \cap U| - 1)$.
    PROOF: Expand the indicators and use $|T| = |U| = 3$ in a 9-set. □
  ⟨2⟩2. $3(\mathbf{e}_i - \mathbf{e}_j) \cdot t_T = 9(\mathbf{1}_T(j) - \mathbf{1}_T(i))$.
    PROOF: Substitute the i, j entries of t_T. □
  ⟨2⟩3. The signed triple vectors are distinct and disjoint from the pair vectors.
    PROOF: A triple vector has three entries -2 and six entries 1; its negative has different entry values, while a pair vector has seven zero entries. □
  ⟨2⟩4. Q.E.D.
    PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, and ⟨1⟩1 scaled by 3. □
⟨1⟩3. The assertions hold for model (ii).
  ⟨2⟩1. For two parity-even sign vectors u, v, their Hamming distance d is even, and $u \cdot v = 8 - 2d \in \{-8,-4,0,4,8\}$.

PROOF: The product of the eight coordinatewise products is 1, so the number of -1 products is even. □

⟨2⟩2. $2(s\mathbf{e}_i + t\mathbf{e}_j) \cdot \epsilon = 2(s\epsilon(i) + t\epsilon(j)) \in \{-4, 0, 4\}$.
PROOF: Each summand in parentheses is a sign. □

⟨2⟩3. The pair and half-root lists are disjoint and individually injective.
PROOF: Pair roots have six zeros; half roots have none and their first seven signs determine the last by parity. □

⟨2⟩4. Q.E.D.
PROOF: Use the last three steps and ⟨1⟩1; reversing all eight signs preserves parity. □

⟨1⟩4. The assertions hold for model (iii).

⟨2⟩1. The stated block sizes, intersections, and unique-triple property hold.
PROOF: Enumerate the 14 displayed blocks and the 56 triples. These are the existing kernel results `LineSystem.steinerBlock_card`, `LineSystem.steinerBlock_inter_even`, and `LineSystem.existsUnique_subset_steiner348` in `Models.lean`. □

⟨2⟩2. Raw squared norms are 4, and products are even integers of absolute value at most 4.

⟨3⟩1. Two axes have product 0 or ± 4; an axis and a block vector have product 0 or ± 2.
PROOF: The axis has its only nonzero entry equal to ± 2. □

⟨3⟩2. Two block vectors have an even product between -4 and 4.
PROOF: By ⟨2⟩1, the product sums 0, 2, or 4 signs. □

⟨3⟩3. Q.E.D.
PROOF: The two root types are exhausted by ⟨3⟩1 and ⟨3⟩2; each self-product is 4. □

⟨2⟩3. The raw list is injective and closed under negation.
PROOF: Support size distinguishes axes from blocks; support and entries recover every index and sign. All signs are allowed. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩2 and ⟨2⟩3. □

⟨1⟩5. Q.E.D.
PROOF: Apply `LineSystem.ofIntegerRoots` and `LineSystem.inner_integerVector` to the five checked presentations. These declarations are in `LineGraphs/LineSystem/IntegerModel.lean`. □

Definition 3.3. ASSUME: E_8 is model (ii) in $\mathbb{R}^8$.
LET: $\mathbf{r}_0 = \mathbf{e}_7 - \mathbf{e}_8$, $\mathbf{s}_0 = \mathbf{e}_6 - \mathbf{e}_7$, and

$$\Phi(\mathrm{E}_7) = \{\mathbf{x} \in \Phi(\mathrm{E}_8) : x_7 = x_8\}, \qquad \Phi(\mathrm{E}_6) = \{\mathbf{x} \in \Phi(\mathrm{E}_8) : x_6 = x_7 = x_8\}.$$

Thus $\mathrm{E}_7 = \mathrm{E}_8 \cap \mathbf{r}_0^\perp$ and $\mathrm{E}_6 = \mathrm{E}_8 \cap \mathbf{r}_0^\perp \cap \mathbf{s}_0^\perp$. The roots $\mathbf{r}_0, \mathbf{s}_0, -\mathbf{r}_0 - \mathbf{s}_0$ form a star. **[repair]** We fix these choices first, and prove independence of both choices below (Gap 5); no transitivity hypothesis is part of the definition. For the printed counting argument we also name two cuts in model (iii): $T_7 = \mathrm{E}_8^{(\mathrm{iii})} \cap \mathbf{e}_8^\perp$ and $T_6 = T_7 \cap (\mathbf{e}_5 + \mathbf{e}_6 + \mathbf{e}_7 + \mathbf{e}_8)^\perp$. Orthogonal cuts are line systems by restriction of the defining conditions.

Lemma (spans of the models). ASSUME: The models have the parameter ranges in Definitions 3.1–3.3.
PROVE: Their spans and ranks are

A_ℓ	$\{\mathbf{x} \in \mathbb{R}^{\ell+1} : \sum x_i = 0\}$	ℓ
D_ℓ	$\mathbb{R}^\ell$	ℓ
E_8 (i)	$\{\mathbf{x} \in \mathbb{R}^9 : \sum x_i = 0\}$	8
E_8 (ii), (iii)	$\mathbb{R}^8$	8
E_7	$\{\mathbf{x} \in \mathbb{R}^8 : x_7 = x_8\}$	7
E_6	$\{\mathbf{x} \in \mathbb{R}^8 : x_6 = x_7 = x_8\}$	6.

PROOF SKETCH: **[repair]** We supply the dimension inputs deferred by Theorem 2.13 (Gap 4). We will give spanning vectors with explicit independent coordinates.

⟨1⟩1. The A_ℓ span is the stated hyperplane of dimension ℓ.

⟨2⟩1. Every root has coordinate sum zero.

PROOF: Its two nonzero coordinates are 1 and -1. □

⟨2⟩2. The vectors $\mathbf{e}_i - \mathbf{e}_{\ell+1}$, $1 \le i \le \ell$, form a basis of this hyperplane.

PROOF: A vector in the hyperplane is $\sum_{i=1}^{\ell} x_i(\mathbf{e}_i - \mathbf{e}_{\ell+1})$; its first ℓ coordinates make these coefficients unique. □

⟨2⟩3. Q.E.D.

PROOF: The basis roots of ⟨2⟩2 and the containment in ⟨2⟩1 give equality. □

⟨1⟩2. The D_ℓ roots span $\mathbb{R}^\ell$.

PROOF: For $j \neq i$, $\mathbf{e}_i = ((\mathbf{e}_i + \mathbf{e}_j) + (\mathbf{e}_i - \mathbf{e}_j))/2$ lies in the root span. □

⟨1⟩3. Model (i) spans the zero-sum hyperplane in $\mathbb{R}^9$.

PROOF: Its additional roots have sum $9/3 - 3 = 0$, and it contains the A_8 hyperplane of ⟨1⟩1. □

⟨1⟩4. Model (ii) spans $\mathbb{R}^8$.

PROOF: It contains D_8, whose span is given by ⟨1⟩2. □

⟨1⟩5. Model (iii) spans $\mathbb{R}^8$.

PROOF: Its eight axis roots are nonzero scalar multiples of a coordinate basis. □

⟨1⟩6. The E_7 span has basis $\mathbf{e}_1, \dots, \mathbf{e}_6, \mathbf{e}_7 + \mathbf{e}_8$.

⟨2⟩1. These seven vectors lie in the span.

PROOF: The D_6 roots on the first six coordinates give the six axes by ⟨1⟩2; $\mathbf{e}_7 + \mathbf{e}_8$ is itself an E_7 root. □

⟨2⟩2. Q.E.D.

PROOF: The displayed vectors uniquely expand every vector with $x_7 = x_8$, which contains all the roots by Definition 3.3. □

⟨1⟩7. The E_6 span has basis $\mathbf{e}_1, \dots, \mathbf{e}_5, \mathbf{e}_6 + \mathbf{e}_7 + \mathbf{e}_8$.

⟨2⟩1. The first five axes belong to the span.

PROOF: Apply ⟨1⟩2 to the D_5 roots on the first five coordinates. □

⟨2⟩2. The last basis vector belongs to the span.

PROOF: Subtract $\sum_{i=1}^{5} \mathbf{e}_i$ from twice the root $\frac{1}{2}\mathbf{1}$, using ⟨2⟩1. □

⟨2⟩3. Q.E.D.

PROOF: These six vectors uniquely expand every vector with $x_6 = x_7 = x_8$, which contains the roots by Definition 3.3. □

⟨1⟩8. Q.E.D.

PROOF: The six cases exhaust the models. The cited dimension declarations already implement these arguments in `LineGraphs/LineSystem/ModelSpans.lean`. □

Computation C2. ASSUME: The models and the fixed cuts T_7, T_6 are as defined above.
PROVE: Their line counts are $\ell(\ell+1)/2$, $\ell(\ell-1)$, 120 in each E_8 model, 63 for E_7, T_7, and 36 for E_6, T_6.

PROOF SKETCH: We will count injective signed indices and divide by 2.

⟨1⟩1. A_ℓ has $\ell(\ell+1)/2$ lines.

PROOF: There are $(\ell+1)\ell$ ordered distinct pairs in Definition 3.1; injectivity is Lemma (integer realizations). □

⟨1⟩2. D_ℓ has $\ell(\ell-1)$ lines.

PROOF: There are $4\binom{\ell}{2}$ signed indices; divide by 2. □

⟨1⟩3. Model (i) has $36 + 84 = 120$ lines.

PROOF: Its disjoint types contribute $\binom{9}{2}$ and $\binom{9}{3}$ antipodal pairs. □

⟨1⟩4. Model (ii) has $56 + 64 = 120$ lines.

PROOF: Its disjoint types contribute $2\binom{8}{2}$ and $2^7/2$ antipodal pairs. □

⟨1⟩5. Model (iii) has $8 + 8 \cdot 14 = 120$ lines.

PROOF: Each of the 14 supports contributes $2^4/2$ pairs, in addition to the eight axes. □

⟨1⟩6. E_7 has $(60 + 2 + 64)/2 = 63$ lines.

⟨2⟩1. Its pair roots are D_6 and $\pm(\mathbf{e}_7 + \mathbf{e}_8)$.

PROOF: The condition $x_7 = x_8$ permits a support inside [6], or the support $\{7, 8\}$ with equal signs. □

⟨2⟩2. It has $2^5 \cdot 2 = 64$ half roots.

PROOF: The first six signs have even parity; the last two are equal with two choices. □

⟨2⟩3. Q.E.D.

PROOF: Add the two disjoint counts and divide by 2. □

⟨1⟩7. E_6 has $(40 + 32)/2 = 36$ lines.

⟨2⟩1. Its pair roots are exactly D_5 on [5].

PROOF: A two-element support cannot meet the three required equal nonzero coordinates $6, 7, 8$. □

⟨2⟩2. There are $2^5 = 32$ half roots.

PROOF: The first five signs determine the common last sign by the even-parity condition. □

⟨2⟩3. Q.E.D.

PROOF: Add the disjoint counts and divide by 2. □

⟨1⟩8. T_7 has $7 + 8 \cdot 7 = 63$ lines.

PROOF: Exactly seven blocks in Definition 3.2 avoid 8; each contributes eight lines, in addition to seven axes. □

⟨1⟩9. T_6 has $4 + 8 + 4 \cdot 6 = 36$ lines.

⟨2⟩1. The four surviving axes have indices $1, 2, 3, 4$.

PROOF: Impose $x_8 = 0$ and $x_5 + x_6 + x_7 = 0$ on an axis. □

⟨2⟩2. The block 1234 contributes eight lines.

PROOF: Both linear conditions vanish on every sign choice on this block. □

⟨2⟩3. Each of the other six blocks avoiding 8 contributes four lines.

PROOF: Each meets $\{5, 6, 7\}$ twice; those signs must be opposite. There are $2 \cdot 2^2 = 8$ signed roots per block. □

⟨2⟩4. Q.E.D.

PROOF: The three disjoint root types give the stated sum. □

⟨1⟩10. Q.E.D.

PROOF: All counts use the injectivity in Lemma (integer realizations). The common Lean transfer is `LineSystem.IntegerModel.numLines_eq`. The finite cardinal certificates `LineSystem.card_E7Index` and `LineSystem.card_E6Index` already check 126 and 72 in the kernel. □

Computation C3. ASSUME: The three models have the normalization of Definition 3.2.
PROVE: Model (i) is isometric to model (ii) by

$$U(\mathbf{x})_i = d_i(x_i + x_9/2) \quad (1 \le i \le 8), \qquad d_1 = \cdots = d_7 = 1, \quad d_8 = -1,$$

on the zero-sum hyperplane. Model (iii) is isometric to model (ii) by $Q/\sqrt{2}$, where Q is block diagonal with four blocks $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ on the coordinate pairs $12, 34, 56, 78$. Both maps carry the full signed root sets onto one another.

PROOF SKETCH: [**new**] We give explicit maps, so equivalence of the definitions does not use Theorem 3.5. In raw coordinates their formulas are rational, even though the normalization of the Steiner map introduces $\sqrt{2}$.

⟨1⟩1. U preserves inner products on the zero-sum hyperplane.

PROOF: For $\sum x_i = \sum y_i = 0$, expansion gives $\sum_{i=1}^{8}(x_i + x_9/2)(y_i + y_9/2) = \sum_{i=1}^{9} x_i y_i$. The factors $d_i^2 = 1$ do not change this sum. □

⟨1⟩2. U sends every root of model (i) to a root of model (ii).

⟨2⟩1. CASE: $\mathbf{x} = \mathbf{e}_i - \mathbf{e}_j$, with $i, j \le 8$.

PROOF: Its image is $d_i\mathbf{e}_i - d_j\mathbf{e}_j$, a D_8 root. □

⟨2⟩2. CASE: $\mathbf{x} = \mathbf{e}_i - \mathbf{e}_9$, with $i \le 8$.

PROOF: Before the factor d, the image has one entry $1/2$ and seven entries $-1/2$. Negating coordinate 8 makes its sign parity even. □

⟨2⟩3. CASE: $\mathbf{x} = \mathbf{1}/3 - \mathbf{1}_T$ and $9 \notin T$.

PROOF: Before d, the image is $\mathbf{1}/2 - \mathbf{1}_T$, with three negative signs; the coordinate-8 sign change makes their number even. □

⟨2⟩4. CASE: $\mathbf{x} = \mathbf{1}/3 - \mathbf{1}_T$ and $T = \{i, j, 9\}$.

PROOF: Its image is $-d_i\mathbf{e}_i - d_j\mathbf{e}_j$, a D_8 root. □

⟨2⟩5. Negatives of the listed roots also map to roots.

PROOF: U is linear and model (ii) is closed under negation by Lemma (integer realizations). □

⟨2⟩6. Q.E.D.

PROOF: The pair and triple alternatives in Definition 3.2 are exhausted by the five preceding steps. □

⟨1⟩3. $Q/\sqrt{2}$ preserves inner products.

PROOF: Each displayed block has transpose times itself equal to $2I_2$, so $Q^TQ = 2I_8$. □

⟨1⟩4. $Q/\sqrt{2}$ sends the roots of model (iii) into model (ii).

⟨2⟩1. An axis root $\pm\sqrt{2}\mathbf{e}_i$ maps to a D_8 root.

PROOF: The ith column of Q has two entries in $\{1, -1\}$. □

⟨2⟩2. CASE: B is the union of two pairs among $12, 34, 56, 78$.

PROOF: On each occupied pair Q sends two signs to one entry ± 2 and one zero; division by 2 gives a D_8 root. □

⟨2⟩3. CASE: B meets each of the four pairs once.

⟨3⟩1. The number of second positions selected by B is even.

PROOF: Read the eight transversal blocks in Definition 3.2:

$$\begin{array}{cccc} 1357 & 1368 & 1458 & 1467 \\ 2358 & 2367 & 2457 & 2468. \end{array}$$

□

⟨3⟩2. Q.E.D.

PROOF: Selecting a first position gives a pair of signs with product 1; selecting a second gives product -1. Step ⟨3⟩1 makes the total product 1, so $Q\mathbf{v}/2$ is a half root of model (ii). □

⟨2⟩4. Q.E.D.

PROOF: The other six blocks are $1234, 1256, 1278, 3456, 3478, 5678$, the unions in ⟨2⟩2; together with axes these exhaust the raw types. □

⟨1⟩5. Both root maps are onto.

PROOF: Each is injective by its isometry property and maps between two finite sets of 240 roots by C2, so each is surjective. □

⟨1⟩6. Q.E.D.

PROOF: Use ⟨1⟩5 and the span descriptions in Lemma (spans of the models). Equivalently, the induced index bijections satisfy the hypothesis of the existing constructor `LineSystem.Iso.ofIntegerModels`. □

For a kernel certificate the raw maps are $\mathbf{v} \mapsto (d_i(2v_i + v_9)/3)_{i=1}^{8}$ and $\mathbf{v} \mapsto Q\mathbf{v}$. The 240 images in each case are integral. In Lean (`Computations/C3.lean`) the kernel checks that the

lists of images are permutations of the roots of model (ii), and checks the scaled Gram equalities; the script `blueprint/checks/sec3/check_sec3.py` checks the images, their bijections, and the scaled Gram equalities in exact arithmetic.

Lemma (a connected spanning set). ASSUME: S is a line system in a finite-dimensional real inner product space; $B \subseteq \Phi(S)$ is nonempty, spans $\operatorname{span}\Phi(S)$, and its graph joining distinct roots of nonzero inner product is connected.
PROVE: S is indecomposable.

PROOF SKETCH: **[new]** We isolate the spanning criterion for the model verifications. An orthogonal partition cannot separate an edge of B, or a root from every vector of a spanning set.
⟨1⟩1. ASSUME: $P \cup Q = \Phi(S)$ and $P \perp Q$.
PROVE: $P = \varnothing$ or $Q = \varnothing$.
⟨2⟩1. No root belongs to both parts.
PROOF: Such a root would have self-product both 0 and 2. □
⟨2⟩2. The endpoints of every edge of B lie in the same part.
PROOF: Different parts have inner product zero by ⟨1⟩. □
⟨2⟩3. All of B lies in one part.
PROOF: Choose a vertex of the nonempty B and propagate its part along a path to each vertex, using ⟨2⟩2. □
⟨2⟩4. The other part is empty.
⟨3⟩1. Every vector of that part is orthogonal to the whole span.
PROOF: It is orthogonal to B by ⟨1⟩ and ⟨2⟩3; extend by the finite-span lemma in Section 2. □
⟨3⟩2. Q.E.D.
PROOF: A root in that part is in the span, so ⟨3⟩1 would give its self-product as 0 instead of 2. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4, whichever part contains B. □
⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1 is the indecomposability definition in Section 2. □

Computation C4. ASSUME: We use the models in their specified ranges, and $X = \operatorname{Orth}(\Gamma_a)$ has the convention of Hypothesis 2.4: distinct vertices are adjacent when their inner product is 0.
PROVE: Every model is nonempty, star-closed, and indecomposable. The model A_1 has one line and rank 1. For the marked star $\mathbf{a} = \mathbf{e}_1 - \mathbf{e}_2$, $\mathbf{b} = \mathbf{e}_2 - \mathbf{e}_3$ the following data hold:

model	rank	Γ_a or its graph	$\lvert\Gamma_a\rvert$	$\lvert\Delta\rvert/2$
A_ℓ $(\ell \geq 2)$	ℓ	$\{\mathbf{e}_j - \mathbf{e}_3 : 4 \leq j \leq \ell+1\}$	$\ell - 2$	not needed
D_ℓ	ℓ	$\{\mathbf{e}_1 + \mathbf{e}_2\} \cup \{-\mathbf{e}_3 \pm \mathbf{e}_j : 4 \leq j \leq \ell\}$	$2\ell - 5$	not needed
E_6	6	strongly regular $(9, 4, 1, 2)$	9	6
E_7	7	strongly regular $(15, 6, 1, 3)$	15	15
E_8 (ii)	8	strongly regular $(27, 10, 1, 5)$	27	36

The A graph is empty and the D graph is the friendship graph of $\ell - 3$ triangles. The three exceptional rows hold for *every* ordered pair $\mathbf{a}, \mathbf{b}$ of roots with inner product -1 in the corresponding fixed model. The signed permutation σ with $\sigma(\mathbf{e}_1) = \mathbf{e}_2$, $\sigma(\mathbf{e}_2) = -\mathbf{e}_1$, $\sigma(\mathbf{e}_3) = -\mathbf{e}_3$, fixing the other coordinate vectors, is an automorphism of D_ℓ, E_8, E_7 and E_6 that carries $\mathbf{a}, \mathbf{b}$ to the ordered star $\mathbf{a}' = \mathbf{e}_1 + \mathbf{e}_2$, $\mathbf{b}' = -\mathbf{e}_1 + \mathbf{e}_3$. For this star, the graph $\operatorname{Orth}(\Gamma_a)$ of E_6 is the 3×3 lattice graph (adjacency in one common coordinate), and that of E_7 is the disjointness graph on $\binom{[6]}{2}$, the complement of $T(6)$.

Finally, the isometries of each exceptional model are transitive on its ordered stars: for $t \in \{6,7,8\}$ and $\mathbf{v},\mathbf{w} \in \Phi(\mathrm{E}_t)$ with $(\mathbf{v},\mathbf{w}) = -1$ there is a linear isometry g of $\mathbb{R}^8$ with $g(\Phi(\mathrm{E}_t)) = \Phi(\mathrm{E}_t)$, $g(\mathbf{a}') = \mathbf{v}$ and $g(\mathbf{b}') = \mathbf{w}$.

PROOF SKETCH: We will prove the classical assertions parametrically. Exceptional star data are finite tests on the complete integer root lists. Indecomposability has a short spanning-set proof, independent of any classification.

⟨1⟩1. The five default models are star-closed.

⟨2⟩1. D_ℓ is star-closed.

⟨3⟩1. ASSUME: u, v are integral norm-2 roots with $u \cdot v = -1$.
PROVE: $(u+v)\cdot(u+v) = 2$.
PROOF: Bilinearity gives $2 + 2(-1) + 2 = 2$. □

⟨3⟩2. Q.E.D.
PROOF: An integral norm-2 vector has exactly two nonzero entries, each ± 1: this is `LineSystem.exists_dnRoot_of_inner_self`. Apply it to ⟨3⟩1. □

⟨2⟩2. A_ℓ is star-closed.
PROOF: In ⟨2⟩1 the sum also has coordinate sum zero, so its two signs are opposite. The Lean criterion is `LineSystem.exists_anRoot_of_inner_self_of_sum`. □

⟨2⟩3. E_8 is star-closed.
PROOF: The existing certificate `Computations.C4.e8_star_checked` checks `Computations.checkStar` with dimension 8, target -4, the complete doubled root list, and its sound root decoder. Apply `LineSystem.IntegerModel.isStarClosed_of_checkStar`. □

⟨2⟩4. E_7 is star-closed.
PROOF: Apply `LineSystem.IsStarClosed.orthogonalTo` to ⟨2⟩3 and $\mathbf{r}_0$. □

⟨2⟩5. E_6 is star-closed.
PROOF: Apply the same named orthogonal-cut theorem to ⟨2⟩4 and $\mathbf{s}_0$. □

⟨2⟩6. Q.E.D.
PROOF: These five cases exhaust the default models. □

⟨1⟩2. The five default models are nonempty and indecomposable.
LET: $B_m = \{\mathbf{e}_i - \mathbf{e}_m : 1 \le i < m\} \cup \{\mathbf{e}_1 + \mathbf{e}_m\}$ for $m \ge 3$, and $\mathbf{h} = \frac{1}{2}\mathbf{1} \in \mathbb{R}^8$.

⟨2⟩1. A_ℓ has a nonempty connected spanning set.
PROOF: The basis $\{\mathbf{e}_i - \mathbf{e}_{\ell+1} : 1 \le i \le \ell\}$ in Lemma (spans of the models) has product 1 between distinct members; for $\ell = 1$ its graph has one vertex. □

⟨2⟩2. B_m is a connected spanning set in D_m.

⟨3⟩1. Its difference roots form a clique, and $\mathbf{e}_1 + \mathbf{e}_m$ has product -1 with $\mathbf{e}_2 - \mathbf{e}_m$.
PROOF: Compute the products from the coordinate basis, using $m \ge 3$. □

⟨3⟩2. B_m spans $\mathbb{R}^m$.
PROOF: The difference of $\mathbf{e}_1 + \mathbf{e}_m$ and $\mathbf{e}_1 - \mathbf{e}_m$ gives $2\mathbf{e}_m$; adding this recovered axis to each difference gives every other axis. □

⟨3⟩3. Q.E.D.
PROOF: Use ⟨3⟩1 and ⟨3⟩2. □

⟨2⟩3. B_8 is a connected spanning subset of E_8.
PROOF: It lies in $\mathrm{D}_8 \subseteq \mathrm{E}_8$ and has the properties in ⟨2⟩2. □

⟨2⟩4. $B_6 \cup \{\mathbf{e}_7 + \mathbf{e}_8, \mathbf{h}\}$ is a connected spanning subset of E_7.

⟨3⟩1. $\mathbf{h}$ has product 1 with both $\mathbf{e}_1 + \mathbf{e}_6 \in B_6$ and $\mathbf{e}_7 + \mathbf{e}_8$.
PROOF: Each product is $1/2 + 1/2$. □

⟨3⟩2. Q.E.D.
PROOF: Step ⟨2⟩2 spans the first six axes; adding $\mathbf{e}_7 + \mathbf{e}_8$ spans the space in Lemma (spans of the models). Step ⟨3⟩1 connects the parts. □

⟨2⟩5. $B_5 \cup \{\mathbf{h}\}$ is a connected spanning subset of E_6.
PROOF: B_5 spans the first five axes by ⟨2⟩2; $\mathbf{h}$ has nonzero last three equal coordinates and has product 1 with $\mathbf{e}_1 + \mathbf{e}_5$. Use the E_6 span in Lemma (spans of the models). □
⟨2⟩6. Q.E.D.
PROOF: Apply Lemma (a connected spanning set) to each displayed nonempty set. □
⟨1⟩3. The other two E_8 models are nonempty, star-closed, and indecomposable.
PROOF: Transport ⟨1⟩1 and ⟨1⟩2 through C3, using Lemma (transport on spans). □
⟨1⟩4. The marked A_ℓ graph has the asserted vertices and no edges.
⟨2⟩1. The equations $x_1 = x_2$ and $x_2 - x_3 = 1$ select precisely $\mathbf{e}_j - \mathbf{e}_3$ for $4 \le j \le \ell + 1$.
PROOF: A difference root cannot have two equal nonzero entries at 1 and 2, so $x_1 = x_2 = 0$ and $x_3 = -1$. □
⟨2⟩2. Q.E.D.
PROOF: Two distinct selected roots have product 1; their number is $\ell - 2$. □
⟨1⟩5. The marked D_ℓ graph is the asserted friendship graph.
⟨2⟩1. The equations select $\mathbf{e}_1 + \mathbf{e}_2$ and $-\mathbf{e}_3 \pm \mathbf{e}_j$ for $j \ge 4$.
PROOF: Either $x_1 = x_2 \neq 0$, forcing $x_1 = x_2 = 1, x_3 = 0$; or $x_1 = x_2 = 0$, forcing $x_3 = -1$ and one further sign outside [3]. □
⟨2⟩2. The vertex $\mathbf{e}_1 + \mathbf{e}_2$ is adjacent to all the others.
PROOF: Their supports are disjoint. □
⟨2⟩3. Among the other vertices, adjacency pairs the two signs of the same j.
PROOF: The product is 0 for those two signs, and 1 for different indices. □
⟨2⟩4. Q.E.D.
PROOF: Steps ⟨2⟩2 and ⟨2⟩3 give $\ell - 3$ triangles and $1 + 2(\ell - 3)$ vertices. □
⟨1⟩6. **[new]** Every ordered star in each default exceptional model has its stated row. We check all choices to avoid assuming transitivity.
LET: R be the sorted complete doubled integer root list of the model; for $a, b \in R$ with $a \cdot b = -4$, set $L(a,b) = \{v \in R : v \cdot a = 0,\ v \cdot b = 4\}$ and $D(a,b) = \{v \in R : v \cdot a = v \cdot b = 0\}$.
A row test compares the two list lengths with $m, 2\delta$; for each $u \in L$ it counts $v \in L$ with $u \cdot v = 0$ and compares with d; for each distinct $u, v \in L$ it counts $w \in L$ with $u \cdot w = v \cdot w = 0$ and compares with λ when $u \cdot v = 0$, and with μ otherwise.
⟨2⟩1. These lists represent exactly Γ_a and Δ, respectively.
PROOF: Divide raw dot products by $q = 4$ using `LineSystem.inner_integerVector`; the third equation of Γ_a follows from $\mathbf{c} = -\mathbf{a} - \mathbf{b}$. Completeness is the model's `mem_rootList` theorem in `Computations/ModelTables.lean`. □
⟨2⟩2. The complete finite tests give

R	$\#(a,b)$	$\lvert L\rvert$	$\lvert D\rvert$	d	λ	μ
E_6	1440	9	12	4	1	2
E_7	4032	15	30	6	1	3
E_8	13440	27	72	10	1	5.

PROOF: Exact integer evaluation gives `true` for `Computations.checkNoSolution` on $R \times R$, where a bad pair has dot product -4 and fails the specified row test. The lengths of the filtered ordered-pair lists are the three displayed counts. No real arithmetic is evaluated. □
⟨2⟩3. The row holds for every mathematical ordered star.
PROOF: Apply `Computations.checkNoSolution_sound` to ⟨2⟩2, using the complete list identification in ⟨2⟩1. □
⟨2⟩4. Q.E.D.
PROOF: Adjacency is exactly a zero raw product; divide the length of each list $D(a,b)$ by 2 to count the lines of Δ. Thus ⟨2⟩3 gives the asserted parameters. □
⟨1⟩7. **[new]** The map σ is an automorphism of D_ℓ, E_8, E_7 and E_6, and $\sigma(\mathbf{a}) = \mathbf{a}'$, $\sigma(\mathbf{b}) = \mathbf{b}'$.

⟨2⟩1. σ permutes $\Phi(\mathrm{D}_\ell)$ and the pair roots of model (ii).
PROOF: A signed permutation of coordinates carries $s\mathbf{e}_i + t\mathbf{e}_j$ to a vector of the same form, in Definitions 3.1 and 3.2. □
⟨2⟩2. σ permutes the half roots of model (ii).
PROOF: It permutes the entries of $\frac{1}{2}\sum \epsilon(i)\mathbf{e}_i$ and changes two signs, so the product condition of Definition 3.2(ii) is kept. □
⟨2⟩3. σ permutes $\Phi(\mathrm{E}_7)$ and $\Phi(\mathrm{E}_6)$.
PROOF: It fixes the coordinates 6, 7 and 8, so it keeps the equations of Definition 3.3; use ⟨2⟩1 and ⟨2⟩2. □
⟨2⟩4. $\sigma(\mathbf{a}) = \mathbf{a}'$ and $\sigma(\mathbf{b}) = \mathbf{b}'$.
PROOF: $\sigma(\mathbf{e}_1 - \mathbf{e}_2) = \mathbf{e}_2 + \mathbf{e}_1$ and $\sigma(\mathbf{e}_2 - \mathbf{e}_3) = -\mathbf{e}_1 + \mathbf{e}_3$. □
⟨2⟩5. Q.E.D.
PROOF: The map σ is an orthogonal linear map; the four steps give the assertions. □
⟨1⟩8. For the ordered star $\mathbf{a}' = \mathbf{e}_1 + \mathbf{e}_2$, $\mathbf{b}' = -\mathbf{e}_1 + \mathbf{e}_3$, the orthogonality graphs on Γ_a of E_6 and E_7 are the lattice graph and the six-symbol disjointness graph.
LET: A_{E_6} and A_{E_7} be the integer adjacency tables `e6Adjacency` and `e7Adjacency` of `Computations/C4E6Data.lean` and `Computations/C4E7Data.lean`, and let A_{lat} and A_{dis} be the tables of adjacency in one common coordinate on $\{0,1,2\}^2$ and of disjointness on $\binom{[6]}{2}$, in lexicographic order.
⟨2⟩1. The roots $\mathbf{a}', \mathbf{b}'$ are the marked star of the Lean term `LineSystem.E6.hyp24`.
PROOF: These are `LineSystem.E6.hyp24_a` and `LineSystem.E6.hyp24_b`, in zero-based coordinates. □
⟨2⟩2. The roots $\mathbf{a}', \mathbf{b}'$ are the marked star of the Lean term `LineSystem.E7.hyp24`.
PROOF: These are `LineSystem.E7.hyp24_a` and `LineSystem.E7.hyp24_b`, in zero-based coordinates. □
⟨2⟩3. The graph $\mathrm{Orth}(\Gamma_a)$ of E_6 for this star is isomorphic to the graph decoded from A_{E_6}.
PROOF: This is `LineSystem.E6.orthGraphIso`, by ⟨2⟩1. Its target `LineSystem.Hyp24.orthGraph` has adjacency at inner product 0, by `LineSystem.Hyp24.orthGraph_adj`. □
⟨2⟩4. The graph $\mathrm{Orth}(\Gamma_a)$ of E_7 for this star is isomorphic to the graph decoded from A_{E_7}.
PROOF: This is `LineSystem.E7.orthGraphIso`, by ⟨2⟩2, with the same adjacency `LineSystem.Hyp24.orthGraph_adj`. □
⟨2⟩5. The graph decoded from A_{E_6} is isomorphic to the lattice graph on $\{0,1,2\}^2$.
PROOF: Apply `Computations.nonempty_graphOfMatrix_iso` to `Computations.checkGraphIso` on A_{E_6}, A_{lat} and a nine-entry permutation; in Lean it is the labelling `LineSystem.E.latticeLabel` of `Computations/C4Graphs.lean`. □
⟨2⟩6. The graph decoded from A_{E_7} is isomorphic to disjointness on $\binom{[6]}{2}$.
PROOF: Apply `Computations.nonempty_graphOfMatrix_iso` to `Computations.checkGraphIso` on A_{E_7}, A_{dis} and a fifteen-entry permutation; in Lean it is the labelling `LineSystem.E.pairLabel` of the same file. □
⟨2⟩7. The E_6 assertion holds.
PROOF: Compose the isomorphisms of ⟨2⟩3 and ⟨2⟩5. □
⟨2⟩8. The E_7 assertion holds.
PROOF: Compose the isomorphisms of ⟨2⟩4 and ⟨2⟩6. □
⟨2⟩9. Q.E.D.
PROOF: Steps ⟨2⟩7 and ⟨2⟩8 are the two assertions. □
⟨1⟩9. **[new]** For $t \in \{6,7,8\}$ the isometries of $\mathbb{R}^8$ preserving $\Phi(\mathrm{E}_t)$ are transitive on the ordered stars of E_t, as stated.

PROOF: This is `LineSystem.E6.exists_isometry_star` and its analogues for E_7 and E_8, whose fixed star is $\mathbf{a}', \mathbf{b}'$ by `LineSystem.E8.hyp24_a`, `LineSystem.E8.hyp24_b` and their analogues; their trees of reflections are checked by `decide +kernel`. □

⟨1⟩10. The ranks are ℓ, ℓ, 6, 7, 8, and A_1 has one line and rank 1.

PROOF: Read the ranks from Lemma (spans of the models); the line of A_1 is $\mathbf{e}_1 - \mathbf{e}_2$ by Definition 3.1. □

⟨1⟩11. Q.E.D.

PROOF: Combine ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, ⟨1⟩7, ⟨1⟩8, ⟨1⟩9, and ⟨1⟩10. None uses Theorem 2.13 or 3.5. □

The two classical closure declarations are in `LineSystem/ClassicalModels.lean`; the three exceptional closure declarations are in `Computations/E8.lean`. Computation C4 is proved in Lean, for the marked star and for every ordered star of the exceptional models. The indecomposability theorems are in `Computations/C4.lean`, `Computations/C4Classical.lean` and `Computations/C4E6.lean` to `Computations/C4E8.lean`; the marked stars `hyp24`, the isomorphisms `orthGraphIso`, the counts `card_Γa` and `card_Δ_div_two`, and the strongly regular parameters `orthGraph_isSRGWith` are in `Computations/C4An.lean`, `Computations/C4Dn.lean` and `Computations/C4E6.lean` to `Computations/C4E8.lean`. The Lean star of A_ℓ and of D_ℓ is our $\mathbf{a}, \mathbf{b}$ in zero-based coordinates, $\mathbf{e}_0 - \mathbf{e}_1$, $\mathbf{e}_1 - \mathbf{e}_2$. The Lean star `hyp24` of E_6, E_7 and E_8 is $\mathbf{a}', \mathbf{b}'$, the image of our star under σ; our star itself is `hyp24Simple`, and `Computations.C4.marked_star_data` records the table above for our star in all five rows. For every ordered star, a reflection in a root of a star-closed system is an automorphism, `LineSystem.rootReflection_mem_isometryGroup` in `LineSystem/Reflections.lean`, and two reflection trees for each exceptional model, checked by `decide +kernel` in `Computations/C4E6Stars.lean` to `Computations/C4E8Stars.lean`, give transitivity on ordered stars: `LineSystem.E6.exists_isometry_star`, `LineSystem.E7.exists_isometry_star` and `LineSystem.E8.exists_isometry_star`. For every `H : Hyp24` with `H.S` one of the three models, `LineSystem.Hyp24.orthGraph_isSRGWith_of_S_eq_E6`, `LineSystem.Hyp24.card_Γa_of_S_eq_E6` and `LineSystem.Hyp24.card_Δ_of_S_eq_E6`, with their E_7 and E_8 analogues, give the exceptional rows; Lean counts Δ as $12, 30, 72$ roots, twice the entries of the last column. The bundle `Computations.C4.exceptional_star_data` collects them. The graph labels are the isomorphisms `LineSystem.E6.latticeGraphIso`, from the line graph of $K_{3,3}$, and `LineSystem.E7.triangularGraphComplIso`, from the complement of the line graph of K_6, both for the star $\mathbf{a}', \mathbf{b}'$, in `Computations/C4Graphs.lean`; their every-star forms are `LineSystem.Hyp24.nonempty_latticeGraph_iso_of_S_eq_E6` and `LineSystem.Hyp24.nonempty_triangularGraph_compl_iso_of_S_eq_E7`. In Lean the transitivity replaces the transfer by σ. All these isomorphisms are for the orthogonality graph; the bridge `LineSystem.IntegerModel.adjMatrix_eq_dotAdjacency` builds the graph at inner product 1, and we do not use it. The script `blueprint/checks/sec3/check_sec3.py` recomputes the finite data; no Lean proof evaluates a spectrum.

The source lists have the membership theorems `LineSystem.E6.mem_rootList`, `LineSystem.E7.mem_rootList`, and `LineSystem.E8viaD8.mem_rootList`. Sorting does not change membership; the lists R, L, D in C4 contain each raw vector once.

Lemma (independence of the perpendicular choices). ASSUME: E_8 is the default model in $\mathbb{R}^8$; $\mathbf{r} \in \Phi(E_8)$ and, for the star assertion, $\mathbf{s} \in \Phi(E_8)$ with $(\mathbf{r}, \mathbf{s}) = -1$.

PROVE: There is an automorphism of E_8, induced by a linear isometry of $\mathbb{R}^8$, carrying $(\mathbf{r}, \mathbf{s})$ to $(\mathbf{r}_0, \mathbf{s}_0)$. Every root $\mathbf{r}$ also admits such an automorphism carrying it to $\mathbf{r}_0$, without an initially chosen partner. Thus the cuts perpendicular to any line or any star are

isometric to E_7 or E_6, respectively. The same assertions hold in the other two models, and $T_7 \cong E_7$, $T_6 \cong E_6$.

PROOF SKETCH: **[repair]** We fill Gap 5 before classification. C4 and Lemma 2.10 identify the marked graphs; Lemma 2.12 extends their isomorphism to the ordered stars.
⟨1⟩1. Any two ordered stars of E_8 are carried to each other by an automorphism.
⟨2⟩1. Their graphs on Γ_a are isomorphic.
PROOF: C4 gives order 27, degree 10, and star-closure and indecomposability for each instance. Apply Lemma 2.10; the positive degree excludes an empty graph and degree $10 < 26$ excludes a universal vertex. □
⟨2⟩2. Q.E.D.
PROOF: Lemma 2.12 extends this graph isomorphism to a surjective span isometry carrying the ordered stars to each other. □
⟨1⟩2. Every root has a partner of inner product -1.
⟨2⟩1. Some root outside $\{\mathbf{r}, -\mathbf{r}\}$ has nonzero product with $\mathbf{r}$.
PROOF: Otherwise this antipodal pair and its nonempty complement would be an orthogonal partition. This contradicts C4; the complement is nonempty since C2 gives 120 lines. □
⟨2⟩2. Q.E.D.
PROOF: The product in ⟨2⟩1 is 1 or -1 by Section 2; negate the partner when the product is 1. □
⟨1⟩3. The default-model perpendicular cuts have the asserted isomorphisms.
⟨2⟩1. The cut perpendicular to $\mathbf{r}$ maps onto E_7.
PROOF: Use ⟨1⟩2 and ⟨1⟩1 to send $\mathbf{r}$ to $\mathbf{r}_0$; an isometry preserves the equation $(\mathbf{x}, \mathbf{r}) = 0$. □
⟨2⟩2. The cut perpendicular to the ordered star maps onto E_6.
PROOF: Step ⟨1⟩1 preserves the two equations of orthogonality to $\mathbf{r}, \mathbf{s}$; orthogonality to $-\mathbf{r}-\mathbf{s}$ is their sum. □
⟨2⟩3. Q.E.D.
PROOF: These are the two cuts in Definition 3.3. □
⟨1⟩4. In model (iii), T_7 and T_6 are cuts of the required kinds.
PROOF: Take $\mathbf{a} = \sqrt{2}\mathbf{e}_8$ and $\mathbf{b} = -(\mathbf{e}_5 + \mathbf{e}_6 + \mathbf{e}_7 + \mathbf{e}_8)/\sqrt{2}$. Here $\mathbf{a}$ is an axis root and $\mathbf{b}$ is a root by block 5678, and their product is -1. Their two perpendicular equations are exactly those defining T_7, T_6. □
⟨1⟩5. Q.E.D.
PROOF: C3 carries any root or ordered star in either other model to the default model. Apply ⟨1⟩3, including the cuts identified by ⟨1⟩4. □

Lemma 3.4. ASSUME: The model parameters are $\ell \geq 1$ for A_ℓ and $\ell \geq 4$ for D_ℓ; the exceptional systems have the definitions above.
PROVE: **[repair]** With $|A_\ell|$ in place of the printed $|A|$ (Error 13),

$$|A_\ell| = \tfrac{1}{2}\ell(\ell+1), \quad |D_\ell| = \ell(\ell-1), \quad |E_8| = 120, \quad |E_7| = 63, \quad |E_6| = 36.$$

These are counts of lines, independent of the model and of the perpendicular choices.

PROOF SKETCH: We will use the signed counts of C2, and then justify the paper's use of model (iii) for the perpendicular systems.
⟨1⟩1. The first two formulas hold.
PROOF: Apply the first two counts of C2 in the stated parameter ranges. □
⟨1⟩2. All three definitions of E_8 have 120 lines.
PROOF: C2 gives $36 + 84 = 56 + 64 = 8 + 8 \cdot 14 = 120$. □
⟨1⟩3. Every perpendicular-line system has 63 lines.

PROOF: Lemma (independence of the perpendicular choices) identifies it with T_7; C2 gives $7 + 8 \cdot 7 = 63$. □

⟨1⟩4. Every perpendicular-star system has 36 lines.

PROOF: Lemma (independence of the perpendicular choices) identifies it with T_6; C2 gives $4 + 8 + 4 \cdot 6 = 36$. □

⟨1⟩5. Q.E.D.

PROOF: These four steps cover all five types and all allowed choices. The seven named count declarations already exist in `LineSystem/Models.lean`. □

Theorem 3.5. ASSUME: 1. V is a finite-dimensional real inner product space.
2. S is a nonempty line system in V, and $W = \operatorname{span}\Phi(S)$.

PROVE: S is indecomposable and star-closed if and only if it is isomorphic to exactly one of A_ℓ ($\ell \geq 1$), D_ℓ ($\ell \geq 4$), E_6, E_7, E_8. The isomorphism is induced by a surjective linear isometry of spans. All three definitions of E_8 give that same type.

[new] In Lean the list is an index type `LineSystem.RootType`, with constructors for A_ℓ with $1 \leq \ell$ and D_ℓ with $4 \leq \ell$, and three constants for E_6, E_7, E_8. Each type t has an ambient dimension `t.dim`, a model `t.model`, a line system in the Euclidean space of that dimension, and a rank `t.rank`. The Lean statement then says that S is indecomposable and star-closed if and only if there is a unique t with an isomorphism from S to `t.model`. Theorem 3.6 uses the same type, with A_7, A_8 and D_8 excluded.

PROOF SKETCH: The models realize the five possibilities of Theorem 2.13. Its inputs from this section are C2, C4, and the span calculations, which have all been established without either classification theorem. We will include the one-line case. **[new]** The uniqueness assertion of Theorem 2.13 is stated for systems spanning $\mathbb{R}^\ell$. We will restrict S and the realizing model to their spans, map both spans to $\mathbb{R}^\ell$ through orthonormal bases, apply uniqueness there, and compose.

⟨1⟩1. ASSUME: S is indecomposable and star-closed.
PROVE: There are a listed model Y and a surjective linear isometry $F : W \to \operatorname{span}\Phi(Y)$ with $F(\Phi(S)) = \Phi(Y)$.

LET: $\ell = \dim W$, and let $S|_W$ be the restriction of S to W in Lemma (restriction to the span).

⟨2⟩1. $S|_W$ satisfies the hypotheses of Theorem 2.13, with W as the ambient space of dimension ℓ.

PROOF: Lemma (restriction to the span) makes $S|_W$ a line system spanning W, nonempty by ⟨0⟩:2 and indecomposable and star-closed by ⟨1⟩; W is finite-dimensional by ⟨0⟩:1. □

⟨2⟩2. $S|_W$ lies in one of the five cases of Theorem 2.13.

PROOF: Apply Theorem 2.13 to ⟨2⟩1. □

⟨2⟩3. A listed model Y of rank ℓ has the number of lines of that case and, when that case has more than one line, an ordered star with the data on Γ_a and Δ of that case.

⟨3⟩1. CASE: Case 1 of Theorem 2.13.

PROOF: Take $Y = \mathrm{A}_\ell$: C4 gives rank ℓ, the empty graph on $\ell - 2$ vertices for $\ell \geq 2$, and one line for $\ell = 1$; Lemma 3.4 gives $\ell(\ell+1)/2$ lines. □

⟨3⟩2. CASE: Case 2 of Theorem 2.13, so $\ell \geq 4$.

PROOF: Take $Y = \mathrm{D}_\ell$: C4 gives rank ℓ and the friendship graph of $\ell - 3$ triangles; Lemma 3.4 gives $\ell(\ell-1)$ lines. □

⟨3⟩3. CASE: Case 3 of Theorem 2.13, so $\ell = 8$.

PROOF: Take $Y = \mathrm{E}_8$: C4 gives rank 8, $|\Gamma_a| = 27$ and $|\Delta|/2 = 36$; Lemma 3.4 gives 120 lines. □

⟨3⟩4. CASE: Case 4 of Theorem 2.13, so $\ell = 7$.

PROOF: Take $Y = \mathrm{E}_7$: C4 gives rank 7, $|\Gamma_a| = 15$ and $|\Delta|/2 = 15$; Lemma 3.4 gives 63 lines. □

⟨3⟩5. CASE: Case 5 of Theorem 2.13, so $\ell = 6$.
PROOF: Take $Y = \mathrm{E}_6$: C4 gives rank 6, $|\Gamma_a| = 9$ and $|\Delta|/2 = 6$; Lemma 3.4 gives 36 lines. □

⟨3⟩6. Q.E.D.
PROOF: These are the five cases in ⟨2⟩2. □

LET: $W_Y = \operatorname{span}\Phi(Y)$, and let $Y|_{W_Y}$ be the restriction of Y to W_Y in Lemma (restriction to the span).

⟨2⟩4. $Y|_{W_Y}$ satisfies the hypotheses of Theorem 2.13, with W_Y as the ambient space of dimension ℓ.
⟨3⟩1. Y is nonempty, indecomposable and star-closed.
PROOF: This is the first assertion of C4. □
⟨3⟩2. $\dim W_Y = \ell$.
PROOF: The rank of Y is ℓ by ⟨2⟩3. □
⟨3⟩3. Q.E.D.
PROOF: Lemma (restriction to the span) makes $Y|_{W_Y}$ a line system spanning W_Y with the properties of ⟨3⟩1; its dimension is ⟨3⟩2. □

⟨2⟩5. $Y|_{W_Y}$ lies in the same case of Theorem 2.13 as $S|_W$.
PROOF: Lemma (restriction to the span) carries the number of lines and the marked-star data of ⟨2⟩3 from Y to $Y|_{W_Y}$, whose dimension is ℓ by ⟨2⟩4; these are the data of the case of ⟨2⟩2. □

⟨2⟩6. There are surjective linear isometries $\iota : W \to \mathbb{R}^\ell$ and $\iota_Y : W_Y \to \mathbb{R}^\ell$; the sets $\iota(\Phi(S))$ and $\iota_Y(\Phi(Y))$ are the root sets of line systems S' and Y' in $\mathbb{R}^\ell$, which satisfy the hypotheses of Theorem 2.13 and lie in the case of ⟨2⟩2.
⟨3⟩1. The isometries ι and ι_Y exist.
PROOF: Take the representation `OrthonormalBasis.repr` of `stdOrthonormalBasis` of W and of W_Y, reindexed to $\mathrm{Fin}\,\ell$ by `OrthonormalBasis.reindex` along `finCongr`, using $\dim W = \ell$ and ⟨2⟩4. □
⟨3⟩2. S' and Y' are line systems in $\mathbb{R}^\ell$, and ι and ι_Y restrict to bijections of root sets that preserve inner products.
PROOF: A linear isometry is injective, commutes with negation and preserves inner products, by `LinearIsometry.inner_map_map`; the conditions of the line-system definition in Section 2 are of these kinds. □
⟨3⟩3. $\Phi(S')$ and $\Phi(Y')$ span $\mathbb{R}^\ell$.
PROOF: By `Submodule.span_image`, the span of an image is the image of the span, which is the whole of $\mathbb{R}^\ell$ for a surjective map; use ⟨2⟩1 and ⟨2⟩4. □
⟨3⟩4. Q.E.D.
PROOF: Lemma (transport on spans), applied to the bijections of ⟨3⟩2, carries nonemptiness, indecomposability, star-closure, the number of lines and the marked-star data from $S|_W$ and $Y|_{W_Y}$, which have them by ⟨2⟩1, ⟨2⟩2, ⟨2⟩4 and ⟨2⟩5; ⟨3⟩3 gives spanning. □

⟨2⟩7. There is an orthogonal transformation O of $\mathbb{R}^\ell$ with $O(\Phi(S')) = \Phi(Y')$.
PROOF: Apply the uniqueness assertion of Theorem 2.13, for the ambient space $\mathbb{R}^\ell$, to ⟨2⟩6. □

⟨2⟩8. Q.E.D.
PROOF: Put $F = \iota_Y^{-1} \circ O \circ \iota$, a surjective linear isometry $W \to W_Y$ by `LinearIsometryEquiv.symm` and `LinearIsometryEquiv.trans`; ⟨2⟩6 and ⟨2⟩7 give $F(\Phi(S)) = \Phi(Y)$. □

⟨1⟩2. ASSUME: S is isomorphic to a listed model.
PROVE: S is indecomposable and star-closed.

PROOF: C4 gives these properties for the model; apply Lemma (transport on spans). □

⟨1⟩3. Different listed types are not isomorphic.

⟨2⟩1. Isomorphic models have the same rank.

PROOF: Apply Lemma (transport on spans). □

⟨2⟩2. At a common rank $\ell \geq 4$, the D count exceeds the A count by $\ell(\ell-3)/2 > 0$.

PROOF: Subtract the formulas in Lemma 3.4. □

⟨2⟩3. At ranks $6, 7, 8$ the respective $(\mathrm{A}, \mathrm{D}, \mathrm{E})$ counts are $(21, 30, 36)$, $(28, 42, 63)$, $(36, 56, 120)$, with strict inequalities in each row.

PROOF: Substitute those ranks in Lemma 3.4. □

⟨2⟩4. Q.E.D.

PROOF: Line count is an isomorphism invariant by Lemma (transport on spans); ⟨2⟩1–⟨2⟩3 distinguish every pair of listed types. □

⟨1⟩4. The system S cannot be isomorphic to two different listed types.

PROOF: Compose one span isometry with the inverse of the other from Lemma (transport on spans). The resulting isometry between the two models contradicts ⟨1⟩3. □

⟨1⟩5. Q.E.D.

PROOF: Steps ⟨1⟩1, ⟨1⟩2, and ⟨1⟩4 prove the equivalence and uniqueness. C3 identifies the three E_8 models independently. □

Lemma (a maximal extension in the span). ASSUME: S is a nonempty indecomposable line system in a finite-dimensional real inner product space, $W = \operatorname{span}\Phi(S)$, and S is not maximal in W.

PROVE: There is a line system T in W with $\Phi(S) \subsetneq \Phi(T)$, spanning W, that is maximal in W, indecomposable, and star-closed.

PROOF SKETCH: **[repair]** For the second omission in Gap 13 we will bound all candidate roots by their finite signatures against $\Phi(S)$, maximize the size of an extension, and prove indecomposability using the equality of spans.

LET: $K = \{\mathbf{x} \in W : (\mathbf{x},\mathbf{x}) = 2,\ (\mathbf{x},\mathbf{r}) \in \{-2,-1,0,1,2\} \text{ for every } \mathbf{r} \in \Phi(S)\}$.

⟨1⟩1. K is finite.

⟨2⟩1. The signature map $\mathbf{x} \mapsto ((\mathbf{x},\mathbf{r}))_{\mathbf{r}\in\Phi(S)}$ is injective on K.

⟨3⟩1. Equal signatures make $\mathbf{x}-\mathbf{y}$ orthogonal to W.

PROOF: Use bilinearity and the finite-span lemma in Section 2. □

⟨3⟩2. Q.E.D.

PROOF: Since $\mathbf{x}-\mathbf{y} \in W$, ⟨3⟩1 gives zero squared norm, so $\mathbf{x} = \mathbf{y}$ by Section 2. □

⟨2⟩2. Q.E.D.

PROOF: The signatures lie in a set of $5^{|\Phi(S)|}$ functions. Apply `Set.Finite.of_injOn` to ⟨2⟩1. □

⟨1⟩2. Choose a largest-cardinality subset $\Psi \subseteq K$ containing $\Phi(S)$ that is the signed root set of a line system.

PROOF: The eligible subsets form a nonempty finite family: it contains $\Phi(S)$ and is contained in the powerset of the finite K. Apply `Finset.exists_max_image` to their cardinalities. □

LET: T be this line system with root set Ψ.

⟨1⟩3. T is maximal within W.

⟨2⟩1. ASSUME: $\mathbf{x} \in W$ has squared norm 2 and allowed products with Ψ.
PROVE: $\mathbf{x} \in \Psi$.

⟨3⟩1. $\mathbf{x}, -\mathbf{x} \in K$.

PROOF: The candidate products include those with $\Phi(S) \subseteq \Psi$; the allowed product set is closed under negation. □

⟨3⟩2. $\Psi \cup \{\mathbf{x}, -\mathbf{x}\}$ is eligible in ⟨1⟩2.

PROOF: Old-new products are allowed by ⟨2⟩, and the new-new products are ± 2; use ⟨3⟩1 for containment in K. □

⟨3⟩3. Q.E.D.

PROOF: Maximal cardinality in ⟨1⟩2 forbids a proper enlargement by ⟨3⟩2. □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨2⟩1 is Definition 2.2. □

⟨1⟩4. $\operatorname{span} \Psi = W$.

PROOF: $\Phi(S) \subseteq \Psi \subseteq W$ and $\Phi(S)$ spans W. □

⟨1⟩5. T is indecomposable.

⟨2⟩1. ASSUME: $P \cup Q = \Psi$ and $P \perp Q$.
PROVE: $P = \varnothing$ or $Q = \varnothing$.

⟨3⟩1. All roots of S lie in one of P, Q.

PROOF: Intersect the partition with $\Phi(S)$ and use its indecomposability. □

⟨3⟩2. The other part is empty.

PROOF: Each root there is orthogonal to the spanning $\Phi(S)$, thus to itself by Section 2; its norm would be zero. □

⟨3⟩3. Q.E.D.

PROOF: Use ⟨3⟩2. □

⟨2⟩2. Q.E.D.

PROOF: Apply the indecomposability definition in Section 2 to ⟨2⟩1. □

⟨1⟩6. $\Phi(S) \subsetneq \Psi$.

PROOF: Equality would make S maximal by ⟨1⟩3, contrary to the hypothesis. □

⟨1⟩7. T is star-closed.

PROOF: Lemma 2.3 applies to ⟨1⟩3. □

⟨1⟩8. Q.E.D.

PROOF: Combine ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, and ⟨1⟩7. □

The finiteness and maximum declarations used here are in `Mathlib/Data/Set/Finite/Basic.lean` and `Mathlib/Data/Finset/Max.lean`.

Lemma (marked inclusions). ASSUME: S, T are indecomposable star-closed line systems and a linear isometry of spans sends $\Phi(S)$ into $\Phi(T)$. Choose $\mathbf{a}, \mathbf{b} \in \Phi(S)$ with $(\mathbf{a}, \mathbf{b}) = -1$, and mark T by their images.

PROVE: The isometry induces an injection from the source Γ_a to the target Γ_a that preserves both adjacency and nonadjacency in their orthogonality graphs.

PROOF SKETCH: **[new]** We state the induced-embedding direction needed for C5 separately from the isomorphism assertion of Lemma 2.12. We will retain the same marked star and use the defining inner products.

⟨1⟩1. A source member of Γ_a maps into the target Γ_a.

PROOF: The three defining products of Hypothesis 2.4 are preserved, and the image is a target root. □

⟨1⟩2. The induced map is injective.

PROOF: A linear isometry is injective. □

⟨1⟩3. For two distinct source vertices, the product is zero if and only if the image product is zero.

PROOF: The isometry preserves that product. □

⟨1⟩4. Q.E.D.

PROOF: The three steps give an induced graph embedding. □

Computation C5. ASSUME: All inclusions are isometric inclusions of the signed models, in their spans.

PROVE: There are proper inclusions $A_8 \subset E_8$, $D_8 \subset E_8$, and $A_7 \subset E_7$, all of equal rank. There are no inclusions $A_\ell \subset D_\ell$ for $\ell \geq 4$, $A_6 \subset E_6$, $D_6 \subset E_6$, or $D_7 \subset E_7$.

PROOF SKETCH: We will exhibit the inclusions, then obstruct induced marked graphs using coclique sizes and degrees. The A_ℓ obstruction fills the first omission in Gap 13.

⟨1⟩1. The target marked graphs have the asserted types for every ordered star.

⟨2⟩1. Signed coordinate permutations of D_ℓ are transitive on ordered root pairs of product -1.

⟨3⟩1. Such roots share exactly one support position, with opposite signs there.

PROOF: Equal supports give product $-2, 0, 2$; disjoint supports give 0; a one-position intersection contributes its single sign product. □

⟨3⟩2. Q.E.D.

PROOF: Send the unshared position of the first root to 1, the shared position to 2, and the unshared position of the second to 3. Choose signs to obtain $\mathbf{e}_1 - \mathbf{e}_2, \mathbf{e}_2 - \mathbf{e}_3$; extend the coordinate permutation and signs arbitrarily. □

⟨2⟩2. Every E_6 marked graph is the lattice graph; every E_7 marked graph is the disjointness graph on six symbols.

PROOF: C4 gives constant order and parameters for every star. Lemma 2.10 identifies its graph with the fixed-star graph, whose labelled type is certified in C4. □

⟨2⟩3. Q.E.D.

PROOF: For D_ℓ apply C4 after ⟨2⟩1; for the two exceptional targets use ⟨2⟩2. □

⟨1⟩2. A_8 embeds properly in E_8 with equal rank.

PROOF: It is a subset of model (i); apply C3. Lemma (spans of the models) gives rank 8 on both sides, and Lemma 3.4 gives $36 < 120$ lines. □

⟨1⟩3. D_8 embeds properly in E_8 with equal rank.

PROOF: Definition 3.2(ii) includes these roots. Their ranks are 8 and their counts satisfy $56 < 120$ by the same two lemmas. □

⟨1⟩4. **[new]** A_7 embeds properly in E_7 with equal rank. An explicit coordinate inclusion replaces the paper's embedding of marked graphs.

LET: $\mathbf{h} = \frac{1}{2}\mathbf{1} \in \mathbb{R}^8$.

⟨2⟩1. $A_7 \subset E_8 \cap \mathbf{h}^\perp$.

PROOF: Its roots are $\mathbf{e}_i - \mathbf{e}_j$ in $\mathbb{R}^8$, which are D_8 roots with product zero with $\mathbf{h}$. The all-positive half root $\mathbf{h}$ belongs to E_8. □

⟨2⟩2. An E_8 automorphism carries this inclusion into E_7.

PROOF: Use the root assertion of Lemma (independence of the perpendicular choices) to send $\mathbf{h}$ to $\mathbf{r}_0$. □

⟨2⟩3. The inclusion has rank 7 on both sides and is proper.

PROOF: Lemma (spans of the models) gives the ranks; Lemma 3.4 gives $28 < 63$. □

⟨2⟩4. Q.E.D.

PROOF: Combine ⟨2⟩2 and ⟨2⟩3. □

⟨1⟩5. **[repair]**

ASSUME: $\ell \geq 4$ and an isometric inclusion $A_\ell \subset D_\ell$ exists.

PROVE: FALSE.

⟨2⟩1. The target friendship graph with $\ell - 3$ triangles contains a coclique of size $\ell - 2$.

PROOF: Mark the source by its C4 star and apply Lemma (marked inclusions) and ⟨1⟩1. □

⟨2⟩2. A friendship graph with $t \geq 1$ triangles has no coclique larger than t.

⟨3⟩1. A coclique containing the common vertex has size 1.

PROOF: Every other vertex is adjacent to that vertex. □

⟨3⟩2. A coclique avoiding the common vertex has at most one vertex from each of the t outer pairs.

PROOF: Each outer pair is an edge. □

⟨3⟩3. Q.E.D.

PROOF: These two cases give the bound t, since $t \geq 1$. □

⟨2⟩3. Q.E.D.

PROOF: Steps ⟨2⟩1 and ⟨2⟩2 would give $\ell - 2 \leq \ell - 3$. □

⟨1⟩6. ASSUME: An isometric inclusion $\mathrm{A}_6 \subset \mathrm{E}_6$ exists.
PROVE: FALSE.

⟨2⟩1. The 3×3 lattice graph contains a 4-coclique.

PROOF: C4 gives a 4-coclique for A_6; use Lemma (marked inclusions) and ⟨1⟩1. □

⟨2⟩2. Q.E.D.

PROOF: Among four grid positions two share one of the three rows, so those two are adjacent. This contradicts ⟨2⟩1. □

⟨1⟩7. ASSUME: An isometric inclusion $\mathrm{D}_6 \subset \mathrm{E}_6$ exists.
PROVE: FALSE.

PROOF: The centre in the source marked graph has six neighbours by C4. Its image has at least six distinct neighbours by Lemma (marked inclusions); C4 gives target degree 4. □

⟨1⟩8. ASSUME: An isometric inclusion $\mathrm{D}_7 \subset \mathrm{E}_7$ exists.
PROVE: FALSE.

PROOF: The source centre has eight neighbours by C4; the induced injection of Lemma (marked inclusions) contradicts target degree 6. □

⟨1⟩9. Q.E.D.

PROOF: The three positive inclusions and four obstructions give the assertion. □

The coordinate inclusions already have Lean declarations `LineSystem.An_Φ_subset_E8viaA8` and `LineSystem.Dn_Φ_subset_E8`. C5 is proved in Lean, in `LineSystem/Inclusions.lean`, `Computations/C5Inclusions.lean`, `Computations/C5Dn.lean` and `Computations/C5.lean`. An isometric inclusion is `LineSystem.IsometricInclusion`; the lemma on an equal-rank proper inclusion is `LineSystem.not_isMaximal_of_proper_isometric_inclusion`, and the marked-inclusion lemma is `LineSystem.Hyp24.orthGraph_embedding_of_subset`. Lean proves $\mathrm{A}_7 \subset \mathrm{E}_7$ and $\mathrm{A}_8 \subset \mathrm{E}_8$ by explicit integer coordinate rows, `LineSystem.An.inclusionE7` and `LineSystem.An.inclusionE8`, not through C3; $\mathrm{D}_8 \subset \mathrm{E}_8$ is `LineSystem.Dn.inclusionE8`. Step `s3c5Dtransitive` is `LineSystem.Dn.exists_isometry_star`, with `LineSystem.Hyp24.nonempty_orthGraph_iso_of_S_eq_Dn`. The four obstructions are `LineSystem.not_embeds_An_Dn` (for $\ell \geq 4$), `LineSystem.not_embeds_An_E6`, `LineSystem.not_embeds_Dn_E6` and `LineSystem.not_embeds_Dn_E7`, and `Computations.C5.inclusions_and_obstructions` bundles the three proper inclusions with them. There is no finite search: the grid bound of step `s3c5Asix` is the pigeonhole argument above, and the friendship bound and the two degree contradictions are proofs above as well.

Remark. **[repair]** The paper derives $\mathrm{A}_7 \subset \mathrm{E}_7$ from the statement that a 5-coclique embeds in the triangular graph $T(6)$ (Error 5). With the orthogonality convention of C4, the graph of A_7 is a 5-coclique and the graph of E_7 is the disjointness graph $\overline{T(6)}$; the witness is a 5-coclique of $\overline{T(6)}$, equivalently a 5-clique of $T(6)$, and the five pairs $\{1, j\}$ with $2 \leq j \leq 6$ pairwise intersect. We record the correction only; C5 does not use this witness, since it proves the inclusion by an explicit coordinate inclusion.

Theorem 3.6. ASSUME: S is a nonempty indecomposable line system in a finite-dimensional real inner product space, and $W = \operatorname{span} \Phi(S)$.
PROVE: S is maximal within W if and only if it is isomorphic to one of

$$\mathrm{A}_\ell \ (\ell \geq 1,\ \ell \neq 7, 8), \qquad \mathrm{D}_\ell \ (\ell \geq 4,\ \ell \neq 8), \qquad \mathrm{E}_6,\ \mathrm{E}_7,\ \mathrm{E}_8.$$

[repair] Maximality is within the span (Implicit hypothesis 7), including for $\mathrm{A}_\ell \subset \mathbb{R}^{\ell+1}$.

PROOF SKETCH: A maximal system is star-closed. For the converse we will put any hypothetical proper extension inside an indecomposable star-closed system of the same rank, then exhaust its type using counts and C5.

⟨1⟩1. ASSUME: S is maximal within W.
PROVE: S has a type in the displayed list.

⟨2⟩1. S is star-closed.
PROOF: Apply Lemma 2.3. □

⟨2⟩2. S has a type from Theorem 3.5.
PROOF: Apply Theorem 3.5 to ⟨2⟩1 and the standing hypotheses. □

⟨2⟩3. Its type is not A_7, A_8, or D_8.

⟨3⟩1. A_7 is not maximal within its span.
PROOF: Apply Lemma (an equal-rank proper inclusion) to the proper isometric inclusion $\mathrm{A}_7 \subset \mathrm{E}_7$ of equal rank in C5. □

⟨3⟩2. A_8 is not maximal within its span.
PROOF: Apply Lemma (an equal-rank proper inclusion) to the proper isometric inclusion $\mathrm{A}_8 \subset \mathrm{E}_8$ of equal rank in C5. □

⟨3⟩3. D_8 is not maximal within its span.
PROOF: Apply Lemma (an equal-rank proper inclusion) to the proper isometric inclusion $\mathrm{D}_8 \subset \mathrm{E}_8$ of equal rank in C5. □

⟨3⟩4. Q.E.D.
PROOF: A span isometry from S to one of these models would carry maximality within W, from ⟨1⟩, to the model, by Lemma (transport on spans); this contradicts ⟨3⟩1, ⟨3⟩2 or ⟨3⟩3. □

⟨2⟩4. Q.E.D.
PROOF: Remove the three types of ⟨2⟩3 from ⟨2⟩2. □

⟨1⟩2. Every retained model is maximal in its span.

⟨2⟩1. ASSUME: M is a retained model and is not maximal in its span.
PROVE: FALSE.

⟨3⟩1. There is an indecomposable star-closed proper extension T with the same rank as M.
PROOF: C4 gives that M is nonempty and indecomposable; apply Lemma (a maximal extension in the span) to M in its span. □

⟨3⟩2. There is a proper isometric inclusion of M in a listed model of Theorem 3.5 of the same rank, with strictly more lines.
LET: $W_M = \operatorname{span}\Phi(M)$.

⟨4⟩1. T is a nonempty indecomposable star-closed line system in W_M, and $\operatorname{span}\Phi(T) = W_M$.
PROOF: Lemma (a maximal extension in the span) gives these properties with the ambient space W_M, and $\Phi(M) \subseteq \Phi(T)$ makes T nonempty. □

⟨4⟩2. There are a listed model Y and a surjective linear isometry $G : W_M \to \operatorname{span}\Phi(Y)$ with $G(\Phi(T)) = \Phi(Y)$.
PROOF: Apply the forward direction of Theorem 3.5 to T in the finite-dimensional space W_M, using ⟨4⟩1; its span is W_M. □

⟨4⟩3. G is an isometric inclusion of M in Y with $G(\Phi(M)) \subsetneq \Phi(Y)$.
PROOF: G is defined on $W_M = \operatorname{span}\Phi(M)$, it is injective, and $\Phi(M) \subsetneq \Phi(T)$ by ⟨3⟩1; apply ⟨4⟩2. □

⟨4⟩4. Y has the rank of M.
PROOF: G is a linear bijection $W_M \to \operatorname{span}\Phi(Y)$ by ⟨4⟩2. □

⟨4⟩5. Q.E.D.
PROOF: A proper inclusion of finite antipodal sets strictly increases the number of pairs; use ⟨4⟩3 and ⟨4⟩4. □
⟨3⟩3. CASE: $M = \mathrm{E}_r$ for $r \in \{6, 7, 8\}$.
PROOF: At those ranks the (A, D, E) line counts in Lemma 3.4 are $(21, 30, 36)$, $(28, 42, 63)$, and $(36, 56, 120)$. The E entry is largest in each row, contrary to ⟨3⟩2. □
⟨3⟩4. CASE: $M = \mathrm{A}_\ell$, $\ell \neq 7, 8$.
⟨4⟩1. CASE: $\ell \leq 3$.
PROOF: Theorem 3.5 has only A_ℓ at this rank, so the strict count increase in ⟨3⟩2 is impossible. □
⟨4⟩2. CASE: $\ell \geq 4$.
⟨5⟩1. The target cannot be D_ℓ.
PROOF: C5 excludes $\mathrm{A}_\ell \subset \mathrm{D}_\ell$ for every $\ell \geq 4$. □
⟨5⟩2. The target cannot be exceptional.
PROOF: Rank equality and $\ell \neq 7, 8$ leave only E_6; C5 excludes $\mathrm{A}_6 \subset \mathrm{E}_6$. □
⟨5⟩3. Q.E.D.
PROOF: The only remaining target is A_ℓ, with equal count, contrary to ⟨3⟩2. □
⟨4⟩3. Q.E.D.
PROOF: Steps ⟨4⟩1 and ⟨4⟩2 exhaust positive ranks. □
⟨3⟩5. CASE: $M = \mathrm{D}_\ell$, $\ell \geq 4$, $\ell \neq 8$.
⟨4⟩1. The target cannot be A_ℓ or D_ℓ.
PROOF: Their counts are at most that of D_ℓ, since $\ell(\ell-1) - \ell(\ell+1)/2 = \ell(\ell-3)/2 > 0$; this contradicts ⟨3⟩2. □
⟨4⟩2. The target cannot be exceptional.
PROOF: With equal ranks and $\ell \neq 8$, only $\mathrm{D}_6 \subset \mathrm{E}_6$ and $\mathrm{D}_7 \subset \mathrm{E}_7$ remain; C5 excludes both. □
⟨4⟩3. Q.E.D.
PROOF: Theorem 3.5 has no further target types. □
⟨3⟩6. Q.E.D.
PROOF: The retained model list is exhausted by ⟨3⟩3, ⟨3⟩4, and ⟨3⟩5. □
⟨2⟩2. Q.E.D.
PROOF: Step ⟨2⟩1 rules out nonmaximality of any retained model. □
⟨1⟩3. ASSUME: S is isomorphic to a retained model.
PROVE: S is maximal within W.
PROOF: Transport ⟨1⟩2 by Lemma (transport on spans). □
⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩3 give the equivalence. □

Definition (signings and the matrix convention). ASSUME: S is a line system, $W = \operatorname{span} \Phi(S)$, and I is a finite index set.
[new] We make "one vector per line" an explicit bijection.
LET: A signing is a family $\sigma : I \to \Phi(S)$ for which $(i, \epsilon) \mapsto \epsilon\sigma(i)$ is a bijection $I \times \{1, -1\} \to \Phi(S)$.
Put $N = |I|$, $M(i, j) = (\sigma(i), \sigma(j))$, and $C = M - 2I_N$.
Thus N is the number of lines. The diagonal of C is zero, and its off-diagonal entries lie in $\{0, 1, -1\}$ by Section 2. A change of signs conjugates C by a diagonal matrix of signs, and a reordering conjugates by a permutation matrix. We use I_N for the identity matrix, to distinguish it from the index set I. Every line system has a signing indexed by $\{1, \ldots, N\}$, where N is its number of lines: choose one of the two vectors on each line (`LineSystem.nonempty_signing`). Thus the equations of Theorem 3.7, which hold for every signing, are not vacuous.

Computation C6. ASSUME: R is the complete, duplicate-free list of raw signed roots from Lemma (integer realizations), with the coordinate cuts for $\mathrm{E}_7, \mathrm{E}_6$ from Definition 3.3. Write $K = \sum_{v \in R} vv^T$ and let J_m be the $m \times m$ all-one matrix.
PROVE: The following are exact integer matrix identities:

model	q	K
A_ℓ	1	$2((\ell+1)I_{\ell+1} - J_{\ell+1})$
D_ℓ	1	$4(\ell-1)I_\ell$
E_8 (i)	9	$540I_9 - 60J_9$
E_8 (ii)	4	$240I_8$
E_8 (iii)	2	$120I_8$
E_7	4	$\mathrm{diag}(144I_6, 72J_2)$
E_6	4	$\mathrm{diag}(96I_5, 32J_3)$.

PROOF SKETCH: **[new]** We replace spectral calculations by small coordinate frame matrices. The classical rows are parametric; the exceptional rows use the existing Gram checker on the columns of the raw root table.

⟨1⟩1. The A_ℓ identity holds for every $\ell \geq 1$.
⟨2⟩1. $K(i,i) = 2\ell$.
PROOF: There are ℓ difference roots with entry 1 at i and ℓ with entry -1 there. □
⟨2⟩2. For $i \neq j$, $K(i,j) = -2$.
PROOF: Only $\mathbf{e}_i - \mathbf{e}_j$ and its negative contribute; both products are -1. □
⟨2⟩3. Q.E.D.
PROOF: These entries are those of $2((\ell+1)I - J)$. □
⟨1⟩2. The D_ℓ identity holds for every $\ell \geq 4$.
⟨2⟩1. $K(i,i) = 4(\ell-1)$.
PROOF: Each support $\{i,j\}$ with $j \neq i$ has four sign choices, each contributing 1. □
⟨2⟩2. For $i \neq j$, $K(i,j) = 0$.
PROOF: The four signs on the only contributing support $\{i,j\}$ give products $1, -1, -1, 1$. □
⟨2⟩3. Q.E.D.
PROOF: The entries in the two steps give $4(\ell-1)I$. □
⟨1⟩3. The five exceptional raw identities hold.
LET: $P_i = (v_i)_{v \in R}$ be the ith column of the raw root table, in its list order. The list P has length equal to the ambient dimension.
⟨2⟩1. For model (i), `checkGram` accepts P and $540I_9 - 60J_9$.
PROOF: C6 checks `Computations.checkGram` with vector dimension 240 and nine columns, from the pair and signed-triple list of C2. □
⟨2⟩2. For model (ii), `checkGram` accepts P and $240I_8$.
PROOF: C6 applies that checker with dimension 240 and eight columns of the doubled pair and even-sign list. □
⟨2⟩3. For model (iii), `checkGram` accepts P and $120I_8$.
PROOF: C6 applies that checker with dimension 240 and eight columns of the axis and block-sign list. □
⟨2⟩4. For E_7, `checkGram` accepts P and $\mathrm{diag}(144I_6, 72J_2)$.
PROOF: C6 applies that checker with dimension 126 and eight columns of the doubled list filtered by $v_7 = v_8$. □
⟨2⟩5. For E_6, `checkGram` accepts P and $\mathrm{diag}(96I_5, 32J_3)$.
PROOF: C6 applies that checker with dimension 72 and eight columns of the doubled list filtered by $v_6 = v_7 = v_8$. □
⟨2⟩6. Q.E.D.

PROOF: For each accepted table, `Computations.checkGram_sound` gives $K(i,j) = P_i \cdot P_j = \sum_{v \in R} v_i v_j$, including diagonal entries. □

⟨1⟩4. Q.E.D.

PROOF: The two parametric cases and five column-Gram checks cover the seven rows. □

C6 is proved in Lean, in `LineSystem/Frame.lean` and `Computations/C6Classical.lean`, `Computations/C6E8.lean`, `Computations/C6E7E6.lean` and `Computations/C6.lean`. Lean states C6 as the frame property on the span, `LineSystem.IsFrame`, equivalent to the column-Gram form by `LineSystem.isFrame_iff_gram` and transported by `LineSystem.Iso.isFrame`. The two parametric rows are `Computations.C6.an_frame_matrix` and `Computations.C6.dn_frame_matrix`, giving `LineSystem.isFrame_An` and `LineSystem.isFrame_Dn`; the exceptional matrices are checked by `decide +kernel` over structured indices, not by `Computations.checkGram`, and give `LineSystem.isFrame_E`$_8$, `LineSystem.isFrame_E`$_7$ and `LineSystem.isFrame_E`$_6$. The bundle is `Computations.C6.coordinate_frame_matrices`. The checker `Computations.checkGram` is defined in `Computations/ListData.lean`, and its soundness theorem `Computations.checkGram_sound` is in `Computations/Checkers.lean`; C6 does not use them.
The frame lemma and Theorem 3.7 below are proved in Lean as well, the frame lemma with the rank of the model (`LineSystem.frame_identity`) and with the rank of the span (`LineSystem.frame_identity_span`). The three span lemmas are in Lean: `LineSystem.Iso.spanIsometry` with the transport of the predicates and of the marked sets, `LineSystem.restrictSpan` (with `LineSystem.Iso.restrictSpan` and `LineSystem.Iso.ofSpanIsometry`, the interface of the final step of Theorem 3.5) in `LineSystem/SpanTransport.lean`, and `LineSystem.exists_maximal_extension_span` in `LineSystem/SpanExtension.lean`; `LineSystem/SpanAudit.lean` prints their axioms. The Lean proofs of C5 and C6 transport along `LineSystem.Iso` and use the model's rank.
Theorems 3.5 and 3.6 are in Lean. Theorem 3.5 is `LineSystem.classification` in `LineSystem/Classification.lean`, with its forward direction `LineSystem.exists_iso_rootType`; Theorem 3.6 is `LineSystem.classification_of_isMaximal` in `LineSystem/ClassificationMaximal.lean`, with the form that Section 4 uses, `LineSystem.IsMaximal.exists_iso_rootType`. Both rest on the unconditional Theorem 2.13, `LineSystem.five_possibilities'`, whose inputs from Section 3 are the packaged facts `LineSystem.modelInputs`, proved from C2 and C4 without either classification, in `LineSystem/ClassificationModels.lean`. Lemma (a connected spanning set) is `LineSystem.isIndecomposable_of_connected_spanning` in `LineSystem/ClassificationConnected.lean`, and the perpendicular choices of Lemma (independence of the perpendicular choices) in the default model are `LineSystem.E`$_8$`.nonempty_iso_E`$_7$`_orthogonalTo`, `LineSystem.E`$_8$`.nonempty_iso_E`$_6$`_orthogonalTo` and `LineSystem.E`$_8$`.exists_isometry_of_ordered_star` in `LineSystem/ClassificationChoices.lean`. The Lean departs from the proofs above in three places. The ambient space of Theorems 3.5 and 3.6 need not be finite-dimensional, since the span of the finite root set is, and Theorem 2.13 is applied there. The forward direction of Theorem 3.5 is shorter: the isomorphisms of Theorem 2.13 go directly to the named models and are composed with `LineSystem.Iso.restrictSpan`, so steps `s335coords` and `s335orthogonal` of the proof below are not needed. The retained list of Theorem 3.6 is stated as an explicit disjunction of the five families.
The two remaining pieces are in Lean as well, and their modules are imported by `LineGraphs.lean`. The bundles `Computations.C2.model_presentations` and `Computations.C2.line_counts` of Lemma (integer realizations) and C2 are in `Computations/C2.lean`, with the cuts `LineSystem.T`$_7$ and `LineSystem.T`$_6$ of Definition 3.3; Lean checks their 126 and 72 roots in the kernel, as for E_7 and E_6. The assertions of Lemma (independence of the

perpendicular choices) for the other two E_8 models, with $T_7 \cong \mathrm{E}_7$ and $T_6 \cong \mathrm{E}_6$, are in `LineSystem/ClassificationChoicesModels.lean`. In the final step the Lean departs from the proof above: a C3 isomorphism is a bijection of roots, not a map of the ambient space, so Lean carries the automorphisms of the default model across it through the span isometry of Lemma (transport on spans), extended to the whole space (for model (i), from the zero-sum hyperplane to $\mathbb{R}^9$). The file `Computations/C2ChoicesAudit.lean` prints the axioms of all these declarations.

Lemma (the frame identity within the span). ASSUME: S is one of the listed models, or an isometric copy of it; $W = \operatorname{span}\Phi(S)$ has rank $\ell > 0$ and N lines. Let $\sigma : I \to \Phi(S)$ be any signing.

PROVE: For every $\mathbf{x} \in W$,

$$\sum_{\mathbf{r}\in\Phi(S)} (\mathbf{r},\mathbf{x})\mathbf{r} = \frac{4N}{\ell}\mathbf{x}, \qquad \sum_{i\in I}(\sigma(i),\mathbf{x})\sigma(i) = \frac{2N}{\ell}\mathbf{x}.$$

[repair] These formulas are asserted on W, not on a larger coordinate space (Implicit hypothesis 7 and the model-span correction).

PROOF SKETCH: We will divide C6 by the raw scale q, restrict to the known span, and halve the sum using antipodal pairs.

⟨1⟩1. For a coordinate model, the full signed-root operator is K/q.
PROOF: Substitute $\mathbf{r} = c\mathbf{v}$ in the sum, using $c^2 = 1/q$. The exact no-multiplicity transfer is `LineSystem.IntegerModel.sum_Φ`. □

⟨1⟩2. The operator is scalar on W, with scalar $4N/\ell$.
⟨2⟩1. On the A_ℓ span, $K/q = 2(\ell+1)I$.
PROOF: $J\mathbf{x} = 0$ on the zero-sum span in Lemma (spans of the models); use the first row of C6. □
⟨2⟩2. On the D_ℓ span, $K/q = 4(\ell-1)I$.
PROOF: Its C6 matrix has that value on the whole coordinate space. □
⟨2⟩3. On model (i)'s span, $K/q = 60I$.
PROOF: Its span has sum zero, so $(540I - 60J)\mathbf{x}/9 = 60\mathbf{x}$. □
⟨2⟩4. On model (ii)'s span, $K/q = 60I$.
PROOF: Divide $240I_8$ by $q = 4$. □
⟨2⟩5. On model (iii)'s span, $K/q = 60I$.
PROOF: Divide $120I_8$ by $q = 2$. □
⟨2⟩6. On the E_7 span, $K/q = 36I$.
PROOF: On a last-coordinate pair (t,t), $72J_2(t,t) = 144(t,t)$; the first block is $144I_6$. Divide by 4. □
⟨2⟩7. On the E_6 span, $K/q = 24I$.
PROOF: On (t,t,t), $32J_3(t,t,t) = 96(t,t,t)$; the first block is $96I_5$. Divide by 4. □
⟨2⟩8. Q.E.D.
PROOF: Substitution of the C2 line counts gives $4N/\ell$ in each row. □

⟨1⟩3. The full-root formula holds for isometric copies in their spans.
PROOF: Conjugate the full-root operator through the span isometry of Lemma (transport on spans); reindex by its root bijection. □

⟨1⟩4. The full-root sum equals twice the signing sum.
⟨2⟩1. The roots $\mathbf{r}, -\mathbf{r}$ contribute the same summand.
PROOF: $(-\mathbf{r},\mathbf{x})(-\mathbf{r}) = (\mathbf{r},\mathbf{x})\mathbf{r}$ by bilinearity. □
⟨2⟩2. Q.E.D.

PROOF: Reindex the full-root sum by the signing bijection in Definition (signings and the matrix convention) and use ⟨2⟩1. □

⟨1⟩5. Q.E.D.

PROOF: The full formula follows from ⟨1⟩1–⟨1⟩3; divide the equality in ⟨1⟩4 by 2 for the signing formula. □

Lemma (the Gram polynomial from a frame). ASSUME: 1. S is a line system and $W = \operatorname{span}\Phi(S)$.
2. $\sigma : I \to \Phi(S)$ is a signing.
3. $t \in \mathbb{R}$ satisfies $\sum_i(\sigma(i), \mathbf{x})\sigma(i) = t\mathbf{x}$ for every $\mathbf{x} \in W$.

PROVE: With $M = 2I_N + C$ as in Definition (signings and the matrix convention), $M^2 = tM$ and $(C + 2I_N)(C - (t-2)I_N) = 0$.

PROOF SKETCH: **[new]** We isolate the algebraic passage from a frame to the signed Gram polynomial. We will apply the frame equation to each root and take its product with each other root. This avoids choosing an ambient orthonormal basis.

⟨1⟩1. For all $i, j \in I$, $(M^2)(i, j) = tM(i, j)$.

⟨2⟩1. $(M^2)(i, j) = (\sigma(i), \sum_k(\sigma(k), \sigma(j))\sigma(k))$.

PROOF: Expand matrix multiplication and then bilinearity from Section 2. □

⟨2⟩2. The vector sum in ⟨2⟩1 is $t\sigma(j)$.

PROOF: Apply the assumed frame identity to $\sigma(j) \in \Phi(S) \subseteq W$. □

⟨2⟩3. Q.E.D.

PROOF: Substitute ⟨2⟩2 into ⟨2⟩1 and use scalar linearity. □

⟨1⟩2. $M(M - tI_N) = 0$.

PROOF: Step ⟨1⟩1 gives $M^2 = tM$ entry by entry; distribute multiplication by $M - tI_N$. □

⟨1⟩3. Q.E.D.

PROOF: Substitute $M = C + 2I_N$ in ⟨1⟩2; then $M - tI_N = C - (t-2)I_N$. □

Theorem 3.7. ASSUME: S is A_ℓ with $\ell \geq 1$, D_ℓ with $\ell \geq 4$, or one of $\mathrm{E}_8, \mathrm{E}_7, \mathrm{E}_6$, in any isometric realization. Choose one vector of squared norm 2 on each line, and let $2I_N + C$ be their Gram matrix.

PROVE: For every choice and ordering of those vectors,

$$\begin{aligned}
\mathrm{A}_\ell &: \quad (C + 2I_N)(C - (\ell-1)I_N) &&= 0,\\
\mathrm{D}_\ell &: \quad (C + 2I_N)(C - 2(\ell-2)I_N) &&= 0,\\
\mathrm{E}_8 &: \quad (C + 2I_N)(C - 28I_N) &&= 0,\\
\mathrm{E}_7 &: \quad (C + 2I_N)(C - 16I_N) &&= 0,\\
\mathrm{E}_6 &: \quad (C + 2I_N)(C - 10I_N) &&= 0.
\end{aligned}$$

[repair] The E_8 constant is 28, correcting the printed 18 (Error 1).

PROOF SKETCH: We will use C6's frame identity within the span. The second factor has constant $\theta = 2N/\ell - 2$, with ℓ the rank, not the number of ambient coordinates.

⟨1⟩1. $(C + 2I_N)(C - (2N/\ell - 2)I_N) = 0$.

PROOF: Apply Lemma (the Gram polynomial from a frame) to the signing formula of Lemma (the frame identity within the span). □

⟨1⟩2. For A_ℓ, $2N/\ell - 2 = \ell - 1$.

PROOF: Lemma 3.4 gives $N = \ell(\ell+1)/2$; cancel the positive ℓ. □

⟨1⟩3. For D_ℓ, $2N/\ell - 2 = 2(\ell - 2)$.

PROOF: Substitute $N = \ell(\ell-1)$ and cancel $\ell > 0$. □

⟨1⟩4. For E_8, $2N/\ell - 2 = 28$.

PROOF: The rank and count are 8 and 120 by Lemmas (spans of the models) and 3.4; $240/8 - 2 = 28$. □

⟨1⟩5. For E_7, $2N/\ell - 2 = 16$.
PROOF: Its rank and count are 7 and 63, so $126/7 - 2 = 16$. □
⟨1⟩6. For E_6, $2N/\ell - 2 = 10$.
PROOF: Its rank and count are 6 and 36, so $72/6 - 2 = 10$. □
⟨1⟩7. Q.E.D.
PROOF: Substitute the five constants into ⟨1⟩1. The frame lemma applies to every signing and every isometric copy, so no additional switching or eigenvalue calculation is needed. □

3.1. **The classification behind Remark 3.11.** We will follow the ordinary proof in `cgss/proofs/remark311/remark311.tex`. Its lemma labels are retained. Theorems A, B and C are auxiliary results; Corollary D is the strengthened form of [CGSS, Theorem 3.8]. All spaces below are finite-dimensional real inner product spaces. Coordinate vectors $\mathbf{u}_i$ are orthonormal and start at index 1; simple roots α_i start at index 0. A graph called connected is nonempty. All Lean declarations of ours, as opposed to library declarations, occur in `\lean` metadata.
Every one of these declarations is in Lean. Theorem A and Corollary D are in `LineGraphs/Cartan/`. Theorems B and C are in `LineGraphs/RootSetClassification/`, and Remark (small ranks) and Lemma (a Gram realization) in `LineGraphs/Cartan/SmallRanks.lean` and `LineGraphs/Cartan/GramRealization.lean`. Their modules are imported by `LineGraphs.lean`, and `LineGraphs/RootSetClassification/Audit.lean` prints their axioms.

Definition 3.9. ASSUME: V is a finite-dimensional real inner product space.
A Root System is a finite set $\Phi \subseteq V \setminus \{0\}$ spanning V, such that, for all $\alpha, \beta \in \Phi$,

$$\frac{2(\alpha,\beta)}{(\alpha,\alpha)} \in \mathbb{Z}, \qquad s_\alpha\beta := \beta - \frac{2(\alpha,\beta)}{(\alpha,\alpha)}\alpha \in \Phi,$$

and $t\alpha \in \Phi$ implies $t \in \{1,-1\}$ for every $t \in \mathbb{R}$. It is indecomposable if it is nonempty and has no partition into two nonempty mutually orthogonal parts. A similarity is a linear bijection $f : V \to W$ carrying the root sets onto one another and satisfying $(f\mathbf{x}, f\mathbf{y}) = a(\mathbf{x},\mathbf{y})$ for some $a > 0$ and all $\mathbf{x},\mathbf{y}$. **[repair]** Nonzero roots, nonemptiness in indecomposability, and the use of Φ for the printed S in condition (b) are explicit here.

Lemma (composition and inversion of similarities). ASSUME: Φ, Ψ, Ω are Root Systems in real inner product spaces.
PROVE: The inverse of any similarity $f : \Phi \to \Psi$ with factor $a > 0$ is a similarity with factor a^{-1}. Given also a similarity $g : \Psi \to \Omega$ with factor $b > 0$, $g \circ f$ is a similarity with factor ab.

PROOF SKETCH: Substitute the metric identities and compose or invert the linear bijections and their equalities of root sets.
⟨1⟩1. ASSUME: $f : \Phi \to \Psi$, $g : \Psi \to \Omega$ are similarities with factors $a, b > 0$.
PROVE: $g \circ f$ is a linear bijection, $(gf\mathbf{x}, gf\mathbf{y}) = ab(\mathbf{x},\mathbf{y})$, and $gf(\Phi) = \Omega$.
PROOF: Substitute the two identities of Definition 3.9; $ab > 0$. □
⟨1⟩2. ASSUME: $f : \Phi \to \Psi$ is a similarity with factor $a > 0$.
PROVE: f^{-1} is a linear bijection, $(f^{-1}\mathbf{x}, f^{-1}\mathbf{y}) = a^{-1}(\mathbf{x},\mathbf{y})$, and $f^{-1}(\Psi) = \Phi$.
PROOF: Evaluate the identity for f on the two inverse images and divide by $a > 0$. □
⟨1⟩3. Q.E.D.
PROOF: Composition and inversion of linear bijections, with ⟨1⟩1, ⟨1⟩2, give the assertions. □

Definition (bases and Cartan conventions). ASSUME: Φ is a Root System in V.

A root base is a linearly independent finite family in Φ in which every root has integer coefficients all nonnegative or all nonpositive. Its matrix is

$$M(i,j) = p(i,j) := 2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j). \tag{2}$$

The second root supplies the denominator. A finite integer matrix is finite Cartan if its diagonal is 2, its other entries are nonpositive, its zeros are symmetric, and some positive integer diagonal D makes DM symmetric and positive on all nonzero integer vectors. The permitted models are

$$\begin{gathered} \mathrm{A}_\ell\ (\ell \geq 1), \quad \mathrm{B}_\ell\ (\ell \geq 2), \quad \mathrm{C}_\ell\ (\ell \geq 3), \quad \mathrm{D}_\ell\ (\ell \geq 4), \\ \mathrm{E}_6,\ \mathrm{E}_7,\ \mathrm{E}_8,\ \mathrm{F}_4,\ \mathrm{G}_2. \end{gathered} \tag{3}$$

These mean the actual Mathlib matrices, with rank ℓ for `CartanMatrix.A` despite its docstring. The model predicate includes the displayed rank restrictions, and equality after transport to its index type. Every root base is a real vector-space basis of V: its span contains every root by the coefficient condition, and the roots span V by Definition 3.9; independence is part of the base definition.

Library interface. We use Mathlib at commit `082e2d37e8b0463410cdb532e111cd43d5a66174` (4 September 2026), the version pinned in `lake-manifest.json`. The following are the precise library packages used in the proofs; every vector space in them is over $\mathbb{R}$ and finite dimensional, and all matrix index sets are finite with classically chosen decidable equality.

(1) `Matrix.IsFiniteCartan` and `CartanMatrix.A`, `CartanMatrix.B`, `CartanMatrix.C`, `CartanMatrix.D`, `CartanMatrix.E`, and the matrices $\mathrm{F}_4, \mathrm{G}_2$ are in `Mathlib/LinearAlgebra/Matrix/Cartan.lean`. `Matrix.IsIndecomposable` in `Mathlib/LinearAlgebra/Matrix/Block.lean` excludes a reindexing as $\left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right)$ with both diagonal block sizes positive. It imposes no nonempty-index hypothesis.

(2) `Matrix.posDef_iff_dotProduct_mulVec` in `Mathlib/LinearAlgebra/Matrix/PosDef.lean` converts positivity on finitely supported vectors to the finite quadratic form, over both $\mathbb{Z}$ and $\mathbb{R}$. `Matrix.exists_mulVec_eq_zero_iff` in `Mathlib/LinearAlgebra/Matrix/ToLinearEquiv.lean` identifies a nonzero kernel vector with zero determinant over any commutative domain. `RingHom.map_det` in `Mathlib/LinearAlgebra/Matrix/Determinant/Basic.lean` commutes determinants with casts. In the same file, `Matrix.det_apply'` is the signed permutation expansion used for a leaf deletion.

(3) `isCompact_sphere` in `Mathlib/Topology/MetricSpace/ProperSpace.lean` and `IsCompact.exists_isMaxOn` in `Mathlib/Topology/Order/Compact.lean` give the maximum of a continuous quadratic form on a nonempty unit sphere. Finite-dimensional Euclidean spaces are proper. No Perron–Frobenius classification is a premise. `Rat.denseRange_cast`, in `Mathlib/Topology/Algebra/Order/Archimedean.lean`, gives density of rationals in the reals; `DenseRange.piMap`, in `Mathlib/Topology/NhdsWithin.lean`, gives the coordinatewise version.

(4) `SimpleGraph.IsAcyclic.path_unique` in `Mathlib/Combinatorics/SimpleGraph/Acyclic.lean` gives uniqueness of paths in an acyclic graph. Connectedness supplies their existence. The graph definitions are in `Connectivity/Connected.lean`, `Finite.lean`, and `AdjMatrix.lean` under `Mathlib/Combinatorics/SimpleGraph/`.

(5) `RootPairing` and `RootPairing.IsRootSystem` are in `Mathlib/LinearAlgebra/RootSystem/Defs.lean`. Crystallographic, reduced, and irreducible are defined respectively in `IsValuedIn.lean`, `Reduced.lean`, and `Irreducible.lean` in that directory. Irreducibility requires nontrivial root and coroot

spaces and no nonzero proper reflection-invariant submodule in either. The real bilinear inner product `inner`$_1$ is in `Mathlib/Analysis/InnerProductSpace/Basic.lean`. `LinearMap.IsPerfPair.of_injective` in `Mathlib/LinearAlgebra/PerfectPairing/Basic.lean` makes it perfect from injectivity in both arguments and finite dimensionality.

(6) `RootPairing.nonempty_base` and `RootPairing.Base.mk'` in `Mathlib/LinearAlgebra/RootSystem/BaseExists.lean` require a field of characteristic zero, finitely many roots, spanning roots and coroots, crystallographic pairings, and reducedness. The second constructor needs root independence and membership of each root or its negative in the additive monoid of the selected roots. `Submodule.span_nat_eq_addSubmonoidClosure` in `Mathlib/LinearAlgebra/Span/Defs.lean` identifies this monoid with nonnegative integer combinations.

(7) `RootPairing.Base.support_nonempty` in `Mathlib/LinearAlgebra/RootSystem/Base.lean` needs nonempty roots and $2 \neq 0$; `RootPairing.Base.toWeightBasis` and `RootPairing.Base.toWeightBasis_apply` need spanning. `Module.Basis.equiv` and `Module.Basis.equiv_apply` in `Mathlib/LinearAlgebra/Basis/Defs.lean` extend a bijection of basis indices to a linear equivalence with the stated values.

(8) In `Mathlib/LinearAlgebra/RootSystem/CartanMatrix.lean`, `RootPairing.Base.cartanMatrix_isFiniteCartan` needs a characteristic zero domain, finite roots, crystallographic pairings, decidable equality, and spanning. `RootPairing.Base.cartanMatrix_isIndecomposable` needs a characteristic zero domain, finite roots, crystallographic pairings, reducedness, and irreducibility. `RootPairing.Base.apply_mem_range_root_of_cartanMatrixEq` needs finite, spanning, reduced, crystallographic source roots over a characteristic zero domain; the target needs crystallographic pairings and a base. Given equality of base matrices and a linear equivalence taking one base to the other, it maps source roots into target roots. `RootPairing.Base.equivOfCartanMatrixEq` packages both directions with the basis equivalence above; it additionally requires finite, spanning, reduced target roots. `RootPairing.Base.algebraMap_cartanMatrixIn_apply` identifies the integer Cartan entry cast to $\mathbb{R}$ with the real pairing.

The elementary identities of 2 remain available. Arithmetic leaves below are substitutions in displayed finite sums or polynomial identities (the Lean proofs may use `ring`, `norm_num`, or `linarith`); they do not appeal to Python output.

Lemma (reflection identities). ASSUME: V is a real inner product space and $\alpha \neq 0$.
PROVE: s_α is an orthogonal linear involution; $s_\alpha \alpha = -\alpha$; its -1 eigenspace is $\mathbb{R}\alpha$; $s_{-\alpha} = s_\alpha$; and for every linear isometry f onto W, $s_{f\alpha} = f s_\alpha f^{-1}$. In a Root System these reflections permute the roots and $\Phi = -\Phi$.

PROOF SKETCH: We resolve a vector into its component along α and its orthogonal component.
⟨1⟩1. For $t = (\alpha, \mathbf{x})/(\alpha, \alpha)$ and $\mathbf{y} = \mathbf{x} - t\alpha$, $(\alpha, \mathbf{y}) = 0$.
PROOF: Substitute t using bilinearity in 2. □
⟨1⟩2. $s_\alpha \mathbf{x} = -t\alpha + \mathbf{y}$.
PROOF: Substitute ⟨1⟩1 in Definition 3.9. □
⟨1⟩3. s_α is linear.
PROOF: The formula in Definition 3.9 is a difference of linear maps, by 2. □
⟨1⟩4. $s_\alpha(s_\alpha \mathbf{x}) = \mathbf{x}$.
PROOF: Apply ⟨1⟩2 to the components $-t, \mathbf{y}$. □

⟨1⟩5. $(s_\alpha \mathbf{x}, s_\alpha \mathbf{w}) = (\mathbf{x}, \mathbf{w})$.
PROOF: Expand both sides by ⟨1⟩1, ⟨1⟩2 and 2; both are $tu(\alpha,\alpha)+(\mathbf{y},\mathbf{v})$ for the decompositions $\mathbf{x} = t\alpha + \mathbf{y}$, $\mathbf{w} = u\alpha + \mathbf{v}$. □
⟨1⟩6. $s_\alpha \alpha = -\alpha$ and $s_\alpha \mathbf{x} = -\mathbf{x} \iff \mathbf{x} \in \mathbb{R}\alpha$.
PROOF: In ⟨1⟩2, equality to $-\mathbf{x}$ is exactly $2\mathbf{y} = 0$. □
⟨1⟩7. $s_{-\alpha} = s_\alpha$.
PROOF: Substitute $-\alpha$ into Definition 3.9. □
⟨1⟩8. $s_{f\alpha}(f\mathbf{x}) = f(s_\alpha \mathbf{x})$.
PROOF: Substitute the isometry identities into Definition 3.9. □
⟨1⟩9. ASSUME: Φ is a Root System and $\alpha \in \Phi$.
PROVE: s_α permutes Φ and $-\alpha \in \Phi$.
PROOF: Restrict ⟨1⟩4 and ⟨1⟩6 using reflection closure in Definition 3.9. □
⟨1⟩10. Q.E.D.
PROOF: Combine ⟨1⟩3–⟨1⟩9. □

Lemma (a null vector of a nonnegative form). ASSUME: 1. Q is real symmetric on a finite index set, and $q(\mathbf{x}) := \mathbf{x}^T Q \mathbf{x} \geq 0$ for every real vector $\mathbf{x}$.
2. $q(\mathbf{y}) = 0$.
PROVE: $Q\mathbf{y} = 0$.

PROOF SKETCH: A quadratic polynomial nonnegative on the line cannot have a nonzero linear coefficient at a zero.
⟨1⟩1. For every $\mathbf{z}, t$, $0 \leq 2t\mathbf{z}^T Q\mathbf{y} + t^2 q(\mathbf{z})$.
PROOF: Expand $q(\mathbf{y}+t\mathbf{z})$ using ⟨0⟩:1, ⟨0⟩:2. □
⟨1⟩2. For every $\mathbf{z}$, $\mathbf{z}^T Q\mathbf{y} = 0$.
LET: $c = \mathbf{z}^T Q\mathbf{y}$, $d = q(\mathbf{z}) \geq 0$.
⟨2⟩1. ASSUME: $c \neq 0$.
PROVE: For $t = -c/(d+1)$, $2tc + t^2 d = -c^2(d+2)/(d+1)^2 < 0$.
PROOF: Substitute t, collect the common denominator, and use $c^2 > 0$, $d+2 > 0$, $(d+1)^2 > 0$. □
⟨2⟩2. Q.E.D.
PROOF: The inequality in ⟨2⟩1 contradicts ⟨1⟩1. □
⟨1⟩3. Q.E.D.
PROOF: Apply ⟨1⟩2 to each coordinate unit vector. □

Lemma (integer positivity). ASSUME: 1. I is finite and $H : I \times I \to \mathbb{Z}$ is symmetric.
2. $\mathbf{x}^T H \mathbf{x} > 0$ for every nonzero $\mathbf{x} \in \mathbb{Z}^I$.
PROVE: The entrywise real cast of H satisfies $\mathbf{y}^T H \mathbf{y} > 0$ for every nonzero $\mathbf{y} \in \mathbb{R}^I$.

PROOF SKETCH: A negative real value would yield a negative integer value by density and scaling. A real null vector would force an integer null vector through the determinant. Only the integer matrix is needed here.
⟨1⟩1. The form is nonnegative on real vectors.
⟨2⟩1. ASSUME: Some real vector has negative quadratic value.
PROVE: FALSE.
⟨3⟩1. The negative-value set of $\mathbf{x} \mapsto \mathbf{x}^T H \mathbf{x}$ is nonempty and open.
PROOF: A finite sum of products of coordinates is continuous; use ⟨2⟩ for nonemptiness. □
⟨3⟩2. PICK a rational vector $\mathbf{v}$ with $\mathbf{v}^T H \mathbf{v} < 0$.
PROOF: Apply `DenseRange.piMap` to `Rat.denseRange_cast` and ⟨3⟩1. □
⟨3⟩3. PICK an integer $k > 0$ with $k\mathbf{v} \in \mathbb{Z}^I$.

PROOF: The product of the finitely many positive coordinate denominators of $\mathbf{v}$ is such a k. □

⟨3⟩4. $(k\mathbf{v})^T H(k\mathbf{v}) = k^2 \mathbf{v}^T H\mathbf{v} < 0$.

PROOF: Expand the quadratic form and use ⟨3⟩2, ⟨3⟩3. □

⟨3⟩5. Q.E.D.

PROOF: The vector in ⟨3⟩4 is nonzero and integral, contradicting ⟨0⟩:2 after the order-preserving integer cast. □

⟨2⟩2. Q.E.D.

PROOF: The contradiction ⟨2⟩1 excludes every negative value. □

⟨1⟩2. ASSUME: $\mathbf{y} \neq 0$ and $\mathbf{y}^T H\mathbf{y} = 0$.

PROVE: FALSE.

⟨2⟩1. $H\mathbf{y} = 0$.

PROOF: Apply Lemma (a null vector of a nonnegative form) to ⟨1⟩1 and ⟨1⟩. □

⟨2⟩2. $\det H = 0$ over $\mathbb{R}$.

PROOF: Apply `Matrix.exists_mulVec_eq_zero_iff` to ⟨2⟩1 and $\mathbf{y} \neq 0$ from ⟨1⟩. □

⟨2⟩3. $\det H = 0$ over $\mathbb{Z}$.

PROOF: Use `RingHom.map_det` and injectivity of the integer cast in ⟨2⟩2. □

⟨2⟩4. PICK a nonzero $\mathbf{v} \in \mathbb{Z}^I$ with $H\mathbf{v} = 0$.

PROOF: Apply `Matrix.exists_mulVec_eq_zero_iff` over $\mathbb{Z}$ to ⟨2⟩3. □

⟨2⟩5. Q.E.D.

PROOF: The zero value $\mathbf{v}^T H\mathbf{v} = 0$ from ⟨2⟩4 contradicts ⟨0⟩:2. □

⟨1⟩3. Q.E.D.

PROOF: Combine nonnegativity ⟨1⟩1 and the exclusion of nonzero null vectors ⟨1⟩2. □

Lemma (connected support). ASSUME: 1. I is finite and nonempty, R is a ring, and $M : I \times I \to R$.

2. $M(i,j) = 0 \iff M(j,i) = 0$ for every i, j.

PROVE: The simple graph $i \sim j \iff i \neq j$ and $M(i,j) \neq 0$ is connected if and only if M is Mathlib-indecomposable.

PROOF SKETCH: A disconnected component gives two blocks; a block partition cannot be crossed by a path.

⟨1⟩1. ASSUME: the support is disconnected.

PROVE: M is decomposable.

⟨2⟩1. There is a nonempty proper reachable component $P \subset I$.

PROOF: Unfold disconnectedness on the nonempty type I. □

⟨2⟩2. $M(i,j) = 0$ whenever $i \in P$ and $j \notin P$.

PROOF: A nonzero entry would give an edge from i to j, contrary to the definition of P in ⟨2⟩1. □

⟨2⟩3. Enumerating P and $I \setminus P$ gives a block diagonal matrix with both block sizes positive.

PROOF: Use ⟨2⟩2 and ⟨0⟩:2 in the four blocks of the two-set partition. □

⟨2⟩4. Q.E.D.

PROOF: The decomposition in ⟨2⟩3 violates `Matrix.IsIndecomposable`. □

⟨1⟩2. ASSUME: M has a nontrivial upper block triangular decomposition.

PROVE: Its support is disconnected.

⟨2⟩1. Both cross blocks vanish.

PROOF: The lower block vanishes by hypothesis and the upper one vanishes by ⟨0⟩:2. □

⟨2⟩2. Every path starting in the first block stays in that block.

PROOF: Induct on path length using ⟨2⟩1. □

⟨2⟩3. Q.E.D.

PROOF: Choose one vertex in each positive-size block and apply ⟨2⟩2. □
⟨1⟩3. Q.E.D.
PROOF: Contraposition of ⟨1⟩1 and ⟨1⟩2 gives the equivalence. □

Definition (weighted graphs). ASSUME: I is finite and nonempty.

A weighted graph on I has a real symmetric matrix B, zero diagonal, and nonnegative entries. Its support joins i, j when $B(i,j) > 0$. For any real symmetric matrix on I, write $\lambda(B) = \max_{(\mathbf{x},\mathbf{x})=1} \mathbf{x}^T B\mathbf{x}$. For weighted graphs Y, X, write $Y \preccurlyeq X$ if an injection e satisfies $B_Y(i,j) \leq B_X(e(i), e(j))$ for every i, j. This specified containment is proper when e is not onto or one inequality is strict. Edges may be omitted and weights decreased.

Lemma (edge-constant functions). ASSUME: 1. X is a connected simple graph on a vertex set I.
2. S is any set and $f : I \to S$ satisfies $f(u) = f(v)$ whenever $u \sim v$.
PROVE: For every $u, v \in I$, $f(u) = f(v)$. In particular there is $c \in S$ with $f(v) = c$ for every v.

PROOF SKETCH: Induct on the walk supplied by connectedness.
⟨1⟩1. For every walk from u to v, $f(u) = f(v)$.
⟨2⟩1. A walk of length zero has equal endpoint values.
PROOF: Its endpoints coincide by the definition of a walk. □
⟨2⟩2. An edge $u \sim w$ followed by a walk from w to v with $f(w) = f(v)$ gives $f(u) = f(v)$.
PROOF: Compose the edge equality in ⟨0⟩:2 with the equality for the remaining walk. □
⟨2⟩3. Q.E.D.
PROOF: Walk induction using ⟨2⟩1, ⟨2⟩2. □
⟨1⟩2. For every u, v, $f(u) = f(v)$.
PROOF: Connectedness in ⟨0⟩:1 supplies a walk; apply ⟨1⟩1. □
⟨1⟩3. Q.E.D.
PROOF: Choose u by nonemptiness in connectedness and set $c = f(u)$ in ⟨1⟩2. □

Lemma (propagation on a connected support). ASSUME: 1. B is a nonnegative real symmetric matrix on a finite nonempty set. The simple graph $i \sim j \iff i \neq j$ and $B(i,j) > 0$ is connected.
2. $B\mathbf{x} = t\mathbf{x}$, all $\mathbf{x}(i) \geq 0$, and $\mathbf{x}(i_0) = 0$.
PROVE: $\mathbf{x} = 0$.

PROOF SKETCH: Show that having value zero is constant across every edge.
⟨1⟩1. ASSUME: $\mathbf{x}(i) = 0$ and $B(i,j) > 0$.
PROVE: $\mathbf{x}(j) = 0$.
⟨2⟩1. $\sum_k B(i,k)\mathbf{x}(k) = 0$ with every summand nonnegative.
PROOF: Evaluate ⟨0⟩:2 at i. □
⟨2⟩2. $B(i,j)\mathbf{x}(j) = 0$.
PROOF: A nonnegative summand of the zero sum ⟨2⟩1 is zero. □
⟨2⟩3. Q.E.D.
PROOF: Cancel $B(i,j) > 0$ in ⟨2⟩2. □
⟨1⟩2. Every vertex has value zero.
PROOF: By ⟨1⟩1 in both directions, the indicator of $\{i : \mathbf{x}(i) = 0\}$ agrees across every support edge. Lemma (edge-constant functions) makes it constant; its value at i_0 is 1 by ⟨0⟩:2. □
⟨1⟩3. Q.E.D.
PROOF: Function extensionality applied to ⟨1⟩2. □

Lemma (Rayleigh quotients). ASSUME: B is real symmetric on a finite nonempty set I.

PROVE: For every real vector $\mathbf{y}$, $\mathbf{y}^T B\mathbf{y} \le \lambda(B)(\mathbf{y},\mathbf{y})$. $\lambda(B)$ is an eigenvalue bounding every real eigenvalue. A nonzero vector attains its Rayleigh quotient exactly when it is a $\lambda(B)$-eigenvector. If $B \ge 0$ there is a nonnegative unit maximizer. If its support is also connected, this vector is strictly positive and the $\lambda(B)$-eigenspace is one-dimensional.

PROOF SKETCH: Use a nonnegative form at the maximum, then absolute values and the propagation lemma. The nonnegative conclusion does not require connectedness and is stated for use with a disconnected smaller graph.

⟨1⟩1. PICK a unit vector $\mathbf{x}$ attaining $\lambda = \lambda(B)$.
PROOF: Apply the compact maximum facts of 3.1 to the continuous quadratic form. □

⟨1⟩2. For every $\mathbf{y}$, $\mathbf{y}^T B\mathbf{y} \le \lambda(\mathbf{y},\mathbf{y})$.
⟨2⟩1. CASE: $\mathbf{y} = 0$.
PROOF: Both sides vanish by substitution. □
⟨2⟩2. CASE: $\mathbf{y} \ne 0$.
PROOF: Apply ⟨1⟩1 to $\mathbf{y}/\|\mathbf{y}\|$ and multiply by $\|\mathbf{y}\|^2 > 0$. □
⟨2⟩3. Q.E.D.
PROOF: The cases ⟨2⟩1, ⟨2⟩2 exhaust all vectors. □

⟨1⟩3. For nonzero $\mathbf{y}$, equality in ⟨1⟩2 holds if and only if $B\mathbf{y} = \lambda\mathbf{y}$.
⟨2⟩1. Equality implies $(\lambda I - B)\mathbf{y} = 0$.
PROOF: Apply Lemma (a null vector of a nonnegative form) to the nonnegative form ⟨1⟩2. □
⟨2⟩2. An eigenvector satisfies $\mathbf{y}^T B\mathbf{y} = \lambda(\mathbf{y},\mathbf{y})$.
PROOF: Substitute the eigenvector equation. □
⟨2⟩3. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩2. □

⟨1⟩4. λ is an eigenvalue and every real eigenvalue is at most λ.
⟨2⟩1. $B\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \ne 0$.
PROOF: Apply ⟨1⟩3 to ⟨1⟩1. □
⟨2⟩2. ASSUME: $B\mathbf{v} = t\mathbf{v}$ and $\mathbf{v} \ne 0$.
PROVE: $t \le \lambda$.
PROOF: Substitute in ⟨1⟩2 and cancel $(\mathbf{v},\mathbf{v}) > 0$. □
⟨2⟩3. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩2. □

⟨1⟩5. ASSUME: $B \ge 0$.
PROVE: $|\mathbf{x}|$ is a nonnegative unit maximizer.
⟨2⟩1. $(|\mathbf{x}|,|\mathbf{x}|) = 1$.
PROOF: Use $|x|^2 = x^2$ in the coordinate sum of ⟨1⟩1. □
⟨2⟩2. $|\mathbf{x}|^T B|\mathbf{x}| \ge \mathbf{x}^T B\mathbf{x}$.
PROOF: Sum $B(i,j)x(i)x(j) \le B(i,j)|x(i)||x(j)|$ over ordered pairs, using $B(i,j) \ge 0$. □
⟨2⟩3. Q.E.D.
PROOF: Sandwich ⟨2⟩2 between the maximum in ⟨1⟩1 and the bound ⟨1⟩2, using ⟨2⟩1. □

⟨1⟩6. ASSUME: $B \ge 0$, its support is connected, and $\mathbf{w} \ge 0$ is a nonzero maximizing eigenvector.
PROVE: $\mathbf{w}$ is positive everywhere.
PROOF: A zero coordinate would force $\mathbf{w} = 0$ by Lemma (propagation on a connected support). □

⟨1⟩7. ASSUME: $B \ge 0$ has connected support, $\mathbf{w} > 0$, and $B\mathbf{w} = \lambda\mathbf{w}$.
PROVE: Every λ-eigenvector is a multiple of $\mathbf{w}$.
LET: For a λ-eigenvector $\mathbf{v}$, let $t = \max_i \mathbf{v}(i)/\mathbf{w}(i)$.
⟨2⟩1. $t\mathbf{w} - \mathbf{v}$ is nonnegative and has a zero coordinate.
PROOF: Use the attained finite maximum defining t and ⟨1⟩6. □

⟨2⟩2. $B(t\mathbf{w} - \mathbf{v}) = \lambda(t\mathbf{w} - \mathbf{v})$.
PROOF: Linearity and the two eigenvector equations. □
⟨2⟩3. Q.E.D.
PROOF: Lemma (propagation on a connected support) applied to ⟨2⟩1, ⟨2⟩2 gives $\mathbf{v} = t\mathbf{w}$. □
⟨1⟩8. Q.E.D.
PROOF: Combine ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, ⟨1⟩7. □

Lemma (Rayleigh quotients under reindexing). ASSUME: B_I, B_J are real symmetric matrices on finite nonempty sets I, J, and $e : I \to J$ is a bijection with $B_J(e(i), e(j)) = B_I(i, j)$ for all i, j.
PROVE: $\lambda(B_I) = \lambda(B_J)$. In particular isomorphic weighted graphs have equal largest eigenvalues.

PROOF SKETCH: Reindex the unit sphere and the quadratic form.
⟨1⟩1. The bijection $\mathbf{x} \mapsto \mathbf{x} \circ e$ preserves squared norms.
PROOF: Reindex the finite sum of squares along e. □
⟨1⟩2. $(\mathbf{x} \circ e)^T B_I (\mathbf{x} \circ e) = \mathbf{x}^T B_J \mathbf{x}$.
PROOF: Reindex the double sum along $e \times e$ using the entry equality in ⟨0⟩. □
⟨1⟩3. Q.E.D.
PROOF: The sets of values maximized in Definition (weighted graphs) coincide by ⟨1⟩1, ⟨1⟩2. □

Lemma (strict monotonicity). ASSUME: 1. X is a finite connected weighted graph with nonnegative weights.
2. Y is a finite nonempty weighted graph with nonnegative weights, and e witnesses $Y \preccurlyeq X$.
PROVE: $\lambda(Y) \leq \lambda(X)$, strictly for proper containment. For any connected nonnegative real symmetric matrix, a strictly positive eigenvector has eigenvalue equal to its largest eigenvalue.

PROOF SKETCH: Extend a nonnegative maximizer by zero; equality precludes both missing vertices and increased entries.
⟨1⟩1. PICK a nonnegative unit maximizer $\mathbf{v}$ of Y.
PROOF: Apply Lemma (Rayleigh quotients) to B_Y. □
LET: $\mathbf{x}(e(i)) = \mathbf{v}(i)$ and $\mathbf{x}(j) = 0$ off the image of e.
⟨1⟩2. $(\mathbf{x}, \mathbf{x}) = 1$.
PROOF: Reindex the sum of squares by the injection e in ⟨0⟩:2, using ⟨1⟩1. □
⟨1⟩3. $\lambda(Y) \leq \mathbf{x}^T B_X \mathbf{x} \leq \lambda(X)$.
⟨2⟩1. $\lambda(Y) \leq \mathbf{x}^T B_X \mathbf{x}$.
PROOF: Sum the entrywise inequalities in ⟨0⟩:2 multiplied by the nonnegative $\mathbf{v}(i)\mathbf{v}(j)$ from ⟨1⟩1. □
⟨2⟩2. $\mathbf{x}^T B_X \mathbf{x} \leq \lambda(X)$.
PROOF: Apply the definition of the maximum to ⟨1⟩2. □
⟨2⟩3. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩2. □
⟨1⟩4. ASSUME: $\lambda(Y) = \lambda(X)$.
PROVE: The specified containment is not proper.
⟨2⟩1. $\mathbf{x}$ is a maximizing eigenvector of X.
PROOF: Apply Lemma (Rayleigh quotients) to equality in ⟨1⟩3. □
⟨2⟩2. $\mathbf{x}(j) > 0$ for every vertex j of X.
PROOF: Apply Lemma (propagation on a connected support) to ⟨2⟩1, ⟨1⟩2 and nonnegativity of $\mathbf{x}$. □

⟨2⟩3. e is onto.
PROOF: A vertex outside the image has $\mathbf{x}(j) = 0$ by definition, contradicting ⟨2⟩2. □
⟨2⟩4. All entry inequalities are equalities.
PROOF: In the zero sum $\sum_{i,j}(B_X(e(i), e(j)) - B_Y(i, j))\mathbf{v}(i)\mathbf{v}(j) = 0$ from ⟨1⟩3, every summand is nonnegative and both vector factors are positive by ⟨2⟩2. □
⟨2⟩5. Q.E.D.
PROOF: Use the definition of properness with ⟨2⟩3, ⟨2⟩4. □
⟨1⟩5. ASSUME: B is real symmetric, nonnegative, and has connected off-diagonal positive support; $B\mathbf{v} = t\mathbf{v}$ and $\mathbf{v} > 0$.
PROVE: $t = \lambda(B)$.
⟨2⟩1. PICK $\mathbf{w} > 0$ with $B\mathbf{w} = \lambda(B)\mathbf{w}$.
PROOF: Apply Lemma (Rayleigh quotients) to this matrix B. □
⟨2⟩2. $(t - \lambda(B))(\mathbf{v}, \mathbf{w}) = 0$.
PROOF: Use symmetry of B in $(B\mathbf{v}, \mathbf{w}) = (\mathbf{v}, B\mathbf{w})$ and ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Cancel the positive sum $(\mathbf{v}, \mathbf{w})$ in ⟨2⟩2. □
⟨1⟩6. Q.E.D.
PROOF: Combine ⟨1⟩3, the contrapositive of ⟨1⟩4, and ⟨1⟩5. □

Lemma (symmetrization). ASSUME: 1. I is finite and nonempty, $M : I \times I \to \mathbb{Z}$ is finite Cartan and indecomposable.
2. $d : I \to \mathbb{Z}_{>0}$ symmetrizes M as in its finite Cartan hypothesis, and $D = \mathrm{diag}(d)$.
PROVE: $S = D^{1/2}MD^{-1/2}$ is symmetric positive definite and equals $2I - B$, where $B(i, j) = \sqrt{M(i, j)M(j, i)}$ off the diagonal and $B(i, i) = 0$. The weighted graph of B is connected, has edge weights $1, \sqrt{2}, \sqrt{3}$, and $\lambda(B) < 2$.

PROOF SKETCH: Congruence transfers positivity; a two-coordinate test bounds the integer edge products.
⟨1⟩1. DM is positive definite over $\mathbb{R}$.
PROOF: Apply Lemma (integer positivity) and the quadratic-form interface in 3.1. □
⟨1⟩2. $S = D^{-1/2}(DM)D^{-1/2}$ is symmetric positive definite.
PROOF: Substitute the invertible diagonal change of coordinates in the form of ⟨1⟩1. □
⟨1⟩3. $S(i, i) = 2$ for every i.
PROOF: The diagonal factors cancel and $M(i, i) = 2$ by ⟨0⟩:1. □
⟨1⟩4. For $i \neq j$, $S(i, j) \leq 0$ and $S(i, j)^2 = M(i, j)M(j, i)$.
PROOF: Substitute $S(i, j) = \sqrt{d(i)/d(j)}M(i, j)$ and $d(i)M(i, j) = d(j)M(j, i)$ from ⟨0⟩:2. □
⟨1⟩5. For $i \neq j$, $S(i, j)^2 < 4$.
LET: $b = S(i, j)$ and $\mathbf{x} = \mathbf{u}_i - (b/2)\mathbf{u}_j \neq 0$.
⟨2⟩1. $\mathbf{x}^T S\mathbf{x} = 2 - b^2/2$.
PROOF: Expand the two coordinates using ⟨1⟩3 and symmetry in ⟨1⟩2. □
⟨2⟩2. Q.E.D.
PROOF: Positivity in ⟨1⟩2 and ⟨2⟩1 give $2 - b^2/2 > 0$. □
⟨1⟩6. $M(i, j)M(j, i) \in \{0, 1, 2, 3\}$.
PROOF: Apply integer interval arithmetic to ⟨1⟩4, ⟨1⟩5. □
⟨1⟩7. $B = 2I - S$ has precisely the required entries and weights.
PROOF: The nonpositive square root is unique in ⟨1⟩4; use ⟨1⟩3, ⟨1⟩6. □
⟨1⟩8. Its support is connected.
PROOF: Apply Lemma (connected support) to the identical nonzero support in ⟨1⟩7. □
⟨1⟩9. $\lambda(B) < 2$.

⟨2⟩1. PICK a unit maximizer $\mathbf{v}$ for B.
PROOF: Apply Lemma (Rayleigh quotients). □
⟨2⟩2. Q.E.D.
PROOF: At ⟨2⟩1, $\lambda(B) = 2 - \mathbf{v}^T S\mathbf{v} < 2$ by ⟨1⟩2, ⟨1⟩7. □
⟨1⟩10. Q.E.D.
PROOF: Combine ⟨1⟩2, ⟨1⟩7, ⟨1⟩8, ⟨1⟩9. □

3.1.1. *Affine obstructions and labelled trees.*

Definition (affine obstructions). ASSUME: All lengths below are integers in the displayed ranges.
The tree $T(a,b,c)$, $a,b,c \geq 1$, has vertex set

$$V_{a,b,c} = \{*\} \sqcup (\{1\} \times [1,a]) \sqcup (\{2\} \times [1,b]) \sqcup (\{3\} \times [1,c]).$$

Its only edges are $* \sim (r,1)$ and consecutive positions on an arm. The affine obstructions are the following weighted graphs:

(1) Unit cycles Cyc_m, $m \geq 3$.
(2) $\widetilde{\mathrm{D}}_4 = K_{1,4}$; and $\widetilde{\mathrm{D}}_\ell$, $\ell \geq 5$, a unit path of $\ell - 4$ edges with two new leaves at each end.
(3) $\widetilde{\mathrm{E}}_6 = T(2,2,2)$, $\widetilde{\mathrm{E}}_7 = T(1,3,3)$, $\widetilde{\mathrm{E}}_8 = T(1,2,5)$, all with unit weights.
(4) X_m^{end}, $m \geq 3$: a path on m vertices with first and last weights $\sqrt{2}$ and all intervening weights 1.
(5) X_k^{fork}, $k \geq 1$: a path $v_0, \ldots, v_k$ with two unit leaves at v_0, last path weight $\sqrt{2}$, and other path weights 1.
(6) X^{mid}: the path with weights $1, 1, \sqrt{2}, 1$.
(7) X^{triple}: the path with weights $\sqrt{3}, 1$.

Thus both edges of X_3^{end} have weight $\sqrt{2}$, and the weighted edge of X_1^{fork} meets the branch vertex. The project model `smithA` at parameter $\ell \geq 2$ means $\mathrm{Cyc}_{\ell+1}$.

Lemma (connected labelled graphs). ASSUME: X is a finite weighted graph whose support admits a nonempty path labelling, an arm labelling $T(a,b,c)$ with $a,b,c \geq 1$, or is the support of an obstruction in Definition (affine obstructions).
PROVE: The support of X is connected.

PROOF SKETCH: Give a walk from every vertex to one chosen vertex.
⟨1⟩1. CASE: The support has a path labelling.
PROOF: Consecutive positions $i, i-1, \ldots, 0$ give a walk from every vertex to position 0, including the singleton path. □
⟨1⟩2. CASE: The support has an arm labelling.
PROOF: From (r,t) follow $(r,t-1), \ldots, (r,1), *$; the centre has the length-zero walk to itself. □
⟨1⟩3. CASE: X has the support of Cyc_m.
PROOF: Follow consecutive cycle positions down to 0; each is an edge by Definition (affine obstructions). □
⟨1⟩4. CASE: X has the support of $\widetilde{\mathrm{D}}_4$.
PROOF: Every leaf has an edge to the centre of $K_{1,4}$. □
⟨1⟩5. CASE: X has the support of $\widetilde{\mathrm{D}}_\ell$, $\ell \geq 5$.
PROOF: A leaf first reaches its attached endpoint; from any path position, consecutive path edges reach the first endpoint. □
⟨1⟩6. CASE: X has the support of X_k^{fork}.
PROOF: Its extra leaves reach v_0 in one edge; v_i reaches it along $v_i, v_{i-1}, \ldots, v_0$. □
⟨1⟩7. Q.E.D.

PROOF: The $\widetilde{\mathrm{E}}$ supports are arm trees; $X^{\mathrm{end}}, X^{\mathrm{mid}}, X^{\mathrm{triple}}$ have path supports. Thus ⟨1⟩1–⟨1⟩6 exhaust Definition (affine obstructions) and the two other hypotheses. □

Lemma (unit containment and support embeddings). ASSUME: X is a finite weighted graph with $B_X(u,v) \geq 1$ on every support edge, and Y is a finite nonempty simple graph, given unit edge weights.
PROVE: $Y \preccurlyeq X$ if and only if there is an injection $e : V(Y) \to V(X)$ sending every edge of Y to a support edge of X. The same injection witnesses either assertion; nonedges need not be preserved.

PROOF SKETCH: Check the containment inequality on edges and nonedges.
⟨1⟩1. ASSUME: e witnesses $Y \preccurlyeq X$.
PROVE: e preserves edges into the support.
PROOF: On an edge the inequality is $1 \leq B_X(e(u), e(v))$, so that entry is positive. □
⟨1⟩2. ASSUME: e is an injective edge-preserving map to the support.
PROVE: e witnesses $Y \preccurlyeq X$.
⟨2⟩1. On an edge, $B_Y(u,v) = 1 \leq B_X(e(u), e(v))$.
PROOF: Apply the weight lower bound in ⟨0⟩ to the support edge supplied by ⟨1⟩. □
⟨2⟩2. On a nonedge, $B_Y(u,v) = 0 \leq B_X(e(u), e(v))$.
PROOF: Apply nonnegativity from Definition (weighted graphs). □
⟨2⟩3. Q.E.D.
PROOF: The cases ⟨2⟩1, ⟨2⟩2 give every inequality of containment. □
⟨1⟩3. Q.E.D.
PROOF: Combine ⟨1⟩1, ⟨1⟩2. □

Lemma (a cycle gives a cycle subgraph). ASSUME: Z is a simple graph and p is a closed walk of length m which is a cycle.
PROVE: $m \geq 3$, and the map sending $i \in [0, m-1]$ to the ith vertex of p is an injection preserving every edge of Cyc_m.

PROOF SKETCH: Omit the repeated last vertex from the cycle walk.
⟨1⟩1. $m \geq 3$ and the first m vertices of p are distinct.
⟨2⟩1. $m \geq 3$.
PROOF: Apply `SimpleGraph.Walk.isCycle_iff_isPath_tail_and_le_length` in `Mathlib/Combinatorics/SimpleGraph/Paths.lean`. □
⟨2⟩2. The first m vertices are distinct.
PROOF: Apply `SimpleGraph.Walk.IsCycle.getVert_injOn'` in that file to the positions $0, \ldots, m-1$. □
⟨2⟩3. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩2. □
⟨1⟩2. Successive positions and the pair $m-1, 0$ map to edges.
PROOF: These are the walk edges of p, including its last edge back to its start. □
⟨1⟩3. Q.E.D.
PROOF: Apply ⟨1⟩1, ⟨1⟩2 to the cycle definition in Definition (affine obstructions). □

Lemma (affine eigenvectors). ASSUME: X is one of the graphs in Definition (affine obstructions).
PROVE: $\lambda(X) = 2$. The following positive vectors $\mathbf{w}$ satisfy $B\mathbf{w} = 2\mathbf{w}$: 1 on a cycle; 2 on each nonleaf and 1 on each leaf of $\widetilde{\mathrm{D}}$; centre $h = 3, 4, 6$ respectively on the three $\widetilde{\mathrm{E}}$ trees and value $h(a+1-t)/(a+1)$ at position t of an arm of length a; endpoints 1 and interiors $\sqrt{2}$ for X^{end}; values $2, \ldots, 2, \sqrt{2}$ on the path and 1 on the two extra leaves for X^{fork}; $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$ for X^{mid}; $(\sqrt{3}, 2, 1)$ for X^{triple}.

PROOF SKETCH: Check each vertex type. This proves the unbounded families, including their shortest instances.
LET: $\mathbf{w}$ is the vector specified in the statement for the graph at hand.
⟨1⟩1. On a cycle, each weighted neighbour sum is 2.
PROOF: Each vertex has two unit neighbours by Definition (affine obstructions). □
⟨1⟩2. The displayed vector solves the equations on $\widetilde{\mathrm{D}}$.
⟨2⟩1. At a leaf the sum is $2 = 2 \cdot 1$.
PROOF: Its unique neighbour has value 2 by the stated vector. □
⟨2⟩2. At the centre of $K_{1,4}$ the sum is $4 = 2 \cdot 2$.
PROOF: Add its four leaf values from the definition of $\mathbf{w}$. □
⟨2⟩3. At a fork for $\ell \geq 5$ the sum is $2 + 1 + 1 = 2 \cdot 2$.
PROOF: Use its one path neighbour and two leaves in Definition (affine obstructions). □
⟨2⟩4. At a path interior the sum is $2 + 2 = 2 \cdot 2$.
PROOF: Use its two path neighbours in Definition (affine obstructions). □
⟨2⟩5. Q.E.D.
PROOF: The leaf, centre, fork and interior cases ⟨2⟩1–⟨2⟩4 exhaust the vertices. □
⟨1⟩3. The displayed vectors solve the equations on $\widetilde{\mathrm{E}}_6, \widetilde{\mathrm{E}}_7, \widetilde{\mathrm{E}}_8$.
⟨2⟩1. At a noncentral vertex the sum is twice its value.
PROOF: The arm formula is an arithmetic progression with value h at position 0 and virtual value 0 at position $a + 1$. □
⟨2⟩2. For arms a, b, c and centre value $h > 0$, the central eigenvector equation is equivalent to $1/(a+1) + 1/(b+1) + 1/(c+1) = 1$.
PROOF: Its neighbour sum is $h(a/(a+1) + b/(b+1) + c/(c+1))$; cancel h from equality to $2h$. □
⟨2⟩3. The central equation holds for $(2, 2, 2)$.
PROOF: Substitute $1/3 + 1/3 + 1/3 = 1$ in ⟨2⟩2. □
⟨2⟩4. The central equation holds for $(1, 3, 3)$.
PROOF: Substitute $1/2 + 1/4 + 1/4 = 1$ in ⟨2⟩2. □
⟨2⟩5. The central equation holds for $(1, 2, 5)$.
PROOF: Substitute $1/2 + 1/3 + 1/6 = 1$ in ⟨2⟩2. □
⟨2⟩6. Q.E.D.
PROOF: Combine ⟨2⟩1 with the three central cases ⟨2⟩3–⟨2⟩5. □
⟨1⟩4. The equations hold on X_m^{end}.
⟨2⟩1. At either endpoint the sum is $\sqrt{2}\sqrt{2} = 2$.
PROOF: Substitute the endpoint edge and neighbouring value. □
⟨2⟩2. At an interior vertex each of its two sides contributes $\sqrt{2}$.
⟨3⟩1. A neighbouring endpoint contributes $\sqrt{2} \cdot 1 = \sqrt{2}$.
PROOF: Substitute the endpoint value and edge weight. □
⟨3⟩2. A neighbouring interior contributes $1 \cdot \sqrt{2} = \sqrt{2}$.
PROOF: Substitute the interior value and edge weight. □
⟨3⟩3. Q.E.D.
PROOF: Every neighbour is an endpoint or an interior vertex, so ⟨3⟩1, ⟨3⟩2 apply to each side. □
⟨2⟩3. Q.E.D.
PROOF: Add the two contributions in ⟨2⟩2, together with ⟨2⟩1. □
⟨1⟩5. The equations hold on X_k^{fork}.
⟨2⟩1. An extra leaf has sum $2 = 2 \cdot 1$.
PROOF: Its neighbour is v_0 with value 2. □

⟨2⟩2. At v_k the sum is $\sqrt{2}\cdot 2 = 2\sqrt{2}$.
PROOF: Substitute its unique edge. □
⟨2⟩3. For $k = 1$, the sum at v_0 is $1 + 1 + \sqrt{2}\sqrt{2} = 4$.
PROOF: Substitute all three incident edges from Definition (affine obstructions). □
⟨2⟩4. For $k > 1$, the sum at v_0 is $1 + 1 + 2 = 4$.
PROOF: The edge v_0v_1 has weight 1 and v_1 has value 2. □
⟨2⟩5. For $k > 1$, the sum at v_{k-1} is $2 + \sqrt{2}\sqrt{2} = 4$.
PROOF: Substitute the two incident path edges. □
⟨2⟩6. At v_i with $0 < i < k - 1$, the sum is $2 + 2 = 4$.
PROOF: Both neighbours have value 2 and both edges weight 1. □
⟨2⟩7. Q.E.D.
PROOF: The six cases ⟨2⟩1–⟨2⟩6 exhaust the vertices for $k \geq 1$. □
⟨1⟩6. The equations hold on $X^{\mathrm{mid}}, X^{\mathrm{triple}}$.
⟨2⟩1. At position 0 of X^{mid} the sum is $2 = 2 \cdot 1$.
PROOF: Substitute its sole neighbour. □
⟨2⟩2. At position 1 the sum is $1 + 3 = 2 \cdot 2$.
PROOF: Substitute its two unit edges. □
⟨2⟩3. At position 2 the sum is $2 + \sqrt{2}(2\sqrt{2}) = 2 \cdot 3$.
PROOF: Substitute its unit and double-weight edges. □
⟨2⟩4. At position 3 the sum is $3\sqrt{2} + \sqrt{2} = 2(2\sqrt{2})$.
PROOF: Substitute its double-weight and unit edges. □
⟨2⟩5. At position 4 the sum is $2\sqrt{2} = 2 \cdot \sqrt{2}$.
PROOF: Substitute its sole neighbour. □
⟨2⟩6. At position 0 of X^{triple} the sum is $2\sqrt{3}$.
PROOF: Its neighbour has value 2 and its edge weight is $\sqrt{3}$. □
⟨2⟩7. At position 1 the sum is $\sqrt{3}\sqrt{3} + 1 = 2 \cdot 2$.
PROOF: Substitute its two neighbours. □
⟨2⟩8. At position 2 the sum is $2 = 2 \cdot 1$.
PROOF: Substitute its sole neighbour. □
⟨2⟩9. Q.E.D.
PROOF: Positions ⟨2⟩1–⟨2⟩8 exhaust both vertex sets. □
⟨1⟩7. Every displayed entry is positive.
PROOF: The constants are positive; on an arm $1 \leq t \leq a$ gives $a + 1 - t \geq 1$ in the displayed formula. □
⟨1⟩8. Q.E.D.
PROOF: Apply the positive-eigenvector assertion of Lemma (strict monotonicity) to ⟨1⟩1–⟨1⟩6 and ⟨1⟩7, with connectedness from Lemma (connected labelled graphs). □

Lemma (path labelling). ASSUME: X is a finite nonempty connected acyclic simple graph on m vertices and every degree is at most 2.
PROVE: There is a bijection $e : [0, m-1] \to V(X)$ with $e(i) \sim e(j) \iff |i - j| = 1$. If w is any specified vertex of degree at most 1, such a bijection can be chosen with $e(0) = w$. For every path labelling e, its reversal $i \mapsto e(m - 1 - i)$ is also a path labelling.

PROOF SKETCH: Take a longest simple path; no edge can leave its support.
⟨1⟩1. PICK a simple path $v_0, \ldots, v_k$ of maximum length.
PROOF: Distinct-vertex lists in a finite set form a finite set; singleton lists give a nonempty set of paths. □
⟨1⟩2. No neighbour of v_0 lies off this path.
PROOF: Prepending that neighbour would extend ⟨1⟩1 as a simple path. □

⟨1⟩3. No neighbour of v_k lies off this path.
PROOF: Appending that neighbour would extend ⟨1⟩1 as a simple path. □
⟨1⟩4. No neighbour of any v_i, $0 < i < k$, lies off the path.
PROOF: Such a vertex already has the two distinct neighbours v_{i-1}, v_{i+1}, so a third violates the degree bound in ⟨0⟩. □
⟨1⟩5. Every vertex belongs to the path.
⟨2⟩1. A path from a vertex outside the support to v_0 would have an edge with one endpoint inside and one outside.
PROOF: Take the first entry into the support in its finite vertex sequence. □
⟨2⟩2. Q.E.D.
PROOF: Connectedness and ⟨2⟩1 contradict ⟨1⟩2, ⟨1⟩3, ⟨1⟩4. □
⟨1⟩6. $k + 1 = m$ and $e(i) = v_i$ is a bijection.
PROOF: The path is injective by ⟨1⟩1 and surjective by ⟨1⟩5. □
⟨1⟩7. The only edges are consecutive pairs.
⟨2⟩1. Every consecutive pair is adjacent.
PROOF: Unfold the path in ⟨1⟩1. □
⟨2⟩2. No two nonconsecutive positions are adjacent.
PROOF: Their path segment and their single edge would be distinct simple paths, contradicting `SimpleGraph.IsAcyclic.path_unique`. □
⟨2⟩3. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩2 and the no-loop condition. □
⟨1⟩8. The reversal of every path labelling is a path labelling.
PROOF: The involution $i \mapsto m - 1 - i$ preserves $|i - j| = 1$, so composing with it preserves the defining edge equivalence. □
⟨1⟩9. A prescribed vertex w of degree at most 1 can be first.
⟨2⟩1. CASE: $m = 1$.
PROOF: The bijection ⟨1⟩6 has sole value w. □
⟨2⟩2. CASE: $m \geq 2$.
⟨3⟩1. $w = v_0$ or $w = v_k$.
PROOF: All other positions have two neighbours by ⟨1⟩7. □
⟨3⟩2. ASSUME: $w = v_0$.
PROVE: e is a path labelling with $e(0) = w$.
PROOF: Its first value is w by substitution. □
⟨3⟩3. ASSUME: $w = v_k$.
PROVE: $i \mapsto e(m - 1 - i)$ is a path labelling whose first value is w.
PROOF: Use ⟨1⟩8; the reversed first value is $e(m - 1) = v_k = w$. □
⟨3⟩4. Q.E.D.
PROOF: Use ⟨3⟩1 to select ⟨3⟩2 or ⟨3⟩3. □
⟨2⟩3. Q.E.D.
PROOF: Nonemptiness gives $m = 1$ or $m \geq 2$, the cases ⟨2⟩1, ⟨2⟩2. □
⟨1⟩10. Q.E.D.
PROOF: Combine ⟨1⟩6, ⟨1⟩7, ⟨1⟩9, ⟨1⟩8. □

Lemma (three-arm labelling). ASSUME: X is a finite connected acyclic simple graph, v has degree 3, and all its other vertices have degree at most 2.
PROVE: There are positive integers $a \leq b \leq c$ and a bijection $e : V_{a,b,c} \to V(X)$ with $e(*) = v$ whose edges are exactly the centre edges and the consecutive arm edges of Definition (affine obstructions).

PROOF SKETCH: Partition the other vertices by the first edge of the unique path from v, then apply the endpoint version of path labelling to each part.

⟨1⟩1. There is a unique simple path from v to each vertex.
PROOF: Combine connectedness with `SimpleGraph.IsAcyclic.path_unique`. □

LET: Enumerate the neighbours of v as w_1, w_2, w_3. Let P_r consist of vertices whose path from v starts with vw_r.

⟨1⟩2. P_1, P_2, P_3 are disjoint nonempty sets partitioning $V(X) \setminus \{v\}$.
PROOF: Each nonconstant path in ⟨1⟩1 has one first edge, and $w_r \in P_r$. □

⟨1⟩3. For every r and $z \in P_r$, the segment from w_r to z stays inside P_r.
PROOF: Each prefix is the unique path to its endpoint by ⟨1⟩1. □

⟨1⟩4. There is no edge between distinct P_r, P_s.
PROOF: Such an edge would give two paths from v to an endpoint, contradicting ⟨1⟩1; their prefixes cannot meet after v by that same uniqueness. □

⟨1⟩5. The only neighbour of v in P_r is w_r.
PROOF: The unique path to any neighbour of v is its single edge, by ⟨1⟩1. □

⟨1⟩6. For each $r \in \{1, 2, 3\}$ the induced graph on P_r admits a path labelling starting at w_r.
⟨2⟩1. The induced graph on P_r is connected and acyclic.
PROOF: Use ⟨1⟩3 for reachability; an induced cycle would be a cycle in X. □
⟨2⟩2. All its degrees are at most 2, and w_r has degree at most 1 within it.
PROOF: Deleting v removes the edge vw_r from a vertex with degree at most 2; every remaining vertex also differs from v. □
⟨2⟩3. Q.E.D.
PROOF: Apply Lemma (path labelling) to ⟨2⟩1, ⟨2⟩2, and $w_r \in P_r$. □

⟨1⟩7. These three bijections and $* \mapsto v$ give an arm bijection with all the asserted edges.
PROOF: Take their disjoint union using ⟨1⟩2, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6. □

⟨1⟩8. The three positive lengths may be ordered increasingly.
PROOF: Permute the three labelled parts of ⟨1⟩7 by a sorting permutation. □

⟨1⟩9. Q.E.D.
PROOF: Use ⟨1⟩7, ⟨1⟩8. □

Lemma (unweighted trees without obstructions). ASSUME: X is a finite connected simple graph containing no cycle, no $\widetilde{\mathrm{D}}_\ell$ ($\ell \geq 4$), and no $\widetilde{\mathrm{E}}_6, \widetilde{\mathrm{E}}_7, \widetilde{\mathrm{E}}_8$ as a subgraph (the subgraph need not be induced).
PROVE: X has a path labelling, or an arm labelling of $T(1, 1, c)$, $c \geq 1$, or of $T(1, 2, 2)$, $T(1, 2, 3)$, $T(1, 2, 4)$.

PROOF SKETCH: Exclude high degrees and two branch vertices, then truncate the ordered arms.

⟨1⟩1. X is acyclic.
PROOF: A closed walk satisfying the cycle predicate would embed Cyc_m for $m \geq 3$ by Lemma (a cycle gives a cycle subgraph), contrary to ⟨0⟩; this is the definition of acyclicity. □

⟨1⟩2. Every degree is at most 3.
PROOF: Four distinct neighbours of one vertex would embed $\widetilde{\mathrm{D}}_4$ by Definition (affine obstructions). □

⟨1⟩3. ASSUME: There are distinct degree-3 vertices u, v.
PROVE: FALSE.
PROOF SKETCH: [**new**] We may use any two branch vertices: additional edges at internal path vertices are omitted from the obstruction subgraph. PICK a simple path between u, v, by connectedness in ⟨0⟩.
⟨2⟩1. Each endpoint has two distinct neighbours off the path.

PROOF: An additional neighbour on the path would contradict path uniqueness from ⟨1⟩1; the degree is 3. □

⟨2⟩2. The four selected neighbours are mutually distinct.

PROOF: A common neighbour of the endpoints would give a second simple path between them, contradicting ⟨1⟩1. □

⟨2⟩3. The selected path and four incident edges embed $\widetilde{\mathrm{D}}_{k+4}$, where $k \geq 1$ is the path length.

PROOF: Use the injective vertex list from ⟨2⟩1, ⟨2⟩2 in Definition (affine obstructions). □

⟨2⟩4. Q.E.D.

PROOF: Step ⟨2⟩3 contradicts ⟨0⟩. □

⟨1⟩4. CASE: There is no degree-3 vertex.

PROOF: Apply Lemma (path labelling) using ⟨1⟩2, ⟨1⟩1. □

⟨1⟩5. CASE: There is a degree-3 vertex.

⟨2⟩1. PICK an arm bijection with $1 \leq a \leq b \leq c$.

PROOF: Apply Lemma (three-arm labelling) to ⟨1⟩1, ⟨1⟩2, ⟨1⟩3. □

⟨2⟩2. $a = 1$.

PROOF: Otherwise the restriction to the first two positions of each arm in ⟨2⟩1 embeds $T(2,2,2)$, forbidden by ⟨0⟩. □

⟨2⟩3. $b \leq 2$.

PROOF: If $b \geq 3$, restricting the arms to lengths $1, 3, 3$ in ⟨2⟩1 embeds a forbidden $\widetilde{\mathrm{E}}_7$. □

⟨2⟩4. ASSUME: $b = 1$.

PROVE: The labelling is $T(1,1,c)$.

PROOF: Substitute in ⟨2⟩1, using ⟨2⟩2. □

⟨2⟩5. ASSUME: $b = 2$.

PROVE: $c \in \{2,3,4\}$.

PROOF: A value $c \geq 5$ embeds $T(1,2,5)$ by restriction; $2 \leq c \leq 4$ gives the asserted integer alternatives. □

⟨2⟩6. Q.E.D.

PROOF: The positive integer $b \leq 2$ in ⟨2⟩3 is 1 or 2, so ⟨2⟩4, ⟨2⟩5 exhaust this case. □

⟨1⟩6. Q.E.D.

PROOF: The cases ⟨1⟩4, ⟨1⟩5 are exhaustive. □

Theorem (weighted dichotomy). ASSUME: X is a finite connected weighted graph whose edge weights belong to $\{1, \sqrt{2}, \sqrt{3}\}$.

PROVE: Exactly one alternative holds: X contains an affine obstruction; or it admits a weight-preserving path or arm bijection to one of

$$\mathrm{A}_\ell \ (\ell \geq 1), \quad \mathrm{BC}_\ell \ (\ell \geq 2), \quad T(1,1,\ell-3) = \mathrm{D}_\ell \ (\ell \geq 4),$$
$$T(1,2,\ell-4) = \mathrm{E}_\ell \ (\ell = 6,7,8), \quad \mathrm{F}_4, \quad \mathrm{G}_2.$$

Here A is a unit path, BC has one terminal $\sqrt{2}$ edge, D, E have unit edges, F_4 has path weights $1, \sqrt{2}, 1$, and G_2 is one $\sqrt{3}$ edge. Every graph in the second alternative has largest eigenvalue less than 2.

PROOF SKETCH: Retain the labelling of the unit support and specify the injection for every excluded weighted pattern.

⟨1⟩1. ASSUME: X contains no affine obstruction.

PROVE: X admits one of the listed weight-preserving Dynkin labellings.

⟨2⟩1. The support of X has one of the path or arm labellings in Lemma (unweighted trees without obstructions).

PROOF: By Lemma (unit containment and support embeddings), an unweighted obstruction embedded in the support would be contained in X, since all its edge weights are at least 1. Thus $\langle 1\rangle$ supplies all hypotheses of Lemma (unweighted trees without obstructions). □

$\langle 2\rangle 2$. ASSUME: There is a $\sqrt{3}$ edge with endpoints u, v.
PROVE: Those endpoints give a weight-preserving G_2 labelling of X.

$\langle 3\rangle 1$. ASSUME: There is a vertex outside $\{u, v\}$.
PROVE: FALSE.

$\langle 4\rangle 1$. There is a support edge from $\{u, v\}$ to its complement.
PROOF: Take the first exit from $\{u, v\}$ on a connecting walk to the vertex in $\langle 3\rangle$, using connectedness in $\langle 0\rangle$. □

$\langle 4\rangle 2$. $X^{\text{triple}} \preccurlyeq X$.
PROOF: Keep the edge in $\langle 2\rangle$ and lower the edge in $\langle 4\rangle 1$ to 1; their three distinct vertices give the injection in Definition (affine obstructions). □

$\langle 4\rangle 3$. Q.E.D.
PROOF: Step $\langle 4\rangle 2$ contradicts $\langle 1\rangle$. □

$\langle 3\rangle 2$. Q.E.D.
PROOF: Step $\langle 3\rangle 1$ leaves exactly the two distinct endpoints; their edge has weight $\sqrt{3}$ by $\langle 2\rangle$. □

$\langle 2\rangle 3$. ASSUME: There is no $\sqrt{3}$ edge, and the support has an arm labelling from $\langle 2\rangle 1$.
PROVE: This labelling identifies X with a listed D or E graph.

$\langle 3\rangle 1$. ASSUME: There is a $\sqrt{2}$ edge.
PROVE: FALSE.

PICK the arm containing that edge; permute the arm labels to call it arm 1, and let $k \geq 1$ be the outer endpoint position of that edge.

$\langle 4\rangle 1$. The map from $v_0, \ldots, v_k$ to $e(*), e(1,1), \ldots, e(1,k)$, and from the two extra leaves to $e(2,1), e(3,1)$, is injective.
PROOF: These positions are distinct in the arm bijection of $\langle 2\rangle$, including when $k = 1$. □

$\langle 4\rangle 2$. This map witnesses $X_k^{\text{fork}} \preccurlyeq X$.
PROOF: Retain its last $\sqrt{2}$ edge and lower every other selected edge to 1 in $\langle 4\rangle 1$; use the weight bounds in $\langle 0\rangle$. □

$\langle 4\rangle 3$. Q.E.D.
PROOF: Step $\langle 4\rangle 2$ contradicts $\langle 1\rangle$. □

$\langle 3\rangle 2$. All edges have weight 1.
PROOF: The weight alternatives in $\langle 0\rangle$, absence of $\sqrt{3}$ in $\langle 2\rangle$, and $\langle 3\rangle 1$ exclude both other weights. □

$\langle 3\rangle 3$. CASE: The labelling is $T(1,1,c)$, $c \geq 1$.
PROVE: It is a weight-preserving D_{c+3} labelling.
PROOF: Set $\ell = c + 3 \geq 4$ in the theorem's definition of D_ℓ, using $\langle 3\rangle 2$. □

$\langle 3\rangle 4$. CASE: The labelling is $T(1,2,c)$, $c \in \{2,3,4\}$.
PROVE: It is a weight-preserving E_{c+4} labelling.
PROOF: Set $\ell = c + 4 \in \{6,7,8\}$ in the theorem's definition of E_ℓ, using $\langle 3\rangle 2$. □

$\langle 3\rangle 5$. Q.E.D.
PROOF: The two arm alternatives in $\langle 2\rangle 1$ are exactly $\langle 3\rangle 3$, $\langle 3\rangle 4$. □

$\langle 2\rangle 4$. ASSUME: There is no $\sqrt{3}$ edge, and the support has a path labelling $e : [0, \ell-1] \to V(X)$.
PROVE: X has a weight-preserving A, BC or F_4 labelling.

$\langle 3\rangle 1$. There is at most one $\sqrt{2}$ edge.

$\langle 4\rangle 1$. ASSUME: Two such edges have positions $r < s$, where position i denotes the edge $e(i)e(i+1)$.

PROVE: FALSE.
⟨5⟩1. The map $i \mapsto e(r+i)$, $0 \le i \le s+1-r$, is injective.
PROOF: Restrict the path bijection of ⟨2⟩ to this interval. □
⟨5⟩2. This map witnesses $X^{\mathrm{end}}_{s-r+2} \preccurlyeq X$.
PROOF: Retain the first and last $\sqrt{2}$ edges of ⟨4⟩ and lower intervening weights to 1, using ⟨5⟩1; $s-r+2 \ge 3$ includes consecutive multiple edges. □
⟨5⟩3. Q.E.D.
PROOF: Step ⟨5⟩2 contradicts ⟨1⟩. □
⟨4⟩2. Q.E.D.
PROOF: Two distinct multiple edges would have ordered positions forbidden by ⟨4⟩1. □
⟨3⟩2. CASE: There is no $\sqrt{2}$ edge.
PROVE: e is a weight-preserving A_ℓ labelling.
PROOF: Every edge has weight 1 by ⟨0⟩, ⟨2⟩, and this case. □
⟨3⟩3. CASE: There is a $\sqrt{2}$ edge.
PROVE: X has a weight-preserving BC or F_4 labelling.
⟨4⟩1. PICK a path direction in which the two sides of the multiple edge have $1 \le p \le q$ vertices; its position is $p-1, p$.
PROOF: The reversal assertion of Lemma (path labelling) interchanges the two sides, so one of the two directions orders their sizes. □
⟨4⟩2. CASE: $p = 1$.
PROVE: This is a BC_{p+q} labelling.
PROOF: The unique multiple edge is terminal by ⟨4⟩1; all others are unit by ⟨3⟩1, ⟨2⟩. □
⟨4⟩3. CASE: $p = q = 2$.
PROVE: This is a F_4 labelling.
PROOF: Its three weights are $1, \sqrt{2}, 1$ by ⟨3⟩1, ⟨4⟩1, ⟨2⟩. □
⟨4⟩4. CASE: $p \ge 2, q \ge 3$.
PROVE: FALSE.
⟨5⟩1. The positions $p+2, p+1, p, p-1, p-2$ are distinct and lie in $[0, p+q-1]$.
PROOF: The least position satisfies $p-2 \ge 0$, the greatest satisfies $p+2 \le p+q-1$, and consecutive listed positions differ by 1. □
⟨5⟩2. Their path weights in this order are $1, 1, \sqrt{2}, 1$.
PROOF: Use the unique multiple edge at $p-1, p$ from ⟨3⟩1, ⟨4⟩1, with no $\sqrt{3}$ edge by ⟨2⟩. □
⟨5⟩3. Q.E.D.
PROOF: Steps ⟨5⟩1, ⟨5⟩2 embed X^{mid}, contrary to ⟨1⟩. □
⟨4⟩5. Q.E.D.
PROOF: The integer alternatives ⟨4⟩2, ⟨4⟩3, ⟨4⟩4 exhaust $1 \le p \le q$. □
⟨3⟩4. Q.E.D.
PROOF: A $\sqrt{2}$ edge either exists or does not; apply ⟨3⟩3 or ⟨3⟩2. □
⟨2⟩5. Q.E.D.
PROOF: If a $\sqrt{3}$ edge exists, use ⟨2⟩2. Otherwise ⟨2⟩1 supplies an arm or a path labelling, the hypotheses of ⟨2⟩3 or ⟨2⟩4. □
⟨1⟩2. Every listed Dynkin graph has largest eigenvalue less than 2.
⟨2⟩1. A_1 is a proper subgraph of Cyc_3.
PROOF: Map its sole vertex to any cycle vertex. □
⟨2⟩2. For $\ell \ge 2$, A_ℓ is a proper subgraph of $\mathrm{Cyc}_{\ell+1}$.
PROOF: Delete one vertex of that cycle. □
⟨2⟩3. D_4 is a proper subgraph of $\widetilde{\mathrm{D}}_4$.

PROOF: Delete one leaf of $K_{1,4}$. □

⟨2⟩4. For $\ell \geq 5$, D_ℓ is a proper subgraph of $\widetilde{\mathrm{D}}_\ell$.

PROOF: Delete one leaf at one fork; the other fork has arms $1, 1, (\ell-4)+1$ by Definition (affine obstructions). □

⟨2⟩5. E_6 embeds properly in $T(2,2,2)$.

PROOF: Restrict one arm to length 1. □

⟨2⟩6. E_7 embeds properly in $T(1,3,3)$.

PROOF: Restrict one length-3 arm to length 2. □

⟨2⟩7. E_8 embeds properly in $T(1,2,5)$.

PROOF: Restrict the length-5 arm to length 4. □

⟨2⟩8. BC_ℓ embeds properly in $X^{\mathrm{end}}_{\ell+1}$.

PROOF: Delete its first endpoint, retaining the last $\sqrt{2}$ edge; this includes $\ell = 2$. □

⟨2⟩9. F_4 embeds properly in X^{mid}.

PROOF: Delete the first endpoint of the path with weights $1, 1, \sqrt{2}, 1$. □

⟨2⟩10. G_2 embeds properly in X^{triple}.

PROOF: Delete the endpoint incident only with the unit edge. □

⟨2⟩11. Q.E.D.

PROOF: Apply Lemmas (affine eigenvectors), (strict monotonicity) to each containment ⟨2⟩1–⟨2⟩10; the larger obstruction is connected by Lemma (connected labelled graphs). Lemma (Rayleigh quotients under reindexing) transfers the bound along every weight-preserving labelling. □

⟨1⟩3. No graph admitting a listed Dynkin labelling contains an obstruction.

PROOF: Such a containment would give $2 \leq \lambda$ by Lemmas (affine eigenvectors), (strict monotonicity), contradicting ⟨1⟩2 after transfer by Lemma (Rayleigh quotients under reindexing); its labelled support is connected by Lemma (connected labelled graphs). □

⟨1⟩4. Q.E.D.

PROOF: Combine ⟨1⟩1, ⟨1⟩2, ⟨1⟩3 with excluded middle for obstruction containment. □

Definition (the indexed matrices). ASSUME: The rank is in the corresponding range of (3); indices are $0, \ldots, \ell - 1$.

All diagonals are 2 and unspecified entries are zero. Unit edges have both entries -1. For A the edges are consecutive pairs. For B they are consecutive pairs except that the last forward entry is -2; C is its transpose. For D the edges are $\{i, i+1\}$ for $0 \leq i \leq \ell - 3$, and $\{\ell - 3, \ell - 1\}$. For E they are $\{0, 2\}, \{1, 3\}$ and $\{i, i+1\}$ for $2 \leq i \leq \ell - 2$. The two remaining matrices are

$$\mathrm{F}_4 = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}, \qquad \mathrm{G}_2 = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}.$$

The following positive integer diagonal vectors will be used as symmetrizers:

(4)

A, D, E	$(1, \ldots, 1)$
B	$(1, \ldots, 1, 2)$
C	$(2, \ldots, 2, 1)$
F_4	$(1, 1, 2, 2)$
G_2	$(1, 3)$.

These are the defining expressions in the Cartan file in 3.1, not new matrix conventions.

Lemma (supports of the named matrices). ASSUME: N is A_ℓ with $\ell \geq 1$, B_ℓ or C_ℓ with $\ell \geq 2$, D_ℓ with $\ell \geq 4$, E_ℓ with $\ell \geq 5$, F_4, or G_2, using the actual Mathlib definitions on $\mathrm{Fin}\,\ell$.

LET: $W_N(i,i) = 0$ and $W_N(i,j) = \sqrt{N(i,j)N(j,i)}$ for $i \neq j$.

PROVE: W_N is a weighted graph with the same support as the off-diagonal nonzero entries of N. The following maps are weight-preserving bijections onto it, with exactly the displayed edges, and its support is connected:

N	domain and map	successive weights
A_ℓ	$[0,\ell-1],\ i \mapsto i$	$1,\dots,1$
$\mathrm{B}_\ell, \mathrm{C}_\ell$	$[0,\ell-1],\ i \mapsto i$	$1,\dots,1,\sqrt{2}$
F_4	$[0,3],\ i \mapsto i$	$1,\sqrt{2},1$
G_2	$[0,1],\ i \mapsto i$	$\sqrt{3}$
D_ℓ	$T(1,1,\ell-3),\ h_D$	all 1
E_ℓ	$T(1,2,\ell-4),\ h_E$	all 1

Here the arm bijections are

$$\begin{aligned} h_D(*) &= \ell-3, & h_D(1,1) &= \ell-2, & h_D(2,1) &= \ell-1, \\ h_D(3,t) &= \ell-3-t & & & & (1 \leq t \leq \ell-3), \\ h_E(*) &= 3, & h_E(1,1) &= 1, & h_E(2,1) &= 2, \\ h_E(2,2) &= 0, & h_E(3,t) &= 3+t & & (1 \leq t \leq \ell-4). \end{aligned} \tag{5}$$

All diagonals of N are 2, all unit-edge entries are $(-1,-1)$, and all nonedges have zero entries in both directions. In the displayed path orders the exceptional forward/reverse pairs are $(-2,-1)$ on the last B edge, $(-1,-2)$ on the last C edge, $(-2,-1)$ on the middle F edge, and $(-3,-1)$ on the G edge.

PROOF SKETCH: Check the matrix definitions on their own index types. The arm maps are the inverses of the index orders used to recognize arbitrary matrices; this lemma establishes their properties independently.

⟨1⟩1. The identity is the asserted unit-path labelling for A_ℓ.
PROOF: The off-diagonal nonzero clause of `CartanMatrix.A` is $i+1=j$ or $j+1=i$, with entry -1. □

⟨1⟩2. The identity is the asserted path labelling for B_ℓ.
PROOF: In `CartanMatrix.B`, the forward consecutive entry is -2 exactly at $j=\ell-1$; all reverse consecutive entries are -1, and every other off-diagonal entry vanishes. □

⟨1⟩3. The identity is the asserted path labelling for C_ℓ.
PROOF: In `CartanMatrix.C`, the reverse consecutive entry is -2 exactly at $i=\ell-1$; the other consecutive entries are -1, and the other off-diagonal entries vanish. □

⟨1⟩4. The identity is the asserted path labelling for F_4.
PROOF: Read the consecutive pairs $(-1,-1),(-2,-1),(-1,-1)$ and all zero nonedges in $\mathtt{CartanMatrix.F_4}$. □

⟨1⟩5. The identity is the asserted path labelling for G_2.
PROOF: Its only off-diagonal pair is $(-3,-1)$ in $\mathtt{CartanMatrix.G_2}$. □

⟨1⟩6. h_D is the asserted unit-arm bijection.
⟨2⟩1. h_D is bijective.
PROOF: Its centre and short arms cover $\ell-3,\ell-2,\ell-1$; $t \mapsto \ell-3-t$ bijects arm 3 onto $[0,\ell-4]$. These ranges are disjoint and exhaust $\mathrm{Fin}\,\ell$. □
⟨2⟩2. The nonzero off-diagonal unordered pairs of D_ℓ are $\{i,i+1\}$ for $0 \leq i \leq \ell-4$ and $\{\ell-3,\ell-2\},\{\ell-3,\ell-1\}$, all unit pairs.

PROOF: For $i<j$, the clauses of `CartanMatrix.D` reduce to $j=i+1$, $j+2<\ell$, or $i=\ell-3$, $j\in\{\ell-2,\ell-1\}$. The reversed clauses give the reversed entries. □

⟨2⟩3. The three centre edges map to $\{\ell-3,\ell-2\},\{\ell-3,\ell-1\},\{\ell-3,\ell-4\}$.
PROOF: Substitute the centre and first arm positions in (5). □

⟨2⟩4. The remaining arm edges map exactly to $\{i,i+1\}$ for $0\le i$ and $i+5\le\ell$.
PROOF: On arm 3, consecutive $t,t+1$ give consecutive $\ell-3-t,\ell-4-t$; the interval is empty at $\ell=4$. The two length-1 arms have no internal edges. □

⟨2⟩5. Q.E.D.
PROOF: The edge images in ⟨2⟩3, ⟨2⟩4 exhaust ⟨2⟩2; bijectivity ⟨2⟩1 therefore also excludes every other edge. □

⟨1⟩7. h_E is the asserted unit-arm bijection for every $\ell\ge5$.
⟨2⟩1. h_E is bijective.
PROOF: The centre and first two arms cover $3,1,2,0$; $t\mapsto3+t$ bijects arm 3 onto $[4,\ell-1]$, disjoint from those four values. □

⟨2⟩2. The nonzero off-diagonal unordered pairs are $\{0,2\},\{1,3\}$ and $\{i,i+1\}$ for $2\le i\le\ell-2$, all unit pairs.
PROOF: These are exactly the nonzero clauses in `CartanMatrix.E`, valid in the whole assumed range $\ell\ge5$. □

⟨2⟩3. The centre edges map to $\{3,1\},\{3,2\},\{3,4\}$.
PROOF: Substitute the first position on each arm in (5). □

⟨2⟩4. The remaining edges map to $\{2,0\}$ and $\{i,i+1\}$ for $4\le i\le\ell-2$.
PROOF: The two positions of arm 2 map to $2,0$; consecutive positions on arm 3 map to consecutive integers starting at 4. Arm 1 has no internal edge. □

⟨2⟩5. Q.E.D.
PROOF: The edge images in ⟨2⟩3, ⟨2⟩4 exhaust ⟨2⟩2; use ⟨2⟩1 for the reverse edge implication. □

⟨1⟩8. The weight and support assertions follow from these entries.
PROOF: The products of the displayed nonzero pairs are $1,2,3$; every other off-diagonal product is zero. Their nonnegative square roots define W_N. □

⟨1⟩9. Every displayed support is connected.
PROOF: Apply Lemma (connected labelled graphs) to the path or arm bijections in ⟨1⟩1–⟨1⟩7. □

⟨1⟩10. Q.E.D.
PROOF: Combine ⟨1⟩1–⟨1⟩9; the diagonal clause of each named definition is 2. □

Definition (integer matrix invariants). ASSUME: $N : I\times I\to\mathbb{Z}$ is a matrix on a finite set with $N(i,j)=0 \iff N(j,i)=0$.

Let $d_N(i)=|\{j : j\ne i,\ N(i,j)\ne0\}|$ be its support degree. Define the following nonnegative integers, with cardinalities regarded in $\mathbb{Z}$:

$$
\begin{aligned}
n(N) &= |I|,\\
t(N) &= |\{(i,j) : N(i,j)N(j,i)=3\}|,\\
u(N) &= |\{(i,j) : N(i,j)N(j,i)=2\}|,\\
b(N) &= |\{i : d_N(i)=3\}|,\\
a_{\mathrm{in}}(N) &= |\{(i,j) : N(i,j)=-2,\ d_N(j)=1\}|,\\
a_{\mathrm{out}}(N) &= |\{(i,j) : N(i,j)=-2,\ d_N(i)=1\}|,\\
s(N) &= |\{i : d_N(i)=1,\ \exists j,\ N(i,j)\ne0,\ d_N(j)=3\}|.
\end{aligned}
\tag{6}
$$

Write $\mathcal{I}(N)=(n,t,u,b,a_{\mathrm{in}},a_{\mathrm{out}},s)(N)$. Pairs in $t,u,a_{\mathrm{in}},a_{\mathrm{out}}$ are ordered; s counts leaves, not pairs.

Lemma (invariance and values). ASSUME: N is an integer matrix on a finite set I with symmetric zero relation.

PROVE: For every bijection $e : J \to I$, the matrix $N_e(i,j) = N(e(i), e(j))$ satisfies $\mathcal{I}(N_e) = \mathcal{I}(N)$. For every permitted named matrix, the values are:

(7)

N	n	t	u	b	a_{in}	a_{out}	s
A_ℓ $(\ell \geq 1)$	ℓ	0	0	0	0	0	0
B_2	2	0	2	0	1	1	0
B_ℓ $(\ell \geq 3)$	ℓ	0	2	0	1	0	0
C_ℓ $(\ell \geq 3)$	ℓ	0	2	0	0	1	0
D_4	4	0	0	1	0	0	3
D_ℓ $(\ell \geq 5)$	ℓ	0	0	1	0	0	2
E_6	6	0	0	1	0	0	1
E_7	7	0	0	1	0	0	1
E_8	8	0	0	1	0	0	1
F_4	4	0	2	0	0	0	0
G_2	2	2	0	0	0	0	0

PROOF SKETCH: Transport finite counting sets along the bijection, then count on the independently labelled supports of the named matrices.

⟨1⟩1. All seven integers are preserved by reindexing.

⟨2⟩1. $d_{N_e}(i) = d_N(e(i))$ for every i.

PROOF: The map $j \mapsto e(j)$ bijects the neighbour sets in Definition (integer matrix invariants). □

⟨2⟩2. The integers n, b, s are preserved.

PROOF: Transport their defining vertex sets by e, using ⟨2⟩1 and the entry equality defining N_e. □

⟨2⟩3. The integers $t, u, a_{\text{in}}, a_{\text{out}}$ are preserved.

PROOF: The map $(i,j) \mapsto (e(i), e(j))$ bijects each defining pair set, by ⟨2⟩1 and the entry equality. □

⟨2⟩4. Q.E.D.

PROOF: Combine ⟨2⟩2, ⟨2⟩3. □

⟨1⟩2. The A row has only its size nonzero.

PROOF: Lemma (supports of the named matrices) gives a unit path, with no degree-3 vertex or -2 entry; use Definition (integer matrix invariants). □

⟨1⟩3. The B_2 row has $u = 2$ and $(a_{\text{in}}, a_{\text{out}}) = (1,1)$.

PROOF: The sole double edge has two ordered products 2; both endpoints of its sole -2 entry are leaves, by Lemma (supports of the named matrices). □

⟨1⟩4. For $\ell \geq 3$, the B_ℓ row has $u = 2$ and $(a_{\text{in}}, a_{\text{out}}) = (1,0)$.

PROOF: The sole -2 entry points from position $\ell-2$ of degree 2 to the leaf $\ell-1$ in Lemma (supports of the named matrices); the path has no branch vertex. □

⟨1⟩5. For $\ell \geq 3$, the C_ℓ row has $u = 2$ and $(a_{\text{in}}, a_{\text{out}}) = (0,1)$.

PROOF: The sole -2 entry points from the leaf $\ell-1$ to position $\ell-2$ of degree 2 in Lemma (supports of the named matrices); the path has no branch vertex. □

⟨1⟩6. For D_4, $b = 1$ and $s = 3$.

PROOF: The unit tree $T(1,1,1)$ in Lemma (supports of the named matrices) has one degree-3 centre adjacent to all three leaves. □

⟨1⟩7. For D_ℓ, $\ell \geq 5$, $b = 1$ and $s = 2$.

PROOF: In $T(1,1,\ell-3)$ the third arm has length at least 2, so only the first two leaves meet the centre, by Lemma (supports of the named matrices). □

⟨1⟩8. For each of $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$, $b = 1$ and $s = 1$.
PROOF: The arm lengths $1, 2, \ell - 4$ in Lemma (supports of the named matrices) have exactly one length equal to 1; all edges are unit. □
⟨1⟩9. For F_4, $u = 2$ and $(a_{\text{in}}, a_{\text{out}}) = (0, 0)$.
PROOF: The sole -2 entry joins the two degree-2 vertices of the four-vertex path in Lemma (supports of the named matrices). □
⟨1⟩10. For G_2, $n = t = 2$ and $u = b = a_{\text{in}} = a_{\text{out}} = s = 0$.
PROOF: Lemma (supports of the named matrices) gives two vertices, one product-3 edge, no -2 entry and no branch vertex. □
⟨1⟩11. Q.E.D.
PROOF: In the counts ⟨1⟩2–⟨1⟩10, each matrix has its stated size, and the listed supports and entries exclude all uncounted patterns; together with ⟨1⟩1 these give (7). □

Theorem A. ASSUME: I is a finite nonempty set and $M : I \times I \to \mathbb{Z}$ satisfies `Matrix.IsFiniteCartan` and `Matrix.IsIndecomposable`.
PROVE: There are $\ell > 0$ and a bijection $e : [0, \ell - 1] \to I$ such that $M(e(i), e(j))$ is a matrix in (3). Its type and rank are unique; the bijection need not be unique. Conversely every listed matrix is finite Cartan and indecomposable.

[repair] The empty matrix satisfies both library predicates. It is excluded here, or may be added as a separate rank-zero alternative.

PROOF SKETCH: Classify the weighted support, orient its one possible multiple edge, and distinguish the resulting matrices by intrinsic invariants.
⟨1⟩1. The symmetrized support of M has a Dynkin labelling.
⟨2⟩1. The symmetrization has $\lambda < 2$.
PROOF: Apply Lemma (symmetrization). □
⟨2⟩2. It contains no affine obstruction.
PROOF: Lemmas (affine eigenvectors), (strict monotonicity) would give $2 \le \lambda$, contradicting ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Use the no-obstruction alternative of Theorem (weighted dichotomy) with ⟨2⟩2. □
⟨1⟩2. The possible ordered entry pairs are

unit edge	$(-1, -1)$
double edge	$(-1, -2)$ or $(-2, -1)$
triple edge	$(-1, -3)$ or $(-3, -1)$.

⟨2⟩1. Product 1 forces $(-1, -1)$.
PROOF: The two negative integers are negatives of positive divisors of 1. □
⟨2⟩2. Product 2 forces $(-1, -2)$ or $(-2, -1)$.
PROOF: The positive divisors of 2 are 1 and 2. □
⟨2⟩3. Product 3 forces $(-1, -3)$ or $(-3, -1)$.
PROOF: The positive divisors of 3 are 1 and 3. □
⟨2⟩4. Q.E.D.
PROOF: Apply these three arithmetic cases to Lemma (symmetrization). □
⟨1⟩3. There are $\ell > 0$, a listed type of rank ℓ, and a bijection $e : \mathrm{Fin}\,\ell \to I$ such that $M(e(i), e(j))$ equals that named matrix at every i, j.
⟨2⟩1. For a unit path, its path bijection gives A_ℓ.
PROOF: Substitute its edge equivalence and ⟨1⟩2 in Definition (the indexed matrices). □

⟨2⟩2. For $T(1,1,\ell-3)$, use its arm bijection f and set
$$e(\ell-3)=f(*),\quad e(\ell-2)=f(1,1),\quad e(\ell-1)=f(2,1),$$
$$e(i)=f(3,\ell-3-i)\quad(0\le i\le \ell-4).$$
Then $M(e(i),e(j))=\mathrm{D}_\ell(i,j)$.

⟨3⟩1. The formula for e is a bijection.

PROOF: It is $f\circ h_D^{-1}$, a composition of bijections by Lemma (supports of the named matrices). □

⟨3⟩2. The edges become $\{i,i+1\}$ for $0\le i\le \ell-3$ and $\{\ell-3,\ell-1\}$.

PROOF: Transport the edges of the arm labelling f along h_D^{-1} from Lemma (supports of the named matrices). □

⟨3⟩3. Q.E.D.

PROOF: Use ⟨3⟩1, ⟨3⟩2, ⟨1⟩2, and Definition (the indexed matrices). □

⟨2⟩3. For $T(1,2,\ell-4)$, use its arm bijection f and set
$$e(3)=f(*),\quad e(1)=f(1,1),\quad e(2)=f(2,1),\quad e(0)=f(2,2),$$
$$e(i)=f(3,i-3)\quad(4\le i\le \ell-1).$$
Then $M(e(i),e(j))=\mathrm{E}_\ell(i,j)$.

⟨3⟩1. The formula for e is a bijection.

PROOF: It is $f\circ h_E^{-1}$, a composition of bijections by Lemma (supports of the named matrices). □

⟨3⟩2. Its edges become $\{0,2\},\{1,3\}$ and $\{i,i+1\}$ for $2\le i\le \ell-2$.

PROOF: Transport the edges of f along h_E^{-1} from Lemma (supports of the named matrices). □

⟨3⟩3. Q.E.D.

PROOF: Use ⟨3⟩1, ⟨3⟩2, ⟨1⟩2, and Definition (the indexed matrices). □

⟨2⟩4. For a BC_ℓ support, there is a path bijection e for which $M(e(i),e(j))$ is $\mathrm{B}_\ell(i,j)$ or, when $\ell\ge 3$, $\mathrm{C}_\ell(i,j)$.

⟨3⟩1. PICK a path labelling putting the terminal multiple edge between positions $\ell-2,\ell-1$.

PROOF: Reverse the given path when necessary, by the reversal assertion of Lemma (path labelling). □

⟨3⟩2. ASSUME: its last forward entry is -2.
PROVE: The matrix is B_ℓ.

PROOF: Use ⟨1⟩2 and Definition (the indexed matrices). □

⟨3⟩3. ASSUME: that entry is -1 and $\ell\ge 3$.
PROVE: It is C_ℓ.

PROOF: Use ⟨1⟩2 and Definition (the indexed matrices). □

⟨3⟩4. ASSUME: that entry is -1 and $\ell=2$.
PROVE: Swapping the indices gives B_2.

PROOF: Transpose the two-entry pair in ⟨1⟩2. □

⟨3⟩5. Q.E.D.

PROOF: The entry orders and rank alternatives in ⟨3⟩2, ⟨3⟩3, ⟨3⟩4 are exhaustive. □

⟨2⟩5. For a F_4 support, there is a path bijection e with $M(e(i),e(j))=\mathrm{F}_4(i,j)$ for every i,j.

⟨3⟩1. PICK the direction in which the central forward entry is -2.

PROOF: Reversal in Lemma (path labelling) interchanges the two entries of the central edge in ⟨1⟩2. □

⟨3⟩2. Q.E.D.

PROOF: The two unit edges and the central pair in ⟨3⟩1 are exactly the entries in Lemma (supports of the named matrices); diagonals and nonedges agree. □

⟨2⟩6. For a G_2 support, there is a bijection e with $M(e(i),e(j))=\mathrm{G}_2(i,j)$ for every i,j.

⟨3⟩1. PICK the endpoint order with forward entry -3.
PROOF: One of the two orders has that entry by ⟨1⟩2. □
⟨3⟩2. Q.E.D.
PROOF: Step ⟨3⟩1 gives the off-diagonal pair $(-3,-1)$, with diagonal 2, exactly as in Lemma (supports of the named matrices). □
⟨2⟩7. Q.E.D.
PROOF: The alternatives in ⟨1⟩1 are precisely ⟨2⟩1–⟨2⟩6. □
⟨1⟩4. Every listed matrix is finite Cartan and indecomposable.
LET: For a listed matrix M take its positive integer symmetrizer from (4).
⟨2⟩1. On each unit edge $d(i) = d(j)$ and both entries are -1.
PROOF: Read Definition (the indexed matrices) and (4) on the unit edges. □
⟨2⟩2. On the B multiple edge the two products are $1(-2) = 2(-1)$.
PROOF: Substitute its two entries and symmetrizer values. □
⟨2⟩3. On the C multiple edge they are $2(-1) = 1(-2)$.
PROOF: Substitute its two entries and symmetrizer values. □
⟨2⟩4. On the F multiple edge they are $1(-2) = 2(-1)$.
PROOF: Substitute its two entries and symmetrizer values. □
⟨2⟩5. On the G edge they are $1(-3) = 3(-1)$.
PROOF: Substitute its two entries and symmetrizer values. □
⟨2⟩6. On every nonedge both products vanish.
PROOF: Use the unspecified-entry rule in Definition (the indexed matrices). □
⟨2⟩7. DM is symmetric and $D^{1/2}MD^{-1/2} = 2I - B$ for the corresponding Dynkin graph.
PROOF: Combine the entry equalities ⟨2⟩1–⟨2⟩6, including the diagonal 2, with the labelled weighted supports of Lemma (supports of the named matrices). □
⟨2⟩8. $2I - B$ is positive definite over $\mathbb{R}$.
⟨3⟩1. $\lambda(B) < 2$.
PROOF: Lemma (supports of the named matrices) gives a weight-preserving Dynkin labelling; transfer the bound in Theorem (weighted dichotomy) by Lemma (Rayleigh quotients under reindexing). □
⟨3⟩2. For $\mathbf{x} \neq 0$, $\mathbf{x}^T(2I - B)\mathbf{x} \geq (2 - \lambda(B))(\mathbf{x}, \mathbf{x}) > 0$.
PROOF: Use the quadratic-form bound in Lemma (Rayleigh quotients) and ⟨3⟩1. □
⟨3⟩3. Q.E.D.
PROOF: Symmetry from ⟨2⟩7 and ⟨3⟩2 are positive definiteness. □
⟨2⟩9. DM is positive on nonzero integer vectors.
PROOF: Substitute $D^{1/2}\mathbf{x} \neq 0$ in ⟨2⟩8, using ⟨2⟩7. □
⟨2⟩10. The matrix is finite Cartan.
PROOF: Assemble its four defining fields from Definition (the indexed matrices), ⟨2⟩7, ⟨2⟩9 and `Matrix.posDef_iff_dotProduct_mulVec`. □
⟨2⟩11. The matrix is indecomposable.
PROOF: Apply Lemma (connected support) to the symmetric zero relation and connected support in Lemma (supports of the named matrices). □
⟨2⟩12. Q.E.D.
PROOF: Combine ⟨2⟩10, ⟨2⟩11. □
⟨1⟩5. ASSUME: P, Q are two permitted named matrices equal after reindexing by a bijection.
PROVE: Their types and ranks agree.
⟨2⟩1. $\mathcal{I}(P) = \mathcal{I}(Q)$; in particular their ranks agree.
PROOF: Apply Lemma (invariance and values) to the reindexing equality in ⟨1⟩. □
⟨2⟩2. CASE: $t(P) = 2$.

PROVE: P, Q both have type G_2.
PROOF: Only the G_2 row in (7) has $t = 2$; use ⟨2⟩1. □
⟨2⟩3. CASE: $u(P) = 2$ and $(a_{\text{in}}(P), a_{\text{out}}(P)) = (0, 0)$.
PROVE: P, Q both have type F_4.
PROOF: Among the four double-edge rows of (7), only F_4 has this pair; use ⟨2⟩1. □
⟨2⟩4. CASE: $u(P) = 2$ and $(a_{\text{in}}(P), a_{\text{out}}(P)) \neq (0, 0)$.
PROVE: P, Q have the same B or C type and rank.
⟨3⟩1. CASE: $n(P) = 2$.
PROOF: Only B_2 among the permitted double-edge rows has this size in (7); apply ⟨2⟩1. □
⟨3⟩2. CASE: $n(P) \geq 3$ and $a_{\text{in}}(P) = 1$.
PROOF: The pair is $(1, 0)$ only for the B_ℓ row at these ranks in (7); apply ⟨2⟩1. □
⟨3⟩3. CASE: $n(P) \geq 3$ and $a_{\text{in}}(P) = 0$.
PROOF: Under ⟨2⟩, the pair is $(0, 1)$, which singles out C_ℓ in (7); apply ⟨2⟩1. □
⟨3⟩4. Q.E.D.
PROOF: The three eligible rows give exactly ⟨3⟩1–⟨3⟩3. □
⟨2⟩5. CASE: $t(P) = u(P) = b(P) = 0$.
PROVE: P, Q both have type A at the same rank.
PROOF: Only the A_ℓ row has these three zero counts in (7); use ⟨2⟩1. □
⟨2⟩6. CASE: $t(P) = u(P) = 0$ and $b(P) = 1$.
PROVE: P, Q have the same D or E type and rank.
⟨3⟩1. CASE: $s(P) = 3$.
PROOF: Only the D_4 row has $s = 3$ in (7); use ⟨2⟩1. □
⟨3⟩2. CASE: $s(P) = 2$.
PROOF: Only the D_ℓ, $\ell \geq 5$, row has $s = 2$ in (7); size in ⟨2⟩1 fixes ℓ. □
⟨3⟩3. CASE: $s(P) = 1$.
PROOF: The remaining rows in (7) are E_6, E_7, E_8; their sizes $6, 7, 8$ distinguish them by ⟨2⟩1. □
⟨3⟩4. Q.E.D.
PROOF: The branched rows give precisely the three values in ⟨3⟩1–⟨3⟩3. □
⟨2⟩7. Q.E.D.
PROOF: The table exhausts the cases ⟨2⟩2–⟨2⟩6; all parameter equalities follow from the size equality in ⟨2⟩1. □
⟨1⟩6. Q.E.D.
PROOF: Combine ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, composing the two labellings when comparing two models of the original matrix. □

Remark (small ranks). ASSUME: ℓ is a natural number.

Here the symbols $A_\ell, B_\ell, C_\ell, D_\ell, E_\ell$ denote Mathlib's generalized definitions at all natural ranks. An old-index order $(i_0, \ldots, i_{\ell-1})$ lists, for each new index k, its old index i_k: the identification with N means $N(k, l) = M(i_k, i_l)$.

PROVE: $B_1 = C_1 = D_1 = A_1 = [2]$, $D_2 = 2I_2$, and the following old-index orders give the identifications: $C_2 \cong B_2$ by $(1, 0)$; $D_3 \cong A_3$ by $(1, 0, 2)$; $E_1 = A_1$, $E_2 = 2I_2$; $E_3 \cong A_2 \oplus A_1$ by $(0, 2, 1)$; $E_4 \cong A_4$ by $(0, 2, 3, 1)$; $E_5 \cong D_5$ by $(0, 2, 3, 1, 4)$. The matrices E_ℓ, $\ell \geq 9$, are not finite Cartan. All five parameterized definitions at rank 0 give the empty matrix.

PROOF SKETCH: The small identities are entrywise substitutions; the long E arm contains an affine obstruction.
⟨1⟩1. All stated rank-1 matrices equal [2].
PROOF: The only index pair is diagonal in the definitions in 3.1. □
⟨1⟩2. $D_2 = 2I_2$.

PROOF: Apply `CartanMatrix.D_two`. □

⟨1⟩3. The order $(1, 0)$ sends C_2 to B_2.
PROOF: Substitute in `CartanMatrix.B_two`, `CartanMatrix.C_two`. □

⟨1⟩4. The order $(1, 0, 2)$ sends D_3 to A_3.
PROOF: Apply `CartanMatrix.D_three'` in `Mathlib/LinearAlgebra/Matrix/Cartan.lean`; its transposition of indices 0 and 1 is the order $(1, 0, 2)$. □

⟨1⟩5. $E_2 = 2I_2$.
PROOF: None of the off-diagonal edge clauses of `CartanMatrix.E` has indices less than 2. □

⟨1⟩6. At rank 3 its sole edge is $\{0, 2\}$.
PROOF: Evaluate the edge clauses of `CartanMatrix.E`. □

⟨1⟩7. At rank 4 its path order is $(0, 2, 3, 1)$.
PROOF: Evaluate those edge clauses at indices less than 4. □

⟨1⟩8. At rank 5 its edges in order $(0, 2, 3, 1, 4)$ are exactly the D_5 edges.
PROOF: Substitute that order into Definition (the indexed matrices). □

⟨1⟩9. For $\ell \geq 9$ the leading nine vertices contain $T(1, 2, 5)$.
PROOF: Restrict h_E in Lemma (supports of the named matrices) to the first five positions of its long arm and retain both short arms. □

⟨1⟩10. These E_ℓ are not finite Cartan.
⟨2⟩1. Their support is connected.
PROOF: Apply the connectedness assertion of Lemma (supports of the named matrices), which includes every $\ell \geq 9$. □
⟨2⟩2. The matrices are indecomposable.
PROOF: Apply Lemma (connected support) to ⟨2⟩1 and their symmetric entries. □
⟨2⟩3. $\lambda \geq 2$ for their unit support.
PROOF: Apply Lemmas (affine eigenvectors), (strict monotonicity) to ⟨1⟩9. □
⟨2⟩4. Q.E.D.
PROOF: A finite Cartan hypothesis with ⟨2⟩2 would give $\lambda < 2$ by Lemma (symmetrization), contradicting ⟨2⟩3. □

⟨1⟩11. At rank 0 there is only one matrix.
PROOF: Its index type has no elements. □

⟨1⟩12. Q.E.D.
PROOF: The entry checks ⟨1⟩1–⟨1⟩8, ⟨1⟩10, ⟨1⟩11 give the claims. □

Further linear algebra interfaces. `Matrix.IsHermitian.eigenvectorBasis` and `Matrix.IsHermitian.mulVec_eigenvectorBasis`, in `Mathlib/Analysis/Matrix/Spectrum.lean`, give an orthonormal eigenbasis for a real symmetric matrix on a finite index type. `LinearMap.finrank_range_add_finrank_ker`, in `Mathlib/LinearAlgebra/FiniteDimensional/Lemmas.lean`, is rank-nullity; `finrank_span_singleton`, in `Mathlib/LinearAlgebra/FiniteDimensional/Basic.lean`, gives dimension 1 for the span of a nonzero vector. `LinearIndependent.span_eq_top_of_card_eq_finrank'`, in the former file, makes an independent family of the right cardinality spanning. `linearIndependent_of_top_le_span_of_card_eq_finrank`, in `Mathlib/LinearAlgebra/Dimension/OrzechProperty.lean`, proves the converse implication for a spanning family of the right cardinality. All applications below are over a field, with finite-dimensional spaces. `Matrix.inv_def`, in `Mathlib/LinearAlgebra/Matrix/NonsingularInverse.lean`, expresses the inverse as the inverse determinant times the adjugate.

Lemma (a Gram realization). ASSUME: Q is real symmetric on a finite set I and $\mathbf{x}^T Q\mathbf{x} \geq 0$ for every real vector $\mathbf{x}$.
PROVE: There is a finite-dimensional Euclidean space V and spanning vectors $(\mathbf{v}_i)_{i\in I}$ with $(\mathbf{v}_i, \mathbf{v}_j) = Q(i,j)$. For every family $(\mathbf{w}_i)_{i\in I}$ with Gram matrix Q in any real inner product space, its relations are exactly $\ker Q$ and $\dim \operatorname{span}\{\mathbf{w}_i : i \in I\} = |I| - \dim\ker Q$. In particular $\dim V = |I| - \dim\ker Q$ for the constructed realization.

PROOF SKETCH: **[new]** Use the orthonormal eigenbasis to construct the square root map explicitly. This replaces the ordinary proof's quotient inner-product construction.
⟨1⟩1. PICK an orthonormal eigenbasis $(\mathbf{b}_k)$ with eigenvalues t_k.
PROOF: Apply the eigenbasis declarations in 3.1.1. □
⟨1⟩2. $t_k \geq 0$ for every k.
PROOF: Evaluate the nonnegative form at the unit eigenvector $\mathbf{b}_k$ of ⟨1⟩1. □
LET: $T\mathbf{x} = \sum_k \sqrt{t_k}(\mathbf{b}_k, \mathbf{x})\mathbf{b}_k$, $V = \operatorname{range} T$, and $\mathbf{v}_i = T\mathbf{u}_i$ regarded in V.
⟨1⟩3. $(T\mathbf{x}, T\mathbf{y}) = \mathbf{x}^T Q\mathbf{y}$.
PROOF: Expand in the orthonormal basis ⟨1⟩1 and use $(\sqrt{t_k})^2 = t_k$ from ⟨1⟩2. □
⟨1⟩4. $(\mathbf{v}_i, \mathbf{v}_j) = Q(i,j)$.
PROOF: Specialize ⟨1⟩3 to $\mathbf{u}_i, \mathbf{u}_j$. □
⟨1⟩5. The vectors $\mathbf{v}_i$ span V.
PROOF: Apply the linear map T to the coordinate expansion $\mathbf{x} = \sum_i \mathbf{x}(i)\mathbf{u}_i$. □
⟨1⟩6. For any family $(\mathbf{w}_i)$ with Gram matrix Q, $\sum_i x(i)\mathbf{w}_i = 0$ if and only if $Qx = 0$.
⟨2⟩1. $\|\sum_i x(i)\mathbf{w}_i\|^2 = x^T Q x$.
PROOF: Bilinearity in 2 and the Gram entries. □
⟨2⟩2. The sum is zero if and only if $x^T Q x = 0$.
PROOF: Apply definiteness in 2 to ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Lemma (a null vector of a nonnegative form) and substitution of $Qx = 0$ give the equivalence in ⟨2⟩2. □
⟨1⟩7. For every realizing family $(\mathbf{w}_i)$, $\dim \operatorname{span}\{\mathbf{w}_i : i \in I\} = |I| - \dim\ker Q$.
PROOF: Apply rank-nullity from 3.1.1 to $x \mapsto \sum_i x(i)\mathbf{w}_i$: its range is the span by 2, and its kernel is ⟨1⟩6. □
⟨1⟩8. Q.E.D.
PROOF: Combine ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, ⟨1⟩7. □

Theorem 3.8. ASSUME: X is a finite connected simple graph on m vertices and B is its real adjacency matrix.
PROVE: (Corollary D of the ordinary proof.) $\lambda(B) = 2$ if and only if X is isomorphic to a cycle Cyc_s $(s \geq 3)$, $\widetilde{\mathrm{D}}_\ell$ $(\ell \geq 4)$, or one of $\widetilde{\mathrm{E}}_6, \widetilde{\mathrm{E}}_7, \widetilde{\mathrm{E}}_8$. For any such graph, $2I - B$ is the Gram matrix of spanning vectors of norm $\sqrt{2}$ in dimension $m-1$. In every realization, deleting any one vector leaves a basis of their span.

[repair] The ambient space in the deletion assertion is the span. The assertion includes every deletion, thus includes the marked vertices of [CGSS, Figure 1]. Its A_n is our cycle Cyc_{n+1}.

PROOF SKETCH: Strict monotonicity gives equality only at the unweighted obstructions; a positive null vector controls every deletion.
⟨1⟩1. $\lambda(B) = 2$ exactly for graphs isomorphic to the listed unweighted obstructions.
⟨2⟩1. Only unweighted obstructions can be contained in X.
PROOF: A weight $\sqrt{2}$ or $\sqrt{3}$ cannot be at most an adjacency entry 0 or 1, by Definition (weighted graphs). □
⟨2⟩2. ASSUME: $\lambda(B) = 2$.

PROVE: X is isomorphic to a listed unweighted obstruction.

⟨3⟩1. X has no listed Dynkin labelling.

PROOF: Such a labelling would give $\lambda(B) < 2$ by Theorem (weighted dichotomy) and Lemma (Rayleigh quotients under reindexing), contradicting ⟨2⟩. □

⟨3⟩2. PICK an obstruction Y and a containment $e : Y \preccurlyeq X$.

PROOF: Theorem (weighted dichotomy) and ⟨3⟩1 give the obstruction alternative. □

⟨3⟩3. Y is one of the listed unweighted graphs.

PROOF: Apply ⟨2⟩1 to ⟨3⟩2. □

⟨3⟩4. The containment e is not proper.

PROOF: Otherwise Lemmas (affine eigenvectors), (strict monotonicity) give $2 = \lambda(Y) < \lambda(B) = 2$, contradicting ⟨2⟩. □

⟨3⟩5. e is a graph isomorphism from Y onto X.

PROOF: By Definition (weighted graphs), ⟨3⟩4 says e is onto and every entry agrees. Both graphs have unit edge weights by ⟨3⟩3. □

⟨3⟩6. Q.E.D.

PROOF: Use ⟨3⟩3, ⟨3⟩5. □

⟨2⟩3. ASSUME: X is isomorphic to a listed unweighted graph Y.

PROVE: $\lambda(B) = 2$.

⟨3⟩1. $\lambda(Y) = 2$.

PROOF: Apply Lemma (affine eigenvectors). □

⟨3⟩2. $\lambda(B) = \lambda(Y)$.

PROOF: The graph isomorphism in ⟨2⟩ preserves unit adjacency entries; apply Lemma (Rayleigh quotients under reindexing). □

⟨3⟩3. Q.E.D.

PROOF: Combine ⟨3⟩1, ⟨3⟩2. □

⟨2⟩4. Q.E.D.

PROOF: Steps ⟨2⟩2, ⟨2⟩3 are the two directions of the equivalence. □

⟨1⟩2. ASSUME: X is isomorphic to a listed graph.

PROVE: The stated realization and every-deletion basis assertion hold.

⟨2⟩1. PICK $\mathbf{w} > 0$ with $B\mathbf{w} = 2\mathbf{w}$.

PROOF: Apply Lemma (Rayleigh quotients) to the connected adjacency matrix, with $\lambda(B) = 2$ from ⟨1⟩1 and ⟨1⟩. □

LET: $Q = 2I - B$.

⟨2⟩2. For every real vector $\mathbf{x}$,

$$\mathbf{x}^T Q \mathbf{x} = \sum_{\{i,j\} \in E(X)} w(i)w(j) \left(\frac{x(i)}{w(i)} - \frac{x(j)}{w(j)} \right)^2 . \tag{8}$$

⟨3⟩1. The mixed term for each edge is $-2x(i)x(j)$.

PROOF: Expand that edge's square in (8). □

⟨3⟩2. The coefficient of $x(i)^2$ on the right is 2.

PROOF: It is $\sum_{j \sim i} w(j)/w(i) = 2$ by the chosen eigenvector equation. □

⟨3⟩3. Q.E.D.

PROOF: Compare the coefficients in ⟨3⟩1, ⟨3⟩2 with the quadratic form of $2I - B$. □

⟨2⟩3. The form of Q is nonnegative.

PROOF: Every summand in ⟨2⟩2 is a nonnegative square times a positive number. □

⟨2⟩4. $\ker Q = \mathbb{R}\mathbf{w}$.

⟨3⟩1. ASSUME: $Q\mathbf{x} = 0$.

PROVE: $x(i)/w(i) = x(j)/w(j)$ on every edge.

PROOF: Each nonnegative summand in the zero sum ⟨2⟩2 must vanish. □

⟨3⟩2. ASSUME: $Q\mathbf{x} = 0$.
PROVE: $\mathbf{x}$ is a scalar multiple of $\mathbf{w}$.
PROOF: Apply Lemma (edge-constant functions) to $i \mapsto x(i)/w(i)$ on the connected adjacency graph X, using ⟨3⟩1. □
⟨3⟩3. Every multiple of $\mathbf{w}$ is in $\ker Q$.
PROOF: Use $Q\mathbf{w} = 2\mathbf{w} - B\mathbf{w} = 0$. □
⟨3⟩4. Q.E.D.
PROOF: Combine ⟨3⟩2, ⟨3⟩3. □
⟨2⟩5. There is a spanning realization of Q in dimension $m - 1$, and each realizing vector has squared norm 2.
⟨3⟩1. A realization exists with dimension $m - \dim\ker Q$.
PROOF: Apply Lemma (a Gram realization) to ⟨2⟩3. □
⟨3⟩2. $\dim\ker Q = 1$.
PROOF: Use ⟨2⟩4, $\mathbf{w} \neq 0$, and `finrank_span_singleton`. □
⟨3⟩3. Every realizing vector has squared norm 2.
PROOF: Its squared norm is the diagonal entry $Q(i,i) = 2$. □
⟨3⟩4. Q.E.D.
PROOF: Combine ⟨3⟩1, ⟨3⟩2, ⟨3⟩3. □
⟨2⟩6. For any realization and any index i_0, the remaining vectors form a basis of its span.
⟨3⟩1. A relation omitting i_0 has coefficient vector $t\mathbf{w}$ and zero i_0-coordinate.
PROOF: Use Lemma (a Gram realization) and ⟨2⟩4. □
⟨3⟩2. All coefficients of this relation vanish.
PROOF: In ⟨3⟩1, $tw(i_0) = 0$ and $w(i_0) > 0$ force $t = 0$. □
⟨3⟩3. The omitted vector $\mathbf{v}_{i_0}$ is in the span of the remaining vectors.
PROOF: By Lemma (a Gram realization) and $Q\mathbf{w} = 0$, $\sum_i w(i)\mathbf{v}_i = 0$; solving with $w(i_0) > 0$ gives $\mathbf{v}_{i_0} = -\sum_{i\neq i_0}(w(i)/w(i_0))\mathbf{v}_i$. □
⟨3⟩4. Q.E.D.
PROOF: The independent remaining family of ⟨3⟩2 spans the full realizing span by ⟨3⟩3, so it is a basis. □
⟨2⟩7. Q.E.D.
PROOF: Combine ⟨2⟩5, ⟨2⟩6. □
⟨1⟩3. Q.E.D.
PROOF: Combine ⟨1⟩1, ⟨1⟩2. □

3.1.2. *Coordinate systems and their bases.*

Definition (classical coordinate models). ASSUME: $\ell \geq 1$ for A, $\ell \geq 2$ for B, C, and $\ell \geq 4$ for D.
In standard coordinates put

$$\Phi(\mathrm{A}_\ell) = \{\mathbf{u}_i - \mathbf{u}_j : 1 \leq i \neq j \leq \ell+1\}, \quad V(\mathrm{A}_\ell) = \{\mathbf{x} : \sum_i x(i) = 0\},$$
$$\Phi(\mathrm{B}_\ell) = \{\pm\mathbf{u}_i\} \cup \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\}, \quad V(\mathrm{B}_\ell) = \mathbb{R}^\ell,$$
$$\Phi(\mathrm{C}_\ell) = \{\pm 2\mathbf{u}_i\} \cup \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\}, \quad V(\mathrm{C}_\ell) = \mathbb{R}^\ell,$$
$$\Phi(\mathrm{D}_\ell) = \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\}, \quad V(\mathrm{D}_\ell) = \mathbb{R}^\ell.$$

The indices in the last three rows range from 1 to ℓ. The proposed bases are

$$\begin{aligned} \mathrm{A}: &\quad \alpha_i = \mathbf{u}_{i+1} - \mathbf{u}_{i+2} \quad (0 \leq i < \ell); \\ \mathrm{B},\mathrm{C},\mathrm{D}: &\quad \alpha_i = \mathbf{u}_{i+1} - \mathbf{u}_{i+2} \quad (0 \leq i < \ell - 1), \end{aligned} \tag{9}$$

with last root $\mathbf{u}_\ell$, $2\mathbf{u}_\ell$, or $\mathbf{u}_{\ell-1}+\mathbf{u}_\ell$, respectively. We retain the C_2 model for comparison with B_2.

Lemma (signed coordinate reflections). ASSUME: $\mathbf{u}_1,\dots,\mathbf{u}_m$ is the standard orthonormal basis of $\mathbb{R}^m$, $i\neq j$, and $c\neq 0$.
PROVE: Reflection in $\mathbf{u}_i-\mathbf{u}_j$ interchanges coordinates i,j; reflection in $\mathbf{u}_i+\mathbf{u}_j$ interchanges and negates both; reflection in $c\mathbf{u}_i$ negates coordinate i and fixes the others. Signed coordinate permutations preserve each set consisting of all axis roots of a fixed length, all two-coordinate roots with coefficients ±1, or all half-coordinate sign vectors. Coordinate permutations preserve all differences and the zero-sum subspace.

PROOF SKETCH: Evaluate the reflection formula in each coordinate.
⟨1⟩1. The difference reflection has entries $x(j),x(i)$ at i,j and $x(k)$ elsewhere.
PROOF: Substitute $(\mathbf{x},\mathbf{u}_i-\mathbf{u}_j)=x(i)-x(j)$ and $\|\mathbf{u}_i-\mathbf{u}_j\|^2=2$ in Definition 3.9. □
⟨1⟩2. The sum reflection has entries $-x(j),-x(i)$ at i,j and $x(k)$ elsewhere.
PROOF: Substitute $(\mathbf{x},\mathbf{u}_i+\mathbf{u}_j)=x(i)+x(j)$ and squared norm 2 in the reflection formula. □
⟨1⟩3. The axis reflection has entry $-x(i)$ at i and $x(k)$ elsewhere.
PROOF: Substitute $(\mathbf{x},c\mathbf{u}_i)=cx(i)$ and squared norm c^2 in the reflection formula. □
⟨1⟩4. Signed permutations preserve the three asserted coordinate patterns.
PROOF: They preserve support size and the absolute value of every nonzero coordinate. □
⟨1⟩5. Coordinate permutations preserve differences and the zero-sum subspace.
PROOF: Reindex their two nonzero coordinates and the coordinate sum. □
⟨1⟩6. Q.E.D.
PROOF: Combine ⟨1⟩1–⟨1⟩5. □

Lemma (classical Root Systems). ASSUME: The parameters, spaces, root sets and candidate bases are those of Definition (classical coordinate models).
PROVE: Each displayed set is a Root System in its specified space. The displayed family is a root base and has exactly Mathlib's Cartan matrix of that type. The dimension of each space is ℓ.

PROOF SKETCH: Check reflection closure by coordinate operations and prove the base property by nonnegative telescoping expansions.
⟨1⟩1. Each root set is finite, contains no zero, and is reduced.
⟨2⟩1. Each set is finite and excludes zero.
PROOF: Its finite index and sign choices in Definition (classical coordinate models) have a specified nonzero coordinate. □
⟨2⟩2. On an axis the only listed vectors are one sign pair.
PROOF: Read the axis clause in Definition (classical coordinate models). □
⟨2⟩3. On a two-coordinate line the absolute values of both nonzero coordinates are 1.
PROOF: Read the two-coordinate clause in Definition (classical coordinate models). □
⟨2⟩4. Q.E.D.
PROOF: Proportional vectors have identical supports, and ⟨2⟩2, ⟨2⟩3 force the scalar to be 1 or -1; combine with ⟨2⟩1. □
⟨1⟩2. Each set is closed under all its root reflections.
⟨2⟩1. In A each reflection permutes coordinates.
PROOF: Apply the difference case of Lemma (signed coordinate reflections). □
⟨2⟩2. In B every reflection is a signed permutation preserving both its patterns.
PROOF: Apply the axis, sum and difference formulas of Lemma (signed coordinate reflections) to its listed roots. □
⟨2⟩3. In C every reflection is a signed permutation preserving both its patterns.

PROOF: Apply that lemma with axis scalar $c = 2$, and its sum and difference formulas, to the C list. □

⟨2⟩4. In D every reflection is a signed permutation preserving its two-coordinate pattern.
PROOF: Apply the sum and difference formulas of that lemma to the D list. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1–⟨2⟩4 and $s_{-\alpha} = s_\alpha$ from Lemma (reflection identities). □

⟨1⟩3. Every root pairing is integral.

⟨2⟩1. Pairing against a two-coordinate root is an integer inner product.
PROOF: All coordinates are integers and its squared norm is 2 in (2). □

⟨2⟩2. Pairing against $\pm\mathbf{u}_i$ is $\pm 2x(i) \in \mathbb{Z}$.
PROOF: Substitute squared norm 1 in (2). □

⟨2⟩3. Pairing against $\pm 2\mathbf{u}_i$ is $\pm x(i) \in \mathbb{Z}$.
PROOF: Substitute squared norm 4 in (2). □

⟨2⟩4. Q.E.D.
PROOF: The denominator-root cases ⟨2⟩1, ⟨2⟩2, ⟨2⟩3 exhaust all four root lists. □

⟨1⟩4. Every listed root has coefficients of one sign in the proposed family.

⟨2⟩1. For $i < j$ in all four types, $\mathbf{u}_i - \mathbf{u}_j = \sum_{k=i-1}^{j-2} \alpha_k$.
PROOF: Telescope the consecutive differences in (9). □

⟨2⟩2. In B, $\mathbf{u}_i = \sum_{k=i-1}^{\ell-1} \alpha_k$.
PROOF: Telescope to the endpoint $\alpha_{\ell-1} = \mathbf{u}_\ell$. □

⟨2⟩3. In B, for $i < j$, $\mathbf{u}_i + \mathbf{u}_j = \sum_{k=i-1}^{j-2} \alpha_k + 2\sum_{k=j-1}^{\ell-1} \alpha_k$.
PROOF: Add ⟨2⟩1 to twice ⟨2⟩2 at j. □

⟨2⟩4. In C, $2\mathbf{u}_i = 2\sum_{k=i-1}^{\ell-2} \alpha_k + \alpha_{\ell-1}$.
PROOF: Telescope to $\alpha_{\ell-1} = 2\mathbf{u}_\ell$. □

⟨2⟩5. In C, for $i < j$, $\mathbf{u}_i + \mathbf{u}_j = \sum_{k=i-1}^{j-2} \alpha_k + 2\sum_{k=j-1}^{\ell-2} \alpha_k + \alpha_{\ell-1}$.
PROOF: Add ⟨2⟩1 to ⟨2⟩4 at j. □

⟨2⟩6. In D, for $i < j \le \ell - 1$,

$$\mathbf{u}_i + \mathbf{u}_j = \sum_{k=i-1}^{j-2} \alpha_k + 2\sum_{k=j-1}^{\ell-3} \alpha_k + \alpha_{\ell-2} + \alpha_{\ell-1}.$$

PROOF: Telescope using $\alpha_{\ell-2} + \alpha_{\ell-1} = 2\mathbf{u}_{\ell-1}$ from (9). □

⟨2⟩7. In D, for $i < \ell$, $\mathbf{u}_i + \mathbf{u}_\ell = \sum_{k=i-1}^{\ell-3} \alpha_k + \alpha_{\ell-1}$.
PROOF: Telescope using only $\alpha_{\ell-1} = \mathbf{u}_{\ell-1} + \mathbf{u}_\ell$. □

⟨2⟩8. The negative of each displayed root has the negative of its displayed coefficient vector.
PROOF: Negate each linear combination. □

⟨2⟩9. Q.E.D.
PROOF: The root patterns in Definition (classical coordinate models) are exactly the vectors in ⟨2⟩1–⟨2⟩7 and their negatives; all displayed coefficients are nonnegative integers. □

⟨1⟩5. The roots span the specified ℓ-dimensional spaces.

⟨2⟩1. The differences $\mathbf{u}_i - \mathbf{u}_{\ell+1}$, $1 \le i \le \ell$, form a real basis of the zero-sum space.
PROOF: The unique coefficients of a zero-sum vector are its first ℓ coordinates, by coordinate substitution. □

⟨2⟩2. The B axis roots span $\mathbb{R}^\ell$.
PROOF: They include its standard basis. □

⟨2⟩3. The C axis roots span $\mathbb{R}^\ell$.
PROOF: Each standard vector is half a listed axis root. □

⟨2⟩4. The D roots span $\mathbb{R}^\ell$.

PROOF: For each i, choose $j \neq i$ and use $2\mathbf{u}_i = (\mathbf{u}_i + \mathbf{u}_j) + (\mathbf{u}_i - \mathbf{u}_j)$. □

⟨2⟩5. Q.E.D.

PROOF: Combine ⟨2⟩1–⟨2⟩4. □

⟨1⟩6. The proposed family is a root base.

⟨2⟩1. Every proposed simple root belongs to its root list.

PROOF: Substitute the simple-root formulas in Definition (classical coordinate models). □

⟨2⟩2. The ℓ simple roots span their ambient space.

PROOF: Combine ⟨1⟩4 with ⟨1⟩5. □

⟨2⟩3. They are linearly independent.

PROOF: Apply the spanning-cardinality lemma of 3.1.1 to ⟨2⟩2 and dimension ℓ in ⟨1⟩5. □

⟨2⟩4. Q.E.D.

PROOF: Use the base definition with ⟨2⟩1, ⟨2⟩3, ⟨1⟩4. □

⟨1⟩7. The base matrices have the asserted entries.

⟨2⟩1. Consecutive differences have squared norm 2, inner product -1 at consecutive indices, and 0 at nonconsecutive indices.

PROOF: Expand $(\mathbf{u}_{i+1} - \mathbf{u}_{i+2}, \mathbf{u}_{j+1} - \mathbf{u}_{j+2})$ using orthonormality. □

⟨2⟩2. The B last root has squared norm 1, inner product -1 with its predecessor, and zero with the other differences.

PROOF: Substitute $\alpha_{\ell-1} = \mathbf{u}_\ell$. □

⟨2⟩3. The C last root has squared norm 4, inner product -2 with its predecessor, and zero with the other differences.

PROOF: Substitute $\alpha_{\ell-1} = 2\mathbf{u}_\ell$. □

⟨2⟩4. The D last root has squared norm 2, product -1 with $\alpha_{\ell-3}$, and zero with all other simple roots.

PROOF: Expand the product with $\mathbf{u}_{\ell-1} + \mathbf{u}_\ell$ using orthonormality, including $1 - 1 = 0$ at $\alpha_{\ell-2}$. □

⟨2⟩5. Q.E.D.

PROOF: Divide the columns in ⟨2⟩1–⟨2⟩4 by half their squared lengths, as in (2), and compare Definition (the indexed matrices). □

⟨1⟩8. Q.E.D.

PROOF: Assemble Definition 3.9 from ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩5, and add ⟨1⟩6, ⟨1⟩7. □

Definition (exceptional coordinate data). ASSUME: Coordinates have the standard real inner product.

In $\mathbb{R}^8$ define

$$L = \{\mathbf{x} \in \mathbb{Z}^8 : \sum_i x(i) \in 2\mathbb{Z}\} \ \cup \ \{\mathbf{x} \in (\mathbb{Z} + \tfrac{1}{2})^8 : \sum_i x(i) \in 2\mathbb{Z}\}. \tag{10}$$

Put

$$\begin{aligned}
\Phi(\mathrm{E}_8) &= \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\} \\
&\qquad \cup \{\tfrac{1}{2}\sum_{i=1}^{8} \epsilon(i)\mathbf{u}_i : \epsilon(i) \in \{1, -1\}, \ \prod_i \epsilon(i) = 1\}, \\
V(\mathrm{E}_7) &= \{\mathbf{x} : x(7) + x(8) = 0\}, \\
V(\mathrm{E}_6) &= \{\mathbf{x} : x(6) = x(7) = -x(8)\}, \\
\Phi(\mathrm{E}_j) &= \Phi(\mathrm{E}_8) \cap V(\mathrm{E}_j) \quad (j = 6, 7), \qquad V(\mathrm{E}_8) = \mathbb{R}^8.
\end{aligned}$$

The ordered E_8 family is

$$\begin{aligned} \alpha_0 &= \tfrac{1}{2}(\mathbf{u}_1 + \mathbf{u}_8 - \mathbf{u}_2 - \mathbf{u}_3 - \mathbf{u}_4 - \mathbf{u}_5 - \mathbf{u}_6 - \mathbf{u}_7), \\ \alpha_1 &= \mathbf{u}_1 + \mathbf{u}_2, \qquad \alpha_i = \mathbf{u}_i - \mathbf{u}_{i-1} \quad (2 \le i \le 7). \end{aligned} \tag{11}$$

Use its first j vectors for E_j. In $\mathbb{R}^4$ put

$$\Phi(\mathrm{F}_4) = \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\} \cup \{\pm\mathbf{u}_i\} \cup \{\tfrac{1}{2}\sum_{i=1}^{4} \epsilon(i)\mathbf{u}_i : \epsilon(i) \in \{1, -1\}\} \tag{12}$$

with family

$$(\alpha_0, \alpha_1, \alpha_2, \alpha_3) = (\mathbf{u}_2 - \mathbf{u}_3,\ \mathbf{u}_3 - \mathbf{u}_4,\ \mathbf{u}_4,\ \tfrac{1}{2}(\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3 - \mathbf{u}_4)). \tag{13}$$

In the zero-sum plane $V \subset \mathbb{R}^3$ put

$$\Phi(\mathrm{G}_2) = \{\mathbf{u}_i - \mathbf{u}_j : i \ne j\} \cup \{\pm(2\mathbf{u}_i - \mathbf{u}_j - \mathbf{u}_k) : \{i, j, k\} = \{1, 2, 3\}\}, \tag{14}$$

with family

$$\alpha_0 = \mathbf{u}_1 - 2\mathbf{u}_2 + \mathbf{u}_3, \qquad \alpha_1 = \mathbf{u}_2 - \mathbf{u}_3. \tag{15}$$

All sign choices are independent except for the displayed E_8 product condition.

Lemma (the even lattice). ASSUME: $L \subset \mathbb{R}^8$ is the set in (10).
PROVE: L is an additive subgroup, all its inner products are integers, all its squared norms are even integers, and its squared-norm-2 vectors are exactly $\Phi(\mathrm{E}_8)$. Every nonzero element has squared norm at least 2.

PROOF SKETCH: Describe two cosets of the integer even-sum subgroup and minimize the sum of the eight coordinate squares.
LET: $L_0 = \{\mathbf{x} \in \mathbb{Z}^8 : \sum_i x(i) \in 2\mathbb{Z}\}$ and $\mathbf{v} = \frac{1}{2}\sum_i \mathbf{u}_i$.
⟨1⟩1. $L = L_0 \cup (L_0 + \mathbf{v})$ and $2\mathbf{v} \in L_0$.
  PROOF: Subtracting $\mathbf{v}$ changes the coordinate sum by 4; $2\mathbf{v}$ has integer coordinates and sum 8. □
⟨1⟩2. L is an additive subgroup.
  ⟨2⟩1. $L_0 + L_0 \subseteq L_0$.
    PROOF: Addition preserves integer coordinates and even sums. □
  ⟨2⟩2. $L_0 + (L_0 + \mathbf{v}) \subseteq L_0 + \mathbf{v}$.
    PROOF: Translate the inclusion ⟨2⟩1 by $\mathbf{v}$. □
  ⟨2⟩3. $(L_0 + \mathbf{v}) + (L_0 + \mathbf{v}) \subseteq L_0$.
    PROOF: Use ⟨2⟩1 and $2\mathbf{v} \in L_0$ from ⟨1⟩1. □
  ⟨2⟩4. $-L_0 = L_0$.
    PROOF: Negation preserves integer coordinates and even sums. □
  ⟨2⟩5. $-(L_0 + \mathbf{v}) = L_0 + \mathbf{v}$.
    PROOF: Write $-\mathbf{x} - \mathbf{v} = (-\mathbf{x} - 2\mathbf{v}) + \mathbf{v}$ using ⟨1⟩1, ⟨2⟩4. □
  ⟨2⟩6. Q.E.D.
    PROOF: The two-coset cases ⟨2⟩1–⟨2⟩5, commutativity of addition, and $0 \in L_0$ give the subgroup laws. □
⟨1⟩3. All inner products in L are integers.
  ⟨2⟩1. For $\mathbf{x}, \mathbf{y} \in L_0$, $(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}$.
    PROOF: Sum products of their integer coordinates. □
  ⟨2⟩2. For $\mathbf{x} \in L_0$, $(\mathbf{x}, \mathbf{v}) = \sum_i x(i)/2 \in \mathbb{Z}$.
    PROOF: Use the even-sum definition of L_0. □
  ⟨2⟩3. $(\mathbf{v}, \mathbf{v}) = 2 \in \mathbb{Z}$.

PROOF: Sum eight copies of $1/4$. □

⟨2⟩4. Q.E.D.

PROOF: Expand $(\mathbf{x} + a\mathbf{v}, \mathbf{y} + b\mathbf{v})$ for $a, b \in \{0, 1\}$ using ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □

⟨1⟩4. Every squared norm in L is an even integer.

⟨2⟩1. For $\mathbf{x} \in L_0$, $\sum_i x(i)^2$ is even.

PROOF: Each integer satisfies $x(i)^2 \equiv x(i) \pmod 2$, and $\sum_i x(i)$ is even. □

⟨2⟩2. For $\mathbf{x} \in L_0$, $\|\mathbf{x} + \mathbf{v}\|^2 = \sum_i x(i)(x(i) + 1) + 2$ is even.

PROOF: Expand the eight coordinate squares; a product of consecutive integers is even. □

⟨2⟩3. Q.E.D.

PROOF: Use the coset partition ⟨1⟩1 with ⟨2⟩1, ⟨2⟩2. □

⟨1⟩5. A nonzero element of L has squared norm at least 2.

PROOF: A positive even integer from ⟨1⟩4 is at least 2, using definiteness in 2. □

⟨1⟩6. The squared-norm-2 elements of L_0 are precisely the two-coordinate roots $\pm\mathbf{u}_i \pm \mathbf{u}_j$, $i < j$.

⟨2⟩1. Every nonzero integer coordinate contributes at least 1, and any coordinate of absolute value at least 2 contributes at least 4.

PROOF: Square the corresponding integer inequalities. □

⟨2⟩2. A sum of integer squares equal to 2 has exactly two nonzero coordinates, both of absolute value 1.

PROOF: Apply ⟨2⟩1 to the finite sum of squares. □

⟨2⟩3. Q.E.D.

PROOF: The patterns in ⟨2⟩2 have even coordinate sum and squared norm 2 by substitution. □

⟨1⟩7. The squared-norm-2 elements in $L_0 + \mathbf{v}$ are precisely the half-coordinate roots with an even number of minus signs.

⟨2⟩1. Each of their eight absolute coordinates is at least $1/2$.

PROOF: Each coordinate is an integer plus $1/2$. □

⟨2⟩2. Squared norm 2 forces all absolute coordinates to be $1/2$.

PROOF: The eight lower bounds $1/4$ from ⟨2⟩1 already sum to 2. □

⟨2⟩3. For t minus signs, the coordinate sum is $4 - t$.

PROOF: Sum $8 - t$ copies of $1/2$ and t copies of $-1/2$. □

⟨2⟩4. Q.E.D.

PROOF: The sum in ⟨2⟩3 is even exactly for even t; all these patterns have squared norm 2 by ⟨2⟩2. □

⟨1⟩8. Q.E.D.

PROOF: Combine ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, ⟨1⟩7 with the root list in Definition (exceptional coordinate data). □

Lemma (minimal vectors and an obtuse basis). ASSUME: 1. L is an additive subgroup of a finite-dimensional real inner product space, $b > 0$, and every nonzero $\mathbf{x} \in L$ has $(\mathbf{x}, \mathbf{x}) \geq b$.

2. $(\alpha_i)_{i<m}$ is a real basis generating L over $\mathbb{Z}$, and $(\alpha_i, \alpha_j) \leq 0$ for $i \neq j$.

PROVE: Every element of L of squared norm b has integer basis coefficients all nonnegative or all nonpositive.

PROOF SKETCH: Split a coefficient vector into disjoint positive and negative parts.

LET: For $\mathbf{x} = \sum_i c(i)\alpha_i \in L$ with $(\mathbf{x}, \mathbf{x}) = b$, put $\mathbf{v} = \sum_i \max(c(i), 0)\alpha_i$ and $\mathbf{w} = \sum_i \max(-c(i), 0)\alpha_i$.

⟨1⟩1. $\mathbf{x} = \mathbf{v} - \mathbf{w}$, with $\mathbf{v}, \mathbf{w} \in L$.

PROOF: Use the integer generation in ⟨0⟩:2 and $c = \max(c, 0) - \max(-c, 0)$. □

⟨1⟩2. $(\mathbf{v}, \mathbf{w}) \leq 0$.

PROOF: Expand the inner product; same-index coefficients have product zero, and all other summands are nonpositive by ⟨0⟩:2. □

⟨1⟩3. At least one of $\mathbf{v}, \mathbf{w}$ is zero.

PROOF: If both were nonzero, ⟨0⟩:1, ⟨1⟩1, ⟨1⟩2 would give $b = (\mathbf{v},\mathbf{v}) + (\mathbf{w},\mathbf{w}) - 2(\mathbf{v},\mathbf{w}) \geq 2b > b$. □

⟨1⟩4. Q.E.D.

PROOF: Independence in ⟨0⟩:2 makes every coefficient of the zero part in ⟨1⟩3 vanish. □

Lemma (the three E systems). ASSUME: $j \in \{6,7,8\}$, with L, $V(\mathrm{E}_j)$, $\Phi(\mathrm{E}_j)$, and the first j vectors of (11) as in Definition (exceptional coordinate data).

PROVE: $\Phi(\mathrm{E}_j)$ is a Root System in $V(\mathrm{E}_j)$; those j vectors are a root base and their Cartan matrix is `CartanMatrix.E` at rank j. They are also an integer basis for $L \cap V(\mathrm{E}_j)$, and $\dim V(\mathrm{E}_j) = j$.

PROOF SKETCH: Invert the coordinate matrix, then apply the minimum-norm lemma in each subspace.

⟨1⟩1. The coefficients of every $\mathbf{x} \in \mathbb{R}^8$ are

$$\begin{aligned} c(0) &= 2x(8), \qquad c(1) = \tfrac{1}{2}\sum_{i=1}^{8} x(i) + 2x(8), \\ c(2) &= c(1) + x(8) - x(1), \\ c(k) &= \sum_{i=k}^{7} x(i) + (8-k)x(8) \quad (3 \leq k \leq 7). \end{aligned} \tag{16}$$

In particular these formulas are the inverse of the coordinate map.

⟨2⟩1. The eighth coordinate of $\sum_i c(i)\alpha_i$ is $x(8)$.

PROOF: It is $c(0)/2$ by (11); use (16). □

⟨2⟩2. Its first coordinate is $x(1)$.

PROOF: Substitute in $c(0)/2 + c(1) - c(2)$. □

⟨2⟩3. Its second coordinate is $x(2)$.

PROOF: Substitute in $-c(0)/2 + c(1) + c(2) - c(3)$. □

⟨2⟩4. For $3 \leq i \leq 6$, its ith coordinate is $x(i)$.

PROOF: Telescope $-c(0)/2 + c(i) - c(i+1)$ using (16). □

⟨2⟩5. Its seventh coordinate is $x(7)$.

PROOF: Substitute in $-c(0)/2 + c(7)$. □

⟨2⟩6. These coefficients are unique.

PROOF: The same coordinate equations solve successively for $c(0), c(7), c(6), c(5), c(4), c(3)$ and then $c(1), c(2)$. □

⟨2⟩7. Q.E.D.

PROOF: Combine the exhaustive coordinate checks ⟨2⟩1–⟨2⟩5 and ⟨2⟩6. □

⟨1⟩2. For $\mathbf{x} \in L$, all coefficients in (16) are integers.

⟨2⟩1. $c(0)$ is an integer.

PROOF: Twice an integer or half-integer is integral. □

⟨2⟩2. $c(1)$ is an integer.

PROOF: Half the even coordinate sum is integral, and so is $2x(8)$ by ⟨2⟩1. □

⟨2⟩3. $c(2)$ is an integer.

PROOF: Use ⟨2⟩2 and the integer difference $x(8) - x(1)$ of two coordinates in the same coset. □

⟨2⟩4. For $3 \leq k \leq 7$, $c(k)$ is an integer.

PROOF: Write it as $\sum_{i=k}^{7}(x(i) + x(8))$; each summand is an integer in either coset of (10). □

⟨2⟩5. Q.E.D.

PROOF: The index cases ⟨2⟩1–⟨2⟩4 are exhaustive. □

⟨1⟩3. All displayed simple roots lie in L and have squared norm 2. The first j form a real basis of $V(\mathrm{E}_j)$ and generate $L \cap V(\mathrm{E}_j)$ over $\mathbb{Z}$.

⟨2⟩1. $x(7) + x(8) = 0$ if and only if $c(7) = 0$.

PROOF: Substitute $k = 7$ in (16). □

⟨2⟩2. The two E_6 subspace equations hold if and only if $c(6) = c(7) = 0$.

PROOF: Use $c(6) = x(6) + x(7) + 2x(8)$ and ⟨2⟩1. □

⟨2⟩3. Every α_i lies in L and has squared norm 2.

PROOF: For α_0, its six minus signs and eight half-coordinates satisfy (10); each other vector has two unit coordinates and even sum. □

⟨2⟩4. Q.E.D.

PROOF: Restrict the inverse bijection ⟨1⟩1 by ⟨2⟩1, ⟨2⟩2; use ⟨1⟩2, ⟨2⟩3, and subgroup closure for integer generation. □

⟨1⟩4. The simple Gram matrix is E_j.

⟨2⟩1. $(\alpha_0, \alpha_1) = 0$.

PROOF: The contributions in coordinates 1 and 2 are $1/2 - 1/2 = 0$. □

⟨2⟩2. $(\alpha_0, \alpha_2) = -1$.

PROOF: The contributions are $-1/2 - 1/2 = -1$. □

⟨2⟩3. For $3 \le k \le 7$, $(\alpha_0, \alpha_k) = 0$.

PROOF: The two coefficients of α_0 are both $-1/2$ and cancel in the difference α_k. □

⟨2⟩4. $(\alpha_1, \alpha_2) = 0$.

PROOF: Expand $(\mathbf{u}_1 + \mathbf{u}_2, \mathbf{u}_2 - \mathbf{u}_1) = 1 - 1$. □

⟨2⟩5. $(\alpha_1, \alpha_3) = -1$.

PROOF: Expand $(\mathbf{u}_1 + \mathbf{u}_2, \mathbf{u}_3 - \mathbf{u}_2) = -1$. □

⟨2⟩6. For $k \ge 4$, $(\alpha_1, \alpha_k) = 0$.

PROOF: The supports are disjoint. □

⟨2⟩7. For distinct $i, k \ge 2$, the product is -1 when $|i - k| = 1$ and zero otherwise.

PROOF: Expand $(\mathbf{u}_i - \mathbf{u}_{i-1}, \mathbf{u}_k - \mathbf{u}_{k-1})$. □

⟨2⟩8. Q.E.D.

PROOF: Combine ⟨2⟩1–⟨2⟩7, symmetry, and squared norms 2 from ⟨1⟩3 with Definition (the indexed matrices). □

⟨1⟩5. Every norm-2 vector in $L \cap V(\mathrm{E}_j)$ has integer coefficients of one sign.

PROOF: Apply Lemma (minimal vectors and an obtuse basis) with $b = 2$, using Lemma (the even lattice), ⟨1⟩3, ⟨1⟩4. □

⟨1⟩6. For $\alpha, \beta \in \Phi(\mathrm{E}_j)$, $s_\alpha\beta \in \Phi(\mathrm{E}_j)$.

⟨2⟩1. $s_\alpha\beta = \beta - (\alpha, \beta)\alpha \in L$.

PROOF: Use integer inner products and subgroup closure in Lemma (the even lattice), since $(\alpha, \alpha) = 2$. □

⟨2⟩2. $s_\alpha\beta \in V(\mathrm{E}_j)$.

PROOF: The reflection formula is a linear combination of α, β in that subspace. □

⟨2⟩3. $\|s_\alpha\beta\|^2 = 2$.

PROOF: Apply Lemma (reflection identities). □

⟨2⟩4. Q.E.D.

PROOF: Lemma (the even lattice) identifies the vectors in ⟨2⟩1–⟨2⟩3 with the stated root list. □

⟨1⟩7. The displayed set is a Root System.

⟨2⟩1. It is finite and contains no zero.

PROOF: Its finite coordinate patterns all have squared norm 2 by Lemma (the even lattice). □

⟨2⟩2. It is reduced.
PROOF: Equal positive squared norms force a proportionality scalar to satisfy $t^2 = 1$. □
⟨2⟩3. Its pairings are integers.
PROOF: Equation (2) equals the integer inner product of Lemma (the even lattice). □
⟨2⟩4. Q.E.D.
PROOF: Assemble Definition 3.9 using ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨1⟩6, and the spanning roots of ⟨1⟩3. □
⟨1⟩8. Q.E.D.
PROOF: Combine ⟨1⟩7, ⟨1⟩3, ⟨1⟩5, ⟨1⟩4, and denominator 2 in (2). □

Lemma (the F system). ASSUME: $\Phi(\mathrm{F}_4) \subset \mathbb{R}^4$ and $(\alpha_i)_{i<4}$ are the set and family in (12), (13).
PROVE: This is a Root System with the displayed root base and Cartan matrix F_4. Its first two simple roots have squared length 2 and its last two squared length 1.

PROOF SKETCH: Only half-coordinate reflections need a new closure check; an inverse coordinate formula proves the base property.
⟨1⟩1. The set is finite, nonzero, reduced, spans $\mathbb{R}^4$, and its two-coordinate roots have squared length 2, all others length squared 1.
⟨2⟩1. The set is finite and has no zero.
PROOF: Its three finite coordinate patterns have a nonzero coordinate in (12). □
⟨2⟩2. The squared lengths are $1 + 1 = 2$, 1, and $4/4 = 1$.
PROOF: Sum coordinate squares in the three patterns. □
⟨2⟩3. Proportional roots differ only by sign.
PROOF: Support sizes 2, 1, 4 separate the patterns; within each pattern the fixed absolute nonzero coordinates force $|t| = 1$. □
⟨2⟩4. The roots span $\mathbb{R}^4$.
PROOF: The axis pattern contains the standard basis. □
⟨2⟩5. Q.E.D.
PROOF: Combine ⟨2⟩1–⟨2⟩4. □
⟨1⟩2. All root reflections preserve the set.
⟨2⟩1. Reflections in an axis or two-coordinate root preserve it.
PROOF: Apply Lemma (signed coordinate reflections) to the three complete signed patterns in (12). □
LET: $\mathbf{v} = (\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3 + \mathbf{u}_4)/2$.
⟨2⟩2. $s_{\mathbf{v}}\mathbf{x} = \mathbf{x} - (\sum_i x(i))\mathbf{v}$.
PROOF: Substitute $(\mathbf{v}, \mathbf{v}) = 1$ in the reflection formula. □
⟨2⟩3. For an axis root $\epsilon\mathbf{u}_i$, its image is $\epsilon(\mathbf{u}_i - \mathbf{v})$, a half-coordinate root.
PROOF: Substitute its coordinate sum ϵ in ⟨2⟩2. □
⟨2⟩4. For $\epsilon(\mathbf{u}_i - \mathbf{u}_j)$ its image is itself.
PROOF: Its coordinate sum is zero in ⟨2⟩2. □
⟨2⟩5. For $\epsilon(\mathbf{u}_i + \mathbf{u}_j)$ its image is $-\epsilon(\mathbf{u}_k + \mathbf{u}_l)$ on the complementary pair.
PROOF: Substitute its coordinate sum 2ϵ in ⟨2⟩2. □
⟨2⟩6. A half-coordinate root with zero plus signs maps to its negative.
PROOF: It is $-\mathbf{v}$, with sum -2, in ⟨2⟩2. □
⟨2⟩7. A half-coordinate root with one plus sign maps to the positive axis at that sign.
PROOF: Its sum is -1, so ⟨2⟩2 adds $\mathbf{v}$. □
⟨2⟩8. A half-coordinate root with two plus signs is fixed.
PROOF: Its sum is zero in ⟨2⟩2. □
⟨2⟩9. A half-coordinate root with three plus signs maps to the negative axis at its minus sign.
PROOF: Its sum is 1, so ⟨2⟩2 subtracts $\mathbf{v}$. □

⟨2⟩10. A half-coordinate root with four plus signs maps to its negative.
PROOF: It is $\mathbf{v}$, with sum 2, in ⟨2⟩2. □
⟨2⟩11. $s_{\mathbf{v}}$ preserves the whole set.
PROOF: The cases ⟨2⟩3–⟨2⟩10 exhaust (12). □
⟨2⟩12. Reflection in any half-coordinate root preserves the set.
⟨3⟩1. A diagonal sign isometry taking $\mathbf{v}$ to that root permutes $\Phi(\mathrm{F}_4)$.
PROOF: Apply the signed-pattern assertion of Lemma (signed coordinate reflections). □
⟨3⟩2. Q.E.D.
PROOF: Conjugate ⟨2⟩11 by ⟨3⟩1 using Lemma (reflection identities). □
⟨2⟩13. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩12 over the three root patterns. □
⟨1⟩3. Every root pairing is an integer.
⟨2⟩1. Pairing an integer root against a long root is integral.
PROOF: Its value is an inner product of integer coordinates by (2) and ⟨1⟩1. □
⟨2⟩2. Pairing a half-coordinate root against a long root is integral.
PROOF: The value is a sum of two signed halves by (2). □
⟨2⟩3. Pairing any root against an axis root is integral.
PROOF: Its value is twice a signed coordinate by (2); all coordinates are half-integers. □
⟨2⟩4. Pairing an integer root against a half-coordinate root is integral.
PROOF: Twice the inner product is a signed sum of the integer root's coordinates. □
⟨2⟩5. Pairing two half-coordinate roots is integral.
PROOF: If their signs differ in t places, the value is $2(4-2t)/4 = 2-t$. □
⟨2⟩6. Q.E.D.
PROOF: The denominator and numerator patterns in ⟨2⟩1–⟨2⟩5 are exhaustive. □
⟨1⟩4. The inverse coefficients in the displayed family are

$$\begin{aligned} c(0) &= x(1)+x(2), & c(1) &= 2x(1)+x(2)+x(3),\\ c(2) &= 3x(1)+x(2)+x(3)+x(4), & c(3) &= 2x(1). \end{aligned} \tag{17}$$

⟨2⟩1. The first coordinate of the combination is $c(3)/2 = x(1)$.
PROOF: Substitute (13), (17). □
⟨2⟩2. The second is $c(0) - c(3)/2 = x(2)$.
PROOF: Substitute (17). □
⟨2⟩3. The third is $-c(0) + c(1) - c(3)/2 = x(3)$.
PROOF: Substitute (17). □
⟨2⟩4. The fourth is $-c(1) + c(2) - c(3)/2 = x(4)$.
PROOF: Substitute (17). □
⟨2⟩5. Q.E.D.
PROOF: These four triangular equations uniquely solve for $c(3), c(0), c(1), c(2)$, giving the asserted inverse. □
⟨1⟩5. The displayed family is a root base.
⟨2⟩1. Orient a half-coordinate root to have $x(1) = 1/2$. Its coefficients are integers in $[0,1], [0,2], [0,3], \{1\}$.
PROOF: Substitute the three independent signed halves in (17). □
⟨2⟩2. For an integer root with $x(1) \neq 0$, orient it with $x(1) = 1$; all four coefficients are nonnegative integers.
PROOF: In (17) at most one of $x(2), x(3), x(4)$ is nonzero, and it is 1 or -1. □
⟨2⟩3. For $x(1) = 0$, orient the first nonzero coordinate among 2,3,4 positively; the four coefficients are nonnegative integers.
⟨3⟩1. Every prefix sum of coordinates 2,3,4 is nonnegative.

PROOF: It begins with zeros, then 1, and has at most one further nonzero term, bounded below by -1. □

⟨3⟩2. Q.E.D.

PROOF: For $x(1) = 0$, (17) consists of those three prefix sums and zero; apply ⟨3⟩1. □

⟨2⟩4. The four simple roots are linearly independent.

PROOF: Their coordinate map has the inverse ⟨1⟩4. □

⟨2⟩5. Q.E.D.

PROOF: The root patterns in ⟨2⟩1, ⟨2⟩2, ⟨2⟩3 exhaust the roots up to sign; use ⟨2⟩4 and their membership in (12). □

⟨1⟩6. The simple squared lengths are $(2, 2, 1, 1)$ and their Cartan matrix is F_4.

⟨2⟩1. $(\alpha_0, \alpha_1) = -1$.

PROOF: Expand the product of $\mathbf{u}_2 - \mathbf{u}_3$ and $\mathbf{u}_3 - \mathbf{u}_4$. □

⟨2⟩2. $(\alpha_1, \alpha_2) = -1$.

PROOF: Expand the product of $\mathbf{u}_3 - \mathbf{u}_4$ and $\mathbf{u}_4$. □

⟨2⟩3. $(\alpha_2, \alpha_3) = -1/2$.

PROOF: Read the fourth coordinate of α_3. □

⟨2⟩4. $(\alpha_0, \alpha_2) = 0$.

PROOF: Their supports are disjoint. □

⟨2⟩5. $(\alpha_0, \alpha_3) = 0$.

PROOF: The equal second and third coordinates of α_3 cancel. □

⟨2⟩6. $(\alpha_1, \alpha_3) = 0$.

PROOF: The equal third and fourth coordinates of α_3 cancel. □

⟨2⟩7. Q.E.D.

PROOF: Use ⟨1⟩1 for diagonal entries and ⟨2⟩1–⟨2⟩6 in (2); compare Definition (the indexed matrices). □

⟨1⟩7. Q.E.D.

PROOF: Assemble the Root System conditions from ⟨1⟩1, ⟨1⟩2, ⟨1⟩3; add ⟨1⟩5, ⟨1⟩6. □

Lemma (the G system). ASSUME: V is the zero-sum plane in $\mathbb{R}^3$, with root set (14) and family (15).

PROVE: This is a Root System with the displayed root base and Cartan matrix G_2. Its simple roots 0 and 1 have squared lengths 6 and 2, respectively.

PROOF SKETCH: A long-root reflection is a permutation followed by global negation. Six coefficient pairs give one root from every sign pair.

⟨1⟩1. There are six short roots of squared length 2 and six long roots of squared length 6; they are finite, nonzero and reduced and span V.

⟨2⟩1. The six distinct-coordinate differences have squared norm 2.

PROOF: There are $3 \cdot 2$ ordered pairs and each squared norm is $1 + 1$. □

⟨2⟩2. The six signed roots $(2, -1, -1)$ under coordinate permutations have squared norm 6.

PROOF: Choose the doubled coordinate and a global sign; their squared norm is $4 + 1 + 1$. □

⟨2⟩3. Proportional listed vectors differ by sign.

PROOF: Short and long supports have different sizes; in either pattern the ratios of nonzero coordinate magnitudes are fixed. □

⟨2⟩4. The differences span the zero-sum plane.

PROOF: A vector in that plane is $x(1)(\mathbf{u}_1 - \mathbf{u}_3) + x(2)(\mathbf{u}_2 - \mathbf{u}_3)$. □

⟨2⟩5. Q.E.D.

PROOF: Combine ⟨2⟩1–⟨2⟩4. □

LET: $\beta = 2\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3$.

⟨1⟩2. For $\mathbf{x} \in V$, $(\mathbf{x}, \beta) = 3x(1)$ and $s_\beta \mathbf{x} = (-x(1), -x(3), -x(2))$.

⟨2⟩1. $2x(1) - x(2) - x(3) = 3x(1)$.
PROOF: Substitute the zero-sum equation of V. □
⟨2⟩2. Q.E.D.
PROOF: Substitute ⟨2⟩1, $(\beta, \beta) = 6$ in the reflection formula, and use $x(1)+x(2)+x(3) = 0$. □
⟨1⟩3. The set is closed under every root reflection.
⟨2⟩1. Each short reflection preserves it.
PROOF: It is a coordinate interchange by Lemma (signed coordinate reflections); both lists are permutation-invariant. □
⟨2⟩2. s_β preserves it.
PROOF: Its formula in ⟨1⟩2 is a coordinate interchange followed by global negation, preserving both lists. □
⟨2⟩3. Each long reflection preserves it.
PROOF: Every long root is a signed coordinate permutation of β using only a global sign; conjugate ⟨2⟩2 by Lemma (reflection identities). □
⟨2⟩4. Q.E.D.
PROOF: The short and long cases ⟨2⟩1, ⟨2⟩3 exhaust (14). □
⟨1⟩4. All pairings are integral.
⟨2⟩1. Against a short root the pairing is an integer inner product.
PROOF: All root coordinates are integers and its norm squared is 2 by ⟨1⟩1. □
⟨2⟩2. Against β the pairing is $x(1) \in \mathbb{Z}$.
PROOF: Divide twice the inner product in ⟨1⟩2 by 6. □
⟨2⟩3. Against every long root it is integral.
PROOF: Transport ⟨2⟩2 by the global sign and coordinate permutation carrying β to that root. □
⟨2⟩4. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩3. □
⟨1⟩5. The six positive coefficient pairs are
(18) $$(0,1),\ (1,1),\ (1,2),\ (1,0),\ (1,3),\ (2,3).$$
They give all roots up to sign.
⟨2⟩1. $\alpha_1 = \mathbf{u}_2 - \mathbf{u}_3$.
PROOF: Use (15). □
⟨2⟩2. $\alpha_0 + \alpha_1 = \mathbf{u}_1 - \mathbf{u}_2$.
PROOF: Add the coordinates in (15). □
⟨2⟩3. $\alpha_0 + 2\alpha_1 = \mathbf{u}_1 - \mathbf{u}_3$.
PROOF: Add one more α_1 to ⟨2⟩2. □
⟨2⟩4. $\alpha_0 = \mathbf{u}_1 - 2\mathbf{u}_2 + \mathbf{u}_3$.
PROOF: Use (15). □
⟨2⟩5. $\alpha_0 + 3\alpha_1 = \mathbf{u}_1 + \mathbf{u}_2 - 2\mathbf{u}_3$.
PROOF: Substitute (15). □
⟨2⟩6. $2\alpha_0 + 3\alpha_1 = 2\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3$.
PROOF: Substitute (15). □
⟨2⟩7. Q.E.D.
PROOF: The first three vectors and their negatives enumerate the short pattern; the last three and their negatives enumerate the long pattern of (14). □
⟨1⟩6. The simple Gram matrix is $\left(\begin{smallmatrix} 6 & -3 \\ -3 & 2 \end{smallmatrix}\right)$ and the Cartan matrix is G_2.
PROOF: Sum coordinate products in (15) and divide columns by 3 and 1 according to (2). □
⟨1⟩7. The two simple roots are independent.
PROOF: Their Gram determinant from ⟨1⟩6 is $12 - 9 = 3 \neq 0$. □

⟨1⟩8. Q.E.D.
PROOF: Assemble Definition 3.9 from ⟨1⟩1, ⟨1⟩3, ⟨1⟩4; ⟨1⟩5, ⟨1⟩7 give the base, and ⟨1⟩6 gives its matrix. □

Lemma (indecomposability from a base). ASSUME: 1. Φ is a Root System in V with a nonempty root base Δ.
2. The graph on Δ joining distinct nonorthogonal roots is connected.
PROVE: Φ is indecomposable.

PROOF SKETCH: An orthogonal partition puts all simple roots on the same side, leaving a vector orthogonal to a spanning set on the other side.
⟨1⟩1. ASSUME: $\Phi = P \sqcup Q$ is a partition into mutually orthogonal parts.
PROVE: $P = \varnothing$ or $Q = \varnothing$.
⟨2⟩1. Adjacent base roots belong to the same part.
PROOF: Roots in different parts are orthogonal by the partition hypothesis. □
⟨2⟩2. All base roots belong to one part.
PROOF: Apply Lemma (edge-constant functions) to the function recording the part containing a simple root, on the connected nonorthogonality graph of Δ, using ⟨2⟩1. □
⟨2⟩3. The other part is empty.
⟨3⟩1. A root in the other part is orthogonal to every vector in V.
PROOF: Expand any vector in the spanning base, using ⟨2⟩2 and bilinearity. □
⟨3⟩2. Q.E.D.
PROOF: Pair such a root with itself in ⟨3⟩1; definiteness contradicts its nonzero status. □
⟨2⟩4. Q.E.D.
PROOF: Step ⟨2⟩3 makes one part empty. □
⟨1⟩2. Q.E.D.
PROOF: The nonempty base and ⟨1⟩1 give indecomposability in Definition 3.9. □

Lemma (the models are indecomposable). ASSUME: The model is A_ℓ ($\ell \geq 1$), B_ℓ or C_ℓ ($\ell \geq 2$), D_ℓ ($\ell \geq 4$), E_j ($j = 6, 7, 8$), F_4, or G_2, in its specified space.
PROVE: It is indecomposable.

PROOF SKETCH: Apply a single general base-connectivity lemma to each model, so this proof never reuses local steps for a second Root System.
⟨1⟩1. The A base graph is a connected nonempty path.
PROOF: Its Cartan matrix is A_ℓ by Lemma (classical Root Systems); Lemma (supports of the named matrices) identifies its connected support with the base nonorthogonality graph. □
⟨1⟩2. Each B, C base graph is a connected nonempty path.
PROOF: Use the two matrix identities in Lemma (classical Root Systems) and the path supports of Lemma (supports of the named matrices), including C_2. □
⟨1⟩3. The D base graph is the connected tree $T(1, 1, \ell - 3)$.
PROOF: Combine Lemma (classical Root Systems) with h_D and connectedness in Lemma (supports of the named matrices). □
⟨1⟩4. Each E base graph is the connected tree $T(1, 2, j - 4)$.
PROOF: Combine Lemma (the three E systems) with h_E and connectedness in Lemma (supports of the named matrices). □
⟨1⟩5. The F_4 base graph is a connected four-vertex path.
PROOF: Use the Cartan matrix in Lemma (the F system) and its connected support in Lemma (supports of the named matrices). □
⟨1⟩6. The G_2 base graph is a connected single edge.

PROOF: Use the Cartan matrix in Lemma (the G system) and its connected support in Lemma (supports of the named matrices). □

⟨1⟩7. Q.E.D.

PROOF: Apply Lemma (indecomposability from a base) to the nonempty bases and connected graphs in ⟨1⟩1–⟨1⟩6. □

Lemma (root, length and line counts). ASSUME: The coordinate models are in their construction ranges: $\ell \geq 1$ for A, $\ell \geq 2$ for B, C, $\ell \geq 4$ for D, and $j \in \{6, 7, 8\}$ for E_j.

PROVE: The counts are those in the following table. The two-length models have squared-length ratio 2 in types B, C, F and 3 in type G; all other models have one length.

Type	one length	short	long	lines
A_ℓ	$\ell(\ell+1)$			$\ell(\ell+1)/2$
B_ℓ		2ℓ	$2\ell(\ell-1)$	ℓ^2
C_ℓ		$2\ell(\ell-1)$	2ℓ	ℓ^2
D_ℓ	$2\ell(\ell-1)$			$\ell(\ell-1)$
E_6	72			36
E_7	126			63
E_8	240			120
F_4		24	24	24
G_2		6	6	6

PROOF SKETCH: Count disjoint coordinate patterns, then divide by the two roots on each line.

⟨1⟩1. The A_ℓ set has $\ell(\ell+1)$ roots.

PROOF: The ordered distinct-coordinate pair is recovered from its coefficients $1, -1$; count $(\ell+1)\ell$ pairs. □

⟨1⟩2. There are $4\binom{\ell}{2} = 2\ell(\ell-1)$ signed two-coordinate roots.

PROOF: Their support and the two signs give a bijection with two-element coordinate subsets times $\{1, -1\}^2$. □

⟨1⟩3. There are 2ℓ axis roots in B and in C.

PROOF: The coordinate and sign are uniquely recovered from each such root in Definition (classical coordinate models). □

⟨1⟩4. The E_8 integer pattern has 112 roots.

PROOF: Evaluate ⟨1⟩2 at $\ell = 8$. □

⟨1⟩5. Its half-coordinate pattern has 128 roots.

PROOF: Choose seven signs freely; their product determines the eighth in Definition (exceptional coordinate data). □

⟨1⟩6. The E_7 integer pattern has $60 + 2 = 62$ roots.

⟨2⟩1. There are 60 with both coordinates among the first six.

PROOF: Evaluate ⟨1⟩2 at $\ell = 6$. □

⟨2⟩2. There are two using coordinates 7 or 8.

PROOF: The equation $x(7) + x(8) = 0$ forces the pair $\pm(\mathbf{u}_7 - \mathbf{u}_8)$ in a two-coordinate root. □

⟨2⟩3. Q.E.D.

PROOF: The disjoint cases ⟨2⟩1, ⟨2⟩2 exhaust the integer pattern. □

⟨1⟩7. The E_7 half-coordinate pattern has 64 roots.

PROOF: Its last two signs are opposite, the first six have product -1, and five of the first six signs and sign 8 are free. □

⟨1⟩8. The E_6 integer pattern has 40 roots.

⟨2⟩1. Its nonzero coordinates are among the first five.

PROOF: A nonzero coordinate among 6,7,8 would force all three to be nonzero, contrary to support size 2. □

⟨2⟩2. Q.E.D.

PROOF: Apply ⟨1⟩2 at $\ell = 5$ to ⟨2⟩1. □

⟨1⟩9. The E_6 half-coordinate pattern has 32 roots.

PROOF: The first five signs are free; their product is sign 8, and signs 6,7 are its negative. □

⟨1⟩10. There are 24 long F_4 roots.

PROOF: Apply ⟨1⟩2 at $\ell = 4$. □

⟨1⟩11. There are $8 + 16 = 24$ short F_4 roots.

PROOF: The disjoint axis and half-coordinate patterns have $2 \cdot 4$ and 2^4 independent index/sign choices. □

⟨1⟩12. There are six roots of each length in G_2.

PROOF: Count $3 \cdot 2$ ordered distinct differences and $3 \cdot 2$ choices of doubled coordinate and global sign in (14). □

⟨1⟩13. The squared lengths are 2 for A, D, E; 1 and 2 for B, F; 2 and 4 for C; 2 and 6 for G.

PROOF: Sum the coordinate squares in Definitions (classical coordinate models), (exceptional coordinate data). □

⟨1⟩14. Each line contains exactly two roots.

PROOF: Reducedness and closure under negation, from the model lemmas and Lemma (reflection identities), give precisely a sign pair. □

⟨1⟩15. Q.E.D.

PROOF: Add disjoint pattern counts ⟨1⟩1–⟨1⟩12, separate by ⟨1⟩13, and divide by 2 using ⟨1⟩14. □

3.1.3. *The Mathlib bridge and classification.*

Lemma (the finite-root-set pairing). ASSUME: Φ is a Root System in a finite-dimensional real inner product space V.

PROVE: It defines a finite reduced crystallographic root pairing P over $\mathbb{R}$ with root and coroot spaces both V, and P is a root system in Mathlib's sense. Its pairing is (2). If $\Phi \neq \varnothing$, its indecomposability is equivalent to P being Mathlib-irreducible.

PROOF SKETCH: Construct the structure on the prescribed index set Φ. For irreducibility a reflection-invariant subspace assigns each root to itself or to its orthogonal complement.

LET: Use root indices Φ, root embedding $\alpha \mapsto \alpha$, coroot map $\alpha^\vee = 2\alpha/(\alpha, \alpha)$, and bilinear form the real inner product.

⟨1⟩1. This bilinear pairing is perfect.

⟨2⟩1. The map from V to its dual is injective.

PROOF: If $(\mathbf{x} - \mathbf{y}, \mathbf{z}) = 0$ for every $\mathbf{z}$, choose $\mathbf{z} = \mathbf{x} - \mathbf{y}$ and apply definiteness in 2. □

⟨2⟩2. The flipped map is injective.

PROOF: Rewrite it as the same map by real inner-product symmetry and use ⟨2⟩1. □

⟨2⟩3. Q.E.D.

PROOF: Apply `LinearMap.IsPerfPair.of_injective` to ⟨2⟩1, ⟨2⟩2, using finite dimensionality. □

⟨1⟩2. The coroot map is injective.

⟨2⟩1. $2\alpha^\vee/(\alpha^\vee, \alpha^\vee) = \alpha$ for $\alpha \neq 0$.

PROOF: Substitute $(\alpha^\vee, \alpha^\vee) = 4/(\alpha, \alpha)$ using bilinearity and $(\alpha, \alpha) > 0$. □

⟨2⟩2. Q.E.D.

PROOF: A map with the left inverse ⟨2⟩1 is injective. □

⟨1⟩3. $(\alpha, \alpha^\vee) = 2$.

PROOF: Cancel $(\alpha,\alpha) > 0$ in the coroot formula. □

⟨1⟩4. The restricted map s_α is a permutation of the root indices.
PROOF: Apply Lemma (reflection identities) and reflection closure in Definition 3.9. □

⟨1⟩5. $(s_\alpha\beta)^\vee = \beta^\vee - (\alpha,\beta^\vee)\alpha^\vee$.
⟨2⟩1. $(s_\alpha\beta)^\vee = s_\alpha(\beta^\vee)$.
PROOF: The orthogonal linear map of Lemma (reflection identities) commutes with normalization by squared norm. □
⟨2⟩2. Q.E.D.
PROOF: Substitute the reflection formula and $\alpha^\vee = 2\alpha/(\alpha,\alpha)$ in ⟨2⟩1. □

⟨1⟩6. These data define a root pairing P with $P.p(i,j) = 2(\alpha_i,\alpha_j)/(\alpha_j,\alpha_j)$.
PROOF: Fill the structure fields of `RootPairing` using ⟨1⟩1, root inclusion, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5 and the defining root reflection identity. □

⟨1⟩7. The root and coroot reflection at index α are both s_α on every vector of V.
PROOF: The root formula subtracts $(\mathbf{x},\alpha^\vee)\alpha$; the coroot formula subtracts $(\alpha,\mathbf{x})\alpha^\vee$. Both equal $2(\alpha,\mathbf{x})\alpha/(\alpha,\alpha)$ by the coroot definition. □

⟨1⟩8. P is finite, crystallographic, reduced and spanning.
⟨2⟩1. The index type is finite.
PROOF: It is the finite set Φ of Definition 3.9. □
⟨2⟩2. All its pairings are integral.
PROOF: Apply crystallographic integrality in Definition 3.9 with the two roots interchanged to ⟨1⟩6. □
⟨2⟩3. A dependent pair of roots is an equal or opposite pair.
⟨3⟩1. For a dependent pair α,β, choose $(a,b) \neq (0,0)$ with $a\alpha + b\beta = 0$.
PROOF: Unfold linear dependence for the two-vector family. □
⟨3⟩2. $b \neq 0$.
PROOF: Otherwise ⟨3⟩1 and $\alpha \neq 0$ would force $a = 0$. □
⟨3⟩3. $\beta = (-a/b)\alpha$.
PROOF: Solve the relation ⟨3⟩1 using ⟨3⟩2. □
⟨3⟩4. Q.E.D.
PROOF: Apply reducedness in Definition 3.9 to the proportionality ⟨3⟩3. □
⟨2⟩4. The roots span V.
PROOF: This is the spanning condition in Definition 3.9. □
⟨2⟩5. The coroots span V.
PROOF: Each root is a nonzero scalar multiple of its coroot, so ⟨2⟩4 lies in the coroot span. □
⟨2⟩6. Q.E.D.
PROOF: Assemble the Mathlib predicates from ⟨2⟩1–⟨2⟩5 using 3.1. □

⟨1⟩9. ASSUME: Φ is nonempty and indecomposable.
PROVE: P is irreducible.
⟨2⟩1. V is nontrivial.
PROOF: A chosen root is nonzero by Definition 3.9. □
⟨2⟩2. Every reflection-invariant subspace W is 0 or V.
⟨3⟩1. For every root α, either $\alpha \in W$ or $\alpha \in W^\perp$.
⟨4⟩1. ASSUME: some $\mathbf{w} \in W$ has $(\mathbf{w},\alpha) \neq 0$.
PROVE: $\alpha \in W$.
PROOF: The vector $\mathbf{w} - s_\alpha\mathbf{w}$ is a nonzero scalar multiple of α in W. □
⟨4⟩2. ASSUME: no such $\mathbf{w}$ exists.
PROVE: $\alpha \in W^\perp$.
PROOF: Unfold the definition of the orthogonal complement. □

⟨4⟩3. Q.E.D.
PROOF: Excluded middle for the detecting vector gives ⟨4⟩1 or ⟨4⟩2. □
⟨3⟩2. The roots in W and the roots in $W^\perp$ partition Φ into orthogonal parts.
PROOF: Use ⟨3⟩1 and $W \cap W^\perp = \{0\}$, excluding zero roots. □
⟨3⟩3. One of those parts is empty.
PROOF: Apply indecomposability to ⟨3⟩2. □
⟨3⟩4. ASSUME: all roots lie in W.
PROVE: $W = V$.
PROOF: Use root spanning in Definition 3.9. □
⟨3⟩5. ASSUME: all roots lie in $W^\perp$.
PROVE: $W = 0$.
PROOF: Each $\mathbf{w} \in W$ is orthogonal to the spanning root set, thus to itself; use definiteness. □
⟨3⟩6. Q.E.D.
PROOF: Use ⟨3⟩3 to select ⟨3⟩4 or ⟨3⟩5. □
⟨2⟩3. Q.E.D.
PROOF: Both copies of V have the same orthogonal reflection maps by ⟨1⟩7; ⟨2⟩1, ⟨2⟩2 supply both irreducibility fields. □
⟨1⟩10. ASSUME: P is irreducible and $\Phi = A \sqcup B$ is a partition into nonempty mutually orthogonal parts.
PROVE: FALSE.
LET: $W = \operatorname{span} A$.
⟨2⟩1. $W \neq 0$.
PROOF: A member of nonempty A is a nonzero vector in W. □
⟨2⟩2. $W \neq V$.
PROOF: A nonzero root in B is orthogonal to W and cannot be orthogonal to itself. □
⟨2⟩3. A reflection at a root of A preserves W.
PROOF: Its formula subtracts a scalar multiple of an element of A from a vector of W. □
⟨2⟩4. A reflection at a root of B fixes W pointwise.
PROOF: Orthogonality to A extends linearly to W in the reflection formula. □
⟨2⟩5. Q.E.D.
PROOF: The invariant nonzero proper subspace in ⟨2⟩1–⟨2⟩4 contradicts irreducibility. □
⟨1⟩11. Q.E.D.
PROOF: Combine ⟨1⟩6, ⟨1⟩8, ⟨1⟩9, ⟨1⟩10. □

[new] We use the direct structure constructor. `RootPairing.ofBilinear` takes all reflective vectors of a form, and would not select this prescribed finite root set over $\mathbb{R}$.

Lemma (one common similarity factor). ASSUME: 1. V, W are finite-dimensional real inner product spaces and $(\alpha_i)_{i\in I}$, $(\beta_i)_{i\in I}$ are real vector-space bases indexed by the same finite nonempty set I.
2. The matrices $p_\alpha(i,j) = 2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j)$ and $p_\beta(i,j) = 2(\beta_i, \beta_j)/(\beta_j, \beta_j)$ agree, and their common nonzero off-diagonal support is connected.
PROVE: The linear equivalence $f : V \to W$ taking α_i to β_i satisfies $(f\mathbf{x}, f\mathbf{y}) = a(\mathbf{x}, \mathbf{y})$ for a single $a > 0$. For any Root Systems $\Phi \subset V$, $\Psi \subset W$ with $f(\Phi) = \Psi$, this is a similarity of those Root Systems.

PROOF SKETCH: Ratios of squared simple-root lengths propagate along the connected support; bilinearity extends the resulting Gram identity.
LET: $q(i) = (\alpha_i, \alpha_i)$, $r(i) = (\beta_i, \beta_i)$, and $M = p_\alpha = p_\beta$.
⟨1⟩1. $q(i) > 0$ and $r(i) > 0$ for every i.

PROOF: A vector in a linearly independent family is nonzero; use the bases in ⟨0⟩:1 and definiteness. □

⟨1⟩2. $M(i,j)q(j) = M(j,i)q(i)$ and $M(i,j)r(j) = M(j,i)r(i)$.
PROOF: Use symmetry of each inner product in (2). □

⟨1⟩3. On a support edge, $r(i)/q(i) = r(j)/q(j)$.
PROOF: Divide the two equalities in ⟨1⟩2 by their nonzero entries and the positive lengths of ⟨1⟩1. □

⟨1⟩4. There is $a > 0$ with $r(i) = aq(i)$ for all i.
PROOF: Apply Lemma (edge-constant functions) to $i \mapsto r(i)/q(i)$ on the common Cartan support, using ⟨1⟩3 and connectedness in ⟨0⟩:2; ⟨1⟩1 gives a positive value at any vertex. □

⟨1⟩5. $(\beta_i, \beta_j) = a(\alpha_i, \alpha_j)$ for every i, j.
PROOF: Substitute ⟨1⟩4 in $(\beta_i, \beta_j) = M(i,j)r(j)/2$ and (2). □

⟨1⟩6. The basis correspondence extends to a linear equivalence f.
PROOF: Apply `Module.Basis.equiv` to the two real bases and the identity index equivalence. □

⟨1⟩7. $(f\mathbf{x}, f\mathbf{y}) = a(\mathbf{x}, \mathbf{y})$ for all vectors.
PROOF: Expand both vectors in their finite bases and apply ⟨1⟩5 term by term. □

⟨1⟩8. Q.E.D.
PROOF: Combine ⟨1⟩4, ⟨1⟩6, ⟨1⟩7; root-set equality supplies exactly the last clause of the similarity definition. □

Lemma (orthogonal components). ASSUME: Φ is a Root System in V. Join distinct roots when their inner product is nonzero.
PROVE: Each connected component is an indecomposable Root System in its own span. Different component spans are orthogonal, and V is their orthogonal direct sum. In particular, a nonempty Φ is indecomposable if and only if this graph is connected. The empty Root System has $V = 0$ and no components.

PROOF SKETCH: Reflections preserve the component of their reflecting root; orthogonality separates different components and their spans.

⟨1⟩1. Roots in distinct components are orthogonal.
PROOF: A nonzero product would give an edge between the components by the graph definition. □

⟨1⟩2. A component A is closed under reflections in its roots.
LET: Fix $\alpha, \beta \in A$.
⟨2⟩1. ASSUME: $(\alpha, \beta) = 0$.
PROVE: $s_\alpha \beta = \beta \in A$.
PROOF: Substitute zero in the reflection formula. □
⟨2⟩2. ASSUME: $(\alpha, \beta) \neq 0$.
PROVE: $s_\alpha \beta \in A$.
⟨3⟩1. $(\alpha, s_\alpha \beta) = -(\alpha, \beta) \neq 0$.
PROOF: Expand the reflection formula by bilinearity. □
⟨3⟩2. Q.E.D.
PROOF: The reflected root equals α or is joined to it by ⟨3⟩1, so it lies in the same component. □
⟨2⟩3. Q.E.D.
PROOF: The cases ⟨2⟩1, ⟨2⟩2 exhaust the inner-product values. □

⟨1⟩3. Each component is a Root System in its span.
PROOF: Finiteness, nonzero roots, integrality and reducedness are inherited from Φ; ⟨1⟩2 gives closure and its own span gives the spanning field of Definition 3.9. □

⟨1⟩4. Each component is indecomposable.

PROOF: An orthogonal partition could have no graph edge crossing its parts by definition, contrary to connectedness of the component. □

⟨1⟩5. Distinct component spans are orthogonal.
PROOF: Expand finite linear combinations using ⟨1⟩1 and bilinearity. □

⟨1⟩6. The sum of those spans is direct and equals V.
⟨2⟩1. The sum equals V.
PROOF: Every root lies in a component, and Φ spans V. □
⟨2⟩2. ASSUME: $\sum_A \mathbf{x}_A = 0$ with each $\mathbf{x}_A$ in its component span.
PROVE: Every $\mathbf{x}_A = 0$.
PROOF: Pair the relation with $\mathbf{x}_A$; ⟨1⟩5 leaves $(\mathbf{x}_A, \mathbf{x}_A) = 0$, so definiteness applies. □
⟨2⟩3. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩2. □

⟨1⟩7. For nonempty Φ, indecomposability is equivalent to connectedness of its graph.
PROOF: Multiple components give an orthogonal partition by ⟨1⟩1; one component is indecomposable by ⟨1⟩4. □

⟨1⟩8. ASSUME: Φ is empty.
PROVE: $V = 0$.
PROOF: Use its spanning condition and the span of the empty set. □

⟨1⟩9. Q.E.D.
PROOF: Combine ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, ⟨1⟩7, ⟨1⟩8. □

Lemma (indecomposable classification). ASSUME: Φ is a nonempty indecomposable Root System in a finite-dimensional real inner product space V.
PROVE: After one common positive rescaling, Φ is isometric to exactly one coordinate Root System in (3). Exactly $\mathrm{B}, \mathrm{C}, \mathrm{F}, \mathrm{G}$ have two lengths. The additional C_2 coordinate model is similar to B_2.

PROOF SKETCH: Apply Theorem A to a Mathlib base. Construct the basis map and use the library equivalence of root pairings. Counts distinguish the models without a conjugacy theorem for bases.

⟨1⟩1. The pairing P constructed from Φ is finite, spanning, reduced, crystallographic and irreducible over $\mathbb{R}$.
PROOF: Apply Lemma (the finite-root-set pairing) to ⟨0⟩. □

⟨1⟩2. PICK a Mathlib base b for P.
PROOF: Apply `RootPairing.nonempty_base` to ⟨1⟩1; $\mathbb{R}$ is a characteristic zero field. □

⟨1⟩3. The finite base support is nonempty.
PROOF: Apply `RootPairing.Base.support_nonempty` to ⟨1⟩2, the nonempty root set and $2 \neq 0$ in $\mathbb{R}$. □

⟨1⟩4. Its integer Cartan matrix is finite Cartan.
PROOF: Apply `RootPairing.Base.cartanMatrix_isFiniteCartan` using ⟨1⟩1 and classically chosen decidable equality. □

⟨1⟩5. That matrix is indecomposable.
PROOF: Apply `RootPairing.Base.cartanMatrix_isIndecomposable` using ⟨1⟩1. □

⟨1⟩6. PICK a type, rank ℓ, and reindexing from its model indices to the support of b.
PROOF: Apply Theorem A to ⟨1⟩3, ⟨1⟩4, ⟨1⟩5. □

⟨1⟩7. There is a coordinate Root System Ψ with a root base having precisely this reindexed matrix.
PROOF: Select the matching case of Lemmas (classical Root Systems), (the three E systems), (the F system), (the G system), using the rank restrictions in ⟨1⟩6. □

⟨1⟩8. Its pairing P_2 is finite, spanning, reduced and crystallographic over $\mathbb{R}$.

PROOF: Apply Lemma (the finite-root-set pairing) to the Root System in ⟨1⟩7. □

⟨1⟩9. The displayed root base gives a Mathlib base b_2.

⟨2⟩1. Every model root or its negative lies in the additive monoid of the selected simple roots.

PROOF: Use the nonnegative integer expansions in ⟨1⟩7 and `Submodule.span_nat_eq_addSubmonoidClosure`. □

⟨2⟩2. Q.E.D.

PROOF: Apply `RootPairing.Base.mk'` using ⟨1⟩8, root independence in ⟨1⟩7, and ⟨2⟩1. □

LET: Let $e : b.\text{support} \to b_2.\text{support}$ be the index bijection induced by ⟨1⟩6 and the model base parametrization.

⟨1⟩10. For all base indices i, j, $b_2.\text{cartanMatrix}(e(i), e(j)) = b.\text{cartanMatrix}(i, j)$.

⟨2⟩1. The real pairings of the two bases agree through e.

PROOF: The entry equality in ⟨1⟩6 and the model pairing matrix in ⟨1⟩7 identify both real pairings with the same cast named matrix, using `RootPairing.Base.algebraMap_cartanMatrixIn_apply`. □

⟨2⟩2. Q.E.D.

PROOF: Rewrite both real pairings in ⟨2⟩1 as casts of integer Cartan entries by `RootPairing.Base.algebraMap_cartanMatrixIn_apply`; injectivity of $\mathbb{Z} \to \mathbb{R}$ gives the equality. □

⟨1⟩11. PICK the linear equivalence $f = b.\text{toWeightBasis.equiv}\ b_2.\text{toWeightBasis}\ e$.

PROOF: Use the spanning pairings ⟨1⟩1, ⟨1⟩8 in the weight-basis and basis-equivalence declarations of 3.1. □

⟨1⟩12. $f(\alpha_i) = \beta_{e(i)}$ for every simple root.

PROOF: Apply `Module.Basis.equiv_apply` and `RootPairing.Base.toWeightBasis_apply` to ⟨1⟩11. □

⟨1⟩13. There is a root-pairing equivalence $P \simeq P_2$ whose map on root spaces is f.

PROOF: Apply `RootPairing.Base.equivOfCartanMatrixEq` to b, b_2, e and ⟨1⟩10, using the source hypotheses in ⟨1⟩1 and finite, spanning, reduced, crystallographic target hypotheses in ⟨1⟩8. Its defining root-space map is exactly ⟨1⟩11. □

⟨1⟩14. $f(\Phi) = \Psi$.

PROOF: The index equivalence and root compatibility of ⟨1⟩13 give a bijection of the root sets. □

⟨1⟩15. The common base matrix has connected nonzero support.

PROOF: Apply Lemma (connected support) to ⟨1⟩3, ⟨1⟩4, ⟨1⟩5. □

⟨1⟩16. f is a similarity using one positive factor.

⟨2⟩1. The families $\alpha_i = b.\text{toWeightBasis}(i)$ and $\widehat{\beta}_i = b_2.\text{toWeightBasis}(e(i))$, indexed by $b.\text{support}$, are real vector-space bases.

PROOF: Reindex the second weight basis along e; the first is the weight basis from ⟨1⟩11. □

⟨2⟩2. For some $a > 0$, $(f\mathbf{x}, f\mathbf{y}) = a(\mathbf{x}, \mathbf{y})$ for all vectors.

PROOF: Apply Lemma (one common similarity factor) to ⟨2⟩1, ⟨1⟩10 cast to real pairings, and ⟨1⟩15; ⟨1⟩12 identifies its basis map with f. □

⟨2⟩3. Q.E.D.

PROOF: Steps ⟨2⟩2, ⟨1⟩14 are the similarity conditions of Definition 3.9. □

⟨1⟩17. No two distinct models in (3) are similar.

⟨2⟩1. Similarity preserves rank, root count, number of lengths, their ratio, and the numbers of short and long roots.

PROOF: A bijection scaling every squared length by the same $a > 0$ preserves these definitions. □

⟨2⟩2. One-length and two-length models are distinguished.

PROOF: Use the lengths in Lemma (root, length and line counts) with ⟨2⟩1. □

⟨2⟩3. At equal rank, the A and D root counts differ throughout the permitted D range.
PROOF: The equality $\ell(\ell+1) = 2\ell(\ell-1)$ with $\ell > 0$ implies $\ell = 3$, excluded by $\ell \geq 4$. □
⟨2⟩4. At rank 6, the counts $42, 60, 72$ distinguish $\mathrm{A}, \mathrm{D}, \mathrm{E}$.
PROOF: Evaluate Lemma (root, length and line counts) at rank 6. □
⟨2⟩5. At rank 7, the counts $56, 84, 126$ distinguish these types.
PROOF: Evaluate Lemma (root, length and line counts) at rank 7. □
⟨2⟩6. At rank 8, the counts $72, 112, 240$ distinguish these types.
PROOF: Evaluate Lemma (root, length and line counts) at rank 8. □
⟨2⟩7. Squared-length ratio 3 distinguishes G_2.
PROOF: All other two-length models have ratio 2 by Lemma (root, length and line counts). □
⟨2⟩8. At rank 4, F_4 has 48 roots whereas $\mathrm{B}_4, \mathrm{C}_4$ have 32.
PROOF: Evaluate Lemma (root, length and line counts). □
⟨2⟩9. For $\ell \geq 3$, $\mathrm{B}_\ell, \mathrm{C}_\ell$ have different numbers of short roots.
PROOF: Their counts 2ℓ and $2\ell(\ell-1)$ from Lemma (root, length and line counts) differ when $\ell \geq 3$. □
⟨2⟩10. Q.E.D.
PROOF: The cases ⟨2⟩2–⟨2⟩9 separate every pair at equal rank, and ⟨2⟩1 separates unequal ranks. □
⟨1⟩18. The map with matrix $\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ is a similarity from B_2 onto C_2.
⟨2⟩1. $\mathbf{u}_1$ maps to $\mathbf{u}_1 - \mathbf{u}_2$.
PROOF: Read column 1 of the displayed matrix. □
⟨2⟩2. $\mathbf{u}_2$ maps to $\mathbf{u}_1 + \mathbf{u}_2$.
PROOF: Read column 2 of the displayed matrix. □
⟨2⟩3. $\mathbf{u}_1 + \mathbf{u}_2$ maps to $2\mathbf{u}_1$.
PROOF: Add ⟨2⟩1, ⟨2⟩2. □
⟨2⟩4. $\mathbf{u}_1 - \mathbf{u}_2$ maps to $-2\mathbf{u}_2$.
PROOF: Subtract ⟨2⟩2 from ⟨2⟩1. □
⟨2⟩5. The transpose times the matrix is $2I$.
PROOF: Compute each entry of $T^T T$ as the dot product of two columns of the displayed matrix T. □
⟨2⟩6. Q.E.D.
PROOF: The four images and their negatives exhaust $\Phi(\mathrm{C}_2)$, while ⟨2⟩5 gives scale 2 and invertibility. □
⟨1⟩19. Q.E.D.
PROOF: Use ⟨1⟩16, ⟨1⟩17, ⟨1⟩18 and the length assertion of Lemma (root, length and line counts). Two similarities from Φ to models compose, after inversion of the first, to a similarity between the models by Lemma (composition and inversion of similarities); ⟨1⟩17 then forces the same model. □

Theorem B. ASSUME: Φ is a Root System in a finite-dimensional real inner product space V.
PROVE: If Φ is nonempty and indecomposable, it is similar, by one common positive scale, to exactly one model in (3). Exactly the $\mathrm{B}, \mathrm{C}, \mathrm{F}, \mathrm{G}$ models have two lengths; C_2 is similar to B_2 and is omitted from that list. In general V is the orthogonal direct sum of the component spans, and each nonzero component is similar to a listed indecomposable model, with an independent scale on each component. Φ is empty exactly when V is the zero space, and then there are no components.

PROOF SKETCH: **[new]** The indecomposable assertion is a separate lemma so that applying it to each component does not reuse local proof steps.
⟨1⟩1. The nonempty indecomposable assertion and the model length and rank-2 assertions hold.

PROOF: Apply Lemma (indecomposable classification). □

⟨1⟩2. Every nonempty component is an indecomposable Root System in its span, and these spans form an orthogonal direct sum.

PROOF: Apply Lemma (orthogonal components) to Φ. □

⟨1⟩3. Each component is similar to a unique listed model using a single positive scale for that component.

PROOF: Apply Lemma (indecomposable classification) separately to each Root System supplied by ⟨1⟩2. □

⟨1⟩4. All listed models are indecomposable.

PROOF: Apply Lemma (the models are indecomposable). □

⟨1⟩5. Q.E.D.

PROOF: Combine ⟨1⟩1–⟨1⟩4; the empty case is the final clause of Lemma (orthogonal components), and conversely the zero space has no nonzero vector, so no root. □

3.1.4. *Angles, lines and the normalized matrix.*

Angle convention. For two distinct lines the angle lies in $(0, 90°]$ and its cosine is $|(\alpha, \beta)|/(\|\alpha\|\|\beta\|)$ for nonzero representatives. We use `Real.cos_pi_div_three`, `Real.cos_pi_div_four`, `Real.cos_pi_div_six`, and `Real.injOn_cos`, all in `Mathlib/Analysis/SpecialFunctions/Trigonometric/Basic.lean`. Their values and injectivity on $[0, \pi]$ identify squared cosines $1/4, 1/2, 3/4$ with $60°, 45°, 30°$, respectively.

Lemma (angles and fixed-length subsystems). ASSUME: Φ is a Root System in a finite-dimensional real inner product space.

PROVE: For every $\alpha, \beta \in \Phi$ which are neither perpendicular nor proportional,

$$\frac{4(\alpha, \beta)^2}{(\alpha, \alpha)\,(\beta, \beta)} \in \{1, 2, 3\}.$$

The longer-to-shorter length ratio and acute line angle are exactly one of $(1, 60°)$, $(\sqrt{2}, 45°)$, $(\sqrt{3}, 30°)$. For every occurring length $a > 0$, the roots of length a form a Root System in their own span, and it is an orthogonal sum of systems similar to A_ℓ $(\ell \geq 1)$, D_ℓ $(\ell \geq 4)$, $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$.

[repair] The hypotheses exclude proportional pairs; fixed-length subsystems need not be indecomposable or span the original space.

PROOF SKETCH: Bound the product of two nonzero integer pairings by 4. Reflections preserve length, then Theorem B classifies each component.

⟨1⟩1. ASSUME: $\alpha, \beta \in \Phi$ are neither perpendicular nor proportional.

PROVE: Their pairing product, length ratio and angle have the stated alternatives.

LET: $q = (\alpha, \alpha) > 0$, $r = (\beta, \beta) > 0$, $h = (\alpha, \beta) \neq 0$, $u = 2h/q$, and $v = 2h/r$.

⟨2⟩1. u, v are nonzero integers of the same sign.

PROOF: Use the two integrality conditions in Definition 3.9 and the positive denominators q, r. □

⟨2⟩2. $h^2 < qr$.

⟨3⟩1. $\beta - (h/q)\alpha \neq 0$.

PROOF: Otherwise β would be proportional to α, contrary to ⟨1⟩. □

⟨3⟩2. $\|\beta - (h/q)\alpha\|^2 = r - h^2/q > 0$.

⟨4⟩1. $\|\beta - (h/q)\alpha\|^2 = r - h^2/q$.

PROOF: Expand by bilinearity in 2 and substitute the definitions of q, r, h. □

⟨4⟩2. $\|\beta - (h/q)\alpha\|^2 > 0$.

PROOF: Apply definiteness in 2 to ⟨3⟩1. □

⟨4⟩3. Q.E.D.
PROOF: Combine ⟨4⟩1, ⟨4⟩2. □
⟨3⟩3. Q.E.D.
PROOF: Multiply ⟨3⟩2 by $q > 0$. □
⟨2⟩3. $uv = 4h^2/(qr) \in \{1, 2, 3\}$.
PROOF: It is a positive integer less than 4 by ⟨2⟩1, ⟨2⟩2. □
⟨2⟩4. The length and angle alternatives follow.
⟨3⟩1. ASSUME: $uv = 1$.
PROVE: $|u| = |v| = 1$, $q = r$, and the angle is 60°.
⟨4⟩1. $|u| = |v| = 1$.
PROOF: A product of two positive integers is 1 only when both are 1, applied to ⟨2⟩1 and $uv = 1$. □
⟨4⟩2. $q = r$.
PROOF: Substitute ⟨4⟩1 in $u/v = r/q$, using the equal signs in ⟨2⟩1. □
⟨4⟩3. The angle is 60°.
PROOF: The squared cosine is $h^2/(qr) = uv/4 = 1/4$; apply 3.1.4. □
⟨4⟩4. Q.E.D.
PROOF: Combine ⟨4⟩1, ⟨4⟩2, ⟨4⟩3. □
⟨3⟩2. ASSUME: $uv = 2$.
PROVE: $\{|u|, |v|\} = \{1, 2\}$, the squared-length ratio is 2, and the angle is 45°.
⟨4⟩1. $(|u|, |v|)$ is $(1, 2)$ or $(2, 1)$.
PROOF: The only positive divisors of 2 are 1 and 2, applied to ⟨2⟩1 and $uv = 2$. □
⟨4⟩2. $\max(q, r)/\min(q, r) = 2$.
PROOF: Substitute the two alternatives ⟨4⟩1 in $u/v = r/q$, with $q, r > 0$. □
⟨4⟩3. The angle is 45°.
PROOF: The squared cosine is $h^2/(qr) = uv/4 = 1/2$; apply 3.1.4. □
⟨4⟩4. Q.E.D.
PROOF: Combine ⟨4⟩1, ⟨4⟩2, ⟨4⟩3. □
⟨3⟩3. ASSUME: $uv = 3$.
PROVE: $\{|u|, |v|\} = \{1, 3\}$, the squared-length ratio is 3, and the angle is 30°.
⟨4⟩1. $(|u|, |v|)$ is $(1, 3)$ or $(3, 1)$.
PROOF: The only positive divisors of 3 are 1 and 3, applied to ⟨2⟩1 and $uv = 3$. □
⟨4⟩2. $\max(q, r)/\min(q, r) = 3$.
PROOF: Substitute the two alternatives ⟨4⟩1 in $u/v = r/q$, with $q, r > 0$. □
⟨4⟩3. The angle is 30°.
PROOF: The squared cosine is $h^2/(qr) = uv/4 = 3/4$; apply 3.1.4. □
⟨4⟩4. Q.E.D.
PROOF: Combine ⟨4⟩1, ⟨4⟩2, ⟨4⟩3. □
⟨3⟩4. Q.E.D.
PROOF: Take positive square roots of the squared-length ratios in the exhaustive cases ⟨3⟩1–⟨3⟩3. □
⟨2⟩5. Q.E.D.
PROOF: Combine ⟨2⟩3, ⟨2⟩4. □
⟨1⟩2. For each occurring $a > 0$, the set $\Psi = \{\gamma \in \Phi : \|\gamma\| = a\}$ is a Root System in its span.
⟨2⟩1. Ψ is finite, nonzero, reduced, integral, and spans its own span.
PROOF: Restrict the corresponding conditions of Definition 3.9 to Ψ. □
⟨2⟩2. For $\gamma, \delta \in \Psi$, $s_\gamma \delta \in \Phi$.
PROOF: Apply reflection closure of Φ. □

⟨2⟩3. $\|s_\gamma \delta\| = a$.
PROOF: Apply Lemma (reflection identities) and $\|\delta\| = a$. □
⟨2⟩4. $s_\gamma \delta$ lies in span Ψ.
PROOF: Its defining formula is a linear combination of $\gamma, \delta \in \Psi$. □
⟨2⟩5. Q.E.D.
PROOF: Assemble Definition 3.9 from ⟨2⟩1–⟨2⟩4. □
⟨1⟩3. Every nonempty component of this subsystem has type A, D or E.
PROOF: Apply Theorem B to ⟨1⟩2; its other models have two lengths by Lemma (root, length and line counts), impossible in the fixed-length subset. □
⟨1⟩4. Q.E.D.
PROOF: Combine ⟨1⟩1, ⟨1⟩2, ⟨1⟩3. □

Lemma (unit-line orthogonal projections). ASSUME: V is a finite-dimensional real inner product space, $\mathbf{v}$ is a unit vector, and $P_\mathbf{v}\mathbf{x} = (\mathbf{v}, \mathbf{x})\mathbf{v}$ is the rank-one orthogonal projection onto $\mathbb{R}\mathbf{v}$.
PROVE: $P_{-\mathbf{v}} = P_\mathbf{v}$; $P_\mathbf{v}$ is symmetric; its trace is 1; and for every orthogonal linear bijection f, $f P_\mathbf{v} f^{-1} = P_{f\mathbf{v}}$.

PROOF SKETCH: Expand the rank-one operator and its diagonal.
⟨1⟩1. $P_{-\mathbf{v}}\mathbf{x} = P_\mathbf{v}\mathbf{x}$.
PROOF: The two minus signs cancel by bilinearity. □
⟨1⟩2. $(P_\mathbf{v}\mathbf{x}, \mathbf{y}) = (\mathbf{x}, P_\mathbf{v}\mathbf{y})$.
PROOF: Both sides are $(\mathbf{v}, \mathbf{x})(\mathbf{v}, \mathbf{y})$ by 2. □
⟨1⟩3. In an orthonormal basis the trace is $\sum_i v(i)^2 = 1$.
PROOF: The ith diagonal entry is $(\mathbf{v}, \mathbf{u}_i)^2$; sum and use $\|\mathbf{v}\| = 1$. □
⟨1⟩4. $f P_\mathbf{v} f^{-1}\mathbf{x} = (f\mathbf{v}, \mathbf{x}) f\mathbf{v}$.
PROOF: Substitute the orthogonal projection formula and the isometry identity $(\mathbf{v}, f^{-1}\mathbf{x}) = (f\mathbf{v}, \mathbf{x})$. □
⟨1⟩5. Q.E.D.
PROOF: Operator extensionality applied to ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4 gives all four identities. □

Lemma (the line frame). ASSUME: 1. Φ is a nonempty indecomposable Root System in an n-dimensional real inner product space V.
2. Its line set is $\mathcal{L}$, $|\mathcal{L}| = N$, and $\mathbf{v}_L$ is any unit representative of each line.
PROVE: $n > 0$ and

$$\mathcal{F} := \sum_{L \in \mathcal{L}} P_{\mathbf{v}_L} = \frac{N}{n} I_V. \tag{19}$$

The identity is independent of all representative signs.

PROOF SKETCH: The symmetric frame operator commutes with every root reflection. It acts by one scalar on all roots, and its trace determines it.
⟨1⟩1. $n > 0$.
PROOF: The space contains a nonzero root from ⟨0⟩:1. □
⟨1⟩2. The operator $\mathcal{F}$ is independent of signs and symmetric.
PROOF: Sum the sign and symmetry identities of Lemma (unit-line orthogonal projections). □
⟨1⟩3. $\mathcal{F}$ commutes with each root reflection s_α.
⟨2⟩1. s_α permutes $\mathcal{L}$.
PROOF: It is an orthogonal root permutation by Lemma (reflection identities). □
⟨2⟩2. $s_\alpha \mathcal{F} s_\alpha^{-1} = \mathcal{F}$.

PROOF: Reindex the sum by ⟨2⟩1, using projector conjugation and sign independence in Lemma (unit-line orthogonal projections). □

⟨2⟩3. Q.E.D.

PROOF: Multiply ⟨2⟩2 on the right by s_α. □

⟨1⟩4. For every root α there is a unique scalar t_α with $\mathcal{F}\alpha = t_\alpha\alpha$.

⟨2⟩1. $s_\alpha(\mathcal{F}\alpha) = -\mathcal{F}\alpha$.

PROOF: Use ⟨1⟩3 and $s_\alpha\alpha = -\alpha$ from Lemma (reflection identities). □

⟨2⟩2. Q.E.D.

PROOF: The -1 eigenspace in Lemma (reflection identities) is $\mathbb{R}\alpha$; uniqueness uses $\alpha \neq 0$. □

⟨1⟩5. Nonorthogonal roots have the same scalar.

PROOF: Symmetry in ⟨1⟩2 gives $(t_\alpha - t_\beta)(\alpha, \beta) = 0$ by ⟨1⟩4; cancel the nonzero inner product. □

⟨1⟩6. There is a single real t with $t_\alpha = t$ for all roots.

PROOF: Apply Lemma (edge-constant functions) to $\alpha \mapsto t_\alpha$ on the nonorthogonality graph of all roots, connected by Lemma (orthogonal components), using ⟨1⟩5. □

⟨1⟩7. $\mathcal{F} = tI_V$.

PROOF: Expand a vector in the spanning roots and use ⟨1⟩4, ⟨1⟩6. □

⟨1⟩8. $\operatorname{tr}\mathcal{F} = N$.

PROOF: Sum the diagonal entries and the unit traces of Lemma (unit-line orthogonal projections). □

⟨1⟩9. $t = N/n$.

PROOF: The trace of ⟨1⟩7 is nt; use ⟨1⟩8 and divide by $n > 0$ from ⟨1⟩1. □

⟨1⟩10. Q.E.D.

PROOF: Substitute ⟨1⟩9 in ⟨1⟩7 and use ⟨1⟩1, ⟨1⟩2. □

Lemma (the normalized matrix convention). ASSUME: 1. N lines span an n-dimensional real inner product space, $n > 0$.

2. In an orthonormal coordinate basis, U has a chosen unit vector on each line as its columns, $G = U^TU$, and $\theta \in \mathbb{R}$.

LET:

$$C = 2G - 2I_N. \tag{20}$$

PROVE: $(C + 2I_N)(C - \theta I_N) = 0$ if and only if $UU^T = (N/n)I_n$ and $\theta = 2N/n - 2$. Both conditions are independent of the choices of signs.

[repair] This extends the squared-norm-2 representative convention preceding Theorem 3.7. Unequal-length roots are normalized line by line for this matrix; that operation is not asserted to preserve the Root System. Entries may be $\pm\sqrt{2}, \pm\sqrt{3}$.

PROOF SKETCH: Reduce the polynomial to $G^2 = tG$, then use surjectivity of U to cancel on both sides and use the trace.

LET: $t = (\theta + 2)/2$.

⟨1⟩1. $(C + 2I)(C - \theta I) = 4(G^2 - tG)$.

PROOF: Expand using (20) and the definition of t. □

⟨1⟩2. $G^2 - tG = U^T(UU^T - tI)U$.

PROOF: Substitute $G = U^TU$ and use matrix associativity. □

⟨1⟩3. U is surjective onto the ambient coordinate space.

PROOF: Its columns span by ⟨0⟩:1. □

⟨1⟩4. ASSUME: $G^2 - tG = 0$.

PROVE: $UU^T = tI$.

⟨2⟩1. For any $\mathbf{x}, \mathbf{y}$ there exist $\mathbf{v}, \mathbf{w}$ with $U\mathbf{v} = \mathbf{x}$, $U\mathbf{w} = \mathbf{y}$.

PROOF: Apply ⟨1⟩3 to each of $\mathbf{x}, \mathbf{y}$. □

⟨2⟩2. $\mathbf{x}^T(UU^T - tI)\mathbf{y} = 0$ for every $\mathbf{x}, \mathbf{y}$.
PROOF: Substitute ⟨2⟩1 into ⟨1⟩2 under $G^2 - tG = 0$. □
⟨2⟩3. Q.E.D.
PROOF: Evaluate ⟨2⟩2 on every pair of coordinate unit vectors to obtain each zero entry. □
⟨1⟩5. ASSUME: $UU^T = tI$.
PROVE: $G^2 - tG = 0$.
PROOF: Substitute in ⟨1⟩2. □
⟨1⟩6. ASSUME: $UU^T = tI$.
PROVE: $t = N/n$.
⟨2⟩1. $\mathrm{tr}(UU^T) = N$.
PROOF: Sum the squared coordinates of each unit column in ⟨0⟩:2, using Lemma (unit-line orthogonal projections). □
⟨2⟩2. Q.E.D.
PROOF: The assumed equality gives $nt = N$ by ⟨2⟩1; cancel $n > 0$. □
⟨1⟩7. Changing column signs leaves UU^T unchanged and conjugates both G and C by a diagonal sign matrix.
LET: $D_\epsilon = \mathrm{diag}(\epsilon)$ with $\epsilon(i)^2 = 1$.
⟨2⟩1. $D_\epsilon^T = D_\epsilon$ and $D_\epsilon^2 = I$.
PROOF: Multiply diagonal entries. □
⟨2⟩2. $(UD_\epsilon)(UD_\epsilon)^T = UU^T$.
PROOF: Apply associativity and ⟨2⟩1. □
⟨2⟩3. $(UD_\epsilon)^T(UD_\epsilon) = D_\epsilon G D_\epsilon$.
PROOF: Use transpose of a product and $G = U^T U$. □
⟨2⟩4. Q.E.D.
PROOF: Substitute ⟨2⟩3 in (20), and use ⟨2⟩1, ⟨2⟩2. □
⟨1⟩8. Q.E.D.
PROOF: Combine ⟨1⟩1, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, $t = (\theta + 2)/2$, and the conjugation invariance in ⟨1⟩7. □

Lemma (three lines at forty-five degrees). ASSUME: $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are orthonormal in $\mathbb{R}^3$ and $c = 1/\sqrt{2}$.
LET: $\mathbf{v}_1 = \mathbf{u}_1$, $\mathbf{v}_2 = c\mathbf{u}_1 + c\mathbf{u}_2$, and $\mathbf{v}_3 = c\mathbf{u}_1 + (1-c)\mathbf{u}_2 + \sqrt{2c-1}\mathbf{u}_3$.
PROVE: These are unit vectors on three distinct lines pairwise at 45°. There are no Root System Φ in any Euclidean space and $\alpha, \beta, \gamma \in \Phi$ satisfying all three equations

$$4(\alpha,\beta)^2 = 2(\alpha,\alpha)\,(\beta,\beta),$$
$$4(\alpha,\gamma)^2 = 2(\alpha,\alpha)\,(\gamma,\gamma),$$
$$4(\beta,\gamma)^2 = 2(\beta,\beta)\,(\gamma,\gamma).$$

PROOF SKETCH: Check their Gram matrix, then choose a shortest root on the three lines in a putative containing Root System.
⟨1⟩1. $c^2 = 1/2$, $1/2 < c < 1$, and $2c - 1 > 0$.
PROOF: Use $1 < \sqrt{2} < 2$ and $(\sqrt{2})^2 = 2$ in the definition of c. □
⟨1⟩2. $(\mathbf{v}_1, \mathbf{v}_1) = 1$.
PROOF: Use orthonormality. □
⟨1⟩3. $(\mathbf{v}_2, \mathbf{v}_2) = 2c^2 = 1$.
PROOF: Sum its coordinate squares using ⟨1⟩1. □
⟨1⟩4. $(\mathbf{v}_3, \mathbf{v}_3) = c^2 + (1-c)^2 + (2c-1) = 1$.
PROOF: Sum its coordinate squares and use ⟨1⟩1. □
⟨1⟩5. $(\mathbf{v}_1, \mathbf{v}_2) = c$.

PROOF: Read the first coordinate of $\mathbf{v}_2$. □

⟨1⟩6. $(\mathbf{v}_1, \mathbf{v}_3) = c$.

PROOF: Read the first coordinate of $\mathbf{v}_3$. □

⟨1⟩7. $(\mathbf{v}_2, \mathbf{v}_3) = c^2 + c(1-c) = c$.

PROOF: Sum their coordinate products. □

⟨1⟩8. The three lines are distinct and pairwise at 45°.

PROOF: Use the unit norms ⟨1⟩2–⟨1⟩4, the inner products ⟨1⟩5–⟨1⟩7, and $0 < c < 1$ in 3.1.4. □

⟨1⟩9. ASSUME: A Root System Φ and roots $\alpha, \beta, \gamma \in \Phi$ satisfy the three equations in the statement.

PROVE: FALSE.

⟨2⟩1. The three root pairs are nonperpendicular, nonproportional, and have line angle 45°.

PROOF: Their squared cosines are 1/2 by ⟨1⟩ and the positive squared norms of nonzero roots in Definition 3.9; use 3.1.4. □

⟨2⟩2. PICK a permutation of the three roots with α of minimum squared length $b = (\alpha, \alpha) > 0$.

PROOF: A finite nonempty set of three squared lengths has a minimum; the three equations are unchanged by the permutation. □

⟨2⟩3. $(\beta, \beta) = 2b$.

PROOF: Apply Lemma (angles and fixed-length subsystems) to the 45° pair α, β from ⟨2⟩1 and minimality in ⟨2⟩2. □

⟨2⟩4. $(\gamma, \gamma) = 2b$.

PROOF: Apply Lemma (angles and fixed-length subsystems) to the 45° pair α, γ from ⟨2⟩1 and minimality in ⟨2⟩2. □

⟨2⟩5. Q.E.D.

PROOF: The equal lengths in ⟨2⟩3, ⟨2⟩4 contradict Lemma (angles and fixed-length subsystems) for the 45° pair β, γ. □

⟨1⟩10. Q.E.D.

PROOF: Combine ⟨1⟩2–⟨1⟩4, ⟨1⟩8, ⟨1⟩9. □

Theorem C. ASSUME: 1. Root Systems are as in Definition 3.9; every line matrix is $C = 2U^T U - 2I$ with unit columns.

2. For the classical line assertions $n \geq 2$, and the models are those of Definitions (classical coordinate models), (exceptional coordinate data).

PROVE: Nonperpendicular, nonproportional root pairs have exactly the length-ratio/angle alternatives in Lemma (angles and fixed-length subsystems). Each occurring fixed-length subset is a Root System in its own span and an orthogonal sum of systems similar to A_ℓ ($\ell \geq 1$), D_ℓ ($\ell \geq 4$), $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$. The $\mathrm{B}_n, \mathrm{C}_n$ models have the same n^2 lines in dimension n, with allowed angles $90°, 60°, 45°$, and $(C + 2I)(C - 2(n-1)I) = 0$. Type F_4 has 24 lines in dimension 4, with those three angles, and $(C + 2I)(C - 10I) = 0$. Type G_2 has 6 lines in dimension 2, with angles $90°, 60°, 30°$, and $(C + 2I)(C - 4I) = 0$. For $n = 2$ the 60° classical angle does not occur; for $n \geq 3$ all three classical angles occur. All listed angles occur in F_4 and G_2. Three lines pairwise at 45° exist in $\mathbb{R}^3$ and are contained in no Root System in any ambient dimension.

In Lean the matrix equations hold for every choice of one root on each line, with a unit vector on the same line. Such a choice of roots exists, indexed by $\{1, \ldots, N\}$ for the number N of lines (`IsRootSet.exists_rootRepresentatives`), so the equations are not vacuous.

PROOF SKETCH: Use the common line frame for the matrix equations, and give witness pairs for each claimed occurring angle.

⟨1⟩1. The root-pair and fixed-length assertions hold.

PROOF: Apply Lemma (angles and fixed-length subsystems). □

⟨1⟩2. The B_n and C_n line sets coincide.
⟨2⟩1. Their axis roots lie on exactly the same axes.
PROOF: The vectors $\mathbf{u}_i$ and $2\mathbf{u}_i$ are nonzero multiples in Definition (classical coordinate models). □
⟨2⟩2. Their two-coordinate roots are identical.
PROOF: Compare their second clauses in Definition (classical coordinate models). □
⟨2⟩3. Q.E.D.
PROOF: The two patterns in ⟨2⟩1, ⟨2⟩2 exhaust both root sets. □
⟨1⟩3. The line counts and dimensions are (n^2, n), $(24, 4)$, $(6, 2)$ in the classical, F, and G cases.
PROOF: Apply Lemma (root, length and line counts) and the coordinate-space definitions. □
⟨1⟩4. The allowed angles are the asserted ones.
⟨2⟩1. In B, C, F a nonorthogonal distinct-line pair has angle $60°$ or $45°$.
PROOF: Lemma (root, length and line counts) gives equal lengths or ratio $\sqrt{2}$; apply Lemma (angles and fixed-length subsystems). □
⟨2⟩2. In G such a pair has angle $60°$ or $30°$.
PROOF: Lemma (root, length and line counts) gives equal lengths or ratio $\sqrt{3}$; apply Lemma (angles and fixed-length subsystems). □
⟨2⟩3. Q.E.D.
PROOF: Adjoin the perpendicular case $90°$ to ⟨2⟩1, ⟨2⟩2. □
⟨1⟩5. All claimed angles occur, with the stated rank-2 exception.
⟨2⟩1. The axes $\mathbb{R}\mathbf{u}_1, \mathbb{R}\mathbf{u}_2$ give $90°$ in B_n, C_n, F_4.
PROOF: Their product is zero, and both lines occur in the coordinate lists. □
⟨2⟩2. The lines of $\mathbf{u}_1, \mathbf{u}_1 + \mathbf{u}_2$ give $45°$ in those models.
PROOF: Their squared cosine is $1/(1 \cdot 2) = 1/2$; use 3.1.4. □
⟨2⟩3. For $n \geq 3$, the lines of $\mathbf{u}_1 + \mathbf{u}_2, \mathbf{u}_1 + \mathbf{u}_3$ give $60°$ in B_n, C_n and also in F_4.
PROOF: Their squared cosine is $1/(2 \cdot 2) = 1/4$. □
⟨2⟩4. For $n = 2$ the classical models have no $60°$ pair.
⟨3⟩1. The two axes are perpendicular.
PROOF: Their inner product is zero. □
⟨3⟩2. The two diagonal lines are perpendicular.
PROOF: $(\mathbf{u}_1 + \mathbf{u}_2, \mathbf{u}_1 - \mathbf{u}_2) = 1 - 1 = 0$. □
⟨3⟩3. Every axis-diagonal pair has squared cosine $1/2$.
PROOF: Its absolute inner product is 1 and its squared norms are 1 and 2. □
⟨3⟩4. Q.E.D.
PROOF: These three pair types exhaust the four lines in Definition (classical coordinate models). □
⟨2⟩5. The short roots $\mathbf{u}_1 - \mathbf{u}_2, \mathbf{u}_1 - \mathbf{u}_3$ give $60°$ in G_2.
PROOF: Their squared cosine is $1/(2 \cdot 2) = 1/4$. □
⟨2⟩6. The roots $2\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3, \mathbf{u}_2 - \mathbf{u}_3$ give $90°$.
PROOF: Their inner product is $-1 + 1 = 0$. □
⟨2⟩7. The roots $2\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3, \mathbf{u}_1 - \mathbf{u}_2$ give $30°$.
PROOF: Their squared cosine is $3^2/(6 \cdot 2) = 3/4$. □
⟨2⟩8. Q.E.D.
PROOF: Combine ⟨2⟩1–⟨2⟩7 using 3.1.4. □
⟨1⟩6. All three asserted matrix equations hold.
⟨2⟩1. The frame identity holds for each model.
PROOF: Apply Lemma (the line frame) to each indecomposable Root System of Lemma (the models are indecomposable). □

⟨2⟩2. For B_n and C_n, $\theta = 2(n-1)$.
PROOF: Substitute $N = n^2$ from ⟨1⟩3 and the frame ⟨2⟩1 in Lemma (the normalized matrix convention). □
⟨2⟩3. For F_4, $\theta = 10$.
PROOF: Substitute $N = 24$, dimension 4, and ⟨2⟩1 in Lemma (the normalized matrix convention). □
⟨2⟩4. For G_2, $\theta = 4$.
PROOF: Substitute $N = 6$, dimension 2, and ⟨2⟩1 in Lemma (the normalized matrix convention). □
⟨2⟩5. Q.E.D.
PROOF: Combine ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩7. The asserted three-line example exists and cannot lie in any Root System.
PROOF: Apply Lemma (three lines at forty-five degrees). □
⟨1⟩8. Q.E.D.
PROOF: Combine ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, ⟨1⟩7. □

Corollary (the equal-length matrix constants). ASSUME: The model is A_n ($n \geq 1$), D_n ($n \geq 4$), or E_6, E_7, E_8, in its span; $C = 2U^TU - 2I$ uses unit representatives of its lines.
PROVE: $(C + 2I)(C - \theta I) = 0$, with respective constants $n-1, 2(n-2), 10, 16, 28$.

PROOF SKETCH: **[repair]** The last constant is 28, correcting the printed 18 in Theorem 3.7.
⟨1⟩1. Each model satisfies the line frame identity.
PROOF: Apply Lemma (the line frame) using Lemma (the models are indecomposable). □
⟨1⟩2. For A_n, the second constant is $n-1$.
PROOF: Use $N = n(n+1)/2$ from Lemma (root, length and line counts) in Lemma (the normalized matrix convention) with ⟨1⟩1. □
⟨1⟩3. For D_n, it is $2(n-2)$.
PROOF: Substitute $N = n(n-1)$ from Lemma (root, length and line counts) in Lemma (the normalized matrix convention) with ⟨1⟩1. □
⟨1⟩4. For E_6, it is $2(36/6) - 2 = 10$.
PROOF: Substitute the E_6 count in Lemma (the normalized matrix convention) with ⟨1⟩1. □
⟨1⟩5. For E_7, it is $2(63/7) - 2 = 16$.
PROOF: Substitute the E_7 count in Lemma (the normalized matrix convention) with ⟨1⟩1. □
⟨1⟩6. For E_8, it is $2(120/8) - 2 = 28$.
PROOF: Substitute the E_8 count in Lemma (the normalized matrix convention) with ⟨1⟩1. □
⟨1⟩7. Q.E.D.
PROOF: The cases ⟨1⟩2–⟨1⟩6 exhaust the listed models. □

3.1.5. *Finite certificates from the ordinary proof.*

Lemma (leaf determinant expansion). ASSUME: M is a square matrix over a commutative ring, on a finite index set; $v \neq w$, $M(v,v) = 2$, and every off-diagonal entry in row or column v is zero except possibly the entry at w.
PROVE: Writing $p = M(v,w)M(w,v)$ and subscripts for principal deletions,

$$\det M = 2 \det M_{-v} - p \det M_{-v,-w}. \tag{21}$$

PROOF SKETCH: Separate permutations fixing v from those interchanging v, w. This applies to a leaf of any support, not only a path.
⟨1⟩1. Permutation terms fixing v sum to $2 \det M_{-v}$.
PROOF: Restrict those permutations to the complement of v in `Matrix.det_apply'`; its diagonal factor is 2. □

⟨1⟩2. A nonzero permutation term not fixing v interchanges v and w.
⟨2⟩1. Column v must choose row w.
PROOF: All other available off-diagonal entries of column v vanish by ⟨0⟩. □
⟨2⟩2. Row v must choose column w.
PROOF: Column v is already used by row w in ⟨2⟩1, and all remaining entries of row v except at w vanish. □
⟨2⟩3. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩2. □
⟨1⟩3. The moving terms sum to $-p \det M_{-v,-w}$.
PROOF: Delete the transposition in ⟨1⟩2 from each permutation; it contributes sign -1 and entry product p in `Matrix.det_apply'`. □
⟨1⟩4. Q.E.D.
PROOF: Add the exhaustive permutation classes ⟨1⟩1, ⟨1⟩3. □

Lemma (leading determinants). ASSUME: M is a named matrix in (3); Δ_k is its leading k-row principal determinant and $\Delta_0 = 1$.
PROVE: The complete lists are

(22)

A_ℓ	$\Delta_k = k+1$
B_ℓ, C_ℓ	$\Delta_k = k+1$ $(k<\ell)$, $\Delta_\ell = 2$
D_ℓ	$\Delta_k = k+1$ $(k<\ell)$, $\Delta_\ell = 4$
E_6	$(2,4,6,5,4,3)$
E_7	$(2,4,6,5,4,3,2)$
E_8	$(2,4,6,5,4,3,2,1)$
F_4	$(2,3,2,1)$
G_2	$(2,1)$.

PROOF SKETCH: Apply the leaf formula with the actual deleted neighbour, then solve the path recurrence.
⟨1⟩1. For a path extension with last edge product p, $\Delta_k = 2\Delta_{k-1} - p\Delta_{k-2}$.
PROOF: Apply Lemma (leaf determinant expansion) to its last vertex. □
⟨1⟩2. The unit-path determinants are $k+1$.
⟨2⟩1. $\Delta_0 = 1$ and $\Delta_1 = 2$.
PROOF: Evaluate the empty and one-entry determinants. □
⟨2⟩2. ASSUME: $\Delta_{k-1} = k$ and $\Delta_{k-2} = k-1$.
PROVE: $\Delta_k = k+1$.
PROOF: Substitute in ⟨1⟩1 with $p = 1$. □
⟨2⟩3. Q.E.D.
PROOF: Induction using ⟨2⟩1, ⟨2⟩2. □
⟨1⟩3. The B, C final determinants are 2.
PROOF: Their last edge product is 2, so ⟨1⟩1, ⟨1⟩2 give $2\ell - 2(\ell-1) = 2$. □
⟨1⟩4. The D_ℓ final determinant is 4.
⟨2⟩1. Deleting $v = \ell - 1$ leaves a unit path of determinant ℓ.
PROOF: Use Definition (the indexed matrices) and ⟨1⟩2. □
⟨2⟩2. Deleting also $w = \ell-3$ leaves a path of $\ell-3$ vertices and an isolated vertex, of determinant $2(\ell-2)$.
PROOF: Factor the isolated diagonal entry 2 in the determinant expansion and apply ⟨1⟩2. □
⟨2⟩3. Q.E.D.
PROOF: Lemma (leaf determinant expansion) with $p = 1$, ⟨2⟩1, ⟨2⟩2 gives $2\ell - 2(\ell-2) = 4$. □
⟨1⟩5. The E prefix determinants are $2, 4, 6, 5, 4, 3, 2, 1$.

⟨2⟩1. The first determinant is 2.
PROOF: Evaluate the single diagonal entry. □
⟨2⟩2. The second is 4.
PROOF: The leading two vertices are isolated, giving $2 \cdot 2$. □
⟨2⟩3. The third is 6.
PROOF: Its single unit edge contributes determinant 3 and its isolated vertex contributes 2. □
⟨2⟩4. The fourth is 5.
PROOF: The order in Remark (small ranks) makes it a four-vertex unit path; apply ⟨1⟩2. □
⟨2⟩5. The next four determinants are $4, 3, 2, 1$.
PROOF: Its leading k-row matrix is exactly E_k by `CartanMatrix.E`; for $k = 5, 6, 7, 8$, apply `CartanMatrix.E_det`, which gives $9 - k$ for every $k \geq 3$. □
⟨2⟩6. Q.E.D.
PROOF: Concatenate ⟨2⟩1–⟨2⟩5. □
⟨1⟩6. The F determinants are $2, 3, 2, 1$.
⟨2⟩1. The first three determinants are $2, 3, 2$.
PROOF: Substitute the edge products $1, 2$ in ⟨1⟩1, starting from $\Delta_0 = 1, \Delta_1 = 2$. □
⟨2⟩2. The fourth determinant is 1.
PROOF: Apply `CartanMatrix.F4_det` in `Mathlib/LinearAlgebra/Matrix/Cartan.lean`. □
⟨2⟩3. Q.E.D.
PROOF: Combine ⟨2⟩1, ⟨2⟩2. □
⟨1⟩7. The G determinants are $2, 1$.
PROOF: The first diagonal entry is 2; its final determinant is 1 by `CartanMatrix.G2_det` in `Mathlib/LinearAlgebra/Matrix/Cartan.lean`. □
⟨1⟩8. Q.E.D.
PROOF: Proper classical prefixes are unit paths; combine ⟨1⟩2–⟨1⟩7. □

Lemma (rational positivity certificates). ASSUME: M is $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8, \mathrm{F}_4$ or G_2, and $H = \mathrm{diag}(d)M$ with d from (4).
PROVE: There is a lower triangular L with diagonal 1 such that

$$H = L\,\mathrm{diag}(q)L^T, \qquad \mathbf{x}^T H \mathbf{x} = \sum_i q(i)((L^T\mathbf{x})(i))^2, \tag{23}$$

with every $q(i) > 0$. The certificate data are:

E_8	$q = (2, 2, 3/2, 5/6, 4/5, 3/4, 2/3, 1/2)$
F_4	$q = (2, 3/2, 4/3, 1)$
G_2	$q = (2, 3/2)$.

For E_8 the nonzero strict lower entries are

$$L(2,0) = -1/2, \quad L(3,1) = -1/2, \quad L(3,2) = -2/3,$$
$$L(4,3) = -6/5, \quad L(5,4) = -5/4, \quad L(6,5) = -4/3, \quad L(7,6) = -3/2.$$

Use the leading six or seven entries and rows for $\mathrm{E}_6, \mathrm{E}_7$. For F_4 use $L(1,0) = -1/2$, $L(2,1) = -4/3$, $L(3,2) = -3/2$. For G_2 use $L(1,0) = -3/2$; all unspecified strict lower entries are zero. These data prove H positive definite.

PROOF SKETCH: These are bounded rational matrix identities, followed by back substitution in a triangular matrix.
⟨1⟩1. The eight-row data satisfy $H(i,j) = \sum_k L(i,k)q(k)L(j,k)$ for all $i, j < 8$.
PROOF: Substitute the displayed rational entries and the E_8 entries of Definition (the indexed matrices) in that finite sum. □

⟨1⟩2. The same equality holds for its six- and seven-row prefixes.
PROOF: Lower triangularity makes every omitted summand zero in the prefix entries of ⟨1⟩1. □
⟨1⟩3. The equality holds for the four-row F data.
PROOF: Substitute those rational entries and $d = (1, 1, 2, 2)$ in the finite entry sum. □
⟨1⟩4. The equality holds for the two-row G data.
PROOF: Substitute those rational entries and $d = (1, 3)$ in the finite entry sum. □
⟨1⟩5. The quadratic identity in (23) holds.
PROOF: Multiply the identities ⟨1⟩1–⟨1⟩4 on the left and right by $\mathbf{x}^T, \mathbf{x}$. □
⟨1⟩6. ASSUME: $L^T\mathbf{x} = 0$.
PROVE: $\mathbf{x} = 0$.
PROOF: Backward induction on the coordinates of the upper triangular matrix L^T with diagonal 1. □
⟨1⟩7. Q.E.D.
PROOF: For $\mathbf{x} \neq 0$, ⟨1⟩6 makes at least one square in ⟨1⟩5 positive; every displayed $q(i)$ is positive, so the sum is positive. □

Lemma (rational coordinate frame). ASSUME: A coordinate model from the construction ranges has N root lines and rank n; the matrix A has its simple roots as columns in the given coordinate space, which may be larger than its span.
PROVE:

$$\frac{1}{2}\sum_{\alpha\in\Phi}\frac{\alpha\alpha^T}{(\alpha,\alpha)} = \frac{N}{n}\,A(A^TA)^{-1}A^T. \tag{24}$$

All entries in this equation are rational.

PROOF SKETCH: Both sides act by N/n on the root span and vanish on its orthogonal complement.
⟨1⟩1. $P = A(A^TA)^{-1}A^T$ is the orthogonal projection onto the column span.
⟨2⟩1. A^TA is invertible.
PROOF: A null coefficient vector has combination of squared norm zero, thus zero; the model bases are independent by Lemmas (classical Root Systems), (the three E systems), (the F system), (the G system). □
⟨2⟩2. $P(Ax) = Ax$.
PROOF: Cancel $(A^TA)^{-1}(A^TA)$ using ⟨2⟩1. □
⟨2⟩3. ASSUME: $A^T\mathbf{y} = 0$.
PROVE: $P\mathbf{y} = 0$.
PROOF: Substitute in the definition of P. □
⟨2⟩4. Q.E.D.
PROOF: The column span and its orthogonal complement are the two summands tested in ⟨2⟩2, ⟨2⟩3. □
⟨1⟩2. The left side is the sum of the unit-line orthogonal projections.
PROOF: Each line has exactly its two opposite roots by Lemma (root, length and line counts); both summands are its orthogonal projection by Lemma (unit-line orthogonal projections). □
⟨1⟩3. The two operators agree on the ambient coordinate space.
PROOF: Lemma (the line frame) gives $(N/n)I$ on the root span for ⟨1⟩2, and every root projection annihilates the orthogonal complement; use ⟨1⟩1. □
⟨1⟩4. All entries are rational.
PROOF: The model roots have rational coordinates and positive rational squared norms; an invertible rational matrix has rational inverse by `Matrix.inv_def`. □
⟨1⟩5. Q.E.D.
PROOF: Combine ⟨1⟩3, ⟨1⟩4. □

Remark 3.11. Theorems A, B and C provide the finite-type classification, its passage to Euclidean Root Systems, and all the unequal-length line assertions of this remark. Corollary (the equal-length matrix constants) also recovers the equal-length equations with the corrected E_8 constant. The classification does not use [CGSS, Theorem 3.10] as a premise. Theorem 3.12 is proved separately (Section 3.2).

3.2. **Lines contained in a Root System.** We will follow the ordinary proof in `cgss/proofs/thm312/thm312.tex`. The construction is the explicit set $\Psi(T)$ of Definition (the integer-span construction). All spaces are real inner product spaces; finite dimension is stated where needed. Entries of a vector or a matrix are function values. The reflection $\tau_{\mathbf{r}}$ is $s_{\mathbf{r}}$ of Definition 3.9. We will write $q(\mathbf{x}) = (\mathbf{x}, \mathbf{x})$ and $\mathbb{R}\mathbf{x} = \mathrm{span}_{\mathbb{R}}\{\mathbf{x}\}$. For a real scalar a, $a \in \mathbb{Z}$ and $a \in 2\mathbb{Z}$ mean $a = z$ and $a = 2z$, respectively, for some $z \in \mathbb{Z}$, with the integer cast into $\mathbb{R}$ understood.

The proof format is that of [L95]. Numbered assumptions of a result are cited as $\langle 0\rangle$:1, and assumptions of an enclosing step as $\langle 1\rangle$, $\langle 2\rangle$, and so on. Each proof ends with a QED step; each leaf gives its facts. The tags **[new]** and **[repair]** identify new arguments and repairs to the printed paper. The metadata `\lean` and `\uses` record the Lean declarations and the dependencies.

3.2.1. *Definitions and library facts.*

Definition (angles and compatible colourings). ASSUME: V is a real inner product space, S is a set of real subspaces of V, and ℓ, m are real subspaces.

For $a \in \mathbb{R}$, let HasSquaredCosine(ℓ, m, a) mean

$$\forall \mathbf{x} \in \ell\ \forall \mathbf{y} \in m, \qquad (\mathbf{x}, \mathbf{y})^2 = a\, q(\mathbf{x}) q(\mathbf{y}).$$

Let HasAllowedAngles(S) mean that for all distinct $\ell, m \in S$ there is $a \in \{0, 1/4, 1/2\}$ with this property. Let HasBipartition(S) mean that there is a function $b : S \to \mathrm{Bool}$ such that, for all distinct $\ell, m \in S$,

$$\begin{aligned} b(\ell) = b(m) &\implies \text{HasSquaredCosine}(\ell, m, 0) \ \vee\ \text{HasSquaredCosine}(\ell, m, 1/4), \\ b(\ell) \neq b(m) &\implies \text{HasSquaredCosine}(\ell, m, 0) \ \vee\ \text{HasSquaredCosine}(\ell, m, 1/2). \end{aligned}$$

Neither colour fibre is required to be nonempty.

Definition (lines of a vector set). ASSUME: V is a real inner product space and $T \subseteq V$.

Define

$$\mathcal{L}(T) := \{\ell : \exists \mathbf{r} \in T,\ \ell = \mathbb{R}\mathbf{r}\}.$$

This is `LineSystem.rootLines`. When $0 \notin T$, each member has dimension one.

Definition (the integer-span construction). ASSUME: V is a real inner product space and $T \subseteq V$.

Put $L(T) = \mathrm{span}_{\mathbb{Z}}(T)$ and

$$\begin{aligned} \Psi(T) = \{&\mathbf{x} \in L(T) : q(\mathbf{x}) = 2\} \\ &\cup \{\mathbf{x} \in L(T) : q(\mathbf{x}) = 4 \text{ and } (\mathbf{x}, \mathbf{y}) \in 2\mathbb{Z} \text{ for every } \mathbf{y} \in L(T)\}. \end{aligned}$$

This is `LineSystem.integralSpanRoots`. The evenness condition quantifies over the entire fixed integer span.

Inner products and square roots. All library citations were checked in Mathlib at commit `082e2d37e8b0463410cdb532e111cd43d5a66174` (4 September 2026), the version pinned in `lake-manifest.json`. We will use Elementary identities, together with `real_inner_self_eq_norm_sq` and `abs_real_inner_le_norm` in `Mathlib/Analysis/InnerProductSpace/Basic.lean`. These give $(\mathbf{x}, \mathbf{y})^2 \leq q(\mathbf{x}) q(\mathbf{y})$. For positive real a, $\sqrt{a} > 0$ and $(\sqrt{a})^2 = a$ are `Real.sqrt_pos` and `Real.sq_sqrt` in `Mathlib/Analysis/Real/Sqrt.lean`.

Spans and finite-dimensional linear algebra. `finrank_eq_one_iff'` in `Mathlib/LinearAlgebra/Dimension/FreeAndStrongRankCondition.lean` gives a nonzero generator of a one-dimensional subspace. `eq_span_singleton_of_mem_of_finrank_eq_one` and `finrank_span_singleton` in `Mathlib/LinearAlgebra/FiniteDimensional/Basic.lean` identify its span and give the converse dimension assertion. `Submodule.mem_span_singleton` in `Mathlib/LinearAlgebra/Span/Defs.lean` expresses membership as a scalar multiple. `Submodule.subset_span`, `Submodule.span_mono`, and `Submodule.span_le` in the same file give membership of generators, monotonicity, and the criterion for a span to lie in a subspace. `Submodule.span_span_coe_preimage` there says that the preimage of a generating set under the inclusion of its span spans the whole subtype. `exists_linearIndependent` in `Mathlib/LinearAlgebra/LinearIndependent/Lemmas.lean` chooses an independent subset with the same real span. `LinearIndependent.finite` in `Mathlib/LinearAlgebra/Dimension/Finite.lean` makes that subset finite in finite dimension. `Submodule.span_induction` and `Submodule.span_induction₂` in `Mathlib/LinearAlgebra/Span/Defs.lean` give induction on a span, also over $\mathbb{Z}$. `Submodule.mem_span_range_iff_exists_fun` in `Mathlib/LinearAlgebra/Finsupp/LinearCombination.lean` expresses membership in a finite span by a finite sum. `Int.cast_smul_eq_zsmul` in `Mathlib/Algebra/Module/NatInt.lean` identifies multiplication by a real-cast integer with integer scalar multiplication.
Finite sets and cardinalities. `Set.Finite.of_injOn`, `Set.Finite.image`, `Set.Finite.subset`, and `Set.Finite.preimage` in `Mathlib/Data/Set/Finite/Basic.lean` give finiteness from an injection into a finite set, finiteness of images and subsets, and finiteness of a preimage when the map is injective on that preimage. A function set A^I is finite when A and I are finite.
Integer arithmetic and logarithms. Finite sums of integers are integers; finite sums of even integers are even. We will use field arithmetic, factorization of differences of squares, and integer inequalities in the arithmetic leaves. `zpow_add_one₀` in `Mathlib/Algebra/GroupWithZero/Basic.lean` gives the successor power identity for a nonzero base, including base 2. `Int.log` is the integer logarithm, and `Int.zpow_log_le_self`, `Int.lt_zpow_succ_log_self`, `Int.zpow_le_iff_le_log`, and `Int.lt_zpow_iff_log_lt` in `Mathlib/Data/Int/Log.lean` apply to base $2 > 1$ and $x > 0$. `Int.odd_iff` and `Int.not_even_iff_odd` in `Mathlib/Algebra/Ring/Int/Parity.lean` identify the two parities, also for negative integers. In particular $k + 1$ and $k - 1$ have parity opposite to k, by reduction modulo 2.
Orthogonal complements. `stdOrthonormalBasis` in `Mathlib/Analysis/InnerProductSpace/PiL2.lean` gives an orthonormal basis indexed by the finite dimension. `Submodule.inner_right_of_mem_orthogonal` in `Mathlib/Analysis/InnerProductSpace/Orthogonal.lean` gives $(\mathbf{x}, \mathbf{y}) = 0$ for $\mathbf{x} \in W$ and $\mathbf{y} \in W^\perp$. `Orthonormal.inner_eq_zero` in `Mathlib/Analysis/InnerProductSpace/Orthonormal.lean` gives zero inner products at distinct indices of an orthonormal family. `Submodule.isCompl_orthogonal` and `Submodule.HasOrthogonalProjection.ofCompleteSpace` in `Mathlib/Analysis/InnerProductSpace/Projection/Basic.lean` give $W + W^\perp = V$. Here W is complete by the theorem `FiniteDimensional.complete` in `Mathlib/Topology/Algebra/Module/FiniteDimension.lean`.

These facts apply to the induced inner product on a subspace, including the zero subspace.

3.2.2. *Representatives and finiteness.*

Lemma (representatives of a line). ASSUME: V is a real inner product space, ℓ is a real subspace, $\mathbf{r} \in \ell \setminus \{0\}$, and $h > 0$.

PROVE: There is $\mathbf{t} \in \ell$ with $q(\mathbf{t}) = h$.

PROOF SKETCH: We will rescale the given nonzero vector.
⟨1⟩1. $q(\mathbf{r}) > 0$.
PROOF: Positive definiteness in the inner-product facts, ⟨0⟩. □
LET: $\mathbf{t} = (\sqrt{h}/\sqrt{q(\mathbf{r})})\mathbf{r}$.
⟨1⟩2. $\mathbf{t} \in \ell$.
PROOF: The subspace ℓ contains $\mathbf{r}$ by ⟨0⟩ and is closed under real scalar multiplication. □
⟨1⟩3. $q(\mathbf{t}) = h$.
PROOF: Bilinearity and the square-root facts give $q(\mathbf{t}) = h\,q(\mathbf{r})/q(\mathbf{r}) = h$, using $h > 0$ in ⟨0⟩ and ⟨1⟩1. □
⟨1⟩4. Q.E.D.
PROOF: The witness $\mathbf{t}$ satisfies ⟨1⟩2 and ⟨1⟩3. □

Lemma (squared cosine from representatives). ASSUME: 1. ℓ and m are subspaces of dimension one in the real inner product space V.
2. $\mathbf{r} \in \ell \setminus \{0\}$ and $\mathbf{s} \in m \setminus \{0\}$.
PROVE: For every $a \in \mathbb{R}$,

$$\text{HasSquaredCosine}(\ell, m, a) \iff (\mathbf{r}, \mathbf{s})^2 = a\, q(\mathbf{r})q(\mathbf{s}).$$

For all $a, a' \in \mathbb{R}$, if HasSquaredCosine(ℓ, m, a) and HasSquaredCosine(ℓ, m, a'), then $a = a'$.

PROOF SKETCH: We will write vectors on the two lines as scalar multiples of their representatives, then cancel their positive squared norms.
⟨1⟩1. $q(\mathbf{r})q(\mathbf{s}) > 0$.
PROOF: Positive definiteness and ⟨0⟩:2. □
⟨1⟩2. ASSUME: $a \in \mathbb{R}$ and HasSquaredCosine(ℓ, m, a).
PROVE: $(\mathbf{r}, \mathbf{s})^2 = a\, q(\mathbf{r})q(\mathbf{s})$.
PROOF: Substitute ⟨0⟩:2 in Definition (angles and compatible colourings). □
⟨1⟩3. ASSUME: $a \in \mathbb{R}$ and $(\mathbf{r}, \mathbf{s})^2 = a\, q(\mathbf{r})q(\mathbf{s})$.
PROVE: HasSquaredCosine(ℓ, m, a).
⟨2⟩1. For every $\mathbf{x} \in \ell$ and $\mathbf{y} \in m$ there are $u, v \in \mathbb{R}$ with $\mathbf{x} = u\mathbf{r}$ and $\mathbf{y} = v\mathbf{s}$.
PROOF: Apply the singleton-span facts to each nonzero representative in ⟨0⟩:2. □
⟨2⟩2. Q.E.D.
PROOF: Multiply ⟨1⟩ by u^2v^2 and use bilinearity with ⟨2⟩1; this includes $u = 0$ or $v = 0$. □
⟨1⟩4. ASSUME: $a, a' \in \mathbb{R}$, HasSquaredCosine(ℓ, m, a) and HasSquaredCosine(ℓ, m, a').
PROVE: $a = a'$.
PROOF: Apply ⟨1⟩2 at a and a'. The resulting equalities give $(a - a')q(\mathbf{r})q(\mathbf{s}) = 0$; cancel the positive product in ⟨1⟩1. □
⟨1⟩5. Q.E.D.
PROOF: Steps ⟨1⟩2 and ⟨1⟩3 give the equivalence; ⟨1⟩4 gives uniqueness. □

For lines, HasSquaredCosine(ℓ, m, a) with $a = 0, 1/4, 1/2$ means the unoriented angle is $90°, 60°, 45°$, respectively. We will use the division-free predicate and Lemma (squared cosine from representatives); no trigonometric fact is a premise below.

Lemma (inner-product coordinates). ASSUME: 1. V is a real inner product space, W a real subspace, and $(\mathbf{b}_i)_{i \in I}$ a family spanning W.
2. $\mathbf{x}, \mathbf{y} \in W$ and $(\mathbf{x}, \mathbf{b}_i) = (\mathbf{y}, \mathbf{b}_i)$ for every $i \in I$.
PROVE: $\mathbf{x} = \mathbf{y}$. Equivalently, $\mathbf{x} \mapsto ((\mathbf{x}, \mathbf{b}_i))_{i \in I}$ is injective on W.

PROOF SKETCH: We will show that the difference is orthogonal to its own span.

⟨1⟩1. $(\mathbf{x}-\mathbf{y},\mathbf{b}_i)=0$ for every $i\in I$.
PROOF: Subtract the equalities of ⟨0⟩:2 by bilinearity. □
⟨1⟩2. $(\mathbf{x}-\mathbf{y},\mathbf{w})=0$ for every $\mathbf{w}\in W$.
PROOF: Real span induction from ⟨1⟩1: the assertion holds at zero and is preserved by addition and real scalar multiplication, by bilinearity. □
⟨1⟩3. $q(\mathbf{x}-\mathbf{y})=0$.
PROOF: Substitute $\mathbf{w}=\mathbf{x}-\mathbf{y}\in W$ in ⟨1⟩2. □
⟨1⟩4. Q.E.D.
PROOF: Positive definiteness applied to ⟨1⟩3. □

Lemma (finitely many lines). ASSUME: 1. V is a finite-dimensional real inner product space and S is a set of subspaces of dimension one.
2. HasAllowedAngles(S).
PROVE: S is finite.

PROOF SKETCH: We will encode all unit representatives by their inner products with a finite spanning family chosen from those representatives.
LET:

$$U=\{\mathbf{x}:\exists \ell\in S,\ \mathbf{x}\in\ell \text{ and } q(\mathbf{x})=1\},\qquad W=\operatorname{span}_{\mathbb{R}} U,$$

$$\Theta=\{-1,-1/\sqrt{2},-1/2,0,1/2,1/\sqrt{2},1\}.$$

⟨1⟩1. PICK a finite family $(\mathbf{b}_i)_{i\in I}$ spanning W with every $\mathbf{b}_i\in U$.
PROOF: The independent-subset and finite-dimensional facts in Spans and finite-dimensional linear algebra, applied to U, give this family. The index set is empty if U is empty. □
⟨1⟩2. For every $\mathbf{x}\in U$ and $i\in I$, $(\mathbf{x},\mathbf{b}_i)\in\Theta$.
⟨2⟩1. PICK $\ell,m\in S$ with $\mathbf{x}\in\ell$, $\mathbf{b}_i\in m$, and $q(\mathbf{x})=q(\mathbf{b}_i)=1$.
PROOF: The definition of U and ⟨1⟩1. □
LET: $p=(\mathbf{x},\mathbf{b}_i)$.
⟨2⟩2. CASE: $\ell=m$.
PROVE: $p\in\{-1,1\}$.
⟨3⟩1. There is $t\in\mathbb{R}$ with $\mathbf{x}=t\mathbf{b}_i$ and $t^2=1$.
PROOF: The singleton-span facts give the scalar; the two unit norms give its square by bilinearity. □
⟨3⟩2. Q.E.D.
PROOF: Factor t^2-1 in ⟨3⟩1; then $p=t\in\{-1,1\}$. □
⟨2⟩3. CASE: $\ell\neq m$.
PROVE: $p\in\Theta$.
⟨3⟩1. $p^2\in\{0,1/4,1/2\}$.
PROOF: Substitute the unit representatives in ⟨0⟩:2, using Definition (angles and compatible colourings). □
⟨3⟩2. CASE: $p^2=0$.
PROVE: $p=0$.
PROOF: A real square is zero only at zero. □
⟨3⟩3. CASE: $p^2=1/4$.
PROVE: $p\in\{-1/2,1/2\}$.
PROOF: Factor $(p-1/2)(p+1/2)=0$. □
⟨3⟩4. CASE: $p^2=1/2$.
PROVE: $p\in\{-1/\sqrt{2},1/\sqrt{2}\}$.
PROOF: The square-root facts give $(1/\sqrt{2})^2=1/2$; factor $(p-1/\sqrt{2})(p+1/\sqrt{2})=0$. □
⟨3⟩5. Q.E.D.
PROOF: Step ⟨3⟩1 makes ⟨3⟩2, ⟨3⟩3, and ⟨3⟩4 exhaustive. □

⟨2⟩4. Q.E.D.
PROOF: The lines are equal or distinct; apply ⟨2⟩2 and ⟨2⟩3. □
⟨1⟩3. U is finite.
PROOF: By Lemma (inner-product coordinates) and ⟨1⟩1, the coordinate map is injective on $U \subseteq W$. Step ⟨1⟩2 puts its image in the finite set Θ^I; use the finite-set facts in Finite sets and cardinalities. □
⟨1⟩4. $\{\mathbb{R}\mathbf{x} : \mathbf{x} \in U\} = S$.
⟨2⟩1. For $\mathbf{x} \in U$, $\mathbb{R}\mathbf{x} \in S$.
PROOF: The witness line in the definition of U equals $\mathbb{R}\mathbf{x}$ by the singleton-span facts, since $q(\mathbf{x}) = 1$. □
⟨2⟩2. For $\ell \in S$, some $\mathbf{x} \in U$ satisfies $\ell = \mathbb{R}\mathbf{x}$.
PROOF: A nonzero generator exists by the dimension-one facts. Lemma (representatives of a line), with $h = 1$, normalizes it; the singleton-span facts identify its line. □
⟨2⟩3. Q.E.D.
PROOF: The two inclusions are ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩5. Q.E.D.
PROOF: The image of the finite set in ⟨1⟩3 is finite; use ⟨1⟩4. □

Lemma (a bounded part of an integer span). ASSUME: 1. V is a finite-dimensional real inner product space and $T \subseteq V$ is finite.
2. $q(\mathbf{t}) \in \{2, 4\}$ for $\mathbf{t} \in T$, and $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for $\mathbf{t}, \mathbf{u} \in T$.
LET: $L = \operatorname{span}_{\mathbb{Z}} T$, $W = \operatorname{span}_{\mathbb{R}} T$, $B = \{\mathbf{x} \in L : q(\mathbf{x}) \leq 4\}$.
PROVE: 1. $L \subseteq W$, and $(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}$ for all $\mathbf{x}, \mathbf{y} \in L$.
2. For every $\mathbf{t} \in T$ such that $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for all $\mathbf{u} \in T$, we have $(\mathbf{t}, \mathbf{x}) \in 2\mathbb{Z}$ for all $\mathbf{x} \in L$.
3. B is finite.

PROOF SKETCH: **[new]** We will expand integer combinations and use bounded inner-product coordinates against T itself. This simplifies the ordinary draft's basis argument; no independence of T is assumed.
⟨1⟩1. Every $\mathbf{x} \in L$ has an expression $\mathbf{x} = \sum_{\mathbf{t} \in T} a(\mathbf{t})\mathbf{t}$ with $a : T \to \mathbb{Z}$.
PROOF: Apply the finite-span formula in Spans and finite-dimensional linear algebra over $\mathbb{Z}$ to the inclusion $T \to V$. □
⟨1⟩2. $L \subseteq W$.
PROOF: Cast the integer coefficients in ⟨1⟩1 to real coefficients. □
⟨1⟩3. For all $\mathbf{x}, \mathbf{y} \in L$, $(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}$.
⟨2⟩1. For coefficients $a, b : T \to \mathbb{Z}$,

$$\left(\sum_{\mathbf{t} \in T} a(\mathbf{t})\mathbf{t}, \sum_{\mathbf{u} \in T} b(\mathbf{u})\mathbf{u}\right) = \sum_{\mathbf{t}, \mathbf{u} \in T} a(\mathbf{t})b(\mathbf{u})(\mathbf{t}, \mathbf{u}).$$

PROOF: Bilinearity, iterated over the two finite sums. □
⟨2⟩2. Q.E.D.
PROOF: Each summand in ⟨2⟩1 is an integer by ⟨0⟩:2; use ⟨1⟩1 and closure under finite sums. □
⟨1⟩4. ASSUME: $\mathbf{t} \in T$ and $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for every $\mathbf{u} \in T$.
PROVE: $(\mathbf{t}, \mathbf{x}) \in 2\mathbb{Z}$ for every $\mathbf{x} \in L$.
PROOF: Bilinearity and ⟨1⟩1 give $(\mathbf{t}, \mathbf{x}) = \sum_{\mathbf{u} \in T} a(\mathbf{u})(\mathbf{t}, \mathbf{u})$; every summand is even by ⟨1⟩, so the sum is even. □
⟨1⟩5. For $\mathbf{x} \in B$ and $\mathbf{t} \in T$, $(\mathbf{x}, \mathbf{t}) \in \{-4, -3, \ldots, 4\}$.
⟨2⟩1. $(\mathbf{x}, \mathbf{t})^2 \leq 16$.

PROOF: Cauchy–Schwarz in Inner products and square roots gives $(\mathbf{x},\mathbf{t})^2 \le q(\mathbf{x})q(\mathbf{t}) \le 4\cdot 4$, by the definition of B, $\mathbf{t}\in T$, and ⟨0⟩:2. □

⟨2⟩2. $(\mathbf{x},\mathbf{t})\in\mathbb{Z}$.

PROOF: Step ⟨1⟩3, since $\mathbf{x}\in L$ and $\mathbf{t}\in T\subseteq L$ by `Submodule.subset_span`. □

⟨2⟩3. Q.E.D.

PROOF: An integer whose square is at most 16 lies between -4 and 4; apply ⟨2⟩1 and ⟨2⟩2. □

⟨1⟩6. B is finite.

PROOF: The family indexed by T spans W by definition. Lemma (inner-product coordinates) makes $\mathbf{x}\mapsto((\mathbf{x},\mathbf{t}))_{\mathbf{t}\in T}$ injective on W, which contains B by ⟨1⟩2. Step ⟨1⟩5 puts its image in the finite function set $\{-4,-3,\dots,4\}^T$, since T is finite by ⟨0⟩:1. Apply `Set.Finite.of_injOn`. □

⟨1⟩7. Q.E.D.

PROOF: Steps ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, and ⟨1⟩6. For $T=\varnothing$ the same proof uses the empty coordinate family and gives $B=\{0\}$. □

3.2.3. *The explicit construction.*

Lemma (reflection adjointness). ASSUME: V is a real inner product space and $\mathbf{r}\in V\setminus\{0\}$.
PROVE: For all $\mathbf{x},\mathbf{y}\in V$, $(\tau_{\mathbf{r}}\mathbf{x},\mathbf{y})=(\mathbf{x},\tau_{\mathbf{r}}\mathbf{y})$.

PROOF SKETCH: We will use the metric identity and involution in Lemma (reflection identities).

⟨1⟩1. $(\tau_{\mathbf{r}}\mathbf{x},\mathbf{y})=(\mathbf{x},\tau_{\mathbf{r}}\mathbf{y})$ for all $\mathbf{x},\mathbf{y}\in V$.

PROOF: Apply the metric identity in Lemma (reflection identities) to $\mathbf{x},\tau_{\mathbf{r}}\mathbf{y}$; its involution identity changes the left side to $(\tau_{\mathbf{r}}\mathbf{x},\mathbf{y})$. Its nonzero hypothesis is ⟨0⟩. □

⟨1⟩2. Q.E.D.

PROOF: Step ⟨1⟩1. □

Lemma (integral proportional vectors). ASSUME: 1. V is a real inner product space, $\mathbf{r},\mathbf{s}\in V$, and $q(\mathbf{r}),q(\mathbf{s})\in\{2,4\}$.
2. $(\mathbf{r},\mathbf{s})\in\mathbb{Z}$ and $\mathbf{s}=t\mathbf{r}$ for $t\in\mathbb{R}$.
PROVE: $t=1$ or $t=-1$.

PROOF SKETCH: **[new]** We will use $t^2=1$ for equal lengths. For unequal lengths the integer inner product would have square 8, simplifying the ordinary draft's irrational-square-root argument.

⟨1⟩1. $q(\mathbf{s})=t^2q(\mathbf{r})$.

PROOF: Bilinearity and $\mathbf{s}=t\mathbf{r}$ in ⟨0⟩:2. □

⟨1⟩2. CASE: $q(\mathbf{r})=q(\mathbf{s})$.

PROVE: $t\in\{1,-1\}$.

PROOF: Divide ⟨1⟩1 by $q(\mathbf{r})>0$ and factor t^2-1. □

⟨1⟩3. $(\mathbf{r},\mathbf{s})^2=q(\mathbf{r})q(\mathbf{s})$.

PROOF: Bilinearity gives $(\mathbf{r},\mathbf{s})=tq(\mathbf{r})$. Square this equality and substitute ⟨1⟩1. □

⟨1⟩4. CASE: $q(\mathbf{r})\neq q(\mathbf{s})$.

PROVE: False.

⟨2⟩1. $(\mathbf{r},\mathbf{s})^2=8$.

PROOF: The two unequal members of $\{2,4\}$ in ⟨0⟩:1 have product 8; substitute in ⟨1⟩3. □

⟨2⟩2. PICK $z\in\mathbb{Z}$ with $(\mathbf{r},\mathbf{s})=z$.

PROOF: Integrality in ⟨0⟩:2. □

⟨2⟩3. Q.E.D.

PROOF: Steps ⟨2⟩1 and ⟨2⟩2 give $z^2=8$. Thus $4<|z|^2<9$, so $2<|z|<3$, impossible for an integer. □

⟨1⟩5. Q.E.D.

PROOF: Equality or inequality of the squared lengths gives ⟨1⟩2 or ⟨1⟩4. The unequal-length case is impossible. □

Proposition (the integer-span Root System). ASSUME: 1. V is a finite-dimensional real inner product space, $T \subseteq V$ is finite, and $\operatorname{span}_{\mathbb{R}} T = V$.
2. $q(\mathbf{t}) \in \{2, 4\}$ for every $\mathbf{t} \in T$.
3. $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for every $\mathbf{t}, \mathbf{u} \in T$.
4. If $\mathbf{t} \in T$ and $q(\mathbf{t}) = 4$, then $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for every $\mathbf{u} \in T$.
PROVE: IsRootSet($\Psi(T)$), $T \subseteq \Psi(T)$, and $q(\mathbf{r}) \in \{2, 4\}$ for every $\mathbf{r} \in \Psi(T)$.

PROOF SKETCH: **[repair]** This fills part of Gap 11. The explicit set in Definition (the integer-span construction) replaces a reflection closure, which would need stages, a union and a leastness argument. We will check each Root System requirement, keeping the evenness condition invariant under reflection.
LET: $L = \operatorname{span}_{\mathbb{Z}} T$ and $\Psi = \Psi(T)$.
⟨1⟩1. $\Psi \subseteq L$, every $\mathbf{r} \in \Psi$ has $q(\mathbf{r}) \in \{2, 4\}$, and $0 \notin \Psi$.
PROOF: Definition (the integer-span construction) and $q(0) = 0$. □
⟨1⟩2. $T \subseteq \Psi$.
⟨2⟩1. ASSUME: $\mathbf{t} \in T$ and $q(\mathbf{t}) = 2$.
PROVE: $\mathbf{t} \in \Psi$.
PROOF: By `Submodule.subset_span`, $T \subseteq L$. The membership and length in ⟨2⟩ give the first clause of Definition (the integer-span construction). □
⟨2⟩2. ASSUME: $\mathbf{t} \in T$ and $q(\mathbf{t}) = 4$.
PROVE: $\mathbf{t} \in \Psi$.
⟨3⟩1. $(\mathbf{t}, \mathbf{x}) \in 2\mathbb{Z}$ for every $\mathbf{x} \in L$.
PROOF: Lemma (a bounded part of an integer span), part 2, applied with ⟨0⟩:1, ⟨0⟩:2, and ⟨0⟩:3; the vector belongs to T and has squared norm 4 by ⟨2⟩, so ⟨0⟩:4 supplies its even pairings with T. □
⟨3⟩2. Q.E.D.
PROOF: The second clause of Definition (the integer-span construction) follows from $\mathbf{t} \in L$, the case length, and ⟨3⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Hypothesis ⟨0⟩:2 makes ⟨2⟩1 and ⟨2⟩2 exhaustive. □
⟨1⟩3. Ψ is finite.
PROOF: By ⟨1⟩1, it is contained in $\{\mathbf{x} \in L : q(\mathbf{x}) \leq 4\}$, finite by part 3 of Lemma (a bounded part of an integer span) with ⟨0⟩:1, ⟨0⟩:2, and ⟨0⟩:3. Apply `Set.Finite.subset`. □
⟨1⟩4. $\operatorname{span}_{\mathbb{R}} \Psi = V$.
PROOF: `Submodule.span_mono`, ⟨1⟩2, and ⟨0⟩:1. □
⟨1⟩5. For $\mathbf{r} \in \Psi$ and $\mathbf{x} \in L$, $2(\mathbf{r}, \mathbf{x})/q(\mathbf{r}) \in \mathbb{Z}$.
⟨2⟩1. CASE: $q(\mathbf{r}) = 2$.
PROOF: The coefficient is $(\mathbf{r}, \mathbf{x})$, an integer by Lemma (a bounded part of an integer span) and ⟨1⟩1. □
⟨2⟩2. CASE: $q(\mathbf{r}) = 4$.
PROOF: The coefficient is $(\mathbf{r}, \mathbf{x})/2$, an integer by the evenness clause of Definition (the integer-span construction). □
⟨2⟩3. Q.E.D.
PROOF: Step ⟨1⟩1 makes ⟨2⟩1 and ⟨2⟩2 exhaustive. □
⟨1⟩6. For $\mathbf{r} \in \Psi$ and $\mathbf{x} \in L$, $\tau_{\mathbf{r}}\mathbf{x} \in L$.
PROOF: Step ⟨1⟩5 gives an integer z equal to the reflection coefficient. By `Int.cast_smul_eq_zsmul`, the term $(z : \mathbb{R})\mathbf{r}$ is the integer multiple $z\mathbf{r}$. Step ⟨1⟩1 puts $\mathbf{r}$ in L; the integer submodule L is closed under integer multiples and subtraction. □
⟨1⟩7. For $\mathbf{r}, \mathbf{s} \in \Psi$, $\tau_{\mathbf{r}}\mathbf{s} \in \Psi$.

LET: $\mathbf{u} = \tau_{\mathbf{r}}\mathbf{s}$.
⟨2⟩1. $\mathbf{u} \in L$ and $q(\mathbf{u}) = q(\mathbf{s})$.
PROOF: Step ⟨1⟩6 and the metric identity of Lemma (reflection identities) at $\mathbf{s}, \mathbf{s}$, using the nonzero assertion in ⟨1⟩1. □
⟨2⟩2. CASE: $q(\mathbf{s}) = 2$.
PROVE: $\mathbf{u} \in \Psi$.
PROOF: The first clause of Definition (the integer-span construction), with ⟨2⟩1. □
⟨2⟩3. CASE: $q(\mathbf{s}) = 4$.
PROVE: $\mathbf{u} \in \Psi$.
⟨3⟩1. For every $\mathbf{x} \in L$, $(\mathbf{u}, \mathbf{x}) \in 2\mathbb{Z}$.
PROOF: Lemma (reflection adjointness) gives $(\mathbf{u}, \mathbf{x}) = (\mathbf{s}, \tau_{\mathbf{r}}\mathbf{x})$. This is even by the long-root clause for $\mathbf{s}$ and ⟨1⟩6. □
⟨3⟩2. Q.E.D.
PROOF: Apply the second clause of Definition (the integer-span construction) to ⟨2⟩1 and ⟨3⟩1. □
⟨2⟩4. Q.E.D.
PROOF: Step ⟨1⟩1 makes ⟨2⟩2 and ⟨2⟩3 exhaustive. □
⟨1⟩8. For $\mathbf{r}, t\mathbf{r} \in \Psi$, $t \in \{1, -1\}$.
PROOF: Apply Lemma (integral proportional vectors); its length hypotheses are ⟨1⟩1, and its inner-product hypothesis is part 1 of Lemma (a bounded part of an integer span), applied with ⟨0⟩:1, ⟨0⟩:2, and ⟨0⟩:3 and the inclusion $\Psi \subseteq L$ from ⟨1⟩1. □
⟨1⟩9. Q.E.D.
PROOF: Definition 3.9 has six requirements: ⟨1⟩3, ⟨1⟩1, ⟨1⟩4, ⟨1⟩5 restricted to $\Psi \subseteq L$, ⟨1⟩8, and ⟨1⟩7 supply them. The additional conclusions are ⟨1⟩2 and ⟨1⟩1. □

Lemma (completion of the seeds). ASSUME: 1. V is a finite-dimensional real inner product space and $T \subseteq V$ is finite.
2. $q(\mathbf{t}) \in \{2, 4\}$ for every $\mathbf{t} \in T$.
3. $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for every $\mathbf{t}, \mathbf{u} \in T$.
4. If $\mathbf{t} \in T$ and $q(\mathbf{t}) = 4$, then $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for every $\mathbf{u} \in T$.
PROVE: There is a finite set $T' \supseteq T$ spanning V with $q(\mathbf{t}) \in \{2, 4\}$ for $\mathbf{t} \in T'$, all inner products in T' integral, and $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ whenever $\mathbf{t}, \mathbf{u} \in T'$ and $q(\mathbf{t}) = 4$.

PROOF SKETCH: We will add scaled orthonormal vectors of the orthogonal complement. The diagonal inner product of a new vector is 2.
LET: $W = \operatorname{span}_{\mathbb{R}} T$.
⟨1⟩1. PICK a finite orthonormal basis $(\mathbf{e}_i)_{i \in I}$ of $W^{\perp}$, viewed in V.
PROOF: The orthogonal-complement facts in Orthogonal complements. □
LET: $\mathbf{u}_i = \sqrt{2}\mathbf{e}_i$, $E = \{\mathbf{u}_i : i \in I\}$, and $T' = T \cup E$.
⟨1⟩2. $q(\mathbf{u}_i) = 2$ for every $i \in I$.
PROOF: Orthonormality in ⟨1⟩1 and bilinearity with $(\sqrt{2})^2 = 2$. □
⟨1⟩3. All inner products in T' are integers.
⟨2⟩1. ASSUME: $\mathbf{t}, \mathbf{u} \in T$.
PROVE: $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$.
PROOF: Hypothesis ⟨0⟩:3. □
⟨2⟩2. ASSUME: $\mathbf{t} \in T$ and $i \in I$.
PROVE: $(\mathbf{t}, \mathbf{u}_i) = 0$.
PROOF: Here $\mathbf{t} \in W$ and $\mathbf{e}_i \in W^{\perp}$ by ⟨1⟩1; `Submodule.subset_span` gives the first membership. Apply `Submodule.inner_right_of_mem_orthogonal` and multiply the zero inner product by $\sqrt{2}$. □

⟨2⟩3. ASSUME: $\mathbf{t} \in T$ and $i \in I$.
PROVE: $(\mathbf{u}_i, \mathbf{t}) = 0$.
PROOF: Symmetry and ⟨2⟩2. □
⟨2⟩4. ASSUME: $i \in I$.
PROVE: $(\mathbf{u}_i, \mathbf{u}_i) = 2$.
PROOF: Step ⟨1⟩2. □
⟨2⟩5. ASSUME: $i, j \in I$ and $i \neq j$.
PROVE: $(\mathbf{u}_i, \mathbf{u}_j) = 0$.
PROOF: Apply `Orthonormal.inner_eq_zero` to ⟨1⟩1 and $i \neq j$ in ⟨2⟩; bilinearity preserves zero after scaling. □
⟨2⟩6. Q.E.D.
PROOF: Membership in $T \cup E$ gives the old/old, old/new, new/old, and new/new possibilities. The last splits into equal and distinct indices. The preceding five steps give integers in every case. □
⟨1⟩4. For $\mathbf{t}, \mathbf{u} \in T'$ with $q(\mathbf{t}) = 4$, $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$.
⟨2⟩1. $\mathbf{t} \in T$.
PROOF: Step ⟨1⟩2 excludes $\mathbf{t} \in E$; use $\mathbf{t} \in T \cup E$. □
⟨2⟩2. CASE: $\mathbf{u} \in T$.
PROOF: Hypothesis ⟨0⟩:4 and ⟨2⟩1. □
⟨2⟩3. CASE: $\mathbf{u} \in E$.
PROOF: Step ⟨2⟩1 and `Submodule.subset_span` put $\mathbf{t}$ in W. The definition of E puts $\mathbf{u}$ in $W^{\perp}$. The inner product is zero by `Submodule.inner_right_of_mem_orthogonal`. □
⟨2⟩4. Q.E.D.
PROOF: The two cases cover $\mathbf{u} \in T \cup E$. □
⟨1⟩5. $\operatorname{span}_{\mathbb{R}} T' = V$.
⟨2⟩1. $W \subseteq \operatorname{span}_{\mathbb{R}} T'$.
PROOF: `Submodule.span_mono` and $T \subseteq T'$. □
⟨2⟩2. $W^{\perp} \subseteq \operatorname{span}_{\mathbb{R}} T'$.
PROOF: Since $\sqrt{2} \neq 0$, each basis vector is $\mathbf{e}_i = (1/\sqrt{2})\mathbf{u}_i \in \operatorname{span}_{\mathbb{R}} T'$. The basis in ⟨1⟩1 spans $W^{\perp}$. □
⟨2⟩3. Q.E.D.
PROOF: The orthogonal decomposition $W + W^{\perp} = V$, with ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩6. Q.E.D.
PROOF: The set E is a finite image of I, so T' is finite and contains T. The lengths follow from ⟨0⟩:2 and ⟨1⟩2; the other properties are ⟨1⟩3, ⟨1⟩4, and ⟨1⟩5. The construction allows an empty basis. □

Lemma (seeds from a colouring). ASSUME: 1. V is a finite-dimensional real inner product space, and S is a set of subspaces of dimension one.
2. HasBipartition(S).
LET: $W = \operatorname{span}_{\mathbb{R}}(\bigcup_{\ell \in S} \ell)$.
PROVE: There is a finite set $T \subseteq V$ such that
1. $\mathcal{L}(T) = S$ and $\operatorname{span}_{\mathbb{R}} T = W$;
2. $q(\mathbf{t}) \in \{2, 4\}$ for every $\mathbf{t} \in T$;
3. $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for every $\mathbf{t}, \mathbf{u} \in T$; and
4. $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ whenever $\mathbf{t}, \mathbf{u} \in T$ and $q(\mathbf{t}) = 4$.

PROOF SKETCH: We will give the two colour classes squared lengths 2 and 4 and check each ordered pair of colours.
⟨1⟩1. PICK a compatible colouring $b : S \to \text{Bool}$.

PROOF: Hypothesis ⟨0⟩:2 and Definition (angles and compatible colourings). □

⟨1⟩2. HasAllowedAngles(S) and S is finite.

⟨2⟩1. ASSUME: $\ell, m \in S$, $\ell \neq m$ and $b(\ell) = b(m)$.
PROVE: There is $a \in \{0, 1/4, 1/2\}$ with HasSquaredCosine(ℓ, m, a).
PROOF: The equal-colour clause of ⟨1⟩1 gives a parameter in $\{0, 1/4\}$ with the required predicate. □

⟨2⟩2. ASSUME: $\ell, m \in S$, $\ell \neq m$ and $b(\ell) \neq b(m)$.
PROVE: There is $a \in \{0, 1/4, 1/2\}$ with HasSquaredCosine(ℓ, m, a).
PROOF: The different-colour clause of ⟨1⟩1 gives a parameter in $\{0, 1/2\}$ with the required predicate. □

⟨2⟩3. Q.E.D.
PROOF: Equality or inequality of the colours exhausts all pairs. Definition (angles and compatible colourings) gives allowed angles, and Lemma (finitely many lines) gives finiteness. □

⟨1⟩3. PICK $\mathbf{t}_\ell \in \ell$ for every $\ell \in S$, with $q(\mathbf{t}_\ell) = 2$ when $b(\ell) =$ false and $q(\mathbf{t}_\ell) = 4$ when $b(\ell) =$ true.
PROOF: Choose a nonzero generator by the dimension-one facts, then apply Lemma (representatives of a line) with $h = 2$ or $h = 4$. Choice on the empty set gives the empty family. □

LET: $T = \{\mathbf{t}_\ell : \ell \in S\}$.

⟨1⟩4. T is finite, $\mathcal{L}(T) = S$, and $\operatorname{span}_{\mathbb{R}} T = W$.

⟨2⟩1. T is finite.
PROOF: It is an image of the finite set in ⟨1⟩2. □

⟨2⟩2. $\mathbb{R}\mathbf{t}_\ell = \ell$ for every $\ell \in S$.
PROOF: Step ⟨1⟩3 gives a nonzero member; use the singleton-span facts in Spans and finite-dimensional linear algebra. □

⟨2⟩3. $\operatorname{span}_{\mathbb{R}} T \subseteq W$.
PROOF: Every seed belongs to its line in S by ⟨1⟩3; use the definition of W and `Submodule.span_mono`. □

⟨2⟩4. $W \subseteq \operatorname{span}_{\mathbb{R}} T$.
PROOF: Step ⟨2⟩2 puts every line in S in the span of T, by `Submodule.subset_span` and `Submodule.span_le`. Apply `Submodule.span_le` again to their union. □

⟨2⟩5. Q.E.D.
PROOF: Definition (lines of a vector set) and ⟨2⟩2 give $\mathcal{L}(T) = S$. The other assertions are ⟨2⟩1, ⟨2⟩3, and ⟨2⟩4. □

⟨1⟩5. For $\ell, m \in S$, $(\mathbf{t}_\ell, \mathbf{t}_m)$ is an integer, and is even if $q(\mathbf{t}_\ell) = 4$.
LET: $p = (\mathbf{t}_\ell, \mathbf{t}_m)$.

⟨2⟩1. CASE: $\ell = m$.
PROOF: Step ⟨1⟩3 gives $p = 2$ or $p = 4$; both are even integers. □

⟨2⟩2. CASE: $\ell \neq m$ and $b(\ell) = b(m) =$ false.

⟨3⟩1. $p^2 \in \{0, 1\}$.
PROOF: The norms in ⟨1⟩3 have product 4; substitute them in the equal-colour clause of ⟨1⟩1. □

⟨3⟩2. Q.E.D.
PROOF: Factoring the two possible square equations in ⟨3⟩1 gives $p \in \{0, 1, -1\}$. The evenness implication has false hypothesis, since $q(\mathbf{t}_\ell) = 2$. □

⟨2⟩3. CASE: $\ell \neq m$ and $b(\ell) = b(m) =$ true.

⟨3⟩1. $p^2 \in \{0, 4\}$.
PROOF: The norms in ⟨1⟩3 have product 16; use the equal-colour clause of ⟨1⟩1. □

⟨3⟩2. Q.E.D.
PROOF: Factoring the square equations in ⟨3⟩1 gives $p \in \{0, 2, -2\}$, all even integers. □

⟨2⟩4. CASE: $\ell \neq m$, $b(\ell) =$ false and $b(m) =$ true.
⟨3⟩1. $p^2 \in \{0, 4\}$.
PROOF: The norms have product 8 by ⟨1⟩3; use the different-colour clause of ⟨1⟩1. □
⟨3⟩2. Q.E.D.
PROOF: Factoring ⟨3⟩1 gives $p \in \{0, 2, -2\}$, all even integers. □
⟨2⟩5. CASE: $\ell \neq m$, $b(\ell) =$ true and $b(m) =$ false.
⟨3⟩1. $p^2 \in \{0, 4\}$.
PROOF: The norms have product 8 by ⟨1⟩3; use the different-colour clause of ⟨1⟩1. □
⟨3⟩2. Q.E.D.
PROOF: Factoring ⟨3⟩1 gives $p \in \{0, 2, -2\}$, all even integers. □
⟨2⟩6. Q.E.D.
PROOF: The diagonal and the four ordered Boolean pairs exhaust the possibilities, as in ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4, and ⟨2⟩5. □
⟨1⟩6. Q.E.D.
PROOF: Step ⟨1⟩4 supplies part 1 and finiteness; ⟨1⟩3 supplies part 2; and ⟨1⟩5 supplies parts 3 and 4 on the image T. □

Corollary (normalized containment). ASSUME: 1. V is a finite-dimensional real inner product space, and S is a set of subspaces of dimension one.
2. HasBipartition(S).
LET: $W = \operatorname{span}_{\mathbb{R}}(\bigcup_{\ell \in S} \ell)$, and let $\iota : W \to V$ be the inclusion.
PROVE: 1. There is $\Phi \subseteq V$ with IsRootSet(Φ), $S \subseteq \mathcal{L}(\Phi)$, and $q(\mathbf{r}) \in \{2, 4\}$ for $\mathbf{r} \in \Phi$.
2. There is $\Phi_W \subseteq W$ with IsRootSet(Φ_W) in the induced inner product on W, $S \subseteq \mathcal{L}(\iota(\Phi_W))$, and $q(\mathbf{r}) \in \{2, 4\}$ for $\mathbf{r} \in \Phi_W$.

PROOF SKETCH: We will apply the construction in V after completing seeds. For the span assertion we will apply the same proposition to the seeds regarded as vectors in the separate ambient type W.
⟨1⟩1. PICK a finite set T such that $\mathcal{L}(T) = S$, $\operatorname{span}_{\mathbb{R}} T = W$, each squared length in T is 2 or 4, all inner products in T are integers, and each length-4 vector in T pairs evenly with every vector in T.
PROOF: Lemma (seeds from a colouring) applies to ⟨0⟩:1 and ⟨0⟩:2. □
⟨1⟩2. There is a set Φ satisfying part 1.
⟨2⟩1. PICK a finite $T' \supseteq T$ spanning V, with every squared length 2 or 4, all inner products integral, and every length-4 vector pairing evenly with all of T'.
PROOF: Apply Lemma (completion of the seeds) to the conditions in ⟨1⟩1 and the finite dimension in ⟨0⟩:1. □
⟨2⟩2. $\Phi = \Psi(T')$ is a Root System in V, contains T', and has squared lengths 2 or 4.
PROOF: Proposition (the integer-span Root System) with ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Definition (lines of a vector set) sends $T \subseteq T' \subseteq \Phi$ to $S = \mathcal{L}(T) \subseteq \mathcal{L}(\Phi)$, by ⟨1⟩1 and ⟨2⟩2. □
⟨1⟩3. There is a set Φ_W satisfying part 2.
LET: $T_W = \{\mathbf{w} \in W : \iota\mathbf{w} \in T\}$.
⟨2⟩1. T_W is finite and $\iota(T_W) = T$.
PROOF: The inclusion is injective, and every member of T lies in W by ⟨1⟩1; it restricts to a bijection $T_W \to T$. Finiteness follows from `Set.Finite.preimage`, applied to the finite T and the injectivity of ι on T_W. □
⟨2⟩2. $\operatorname{span}_{\mathbb{R}} T_W = W$ as a subspace of W.

PROOF: Rewrite $W = \operatorname{span}_{\mathbb{R}} T$ using ⟨1⟩1. The assertion is `Submodule.span_span_coe_preimage` for the inclusion ι. □

⟨2⟩3. Every squared length in T_W is 2 or 4, all its inner products are integers, and each length-4 vector pairs evenly with all of T_W.
PROOF: The induced inner product on W is the ambient inner product; substitute ι in the three conditions for T in ⟨1⟩1. □

⟨2⟩4. $\Phi_W = \Psi(T_W)$ is a Root System in W, contains T_W, and has squared lengths 2 or 4.
PROOF: Apply Proposition (the integer-span Root System) in the finite-dimensional inner product space W, with ⟨2⟩1, ⟨2⟩2, and ⟨2⟩3. □

⟨2⟩5. Q.E.D.
PROOF: By ⟨2⟩1 and ⟨2⟩4, $T \subseteq \iota(\Phi_W)$. Definition (lines of a vector set) and ⟨1⟩1 give $S \subseteq \mathcal{L}(\iota(\Phi_W))$. □

⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩2 and ⟨1⟩3. □

3.2.4. *Lengths and the necessary colouring.*

Lemma (lengths on root lines). ASSUME: 1. V is a finite-dimensional real inner product space and IsRootSet(Φ).
2. $\mathbf{r}, \mathbf{s} \in \Phi$.
PROVE: 1. If $4(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$, then $q(\mathbf{r}) = q(\mathbf{s})$.
2. If $2(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$, then $q(\mathbf{s}) = 2q(\mathbf{r})$ or $q(\mathbf{r}) = 2q(\mathbf{s})$.

PROOF SKETCH: We will apply the pairwise conclusion of Lemma (angles and fixed-length subsystems). Only that conclusion is used; the fixed-length classification in the same lemma is not needed. We retain the division-free interfaces above.

⟨1⟩1. $q(\mathbf{r}), q(\mathbf{s}) > 0$.
PROOF: Nonzero roots in Definition 3.9, ⟨0⟩:2, and positive definiteness in Inner products and square roots. □

⟨1⟩2. ASSUME: $k \in \{2, 4\}$ and $k(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$.
PROVE: $\mathbf{r}, \mathbf{s}$ are neither perpendicular nor proportional.

⟨2⟩1. $(\mathbf{r},\mathbf{s}) \neq 0$.
PROOF: The right side of ⟨1⟩ is positive by ⟨1⟩1; its left side cannot vanish. □

⟨2⟩2. ASSUME: $\mathbf{s} = t\mathbf{r}$ for a real t.
PROVE: False.

⟨3⟩1. $(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$.
PROOF: Bilinearity in Inner products and square roots gives $(\mathbf{r},\mathbf{s}) = tq(\mathbf{r})$ and $q(\mathbf{s}) = t^2 q(\mathbf{r})$ from ⟨2⟩. □

⟨3⟩2. Q.E.D.
PROOF: Substitute ⟨3⟩1 into ⟨1⟩ and cancel $q(\mathbf{r})q(\mathbf{s}) > 0$ from ⟨1⟩1. This gives $k = 1$, contrary to $k \in \{2, 4\}$ in ⟨1⟩. □

⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2 give the two exclusions. □

⟨1⟩3. ASSUME: $4(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$.
PROVE: $q(\mathbf{r}) = q(\mathbf{s})$.

⟨2⟩1. The acute angle between their lines is 60°.
PROOF: Divide ⟨1⟩ by the positive product in ⟨1⟩1. The squared cosine is 1/4; use the Angle convention. □

⟨2⟩2. The two roots have equal length.

PROOF: Apply Lemma (angles and fixed-length subsystems) to ⟨0⟩:1, ⟨0⟩:2, and ⟨1⟩2 with $k = 4$ and ⟨1⟩. Step ⟨2⟩1 selects the length ratio 1. □

⟨2⟩3. Q.E.D.

PROOF: Square the equality of lengths in ⟨2⟩2, using Inner products and square roots. □

⟨1⟩4. ASSUME: $2(\mathbf{r}, \mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$.

PROVE: $q(\mathbf{s}) = 2q(\mathbf{r})$ or $q(\mathbf{r}) = 2q(\mathbf{s})$.

⟨2⟩1. The acute angle between their lines is 45°.

PROOF: Divide ⟨1⟩ by the positive product in ⟨1⟩1. The squared cosine is 1/2; use the Angle convention. □

⟨2⟩2. The longer-to-shorter length ratio is $\sqrt{2}$.

PROOF: Apply Lemma (angles and fixed-length subsystems) to ⟨0⟩:1, ⟨0⟩:2, and ⟨1⟩2 with $k = 2$ and ⟨1⟩. Step ⟨2⟩1 selects the length ratio $\sqrt{2}$. □

⟨2⟩3. Q.E.D.

PROOF: Square the ratio in ⟨2⟩2. The identity $(\sqrt{2})^2 = 2$ in Inner products and square roots gives the two stated alternatives according to which root is longer. □

⟨1⟩5. Q.E.D.

PROOF: Steps ⟨1⟩3 and ⟨1⟩4 give the two conclusions. □

Lemma (integer logarithm shifts). ASSUME: $x \in \mathbb{R}$ and $x > 0$.
LET: $\eta(y) = \mathrm{Int.log}(2, y)$ for real $y > 0$.
PROVE: $\eta(2x) = \eta(x) + 1$ and $\eta(x/2) = \eta(x) - 1$.

PROOF SKETCH: We will multiply the defining interval by 2, then apply the doubling identity at $x/2$.

⟨1⟩1. For every real $y > 0$, $\eta(2y) = \eta(y) + 1$.

⟨2⟩1. $2^{\eta(y)} \le y < 2^{\eta(y)+1}$.

PROOF: `Int.zpow_log_le_self` and `Int.lt_zpow_succ_log_self` with base $2 > 1$ and $y > 0$ in ⟨1⟩1. □

⟨2⟩2. $2^{\eta(y)+1} \le 2y < 2^{(\eta(y)+1)+1}$.

PROOF: Multiply ⟨2⟩1 by $2 > 0$ and apply $\texttt{zpow_add_one}_0$ at base $2 \neq 0$. □

⟨2⟩3. $\eta(y) + 1 \le \eta(2y)$.

PROOF: Apply `Int.zpow_le_iff_le_log` at $2y > 0$ to the lower bound of ⟨2⟩2. □

⟨2⟩4. $\eta(2y) < \eta(y) + 2$.

PROOF: Apply `Int.lt_zpow_iff_log_lt` at $2y > 0$ to the upper bound of ⟨2⟩2. □

⟨2⟩5. Q.E.D.

PROOF: The adjacent integer bounds in ⟨2⟩3 and ⟨2⟩4. □

⟨1⟩2. $\eta(x/2) = \eta(x) - 1$.

PROOF: Apply ⟨1⟩1 to $x/2 > 0$, using ⟨0⟩: $\eta(x) = \eta(2(x/2)) = \eta(x/2) + 1$. Rearrange this integer equality. □

⟨1⟩3. Q.E.D.

PROOF: Step ⟨1⟩1 at $x > 0$ and ⟨1⟩2. □

The two logarithm lemmas are declarations in namespace `Int`; they concern positive real numbers, independently of line systems.

Lemma (colouring selected root lines). ASSUME: 1. V is a finite-dimensional real inner product space, IsRootSet(Φ), and $S \subseteq \mathcal{L}(\Phi)$.
2. HasAllowedAngles(S).
PROVE: HasBipartition(S).

PROOF SKETCH: We will colour a line by the parity of the integer logarithm of its squared root length. Orthogonal pairs impose no constraint.

⟨1⟩1. PICK $\mathbf{r}_\ell \in \Phi$ with $\ell = \mathbb{R}\mathbf{r}_\ell$ for every $\ell \in S$.
PROOF: Containment in ⟨0⟩:1 and Definition (lines of a vector set) give the witnesses. □
LET: $\delta(\ell) = q(\mathbf{r}_\ell)$, $\eta(x) = \mathrm{Int.log}(2, x)$, and $b(\ell) =$ true exactly when $\eta(\delta(\ell))$ is odd.
⟨1⟩2. Every $\ell \in S$ has dimension one, and $\delta(\ell) > 0$.
⟨2⟩1. $\mathbf{r}_\ell \neq 0$ and $\delta(\ell) > 0$.
PROOF: Definition 3.9, ⟨1⟩1, and positive definiteness. □
⟨2⟩2. $\dim \ell = 1$.
PROOF: Apply `finrank_span_singleton` to ⟨2⟩1 and ⟨1⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩3. ASSUME: $\ell, m \in S$, $\ell \neq m$, and HasSquaredCosine$(\ell, m, 1/4)$.
PROVE: $b(\ell) = b(m)$.
⟨2⟩1. $\delta(\ell) = \delta(m)$.
PROOF: Lemma (squared cosine from representatives), applied to ⟨1⟩ and the nonzero representatives from ⟨1⟩1 and ⟨1⟩2, gives $4(\mathbf{r}_\ell, \mathbf{r}_m)^2 = \delta(\ell)\delta(m)$. Apply part 1 of Lemma (lengths on root lines) to these roots and the Root System in ⟨0⟩:1. □
⟨2⟩2. Q.E.D.
PROOF: Equal inputs in ⟨2⟩1 give equal integer logarithms and equal Boolean colours. □
⟨1⟩4. ASSUME: $\ell, m \in S$, $\ell \neq m$, and HasSquaredCosine$(\ell, m, 1/2)$.
PROVE: $b(\ell) \neq b(m)$.
⟨2⟩1. $\delta(m) = 2\delta(\ell)$ or $\delta(m) = \delta(\ell)/2$.
PROOF: Lemma (squared cosine from representatives), applied to ⟨1⟩ and the representatives in ⟨1⟩1 and ⟨1⟩2, gives $2(\mathbf{r}_\ell, \mathbf{r}_m)^2 = \delta(\ell)\delta(m)$. Part 2 of Lemma (lengths on root lines), using ⟨0⟩:1, gives $\delta(m) = 2\delta(\ell)$ or $\delta(\ell) = 2\delta(m)$. Divide the second equality by 2. □
⟨2⟩2. CASE: $\delta(m) = 2\delta(\ell)$.
PROVE: $b(\ell) \neq b(m)$.
⟨3⟩1. $\eta(\delta(m)) = \eta(\delta(\ell)) + 1$.
PROOF: The doubling identity in Lemma (integer logarithm shifts) and positivity in ⟨1⟩2. □
⟨3⟩2. Q.E.D.
PROOF: Consecutive integers have opposite parities by Integer arithmetic and logarithms; apply ⟨3⟩1 and the definition of b. □
⟨2⟩3. CASE: $\delta(m) = \delta(\ell)/2$.
PROVE: $b(\ell) \neq b(m)$.
⟨3⟩1. $\eta(\delta(m)) = \eta(\delta(\ell)) - 1$.
PROOF: The halving identity in Lemma (integer logarithm shifts) and positivity in ⟨1⟩2. □
⟨3⟩2. Q.E.D.
PROOF: Integers differing by 1 have opposite parities by Integer arithmetic and logarithms; apply ⟨3⟩1 and the definition of b. □
⟨2⟩4. Q.E.D.
PROOF: Step ⟨2⟩1 makes ⟨2⟩2 and ⟨2⟩3 exhaustive. □
⟨1⟩5. ASSUME: $\ell, m \in S$, $\ell \neq m$, and $b(\ell) = b(m)$.
PROVE: HasSquaredCosine$(\ell, m, 0)$ or HasSquaredCosine$(\ell, m, 1/4)$.
PROOF: Hypothesis ⟨0⟩:2 and Definition (angles and compatible colourings) give a parameter in $\{0, 1/4, 1/2\}$ satisfying the predicate. Step ⟨1⟩4 and the colour equality in ⟨1⟩ exclude the predicate at $1/2$. The other two parameters give the stated disjunction. □
⟨1⟩6. ASSUME: $\ell, m \in S$, $\ell \neq m$, and $b(\ell) \neq b(m)$.
PROVE: HasSquaredCosine$(\ell, m, 0)$ or HasSquaredCosine$(\ell, m, 1/2)$.

PROOF: Hypothesis ⟨0⟩:2 and Definition (angles and compatible colourings) give a parameter in $\{0, 1/4, 1/2\}$ satisfying the predicate. Step ⟨1⟩3 and the colour inequality in ⟨1⟩ exclude the predicate at $1/4$. The other two parameters give the stated disjunction. □

⟨1⟩7. Q.E.D.

PROOF: Steps ⟨1⟩5 and ⟨1⟩6 give the two clauses of Definition (angles and compatible colourings); b is the required witness. For $S = \varnothing$ it is the empty function. □

Theorem 3.12. ASSUME: 1. V is a finite-dimensional real inner product space. The set S consists of real subspaces of V, with $\dim \ell = 1$ for every $\ell \in S$.
2. HasAllowedAngles(S).

PROVE:

$$\big(\exists \Phi \subseteq V, \quad \text{IsRootSet}(\Phi) \text{ and } S \subseteq \mathcal{L}(\Phi)\big) \quad \Longleftrightarrow \quad \text{HasBipartition}(S).$$

PROOF SKETCH: **[repair]** The paper gives no proof (Gap 11). We will use the normalized construction for sufficiency and the integer-logarithm colouring for necessity.

⟨1⟩1. ASSUME: There is a Root System Φ in V with $S \subseteq \mathcal{L}(\Phi)$.
PROVE: HasBipartition(S).

PROOF: Lemma (colouring selected root lines) with the finite dimension in ⟨0⟩:1, the containment and Root System in ⟨1⟩, and the angle hypothesis ⟨0⟩:2. □

⟨1⟩2. ASSUME: HasBipartition(S).
PROVE: There is a Root System Φ in V with $S \subseteq \mathcal{L}(\Phi)$.

PROOF: Part 1 of Corollary (normalized containment), with ⟨0⟩:1 and ⟨1⟩. □

⟨1⟩3. Q.E.D.

PROOF: Steps ⟨1⟩1 and ⟨1⟩2 are the two implications. □

Conventions in the statement. **[repair]** A bipartition means the two fibres of a Boolean colouring; either fibre may be empty. This is required even for a single line, or for two lines at 60°, which must have the same colour. The ordinary proof records this as Implicit hypothesis 8 of `cgss/errata.md`. The word *and* between the two cross-part angles in the printed statement is read as *or* (Error 13, item 3). Finiteness is a conclusion of Lemma (finitely many lines); spanning and nonemptiness of S are not hypotheses. Corollary (normalized containment) separately states the stronger normalized construction, in both the span and the given ambient space. Allowing a containing Root System in a larger finite-dimensional inner product space gives the same condition: apply Lemma (colouring selected root lines) in that space to the isometric images of the selected lines and pull back its colouring.

The Lean declaration has argument `S:Set(SubmoduleRealV)`, not an argument of the bundled type `LineSystemV`, which permits only the 60° and 90° cases. Its ambient assumptions are `NormedAddCommGroupV`, `InnerProductSpaceRealV`, and `FiniteDimensionalRealV`; the line hypothesis is `Module.finrankReal` equal to 1 on each member of S. The docstring records the empty-fibre convention and the errata references above. The names in the metadata are the settled ones from `cgss/notation.md`, Section 10, supplemented by the separately stated helper lemmas.

3.2.5. *The bridge from Section 2.*

Lemma (the lines of a line system). ASSUME: V is a finite-dimensional real inner product space and R : LineSystem(V) is as in Definition (line systems of Section 2).
PROVE: HasAllowedAngles($\mathcal{L}(R.\Phi)$) and HasBipartition($\mathcal{L}(R.\Phi)$), witnessed by the constant colouring with value false.

PROOF SKETCH: We will exclude the two endpoint inner products on distinct lines and then use the constant colouring.

⟨1⟩1. Every $\ell \in \mathcal{L}(R.\Phi)$ has dimension one.
PROOF: A representative has squared norm 2 by Definition (line systems of Section 2), so is nonzero; apply `finrank_span_singleton` and Definition (lines of a vector set). □
⟨1⟩2. ASSUME: $\ell, m \in \mathcal{L}(R.\Phi)$ and $\ell \neq m$.
PROVE: HasSquaredCosine$(\ell, m, 0)$ or HasSquaredCosine$(\ell, m, 1/4)$.
⟨2⟩1. PICK $\mathbf{r}, \mathbf{s} \in R.\Phi$ with $\ell = \mathbb{R}\mathbf{r}$, $m = \mathbb{R}\mathbf{s}$.
PROOF: Definition (lines of a vector set). □
⟨2⟩2. ASSUME: $(\mathbf{r}, \mathbf{s}) = 2$.
PROVE: False.
⟨3⟩1. $q(\mathbf{r} - \mathbf{s}) = 0$ and $\mathbf{r} = \mathbf{s}$.
PROOF: Bilinearity and the squared norms in Definition (line systems of Section 2) give $q(\mathbf{r} - \mathbf{s}) = 2 + 2 - 2 \cdot 2 = 0$; positive definiteness gives the equality. □
⟨3⟩2. Q.E.D.
PROOF: Step ⟨3⟩1 makes the spans in ⟨2⟩1 equal, contrary to ⟨1⟩. □
⟨2⟩3. ASSUME: $(\mathbf{r}, \mathbf{s}) = -2$.
PROVE: False.
⟨3⟩1. $q(\mathbf{r} + \mathbf{s}) = 0$ and $\mathbf{r} = -\mathbf{s}$.
PROOF: Bilinearity gives $q(\mathbf{r} + \mathbf{s}) = 2 + 2 + 2(-2) = 0$; positive definiteness gives the equality. □
⟨3⟩2. Q.E.D.
PROOF: Multiplication by -1 does not change a real span; ⟨3⟩1 contradicts the distinct spans in ⟨1⟩. □
⟨2⟩4. $(\mathbf{r}, \mathbf{s}) \in \{-1, 0, 1\}$.
PROOF: Exclude the two endpoint values in Definition (line systems of Section 2) by ⟨2⟩2 and ⟨2⟩3. □
⟨2⟩5. Q.E.D.
PROOF: Step ⟨2⟩4 gives $(\mathbf{r}, \mathbf{s})^2 \in \{0, 1\}$. The squared norms are 2 by Definition (line systems of Section 2). Lemma (squared cosine from representatives), using ⟨1⟩1 and ⟨2⟩1, gives the predicate at 0 or $1/4$. □
⟨1⟩3. HasAllowedAngles$(\mathcal{L}(R.\Phi))$.
PROOF: Step ⟨1⟩2 supplies an allowed parameter in Definition (angles and compatible colourings). □
⟨1⟩4. The constant function $b(\ell) =$ false is compatible.
⟨2⟩1. For distinct ℓ, m, the equal-colour angle clause holds.
PROOF: Step ⟨1⟩2. □
⟨2⟩2. There are no ℓ, m with $b(\ell) \neq b(m)$.
PROOF: The definition of the constant function. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2 give the two clauses of Definition (angles and compatible colourings). □
⟨1⟩5. Q.E.D.
PROOF: Steps ⟨1⟩3 and ⟨1⟩4. □

The exact computational examples remain in the ordinary proof and its `blueprint/checks/thm312/check_thm312.py`. They are not premises of any result here. The construction and colouring need no two-length assertion for a reducible Root System. The colouring uses only the pairwise angle and length conclusion of Lemma (angles and fixed-length subsystems); that conclusion is proved there from integrality, before the application of classification to fixed-length

subsystems. The proof permits unrelated scales on orthogonal components, and restricts angles only on the selected set of lines.

4. Graphs with least eigenvalue -2

We will follow Section 4 of [CGSS]. The paper's graphs G and H are our graphs X and Y. All graphs are finite and simple. For a graph X we write $A = A(X)$ for its adjacency matrix, J for the all-ones matrix and $\lambda_{\min}(X)$ for its least eigenvalue; the least eigenvalue of X is at least -2 exactly when $2I + A$ is positive semidefinite, and we will state the results of this section in that form, as the Lean statements do. The coordinates $\mathbf{e}_1, \dots, \mathbf{e}_\ell$ are those of Definition 3.1. For a vector $\mathbf{x} = \sum_i \mathbf{x}(i)\mathbf{e}_i$ the *support* of $\mathbf{x}$ is $\operatorname{supp}(\mathbf{x}) = \{i : \mathbf{x}(i) \neq 0\}$. For a graph Y, the line graph $L(Y)$ has the edges of Y as its vertices, two of them adjacent when they are distinct and share an end.

Facts from Mathlib and the project. We will use the following declarations of Mathlib (commit `082e2d37`): `Matrix.posSemidef_iff_eq_sum_vecMulVec` (a positive semidefinite real matrix is a sum of matrices $\mathbf{v}\mathbf{v}^T$), `Matrix.posSemidef_gram` (a Gram matrix is positive semidefinite), `Matrix.rank_eq_finrank_span_cols`, `Submodule.map_span`, `LinearMap.finrank_range_of_inj`, `Submodule.finrank_le`, `stdOrthonormalBasis` with its field `OrthonormalBasis.repr`, `Matrix.det_gram_ne_zero_iff_linearIndependent`, `Matrix.posDef_gram_iff_linearIndependent`, `LinearIndependent.fintype_card_le_finrank`, `finrank_euclideanSpace_fin`, the definition `Matrix.rank` (the dimension of the range), `Submodule.span_mono`, `Submodule.finrank_mono`, `Submodule.finrank_sup_add_finrank_inf_eq`, `finrank_span_singleton`, `SimpleGraph.adjMatrix_mulVec_const_apply_of_regular`, for a real symmetric matrix its eigenvalues `Matrix.IsHermitian.eigenvalues` with `Matrix.IsHermitian.spectral_theorem`, `Matrix.IsHermitian.trace_eq_sum_eigenvalues` and `Matrix.IsHermitian.rank_eq_card_non_zero_eigs`, `Matrix.trace_mul_comm`, the Cauchy–Schwarz inequality `Finset.sum_mul_sq_le_sq_mul_sq`, the predicate `SimpleGraph.IsBipartite` (a 2-colouring) with `SimpleGraph.two_colorable_iff_forall_loop_even` and `SimpleGraph.Coloring.even_length_iff_congr`, and the structure `SimpleGraph.IsSRGWith`. From the project we use the bridge `SimpleGraph.posSemidef_two_add_adjMatrix_iff` ($2I + A$ is positive semidefinite exactly when every eigenvalue of X is at least -2) and `SimpleGraph.leastEigenvalue_eq_neg_two_iff` (for a graph with a vertex, $\lambda_{\min}(X) = -2$ exactly when $2I + A$ is positive semidefinite and singular), both in `LineGraphs/Spectral/Positivity.lean`; and the sets $N_{\mathbf{r}} = \{\mathbf{x} \in \Phi : (\mathbf{x}, \mathbf{r}) = 1\}$ of `LineGraphs/Computations/NeighborCertificates.lean`.

4.1. **Theorem 4.1.**

Definition (representations). Assume: X is a graph, V is a real inner product space and S is a line system in V.

Let: A *representation* of X in V is a family $(\mathbf{x}_u)_{u \in V(X)}$ of vectors of V whose Gram matrix is $2I + A(X)$; that is, $(\mathbf{x}_u, \mathbf{x}_u) = 2$, $(\mathbf{x}_u, \mathbf{x}_w) = 1$ when u and w are adjacent, and $(\mathbf{x}_u, \mathbf{x}_w) = 0$ when $u \neq w$ are not adjacent. The graph X is *represented in* S if it has a representation with $\mathbf{x}_u \in \Phi(S)$ for every u.

[new] The paper speaks of a set of vectors and of the graph it represents; we will index the vectors by the vertices, so that the graph is given first, as in Lean. For $S = \mathrm{E}_8$ we take model (ii) of Definition 3.2 in $\mathbb{R}^8$.

Lemma (Gram factorization). ASSUME: M is a real symmetric matrix indexed by a finite set V, and $r \geq 0$.
PROVE: 1. If M is the Gram matrix of a family $(\mathbf{x}_u)_{u \in V}$ in a real inner product space, then $\operatorname{rk} M$ is the dimension of the span of the $\mathbf{x}_u$.
2. If M is positive semidefinite and $\operatorname{rk} M \leq r$, then M is the Gram matrix of a family in $\mathbb{R}^r$, that is, in `EuclideanSpace ℝ (Fin r)`.

[new] Mathlib factors a positive semidefinite matrix through a sum of an unspecified number of terms; the sections below need the dimension.

PROOF SKETCH: We will compare the columns of M with the span, and then move the span into $\mathbb{R}^r$ by an isometry.

⟨1⟩1. Assertion 1 holds.
PROOF: Let W be the span of the $\mathbf{x}_u$ and $T\mathbf{y} = ((\mathbf{x}_w, \mathbf{y}))_{w \in V}$ for $\mathbf{y} \in W$. The column u of M is $T\mathbf{x}_u$, so the span of the columns is $T(W)$, and T is injective on W, since $T\mathbf{y} = 0$ gives $(\mathbf{y}, \mathbf{y}) = 0$. So $\dim T(W) = \dim W$ by `Submodule.map_span` and `LinearMap.finrank_range_of_inj`; apply `Matrix.rank_eq_finrank_span_cols`. □

⟨1⟩2. ASSUME: M is positive semidefinite.
PROVE: M is the Gram matrix of a family $(\mathbf{y}_u)$ in $\mathbb{R}^m$ for some m.
PROOF: By `Matrix.posSemidef_iff_eq_sum_vecMulVec`, $M = \sum_{k=1}^{m} \mathbf{v}_k \mathbf{v}_k^T$ for some m and vectors $\mathbf{v}_k \in \mathbb{R}^V$. Put $\mathbf{y}_u = \sum_k \mathbf{v}_k(u)\mathbf{e}_k$; then $(\mathbf{y}_u, \mathbf{y}_w) = \sum_k \mathbf{v}_k(u)\mathbf{v}_k(w)$ is the uw-entry of M. □

⟨1⟩3. ASSUME: M is positive semidefinite.
PROVE: M is the Gram matrix of a family in $\mathbb{R}^{\dim W}$, where W is the span of the $\mathbf{y}_u$ of ⟨1⟩2.
PROOF: The coordinates in the orthonormal basis `stdOrthonormalBasis` of W, that is, its `OrthonormalBasis.repr`, form a linear isometry of W onto $\mathbb{R}^{\dim W}$; they preserve inner products. □

⟨1⟩4. ASSUME: $k \leq r$.
PROVE: Padding with $r-k$ zero coordinates maps $\mathbb{R}^k$ into $\mathbb{R}^r$ and preserves inner products.
PROOF: The padded coordinates contribute 0 to each inner product. □

⟨1⟩5. Q.E.D.
PROOF: Assertion 1 is ⟨1⟩1. For assertion 2, $\dim W = \operatorname{rk} M \leq r$ by ⟨1⟩1; apply ⟨1⟩3 and then ⟨1⟩4 with $k = \dim W$. □

Lemma (the Gram matrix). ASSUME: X is a graph on n vertices.
PROVE: 1. X has a representation in $\mathbb{R}^n$ if and only if $2I + A(X)$ is positive semidefinite, that is, if and only if every eigenvalue of X is at least -2.
2. If $n \geq 1$ and X has a representation in a space of dimension less than n, then $\lambda_{\min}(X) = -2$.

PROOF SKETCH: We will factor the matrix into a Gram matrix, and then count independent vectors.

⟨1⟩1. ASSUME: $(\mathbf{x}_u)$ is a representation of X in a real inner product space V.
PROVE: $2I + A(X)$ is positive semidefinite.
PROOF: $2I + A(X)$ is the Gram matrix of $(\mathbf{x}_u)$ by ⟨1⟩ and Definition (representations); apply `Matrix.posSemidef_gram`. □

⟨1⟩2. ASSUME: $2I + A(X)$ is positive semidefinite.
PROVE: X has a representation in $\mathbb{R}^n$.
PROOF: $2I + A(X)$ has rank at most n; apply Lemma (Gram factorization), assertion 2, with $r = n$, and Definition (representations). □

⟨1⟩3. Assertion 1 holds.

PROOF: ⟨1⟩1 and ⟨1⟩2 give the first equivalence, and `SimpleGraph.posSemidef_two_add_adjMatrix_iff` the second. □

⟨1⟩4. ASSUME: $n \geq 1$ and $(\mathbf{x}_u)$ is a representation of X in a space V of dimension less than n.
PROVE: $\det(2I + A(X)) = 0$.
PROOF: The family $(\mathbf{x}_u)$ has n members in a space of dimension less than n, so it is not linearly independent by `LinearIndependent.fintype_card_le_finrank`. Its Gram matrix $2I + A(X)$ is singular by `Matrix.det_gram_ne_zero_iff_linearIndependent`. □

⟨1⟩5. Q.E.D.
PROOF: Assertion 1 is ⟨1⟩3. For assertion 2, $2I + A(X)$ is positive semidefinite by ⟨1⟩1 and singular by ⟨1⟩4; apply `SimpleGraph.least Eigenvalue_eq_neg_two_iff`. □

Lemma (the lines of a representation). ASSUME: $(\mathbf{x}_u)_{u \in V(X)}$ is a representation of a graph X in a finite-dimensional real inner product space V, and $\Phi = \{\pm \mathbf{x}_u : u \in V(X)\}$.
PROVE: 1. $u \mapsto \mathbf{x}_u$ is injective, and $\mathbf{x}_u \neq -\mathbf{x}_w$ for all u, w.
2. Φ is a line system $S_{\mathbf{x}}$ in V, with $|V(X)|$ lines.
3. If X is connected, then $S_{\mathbf{x}}$ is indecomposable.

PROOF SKETCH: We will compare inner products with the squared norm 2.

⟨1⟩1. Assertion 1 holds.
PROOF: For $u \neq w$, $(\mathbf{x}_u, \mathbf{x}_w) \in \{0, 1\}$ by Definition (representations), while $(\mathbf{x}_u, \mathbf{x}_u) = 2$ and $(\mathbf{x}_u, -\mathbf{x}_u) = -2$; so $\mathbf{x}_w \neq \mathbf{x}_u$ and $\mathbf{x}_w \neq -\mathbf{x}_u$. For $u = w$, $\mathbf{x}_u = -\mathbf{x}_u$ would give $\mathbf{x}_u = 0$, against $(\mathbf{x}_u, \mathbf{x}_u) = 2$. □

⟨1⟩2. Assertion 2 holds.
PROOF: $\Phi = -\Phi$, every member has squared norm 2, and the inner products of members are $\pm$ those of the $\mathbf{x}_u$, which lie in $\{0, 1, 2\}$; this is Definition 2. The lines are the pairs $\{\mathbf{x}_u, -\mathbf{x}_u\}$, which are distinct by ⟨1⟩1. □

⟨1⟩3. Q.E.D.
PROOF: Take $B = \{\mathbf{x}_u\}$ in Lemma (a connected spanning set): B spans the span of Φ, and distinct members of B have nonzero inner product exactly when their vertices are adjacent, so the graph of that lemma is a copy of X, which is connected. □

Lemma (transport of representations). ASSUME: S and T are line systems, $f : \Phi(S) \to \Phi(T)$ preserves inner products (for instance an isomorphism, or the inclusion of $\Phi(S)$ in $\Phi(T)$ when $\Phi(S) \subseteq \Phi(T)$), and X is represented in S by $(\mathbf{x}_u)$.
PROVE: X is represented in T by $(f(\mathbf{x}_u))$.

PROOF SKETCH: The Gram matrix does not change.

⟨1⟩1. Q.E.D.
PROOF: $(f(\mathbf{x}_u), f(\mathbf{x}_w)) = (\mathbf{x}_u, \mathbf{x}_w)$ for all u, w by ⟨0⟩, and $f(\mathbf{x}_u) \in \Phi(T)$; apply Definition (representations). □

Lemma (subfamilies of a representation). ASSUME: $(\mathbf{x}_u)_{u \in V(X)}$ is a representation of a graph X in V, and $U \subseteq V(X)$.
PROVE: $(\mathbf{x}_u)_{u \in U}$ is a representation in V of the subgraph $X[U]$ induced on U.

PROOF SKETCH: The Gram matrix of the subfamily is a principal submatrix.

⟨1⟩1. Q.E.D.
PROOF: Two vertices of U are adjacent in $X[U]$ exactly when they are adjacent in X, so $(\mathbf{x}_u, \mathbf{x}_w)$ for $u, w \in U$ is the uw-entry of $2I + A(X[U])$; apply Definition (representations). □

Lemma (a root system containing the lines). ASSUME: X is a connected graph with a representation $(\mathbf{x}_u)$ in a finite-dimensional real inner product space V, and the span of the $\mathbf{x}_u$ has dimension k.

PROVE: 1. X is represented in the model of one of A_k, D_k ($k \geq 4$), E_6, E_7 and E_8, and the rank of that model is k.
2. X is represented in A_ℓ for some $\ell \geq 1$, in D_ℓ for some $\ell \geq 4$, or in E_8.

PROOF SKETCH: This is the paper's "Such a set of lines is contained in a root system. If the graph is connected, then the root system is indecomposable." **[repair]** The paper does not say why a set that is not maximal lies in an indecomposable star-closed set of the same dimension; we will use the fill of Gap 13 (errata), which is Lemma (a maximal extension in the span). We will then place E_6 and E_7 inside E_8. The first of the two sentences holds without connectedness: the vectors of squared norm 2 in the integral span of the $\mathbf{x}_u$ form a star-closed line system in which X is represented. We use this system for Remark 4.14; in Lean it is `SimpleGraph.IsRepresentation.latticeRoots`, with the lemmas `isStarClosed_latticeRoots` and `isRepresentedIn_latticeRoots` in the same namespace. The claim of the introduction of [CGSS], that the lines lie in one of the root systems, needs connectedness (errata, Implicit hypothesis 1).

LET: $S = S_{\mathbf{x}}$ be the line system of Lemma (the lines of a representation) and W the span of $\Phi(S)$, which has dimension k.

⟨1⟩1. S is nonempty and indecomposable.
PROOF: A connected graph has a vertex, and Lemma (the lines of a representation), assertion 3, applies by ⟨0⟩. □

⟨1⟩2. There is an indecomposable line system T in W, maximal in W and spanning W, with $\Phi(S) \subseteq \Phi(T)$.
⟨2⟩1. CASE: S is maximal in W.
PROVE: $T = S$ will do.
PROOF: S is indecomposable by ⟨1⟩1, maximal by ⟨2⟩, and spans W by the definition of W. □
⟨2⟩2. CASE: S is not maximal in W.
PROVE: Such a T exists.
PROOF: Apply Lemma (a maximal extension in the span) to S, with ⟨1⟩1 and ⟨2⟩. □
⟨2⟩3. Q.E.D.
PROOF: The two cases are exhaustive. □

⟨1⟩3. T is isomorphic to the model of A_ℓ ($\ell \geq 1$), D_ℓ ($\ell \geq 4$), E_6, E_7 or E_8, of rank k.
PROOF: Apply Theorem 3.6 to T of ⟨1⟩2; its list is part of this one. The isomorphism preserves rank by Lemma (transport on spans), and T has rank k since it spans W. □

⟨1⟩4. X is represented in T by $(\mathbf{x}_u)$.
PROOF: Lemma (transport of representations), with the inclusion $\Phi(S) \subseteq \Phi(T)$ of ⟨1⟩2. □

⟨1⟩5. Assertion 1 holds.
PROOF: Apply Lemma (transport of representations) to ⟨1⟩4 and the isomorphism of ⟨1⟩3. The model of A_ℓ or D_ℓ has rank ℓ by Lemma (spans of the models), so $\ell = k$. □

⟨1⟩6. $\Phi(\mathrm{E}_6) \subseteq \Phi(\mathrm{E}_8)$ and $\Phi(\mathrm{E}_7) \subseteq \Phi(\mathrm{E}_8)$.
PROOF: Definition 3.3. □

⟨1⟩7. Q.E.D.
PROOF: Assertion 1 is ⟨1⟩5. For assertion 2, in the cases E_6 and E_7 apply Lemma (transport of representations) once more to the inclusions of ⟨1⟩6. □

Lemma (the line graph). ASSUME: Y is a graph with vertex set $\{1, \ldots, n\}$.
PROVE: The vectors $\mathbf{e}_i + \mathbf{e}_j$ of $\mathbb{R}^n$, for the edges $\{i, j\}$ of Y, form a representation of $L(Y)$ in $\mathbb{R}^n$.

PROOF SKETCH: The inner product counts common ends.

⟨1⟩1. For edges e, f of Y, $(\sum_{i \in e} \mathbf{e}_i, \sum_{j \in f} \mathbf{e}_j) = |e \cap f|$.

PROOF: Expand; $(\mathbf{e}_i, \mathbf{e}_j)$ is 1 if $i = j$ and 0 otherwise. □

⟨1⟩2. Q.E.D.

PROOF: $|e \cap e| = 2$; for $e \neq f$, $|e \cap f|$ is 1 when e and f share an end, that is, when they are adjacent in $L(Y)$, and 0 otherwise. Apply ⟨1⟩1. □

Theorem 4.1. ASSUME: X is a graph and $n \geq 1$.

PROVE: X is represented in A_n if and only if $X \cong L(Y)$ for a bipartite graph Y on the vertex set $\{1, \ldots, n+1\}$.

[new] The paper's "any such graph can be represented by a subset of A_n" needs Y to have at most $n+1$ vertices; we will state it with that bound, as the Lean statement does.

PROOF SKETCH: We will read off Y from the supports of the vectors, and show that each coordinate has one sign.

⟨1⟩1. ASSUME: X is represented in A_n by $(\mathbf{x}_u)$.
PROVE: $X \cong L(Y)$ for a bipartite graph Y on $\{1, \ldots, n+1\}$.

⟨2⟩1. PICK $p(u) \neq q(u)$ in $\{1, \ldots, n+1\}$ with $\mathbf{x}_u = \mathbf{e}_{p(u)} - \mathbf{e}_{q(u)}$ for each vertex u.

PROOF: $\mathbf{x}_u \in \Phi(\mathrm{A}_n)$ by ⟨1⟩; apply Definition 3.1. □

⟨2⟩2. For all vertices u, w, $p(u) \neq q(w)$.

PROOF: If $p(u) = q(w) = i$, then $u \neq w$, and $(\mathbf{x}_u, \mathbf{x}_w) = (\mathbf{e}_i - \mathbf{e}_{q(u)}, \mathbf{e}_{p(w)} - \mathbf{e}_i) = -1 - [q(u) = p(w)] < 0$, since $p(w) \neq i$ and $q(u) \neq i$; this contradicts Definition (representations). □

⟨2⟩3. For $u \neq w$, $(\mathbf{x}_u, \mathbf{x}_w) = [p(u) = p(w)] + [q(u) = q(w)]$.

PROOF: Expand $(\mathbf{e}_{p(u)} - \mathbf{e}_{q(u)}, \mathbf{e}_{p(w)} - \mathbf{e}_{q(w)})$; the two cross terms vanish by ⟨2⟩2. □

⟨2⟩4. The supports $\{p(u), q(u)\}$ of distinct vertices are distinct.

PROOF: If $u \neq w$ have the same support, then $p(u) = p(w)$ and $q(u) = q(w)$ by ⟨2⟩2, and $(\mathbf{x}_u, \mathbf{x}_w) = 2$ by ⟨2⟩3, against Definition (representations). □

LET: Y be the graph on $\{1, \ldots, n+1\}$ with edges $\{p(u), q(u)\}$, and $P = \{p(u) : u \in V(X)\}$.

⟨2⟩5. Y is bipartite, with parts P and its complement.

PROOF: Each edge $\{p(u), q(u)\}$ has one end $p(u)$ in P and the other end $q(u)$ outside P, by ⟨2⟩2. □

⟨2⟩6. $u \mapsto \{p(u), q(u)\}$ is an isomorphism from X to $L(Y)$.

PROOF: It is a bijection onto the edges of Y by ⟨2⟩4. For $u \neq w$, the edges share an end exactly when $p(u) = p(w)$ or $q(u) = q(w)$, by ⟨2⟩2, that is, when $(\mathbf{x}_u, \mathbf{x}_w) = 1$ by ⟨2⟩3, which is adjacency of u and w. □

⟨2⟩7. Q.E.D.

PROOF: ⟨2⟩5 and ⟨2⟩6. □

⟨1⟩2. ASSUME: Y is bipartite on $\{1, \ldots, n+1\}$, with parts P and Q, and $X \cong L(Y)$.
PROVE: X is represented in A_n.

LET: $\mathbf{x}_e = \mathbf{e}_p - \mathbf{e}_q$ for the edge $e = \{p, q\}$ of Y with $p \in P$ and $q \in Q$.

⟨2⟩1. For edges $e \neq f$, $(\mathbf{x}_e, \mathbf{x}_f)$ is 1 when e and f share an end, and 0 otherwise; and $(\mathbf{x}_e, \mathbf{x}_e) = 2$.

PROOF: Write $e = \{p, q\}$ and $f = \{p', q'\}$ with $p, p' \in P$. Then $p \neq q'$ and $q \neq p'$, since $P \cap Q = \varnothing$, and $(\mathbf{x}_e, \mathbf{x}_f) = [p = p'] + [q = q']$, which is 2 for $e = f$ and at most 1 for $e \neq f$. □

⟨2⟩2. Q.E.D.

PROOF: $\mathbf{x}_e \in \Phi(\mathrm{A}_n)$ by Definition 3.1, and $(\mathbf{x}_e)$ represents $L(Y)$ by ⟨2⟩1; compose with the isomorphism of ⟨1⟩. □

⟨1⟩3. Q.E.D.

PROOF: ⟨1⟩1 and ⟨1⟩2. □

Remark (after Theorem 4.1). ASSUME: $n \geq 1$ and $N = n + 1$. A set $\Sigma \subseteq \Phi(\mathrm{A}_n)$ has *non-negative inner products* if $(\mathbf{x}, \mathbf{y}) \geq 0$ for all $\mathbf{x}, \mathbf{y} \in \Sigma$, and is *maximal* if no subset of

$\Phi(\mathrm{A}_n)$ that properly contains it has nonnegative inner products. For $P \subseteq \{1,\dots,N\}$ let $\Sigma_P = \{\mathbf{e}_p - \mathbf{e}_q : p \in P,\ q \notin P\}$.

PROVE: 1. A set Σ with nonnegative inner products is maximal if and only if $\Sigma = \Sigma_P$ for some P with $1 \le |P| \le n$.
2. For such P, with $m = |P|$, the vectors $\mathbf{e}_p - \mathbf{e}_q$, indexed by the edges $\{p,q\}$ of the complete bipartite graph Y_P with parts P and $\{1,\dots,N\} \setminus P$, represent $L(Y_P) \cong L(K_{m,N-m})$.
3. Every Σ with nonnegative inner products has $|\Sigma| \le \lfloor \frac{1}{4}N^2 \rfloor$, and $|\Sigma_P| = \lfloor \frac{1}{4}N^2 \rfloor$ for $|P| = \lfloor \frac{1}{2}N \rfloor$.

PROOF SKETCH: **[new]** The paper states the remark without proof. We will show, as in the proof of Theorem 4.1, that each coordinate occurs with one sign in Σ, and then count.

LET: $\mathbf{x} = \mathbf{e}_{p(\mathbf{x})} - \mathbf{e}_{q(\mathbf{x})}$ for $\mathbf{x} \in \Phi(\mathrm{A}_n)$, by Definition 3.1.

⟨1⟩1. ASSUME: $P \subseteq \{1,\dots,N\}$, $p,p' \in P$ and $q,q' \notin P$.
PROVE: $(\mathbf{e}_p - \mathbf{e}_q, \mathbf{e}_{p'} - \mathbf{e}_{q'}) = [p=p'] + [q=q']$; in particular Σ_P has nonnegative inner products.
PROOF: Expand; the cross terms vanish since $p \ne q'$ and $q \ne p'$. □

⟨1⟩2. ASSUME: Σ has nonnegative inner products.
PROVE: $\Sigma \subseteq \Sigma_P$ for $P = \{p(\mathbf{x}) : \mathbf{x} \in \Sigma\}$.
PROOF: We need $q(\mathbf{x}) \notin P$ for $\mathbf{x} \in \Sigma$. If $q(\mathbf{x}) = p(\mathbf{y}) = i$ with $\mathbf{y} \in \Sigma$, then $p(\mathbf{x}) \ne i$ and $q(\mathbf{y}) \ne i$, so $(\mathbf{x},\mathbf{y}) = (\mathbf{e}_{p(\mathbf{x})} - \mathbf{e}_i, \mathbf{e}_i - \mathbf{e}_{q(\mathbf{y})}) = -1 - [p(\mathbf{x}) = q(\mathbf{y})] < 0$, against ⟨1⟩. □

⟨1⟩3. ASSUME: Σ is maximal with nonnegative inner products.
PROVE: $\Sigma = \Sigma_P$ for some P with $1 \le |P| \le n$.
⟨2⟩1. Σ is nonempty.
PROOF: $\{\mathbf{e}_1 - \mathbf{e}_2\}$ has nonnegative inner products, as $N \ge 2$, and properly contains $\varnothing$; apply ⟨1⟩. □
⟨2⟩2. $\Sigma = \Sigma_P$ for $P = \{p(\mathbf{x}) : \mathbf{x} \in \Sigma\}$.
PROOF: $\Sigma \subseteq \Sigma_P$ by ⟨1⟩2, and Σ_P has nonnegative inner products by ⟨1⟩1; apply ⟨1⟩. □
⟨2⟩3. Q.E.D.
PROOF: By ⟨2⟩1, P of ⟨2⟩2 is nonempty, and its complement contains $q(\mathbf{x})$ for $\mathbf{x} \in \Sigma = \Sigma_P$; so $1 \le |P| \le N-1 = n$. □

⟨1⟩4. ASSUME: $1 \le |P| \le n$.
PROVE: Σ_P is maximal with nonnegative inner products.
PROOF: Σ_P has nonnegative inner products by ⟨1⟩1. A root $\mathbf{z} = \mathbf{e}_a - \mathbf{e}_b \notin \Sigma_P$ has $b \in P$ or $a \notin P$; pick $p \in P$ and $q \notin P$, by ⟨1⟩. If $b \in P$, then $(\mathbf{z}, \mathbf{e}_b - \mathbf{e}_q) = -1 - [a=q] < 0$; if $a \notin P$, then $(\mathbf{z}, \mathbf{e}_p - \mathbf{e}_a) = -1 - [b=p] < 0$. Thus no larger set has nonnegative inner products. □

⟨1⟩5. Assertion 2 holds.
PROOF: For edges $e = \{p,q\}$ and $f = \{p',q'\}$ of Y_P with $p,p' \in P$, ⟨1⟩1 gives the inner product $[p=p'] + [q=q'] = |e \cap f|$, which is 2 for $e = f$, and for $e \ne f$ is 1 when e and f share an end and 0 otherwise; apply Definition (representations). $Y_P \cong K_{m,N-m}$. □

⟨1⟩6. $|\Sigma_P| = |P|(N - |P|)$.
PROOF: $(p,q) \mapsto \mathbf{e}_p - \mathbf{e}_q$ is injective on $P \times (\{1,\dots,N\} \setminus P)$. □

⟨1⟩7. ASSUME: Σ has nonnegative inner products.
PROVE: $|\Sigma| \le \lfloor \frac{1}{4}N^2 \rfloor$.
LET: P be as in ⟨1⟩2, and $m = |P|$.
⟨2⟩1. $|\Sigma| \le m(N-m)$.
PROOF: $\Sigma \subseteq \Sigma_P$ by ⟨1⟩ and ⟨1⟩2, and $|\Sigma_P| = m(N-m)$ by ⟨1⟩6. □
⟨2⟩2. $m(N-m) \le \lfloor \frac{1}{4}N^2 \rfloor$.
PROOF: $4m(N-m) = N^2 - (N-2m)^2 \le N^2$, and $m(N-m)$ is an integer. □

⟨2⟩3. Q.E.D.
PROOF: ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩8. ASSUME: $m = \lfloor \frac{1}{2}N \rfloor$.
PROVE: $m(N-m) = \lfloor \frac{1}{4}N^2 \rfloor$.
PROOF: For $N = 2k$ both sides are k^2; for $N = 2k+1$ both are $k^2 + k$. □
⟨1⟩9. Q.E.D.
PROOF: Assertion 1 is ⟨1⟩3 and ⟨1⟩4, and assertion 2 is ⟨1⟩5. Assertion 3 is ⟨1⟩7, with ⟨1⟩6 and ⟨1⟩8 for $|P| = \lfloor \frac{1}{2}N \rfloor$, which lies between 1 and n as $N \geq 2$. □

The paper states assertions 1 to 3 in the form "The maximal subsets of A_n with nonnegative inner products represent the graphs $L(K_{m,n+1-m})$, and have maximum cardinality $\lfloor \frac{1}{4}(n+1)^2 \rfloor$, attained for $m = \lfloor \frac{1}{2}(n+1) \rfloor$." A set Σ with nonnegative inner products lies in Σ_P for $P = \{p(\mathbf{x}) : \mathbf{x} \in \Sigma\}$, by the second step of the proof. Thus, by assertion 2 and Lemma (subfamilies of a representation), the graph that Σ represents is $L(Y)$, where Y is the subgraph of Y_P whose edges are the $\{p, q\}$ with $\mathbf{e}_p - \mathbf{e}_q \in \Sigma$, and assertion 3 bounds its number of vertices.

4.2. **Theorems 4.2 to 4.9.**

Definition (cocktail party graphs). ASSUME: $m \geq 0$ is an integer.
LET: The *cocktail party graph* $CP(m)$ has vertex set $\{1, \ldots, m\} \times \{0, 1\}$, and (i, s) is adjacent to (j, t) exactly when $i \neq j$.
Thus $CP(m)$ is K_{2m} with a perfect matching deleted, and it is $(2m-2)$-regular on $2m$ vertices; $CP(0)$ has no vertices.

Definition (generalized line graphs). ASSUME: Y is a graph with vertex set $\{1, \ldots, m\}$ and $a_1, \ldots, a_m \geq 0$ are integers.
LET: The *generalized line graph* $L(Y; a_1, \ldots, a_m)$ has as vertices the edges of Y and the triples (i, α, s) with $1 \leq i \leq m$, $1 \leq \alpha \leq a_i$ and $s \in \{0, 1\}$. Two edges are adjacent when they are adjacent in $L(Y)$; (i, α, s) and (j, β, t) are adjacent when $i = j$ and $\alpha \neq \beta$; and (i, α, s) is adjacent to the edge e when $i \in e$. A graph is a *generalized line graph* if it is isomorphic to some $L(Y; a_1, \ldots, a_m)$.
For each i, the triples (i, α, s) span a copy of $CP(a_i)$, and $L(Y; 0, \ldots, 0) \cong L(Y)$.

Lemma (the representation of a generalized line graph). ASSUME: Y and $a_1, \ldots, a_m$ are as in Definition (generalized line graphs), $n = \sum_{i=1}^{m}(1 + a_i)$, and the orthonormal basis of $\mathbb{R}^n$ is written $\mathbf{e}_{i,\alpha}$, for $1 \leq i \leq m$ and $0 \leq \alpha \leq a_i$.
PROVE: The vectors $\mathbf{x}_e = \mathbf{e}_{i,0} + \mathbf{e}_{j,0}$ for the edges $e = \{i, j\}$ of Y, and $\mathbf{x}_{(i,\alpha,s)} = \mathbf{e}_{i,0} + (-1)^s \mathbf{e}_{i,\alpha}$, lie in $\Phi(\mathrm{D}_n)$ and represent $L(Y; a_1, \ldots, a_m)$.

PROOF SKETCH: **[repair]** We read the paper's "$1 \leq \alpha \leq \alpha_i$" as $1 \leq \alpha \leq a_i$ (errata, Error 13, item 5). We will compute the three kinds of inner products.
⟨1⟩1. For edges e, f, $(\mathbf{x}_e, \mathbf{x}_f) = |e \cap f|$.
PROOF: Lemma (the line graph), in the coordinates $\mathbf{e}_{i,0}$, gives 2 for $e = f$, 1 when $e \neq f$ share an end, and 0 otherwise; this is $|e \cap f|$. □
⟨1⟩2. $(\mathbf{x}_{(i,\alpha,s)}, \mathbf{x}_e)$ is 1 if $i \in e$ and 0 otherwise.
PROOF: $\mathbf{e}_{i,\alpha}$ with $\alpha \geq 1$ does not occur in $\mathbf{x}_e$, and $(\mathbf{e}_{i,0}, \mathbf{x}_e) = [i \in e]$. □
⟨1⟩3. $(\mathbf{x}_{(i,\alpha,s)}, \mathbf{x}_{(j,\beta,t)})$ is 2 if the triples are equal, 0 if $i = j$ and $\alpha = \beta$ and $s \neq t$, 1 if $i = j$ and $\alpha \neq \beta$, and 0 if $i \neq j$.
PROOF: The inner product is $[i = j]\big(1 + (-1)^{s+t}[\alpha = \beta]\big)$. □
⟨1⟩4. Q.E.D.

PROOF: Each vector is a sum or difference of two distinct basis vectors, so lies in $\Phi(\mathrm{D}_n)$ by Definition 3.1. By ⟨1⟩1, ⟨1⟩2 and ⟨1⟩3, the Gram matrix is $2I$ plus the adjacency matrix of Definition (generalized line graphs). □

Lemma (the normal form in D_n). ASSUME: $n \geq 1$, and $(\mathbf{x}_u)_{u \in V(X)}$ represents a graph X in D_n.

PROVE: There are a set $C \subseteq \{1, \ldots, n\}$, integers $a_c \geq 0$ for $c \in C$, distinct coordinates $p_{c,\alpha}$ for $c \in C$ and $1 \leq \alpha \leq a_c$, which are the members of $\{1, \ldots, n\} \setminus C$, and signs $\epsilon_1, \ldots, \epsilon_n \in \{1, -1\}$ such that, with D the diagonal isometry $\mathbf{e}_i \mapsto \epsilon_i \mathbf{e}_i$ and $\mathbf{y}_u = D\mathbf{x}_u$:

1. $(\mathbf{y}_u)$ represents X in D_n;
2. for each $c \in C$, $1 \leq \alpha \leq a_c$ and $s \in \{0, 1\}$ exactly one vertex u has $\mathbf{y}_u = \mathbf{e}_c + (-1)^s \mathbf{e}_{p_{c,\alpha}}$, and every vertex u not of this form has $\mathbf{y}_u = \mathbf{e}_i + \mathbf{e}_j$ with $i \neq j$ in C, distinct such vertices having distinct supports;
3. $X \cong L(Y; (a_c)_{c \in C})$, where Y is the graph on C whose edges are the supports of the vertices of assertion 2 not of this form.

In particular $|C| + \sum_{c \in C} a_c = n$.

[new] This is the proof of assertion 1 of Theorem 4.2 below, with the coordinates kept; the remark after that theorem counts them.

PROOF SKETCH: We will follow the paper: we will find the pairs $\mathbf{e}_c \pm \mathbf{e}_p$, then choose the signs of the other coordinates. For $n < 4$, D_n is the coordinate model $\{\pm\mathbf{e}_i \pm \mathbf{e}_j : i \neq j\}$, as in the remark of Definition 3.1. **[repair]** The paper leaves both steps as assertions, among them that the set of pairs is unique and maximal (errata, Gap 19); we will prove what we need.

⟨1⟩1. Each $\mathbf{x}_u$ is $\sigma\mathbf{e}_i + \tau\mathbf{e}_j$ with $i \neq j$ and $\sigma, \tau \in \{1, -1\}$; its support has two members.
PROOF: ⟨0⟩ and Definition 3.1. □

⟨1⟩2. ASSUME: $u \neq w$ and $\mathrm{supp}(\mathbf{x}_u) = \mathrm{supp}(\mathbf{x}_w) = \{i, j\}$.
PROVE: $(\mathbf{x}_u, \mathbf{x}_w) = 0$, and $\mathbf{x}_u, \mathbf{x}_w$ agree in exactly one coordinate.
PROOF: The inner product is $\sigma\sigma' + \tau\tau' \in \{-2, 0, 2\}$ by ⟨1⟩1, and lies in $\{0, 1\}$ by ⟨0⟩ and Definition (representations); so it is 0, and exactly one of $\sigma\sigma'$, $\tau\tau'$ is 1. □

LET: A *pair* is a set $\{u, w\}$ as in ⟨1⟩2. Its *centre* is the coordinate in which $\mathbf{x}_u$ and $\mathbf{x}_w$ agree and its *petal* the other one; let Π be the set of petals of all pairs.

⟨1⟩3. ASSUME: $p \in \Pi$ is the petal of the pair $\{u, w\}$, with centre c.
PROVE: u and w are the only vertices z with $p \in \mathrm{supp}(\mathbf{x}_z)$.

⟨2⟩1. ASSUME: $p \in \mathrm{supp}(\mathbf{x}_z)$.
PROVE: $\mathrm{supp}(\mathbf{x}_z) = \{c, p\}$.
PROOF: Otherwise $\mathbf{x}_z = \rho\mathbf{e}_p + \ldots$ has no $\mathbf{e}_c$ term, so $(\mathbf{x}_z, \mathbf{x}_u) = \rho\tau$ and $(\mathbf{x}_z, \mathbf{x}_w) = -\rho\tau$, where $\pm\tau$ are the coefficients of $\mathbf{e}_p$ in $\mathbf{x}_u, \mathbf{x}_w$; one of them is -1, against ⟨0⟩ and Definition (representations). □

⟨2⟩2. Q.E.D.
PROOF: If $z \neq u, w$, then z, u and w have pairwise the support $\{c, p\}$ by ⟨2⟩1, so pairwise agree in exactly one of two coordinates by ⟨1⟩2. Their three sign vectors in $\{1, -1\}^2$ would then be pairwise adjacent in the graph on $\{1, -1\}^2$ joining vectors that differ in exactly one coordinate; that graph is C_4, which has no triangle. □

⟨1⟩4. No centre lies in Π, and every petal is the petal of exactly one pair.
PROOF: The pair of a petal is the set of vertices whose support contains it, by ⟨1⟩3. If the centre c of the pair $\{u, w\}$ were the petal of a pair, that pair would be $\{u, w\}$ by the same step, whose petal is not c. □

LET: $C = \{1, \ldots, n\} \setminus \Pi$.

⟨1⟩5. ASSUME: $i \in C$.
PROVE: All $\mathbf{x}_u$ with $i \in \mathrm{supp}(\mathbf{x}_u)$ have the same coefficient ϵ_i at $\mathbf{e}_i$.

PROOF: Suppose $\mathbf{x}_u$ and $\mathbf{x}_w$ have coefficients 1 and -1 at $\mathbf{e}_i$. If their supports are equal, $\{u, w\}$ is a pair that disagrees at i, so $i \in \Pi$ by the definition of petals; otherwise $(\mathbf{x}_u, \mathbf{x}_w) = -1$. The first contradicts ⟨1⟩, and the second ⟨0⟩ and Definition (representations). □

LET: $\epsilon_i = 1$ for the $i \in C$ in no support, and $\epsilon_p = 1$ for $p \in \Pi$; let D be the diagonal isometry $\mathbf{e}_i \mapsto \epsilon_i \mathbf{e}_i$, and $\mathbf{y}_u = D\mathbf{x}_u$.

⟨1⟩6. $(\mathbf{y}_u)$ represents X in D_n; every $\mathbf{y}_u$ is $\mathbf{e}_c \pm \mathbf{e}_p$ for a pair with centre c and petal p, or $\mathbf{e}_i + \mathbf{e}_j$ with $i, j \in C$.

PROOF: D is an isometry that preserves $\Phi(\mathrm{D}_n)$, so Lemma (transport of representations) applies. A vertex in a pair has support $\{c, p\}$, and $c \in C$ by ⟨1⟩4, so its $\mathbf{e}_c$ coefficient becomes 1 by ⟨1⟩5. Every other support lies in C by ⟨1⟩3, and both coefficients become 1 by ⟨1⟩5. □

LET: a_c be the number of pairs with centre c, for $c \in C$; number the petals of centre c as $p_{c,1}, \ldots, p_{c,a_c}$; and let Y be the graph on C whose edges are the supports of the vertices in no pair.

⟨1⟩7. The $p_{c,\alpha}$ are distinct, and they are the members of $\{1, \ldots, n\} \setminus C$.

PROOF: Each petal has one pair by ⟨1⟩4, and so one centre, which lies in C by the same step; and $\{1, \ldots, n\} \setminus C = \Pi$. □

⟨1⟩8. Assertion 2 holds.

PROOF: The vertices of the pair with petal $p_{c,\alpha}$ have $\mathbf{y}_u = \mathbf{e}_c \pm \mathbf{e}_{p_{c,\alpha}}$ by ⟨1⟩6, one with each sign by ⟨1⟩2, and no other vertex has $p_{c,\alpha}$ in its support by ⟨1⟩3. The vertices in no pair have $\mathbf{y}_u = \mathbf{e}_i + \mathbf{e}_j$ with $i, j \in C$ by ⟨1⟩6, and distinct ones have distinct supports, since equal supports make a pair. □

⟨1⟩9. The map ψ sending a vertex in no pair to its support, and the vertex u with $\mathbf{y}_u = \mathbf{e}_c + (-1)^s \mathbf{e}_{p_{c,\alpha}}$ to (c, α, s), is a bijection from $V(X)$ to the vertex set of $L(Y; (a_c)_{c \in C})$.

PROOF: By ⟨1⟩8, every triple (c, α, s) is the image of exactly one vertex, and the other vertices map injectively onto the edges of Y, by the definition of Y. □

⟨1⟩10. Assertion 3 holds.

PROOF: Renumber C as $\{1, \ldots, m\}$. The vectors $\mathbf{y}_u$ are those of Lemma (the representation of a generalized line graph) for $\psi(u)$, with $\mathbf{e}_{c,0} = \mathbf{e}_c$ and $\mathbf{e}_{c,\alpha} = \mathbf{e}_{p_{c,\alpha}}$, by ⟨1⟩8 and ⟨1⟩7. Both families have Gram matrix $2I$ plus the adjacency matrix, of X by ⟨1⟩6 and of $L(Y; a)$ by that lemma; so ψ of ⟨1⟩9 preserves adjacency. □

⟨1⟩11. Q.E.D.

PROOF: Assertion 1 is part of ⟨1⟩6, and assertions 2 and 3 are ⟨1⟩8 and ⟨1⟩10. The last sentence counts $\{1, \ldots, n\}$ by ⟨1⟩7. □

Theorem 4.2. ASSUME: X is a graph and $n \geq 1$.
PROVE: 1. If X is represented in D_n, then X is a generalized line graph.
2. $L(Y; a_1, \ldots, a_m)$ is represented in D_n for $n = \sum_i (1 + a_i)$.

PROOF SKETCH: Assertion 2 is Lemma (the representation of a generalized line graph). For assertion 1 we will follow the paper: we will find the pairs $\mathbf{e}_c \pm \mathbf{e}_p$, then choose the signs of the other coordinates; this is Lemma (the normal form in D_n), which also fills Gap 19 (errata). **[repair]** The paper's "whose lines from a subset" is "form" (errata, Error 13, item 4).

⟨1⟩1. Assertion 2 holds.

PROOF: Lemma (the representation of a generalized line graph). □

⟨1⟩2. ASSUME: $(\mathbf{x}_u)$ represents X in D_n.
PROVE: X is a generalized line graph.

PROOF: Lemma (the normal form in D_n), assertion 3, with ⟨1⟩, and $n \geq 1$ by ⟨0⟩; then Definition (generalized line graphs). □

⟨1⟩3. Q.E.D.

PROOF: ⟨1⟩2 and ⟨1⟩1. □

Remark (after Theorem 4.2). ASSUME: $n \geq 1$, and $\Sigma \subseteq \Phi(\mathrm{D}_n)$ has nonnegative inner products, in the sense of the remark after Theorem 4.1, with D_n for A_n.

PROVE: 1. If Σ is maximal, then there are $m \geq 1$ and $a_1, \ldots, a_m \geq 0$ with $\sum_{i=1}^{m}(1+a_i) = n$ such that the vectors of Σ, suitably indexed, represent $L(K_m; a_1, \ldots, a_m)$.

2. If $n \geq 4$, then $|\Sigma| \leq \frac{1}{2}n(n-1)$; the set $\{\mathbf{e}_i + \mathbf{e}_j : i < j\}$ has nonnegative inner products and $\frac{1}{2}n(n-1)$ members, and represents $L(K_n)$.

PROOF SKETCH: **[new]** The paper states the remark without proof. We will apply Lemma (the normal form in D_n), which is the proof of Theorem 4.2 with the coordinates kept, and count. **[new]** The bound fails for $n = 2$ and $n = 3$, where the paper does not define D_n: the sets $\{\mathbf{e}_1 \pm \mathbf{e}_2\}$ and $\{\mathbf{e}_1 \pm \mathbf{e}_2, \mathbf{e}_1 \pm \mathbf{e}_3\}$ are larger than $\frac{1}{2}n(n-1)$.

⟨1⟩1. The family $(\mathbf{x})_{\mathbf{x}\in\Sigma}$ represents in D_n the graph X on Σ in which $\mathbf{x} \sim \mathbf{y}$ when $(\mathbf{x}, \mathbf{y}) = 1$.

PROOF: For $\mathbf{x} \neq \mathbf{y}$ in Σ, $(\mathbf{x}, \mathbf{y})$ is an integer at least 0 by ⟨0⟩ and Definition 3.1, and $(\mathbf{x}, \mathbf{y}) < 2$ since $(\mathbf{x}-\mathbf{y}, \mathbf{x}-\mathbf{y}) = 4 - 2(\mathbf{x}, \mathbf{y}) > 0$; so it is 0 or 1. Apply Definition (representations). □

⟨1⟩2. PICK C, (a_c), $(p_{c,\alpha})$, D, $(\mathbf{y}_\mathbf{x})$ and Y as in Lemma (the normal form in D_n) for the representation of ⟨1⟩1, and let $m = |C|$. Then $m + \sum_c a_c = n$ and $m \geq 1$.

PROOF: The equation is the last sentence of the lemma. If $m = 0$, then $C = \varnothing$, so the equation gives $n = 0$, against ⟨0⟩. □

⟨1⟩3. $|\Sigma| = |E(Y)| + 2\sum_c a_c \leq \frac{1}{2}m(m-1) + 2(n-m)$.

PROOF: Lemma (the normal form in D_n), assertion 2, gives two members of Σ for each $p_{c,\alpha}$ and one for each edge of Y, a graph on the m vertices of C; and $\sum_c a_c = n - m$ by ⟨1⟩2. □

⟨1⟩4. ASSUME: Σ is maximal.

PROVE: Y is complete.

⟨2⟩1. ASSUME: $i \neq j$ lie in C.

PROVE: $\mathbf{z} = D(\mathbf{e}_i + \mathbf{e}_j)$ lies in $\Phi(\mathrm{D}_n)$, and $(\mathbf{z}, \mathbf{x}) \geq 0$ for every $\mathbf{x} \in \Sigma$.

⟨3⟩1. $\mathbf{z} = \epsilon_i\mathbf{e}_i + \epsilon_j\mathbf{e}_j$, and $\mathbf{z} \in \Phi(\mathrm{D}_n)$.

PROOF: D is the diagonal isometry $\mathbf{e}_k \mapsto \epsilon_k\mathbf{e}_k$, and $i \neq j$ by ⟨2⟩; apply Definition 3.1. □

⟨3⟩2. ASSUME: $\mathbf{x} \in \Sigma$.

PROVE: $(\mathbf{z}, \mathbf{x}) = (\mathbf{e}_i + \mathbf{e}_j, \mathbf{y}_\mathbf{x})$.

PROOF: D is an isometry with $D^2 = I$, so $(D(\mathbf{e}_i + \mathbf{e}_j), \mathbf{x}) = (\mathbf{e}_i + \mathbf{e}_j, D\mathbf{x})$, and $D\mathbf{x} = \mathbf{y}_\mathbf{x}$. □

⟨3⟩3. ASSUME: $\mathbf{x} \in \Sigma$.

PROVE: $(\mathbf{e}_i + \mathbf{e}_j, \mathbf{y}_\mathbf{x}) \geq 0$.

PROOF: By Lemma (the normal form in D_n), assertion 2, $\mathbf{y}_\mathbf{x}$ is $\mathbf{e}_c \pm \mathbf{e}_{p_{c,\alpha}}$ with $p_{c,\alpha} \notin C$, which gives $[c \in \{i, j\}]$ by ⟨2⟩, or $\mathbf{e}_k + \mathbf{e}_l$, which gives $|\{i, j\} \cap \{k, l\}|$. □

⟨3⟩4. Q.E.D.

PROOF: ⟨3⟩1, ⟨3⟩2 and ⟨3⟩3. □

⟨2⟩2. Q.E.D.

PROOF: For $i \neq j$ in C, $\mathbf{z} \in \Sigma$ by ⟨2⟩1 and ⟨1⟩. Its image $\mathbf{y}_\mathbf{z} = \mathbf{e}_i + \mathbf{e}_j$ has no petal $p_{c,\alpha}$ in its support, so it is of the second kind of Lemma (the normal form in D_n), assertion 2, and $\{i, j\}$ is an edge of Y. □

⟨1⟩5. Assertion 1 holds.

PROOF: Renumber C as $\{1, \ldots, m\}$. By Lemma (the normal form in D_n), assertion 3, and ⟨1⟩4, $X \cong L(K_m; a_1, \ldots, a_m)$, and $\sum_i(1 + a_i) = n$ by ⟨1⟩2; index Σ by this isomorphism. □

⟨1⟩6. ASSUME: $n \geq 4$ and $1 \leq m \leq n$.

PROVE: $\frac{1}{2}m(m-1) + 2(n-m) \leq \frac{1}{2}n(n-1)$.

PROOF: The difference of the two sides is $\frac{1}{2}(n-m)(n+m-5)$, and $n + m \geq 5$ by ⟨1⟩. □

⟨1⟩7. $\{\mathbf{e}_i + \mathbf{e}_j : i < j\}$ has $\frac{1}{2}n(n-1)$ members and nonnegative inner products, and represents $L(K_n)$.

PROOF: Lemma (the line graph) for $Y = K_n$, whose Gram matrix $2I + A$ has no negative entry; the vectors lie in $\Phi(\mathrm{D}_n)$ by Definition 3.1. □

⟨1⟩8. Q.E.D.

PROOF: Assertion 1 is ⟨1⟩5. For assertion 2, ⟨1⟩3 and ⟨1⟩6, with $1 \le m \le n$ by ⟨1⟩2, give the bound, and ⟨1⟩7 the example. □

The paper states the remark as "A maximal subset of D_n with nonnegative inner products represents $L(K_m; \mathbf{a})$ for some m and $\mathbf{a}$; the maximum cardinality is $\frac{1}{2}n(n-1)$, attained by $L(K_n)$." Assertion 1 holds for every $n \ge 1$. For $n = 4$ the maximum 6 is attained also by sets with $(m; a_1) = (1; 3)$; these represent $L(K_1; 3) = CP(3)$, which is isomorphic to $L(K_4)$, so what differs is the set and its parameters, not the graph. For $n \ge 5$ the maximum is attained only by sets with $m = n$, which represent $L(K_n)$, since $\frac{1}{2}(n-m)(n+m-5) > 0$ for $1 \le m < n$.

Lemma (the symmetric square). ASSUME: $\ell \ge 1$, and for $\mathbf{v} \in \mathbb{R}^\ell$ the vector $\sigma(\mathbf{v}) \in \mathbb{R}^{\ell(\ell+1)/2}$ has the coordinates $\mathbf{v}(i)^2$ for $1 \le i \le \ell$ and $\sqrt{2}\,\mathbf{v}(i)\mathbf{v}(j)$ for $1 \le i < j \le \ell$.

PROVE: $(\sigma(\mathbf{v}), \sigma(\mathbf{w})) = (\mathbf{v}, \mathbf{w})^2$ for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^\ell$.

[new] The paper uses the projections onto the lines, in the space of symmetric maps with the trace inner product; the coordinates of that space give σ, and $\mathbb{R}^{36}$ is easier in Lean.

PROOF SKETCH: We will expand the square.

⟨1⟩1. Q.E.D.

PROOF: $(\mathbf{v}, \mathbf{w})^2 = \left(\sum_i \mathbf{v}(i)\mathbf{w}(i)\right)^2 = \sum_i \mathbf{v}(i)^2\mathbf{w}(i)^2 + 2\sum_{i<j} \mathbf{v}(i)\mathbf{v}(j)\mathbf{w}(i)\mathbf{w}(j)$, which is $(\sigma(\mathbf{v}), \sigma(\mathbf{w}))$. □

Lemma (valencies from the sets $N_\mathbf{r}$). ASSUME: 1. S is a star-closed line system, $b \ge 0$, and $|N_\mathbf{r}| \le 2b$ for every $\mathbf{r} \in \Phi(S)$, where $N_\mathbf{r} = \{\mathbf{x} \in \Phi(S) : (\mathbf{x}, \mathbf{r}) = 1\}$.
2. X is represented in S by $(\mathbf{x}_u)$, and u is a vertex of X.

PROVE: u has at most b neighbours.

[repair] The paper cites "Lemma 3.7" for such bounds (errata, Error 6), and Theorem 3.7 gives only the number $|N_\mathbf{r}|$; the pairing below is the fill of Gap 14 (errata), with LN6 of `cgss/prerequisites.md`.

PROOF SKETCH: We will pair the vectors with inner product 1 with $\mathbf{r} = \mathbf{x}_u$.

LET: $\mathbf{r} = \mathbf{x}_u$.

⟨1⟩1. ASSUME: $\mathbf{x} \in N_\mathbf{r}$.
PROVE: $\mathbf{r} - \mathbf{x} \in N_\mathbf{r}$, $\mathbf{r} - \mathbf{x} \ne \mathbf{x}$ and $(\mathbf{x}, \mathbf{r} - \mathbf{x}) = -1$.

PROOF: Since $(\mathbf{x}, -\mathbf{r}) = -1$ and S is star-closed by ⟨0⟩:1, the vector $\mathbf{x} - \mathbf{r}$ and its negative lie in $\Phi(S)$. Further $(\mathbf{r} - \mathbf{x}, \mathbf{r}) = 2 - 1 = 1$ and $(\mathbf{x}, \mathbf{r} - \mathbf{x}) = 1 - 2 = -1$, which also gives $\mathbf{r} - \mathbf{x} \ne \mathbf{x}$. □

⟨1⟩2. $w \mapsto \mathbf{x}_w$ maps the neighbours of u injectively into $N_\mathbf{r}$, and its image meets each pair $\{\mathbf{x}, \mathbf{r} - \mathbf{x}\}$ at most once.

PROOF: $(\mathbf{x}_w, \mathbf{r}) = 1$ for a neighbour w by Definition (representations), and the map is injective by Lemma (the lines of a representation), assertion 1. Two neighbours have inner product 0 or 1, not the -1 of ⟨1⟩1. □

⟨1⟩3. Q.E.D.

PROOF: By ⟨1⟩1, $\mathbf{x} \mapsto \mathbf{r} - \mathbf{x}$ is an involution of $N_\mathbf{r}$ without fixed points, so $N_\mathbf{r}$ is the union of at most b disjoint pairs by ⟨0⟩:1; apply ⟨1⟩2. □

Computation C20. ASSUME: $\mathbf{r}$ is a vector of the model of A_ℓ, D_ℓ, E_6, E_7 or E_8, and X is represented in that model.

PROVE: $|N_\mathbf{r}|$ is $2(\ell-1)$, $4(\ell-2)$, 20, 32 or 56 respectively, and every vertex of X has at most $\ell - 1$, $2(\ell-2)$, 10, 16 or 28 neighbours respectively.

PROOF SKETCH: The counts are LN6 of `cgss/prerequisites.md`. We will cite their Lean proofs, which are parametric in ℓ for A_ℓ and D_ℓ and use `decide` for the three exceptional models.

⟨1⟩1. The counts of $|N_{\mathbf{r}}|$ hold.

PROOF: `LineSystem.An.card_N`, `LineSystem.Dn.card_N`, and `LineSystem.E6.card_N` and its analogues for E_7 and E_8. □

⟨1⟩2. Q.E.D.

PROOF: The models are star-closed by Theorem 3.5; apply Lemma (valencies from the sets $N_{\mathbf{r}}$) with half of each count. □

Computation C7. ASSUME: In $\mathbb{R}^9$, let $\mathbf{c} = \frac{1}{3}\sum_{l=1}^{9}\mathbf{e}_l$, $\mathbf{y}_h = \mathbf{e}_h - \mathbf{e}_9$ for $1 \le h \le 8$, and $\mathbf{z}_{ij} = \mathbf{c} - \mathbf{e}_i - \mathbf{e}_j - \mathbf{e}_9$ for $1 \le i < j \le 8$.

PROVE: These 36 vectors lie in model (i) of E_8 and represent a graph X_{36} on 36 vertices in which $\mathbf{y}_1$ has 28 neighbours.

PROOF SKETCH: We will compute the inner products; the Lean proof is by `decide` on the integer vectors $3\mathbf{y}_h$ and $3\mathbf{z}_{ij}$.

⟨1⟩1. The vectors lie in model (i).

PROOF: $\mathbf{y}_h \in \Phi(\mathrm{A}_8)$, and $\mathbf{z}_{ij}$ is $\frac{1}{3}\sum_l \mathbf{e}_l - \sum_{l\in T}\mathbf{e}_l$ with $T = \{i, j, 9\}$; apply Definition 3.2, (i). □

⟨1⟩2. $(\mathbf{y}_h, \mathbf{y}_k) = 1$ and $(\mathbf{z}_{ij}, \mathbf{z}_{kl}) = |\{i,j\}\cap\{k,l\}|$ for distinct vectors, and $(\mathbf{y}_h, \mathbf{z}_{ij})$ is 0 if $h \in \{i, j\}$ and 1 otherwise.

PROOF: $(\mathbf{y}_h, \mathbf{c}) = 0$ and $(\mathbf{c}, \mathbf{c}) = 1$, and $(\mathbf{c}, \mathbf{e}_l) = \frac{1}{3}$. Then $(\mathbf{y}_h, \mathbf{y}_k) = 1 + [h = k]$, $(\mathbf{y}_h, \mathbf{z}_{ij}) = 1 - [h \in \{i, j\}]$ and $(\mathbf{z}_{ij}, \mathbf{z}_{kl}) = 1 - 2 + |\{i, j, 9\} \cap \{k, l, 9\}|$. □

⟨1⟩3. Q.E.D.

PROOF: Each vector has squared norm 2 by ⟨1⟩2, and the other inner products lie in $\{0, 1\}$; so they represent a graph by Definition (representations). The neighbours of $\mathbf{y}_1$ are the 7 vectors $\mathbf{y}_k$ and the $\binom{7}{2} = 21$ vectors $\mathbf{z}_{ij}$ with $1 \notin \{i, j\}$. □

Theorem 4.3. ASSUME: X is represented in E_8 by $(\mathbf{x}_u)$.

PROVE:
1. X has at most 36 vertices.
2. Every vertex of X has at most 28 neighbours.
3. There is a graph represented in E_8 with 36 vertices and a vertex with 28 neighbours.

PROOF SKETCH: We will follow the paper: the Gram matrix of the symmetric squares is nonsingular. **[repair]** The bound on the valency needs the pairing of Gap 14 (errata).

⟨1⟩1. $4I + A(X)$ is positive definite.

PROOF: $2I + A(X)$ is the Gram matrix of $(\mathbf{x}_u)$, so positive semidefinite by `Matrix.posSemidef_gram`; for $\mathbf{z} \neq 0$, $\mathbf{z}^T(4I + A(X))\mathbf{z} = 2|\mathbf{z}|^2 + \mathbf{z}^T(2I + A(X))\mathbf{z} > 0$. □

⟨1⟩2. The Gram matrix of $(\sigma(\mathbf{x}_u))$ in $\mathbb{R}^{36}$ is $4I + A(X)$.

PROOF: Lemma (the symmetric square) with $\ell = 8$: the entries are $(\mathbf{x}_u, \mathbf{x}_w)^2$, which are 4, 1 and 0 by Definition (representations). □

⟨1⟩3. Assertion 1 holds.

PROOF: The family $(\sigma(\mathbf{x}_u))$ is linearly independent by ⟨1⟩1, ⟨1⟩2 and `Matrix.posDef_gram_iff_linearIndependent`; apply `LinearIndependent.fintype_card_le_finrank` and `finrank_euclideanSpace_fin`. □

⟨1⟩4. Assertion 2 holds.

PROOF: Computation C20, for E_8. □

⟨1⟩5. Assertion 3 holds.

PROOF: The graph X_{36} of Computation C7 is represented in model (i); models (i) and (ii) are isomorphic by Theorem 3.5, so it is represented in model (ii) by Lemma (transport of representations). □

⟨1⟩6. Q.E.D.

PROOF: ⟨1⟩3, ⟨1⟩4 and ⟨1⟩5. □

Lemma (trace and rank). ASSUME: N is a real symmetric matrix indexed by a finite set V.
PROVE: $(\operatorname{tr} N)^2 \le \operatorname{rk} N \cdot \sum_{u,w\in V} N(u,w)^2$.
[new] This is the inequality behind [LS73] Theorem 3.6; we will name it, since Lemma EQ1 and the remark after Theorem 4.4 both use it. **[new]** In Lean the entrywise form stated here is `Matrix.IsHermitian.trace_sq_le_rank_mul_sum_entry_sq`, in `LineGraphs/Representation/Theorem44.lean`, which Theorem 4.4 and the remark after it use. The eigenvalue form $(\operatorname{tr} N)^2 \le \operatorname{rk} N \cdot \sum_{u\in V} \lambda_u^2$ of the steps below is `Matrix.IsHermitian.trace_sq_le_rank_mul_sum_sq`, in `LineGraphs/Spectral/Applications.lean`; the name of the entrywise form differs, since that one was taken first.

PROOF SKETCH: We will apply the Cauchy–Schwarz inequality to the nonzero eigenvalues.
LET: λ_u, for $u \in V$, be the eigenvalues `Matrix.IsHermitian.eigenvalues` of N, and $Z = \{u \in V : \lambda_u \neq 0\}$.
⟨1⟩1. $\operatorname{tr} N = \sum_{u\in Z} \lambda_u$ and $\operatorname{rk} N = |Z|$.
PROOF: `Matrix.IsHermitian.trace_eq_sum_eigenvalues`, with the zero terms dropped, and `Matrix.IsHermitian.rank_eq_card_non_zero_eigs`. □
⟨1⟩2. $\sum_{u,w} N(u,w)^2 = \sum_{u\in Z} \lambda_u^2$.
PROOF: By `Matrix.IsHermitian.spectral_theorem`, $N = UDU^T$ with U orthogonal and D diagonal with the entries λ_u. As N is symmetric, $\sum_{u,w} N(u,w)^2 = \operatorname{tr}(NN^T) = \operatorname{tr}(UD^2U^T) = \operatorname{tr}(D^2U^TU) = \sum_u \lambda_u^2$, where the third equality is `Matrix.trace_mul_comm`; drop the zero terms. □
⟨1⟩3. Q.E.D.
PROOF: By `Finset.sum_mul_sq_le_sq_mul_sq` for the functions λ and 1 on Z, $\left(\sum_{u\in Z} \lambda_u\right)^2 \le |Z| \sum_{u\in Z} \lambda_u^2$; apply ⟨1⟩1 and ⟨1⟩2. □

Lemma (the relative bound, EQ1). ASSUME: The unit vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of a real inner product space span a subspace of dimension k, where $1 \le k \le 7$, and $(\mathbf{v}_i, \mathbf{v}_j) = \pm\frac{1}{3}$ for $i \neq j$.
PROVE: $n \le k \cdot \frac{8}{9}/(1 - \frac{k}{9})$, and in particular $n \le 28$.

PROOF SKETCH: This is [LS73] Theorem 3.6, which [CGSS] cites as [15]; we will prove it from the trace of the Gram matrix.
LET: M be the Gram matrix of the $\mathbf{v}_i$.
⟨1⟩1. $\operatorname{tr} M = n$, $\operatorname{tr} M^2 = n + \frac{1}{9}n(n-1)$ and $\operatorname{rk} M = k$.
PROOF: The diagonal entries of M are 1 and the others $\pm\frac{1}{3}$, and $\operatorname{tr} M^2 = \sum_{i,j} (\mathbf{v}_i, \mathbf{v}_j)^2$; the rank is Lemma (Gram factorization), assertion 1. □
⟨1⟩2. $n^2 \le k\big(n + \frac{1}{9}n(n-1)\big)$.
PROOF: Lemma (trace and rank) for M, where $\sum_{i,j} M(i,j)^2 = \operatorname{tr} M^2$ as M is symmetric; apply ⟨1⟩1. □
⟨1⟩3. Q.E.D.
PROOF: Divide ⟨1⟩2 by n: $n(1 - \frac{k}{9}) \le \frac{8}{9}k$, with $1 - \frac{k}{9} > 0$. The bound $\frac{8}{9}k/(1 - \frac{k}{9}) = 8k/(9-k)$ increases with k, and is 28 for $k = 7$. □

Lemma (lines spanning eight dimensions, EQ2). ASSUME: U is a real inner product space of dimension 8, spanned by unit vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with $(\mathbf{v}_i, \mathbf{v}_j) = \pm\frac{1}{3}$ for $i \neq j$.
PROVE: $n \le 14$.
[repair] This is the cone construction of `cgss/prerequisites.md` (EQ2), which fills the unstated part of [LS73] (errata, Gap 7).

PROOF SKETCH: We will put the lines on a cone around a root $\mathbf{r}$ in $U \oplus \mathbb{R}\mathbf{e}$ and count the pairs of $N_{\mathbf{r}}$.
LET: $\mathbf{r} = \sqrt{2}\,\mathbf{e}$ and $\mathbf{u}_i = \sqrt{3/2}\,\mathbf{v}_i + \mathbf{e}/\sqrt{2}$ in $U \oplus \mathbb{R}\mathbf{e}$, where $\mathbf{e}$ is a unit vector orthogonal to U.
⟨1⟩1. $(\mathbf{r},\mathbf{r}) = (\mathbf{u}_i,\mathbf{u}_i) = 2$, $(\mathbf{u}_i,\mathbf{r}) = 1$, and $(\mathbf{u}_i,\mathbf{u}_j) = \frac{3}{2}(\mathbf{v}_i,\mathbf{v}_j) + \frac{1}{2} \in \{0,1\}$ for $i \neq j$.
PROOF: Expand, with $(\mathbf{v}_i,\mathbf{e}) = 0$. □
LET: X' be the graph on $\{0,1,\dots,n\}$ in which 0 is adjacent to every i, and $i \neq j$ are adjacent when $(\mathbf{u}_i,\mathbf{u}_j) = 1$.
⟨1⟩2. $\mathbf{r},\mathbf{u}_1,\dots,\mathbf{u}_n$ represent the connected graph X', and span a space of dimension 9.
PROOF: Definition (representations) and ⟨1⟩1; 0 is adjacent to every other vertex. The span contains $\mathbf{e} = \mathbf{r}/\sqrt{2}$ and each $\mathbf{v}_i = (\mathbf{u}_i - \mathbf{e}/\sqrt{2})/\sqrt{3/2}$, so it is $U \oplus \mathbb{R}\mathbf{e}$. □
⟨1⟩3. X' is represented in A_9 or in D_9.
PROOF: Lemma (a root system containing the lines), assertion 1, with $k = 9$; the models of E_6, E_7, E_8 have ranks 6, 7, 8 by Lemma (spans of the models). □
⟨1⟩4. Q.E.D.
PROOF: The vertex 0 has n neighbours, at most $9 - 1 = 8$ in A_9 and at most $2(9-2) = 14$ in D_9, by Computation C20 and ⟨1⟩3. □

Computation C8. ASSUME: $n \leq 28$ and $d \leq n-1$ are nonnegative integers with $n < 2d+4$ and $0 \leq (d - 2n + 16) + 14(2n - 2d - 4)/(2d + 4 - n)$.
PROVE: $d \leq 16$, or $n = 28$ and $d = 17$.

PROOF SKETCH: We will clear the positive denominator; the Lean proof is `decide` over the finitely many pairs.
⟨1⟩1. Q.E.D.
PROOF: Multiply by $2d + 4 - n > 0$: $(d - 2n + 16)(2d + 4 - n) + 14(2n - 2d - 4) \geq 0$. Among the pairs (n,d) with $n \leq 28$, $d \leq n-1$ and $n < 2d+4$, this holds only for $d \leq 16$ and for $(28,17)$; this is a finite check of the 406 pairs with $d < n \leq 28$. □

Lemma (the matrices of a regular graph). ASSUME: 1. X is d-regular on $n \geq 1$ vertices, $Q = 2I + A(X)$ is positive semidefinite, and $\mathbf{j}$ is the all-ones vector.
2. $t = (d+2)/(2n)$, $M = \frac{1}{2}Q - tJ$ and $E = \frac{2}{3}Q - \frac{1}{3}J$.
PROVE: 1. $Q\mathbf{j} = (d+2)\mathbf{j}$, and $J\mathbf{z} = 0$ and $(Q\mathbf{z},\mathbf{j}) = 0$ for every $\mathbf{z} \perp \mathbf{j}$.
2. $M\mathbf{j} = 0$ and $M\mathbf{z} = \frac{1}{2}Q\mathbf{z}$ for $\mathbf{z} \perp \mathbf{j}$; M is positive semidefinite, and $\operatorname{rk} M = \operatorname{rk} Q - 1$.
3. If $n < 2d+4$, then $E\mathbf{j} = \frac{1}{3}(2d + 4 - n)\mathbf{j}$ and $E\mathbf{z} = \frac{2}{3}Q\mathbf{z}$ for $\mathbf{z} \perp \mathbf{j}$; E is positive semidefinite, and $\operatorname{rk} E = \operatorname{rk} Q$.
[new] The paper reads these facts off the eigenvalues of A; we will compare ranges instead, and state the facts once, for Theorem 4.4 and for the remark after it.

PROOF SKETCH: We will split $\mathbb{R}^n$ as $\mathbb{R}\mathbf{j} \oplus \mathbf{j}^{\perp}$.
LET: $W = Q(\mathbf{j}^{\perp})$ be the image of $\mathbf{j}^{\perp}$ under Q.
⟨1⟩1. $A\mathbf{j} = d\mathbf{j}$, $Q\mathbf{j} = (d+2)\mathbf{j}$ and $J\mathbf{j} = n\mathbf{j}$.
PROOF: `SimpleGraph.adjMatrix_mulVec_const_apply_of_regular` with ⟨0⟩:1; then $Q\mathbf{j} = 2\mathbf{j} + A\mathbf{j}$, and the rows of J sum to n. □
⟨1⟩2. ASSUME: $\mathbf{z} \perp \mathbf{j}$.
PROVE: $J\mathbf{z} = 0$ and $(Q\mathbf{z},\mathbf{j}) = 0$.
PROOF: Every entry of $J\mathbf{z}$ is $(\mathbf{z},\mathbf{j}) = 0$. Further Q is symmetric, so $(Q\mathbf{z},\mathbf{j}) = (\mathbf{z},Q\mathbf{j}) = (d+2)(\mathbf{z},\mathbf{j}) = 0$ by ⟨1⟩1. □
⟨1⟩3. Every $\mathbf{y} \in \mathbb{R}^n$ is $c\mathbf{j} + \mathbf{z}$ with $c = (\mathbf{y},\mathbf{j})/n$ and $\mathbf{z} \perp \mathbf{j}$.
PROOF: $(\mathbf{y} - c\mathbf{j},\mathbf{j}) = (\mathbf{y},\mathbf{j}) - cn = 0$, as $(\mathbf{j},\mathbf{j}) = n \geq 1$ by ⟨0⟩:1. □
⟨1⟩4. The range of Q is $\mathbb{R}\mathbf{j} + W$.
PROOF: By ⟨1⟩3 and ⟨1⟩1, $Q(c\mathbf{j} + \mathbf{z}) = c(d+2)\mathbf{j} + Q\mathbf{z}$ for $\mathbf{z} \perp \mathbf{j}$, and $d + 2 \neq 0$. □

⟨1⟩5. $\mathbb{R}\mathbf{j} \cap W = 0$.
PROOF: If $c\mathbf{j} \in W$, then $cn = (c\mathbf{j}, \mathbf{j}) = 0$ by ⟨1⟩2, and $n \geq 1$ by ⟨0⟩:1; so $c = 0$. □
⟨1⟩6. $\operatorname{rk} Q = 1 + \dim W$.
PROOF: By the definition `Matrix.rank`, $\operatorname{rk} Q$ is the dimension of the range of Q, which is $\mathbb{R}\mathbf{j}+W$ by ⟨1⟩4. Apply `Submodule.finrank_sup_add_finrank_inf_eq` with ⟨1⟩5, and `finrank_span_singleton`, as $\mathbf{j} \neq 0$. □
⟨1⟩7. Assertion 2 holds.
⟨2⟩1. $M\mathbf{j} = 0$, and $M\mathbf{z} = \frac{1}{2}Q\mathbf{z}$ for $\mathbf{z} \perp \mathbf{j}$.
PROOF: $M\mathbf{j} = \frac{1}{2}(d+2)\mathbf{j} - tn\mathbf{j} = 0$ by ⟨1⟩1, since $tn = \frac{1}{2}(d+2)$ by ⟨0⟩:2; and $J\mathbf{z} = 0$ by ⟨1⟩2. □
⟨2⟩2. M is positive semidefinite.
PROOF: M is symmetric, as Q and J are. For $\mathbf{y} = c\mathbf{j}+\mathbf{z}$ as in ⟨1⟩3, $(\mathbf{y}, M\mathbf{y}) = (c\mathbf{j}+\mathbf{z}, \frac{1}{2}Q\mathbf{z}) = \frac{1}{2}(\mathbf{z}, Q\mathbf{z}) \geq 0$, by ⟨2⟩1, ⟨1⟩2 and ⟨0⟩:1. □
⟨2⟩3. $\operatorname{rk} M = \operatorname{rk} Q - 1$.
PROOF: By ⟨1⟩3 and ⟨2⟩1, $M(c\mathbf{j} + \mathbf{z}) = \frac{1}{2}Q\mathbf{z}$, so the range of M is W; apply ⟨1⟩6 and the definition `Matrix.rank`. □
⟨2⟩4. Q.E.D.
PROOF: ⟨2⟩1, ⟨2⟩2 and ⟨2⟩3. □
⟨1⟩8. ASSUME: $n < 2d + 4$.
PROVE: The conclusion of assertion 3 holds.
LET: $b = \frac{1}{3}(2d + 4 - n)$, which is positive by ⟨1⟩.
⟨2⟩1. $E\mathbf{j} = b\mathbf{j}$, and $E\mathbf{z} = \frac{2}{3}Q\mathbf{z}$ for $\mathbf{z} \perp \mathbf{j}$.
PROOF: $E\mathbf{j} = \frac{2}{3}(d+2)\mathbf{j} - \frac{1}{3}n\mathbf{j}$ by ⟨1⟩1, and $J\mathbf{z} = 0$ by ⟨1⟩2. □
⟨2⟩2. E is positive semidefinite.
PROOF: E is symmetric. For $\mathbf{y} = c\mathbf{j} + \mathbf{z}$ as in ⟨1⟩3, $(\mathbf{y}, E\mathbf{y}) = (c\mathbf{j} + \mathbf{z}, cb\mathbf{j} + \frac{2}{3}Q\mathbf{z}) = c^2bn + \frac{2}{3}(\mathbf{z}, Q\mathbf{z}) \geq 0$, by ⟨2⟩1 and ⟨1⟩2, with $(\mathbf{z}, \mathbf{j}) = 0$, $b > 0$ and ⟨0⟩:1. □
⟨2⟩3. $\operatorname{rk} E = \operatorname{rk} Q$.
PROOF: By ⟨1⟩3 and ⟨2⟩1, $E(c\mathbf{j} + \mathbf{z}) = cb\mathbf{j} + \frac{2}{3}Q\mathbf{z}$ with $b \neq 0$, so the range of E is $\mathbb{R}\mathbf{j} + W$, which is the range of Q by ⟨1⟩4. □
⟨2⟩4. Q.E.D.
PROOF: ⟨2⟩1, ⟨2⟩2 and ⟨2⟩3. □
⟨1⟩9. Q.E.D.
PROOF: Assertion 1 is ⟨1⟩1 and ⟨1⟩2; assertions 2 and 3 are ⟨1⟩7 and ⟨1⟩8. □

Theorem 4.4. ASSUME: 1. X is d-regular on $n \geq 1$ vertices.
2. X is represented in E_8 by $(\mathbf{x}_u)$.
PROVE: $n \leq 28$ and $d \leq 16$.
[new] In Lean the theorem is proved as `SimpleGraph.IsRepresentedIn.card_le_28_of_regular_of_classification` and `SimpleGraph.IsRepresentedIn.degree_le_16_of_regular_of_classification`, in `LineGraphs/Representation/Theorem44.lean`, with the classification of the maximal indecomposable line systems of the nine-dimensional space of Lemma (lines spanning eight dimensions, EQ2) as an explicit hypothesis. The settled names above, without the hypothesis, are proved in `LineGraphs/Representation/Theorem44Unconditional.lean`, by discharging it with Theorem 3.6 (`LineSystem.IsMaximal.exists_iso_rootType`).

PROOF SKETCH: We will follow the paper, with these repairs. The equiangular facts come from Lemmas EQ1 and EQ2 in place of [15] (errata, Error 8 and Gap 7, decision 1 of `cgss/prerequisites.md`). **[repair]** The case $n = 28$, $d = 12$ needs its own treatment (errata, Error

15), and the final valency bound needs the pairing of Gap 14 (errata). The printed Gram matrix is that of the vectors $\sigma(\mathbf{q}_u)$, not of projections (errata, Error 13, item 6), and "each eigenvalue $\alpha \neq d$" concerns the action on $\mathbf{j}^\perp$ (item 7); we will use that action through quadratic forms, without counting eigenvalues.

LET: $Q = 2I + A(X)$, $\mathbf{j}$ the all-ones vector, $t = (d+2)/(2n)$, $M = \frac{1}{2}Q - tJ$ and $E = \frac{2}{3}Q - \frac{1}{3}J$, as in Lemma (the matrices of a regular graph).

⟨1⟩1. Q is positive semidefinite of rank at most 8.

PROOF: Q is the Gram matrix of n vectors of $\mathbb{R}^8$ by ⟨0⟩:2, so it is positive semidefinite by `Matrix.posSemidef_gram`, and of rank at most 8 by Lemma (Gram factorization), assertion 1, since a subspace of $\mathbb{R}^8$ has dimension at most 8 (`Submodule.finrank_le`). □

⟨1⟩2. CASE: $n \leq 17$.

PROVE: $n \leq 28$ and $d \leq 16$.

PROOF: $d \leq n - 1 \leq 16$ by ⟨0⟩:1 and ⟨1⟩. □

⟨1⟩3. M is the Gram matrix of vectors $\mathbf{q}_u \in \mathbb{R}^7$, with $(\mathbf{q}_u, \mathbf{q}_u) = 1 - t$, $(\mathbf{q}_u, \mathbf{q}_w) = \frac{1}{2} - t$ for adjacent u, w, and $-t$ otherwise.

PROOF: By Lemma (the matrices of a regular graph), assertion 2, with ⟨0⟩:1 and ⟨1⟩1, M is positive semidefinite of rank $\operatorname{rk} Q - 1 \leq 7$. Apply Lemma (Gram factorization), assertion 2, with $r = 7$; the entries are those of $\frac{1}{2}Q - tJ$. □

⟨1⟩4. $P = (1-t)^2 I + (\frac{1}{2}-t)^2 A + t^2(J - I - A)$ is positive semidefinite, and $P\mathbf{z} = (1-2t)\mathbf{z} + (\frac{1}{4}-t)A\mathbf{z}$ for $\mathbf{z} \perp \mathbf{j}$.

PROOF: By Lemma (the symmetric square) with $\ell = 7$ and ⟨1⟩3, P is the Gram matrix of the $\sigma(\mathbf{q}_u) \in \mathbb{R}^{28}$; apply `Matrix.posSemidef_gram`. For $\mathbf{z} \perp \mathbf{j}$, $J\mathbf{z} = 0$ by Lemma (the matrices of a regular graph), assertion 1, and $(1-t)^2 - t^2 = 1 - 2t$, $(\frac{1}{2} - t)^2 - t^2 = \frac{1}{4} - t$. □

⟨1⟩5. CASE: $n \geq 18$ and $n \geq 2d + 4$.

PROVE: $n \leq 28$ and $d \leq 12$.

⟨2⟩1. P is positive definite.

PROOF: For $\mathbf{z} \perp \mathbf{j}$, $(\mathbf{z}, A\mathbf{z}) = (\mathbf{z}, Q\mathbf{z}) - 2|\mathbf{z}|^2 \geq -2|\mathbf{z}|^2$ by ⟨1⟩1, and $\frac{1}{4} - t \geq 0$ by ⟨1⟩; so by ⟨1⟩4, $(\mathbf{z}, P\mathbf{z}) \geq (1 - 2t - 2(\frac{1}{4} - t))|\mathbf{z}|^2 = \frac{1}{2}|\mathbf{z}|^2$. Further $P\mathbf{j} = p\mathbf{j}$ with $p = (1-t)^2 + (\frac{1}{2}-t)^2 d + t^2(n-1-d) > 0$, as $1 - t \geq \frac{3}{4}$ since $t \leq \frac{1}{4}$. P is symmetric, being a combination of I, A and J; so $(P\mathbf{z}, \mathbf{j}) = (\mathbf{z}, P\mathbf{j}) = 0$ for $\mathbf{z} \perp \mathbf{j}$, and P maps $\mathbf{j}^\perp$ into itself. So $(\mathbf{y}, P\mathbf{y}) = pc^2 n + (\mathbf{z}, P\mathbf{z}) > 0$ for $\mathbf{y} = c\mathbf{j} + \mathbf{z} \neq 0$. □

⟨2⟩2. Q.E.D.

PROOF: The n vectors $\sigma(\mathbf{q}_u)$ are linearly independent in $\mathbb{R}^{28}$ by ⟨2⟩1 and `Matrix.posDef_gram_iff_linearIndependent`, so $n \leq 28$ by `LinearIndependent.fintype_card_le_finrank`; and $d \leq (n-4)/2 \leq 12$ by ⟨1⟩. □

⟨1⟩6. CASE: $n \geq 18$ and $n < 2d + 4$.

PROVE: $n \leq 28$ and $d \leq 16$.

LET: $B = 2(2n - 2d - 4)/(2d + 4 - n)$, and $\mathbf{b}_k = (\sqrt{2}\,\mathbf{q}_u(k))_{u \in V(X)} \in \mathbb{R}^{V(X)}$ for $1 \leq k \leq 7$, with the $\mathbf{q}_u$ of ⟨1⟩3.

⟨2⟩1. $(\mathbf{z}, Q\mathbf{z}) \leq (B+2)|\mathbf{z}|^2$ for every $\mathbf{z} \perp \mathbf{j}$.

PROOF: By ⟨1⟩4, $0 \leq (\mathbf{z}, P\mathbf{z}) = (1-2t)|\mathbf{z}|^2 + (\frac{1}{4}-t)(\mathbf{z}, A\mathbf{z})$, where $\frac{1}{4} - t = (n-2d-4)/(4n) < 0$ by ⟨1⟩. Divide by it: $(\mathbf{z}, A\mathbf{z}) \leq (1-2t)/(t - \frac{1}{4}) \cdot |\mathbf{z}|^2 = B|\mathbf{z}|^2$, since $1 - 2t = (n - d - 2)/n$; add $2|\mathbf{z}|^2$. □

⟨2⟩2. $2M = \sum_{k=1}^{7} \mathbf{b}_k \mathbf{b}_k^T$, and $\mathbf{b}_k \perp \mathbf{j}$ for every k.

PROOF: The uw-entry of $2M$ is $2(\mathbf{q}_u, \mathbf{q}_w) = \sum_k \mathbf{b}_k(u)\mathbf{b}_k(w)$ by ⟨1⟩3; and $\sum_k (\mathbf{b}_k, \mathbf{j})^2 = (\mathbf{j}, 2M\mathbf{j}) = 0$ by Lemma (the matrices of a regular graph), assertion 2. □

⟨2⟩3. $|\mathbf{b}_k|^2 \leq B + 2$ for every k.

⟨3⟩1. $(\mathbf{b}_k, 2M\mathbf{b}_k) = \sum_l (\mathbf{b}_l, \mathbf{b}_k)^2 \geq |\mathbf{b}_k|^4$.

PROOF: ⟨2⟩2; drop the terms $l \neq k$. □

⟨3⟩2. $(\mathbf{b}_k, 2M\mathbf{b}_k) = (\mathbf{b}_k, Q\mathbf{b}_k) \leq (B+2)|\mathbf{b}_k|^2$.

PROOF: $2M\mathbf{b}_k = Q\mathbf{b}_k$ by Lemma (the matrices of a regular graph), assertion 2, as $\mathbf{b}_k \perp \mathbf{j}$ by ⟨2⟩2; then ⟨2⟩1. □

⟨3⟩3. Q.E.D.

PROOF: If $\mathbf{b}_k = 0$ there is nothing to prove. Otherwise divide $|\mathbf{b}_k|^4 \leq (B+2)|\mathbf{b}_k|^2$, from ⟨3⟩1 and ⟨3⟩2, by $|\mathbf{b}_k|^2 > 0$. □

⟨2⟩4. $0 \leq (d - 2n + 16) + 7B$.

PROOF: The diagonal entries of $2M$ are $2(1-t)$ by ⟨1⟩3, so $\sum_k |\mathbf{b}_k|^2 = \operatorname{tr}(2M) = 2n - d - 2$ by ⟨2⟩2; this is at most $7(B+2)$ by ⟨2⟩3. **[new]** The Lean proof of this step and of ⟨2⟩1, in `LineGraphs/Representation/Theorem44.lean`, is shorter: it bounds the nonzero eigenvalues of M, at most 7 of them, through the positive semidefinite P of ⟨1⟩4, in place of the columns $\mathbf{b}_k$. □

⟨2⟩5. E is the Gram matrix of n unit vectors $\mathbf{w}_u$ in $\mathbb{R}^8$ with $(\mathbf{w}_u, \mathbf{w}_w) = \pm\frac{1}{3}$ for $u \neq w$.

PROOF: By Lemma (the matrices of a regular graph), assertion 3, with ⟨0⟩:1, ⟨1⟩1 and ⟨1⟩, E is positive semidefinite of rank $\operatorname{rk} Q \leq 8$. Apply Lemma (Gram factorization), assertion 2, with $r = 8$; the entries of E are 1 on the diagonal and $\frac{2}{3} - \frac{1}{3}$ or $-\frac{1}{3}$ elsewhere. □

⟨2⟩6. $n \leq 28$.

PROOF: The span of the $\mathbf{w}_u$ of ⟨2⟩5 has dimension $k = \operatorname{rk} E \leq 8$ by Lemma (Gram factorization), assertion 1. If $k = 8$, then $n \leq 14$ by Lemma (lines spanning eight dimensions, EQ2); otherwise $k \leq 7$ and $n \leq 28$ by Lemma (the relative bound, EQ1). □

⟨2⟩7. $d \leq 16$, or $n = 28$ and $d = 17$.

PROOF: Computation C8, with ⟨2⟩4, ⟨2⟩6 and ⟨1⟩. □

⟨2⟩8. ASSUME: $n = 28$.

PROVE: The $\mathbf{w}_u$ of ⟨2⟩5 span a space of dimension at most 7.

PROOF: **[repair]** This is the split at $d = 12$ of Error 15 (errata): here $28 < 2d + 4$ by ⟨1⟩ and ⟨2⟩, so $d > 12$, and the case $d = 12$ is in ⟨1⟩5. The span lies in $\mathbb{R}^8$, so its dimension is at most 8 by `Submodule.finrank_le`; it is not 8, since then Lemma (lines spanning eight dimensions, EQ2) would give $n \leq 14$, against ⟨2⟩. □

⟨2⟩9. ASSUME: $n = 28$.

PROVE: $\operatorname{rk} Q \leq 7$.

PROOF: $\operatorname{rk} Q = \operatorname{rk} E$ by Lemma (the matrices of a regular graph), assertion 3, with ⟨0⟩:1, ⟨1⟩1 and ⟨1⟩; and $\operatorname{rk} E$ is the dimension of the span of the $\mathbf{w}_u$, by Lemma (Gram factorization), assertion 1, and ⟨2⟩5. Apply ⟨2⟩8 with ⟨2⟩. □

⟨2⟩10. ASSUME: $n = 28$.

PROVE: $d \leq 16$.

[new] The Lean proof, `SimpleGraph.not_regular_28_17_of_rank_le_seven`, closes the case $(n, d) = (28, 17)$ left by ⟨2⟩7 without the components: there $t = \frac{19}{56}$, and M has the entries $\frac{37}{56}$ on the diagonal, $\frac{9}{56}$ at the 17 neighbours and $-\frac{19}{56}$ at the other 10 vertices of each row, so $\operatorname{tr} M = \frac{37}{2}$ and $\sum_{u,w} M(u,w)^2 = \frac{227}{4}$. As $\operatorname{rk} M = \operatorname{rk} Q - 1 \leq 6$ by Lemma (the matrices of a regular graph), assertion 2, and ⟨2⟩9, Lemma (trace and rank) would give $(\frac{37}{2})^2 = 342.25 \leq 6 \cdot \frac{227}{4} = 340.5$, which is false. This is the argument of the remark after the theorem, and it needs neither Lemma (a root system containing the lines) nor Computation C20 here; we keep the route of the paper below.

LET: u be a vertex of X, X_u the connected component of X that contains u, and k the dimension of the span of the $\mathbf{x}_w$ with $w \in V(X_u)$.

⟨3⟩1. X_u is connected, the $\mathbf{x}_w$ with $w \in V(X_u)$ represent it in $\mathbb{R}^8$, and u has d neighbours in X_u.

PROOF: A component is connected and is the subgraph induced on its vertices, so Lemma (subfamilies of a representation) applies to ⟨0⟩:2. The component contains every neighbour of u, and u has d neighbours by ⟨0⟩:1. □

⟨3⟩2. $k \le 7$.

PROOF: The span of the $\mathbf{x}_w$ with $w \in V(X_u)$ lies in the span of all the $\mathbf{x}_w$ by `Submodule.span_mono`, and the latter has dimension $\operatorname{rk} Q$ by Lemma (Gram factorization), assertion 1; so $k \le \operatorname{rk} Q$ by `Submodule.finrank_mono`, and $\operatorname{rk} Q \le 7$ by ⟨2⟩9 with ⟨2⟩. □

⟨3⟩3. X_u is represented in the model of A_k, of D_k with $k \ge 4$, of E_6 or of E_7.

PROOF: Lemma (a root system containing the lines), assertion 1, applies by ⟨3⟩1; the model of E_8 is excluded, since it has rank $8 > k$ by Lemma (spans of the models) and ⟨3⟩2. □

⟨3⟩4. Q.E.D.

PROOF: By Computation C20 and ⟨3⟩3, u has at most $k-1$, $2(k-2)$, 10 or 16 neighbours in X_u; these are at most 16 by ⟨3⟩2. Apply ⟨3⟩1. □

⟨2⟩11. Q.E.D.

PROOF: $n \le 28$ by ⟨2⟩6. If $d > 16$, then $n = 28$ by ⟨2⟩7, and ⟨2⟩10 gives $d \le 16$; so $d \le 16$. □

⟨1⟩7. Q.E.D.

PROOF: The cases ⟨1⟩2, ⟨1⟩5 and ⟨1⟩6 are exhaustive. □

Computation C24. ASSUME: X is a 16-regular graph on 28 vertices, and $N = 2I + A(X) - \frac{9}{14}J$.
PROVE: $\operatorname{tr} N = 38$, $\sum_{u,w} N(u,w)^2 = 236$, and $38^2 > 6 \cdot 236$.

[new] Section 12 of `cgss/prerequisites.md` states C24 as $(2m-16)^2 > (27-m)(192-4m)$ for $21 \le m \le 27$, where m is the multiplicity of the eigenvalue -2. We will count entries in its place, which needs no multiplicities. With Lemma (trace and rank), C24 gives only $\operatorname{rk} N \ge 7$. The remark after Theorem 4.4 gets $\operatorname{rk}(2I + A) = 8$ from this, from assertion 2 of Lemma (the matrices of a regular graph), which gives $\operatorname{rk}(2I + A) = \operatorname{rk} N + 1$, and from the bound $\operatorname{rk}(2I + A) \le 8$ of a representation in E_8.

PROOF SKETCH: We will count the entries of each kind; the Lean proof counts them with the degrees and closes with `norm_num`.

⟨1⟩1. Each row of N has the entry $\frac{19}{14}$ on the diagonal, 16 entries $\frac{5}{14}$ and 11 entries $-\frac{9}{14}$.

PROOF: The entries of $2I + A(X)$ in a row are 2 on the diagonal, 1 at the 16 neighbours by ⟨0⟩, and 0 at the other 11 vertices; subtract $\frac{9}{14}$. □

⟨1⟩2. Q.E.D.

PROOF: By ⟨1⟩1, $\operatorname{tr} N = 28 \cdot \frac{19}{14} = 38$ and $\sum_{u,w} N(u,w)^2 = \frac{28}{196}(361 + 16 \cdot 25 + 11 \cdot 81) = \frac{1652}{7} = 236$; and $38^2 = 1444 > 1416 = 6 \cdot 236$. □

Remark (after Theorem 4.4). ASSUME: X is a graph.
PROVE:
1. If X is 16-regular on 28 vertices, then X is not represented in E_8.
2. The bound $n \le 28$ of Theorem 4.4 is attained by $L(K_8)$ and by the Chang graphs I, II and III, which are 12-regular on 28 vertices, and the bound $d \le 16$ by the Schläfli graph, which is 16-regular on 27 vertices; all five are represented in E_8.

PROOF SKETCH: **[repair]** The paper gives no proof of assertion 1; we will follow EQ3 of `cgss/prerequisites.md`, the repair of Gap 7 (errata). The rank of $2I + A$ is 8, so the 28 equiangular lines of the proof of Theorem 4.4 span 8 dimensions, which Lemma EQ2 forbids. **[new]** We will bound the rank by Lemma (trace and rank) and Computation C24, in place of the count of eigenvalues in EQ3. Assertion 2 cites the examples below. **[new]** The Lean proof, `SimpleGraph.not_regular_28_16_isRepresentedIn_E8_of_classification`, runs in the other order: EQ2 first, which gives $\operatorname{rk} Q \le 7$ (`SimpleGraph.IsRepresentedIn.rank_le_seven_of_regular_of_classification`, the argument of the proof of Theorem 4.4 for $n = 28$), so that

$\operatorname{rk} N \le 6$; then C24 and Lemma (trace and rank) give $38^2 \le 6 \cdot 236$, which is false. It carries the same classification hypothesis as Theorem 4.4; the settled name, without it, is proved in `LineGraphs/Representation/Theorem44Unconditional.lean` in the same way.

⟨1⟩1. ASSUME: 1. X is 16-regular on 28 vertices.
2. X is represented in E_8 by $(\mathbf{x}_u)$.
PROVE: FALSE.

LET: $Q = 2I + A(X)$, $M = \frac{1}{2}Q - \frac{9}{28}J$ and $E = \frac{2}{3}Q - \frac{1}{3}J$, the matrices of Lemma (the matrices of a regular graph) for $d = 16$ and $n = 28$, where $t = \frac{18}{56} = \frac{9}{28}$.

⟨2⟩1. Q is positive semidefinite of rank at most 8.
PROOF: Q is the Gram matrix of the $\mathbf{x}_u \in \mathbb{R}^8$ by ⟨1⟩:2, so it is positive semidefinite by `Matrix.posSemidef_gram`, and of rank at most 8 by Lemma (Gram factorization), assertion 1, and `Submodule.finrank_le`. □

⟨2⟩2. $\operatorname{rk} M \ge 7$.
PROOF: $2M$ is the matrix N of Computation C24, by ⟨1⟩:1. By Lemma (trace and rank) and that computation, $38^2 \le \operatorname{rk}(2M) \cdot 236$ and $38^2 > 6 \cdot 236$; so $\operatorname{rk} M = \operatorname{rk}(2M) \ge 7$. □

⟨2⟩3. $\operatorname{rk} Q = 8$.
PROOF: $\operatorname{rk} Q = \operatorname{rk} M + 1 \ge 8$ by Lemma (the matrices of a regular graph), assertion 2, with ⟨1⟩:1, ⟨2⟩1 and ⟨2⟩2; and $\operatorname{rk} Q \le 8$ by ⟨2⟩1. □

⟨2⟩4. E is positive semidefinite of rank 8.
PROOF: As $28 < 2 \cdot 16 + 4$, Lemma (the matrices of a regular graph), assertion 3, applies with ⟨1⟩:1 and ⟨2⟩1; it makes E positive semidefinite with $\operatorname{rk} E = \operatorname{rk} Q$, and $\operatorname{rk} Q = 8$ by ⟨2⟩3. □

⟨2⟩5. E is the Gram matrix of 28 unit vectors $\mathbf{w}_u$ of $\mathbb{R}^8$ with $(\mathbf{w}_u, \mathbf{w}_w) = \pm\frac{1}{3}$ for $u \ne w$.
PROOF: Lemma (Gram factorization), assertion 2 with $r = 8$, applies by ⟨2⟩4; the entries of E are 1 on the diagonal and $\frac{2}{3} - \frac{1}{3}$ or $-\frac{1}{3}$ elsewhere. □

⟨2⟩6. The $\mathbf{w}_u$ of ⟨2⟩5 span a space of dimension 8.
PROOF: Its dimension is $\operatorname{rk} E$ by Lemma (Gram factorization), assertion 1, and ⟨2⟩5; and $\operatorname{rk} E = 8$ by ⟨2⟩4. □

⟨2⟩7. Q.E.D.
PROOF: Lemma (lines spanning eight dimensions, EQ2), for the span of ⟨2⟩6 and the $\mathbf{w}_u$ of ⟨2⟩5, gives $28 \le 14$. □

⟨1⟩2. Assertion 2 holds.

⟨2⟩1. $L(K_8)$ is 12-regular on 28 vertices and is represented in E_8.
PROOF: K_8 has $\binom{8}{2} = 28$ edges, and each has degree $7+7-2 = 12$ in $L(K_8) \cong L(K_8; 0, \dots, 0)$, by Computation C10 and Definition (generalized line graphs). Lemma (the line graph) gives the representation by the $\mathbf{e}_i + \mathbf{e}_j$, which lie in $\Phi(\mathrm{D}_8) \subseteq \Phi(\mathrm{E}_8)$ by Definition 3.2, (ii). □

⟨2⟩2. Q.E.D.
PROOF: Example 4.5 gives the Chang graphs, with parameters $(28, 12, 6, 4)$, and Example 4.6 the Schläfli graph, with parameters $(27, 16, 10, 8)$, each represented in E_8; add ⟨2⟩1. □

⟨1⟩3. Q.E.D.
PROOF: ⟨1⟩1 and ⟨1⟩2. □

The paper introduces the graphs of Examples 4.5 to 4.9 as strongly regular graphs "which are not line graphs". We will prove this in the proof of Remark 4.14, as Corollary (the seven graphs are not line graphs), since the argument uses the properties of the seven graphs that we establish there. We will use model (ii) of Definition 3.2 and the vector $\mathbf{c} = \frac{1}{2}\sum_{k=1}^{8} \mathbf{e}_k$ of $\Phi(\mathrm{E}_8)$. For a graph X and $Y \subseteq V(X)$, the *switched graph* X^Y keeps the pairs inside Y and inside $V(X) \setminus Y$, and complements the pairs between them (SW1 of `cgss/prerequisites.md`).

Lemma (switching by vectors, SW4). ASSUME: 1. X is represented in a star-closed line system S by $(\mathbf{x}_u)$.

2. $\mathbf{h} \in \Phi(S)$, $(\mathbf{h}, \mathbf{x}_u) = 1$ for every u, and $Y \subseteq V(X)$.

PROVE: The family with $\mathbf{h} - \mathbf{x}_u$ for $u \in Y$ and $\mathbf{x}_u$ for $u \notin Y$ represents X^Y in S.

PROOF SKETCH: We will compute the new inner products.

⟨1⟩1. $\mathbf{h} - \mathbf{x}_u \in \Phi(S)$.

PROOF: $(\mathbf{x}_u, -\mathbf{h}) = -1$ by ⟨0⟩:2, so $\mathbf{x}_u - \mathbf{h}$ and its negative lie in $\Phi(S)$, which is star-closed by ⟨0⟩:1. □

⟨1⟩2. For $u, w \in Y$, $(\mathbf{h}-\mathbf{x}_u, \mathbf{h}-\mathbf{x}_w) = (\mathbf{x}_u, \mathbf{x}_w)$; and for $u \in Y$, $w \notin Y$, $(\mathbf{h}-\mathbf{x}_u, \mathbf{x}_w) = 1-(\mathbf{x}_u, \mathbf{x}_w)$.

PROOF: Expand, with $(\mathbf{h}, \mathbf{h}) = 2$ and $(\mathbf{h}, \mathbf{x}_u) = (\mathbf{h}, \mathbf{x}_w) = 1$ by ⟨0⟩:2. □

⟨1⟩3. Q.E.D.

PROOF: By ⟨1⟩2, the Gram matrix keeps the entries inside Y and outside Y, and replaces $0, 1$ by $1, 0$ between them; that is $2I + A(X^Y)$. The vectors lie in $\Phi(S)$ by ⟨1⟩1. □

Computation C9. ASSUME: The graphs of Examples 4.5 to 4.9 are given by the integer vectors $2\mathbf{x}_u$ of their representations. The folded 5-cube $\square_5$ is the graph on the subsets of $\{1, \dots, 5\}$ of even size, adjacent when their symmetric difference has 4 elements, and the double-six graph Q_{27} is the graph on the symbols a_i, b_i and $c_{ij} = c_{ji}$ ($i \neq j$ in $\{1, \dots, 6\}$) with $a_i \sim b_j$ for $i \neq j$, $a_i \sim c_{jl}$ and $b_i \sim c_{jl}$ for $i \in \{j, l\}$, and $c_{ij} \sim c_{lm}$ for disjoint $\{i, j\}$, $\{l, m\}$; these are the folded 5-cube and the double-six graph of the proof of Remark 4.14.

PROVE: The Chang graphs are strongly regular with parameters $(28, 12, 6, 4)$, the Schläfli graph $(27, 16, 10, 8)$, the Shrikhande graph $(16, 6, 2, 2)$, the Clebsch graph $(16, 10, 6, 6)$ and the Petersen graph of Example 4.9 $(10, 3, 0, 1)$; the last is isomorphic to the Kneser graph of the 2-subsets of a 5-set; the Clebsch graph is switching equivalent to $L(K_{4,4})$ and to the Shrikhande graph. The Clebsch graph is isomorphic to $\overline{\square_5}$, and the Schläfli graph to $\overline{Q_{27}}$.

PROOF SKETCH: These are finite checks on at most 28 vertices; the Lean proofs are by `decide` on the adjacency matrices $A(u, w) = \frac{1}{4}(2\mathbf{x}_u, 2\mathbf{x}_w)$, with explicit isomorphisms and switching sets as certificates. The two isomorphisms of the last sentence are the certificates of `cgss/proofs/clebsch/clebsch.tex`, Section 6, found by SageMath; in Lean they are definitions, checked by `decide`.

⟨1⟩1. Q.E.D.

PROOF: For each graph, `decide` checks the degree of every vertex and the number of common neighbours of every pair; for the Petersen graph it checks a given bijection, for the Clebsch graph given switching sets and bijections, and for the Clebsch and Schläfli graphs given bijections onto $\overline{\square_5}$ and $\overline{Q_{27}}$. □

Example 4.5. ASSUME: $T(8) = L(K_8)$ is represented by the vectors $\mathbf{e}_i + \mathbf{e}_j$, and the switching sets are $Y_{\mathrm{I}} = \{\mathbf{e}_i + \mathbf{e}_{4+i} : 1 \le i \le 4\}$, $Y_{\mathrm{II}} = \{\mathbf{e}_i + \mathbf{e}_{i+1} : 1 \le i \le 8\}$ (indices modulo 8) and $Y_{\mathrm{III}} = \{\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_3 + \mathbf{e}_1\} \cup \{\mathbf{e}_4 + \mathbf{e}_5, \mathbf{e}_5 + \mathbf{e}_6, \mathbf{e}_6 + \mathbf{e}_7, \mathbf{e}_7 + \mathbf{e}_8, \mathbf{e}_8 + \mathbf{e}_4\}$.

PROVE: Replacing $\mathbf{e}_i + \mathbf{e}_j$ by $\mathbf{c} - \mathbf{e}_i - \mathbf{e}_j$ on Y_{I}, Y_{II} or Y_{III} gives vectors of $\Phi(\mathrm{E}_8)$ representing the Chang graphs I, II and III, which are strongly regular with parameters $(28, 12, 6, 4)$.

PROOF SKETCH: We will switch with $\mathbf{h} = \mathbf{c}$.

⟨1⟩1. $T(8)$ is represented in E_8 by the $\mathbf{e}_i + \mathbf{e}_j$, and $(\mathbf{c}, \mathbf{e}_i + \mathbf{e}_j) = 1$.

PROOF: Lemma (the line graph); the vectors lie in $\Phi(\mathrm{D}_8) \subseteq \Phi(\mathrm{E}_8)$ by Definition 3.2, (ii). □

⟨1⟩2. Q.E.D.

PROOF: E_8 is star-closed by Theorem 3.5; apply Lemma (switching by vectors, SW4) to ⟨1⟩1, and Computation C9 for the parameters. □

Example 4.6. ASSUME: $\mathbf{x}_{ij} = \mathbf{e}_i + \mathbf{e}_j$ for $1 \le i < j \le 6$, and $\mathbf{y}_i = \mathbf{c} - \mathbf{e}_i - \mathbf{e}_7$ and $\mathbf{z}_i = \mathbf{c} - \mathbf{e}_i - \mathbf{e}_8$ for $1 \le i \le 6$.

PROVE: These 27 vectors lie in $\Phi(\mathrm{E}_8)$ and represent the Schläfli graph, which is strongly regular with parameters $(27, 16, 10, 8)$.

[repair] The paper prints $\frac{1}{3}\sum \mathbf{e}_k$ in the second family, which gives squared norm $14/9$; we use $\mathbf{c}$ (errata, Error 2).

PROOF SKETCH: We will check membership and the inner products.

⟨1⟩1. The vectors lie in $\Phi(\mathrm{E}_8)$.

PROOF: $\mathbf{x}_{ij} \in \Phi(\mathrm{D}_8)$; $\mathbf{y}_i$ and $\mathbf{z}_i$ have entries $\pm\frac{1}{2}$ with exactly two negative ones, so they lie in model (ii) of Definition 3.2. □

⟨1⟩2. For 2-subsets A, B of $\{1, \dots, 8\}$, $(\mathbf{c} - \sum_{a\in A} \mathbf{e}_a, \mathbf{c} - \sum_{b\in B} \mathbf{e}_b) = |A \cap B|$ and $(\mathbf{e}_i + \mathbf{e}_j, \mathbf{c} - \sum_{b\in B} \mathbf{e}_b) = 1 - |\{i, j\} \cap B|$.

PROOF: Expand, with $(\mathbf{c}, \mathbf{c}) = 2$ and $(\mathbf{c}, \mathbf{e}_k) = \frac{1}{2}$. □

⟨1⟩3. Distinct vectors have inner products in $\{0, 1\}$.

PROOF: $(\mathbf{x}_{ij}, \mathbf{x}_{kl}) = |\{i, j\} \cap \{k, l\}|$; by ⟨1⟩2, $(\mathbf{y}_i, \mathbf{y}_k) = (\mathbf{z}_i, \mathbf{z}_k) = 1$ for $i \neq k$, $(\mathbf{y}_i, \mathbf{z}_k) = [i = k]$, and $(\mathbf{x}_{ij}, \mathbf{y}_k) = (\mathbf{x}_{ij}, \mathbf{z}_k) = 1 - [k \in \{i, j\}]$. □

⟨1⟩4. Q.E.D.

PROOF: By ⟨1⟩1 and ⟨1⟩3 the vectors represent a graph (Definition (representations)); Computation C9 gives its parameters. □

Example 4.7. ASSUME: $L(K_{4,4})$ is represented by the vectors $\mathbf{e}_i + \mathbf{e}_j$ with $1 \le i \le 4 < j \le 8$, and $Y = \{\mathbf{e}_i + \mathbf{e}_{4+i} : 1 \le i \le 4\}$.

PROVE: Replacing $\mathbf{e}_i + \mathbf{e}_j$ by $\mathbf{c} - \mathbf{e}_i - \mathbf{e}_j$ on Y gives vectors of $\Phi(\mathrm{E}_8)$ representing the Shrikhande graph, strongly regular with parameters $(16, 6, 2, 2)$; it is an induced subgraph of the Chang graph I.

In Lean the last assertion is `SimpleGraph.shrikhande.nonempty_iso_induce_changI`: the Shrikhande graph is isomorphic to the subgraph of the Chang graph I induced on the sixteen vertices given by `SimpleGraph.shrikhande.changIIndex`, an induced embedding that the kernel checks (`decide +kernel`).

PROOF SKETCH: We will switch with $\mathbf{h} = \mathbf{c}$, as in Example 4.5.

⟨1⟩1. The switched vectors lie in $\Phi(\mathrm{E}_8)$ and represent the Shrikhande graph.

PROOF: Lemma (the line graph) for $K_{4,4}$ gives a representation in $\Phi(\mathrm{D}_8) \subseteq \Phi(\mathrm{E}_8)$ (Definition 3.2) with $(\mathbf{c}, \mathbf{e}_i + \mathbf{e}_j) = 1$; E_8 is star-closed by Theorem 3.5; apply Lemma (switching by vectors, SW4). □

⟨1⟩2. Q.E.D.

PROOF: Computation C9 gives the parameters. The switched vectors are among those of the Chang graph I in Example 4.5, since $Y = Y_{\mathrm{I}}$, and a subfamily of a representation represents the induced subgraph (Lemma (subfamilies of a representation)). □

Example 4.8. ASSUME: The vectors are $\mathbf{e}_i + \mathbf{e}_5$, $\mathbf{e}_i + \mathbf{e}_6$, $\mathbf{c} - \mathbf{e}_i - \mathbf{e}_7$ and $\mathbf{c} - \mathbf{e}_i - \mathbf{e}_8$, for $1 \le i \le 4$.

PROVE: These 16 vectors are the vectors of Example 4.6 with inner product 1 with $\mathbf{e}_5 + \mathbf{e}_6$; they represent the Clebsch graph, strongly regular with parameters $(16, 10, 6, 6)$ and switching equivalent to $L(K_{4,4})$ and to the Shrikhande graph.

In Lean the first two assertions together are `SimpleGraph.clebsch.nonempty_iso_induce_schlafli`: the Clebsch graph is isomorphic to the subgraph of the Schläfli graph induced on the neighbours of the vertex whose vector is $\mathbf{e}_5 + \mathbf{e}_6$, which are the vertices whose vectors have inner product 1 with it. The kernel checks that `SimpleGraph.clebsch.schlafliIndex` is an induced embedding onto these neighbours.

PROOF SKETCH: We will read off the neighbours of $\mathbf{e}_5 + \mathbf{e}_6$ in Example 4.6.

⟨1⟩1. The vectors of Example 4.6 with inner product 1 with $\mathbf{e}_5 + \mathbf{e}_6$ are the 16 listed.

PROOF: Expand: $(\mathbf{e}_5+\mathbf{e}_6, \mathbf{e}_k+\mathbf{e}_l) = |\{5,6\}\cap\{k,l\}|$, which is 1 exactly for $\{k,l\}=\{i,5\}$ or $\{i,6\}$ with $i\le 4$; and $(\mathbf{e}_5+\mathbf{e}_6, \mathbf{c}-\mathbf{e}_k-\mathbf{e}_7) = 1-|\{5,6\}\cap\{k\}|$, which is 1 exactly for $k\le 4$, and the same for $\mathbf{e}_8$. □

⟨1⟩2. Q.E.D.

PROOF: A subfamily of the representation of Example 4.6 represents the induced subgraph (Lemma (subfamilies of a representation)); Computation C9 gives the parameters and the switching equivalences. □

Example 4.9. ASSUME: $L(K_{3,3})$ together with an isolated vertex is represented by $\mathbf{e}_4+\mathbf{e}_8$ and the $\mathbf{e}_i+\mathbf{e}_j$ with $1\le i\le 3$ and $5\le j\le 7$, and $Y=\{\mathbf{e}_i+\mathbf{e}_{4+i} : 1\le i\le 3\}$.

PROVE: Replacing $\mathbf{e}_i+\mathbf{e}_j$ by $\mathbf{c}-\mathbf{e}_i-\mathbf{e}_j$ on Y gives vectors of $\Phi(\mathrm{E}_8)$ representing the Petersen graph, strongly regular with parameters $(10,3,0,1)$.

PROOF SKETCH: We will switch with $\mathbf{h}=\mathbf{c}$, as in Example 4.5.

⟨1⟩1. The ten vectors represent $L(K_{3,3})$ with an isolated vertex, in $\Phi(\mathrm{E}_8)$, and each has inner product 1 with $\mathbf{c}$.

PROOF: Lemma (the line graph) for the graph with the edges $\{i,j\}$ and $\{4,8\}$, whose line graph is $L(K_{3,3})$ with an isolated vertex; the vectors lie in $\Phi(\mathrm{D}_8)\subseteq\Phi(\mathrm{E}_8)$ by Definition 3.2. □

⟨1⟩2. Q.E.D.

PROOF: E_8 is star-closed by Theorem 3.5; apply Lemma (switching by vectors, SW4) to ⟨1⟩1, and Computation C9 for the parameters and the isomorphism with the Petersen graph. □

4.3. Theorems 4.10 and 4.11, and Remark 4.12.

Theorem 4.10. ASSUME: X is a connected graph, and $2I+A(X)$ is positive semidefinite.

PROVE: X is a generalized line graph, or X is represented in E_8.

[repair] The paper does not assume that X is connected, and the theorem is false without it: two disjoint Petersen graphs are neither (errata, Implicit hypothesis 1). We assume only that the least eigenvalue is at least -2; the paper assumes that it is -2.

PROOF SKETCH: "This follows from the Theorems 3.6 and 4.2, and the inclusion $\mathrm{A}_n\subseteq\mathrm{D}_{n+1}$."

⟨1⟩1. X is represented in A_ℓ $(\ell\ge 1)$, in D_ℓ $(\ell\ge 4)$, or in E_8.

PROOF: X has a representation in $\mathbb{R}^{|V(X)|}$ by Lemma (the Gram matrix) and ⟨0⟩; apply Lemma (a root system containing the lines), assertion 2. □

⟨1⟩2. CASE: X is represented in A_ℓ.

PROVE: X is a generalized line graph.

PROOF: $\Phi(\mathrm{A}_\ell)\subseteq\Phi(\mathrm{D}_{\ell+1})$ by Definition 3.1 and its remark on the coordinate model for small ℓ, so X is represented in $\mathrm{D}_{\ell+1}$ by Lemma (transport of representations); apply Theorem 4.2, assertion 1. □

⟨1⟩3. CASE: X is represented in D_ℓ.

PROVE: X is a generalized line graph.

PROOF: Theorem 4.2, assertion 1. □

⟨1⟩4. Q.E.D.

PROOF: The cases of ⟨1⟩1 are exhaustive, and in the third X is represented in E_8. □

Computation C10. ASSUME: Y is a graph on $\{1,\dots,m\}$, $\deg(i)$ is the degree of i in Y, and $a_1,\dots,a_m\ge 0$ are integers.

PROVE: In $L(Y;a_1,\dots,a_m)$ the edge $\{i,j\}$ has degree $2(a_i+a_j)+\deg(i)+\deg(j)-2$, and the vertex (i,α,s) has degree $2(a_i-1)+\deg(i)$.

PROOF SKETCH: We will count the neighbours of each kind; the Lean proof is parametric.

⟨1⟩1. The edge $\{i,j\}$ has $\deg(i)+\deg(j)-2$ neighbours among the edges and $2a_i+2a_j$ among the triples.

PROOF: The edges other than $\{i,j\}$ on i or on j number $(\deg(i)-1)+(\deg(j)-1)$, since no other edge is on both. The triples adjacent to it are the (i,α,s) and the (j,β,t), by Definition (generalized line graphs). □

⟨1⟩2. (i,α,s) has $\deg(i)$ neighbours among the edges and $2(a_i-1)$ among the triples.
PROOF: Its neighbours are the edges on i and the (i,β,t) with $\beta\neq\alpha$, by Definition (generalized line graphs). □

⟨1⟩3. Q.E.D.
PROOF: Add the two counts in ⟨1⟩1 and in ⟨1⟩2. □

Lemma (regular connected generalized line graphs). ASSUME: $X\cong L(Y;a_1,\ldots,a_m)$ is connected and regular, with Y on $\{1,\ldots,m\}$ and $\deg(i)$ the degree of i in Y.
PROVE: $X\cong L(Y')$ for some graph Y', or $X\cong CP(a)$ for some a.

PROOF SKETCH: This is the paper's first paragraph of the proof of Theorem 4.11. **[repair]** The paper says that Y is connected, which need not hold (errata, Gap 18), since vertices i of Y with $\deg(i)=0$ and $a_i=0$ do not appear in X; we will split on the vertices with $a_i>0$ instead.

⟨1⟩1. CASE: $a_i=0$ for every i.
PROVE: $X\cong L(Y)$.
PROOF: $L(Y;0,\ldots,0)\cong L(Y)$ by Definition (generalized line graphs). □

⟨1⟩2. CASE: $a_i>0$ and $\deg(i)=0$ for some i.
PROVE: $X\cong CP(a_i)$.
PROOF: No edge of Y contains i, so by Definition (generalized line graphs) the $2a_i\geq 2$ triples (i,α,s) have all their neighbours among themselves. Since X is connected, they are all its vertices, and they span a copy of $CP(a_i)$. □

⟨1⟩3. CASE: $a_i>0$ and $\deg(i)>0$ for some i.
PROVE: This case does not occur.
PROOF: Pick an edge $\{i,j\}$. By regularity and C10 (Computation C10), $2(a_i+a_j)+\deg(i)+\deg(j)-2=2(a_i-1)+\deg(i)$, so $2a_j+\deg(j)=0$; but $\deg(j)\geq 1$. □

⟨1⟩4. Q.E.D.
PROOF: Either every a_i is 0, or some $a_i>0$, and then $\deg(i)=0$ or $\deg(i)>0$; so the three cases are exhaustive. □

Theorem 4.11. ASSUME: X is a connected regular graph, and $2I+A(X)$ is positive semidefinite.
PROVE: $X\cong L(Y)$ for some graph Y, or $X\cong CP(m)$ for some m, or X is represented in E_8.
[new] As in Theorem 4.10, we assume only that the least eigenvalue is at least -2.

PROOF SKETCH: We will apply Theorem 4.10, and then the lemma on regular generalized line graphs.

⟨1⟩1. X is a generalized line graph, or X is represented in E_8.
PROOF: Theorem 4.10, with ⟨0⟩. □

⟨1⟩2. CASE: $X\cong L(Y;a_1,\ldots,a_m)$.
PROVE: $X\cong L(Y')$ for some Y' or $X\cong CP(a)$ for some a.
PROOF: Lemma (regular connected generalized line graphs), with ⟨0⟩ and ⟨1⟩. □

⟨1⟩3. Q.E.D.
PROOF: The cases of ⟨1⟩1 are exhaustive, by Definition (generalized line graphs). □

Definition (claws). ASSUME: X is a graph and $k\geq 1$.
LET: A *k-claw* of X with *centre* u is a set of k pairwise non-adjacent neighbours of u; with u, they induce a copy of $K_{1,k}$. The graph X is *claw-free* if it has no 3-claw, and *four-claw-free* if it has no 4-claw.

In Lean, `X.IsClaw u T` and `X.IsFourClaw u T` will say that the finite set T is a 3-claw, or a 4-claw, of X with centre u. Following `cgss/notation.md`, the predicates `ClawFree` and

`FourClawFree` will say instead that there is no induced copy of the star graph `SimpleGraph.starGraph` on 4 or 5 vertices; the lemma below proves that the two readings agree. A 4-claw contains a 3-claw with the same centre.

Lemma (claws and stars). ASSUME: X is a graph, $k \geq 1$, and $K_{1,k}$ is the star graph `SimpleGraph.starGraph` on $\{0, \ldots, k\}$ with centre 0.
PROVE: 1. X has an induced copy of $K_{1,k}$ (`SimpleGraph.IsIndContained`) if and only if X has a k-claw.
2. If $X \cong X'$, then X has an induced copy of $K_{1,k}$ if and only if X' has one.

[new] The Lean predicates of Definition (claws) forbid an induced star, while our arguments count claws; the Lean names are the cases $k = 3$ and $k = 4$.

PROOF SKETCH: We will send the centre of the star to the centre of the claw.
⟨1⟩1. ASSUME: f embeds $K_{1,k}$ into X as an induced subgraph.
PROVE: $\{f(1), \ldots, f(k)\}$ is a k-claw of X with centre $f(0)$.
PROOF: By `SimpleGraph.starGraph_adj`, each l with $1 \leq l \leq k$ is adjacent to 0 in $K_{1,k}$, and no two of them are adjacent. As f is injective and preserves adjacency and non-adjacency by ⟨1⟩, the $f(l)$ are k distinct neighbours of $f(0)$, and no two of them are adjacent. □
⟨1⟩2. ASSUME: $T = \{t_1, \ldots, t_k\}$ is a k-claw of X with centre u.
PROVE: The map f with $f(0) = u$ and $f(l) = t_l$ for $1 \leq l \leq k$ embeds $K_{1,k}$ into X as an induced subgraph.
PROOF: $u \notin T$, since u is adjacent to every member of T by ⟨1⟩ and not to itself; so f is injective. By ⟨1⟩ and Definition (claws), $f(0)$ is adjacent to every $f(l)$ with $l \geq 1$, and no two $f(l)$ with $l \geq 1$ are adjacent; these are the adjacencies of $K_{1,k}$, by `SimpleGraph.starGraph_adj`. □
⟨1⟩3. ASSUME: e is an isomorphism from X to X'.
PROVE: Assertion 2 holds.
PROOF: Compose an embedding of $K_{1,k}$ with e or with its inverse, by `SimpleGraph.Iso.isIndContained`, `SimpleGraph.Iso.isIndContained'` and `SimpleGraph.IsIndContained.trans`. □
⟨1⟩4. Q.E.D.
PROOF: Assertion 1 is ⟨1⟩1 and ⟨1⟩2; assertion 2 is ⟨1⟩3. □

Lemma (claws in a generalized line graph). ASSUME: Y is a graph on $\{1, \ldots, m\}$, $\deg(i)$ is the degree of i in Y, $a_1, \ldots, a_m \geq 0$ are integers, and $X = L(Y; a_1, \ldots, a_m)$.

In the Lean names, as in those of Computation C10, a *petal* of X is a vertex (i, α, s) of the cocktail party graph attached at i; it is not the petal of the proof of Theorem 4.2, which is a coordinate.
PROVE: 1. No vertex (i, α, s) of X is the centre of a 3-claw.
2. The edge $\{i, j\}$ of Y is the centre of a 3-claw of X if and only if $a_i \geq 1$ and $a_j + \deg(j) \geq 2$, or $a_j \geq 1$ and $a_i + \deg(i) \geq 2$.
3. The edge $\{i, j\}$ of Y is the centre of a 4-claw of X if and only if $a_i \geq 1$ and $a_j \geq 1$.

PROOF SKETCH: We will split the neighbours of an edge into those at its two ends.
LET: For an edge $e = \{i, j\}$ of Y, $N_i(e)$ is the set of the edges $f \neq e$ of Y with $i \in f$ together with the triples (i, α, s), and $N_j(e)$ is defined in the same way.
⟨1⟩1. The neighbours of $e = \{i, j\}$ in X are the members of $N_i(e) \cup N_j(e)$, and $|N_j(e)| = \deg(j) - 1 + 2a_j$.
PROOF: By Definition (generalized line graphs), e is adjacent to the edges that share an end with it and to the triples (l, α, s) with $l \in e$. The edges $f \neq e$ on j number $\deg(j) - 1$, and the triples (j, α, s) number $2a_j$. □
⟨1⟩2. ASSUME: $T \subseteq N_i(e)$ is a set of pairwise non-adjacent vertices with $|T| \geq 2$.

PROVE: $T = \{(i,\alpha,0),(i,\alpha,1)\}$ for some α; in particular $a_i \geq 1$.
PROOF: By Definition (generalized line graphs), two edges on i are adjacent, a triple (i,α,s) is adjacent to every edge on i, and (i,α,s) and (i,β,r) are adjacent when $\alpha \neq \beta$. So T consists of triples with one value of α, and there are two of these. □

⟨1⟩3. A triple (i,α,s) is adjacent to no member of $N_j(e)$.
PROOF: An edge $f \neq e$ on j does not contain i, since e is the only edge on both i and j; and a triple (j,β,r) has $j \neq i$. Apply Definition (generalized line graphs). □

⟨1⟩4. Assertion 1 holds.
PROOF: By Definition (generalized line graphs), the neighbours of (i,α,s) are the edges on i, each adjacent to all the others, and the triples (i,β,r) with $\beta \neq \alpha$. So a 3-claw would consist of three triples (i,β,r) with one value of β, and there are only two of these. □

⟨1⟩5. Assertion 2 holds.
⟨2⟩1. ASSUME: $a_i \geq 1$ and $a_j + \deg(j) \geq 2$.
PROVE: $e = \{i,j\}$ is the centre of a 3-claw.
PROOF: Since $\deg(j) \geq 1$, ⟨2⟩ gives $\deg(j) - 1 + 2a_j \geq 1$, so $N_j(e)$ has a member z by ⟨1⟩1. The vertices $(i,1,0)$, $(i,1,1)$ and z are neighbours of e by ⟨1⟩1; the first two are not adjacent by Definition (generalized line graphs), and neither is adjacent to z by ⟨1⟩3. □
⟨2⟩2. ASSUME: T is a 3-claw with centre $e = \{i,j\}$.
PROVE: $a_i \geq 1$ and $a_j + \deg(j) \geq 2$, or $a_j \geq 1$ and $a_i + \deg(i) \geq 2$.
PROOF: $T \subseteq N_i(e) \cup N_j(e)$ by ⟨1⟩1, so one of the two meets T in at least two members; by the symmetry of i and j, say $N_i(e)$. By ⟨1⟩2, $a_i \geq 1$ and $|T \cap N_i(e)| = 2$, so $N_j(e)$ meets T. Thus $\deg(j) - 1 + 2a_j \geq 1$ by ⟨1⟩1, which gives $a_j + \deg(j) \geq 2$, as $\deg(j) \geq 1$. □
⟨2⟩3. Q.E.D.
PROOF: ⟨2⟩1, with i and j exchanged for the second alternative, and ⟨2⟩2. □

⟨1⟩6. Assertion 3 holds.
⟨2⟩1. ASSUME: $a_i \geq 1$ and $a_j \geq 1$.
PROVE: $e = \{i,j\}$ is the centre of a 4-claw.
PROOF: The triples $(i,1,0)$, $(i,1,1)$, $(j,1,0)$ and $(j,1,1)$ are neighbours of e by ⟨1⟩1. The two at i are not adjacent, nor are the two at j, by Definition (generalized line graphs); and those at i are not adjacent to those at j by ⟨1⟩3. □
⟨2⟩2. ASSUME: T is a 4-claw with centre $e = \{i,j\}$.
PROVE: $a_i \geq 1$ and $a_j \geq 1$.
PROOF: $T \subseteq N_i(e) \cup N_j(e)$ by ⟨1⟩1, and each of $T \cap N_i(e)$ and $T \cap N_j(e)$ has at most two members by ⟨1⟩2. So each has exactly two, and ⟨1⟩2 gives $a_i \geq 1$ and $a_j \geq 1$. □
⟨2⟩3. Q.E.D.
PROOF: ⟨2⟩1 and ⟨2⟩2. □

⟨1⟩7. Q.E.D.
PROOF: ⟨1⟩4, ⟨1⟩5 and ⟨1⟩6. □

Remark 4.12. ASSUME: X is a graph.
PROVE:
1. Line graphs and cocktail party graphs are claw-free.
2. Let Y be a graph on $\{1,\ldots,m\}$, with $\deg(i)$ the degree of i in Y, and let $a_1,\ldots,a_m \geq 0$ be integers. Then $L(Y;a_1,\ldots,a_m)$ is claw-free if and only if $a_j = 0$ and $\deg(j) = 1$ for every vertex i of Y with $a_i \geq 1$ and every neighbour j of i.
3. If X is a connected generalized line graph, then X is claw-free if and only if $X \cong L(Y')$ for some graph Y' or $X \cong L(K_{1,s};t,0,\ldots,0)$ for some $s,t \geq 0$, where the vertex 1 of $K_{1,s}$ is its centre. The graph $L(K_{1,s};t,0,\ldots,0)$ is K_{s+2t} with t pairwise disjoint edges deleted.

4. If X is connected and claw-free, $2I + A(X)$ is positive semidefinite, and X has more than 36 vertices or a vertex with more than 28 neighbours, then $X \cong L(Y')$ for some graph Y' or $X \cong L(K_{1,s}; t, 0, \ldots, 0)$ for some $s, t \geq 0$.
5. For Y and $a_1, \ldots, a_m$ as in assertion 2, $L(Y; a_1, \ldots, a_m)$ is four-claw-free if and only if $a_i = 0$ or $a_j = 0$ for every edge $\{i, j\}$ of Y.

[new] The paper's $L(K_{1,n}; a, 0, \ldots, 0)$ is our $L(K_{1,s}; t, 0, \ldots, 0)$, since n and a are spoken for (`cgss/notation.md`), and its "sufficiently large" is made precise by Theorem 4.3. The remark on Ray-Chaudhuri's theorem is history, and needs no proof.

PROOF SKETCH: **[repair]** The paper says "it is easy to see"; these are the full arguments that Gap 9 (errata) asks for. We will read everything off the lemma on claws.

⟨1⟩1. Assertion 2 holds.
PROOF: Claw-freeness is the absence of 3-claws, by Definition (claws) and, for the Lean predicate, Lemma (claws and stars), assertion 1. By Lemma (claws in a generalized line graph), assertion 1, only edges of Y can be centres of 3-claws, and by its assertion 2 the edge $\{i, j\}$ is not one exactly when $a_i \geq 1$ implies $a_j + \deg(j) \leq 1$, and the same with i and j exchanged. As $\deg(j) \geq 1$, $a_j + \deg(j) \leq 1$ means that $a_j = 0$ and $\deg(j) = 1$. □

⟨1⟩2. Assertion 1 holds.
PROOF: Claw-freeness is preserved by isomorphisms, by Lemma (claws and stars), assertion 2. A line graph $L(Y')$, with the vertices of Y' numbered, is isomorphic to $L(Y'; 0, \ldots, 0)$ by Definition (generalized line graphs), and the condition of ⟨1⟩1 holds for it trivially. By Definitions (cocktail party graphs) and (generalized line graphs), $CP(t)$ is isomorphic to $L(Y_1; t)$ for the graph Y_1 with one vertex and no edges, and the condition holds trivially again. □

⟨1⟩3. ASSUME: X is connected, and $X \cong L(Y; a_1, \ldots, a_m)$ for Y and $a_1, \ldots, a_m$ as in assertion 2.
PROVE: X is claw-free if and only if $X \cong L(Y')$ for some graph Y' or $X \cong L(K_{1,s}; t, 0, \ldots, 0)$ for some $s, t \geq 0$.

⟨2⟩1. $L(K_{1,s}; t, 0, \ldots, 0)$ is claw-free.
PROOF: The only vertex of $K_{1,s}$ with a nonzero number attached is its centre 1, and every neighbour of 1 has number 0 and degree 1; apply ⟨1⟩1. □

⟨2⟩2. ASSUME: X is claw-free, and $a_i \geq 1$ for some i.
PROVE: $X \cong L(K_{1,s}; t, 0, \ldots, 0)$ with $s = \deg(i)$ and $t = a_i$.
LET: U be the set of the triples (i, α, r) and of the edges of Y on i.

⟨3⟩1. Every neighbour j of i in Y has $a_j = 0$ and $\deg(j) = 1$.
PROOF: X is claw-free by ⟨2⟩ and $X \cong L(Y; a_1, \ldots, a_m)$ by ⟨1⟩, so $L(Y; a_1, \ldots, a_m)$ is claw-free by Lemma (claws and stars), assertion 2. Apply ⟨1⟩1, with ⟨2⟩. □

⟨3⟩2. Every neighbour in $L(Y; a_1, \ldots, a_m)$ of a member of U lies in U.
PROOF: By Definition (generalized line graphs), the neighbours of a triple (i, α, r) are edges on i and triples at i. The neighbours of an edge $\{i, j\}$ are the edges on i or on j and the triples at i or at j; by ⟨3⟩1, the only edge on j is $\{i, j\}$, and there are no triples at j. □

⟨3⟩3. U is the whole vertex set of $L(Y; a_1, \ldots, a_m)$.
PROOF: U is nonempty, as $a_i \geq 1$, and closed under adjacency by ⟨3⟩2; and X is connected by ⟨1⟩. □

⟨3⟩4. Q.E.D.
PROOF: Number the neighbours of i as $j_1, \ldots, j_s$. By ⟨3⟩3, the map sending $\{i, j_k\}$ to the edge $\{1, k+1\}$ of $K_{1,s}$, and (i, α, r) to $(1, \alpha, r)$, is a bijection onto the vertices of $L(K_{1,s}; t, 0, \ldots, 0)$. It preserves adjacency by Definition (generalized line graphs): in both graphs the edges are pairwise adjacent and adjacent to every triple, and two triples are adjacent exactly when their second coordinates differ. □

⟨2⟩3. Q.E.D.

PROOF: If $X \cong L(Y')$, then X is claw-free by ⟨1⟩2; if $X \cong L(K_{1,s}; t, 0, \dots, 0)$, by ⟨2⟩1. Conversely, let X be claw-free. If every a_i is 0, then $X \cong L(Y)$ by Definition (generalized line graphs); otherwise apply ⟨2⟩2. □

⟨1⟩4. $L(K_{1,s}; t, 0, \dots, 0)$ is K_{s+2t} with t pairwise disjoint edges deleted.

PROOF: It has s edges and $2t$ triples, and by Definition (generalized line graphs) two of its vertices are non-adjacent exactly when they are $(1, \alpha, 0)$ and $(1, \alpha, 1)$ for some α; these t pairs are disjoint. □

⟨1⟩5. ASSUME: X is connected and claw-free, $2I + A(X)$ is positive semidefinite, and X has more than 36 vertices or a vertex with more than 28 neighbours.
PROVE: $X \cong L(Y')$ for some graph Y', or $X \cong L(K_{1,s}; t, 0, \dots, 0)$ for some $s, t \geq 0$.

PROOF: By Theorem 4.10, whose hypotheses hold by ⟨1⟩, X is a generalized line graph or is represented in E_8; ⟨1⟩ with Theorem 4.3, assertions 1 and 2, excludes the second. Thus $X \cong L(Y; a_1, \dots, a_m)$ for some Y and $a_1, \dots, a_m$ (Definition (generalized line graphs)), and ⟨1⟩3 applies, as X is connected and claw-free by ⟨1⟩. □

⟨1⟩6. Assertion 5 holds.

PROOF: Four-claw-freeness is the absence of 4-claws, by Lemma (claws and stars), assertion 1. A 4-claw contains a 3-claw with the same centre, so no triple is the centre of a 4-claw, by Lemma (claws in a generalized line graph), assertion 1; apply its assertion 3 to the edges. □

⟨1⟩7. Q.E.D.

PROOF: Assertions 1, 2 and 5 are ⟨1⟩2, ⟨1⟩1 and ⟨1⟩6. Assertion 3 is ⟨1⟩3 and ⟨1⟩4, since a generalized line graph is isomorphic to some $L(Y; a_1, \dots, a_m)$ (Definition (generalized line graphs)). Assertion 4 is ⟨1⟩5. □

4.4. **Theorem 4.13 and Remark 4.14.** A graph X is *strongly regular* if it is regular, any two adjacent vertices have a common neighbours and any two distinct non-adjacent vertices have c common neighbours, for some a and c; this is Mathlib's `SimpleGraph.IsSRGWith`, which allows complete graphs.

Lemma (the line graph case, LG1). ASSUME: Y is a connected graph with at least one edge, and $L(Y)$ is strongly regular.
PROVE: Y is a cycle, or $Y \cong K_n$, $Y \cong K_{h,h}$ or $Y \cong K_{1,h}$ for some n or h.

PROOF SKETCH: **[repair]** We will fill the two steps of the paper named in Gap 9 (errata): the values 0 and 1 of the constant, and the bipartite case, after the proof of [BCN] Theorem 3.12.4. We use the corrected count of Error 7 (errata).

⟨1⟩1. Y is regular, or Y is bipartite and its degree is constant on each part.

⟨2⟩1. ASSUME: j and k are neighbours of a vertex i of Y.
PROVE: $\deg j = \deg k$.

PROOF: The edges $\{i, j\}$ and $\{i, k\}$ have degrees $\deg i + \deg j - 2$ and $\deg i + \deg k - 2$ in $L(Y) \cong L(Y; 0, \dots, 0)$, by Computation C10 and Definition (generalized line graphs); these are equal, as $L(Y)$ is regular by ⟨0⟩. □

⟨2⟩2. ASSUME: P is a walk of even length in Y from u to w.
PROVE: $\deg u = \deg w$.

PROOF: Induction on the length of P: two vertices two apart on P are equal, or are neighbours of the vertex between them and have equal degrees by ⟨2⟩1. □

⟨2⟩3. CASE: Y is not bipartite.
PROVE: Y is regular.

PROOF: By `SimpleGraph.two_colorable_iff_forall_loop_even`, with `SimpleGraph.IsBipartite`, Y has a closed walk C of odd length at some vertex x. Let u, w be vertices; as Y is connected by ⟨0⟩, there are walks P from u to w and R from u to x. The walks P and R, C, R^{-1}, P have lengths of opposite parity, so one is even, and ⟨2⟩2 gives $\deg u = \deg w$. □

⟨2⟩4. Case: Y is bipartite.
Prove: The degree of Y is constant on each part of a 2-colouring of Y.
Proof: `SimpleGraph.IsBipartite` gives a colouring of Y by $\operatorname{Fin} 2$; recolour it by $\operatorname{Fin} 2 \simeq \mathrm{Bool}$ with `SimpleGraph.recolorOfEquiv` and `finTwoEquiv`, which keeps the colour classes, as `SimpleGraph.Coloring.even_length_iff_congr` takes a colouring by Bool. Two vertices of the same colour are joined by a walk, as Y is connected by ⟨0⟩, and it has even length by that lemma; apply ⟨2⟩2. □
⟨2⟩5. Q.E.D.
Proof: The two cases are exhaustive. □
⟨1⟩2. Case: Y is bipartite with degrees $p \neq q$ on its parts.
Prove: $Y \cong K_{1,h}$.
Proof: Two edges on a common vertex of degree p have $p-2$ common neighbours in $L(Y)$, since their other ends lie in one part and are not adjacent; likewise $q-2$. Both kinds of pairs cannot occur, as $p \neq q$; so one degree, say q, is 1, and a connected graph with such a part is a star. □
⟨1⟩3. Case: Y is regular of valency $h \leq 2$.
Prove: $Y \cong K_{1,1}$ or Y is a cycle.
Proof: A connected 1-regular graph is K_2, and a connected 2-regular graph is a cycle. □
⟨1⟩4. Case: Y is regular of valency $h \geq 3$.
Prove: $Y \cong K_n$ or $Y \cong K_{h,h}$.
⟨2⟩1. Two edges $\{i,j\}$, $\{i,k\}$ have $h-1$ common neighbours in $L(Y)$ if $j \sim k$ and $h-2$ otherwise.
Proof: The common neighbours are the $h-2$ other edges on i, and $\{j,k\}$ when it is an edge. □
⟨2⟩2. Case: Some $j \sim k$ with a common neighbour i.
Prove: $Y \cong K_n$.
Proof: By ⟨2⟩1 and the constant a, every path j,i,k closes to a triangle; so neighbourhoods are cliques, and a connected such graph is complete. □
⟨2⟩3. Case: Y is triangle-free.
Prove: $Y \cong K_{h,h}$.
Let: c be the constant of $L(Y)$ for non-adjacent vertices: two disjoint edges are joined by exactly c edges of Y.
⟨3⟩1. Every edge $\{i,j\}$ is disjoint from some edge, and $c \leq 2$.
Proof: Pick $k \sim i$ with $k \neq j$, and $l \sim k$ with $l \neq i$, as $h \geq 3$. Then $l \neq j$, else i,j,k is a triangle; so $\{k,l\}$ is disjoint from $\{i,j\}$. Of the edges joining $\{i,j\}$ to a disjoint edge, at most one leaves i and at most one leaves j, by triangle-freeness. □
⟨3⟩2. $c \neq 0$.
Proof: If $c = 0$, two vertices of $L(Y)$ at distance 2 would have no common neighbour; so the connected graph $L(Y)$ is complete, and Y has no two disjoint edges, against ⟨3⟩1. □
⟨3⟩3. $c \neq 1$.
Proof: Fix an edge $\{i,j\}$. An edge $\{i,k\}$, $k \neq j$, and an edge $\{k,l\}$, $l \neq i$, give a disjoint edge $\{k,l\}$ ($l \neq j$ by triangle-freeness); with the same count at j, there are $2(h-1)^2$ such pairs. If $c = 1$, each disjoint edge arises once, so $|E(Y)| = 1 + 2(h-1) + 2(h-1)^2$, and $|V(Y)| = 2|E(Y)|/h = 4h - 4 + 2/h$ is not an integer for $h \geq 3$. □
⟨3⟩4. Assume: $c = 2$ and $\{i,j\}$ is an edge.
Prove: $Y \cong K_{h,h}$, with parts $N(j)$ and $N(i)$.
Proof: $N(i)$, $N(j)$ are disjoint cocliques, by triangle-freeness. For $x \in N(j) \setminus \{i\}$ and $y \in N(i) \setminus \{j\}$, the disjoint edges $\{x,j\}$ and $\{i,y\}$ are joined by $\{i,j\}$ and one more edge,

which can only be $\{x, y\}$. A vertex $k \notin N(i) \cup N(j)$ has a neighbour $l \notin \{i, j\}$, and the two edges joining $\{k, l\}$ to $\{i, j\}$ put k in $N(i) \cup N(j)$. □

⟨3⟩5. Q.E.D.

PROOF: $c \in \{0, 1, 2\}$ by ⟨3⟩1; apply ⟨3⟩2, ⟨3⟩3 and ⟨3⟩4. □

⟨2⟩4. Q.E.D.

PROOF: Either some path of length 2 closes to a triangle, or Y is triangle-free. □

⟨1⟩5. Q.E.D.

PROOF: By ⟨1⟩1, Y is regular or bipartite with two distinct degrees; the cases ⟨1⟩2, ⟨1⟩3 and ⟨1⟩4 are exhaustive. □

Theorem 4.13. ASSUME: 1. X is a connected strongly regular graph, and $2I + A(X)$ is positive semidefinite.

2. -2 is an eigenvalue of X.

PROVE: $X \cong L(K_n)$, $X \cong L(K_{n,n})$ or $X \cong CP(n)$ for some n, or X is represented in E_8.

[new] The two hypotheses together are the paper's "least eigenvalue -2", read literally, so this is the printed theorem. We state them apart because Theorems 4.10 and 4.11 read the phrase as the lower bound of the first hypothesis alone, and Theorem 4.13 fails under that reading: for $n > 36$, K_n is a counterexample (errata, Implicit hypothesis 2).

PROOF SKETCH: We will apply Theorem 4.11, and then the line graph case LG1.

⟨1⟩1. $X \cong L(Y)$, or $X \cong CP(m)$, or X is represented in E_8.

PROOF: Theorem 4.11, with ⟨0⟩:1. □

⟨1⟩2. CASE: $X \cong L(Y)$.

PROVE: $X \cong L(K_n)$ or $X \cong L(K_{n,n})$.

⟨2⟩1. We may take Y connected, with an edge, and Y is a cycle, K_n, $K_{h,h}$ or $K_{1,h}$.

PROOF: Remove the isolated vertices of Y; since X is connected with a vertex, the rest is connected with an edge. Apply Lemma (the line graph case, LG1). □

⟨2⟩2. X is neither a complete graph nor C_5, and X is not a cycle C_p with $p \geq 6$.

PROOF: $\det(2I+A) \neq 0$ for K_n, since $2I+A = I+J$ is positive definite, and $\det(2I+A(C_5)) = 4$; so neither has the eigenvalue -2 of ⟨0⟩:2, which is a root of $\det(\mu I - A)$. In C_p with $p \geq 6$, vertices at distance 2 and 3 have 1 and 0 common neighbours. □

⟨2⟩3. Q.E.D.

PROOF: $L(K_{1,h}) \cong K_h$ is excluded by ⟨2⟩2. If $Y = C_p$, then $X \cong C_p$, and $p = 4$ by ⟨2⟩2, since C_3 is complete; and $C_4 = L(K_{2,2})$. The rest is $L(K_n)$ or $L(K_{h,h})$, by ⟨2⟩1. □

⟨1⟩3. Q.E.D.

PROOF: The cases of ⟨1⟩1 are exhaustive, and the other two are in the conclusion. □

Remark (after the examples). ASSUME: X is a connected graph, and $2I + A(X)$ is positive semidefinite.

PROVE:

1. If X has more than 36 vertices, or a vertex with more than 28 neighbours, then X is a generalized line graph.
2. If X is d-regular on n vertices, with $n > 28$ or $d > 16$, then $X \cong L(Y)$ for some graph Y or $X \cong CP(m)$ for some m.
3. If X is strongly regular of valency k on n vertices, with $n > 28$ or $k > 16$, and -2 is an eigenvalue of X, then $X \cong L(K_m)$, $X \cong L(K_{m,m})$ or $X \cong CP(m)$ for some m.

[new] The paper places this remark before Theorem 4.10 and speaks of "any subsequent theorem"; we will state it after Theorems 4.10, 4.11 and 4.13, with their hypotheses (errata, Implicit hypotheses 1 and 2).

PROOF SKETCH: We will apply the three theorems, and exclude E_8 by Theorems 4.3 and 4.4.

⟨1⟩1. ASSUME: X has more than 36 vertices, or a vertex with more than 28 neighbours.

PROVE: X is a generalized line graph.
PROOF: By Theorem 4.10 and ⟨0⟩, X is a generalized line graph or is represented in E_8; Theorem 4.3, assertions 1 and 2, with ⟨1⟩, excludes the second. □

⟨1⟩2. ASSUME: X is d-regular on n vertices, and $n > 28$ or $d > 16$.
PROVE: $X \cong L(Y)$ for some graph Y or $X \cong CP(m)$ for some m.
PROOF: By Theorem 4.11, with ⟨0⟩ and ⟨1⟩, $X \cong L(Y)$, $X \cong CP(m)$ or X is represented in E_8; Theorem 4.4 excludes the third, by ⟨1⟩, as $n \geq 1$ for a connected graph, by ⟨0⟩. □

⟨1⟩3. ASSUME: X is strongly regular of valency k on n vertices, $n > 28$ or $k > 16$, and -2 is an eigenvalue of X.
PROVE: $X \cong L(K_m)$, $X \cong L(K_{m,m})$ or $X \cong CP(m)$ for some m.
PROOF: By Theorem 4.13, with ⟨0⟩ and ⟨1⟩, X is one of these or is represented in E_8; Theorem 4.4 excludes the last, as X is k-regular on n vertices by ⟨1⟩, and $n \geq 1$ for a connected graph, by ⟨0⟩. □

⟨1⟩4. Q.E.D.
PROOF: ⟨1⟩1, ⟨1⟩2 and ⟨1⟩3. □

What remains of Section 4.

Remark 4.14 is stated below, and its proof is in Section 4.5. The remarks after Theorems 4.1 and 4.2, the last pieces of the section, are stated and proved after those theorems.

Remark 4.14. ASSUME: X is a connected strongly regular graph with least eigenvalue -2, represented in E_8, and isomorphic to none of $L(K_m)$, $L(K_{m,m})$ and $CP(m)$.
PROVE: X is isomorphic to one of the Chang graphs I, II and III, the Schläfli graph, the Shrikhande graph, the Clebsch graph and the Petersen graph; and these seven graphs are pairwise non-isomorphic and have these properties.

[repair] The paper gives no proof, and "the seven graphs" needs the Chang graphs to be pairwise non-isomorphic (errata, Gap 12). We will give its proof, by the route of `cgss/prerequisites.md` (Section 7.3), in Section 4.5.

4.5. **The proof of Remark 4.14.** We will prove Remark 4.14 by the route of `cgss/prerequisites.md`, Section 7.3, which follows the computer-free part of the proof of [BCN] Theorem 3.12.2 in the language of line systems, and replaces the enumeration of [BCN] Proposition 3.11.2 by three switching arguments. A graph X as in the remark is represented by a family $(\mathbf{x}_u)$ in E_8. We will first pin down its parameters (Computation C18), and then study the line system R of the vectors of squared norm 2 in the $\mathbb{Z}$-span of the $\mathbf{x}_u$. It is of type E_6, E_7 or E_8, and the vector $\mathbf{w} = \frac{1}{k+2}\sum_u \mathbf{x}_u$ has integral inner products with it. The simple roots of Computation C21 then exclude two parameter sets. In the three cases where $(\mathbf{w}, \mathbf{w}) = 2$, $\mathbf{w}$ is a root, and X is a switching of $T(5)$, of $T(6)$ with an isolated vertex, or of $T(8)$, by the edge set of a regular graph; Computation C22 names the result. The two remaining parameter sets are those of the Clebsch and the Schläfli graphs, which are unique. We will write n, k, a, c for the parameters of a strongly regular graph, as in the rest of the blueprint. A result tagged **[new]** departs from Section 7.3 of `cgss/prerequisites.md`, or from the prerequisite that it cites, and the departure is stated after the tag; the untagged results follow it.

Facts from Mathlib and the project. We will use the following declarations of Mathlib (commit `082e2d37`): `SimpleGraph.IsSRGWith` with its fields,
`SimpleGraph.IsSRGWith.matrix_eq` ($A^2 = kI + aA + c\,A(\overline{X})$),
`SimpleGraph.one_add_adjMatrix_add_compl_adjMatrix_eq_of_one` with
`SimpleGraph.compl_adjMatrix_eq_adjMatrix_compl` ($I + A + A(\overline{X}) = J$),
`SimpleGraph.IsSRGWith.param_eq` ($k(k-a-1) = c(n-k-1)$ for $0 < n$, in $\mathbb{N}$ with truncated subtraction), `SimpleGraph.IsSRGWith.compl` (the complement has parameters

$(n, n-k-1, n-2k+c-2, n-2k+a)$), `SimpleGraph.IsSRGWith.card_neighborFinset_union_of_not_adj` ($|N(u) \cup N(w)| = 2k - c$ for distinct non-adjacent u, w), `SimpleGraph.IsRegularOfDegree.compl`, `SimpleGraph.sum_degrees_eq_twice_card_edges`, `SimpleGraph.IsNClique.map` (an embedding maps cliques to cliques), `compl_compl` ($\overline{\overline{X}} = X$), `Matrix.exists_mulVec_eq_zero_iff` (a square matrix is singular exactly when it has a nonzero null vector), `Matrix.inv_def` and `Matrix.mul_adjugate`, `Matrix.det_gram_ne_zero_iff_linearIndependent`, `LinearMap.IsProj.trace` with `LinearMap.isProj_range_iff_isIdempotentElem` (an idempotent has rank equal to its trace), `abs_real_inner_le_norm` (the Cauchy–Schwarz inequality), `exists_linearIndependent` (a spanning set contains a basis) and `Submodule.finrank_le`. From the project we use `SimpleGraph.leastEigenvalue_eq_neg_two_iff` of `LineGraphs/Spectral/Positivity.lean` ($\lambda_{\min}(X) = -2$ exactly when $2I + A$ is positive semidefinite and singular), the checkers `Computations.checkGraphIso`, `Computations.cliqueProfile` and `Computations.cliqueCount` with their soundness theorems `Computations.nonempty_graphOfMatrix_iso`, `Computations.checkNonisomorphic_sound` and `Computations.cliqueCount_sound`, and the invariance `SimpleGraph.Iso.cliqueProfile_eq`, in `LineGraphs/Computations/CliqueCertificates.lean`. Finally, from the Lean of Computation C4 we use the transitivity on ordered stars: for $t \in \{6, 7, 8\}$ and roots $\mathbf{a}, \mathbf{b}$ of E_t with $(\mathbf{a}, \mathbf{b}) = -1$, there is a linear isometry g of $\mathbb{R}^8$ with $g(\Phi(\mathrm{E}_t)) = \Phi(\mathrm{E}_t)$ that carries the fixed star $(\mathbf{a}_0, \mathbf{b}_0)$ of Lean to $(\mathbf{a}, \mathbf{b})$; this is `LineSystem.E8.exists_isometry_star` and its analogues for E_7 and E_6, in `LineGraphs/Computations/C4E8Stars.lean` and its analogues, whose trees of reflections are checked by `decide +kernel`. It replaces LN5 of `cgss/prerequisites.md`.

Throughout, $\mathbf{c} = \frac{1}{2}\sum_{i=1}^{8} \mathbf{e}_i$ is the root of model (ii) of E_8 fixed before Lemma (switching by vectors, SW4), and for a 2-subset $P = \{i, j\}$ of $\{1, \dots, 8\}$ we write $\mathbf{e}_P = \mathbf{e}_i + \mathbf{e}_j$. We will write $\mathcal{E}_6$ for the set of 2-subsets of $\{1, \dots, 5\}$, $\mathcal{E}_7$ for the set of 2-subsets of $\{1, \dots, 6\}$ together with $\varepsilon = \{7, 8\}$, and $\mathcal{E}_8$ for the set of all 2-subsets of $\{1, \dots, 8\}$; Δ_t is the graph on $\{1, \dots, 8\}$ with edge set $\mathcal{E}_t$. Thus $L(\Delta_6) \cong T(5)$, $L(\Delta_7)$ is $T(6)$ together with the isolated vertex ε, and $L(\Delta_8) = T(8)$. For a graph Z and a vertex i we write $\deg_Z(i)$ for its degree, and for a set H of 2-subsets we write $\deg_H(i)$ for the number of members of H that contain i.

Definition (switching). Assume: X is a graph and $U \subseteq V(X)$.
Let: The *switched graph* X^U has vertex set $V(X)$, and distinct u, w are adjacent in X^U when either they are adjacent in X and both or neither lie in U, or they are not adjacent in X and exactly one of them lies in U. For a graph Y and $H \subseteq E(Y)$ we write $L(Y)^H$ for the switching of the line graph $L(Y)$ with respect to H.
This is SW1 of `cgss/prerequisites.md`, with U in place of its Y, since Y is a graph here.

Lemma (switching). Assume: X and X' are graphs, $U \subseteq V(X)$, Y and Y' are graphs and $H \subseteq E(Y)$.
Prove: 1. $X^U = X^{V(X) \setminus U}$.
2. If σ is an isomorphism from X to X', then σ is an isomorphism from X^U to $X'^{\sigma(U)}$.
3. If σ is an isomorphism from Y to Y', then $e \mapsto \sigma(e)$ is an isomorphism from $L(Y)^H$ to $L(Y')^{\sigma(H)}$.

Proof sketch: We will read each assertion off Definition (switching).
⟨1⟩1. Assertion 1 holds.
Proof: A pair has exactly one member in U if and only if it has exactly one member in $V(X) \setminus U$; apply Definition (switching). □

⟨1⟩2. Assertion 2 holds.
PROOF: For distinct u, w, σ preserves adjacency, and $u \in U$ exactly when $\sigma(u) \in \sigma(U)$; so the condition of Definition (switching) for u, w in X and U is that for $\sigma(u), \sigma(w)$ in X' and $\sigma(U)$. □
⟨1⟩3. $e \mapsto \sigma(e)$ is an isomorphism from $L(Y)$ to $L(Y')$.
PROOF: It is a bijection from $E(Y)$ onto $E(Y')$, since σ is an isomorphism, and $|\sigma(e) \cap \sigma(f)| = |e \cap f|$ since σ is injective; two edges are adjacent in the line graph when they are distinct and meet. □
⟨1⟩4. Q.E.D.
PROOF: Assertions 1 and 2 are ⟨1⟩1 and ⟨1⟩2; assertion 3 is assertion 2 applied to the isomorphism of ⟨1⟩3 and the set H. □

Lemma (isomorphisms and complements). ASSUME: X and X' are graphs, and σ is an isomorphism from X to X'.
PROVE: 1. σ is an isomorphism from $\overline{X}$ to $\overline{X'}$.
2. If X is strongly regular with parameters (n, k, a, c), then so is X'.
Mathlib has the complement of an embedding, `SimpleGraph.Embedding.complEquiv`, but neither assertion.

PROOF SKETCH: An isomorphism is a bijection that preserves adjacency and non-adjacency.
⟨1⟩1. Assertion 1 holds.
PROOF: σ is a bijection, and for distinct u, w the vertices $\sigma(u)$, $\sigma(w)$ are distinct and adjacent in X' exactly when u, w are adjacent in X; distinct vertices are adjacent in the complement exactly when they are not adjacent in the graph. □
⟨1⟩2. Assertion 2 holds.
PROOF: $|V(X')| = |V(X)|$, σ maps the neighbours of u onto those of $\sigma(u)$, and the common neighbours of u, w onto those of $\sigma(u), \sigma(w)$. □
⟨1⟩3. Q.E.D.
PROOF: ⟨1⟩1 and ⟨1⟩2. □

Parameters.

Lemma (cocktail party parameters). ASSUME: $m \geq 1$, and X is a $(2m-2)$-regular graph on $2m$ vertices.
PROVE: $X \cong CP(m)$.

PROOF SKETCH: The complement of X is a perfect matching.
⟨1⟩1. The complement $\overline{X}$ is 1-regular.
PROOF: `SimpleGraph.IsRegularOfDegree.compl`: its degree is $2m - 1 - (2m-2) = 1$. □
⟨1⟩2. PICK vertices $p_1, q_1, \ldots, p_m, q_m$, which are all the vertices of X, such that the edges of $\overline{X}$ are the pairs $\{p_i, q_i\}$.
PROOF: By ⟨1⟩1 each vertex lies in exactly one edge of $\overline{X}$, so the edges of $\overline{X}$ partition $V(X)$ into pairs; there are $2m/2 = m$ of them. □
⟨1⟩3. Q.E.D.
PROOF: The map $(i, 0) \mapsto p_i$, $(i, 1) \mapsto q_i$ is a bijection by ⟨1⟩2. Distinct vertices are adjacent in X exactly when they are not an edge of $\overline{X}$, that is, when their indices differ; this is adjacency in $CP(m)$ (Definition (cocktail party graphs)). □

Lemma (strongly regular graphs with eigenvalue −2). ASSUME: 1. X is connected and strongly regular with parameters (n, k, a, c).
2. $\det(2I + A) = 0$, where $A = A(X)$.
PROVE: 1. $1 \leq k \leq n-2$, $a \leq k-1$ and $1 \leq c \leq k$.
2. $2k - c \leq n - 2$.

3. $k - 2a + c = 4$.
4. $(a - c + 4)(\mathrm{rk}(2I + A) - 1) = 2n - k - 2$.

[new] This is SR4 of `cgss/prerequisites.md`, in a form without the spectral theorem: the multiplicity m_1 of the second eigenvalue is replaced by $\mathrm{rk}(2I+A)-1$, through a spectral idempotent. Assertion 2 is the counting condition that the search of C18 needs (`cgss/certificates/c18_parameter_search.sage`).

PROOF SKETCH: We will derive the linear relation from a null vector of $2I + A$, and then find an idempotent multiple of $2I + A - \frac{k+2}{n}J$.
LET: $Q = 2I + A$, and let J and $\mathbf{j}$ be the all-ones matrix and vector.
⟨1⟩1. $k \geq 1$, and X is not complete.
PROOF: If $k = 0$ then $Q = 2I$, and if X is complete then $Q = I + J$, which is positive definite since $(\mathbf{z}, (I + J)\mathbf{z}) = |\mathbf{z}|^2 + (\mathbf{z}, \mathbf{j})^2$; in both cases $\det Q \neq 0$, against ⟨0⟩:2. □
⟨1⟩2. There are an edge uw and distinct non-adjacent vertices u', w'.
PROOF: $k \geq 1$ and X is not complete, by ⟨1⟩1. □
⟨1⟩3. $a \leq k - 1$.
PROOF: The a common neighbours of the edge uw of ⟨1⟩2 lie in $N(u) \setminus \{w\}$, which has $k - 1$ elements. □
⟨1⟩4. $c \leq k$.
PROOF: The c common neighbours of u', w' of ⟨1⟩2 lie in $N(u')$, which has k elements. □
⟨1⟩5. $2k - c \leq n - 2$.
PROOF: $N(u') \cup N(w')$, for u', w' of ⟨1⟩2, has $2k - c$ elements by `SimpleGraph.IsSRGWith.card_neighborFinset_union_of_not_adj`, and lies in $V(X) \setminus \{u', w'\}$, since $u' \not\sim w'$. □
⟨1⟩6. $k \leq n - 2$.
PROOF: $k \leq 2k - c$ by ⟨1⟩4; apply ⟨1⟩5. □
⟨1⟩7. $c \geq 1$.
PROOF: Take non-adjacent distinct u', w' (⟨1⟩1) and a shortest path $u' = u_0, u_1, \ldots, u_d = w'$ in the connected graph X, so $d \geq 2$. Then u_0 and u_2 are distinct and non-adjacent, by minimality, and u_1 is a common neighbour. □
⟨1⟩8. $A(\overline{X}) = J - I - A$.
PROOF: `SimpleGraph.one_add_adjMatrix_add_compl_adjMatrix_eq_of_one` and `SimpleGraph.compl_adjMatrix_eq_adjMatrix_compl`. □
⟨1⟩9. $A^2 = kI + aA + c(J - I - A)$, $AJ = JA = kJ$ and $J^2 = nJ$.
PROOF: `SimpleGraph.IsSRGWith.matrix_eq` gives $A^2 = kI + aA + c\,A(\overline{X})$; apply ⟨1⟩8. $AJ = JA = kJ$ by regularity, and $J^2 = nJ$ since each entry of J^2 is a sum of n ones. □
⟨1⟩10. PICK $\mathbf{z} \neq 0$ with $Q\mathbf{z} = 0$; then $A\mathbf{z} = -2\mathbf{z}$ and $J\mathbf{z} = 0$.
PROOF: `Matrix.exists_mulVec_eq_zero_iff` and ⟨0⟩:2 give $\mathbf{z}$. Then $k(\mathbf{j}, \mathbf{z}) = (A\mathbf{j}, \mathbf{z}) = (\mathbf{j}, A\mathbf{z}) = -2(\mathbf{j}, \mathbf{z})$ by ⟨1⟩9 and the symmetry of A, so $(\mathbf{j}, \mathbf{z}) = 0$, since $k + 2 > 0$; and $J\mathbf{z} = (\mathbf{j}, \mathbf{z})\mathbf{j}$. □
⟨1⟩11. $k - 2a + c = 4$.
PROOF: Apply both sides of ⟨1⟩9 to $\mathbf{z}$ of ⟨1⟩10: $4\mathbf{z} = k\mathbf{z} - 2a\mathbf{z} + c(0 - \mathbf{z} + 2\mathbf{z}) = (k - 2a + c)\mathbf{z}$, and $\mathbf{z} \neq 0$. □
LET: $\eta = a - c + 2$.
⟨1⟩12. $k - c = 2\eta$, $\eta \geq 0$, and $(A - \eta I)Q = cJ$.
PROOF: The first is ⟨1⟩11, and $\eta \geq 0$ follows with $c \leq k$ (⟨1⟩4). By ⟨1⟩9, $(A - \eta I)(A + 2I) = A^2 + (2 - \eta)A - 2\eta I = (k - c - 2\eta)I + (a - c + 2 - \eta)A + cJ$, and both coefficients vanish. □
⟨1⟩13. $cn = (k + 2)(k - \eta)$.
PROOF: Apply ⟨1⟩12 to $\mathbf{j}$: $Q\mathbf{j} = (k+2)\mathbf{j}$ and $(A - \eta I)\mathbf{j} = (k - \eta)\mathbf{j}$ by ⟨1⟩9, so $(k - \eta)(k + 2)\mathbf{j} = cJ\mathbf{j} = cn\mathbf{j}$. □

LET: $E = Q - \frac{k+2}{n}J$.

⟨1⟩14. $PJ = JP = 0$, and $E = Q(I - \frac{1}{n}J)$.

PROOF: $QJ = JQ = (k+2)J$ and $J^2 = nJ$ by ⟨1⟩9. □

⟨1⟩15. $E^2 = (\eta+2)E$.

PROOF:

$$\begin{aligned} E^2 &= PQ - \tfrac{k+2}{n}PJ = PQ = Q^2 - \tfrac{(k+2)^2}{n}J \\ &= (A - \eta I)Q + (\eta+2)Q - \tfrac{(k+2)^2}{n}J = (\eta+2)Q + \Big(c - \tfrac{(k+2)^2}{n}\Big)J \\ &= (\eta+2)Q - (\eta+2)\tfrac{k+2}{n}J = (\eta+2)E, \end{aligned}$$

where the first line uses ⟨1⟩14 and $JQ = (k+2)J$, the second ⟨1⟩12, and the third ⟨1⟩13, which gives $cn - (k+2)^2 = -(k+2)(\eta+2)$. □

⟨1⟩16. $(\eta+2)\operatorname{rk} E = 2n - k - 2$.

PROOF: $\eta + 2 \geq 2$ by ⟨1⟩12, so $E/(\eta+2)$ is idempotent by ⟨1⟩15, and its rank is its trace (`LinearMap.IsProj.trace` and `LinearMap.isProj_range_iff_isIdempotentElem`). Further $\operatorname{tr} E = \operatorname{tr} Q - \frac{k+2}{n}\operatorname{tr} J = 2n - (k+2)$, since $\operatorname{tr} A = 0$. □

⟨1⟩17. $\operatorname{rk} Q = \operatorname{rk} E + 1$.

PROOF: The image of $E = Q(I - \frac{1}{n}J)$ lies in that of Q, and so does $\mathbf{j} = Q\mathbf{j}/(k+2)$; conversely $Q = E + \frac{k+2}{n}J$ has image in $\operatorname{im} E + \mathbb{R}\mathbf{j}$. Finally $\mathbf{j} \notin \operatorname{im} E$: $JP = 0$ (⟨1⟩14) gives $(\mathbf{j}, E\mathbf{y}) = 0$ for every $\mathbf{y}$, while $(\mathbf{j}, \mathbf{j}) = n > 0$. □

⟨1⟩18. Q.E.D.

PROOF: Assertion 1 is ⟨1⟩1, ⟨1⟩6, ⟨1⟩3, ⟨1⟩4 and ⟨1⟩7, assertion 2 is ⟨1⟩5 and assertion 3 is ⟨1⟩11. For assertion 4, $\eta + 2 = a - c + 4$, and ⟨1⟩16 and ⟨1⟩17 give $(\eta+2)(\operatorname{rk} Q - 1) = 2n - k - 2$. □

Computation C18. ASSUME: n, k, a, c, ρ are integers with $1 \leq k \leq n-2$, $n \leq 36$, $k \leq 16$, $0 \leq a \leq k-1$, $1 \leq c \leq k$ and $0 \leq \rho \leq 8$, such that

1. $k(k-a-1) = c(n-k-1)$,
2. $k - 2a + c = 4$,
3. $2k - c \leq n - 2$, and
4. $(a - c + 4)(\rho - 1) = 2n - k - 2$.

PROVE: (n, k, a, c, ρ) is one of the following 15 tuples: $(2m, 2m-2, 2m-4, 2m-2, m+1)$ for $2 \leq m \leq 7$; $(9, 4, 1, 2, 5)$; $(10, 6, 3, 4, 5)$; $(10, 3, 0, 1, 6)$; $(15, 8, 4, 4, 6)$; $(16, 10, 6, 6, 6)$; $(16, 6, 2, 2, 7)$; $(21, 10, 5, 4, 7)$; $(27, 16, 10, 8, 7)$; and $(28, 12, 6, 4, 8)$.

This is Lemma P of `cgss/prerequisites.md` (SR5), with $\rho = 1 + m_1$. The first six are the parameters of $CP(m)$, the next two those of $L_2(3)$ and $T(5)$.

PROOF SKETCH: This is a finite search. The Lean proof is one `Bool` search over the valencies with a soundness lemma, as for the checker `Computations.allSRGParameters` of `LineGraphs/Computations/BoundedSearch.lean`, whose pruning ($a < k$, $0 < c \leq k$, $k + 1 \leq n$) matches our bounds; `interval_cases` would make about $1.5 \cdot 10^5$ goals.

⟨1⟩1. Q.E.D.

PROOF: For each $k \leq 16$, $a < k$ and $1 \leq c \leq k$, condition 1 determines n, and condition 4 determines ρ, since $a - c + 4 = 0$ would give $2n = k + 2$, against $k + 2 \leq n$ and $n \geq 3$; the search keeps the tuples satisfying all four conditions and the bounds. It finds the 15 tuples, and no other (SageMath, `cgss/certificates/c18_parameter_search.sage`). Without condition 3 the tuple $(21, 16, 12, 12, 7)$ would also survive. □

Lemma (strongly regular line graphs). ASSUME: 1. X is a connected graph, strongly regular with parameters (n, k, a, c).

2. $\det(2I + A(X)) = 0$.
3. Y is a graph and $X \cong L(Y)$.

PROVE: $X \cong L(K_m)$ or $X \cong L(K_{m,m})$ for some $m \geq 2$.
[new] This is the line graph case of Theorem 4.13, stated as a result so that we can cite it; its argument is that of the proof of Theorem 4.13, through Lemma LG1.

PROOF SKETCH: We will remove the isolated vertices of Y and apply Lemma LG1; the eigenvalue -2 excludes complete graphs and C_5.
LET: Y' be Y with its isolated vertices removed.
⟨1⟩1. $L(Y') = L(Y)$.
PROOF: Removing isolated vertices keeps the edges and their intersections. □
⟨1⟩2. Y' has an edge.
PROOF: X has a vertex, being connected, so $L(Y) \cong X$ (⟨0⟩:3) has one; it is an edge of Y, and of Y' by ⟨1⟩1. □
⟨1⟩3. Y' is connected.
PROOF: Every vertex of Y' lies on an edge, and a path in the connected graph $L(Y') \cong X$ (⟨1⟩1, ⟨0⟩:3) between two edges gives a walk in Y' between their ends. □
⟨1⟩4. Y' is a cycle, or $Y' \cong K_m$, $Y' \cong K_{q,q}$ or $Y' \cong K_{1,q}$ for some m or q.
PROOF: Lemma (the line graph case, LG1), with ⟨1⟩2 and ⟨1⟩3, since $L(Y')$ is strongly regular by ⟨0⟩:1, ⟨0⟩:3, ⟨1⟩1 and Lemma (isomorphisms and complements), assertion 2. □
⟨1⟩5. X is not complete.
PROOF: If X is complete on $n \geq 1$ vertices, then $2I + A = I + J$, which is positive definite since $(\mathbf{z}, (I+J)\mathbf{z}) = |\mathbf{z}|^2 + (\mathbf{z}, \mathbf{j})^2$; so its determinant is not 0, against ⟨0⟩:2. □
⟨1⟩6. $\det(2I + A(C_5)) = 4$.
PROOF: `decide` on the 5×5 integer matrix. □
⟨1⟩7. $X \ncong C_5$.
PROOF: An isomorphism $X \to C_5$ conjugates $2I + A(X)$ into $2I + A(C_5)$ by a permutation matrix, which keeps the determinant; apply ⟨1⟩6 and ⟨0⟩:2. □
⟨1⟩8. $X \ncong C_p$ for $p \geq 6$.
PROOF: In C_p with $p \geq 6$ there are non-adjacent vertices at distance 2 and at distance 3, with 1 and 0 common neighbours; so C_p is not strongly regular, against ⟨0⟩:1. □
⟨1⟩9. CASE: Y' is a cycle C_p.
PROVE: $X \cong L(K_{2,2})$.
PROOF: $L(C_p) \cong C_p$, and $p \geq 3$. The cases $p = 3$ (complete), $p = 5$ and $p \geq 6$ are excluded by ⟨1⟩5, ⟨1⟩7 and ⟨1⟩8. So $p = 4$, and $C_4 \cong L(C_4) = L(K_{2,2})$. □
⟨1⟩10. CASE: $Y' \cong K_{1,q}$.
PROVE: This case does not occur.
PROOF: Any two edges of $K_{1,q}$ meet, so $L(K_{1,q}) \cong K_q$ is complete, against ⟨1⟩5. □
⟨1⟩11. Q.E.D.
PROOF: By ⟨1⟩4 the cases ⟨1⟩9, ⟨1⟩10, $Y' \cong K_m$ and $Y' \cong K_{q,q}$ are exhaustive. In the last two, $X \cong L(K_m)$ or $L(K_{q,q})$ by ⟨0⟩:3 and ⟨1⟩1. Further $m \geq 2$, since $L(K_m)$ has no vertex for $m \leq 1$ while the connected graph X has one; and $q \geq 2$, since $L(K_{0,0})$ has no vertex and $L(K_{1,1})$ is complete (⟨1⟩5). □

The root system of a representation.

Lemma (the lattice roots). ASSUME: X is a connected graph with a representation $(\mathbf{x}_u)$ in a finite-dimensional real inner product space V; Λ is the $\mathbb{Z}$-span and W the span of the $\mathbf{x}_u$; and $\Phi_R = \{\mathbf{y} \in \Lambda : (\mathbf{y}, \mathbf{y}) = 2\}$.
PROVE: 1. $(\mathbf{y}, \mathbf{z}) \in \mathbb{Z}$ for all $\mathbf{y}, \mathbf{z} \in \Lambda$.
2. Φ_R is the set of vectors of a line system R in V, which is nonempty, star-closed and indecomposable and spans W.

3. X is represented in R by $(\mathbf{x}_u)$.

[new] `cgss/prerequisites.md` (LN3) uses the star closure of the $\mathbf{x}_u$. The larger set Φ_R is star-closed as well, and it lies in Λ by definition, which is what the apex argument needs; we do not need $R \subseteq \mathrm{E}_8$.

PROOF SKETCH: Inner products in Λ are integers, so norms and Cauchy–Schwarz do the rest.

⟨1⟩1. Assertion 1 holds.
PROOF: $\mathbf{y} = \sum_u y_u \mathbf{x}_u$ and $\mathbf{z} = \sum_w z_w \mathbf{x}_w$ with integers y_u, z_w, and $(\mathbf{y}, \mathbf{z}) = \sum_{u,w} y_u z_w (\mathbf{x}_u, \mathbf{x}_w)$, where each $(\mathbf{x}_u, \mathbf{x}_w) \in \{0, 1, 2\}$ by Definition (representations). □

⟨1⟩2. $\Phi_R = -\Phi_R$, and $(\mathbf{y}, \mathbf{z}) \in \{-2, -1, 0, 1, 2\}$ for $\mathbf{y}, \mathbf{z} \in \Phi_R$.
PROOF: $\Lambda = -\Lambda$. By `abs_real_inner_le_norm`, $|(\mathbf{y}, \mathbf{z})| \le |\mathbf{y}||\mathbf{z}| = 2$, and $(\mathbf{y}, \mathbf{z})$ is an integer by ⟨1⟩1. □

⟨1⟩3. $\mathbf{x}_u \in \Phi_R$ for every u, and $\Phi_R \subseteq \Lambda \subseteq W$.
PROOF: $(\mathbf{x}_u, \mathbf{x}_u) = 2$ by Definition (representations); $\Lambda \subseteq W$ since integer combinations are combinations. □

⟨1⟩4. PICK vertices $u_1, \ldots, u_\rho$ such that $\mathbf{x}_{u_1}, \ldots, \mathbf{x}_{u_\rho}$ form a basis of W.
PROOF: `exists_linearIndependent`, since the $\mathbf{x}_u$ span W and W is finite-dimensional. □

⟨1⟩5. The map $\iota : \mathbf{y} \mapsto ((\mathbf{y}, \mathbf{x}_{u_i}))_{i \le \rho}$ is injective on W.
PROOF: It is linear, and a vector of W orthogonal to the basis of ⟨1⟩4 is orthogonal to W, so to itself, and is 0. □

⟨1⟩6. Φ_R is finite.
PROOF: By ⟨1⟩2 and ⟨1⟩3, ι maps $\Phi_R \subseteq W$ into the finite set $\{-2, \ldots, 2\}^\rho$; apply ⟨1⟩5. □

⟨1⟩7. Φ_R is star-closed.
PROOF: For $\mathbf{y}, \mathbf{z} \in \Phi_R$ with $(\mathbf{y}, \mathbf{z}) = -1$, $\mathbf{y} + \mathbf{z} \in \Lambda$ and $(\mathbf{y} + \mathbf{z}, \mathbf{y} + \mathbf{z}) = 2 - 2 + 2 = 2$; this is Definition 2.1. □

⟨1⟩8. Φ_R spans W.
PROOF: By ⟨1⟩3, $\{\mathbf{x}_u\} \subseteq \Phi_R \subseteq W$, and the $\mathbf{x}_u$ span W. □

⟨1⟩9. R is nonempty and indecomposable.
PROOF: Take $B = \{\mathbf{x}_u\}$ in Lemma (a connected spanning set): B is nonempty since X has a vertex, it spans the span of Φ_R by ⟨1⟩8, and distinct members of B have nonzero inner product exactly when their vertices are adjacent (Lemma (the lines of a representation), assertion 1, and Definition (representations)), so its graph is a copy of the connected graph X. □

⟨1⟩10. Q.E.D.
PROOF: Assertion 1 is ⟨1⟩1. By ⟨1⟩2, ⟨1⟩6 and Definition 2, Φ_R is the set of vectors of a line system R; ⟨1⟩7, ⟨1⟩8 and ⟨1⟩9 give the rest of assertion 2. Assertion 3 is ⟨1⟩3 with Definition (representations). □

Lemma (the apex). ASSUME: X is a k-regular graph on $n \ge 1$ vertices with a representation $(\mathbf{x}_u)$ in a real inner product space V; Λ is the $\mathbb{Z}$-span and W the span of the $\mathbf{x}_u$; and $\mathbf{w} = \frac{1}{k+2} \sum_u \mathbf{x}_u$.

PROVE:
1. $(\mathbf{w}, \mathbf{x}_v) = 1$ for every vertex v.
2. $(\mathbf{w}, \mathbf{w}) = n/(k+2)$.
3. $(\mathbf{w}, \mathbf{y}) \in \mathbb{Z}$ for every $\mathbf{y} \in \Lambda$.
4. $\mathbf{w} \in W$.

[new] Step ⟨1⟩5 of `cgss/prerequisites.md`, Section 7.3, asks for integral inner products with the roots of R only; we prove them on the whole $\mathbb{Z}$-span, and add assertion 4, which the switching from the apex needs.

PROOF SKETCH: This is step ⟨1⟩5 of `cgss/prerequisites.md`, Section 7.3; the row sums of $2I + A$ are $k + 2$.

⟨1⟩1. Assertion 1 holds.
PROOF: $(\mathbf{w},\mathbf{x}_v)=\frac{1}{k+2}\sum_u(\mathbf{x}_u,\mathbf{x}_v)=\frac{1}{k+2}(2+k)$, since by Definition (representations) the terms are 2 for $u=v$, 1 for the k neighbours of v and 0 otherwise. □
⟨1⟩2. Assertion 2 holds.
PROOF: $(\mathbf{w},\mathbf{w})=\frac{1}{k+2}\sum_v(\mathbf{w},\mathbf{x}_v)=\frac{n}{k+2}$ by ⟨1⟩1. □
⟨1⟩3. Assertion 3 holds.
PROOF: For $\mathbf{y}=\sum_v z_v\mathbf{x}_v$ with integers z_v, $(\mathbf{w},\mathbf{y})=\sum_v z_v$ by ⟨1⟩1. □
⟨1⟩4. Q.E.D.
PROOF: ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, and $\mathbf{w}$ is a combination of the $\mathbf{x}_u$. □

Lemma (integrality). ASSUME: W is a real inner product space with a basis $\mathbf{b}_1,\dots,\mathbf{b}_t$ whose Gram matrix B has integer entries, and $\mathbf{w}\in W$ has $(\mathbf{w},\mathbf{b}_i)\in\mathbb{Z}$ for every i.
PROVE: $\det(B)\,(\mathbf{w},\mathbf{w})$ is an integer.
[new] Step ⟨1⟩6 of `cgss/prerequisites.md`, Section 7.3, writes $(\mathbf{w},\mathbf{w})=\mathbf{a}^TB^{-1}\mathbf{a}$ and cites LA6, which lists tools only; we state the integrality as a result and prove it with the adjugate.

PROOF SKETCH: We will write $(\mathbf{w},\mathbf{w})$ with the inverse of B, and clear the denominator with the adjugate.
LET: $\mathbf{a}\in\mathbb{Z}^t$ with $\mathbf{a}(i)=(\mathbf{w},\mathbf{b}_i)$, and $\beta\in\mathbb{R}^t$ with $\mathbf{w}=\sum_i\beta(i)\mathbf{b}_i$.
⟨1⟩1. $\mathbf{a}=B\beta$ and $(\mathbf{w},\mathbf{w})=\beta^T\mathbf{a}$.
PROOF: $\mathbf{a}(i)=\sum_j\beta(j)(\mathbf{b}_j,\mathbf{b}_i)$ and $(\mathbf{w},\mathbf{w})=\sum_i\beta(i)(\mathbf{w},\mathbf{b}_i)$, by bilinearity. □
⟨1⟩2. $\det B\neq 0$.
PROOF: The $\mathbf{b}_i$ are linearly independent; apply `Matrix.det_gram_ne_zero_iff_linearIndependent`. □
⟨1⟩3. $\det(B)(\mathbf{w},\mathbf{w})=\mathbf{a}^T\operatorname{adj}(B)\,\mathbf{a}$.
PROOF: $\det(B)(\mathbf{w},\mathbf{w})=\det(B)\,\mathbf{a}^T\beta=\det(B)\,\mathbf{a}^TB^{-1}\mathbf{a}=\mathbf{a}^T\operatorname{adj}(B)\,\mathbf{a}$, by ⟨1⟩1, then $\beta=B^{-1}\mathbf{a}$ with ⟨1⟩2, then $B^{-1}=\det(B)^{-1}\operatorname{adj}(B)$ (`Matrix.inv_def`). □
⟨1⟩4. Q.E.D.
PROOF: The entries of $\operatorname{adj}(B)$ are signed minors of the integer matrix B, and $\mathbf{a}$ is integral; apply ⟨1⟩3. □

Computation C21. ASSUME: $t\in\{6,7,8\}$.
PROVE: There are roots $\mathbf{s}_1,\dots,\mathbf{s}_t\in\Phi(\mathrm{E}_t)$ which form a basis of the span of $\Phi(\mathrm{E}_t)$, and whose Gram matrix B_t has integer entries and determinant 3, 2 and 1 for $t=6,7,8$. Up to a relabelling, B_t is Mathlib's `CartanMatrix.E` t.
This is LN7 of `cgss/prerequisites.md`.

PROOF SKETCH: The roots are those of the SageMath certificate. Lean checks their membership, their Gram matrix and a scaled LDL^T factorization of it by `decide +kernel`; see `LineGraphs/Computations/C21E6.lean` and its analogues.
⟨1⟩1. Q.E.D.
PROOF: `LineSystem.E6.simpleRoot_mem`, `E6.gram_simpleRoot` and `E6.det_gram_simpleRoot` give the roots, their integer Gram matrix and its determinant; the Gram determinant is not 0, so the t roots are independent, and they span since the span has dimension t (Lemma (spans of the models)), which is `E6.simpleBasis`. The same declarations for E_7, E_8 give the other two cases. □

Computation C20 at the apex. ASSUME: $t\in\{6,7,8\}$, and $N_{\mathbf{c}}=\{\mathbf{x}\in\Phi(\mathrm{E}_t):(\mathbf{x},\mathbf{c})=1\}$.
PROVE: $\mathbf{c}\in\Phi(\mathrm{E}_t)$, $N_{\mathbf{c}}=\{\mathbf{e}_P:P\in\mathcal{E}_t\}\cup\{\mathbf{c}-\mathbf{e}_P:P\in\mathcal{E}_t\}$, and $|\mathcal{E}_t|$ is 10, 16 and 28 for $t=6,7,8$.
This is LN6 of `cgss/prerequisites.md`, at the root $\mathbf{c}$.

PROOF SKETCH: Lean identifies the pairs of $N_{\mathbf{c}}$ that Computation C20 lists with the members of $\mathcal{E}_t$, in `LineGraphs/Computations/C20Apex.lean`.
⟨1⟩1. Q.E.D.
PROOF: `LineSystem.E6.c_mem`, `E6.N_c` and `E6.card_CEdge`, and their analogues for E_7 and E_8; in Lean the coordinates are numbered from 0, and $\mathcal{E}_t$ is the subtype `E6.CEdge` of ordered pairs $i < j$. □

Lemma (transitivity on roots). ASSUME: $t \in \{6,7,8\}$ and $\mathbf{r} \in \Phi(\mathrm{E}_t)$.
PROVE: There is a linear isometry ψ of $\mathbb{R}^8$ with $\psi(\Phi(\mathrm{E}_t)) = \Phi(\mathrm{E}_t)$ and $\psi(\mathbf{r}) = \mathbf{c}$.
[new] This replaces LN5 of `cgss/prerequisites.md`, a transitivity by reflections, by the transitivity on ordered stars that the Lean of Computation C4 proves.

PROOF SKETCH: Both $\mathbf{r}$ and $\mathbf{c}$ lie in ordered stars, and the stars form one orbit.
⟨1⟩1. There is $\mathbf{s} \in \Phi(\mathrm{E}_t)$ with $(\mathbf{r},\mathbf{s}) = -1$.
PROOF: E_t is indecomposable (Computation C4) and has more than one line (Lemma 3.4). If every root other than $\pm\mathbf{r}$ were orthogonal to $\mathbf{r}$, then $\{\pm\mathbf{r}\}$ and the other roots would split $\Phi(\mathrm{E}_t)$ orthogonally into nonempty parts (the definition of indecomposability). So $(\mathbf{r},\mathbf{s}') \neq 0$ for some $\mathbf{s}' \neq \pm\mathbf{r}$, and then $(\mathbf{r},\mathbf{s}') = \pm 1$ by (1); take $\mathbf{s} = \mp\mathbf{s}'$. □
⟨1⟩2. $\mathbf{c}$ and $-\mathbf{e}_1 - \mathbf{e}_2$ lie in $\Phi(\mathrm{E}_t)$, and $(\mathbf{c}, -\mathbf{e}_1 - \mathbf{e}_2) = -1$.
PROOF: $\mathbf{c} \in \Phi(\mathrm{E}_t)$ by Computation C20 at the apex; $-\mathbf{e}_1 - \mathbf{e}_2$ lies in $\Phi(\mathrm{D}_8)$ and has $x_6 = x_7 = x_8 = 0$, so it lies in $\Phi(\mathrm{E}_6) \subseteq \Phi(\mathrm{E}_7)$ by Definition 3.3; and $(\mathbf{c}, \mathbf{e}_1 + \mathbf{e}_2) = \frac{1}{2} + \frac{1}{2}$. □
⟨1⟩3. Q.E.D.
PROOF: The transitivity on ordered stars of Computation C4 (in Lean; see the facts at the start of this section), applied to the star $(\mathbf{r},\mathbf{s})$ of ⟨1⟩1 and to the star $(\mathbf{c}, -\mathbf{e}_1 - \mathbf{e}_2)$ of ⟨1⟩2, gives isometries g_1, g_2 preserving $\Phi(\mathrm{E}_t)$ with $g_1(\mathbf{a}_0) = \mathbf{r}$ and $g_2(\mathbf{a}_0) = \mathbf{c}$; take $\psi = g_2 g_1^{-1}$. □

Lemma (switching from the apex). ASSUME: 1. X is a connected k-regular graph on n vertices with a representation $(\mathbf{x}_u)$ in a finite-dimensional real inner product space V.
2. Λ, W and R are as in Lemma (the lattice roots), and $\mathbf{w}$ is as in Lemma (the apex).
3. $t \in \{6,7,8\}$, and $f : \Phi(R) \to \Phi(\mathrm{E}_t)$ is a bijection that preserves inner products.
4. $(\mathbf{w},\mathbf{w}) = 2$ and $n = |\mathcal{E}_t|$.
PROVE: There are a bijection $\pi : V(X) \to \mathcal{E}_t$ and a set $H \subseteq \mathcal{E}_t$ such that π is an isomorphism from X to $L(\Delta_t)^H$.
[new] These are steps ⟨1⟩7 to ⟨1⟩9 of `cgss/prerequisites.md`, Section 7.3; we read the switched graph off the inner products directly, without SW4.

PROOF SKETCH: The apex is a root, the $\mathbf{x}_u$ are a transversal of the pairs of $N_{\mathbf{w}}$, and an isometry takes $\mathbf{w}$ to $\mathbf{c}$, where the pairs are indexed by $\mathcal{E}_t$.
⟨1⟩1. R is maximal in W.
PROOF: R is nonempty and indecomposable and spans W (Lemma (the lattice roots)); it is isomorphic to E_t by ⟨0⟩:3, which is in the list of Theorem 3.6. □
⟨1⟩2. $\mathbf{w} \in \Phi(R)$.
PROOF: Lemma (the apex) applies, since $n = |\mathcal{E}_t| \geq 1$ by ⟨0⟩:4 and Computation C20 at the apex; so $\mathbf{w} \in W$, and $(\mathbf{w},\mathbf{w}) = 2$ by ⟨0⟩:4. For $\mathbf{y} \in \Phi(R) \subseteq \Lambda$, $(\mathbf{w},\mathbf{y})$ is an integer (Lemma (the apex)) with $|(\mathbf{w},\mathbf{y})| \leq |\mathbf{w}||\mathbf{y}| = 2$ (`abs_real_inner_le_norm`); apply ⟨1⟩1 and Definition 2.2. □
⟨1⟩3. There is a bijection $g : \Phi(R) \to \Phi(\mathrm{E}_t)$ that preserves inner products, with $g(\mathbf{w}) = \mathbf{c}$.
PROOF: $f(\mathbf{w}) \in \Phi(\mathrm{E}_t)$ by ⟨1⟩2; Lemma (transitivity on roots) gives ψ with $\psi(f(\mathbf{w})) = \mathbf{c}$, and $g = \psi \circ f$ will do. □
LET: $\mathbf{y}_u = g(\mathbf{x}_u)$.
⟨1⟩4. $(\mathbf{y}_u)$ represents X in E_t, and $(\mathbf{y}_u,\mathbf{c}) = 1$ for every u.
PROOF: X is represented in R by $(\mathbf{x}_u)$ (Lemma (the lattice roots)); apply Lemma (transport of representations) to g. Further $(\mathbf{y}_u,\mathbf{c}) = (g(\mathbf{x}_u), g(\mathbf{w})) = (\mathbf{x}_u,\mathbf{w}) = 1$ by Lemma (the apex). □

⟨1⟩5. PICK $\pi(u) \in \mathcal{E}_t$ for each u, and $D \subseteq V(X)$, such that $\mathbf{y}_u = \mathbf{c} - \mathbf{e}_{\pi(u)}$ for $u \in D$ and $\mathbf{y}_u = \mathbf{e}_{\pi(u)}$ for $u \notin D$.
PROOF: $\mathbf{y}_u \in N_{\mathbf{c}}$ by ⟨1⟩4; apply Computation C20 at the apex. □
⟨1⟩6. For $P, P' \in \mathcal{E}_t$, $(\mathbf{e}_P, \mathbf{e}_{P'}) = (\mathbf{c} - \mathbf{e}_P, \mathbf{c} - \mathbf{e}_{P'}) = |P \cap P'|$ and $(\mathbf{e}_P, \mathbf{c} - \mathbf{e}_{P'}) = 1 - |P \cap P'|$.
PROOF: Expand, with $(\mathbf{c}, \mathbf{c}) = 2$, $(\mathbf{c}, \mathbf{e}_i) = \frac{1}{2}$ and $(\mathbf{e}_i, \mathbf{e}_j) = [i = j]$. □
⟨1⟩7. π is injective.
PROOF: If $u \neq v$ and $\pi(u) = \pi(v) = P$, then $\mathbf{y}_u \neq \mathbf{y}_v$ (Lemma (the lines of a representation), assertion 1), so one is $\mathbf{e}_P$ and the other $\mathbf{c} - \mathbf{e}_P$, and their inner product is $1 - 2 = -1$ by ⟨1⟩6; this is against Definition (representations) and ⟨1⟩4. □
⟨1⟩8. π is a bijection.
PROOF: ⟨1⟩7, and $|V(X)| = n = |\mathcal{E}_t|$ by ⟨0⟩:4. □
LET: $H = \pi(D)$.
⟨1⟩9. Distinct u, v are adjacent in X if and only if $\pi(u)$ and $\pi(v)$ are adjacent in $L(\Delta_t)^H$.
PROOF: By ⟨1⟩7, $|\pi(u) \cap \pi(v)| \leq 1$, and $u \in D$ exactly when $\pi(u) \in H$. By ⟨1⟩4 and ⟨1⟩6, $(\mathbf{y}_u, \mathbf{y}_v) = 1$ exactly when $\pi(u), \pi(v)$ meet and both or neither lie in H, or are disjoint and exactly one lies in H; this is adjacency in $L(\Delta_t)^H$ (Definition (switching)). □
⟨1⟩10. Q.E.D.
PROOF: ⟨1⟩8 and ⟨1⟩9. □

Regular switchings.

Lemma (degrees in a switched line graph). ASSUME: 1. Y is a graph, $M \subseteq V(Y)$ has $m \geq 2$ elements, and every 2-subset of M is an edge of Y; K is the set of these edges.
2. $H \subseteq E(Y)$, $h = |H \cap K|$, and $\deg_{H \cap K}$ is taken on M.
3. $e = \{i, j\} \in K$ and $s = \deg_{H \cap K}(i) + \deg_{H \cap K}(j)$.
PROVE: The number of neighbours of e in $L(Y)^H$ that lie in K is $2s - h - 3 + \binom{m-2}{2}$ if $e \in H$, and $2(m-2) + h - 2s$ if $e \notin H$.
[new] This replaces SW3 of `cgss/prerequisites.md` in step ⟨1⟩10 of its Section 7.3 by a direct count.

PROOF SKETCH: We will count the members of K that meet e and those that miss it, inside and outside H.
⟨1⟩1. The members of $H \cap K$ other than e that meet e number $s - 2$ if $e \in H$ and s if $e \notin H$.
PROOF: No member of K other than e contains both i and j, so the members of $H \cap K$ meeting e number $\deg_{H \cap K}(i) + \deg_{H \cap K}(j)$, counting e twice when $e \in H$. □
⟨1⟩2. The members of K other than e that meet e number $2(m-2)$, and those disjoint from e number $\binom{m-2}{2}$.
PROOF: They are $\{i, l\}$ and $\{j, l\}$ with $l \in M \setminus e$, and the 2-subsets of $M \setminus e$. □
⟨1⟩3. CASE: $e \in H$.
PROVE: The count is $2s - h - 3 + \binom{m-2}{2}$.
PROOF: By Definition (switching), the neighbours of e in K are the $f \in H \cap K$ that meet e, $s - 2$ of them by ⟨1⟩1, and the $f \in K \setminus H$ disjoint from e. Of the $\binom{m-2}{2}$ members of K disjoint from e (⟨1⟩2), $h - 1 - (s - 2)$ lie in H. The total is $(s - 2) + \binom{m-2}{2} - (h - s + 1)$. □
⟨1⟩4. CASE: $e \notin H$.
PROVE: The count is $2(m-2) + h - 2s$.
PROOF: By Definition (switching), the neighbours are the $f \in K \setminus H$ with $f \neq e$ that meet e, which number $2(m-2) - s$ by ⟨1⟩1 and ⟨1⟩2, and the $f \in H \cap K$ disjoint from e, which number $h - s$. □
⟨1⟩5. Q.E.D.
PROOF: The two cases are exhaustive. □

Lemma (constant pair sums). ASSUME: M is a finite set with at least 3 elements, $d : M \to \mathbb{Z}$ and $\sigma \in \mathbb{Z}$, and $d(i) + d(j) = \sigma$ for all distinct $i, j \in M$.
PROVE: $2d(i) = \sigma$ for every $i \in M$.

PROOF SKETCH: Compare two pairs with a common member.
⟨1⟩1. $d(i) = d(j)$ for all $i, j \in M$.
PROOF: For $i \neq j$ pick $l \in M \setminus \{i, j\}$; then $d(i) + d(l) = \sigma = d(j) + d(l)$ by ⟨0⟩. □
⟨1⟩2. Q.E.D.
PROOF: For $i \in M$ pick $j \neq i$; then $\sigma = d(i) + d(j) = 2d(i)$ by ⟨1⟩1. □

Theorem (SE4: regular graphs on 8 vertices). ASSUME: Z is a finite graph.
PROVE: 1. If Z is connected and 2-regular, then Z is a cycle.
2. If Z is d-regular on 8 vertices with $d \leq 3$, then Z is isomorphic to one of the following eleven graphs: for $d = 0$, $\overline{K_8}$; for $d = 1$, $4K_2$; for $d = 2$, C_8, $C_3 + C_5$ or $2C_4$; for $d = 3$, $2K_4$, the cube Q_3, the Wagner graph M_8, or the graphs Y_1, Y_2, Y_4 of `cgss/proofs/reg8/reg8.tex`, Definition 1.1.

This is an input, proved in `cgss/proofs/reg8/reg8.tex` (Lemma 2.1 and Theorem 1.2); in Lean its graphs are `SimpleGraph.fourK2`, `cycleThreeFive`, `twoC4`, `twoK4`, `cubeGraph`, `wagnerGraph`, `truncatedK33Graph`, `twiceTruncatedK4Graph` and `twoDiamondsGraph` on `Fin 8`. Its vertex i is our vertex $i + 1$.

PROOF SKETCH: We cite the structured case split of `cgss/proofs/reg8/reg8.tex`.
⟨1⟩1. Q.E.D.
PROOF: Lemma 2.1 of `cgss/proofs/reg8/reg8.tex` for assertion 1, and its Theorem 1.2 for assertion 2. □

Lemma (2-regular graphs on 5 and 6 vertices). ASSUME: Z is a 2-regular graph on $p \in \{5, 6\}$ vertices.
PROVE: If $p = 5$ then $Z \cong C_5$; if $p = 6$ then $Z \cong C_6$ or $Z \cong C_3 + C_3$.

PROOF SKETCH: The components are cycles of length at least 3.
⟨1⟩1. Each component of Z is a connected 2-regular graph with at least 3 vertices.
PROOF: The neighbours of a vertex lie in its component, so the component is 2-regular; a vertex and its two neighbours are three vertices. □
⟨1⟩2. Each component of Z is a cycle.
PROOF: Theorem (SE4: regular graphs on 8 vertices), assertion 1, with ⟨1⟩1. □
⟨1⟩3. Q.E.D.
PROOF: By ⟨1⟩1 the orders of the components are at least 3 and add up to p: for $p = 5$ this is 5 alone, and for $p = 6$ it is 6 or $3 + 3$. Apply ⟨1⟩2. □

Computation C22. ASSUME: H_5 is the edge set of the 5-cycle $1, 2, 3, 4, 5$; H_6 that of the 6-cycle $1, \ldots, 6$; $H_{3,3} = \{12, 13, 23, 45, 46, 56\}$; the graphs of Theorem (SE4: regular graphs on 8 vertices), assertion 2, are placed on $\{1, \ldots, 8\}$; and $\mathrm{prof}_4(Z)$ is the multiset of the numbers of 4-cliques of Z through its vertices.
PROVE: 1. $L(\Delta_6)^{H_5}$ is isomorphic to the Petersen graph of Example 4.9.
2. $L(\Delta_7)^{H_6}$ is isomorphic to the Shrikhande graph of Example 4.7, and $L(\Delta_7)^{H_{3,3}}$ to $L(K_{4,4})$.
3. For the graphs Z of Theorem (SE4: regular graphs on 8 vertices), assertion 2, $L(\Delta_8)^{E(Z)}$ is isomorphic to $T(8)$ for $Z = \overline{K_8}, 2K_4$; to the Chang graph I of Example 4.5 for $Z = 4K_2, 2C_4, Q_3, Y_4$; to the Chang graph II for $Z = C_8, M_8, Y_2$; and to the Chang graph III for $Z = C_3 + C_5, Y_1$.
4. $\mathrm{prof}_4(T(8)) = [\![40^{28}]\!]$, and prof_4 of the Chang graphs I, II and III is $[\![32^4, 36^{24}]\!]$, $[\![34^{24}, 36^4]\!]$ and $[\![33^{10}, 35^{18}]\!]$.

5. The Shrikhande graph has no 4-clique, and $L(K_{4,4})$ has 8.

These are SE5 and the part of SE6 that we need (`cgss/prerequisites.md`), with the 4-clique profile in place of the maximal cliques (`cgss/certificates/README.md`, C22).

PROOF SKETCH: Each isomorphism is an explicit permutation of at most 28 vertices, found by SageMath (`cgss/certificates/c22_switching.sage`, certificates in `cgss/certificates/data/c22.json`); Lean checks it by `Computations.checkGraphIso` and `decide +kernel`. The profiles are evaluated by `Computations.cliqueProfile`, which enumerates only the 4-subsets that are cliques.

⟨1⟩1. Assertions 1 to 3 hold.
PROOF: For each of the fourteen switched graphs, the adjacency table and the certificate permutation pass `Computations.checkGraphIso`; apply `Computations.nonempty_graphOfMatrix_iso`. For $Z = \overline{K_8}$ the permutation is the identity, since $L(\Delta_8)^{\varnothing} = T(8)$. □

⟨1⟩2. Assertions 4 and 5 hold.
PROOF: `Computations.cliqueProfile` and `Computations.cliqueCount` on the adjacency tables, with their soundness theorems; the profiles of assertion 4 are those of `Computations.Examples.t8_clique_profile` in `LineGraphs/Computations/CliqueBenchmarkT8.lean` and of its analogues for the Chang graphs in `LineGraphs/Computations/CliqueBenchmarkChangI.lean`, `LineGraphs/Computations/CliqueBenchmarkChangII.lean` and `LineGraphs/Computations/CliqueBenchmarkChangIII.lean`. □

⟨1⟩3. Q.E.D.
PROOF: ⟨1⟩1 and ⟨1⟩2. □

Lemma (regular switchings of $T(5)$). ASSUME: H is a subset of $\mathcal{E}_6$, and the graph $L(\Delta_6)^H$ is 3-regular.
PROVE: $L(\Delta_6)^H$ is isomorphic to the Petersen graph of Example 4.9.

PROOF SKETCH: The degree count forces H to be the edge set of a 2-regular graph on $\{1,\dots,5\}$, which is a 5-cycle.

LET: $h = |H|$ and $M = \{1,\dots,5\}$.

⟨1⟩1. $2(\deg_H(i) + \deg_H(j)) = h + 3$ for all distinct $i, j \in M$.
PROOF: Every vertex of $L(\Delta_6)$ lies in $K = \mathcal{E}_6$. Lemma (degrees in a switched line graph) with $m = 5$, $\binom{3}{2} = 3$ and $2(m-2) = 6$ gives degree $2s - h$ if $\{i,j\} \in H$ and $6 + h - 2s$ if not; both equal 3 by ⟨0⟩. □

⟨1⟩2. $4\deg_H(i) = h + 3$ for every $i \in M$.
PROOF: Lemma (constant pair sums), applied to $d = 2\deg_H$ on M with $\sigma = h + 3$, and ⟨1⟩1: $2 \cdot 2\deg_H(i) = h + 3$. □

⟨1⟩3. (M, H) is 2-regular.
PROOF: Summing ⟨1⟩2 over M, with $\sum_i \deg_H(i) = 2h$ (`SimpleGraph.sum_degrees_eq_twice_card_edges`), gives $8h = 5(h+3)$, so $h = 5$ and $\deg_H(i) = 2$. □

⟨1⟩4. $L(\Delta_6)^H \cong L(\Delta_6)^{H_5}$.
PROOF: By ⟨1⟩3 and Lemma (2-regular graphs on 5 and 6 vertices), some bijection σ of M carries H onto H_5; extended by fixing 6, 7, 8, it is an automorphism of Δ_6. Apply Lemma (switching), assertion 3. □

⟨1⟩5. Q.E.D.
PROOF: ⟨1⟩4 and Computation C22, assertion 1. □

Lemma (regular switchings of $T(6)$ with an isolated vertex). ASSUME: H is a subset of $\mathcal{E}_7$, and the graph $L(\Delta_7)^H$ is 6-regular.

PROVE: $L(\Delta_7)^H$ is isomorphic to the Shrikhande graph of Example 4.7 or to $L(K_{4,4})$.

PROOF SKETCH: After complementing H we may assume that $\varepsilon \notin H$; then the degree of ε gives $|H| = 6$, and the degree count makes H 2-regular on $\{1, \ldots, 6\}$.
LET: $M = \{1, \ldots, 6\}$, $K = \mathcal{E}_7 \setminus \{\varepsilon\}$, and $H^* = H$ if $\varepsilon \notin H$, $H^* = \mathcal{E}_7 \setminus H$ if $\varepsilon \in H$.
⟨1⟩1. $\varepsilon \notin H^*$, $H^* \subseteq K$, and $L(\Delta_7)^H = L(\Delta_7)^{H^*}$.
PROOF: The definition of H^*, and Lemma (switching), assertion 1, when $H^* = \mathcal{E}_7 \setminus H$. □
⟨1⟩2. $|H^*| = 6$.
PROOF: ε is disjoint from every other member of $\mathcal{E}_7$, so it is isolated in $L(\Delta_7)$; by Definition (switching) and ⟨1⟩1 its neighbours in $L(\Delta_7)^{H^*}$ are the members of H^*. Its degree is 6 by ⟨0⟩ and ⟨1⟩1. □
⟨1⟩3. $\deg_{H^*}(i) + \deg_{H^*}(j) = 4$ for all distinct $i, j \in M$.
PROOF: Let $e = \{i, j\}$ and $s = \deg_{H^*}(i) + \deg_{H^*}(j)$. By Lemma (degrees in a switched line graph) with $m = 6$, $\binom{4}{2} = 6$, $2(m-2) = 8$ and $h = 6$ (⟨1⟩2), e has $2s - 3$ neighbours in K if $e \in H^*$ and $14 - 2s$ if not. By Definition (switching), ε is a further neighbour exactly when $e \in H^*$. So the degree is $2s - 2$ or $14 - 2s$, and it is 6. □
⟨1⟩4. (M, H^*) is 2-regular.
PROOF: Lemma (constant pair sums) with $\sigma = 4$ and ⟨1⟩3. □
⟨1⟩5. $L(\Delta_7)^{H^*} \cong L(\Delta_7)^{H_6}$ or $L(\Delta_7)^{H^*} \cong L(\Delta_7)^{H_{3,3}}$.
PROOF: By ⟨1⟩4 and Lemma (2-regular graphs on 5 and 6 vertices), some bijection σ of M carries H^* onto H_6 or onto $H_{3,3}$; extended by fixing 7 and 8, it is an automorphism of Δ_7. Apply Lemma (switching), assertion 3. □
⟨1⟩6. Q.E.D.
PROOF: ⟨1⟩1, ⟨1⟩5 and Computation C22, assertion 2. □

Lemma (regular switchings of $T(8)$). ASSUME: H is a subset of $\mathcal{E}_8$, and the graph $L(\Delta_8)^H$ is 12-regular.
PROVE: $L(\Delta_8)^H$ is isomorphic to $T(8)$ or to one of the Chang graphs I, II and III of Example 4.5.

PROOF SKETCH: Here the degree count says only that H is regular; after complementing, its valency is at most 3, and SE4 leaves eleven graphs.
LET: $h = |H|$ and $M = \{1, \ldots, 8\}$.
⟨1⟩1. $2(\deg_H(i) + \deg_H(j)) = h$ for all distinct $i, j \in M$.
PROOF: Every vertex of $L(\Delta_8)$ lies in $K = \mathcal{E}_8$. Lemma (degrees in a switched line graph) with $m = 8$, $\binom{6}{2} = 15$ and $2(m-2) = 12$ gives degree $2s - h + 12$ if $\{i, j\} \in H$ and $12 + h - 2s$ if not; both equal 12 by ⟨0⟩. □
⟨1⟩2. (M, H) is regular of some valency $d \leq 7$.
PROOF: Lemma (constant pair sums), applied to $d' = 2\deg_H$ on M with $\sigma = h$, and ⟨1⟩1 give $4\deg_H(i) = h$ for every $i \in M$; so $d = h/4$, and $d \leq 7$ since $|M| = 8$. □
LET: $H^* = H$ if $d \leq 3$, and $H^* = \mathcal{E}_8 \setminus H$ if $d \geq 4$.
⟨1⟩3. (M, H^*) is regular of valency at most 3, and $L(\Delta_8)^H = L(\Delta_8)^{H^*}$.
PROOF: $\mathcal{E}_8 \setminus H$ is the edge set of the complement of (M, H), which is $(7 - d)$-regular (`SimpleGraph.IsRegularOfDegree.compl`); apply Lemma (switching), assertion 1. □
⟨1⟩4. $L(\Delta_8)^{H^*} \cong L(\Delta_8)^{E(Z)}$ for one of the eleven graphs Z of Theorem (SE4: regular graphs on 8 vertices), assertion 2, placed on M.
PROOF: By ⟨1⟩3 and Theorem (SE4: regular graphs on 8 vertices), some bijection σ of M is an isomorphism from (M, H^*) to such a Z; it is an automorphism of $\Delta_8 = K_8$. Apply Lemma (switching), assertion 3. □
⟨1⟩5. Q.E.D.
PROOF: ⟨1⟩3, ⟨1⟩4 and Computation C22, assertion 3. □

The Schläfli and Clebsch graphs.

Lemma (the triangle property from the parameters, GQ1). ASSUME: Z is strongly regular with parameters $(n, k, 1, c)$, and $n + 3 = 3k$.
PROVE: Z has the triangle property.
[new] This is the converse half of GQ1 of `cgss/prerequisites.md`, with the triangle property of Section 2 in place of a generalized quadrangle whose lines are the triangles; [BCN] Lemma 1.15.1 says only "Straightforward". The hypothesis is $n - 3 = 3(k - 2)$, written without subtraction as in Lean.

PROOF SKETCH: The edges leaving a triangle reach every other vertex exactly once.
⟨1⟩1. Every edge xy has exactly one common neighbour z.
PROOF: The parameter $a = 1$. □
⟨1⟩2. A vertex $v \notin \{x, y, z\}$ is adjacent to at most one of x, y, z.
PROOF: If v were adjacent to two of them, say to x and y, then v and z would be two common neighbours of the edge xy, against ⟨1⟩1; the other pairs are edges of the triangle as well. □
⟨1⟩3. Q.E.D.
PROOF: Each of x, y, z has $k - 2$ neighbours outside $\{x, y, z\}$, so there are $3(k - 2) = n - 3$ edges from the triangle to the other $n - 3$ vertices, by ⟨0⟩; by ⟨1⟩2 no vertex receives two, so every vertex receives exactly one. With ⟨1⟩1, this is Definition (the triangle property). □

Definition (the folded 5-cube). ASSUME: $\Omega_5 = \{1, \ldots, 5\}$, and V_{16} is the set of the subsets of Ω_5 of even size.
LET: The *folded 5-cube* $\square_5$ is the graph on V_{16} in which $s \sim t$ when $|s \triangle t| = 4$.
This is Definition 1.1 of `cgss/proofs/clebsch/clebsch.tex`. V_{16} consists of $\varnothing$, the ten 2-subsets and the five 4-subsets of Ω_5. In Lean the vertex type is {`s : Finset (Fin 5) // Even s.card`}, whose `i` is our $i + 1$. In the complement, distinct s, t are adjacent exactly when $|s \triangle t| = 2$, since $|s \triangle t|$ is even and at most 5.

Definition (the double-six graph). ASSUME: $\Omega_6 = \{1, \ldots, 6\}$, and V_{27} is the set of the 27 symbols a_i, b_i $(i \in \Omega_6)$ and $c_{ij} = c_{ji}$ $(i \neq j$ in $\Omega_6)$.
LET: Q_{27} be the graph on V_{27} in which $a_i \sim b_j$ for $i \neq j$; $a_i \sim c_{jl}$ and $b_i \sim c_{jl}$ for $i \in \{j, l\}$; $c_{ij} \sim c_{lm}$ when $\{i, j\} \cap \{l, m\} = \varnothing$; and there are no other edges.
This is the graph Q of `cgss/proofs/clebsch/clebsch.tex`, Definition 1.2, the collinearity graph of the generalized quadrangle of order $(2, 4)$; we write Q_{27}, since Q is $2I + A$ in this section. In Lean the vertex type is `Fin 2 × Fin 6 ⊕ T` with `T = {s : Finset (Fin 6) // s.card = 2}`; $(0, i)$ is a_{i+1}, $(1, i)$ is b_{i+1}, and $\{i, j\}$ is $c_{i+1,j+1}$. The second Lean name spells out adjacency in $\overline{Q_{27}}$ by the same cases.

Computation (the models of SE2 and SE3). ASSUME: X_{Cl} and X_{Sch} are the Clebsch and Schläfli graphs of Examples 4.8 and 4.6, that is, the graphs of the vectors of their representations (Computation C9).
PROVE: 1. $\square_5$ is strongly regular with parameters $(16, 5, 0, 2)$, and Q_{27} with parameters $(27, 10, 1, 5)$.
2. $X_{\mathrm{Cl}} \cong \overline{\square_5}$ and $X_{\mathrm{Sch}} \cong \overline{Q_{27}}$.
The Clebsch and Schläfli graphs stay the graphs of the vectors; $\square_5$ and Q_{27} are the combinatorial models of `cgss/proofs/clebsch/clebsch.tex`. The isomorphisms of assertion 2 are the certificates of Computation C9, which carry their Lean names there; we repeat them here beside the parameters.

PROOF SKETCH: These are finite checks on at most 27 vertices, as for Computation C9.
⟨1⟩1. Assertion 1 holds.

PROOF: In Lean, `decide` counts the neighbours of every vertex and the common neighbours of every pair, at most 27^3 triples. By hand, Lemmas 3.1 and 4.2 of `cgss/proofs/clebsch/clebsch.tex` count them at one vertex of each orbit of an automorphism group. □

⟨1⟩2. Assertion 2 holds.
PROOF: Computation C9, its last sentence. □

⟨1⟩3. Q.E.D.
PROOF: ⟨1⟩1 and ⟨1⟩2. □

Lemma (the double-six graph from its parameters). ASSUME: Z is a strongly regular graph with parameters $(27, 10, 1, 5)$.
PROVE: $Z \cong Q_{27}$.
[new] SE3 of `cgss/prerequisites.md` identifies Z by GQ2, the classification of the generalized quadrangles of order $(2, 4)$; we use instead the isomorphism from the order of Section 2, as in Theorem 1.4 of `cgss/proofs/clebsch/clebsch.tex`.

PROOF SKETCH: Both graphs have the triangle property, and Section 2 shows that such a graph is determined by its order.

⟨1⟩1. Z and Q_{27} have the triangle property.
PROOF: Lemma (the triangle property from the parameters, GQ1), with ⟨0⟩ and Computation (the models of SE2 and SE3), assertion 1, since $27 + 3 = 3 \cdot 10$. □

⟨1⟩2. Z and Q_{27} have 27 vertices each, an edge, and no vertex adjacent to every other vertex.
PROOF: Both are 10-regular on 27 vertices, by ⟨0⟩ and Computation (the models of SE2 and SE3), assertion 1, and $0 < 10 < 26$. □

⟨1⟩3. Q.E.D.
PROOF: Lemma (isomorphism from the order), with ⟨1⟩1 and ⟨1⟩2. □

Lemma (SE3: the Schläfli graph). ASSUME: X is a strongly regular graph with parameters $(27, 16, 10, 8)$.
PROVE: X is isomorphic to the Schläfli graph of Example 4.6.
[new] SE3 of `cgss/prerequisites.md` goes through GQ1 and GQ2; we go through Lemma (the double-six graph from its parameters) and the certificate of Computation (the models of SE2 and SE3).

PROOF SKETCH: We will pass to the complement, which is the double-six graph.
LET: X_{Sch} be the Schläfli graph of Example 4.6.

⟨1⟩1. $\overline{X}$ is strongly regular with parameters $(27, 10, 1, 5)$.
PROOF: `SimpleGraph.IsSRGWith.compl`: $(27, 27 - 16 - 1, 27 - (32 - 8) - 2, 27 - (32 - 10))$, with no truncated subtraction. □

⟨1⟩2. $\overline{X} \cong Q_{27}$.
PROOF: Lemma (the double-six graph from its parameters) with ⟨1⟩1. □

⟨1⟩3. Q.E.D.
PROOF: Lemma (isomorphisms and complements), assertion 1 (`SimpleGraph.Iso.compl`), applied to ⟨1⟩2, with `compl_compl`, gives $X \cong \overline{Q_{27}}$; and $\overline{Q_{27}} \cong X_{\mathrm{Sch}}$ by Computation (the models of SE2 and SE3), assertion 2. □

Lemma (the folded 5-cube from its parameters). ASSUME: Z is a strongly regular graph with parameters $(16, 5, 0, 2)$.
PROVE: $Z \cong \square_5$.
This is the first half of SE2 of `cgss/prerequisites.md`, after [C17] Theorem 2.2, and Theorem 1.3 of `cgss/proofs/clebsch/clebsch.tex`. [BCN] 3.11 (iii) prints these parameters as $(16, 5, 0, 1)$; they are $(16, 5, 0, 2)$.

PROOF SKETCH: The non-neighbours of a vertex u correspond to the 2-subsets of its neighbourhood, and adjacency among them is disjointness.
LET: u be a vertex of Z, N its neighbourhood and $O = V(Z) \setminus (\{u\} \cup N)$.
⟨1⟩1. N is a coclique of 5 vertices, and $|O| = 10$.
PROOF: $|N| = 5$ and $|O| = 16 - 1 - 5$. Two adjacent vertices of N would have the common neighbour u, against $a = 0$. □
⟨1⟩2. For $w \in O$, $\phi(w) = N(w) \cap N$ has exactly two elements.
PROOF: $w \neq u$ and $w \not\sim u$, so they have $c = 2$ common neighbours, which are the members of $N(w) \cap N$. □
⟨1⟩3. ϕ is a bijection from O onto $\binom{N}{2}$.
PROOF: If $w \neq w'$ in O had $\phi(w) = \phi(w') = \{x, y\}$, the distinct non-adjacent vertices x, y (⟨1⟩1) would have the three common neighbours u, w, w', against $c = 2$. So ϕ is injective, and $|O| = 10 = \binom{5}{2}$. □
⟨1⟩4. ASSUME: $w, w' \in O$ and $\phi(w) \cap \phi(w') \neq \varnothing$.
PROVE: $w \not\sim w'$.
PROOF: Take $x \in \phi(w) \cap \phi(w')$ by ⟨1⟩; an edge ww' would have the common neighbour x, against $a = 0$. □
⟨1⟩5. Each $w \in O$ has exactly 3 neighbours in O.
PROOF: w has 5 neighbours; two lie in N (⟨1⟩2), and u is not one. □
⟨1⟩6. Distinct $w, w' \in O$ are adjacent if and only if $\phi(w)$ and $\phi(w')$ are disjoint.
PROOF: Exactly $\binom{3}{2} = 3$ members of $\binom{N}{2}$ are disjoint from $\phi(w)$, so by ⟨1⟩3 exactly 3 vertices $w' \in O$ have $\phi(w')$ disjoint from $\phi(w)$. By ⟨1⟩4 the 3 neighbours of w in O (⟨1⟩5) are among them; so they are all of them. □
LET: $\beta : N \to \Omega_5$ be a bijection, and $f : V(Z) \to V_{16}$ the map with $f(u) = \varnothing$, $f(x) = \Omega_5 \setminus \{\beta(x)\}$ for $x \in N$ and $f(w) = \beta(\phi(w))$ for $w \in O$.
⟨1⟩7. f is a bijection.
PROOF: f maps u to $\varnothing$, N onto the five 4-subsets, since β is a bijection, and O onto the ten 2-subsets, by ⟨1⟩3; these are all of V_{16} (Definition (the folded 5-cube)). □
⟨1⟩8. For $i \in \Omega_5$ and 2-subsets $P \neq P'$ of Ω_5, $|(\Omega_5 \setminus \{i\}) \triangle P|$ is 4 if $i \in P$ and 2 if $i \notin P$, and $|P \triangle P'| = 4 - 2|P \cap P'|$.
PROOF: $|s \triangle s'| = |s| + |s'| - 2|s \cap s'|$, and $(\Omega_5 \setminus \{i\}) \cap P$ has 1 element if $i \in P$ and 2 if not. □
⟨1⟩9. Distinct vertices p, q of Z are adjacent if and only if $|f(p) \triangle f(q)| = 4$.
⟨2⟩1. CASE: $p = u$ and $q = x \in N$.
PROVE: Both hold.
PROOF: $u \sim x$, and $|\varnothing \triangle (\Omega_5 \setminus \{\beta(x)\})| = 4$. □
⟨2⟩2. CASE: $p = u$ and $q = w \in O$.
PROVE: Neither holds.
PROOF: $w \notin N$, and $|\varnothing \triangle f(w)| = |f(w)| = 2$. □
⟨2⟩3. CASE: $p = x$ and $q = y$ lie in N.
PROVE: Neither holds.
PROOF: $x \not\sim y$ by ⟨1⟩1, and $|f(x) \triangle f(y)| = |\{\beta(x), \beta(y)\}| = 2$. □
⟨2⟩4. CASE: $p = x \in N$ and $q = w \in O$.
PROVE: The two conditions agree.
PROOF: $x \sim w$ exactly when $x \in \phi(w)$, that is, when $\beta(x) \in f(w)$, and by ⟨1⟩8 this is when $|f(x) \triangle f(w)| = 4$. □
⟨2⟩5. CASE: $p = w$ and $q = w'$ lie in O.
PROVE: The two conditions agree.

PROOF: $w \sim w'$ exactly when $f(w) \cap f(w') = \varnothing$, by ⟨1⟩6 and the injectivity of β, and by ⟨1⟩8 this is when $|f(w) \triangle f(w')| = 4$. □

⟨2⟩6. Q.E.D.

PROOF: $V(Z)$ is the disjoint union of $\{u\}$, N and O, so up to the order of p and q the five cases are exhaustive. □

⟨1⟩10. Q.E.D.

PROOF: By ⟨1⟩7 and ⟨1⟩9, f is an isomorphism from Z to $\square_5$ (Definition (the folded 5-cube)). □

Lemma (SE2: the Clebsch graph). ASSUME: X is a strongly regular graph with parameters $(16, 10, 6, 6)$.

PROVE: X is isomorphic to the Clebsch graph of Example 4.8.

This is SE2 of `cgss/prerequisites.md`.

PROOF SKETCH: We will pass to the complement, which is the folded 5-cube.

LET: X_{Cl} be the Clebsch graph of Example 4.8.

⟨1⟩1. $\overline{X}$ is strongly regular with parameters $(16, 5, 0, 2)$.

PROOF: `SimpleGraph.IsSRGWith.compl`: $(16, 16-10-1, 16-(20-6)-2, 16-(20-6))$, with no truncated subtraction. □

⟨1⟩2. $\overline{X} \cong \square_5$.

PROOF: Lemma (the folded 5-cube from its parameters) with ⟨1⟩1. □

⟨1⟩3. Q.E.D.

PROOF: Lemma (isomorphisms and complements), assertion 1 (`SimpleGraph.Iso.compl`), applied to ⟨1⟩2, with `compl_compl`, gives $X \cong \overline{\square_5}$; and $\overline{\square_5} \cong X_{\mathrm{Cl}}$ by Computation (the models of SE2 and SE3), assertion 2. □

The seven graphs.

Proposition (the seven graphs). ASSUME: $\mathcal{S}$ is the list of the seven graphs of Examples 4.5 to 4.9: the Chang graphs I, II and III, the Schläfli graph, the Shrikhande graph, the Clebsch graph and the Petersen graph.

PROVE:
1. Each $Z \in \mathcal{S}$ is connected and strongly regular, has $\lambda_{\min}(Z) = -2$, and is represented in E_8.
2. No $Z \in \mathcal{S}$ is isomorphic to $L(K_m)$, to $L(K_{m,m})$ or to $CP(m)$, for any m.
3. No two graphs of $\mathcal{S}$ are isomorphic.

[repair] "The seven graphs" presupposes that the three Chang graphs are pairwise non-isomorphic, which the paper does not show (errata, Gap 12). We will separate them by their 4-clique profiles (Computation C22), which the kernel can evaluate by extending cliques one vertex at a time; their numbers of maximal cliques, used in SE6 of `cgss/prerequisites.md`, can be enumerated in the same way, but every clique found must then be checked for maximality, which costs the kernel much more.

PROOF SKETCH: The parameters separate all pairs except the ones sharing (n, k), and C22 separates those.

⟨1⟩1. Each $Z \in \mathcal{S}$ is represented in E_8, and is strongly regular with parameters $(28, 12, 6, 4)$ (Chang), $(27, 16, 10, 8)$, $(16, 6, 2, 2)$, $(16, 10, 6, 6)$ and $(10, 3, 0, 1)$.

PROOF: Examples 4.5 to 4.9, and Computation C9. □

⟨1⟩2. Each $Z \in \mathcal{S}$ is connected.

PROOF: In each case $c \geq 1$ by ⟨1⟩1, so any two distinct non-adjacent vertices have a common neighbour. □

⟨1⟩3. $\lambda_{\min}(Z) = -2$ for each $Z \in \mathcal{S}$.

PROOF: Z is represented in $\Phi(\mathrm{E}_8) \subseteq \mathbb{R}^8$ (⟨1⟩1), a space of dimension $8 < 10 \leq |V(Z)|$; apply Lemma (the Gram matrix), assertion 2. □

⟨1⟩4. $L(K_m)$ has $\binom{m}{2}$ vertices and valency $2(m-2)$, $L(K_{m,m})$ has m^2 vertices and valency $2(m-1)$, and $CP(m)$ has $2m$ vertices and valency $2m-2$.
PROOF: An edge $\{i,j\}$ of K_m meets the $2(m-2)$ edges $\{i,l\}$, $\{j,l\}$; an edge of $K_{m,m}$ meets $m-1$ others at each end; and Definition (cocktail party graphs). □
⟨1⟩5. The Schläfli, Clebsch and Petersen graphs are isomorphic to none of $L(K_m)$, $L(K_{m,m})$, $CP(m)$, and the Chang and Shrikhande graphs to no $CP(m)$.
PROOF: Compare ⟨1⟩1 with ⟨1⟩4. For 27 vertices, $\binom{m}{2} = 27$, $m^2 = 27$ and $2m = 27$ have no solution. For 16 vertices, $\binom{m}{2} = 16$ has none, and $m^2 = 16$, $2m = 16$ give valencies 6 and 14, not 10. For 10 vertices, $\binom{m}{2} = 10$ gives valency 6, $m^2 = 10$ has no solution and $2m = 10$ gives 8, not 3. Finally $2m = 28$ and $2m = 16$ give valencies 26 and 14, not 12 and 6. □
⟨1⟩6. The Chang graphs I, II and III and $T(8)$ are pairwise non-isomorphic.
PROOF: Their 4-clique profiles differ (Computation C22, assertion 4), and an isomorphism preserves the profile (`SimpleGraph.Iso.cliqueProfile_eq`). □
⟨1⟩7. The Shrikhande graph is not isomorphic to $L(K_{4,4})$.
PROOF: By Computation C22, assertion 5, $L(K_{4,4})$ has a 4-clique; an isomorphism from $L(K_{4,4})$ to the Shrikhande graph would map it to a 4-clique (`SimpleGraph.IsNClique.map`), which the Shrikhande graph does not have. □
⟨1⟩8. No Chang graph is isomorphic to $L(K_{m,m})$ or to $L(K_m)$ with $m \neq 8$, and the Shrikhande graph is isomorphic to no $L(K_m)$ and to no $L(K_{m,m})$ with $m \neq 4$.
PROOF: By ⟨1⟩1 and ⟨1⟩4: $m^2 = 28$ has no solution, $\binom{m}{2} = 28$ only $m = 8$, $\binom{m}{2} = 16$ none, and $m^2 = 16$ only $m = 4$. □
⟨1⟩9. Q.E.D.
PROOF: Assertion 1 is ⟨1⟩1, ⟨1⟩2 and ⟨1⟩3; assertion 2 is ⟨1⟩5, ⟨1⟩8, ⟨1⟩6 and ⟨1⟩7. For assertion 3, an isomorphism preserves (n,k), and by ⟨1⟩1 only the Chang graphs share it; apply ⟨1⟩6. □

Corollary (the seven graphs are not line graphs). ASSUME: $\mathcal{S}$ is the list of the seven graphs of Proposition (the seven graphs), and Y is a graph, possibly infinite or with isolated vertices.
PROVE: No $Z \in \mathcal{S}$ is isomorphic to $L(Y)$.
This is the claim of the paper, before Example 4.5, that these strongly regular graphs are not line graphs.

PROOF SKETCH: We will apply Lemma (strongly regular line graphs); assertion 2 of Proposition (the seven graphs) excludes both of its alternatives.
⟨1⟩1. ASSUME: $Z \in \mathcal{S}$ and $Z \cong L(Y)$.
PROVE: A contradiction.
⟨2⟩1. Z is connected and strongly regular, and $\det(2I + A(Z)) = 0$.
PROOF: Proposition (the seven graphs), assertion 1, with $Z \in \mathcal{S}$ by ⟨1⟩; since $\lambda_{\min}(Z) = -2$ is an eigenvalue of $A(Z)$, the matrix $2I + A(Z)$ is singular. □
⟨2⟩2. $Z \cong L(K_m)$ or $Z \cong L(K_{m,m})$ for some $m \geq 2$.
PROOF: Lemma (strongly regular line graphs), with ⟨2⟩1 and ⟨1⟩. □
⟨2⟩3. Q.E.D.
PROOF: Proposition (the seven graphs), assertion 2, excludes both alternatives of ⟨2⟩2. □
⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1 applies to every $Z \in \mathcal{S}$. □

The classification.

Remark 4.14, restated. ASSUME: 1. X is a connected graph, strongly regular with parameters (n,k,a,c).

2. $\lambda_{\min}(X) = -2$.
3. X is represented in E_8 by $(\mathbf{x}_u)$.
4. For no m is X isomorphic to $L(K_m)$, to $L(K_{m,m})$ or to $CP(m)$.

PROVE: 1. X is isomorphic to exactly one of the Chang graphs I, II and III, the Schläfli graph, the Shrikhande graph, the Clebsch graph and the Petersen graph.
2. These seven graphs satisfy hypotheses 1 to 4.

[new] The paper cites Seidel [19] and gives no proof. We follow the route of `cgss/prerequisites.md`, Section 7.3 (decision 3 of its Section 13); the departures are tagged in the lemmas above. **[repair]** The proof fills Gap 12 (errata): it proves the classification, and "exactly one" is the non-isomorphism of Proposition (the seven graphs). In Lean, the statement `seidel_seven` of Remark 4.14 is the conjunction of `seidel_seven_classification`, of `seidel_seven_pairwise_not_iso` and of `seidel_seven_properties` (Proposition (the seven graphs)), all four in the namespace `SimpleGraph`; Lean names are unique, so the two results carry different ones.

PROOF SKETCH: We will pin down the parameters, find the type of the lattice roots R, exclude the types A and D by the line graph theorems, exclude two parameter sets by the integrality of the apex, and identify the remaining five cases by switching and by uniqueness.

LET: $A = A(X)$, $Q = 2I + A$, $\rho = \operatorname{rk} Q$, and let Λ, W, R be as in Lemma (the lattice roots) and $\mathbf{w}$ as in Lemma (the apex) for $(\mathbf{x}_u)$.

⟨1⟩1. $n \geq 1$ and $\det Q = 0$.
PROOF: A connected graph has a vertex; ⟨0⟩:2 and `SimpleGraph.leastEigenvalue_eq_neg_two_iff`. □

⟨1⟩2. $n \leq 28$ and $k \leq 16$.
PROOF: Theorem 4.4, with ⟨0⟩:1, ⟨0⟩:3 and ⟨1⟩1. □

⟨1⟩3. $\rho = \dim W \leq 8$.
PROOF: Q is the Gram matrix of $(\mathbf{x}_u)$, so $\rho = \dim W$ by Lemma (Gram factorization), assertion 1; and $W \subseteq \mathbb{R}^8$ (`Submodule.finrank_le`). □

⟨1⟩4. $1 \leq k \leq n-2$, $a \leq k-1$, $1 \leq c \leq k$, and conditions 2 to 4 of Computation C18 hold.
PROOF: Lemma (strongly regular graphs with eigenvalue -2) with ⟨0⟩:1 and ⟨1⟩1. □

⟨1⟩5. $k(k-a-1) = c(n-k-1)$, condition 1 of Computation C18.
PROOF: `SimpleGraph.IsSRGWith.param_eq`, which needs $0 < n$ (⟨1⟩1); it is stated in $\mathbb{N}$ with truncated subtraction, and by ⟨1⟩4 neither $k-a-1$ nor $n-k-1$ is truncated. □

⟨1⟩6. (n, k, a, c, ρ) is one of the 15 tuples of Computation C18.
PROOF: Computation C18: its hypotheses are ⟨1⟩4, ⟨1⟩5, and $n \leq 28 \leq 36$, $k \leq 16$ and $\rho \leq 8$ from ⟨1⟩2 and ⟨1⟩3. □

⟨1⟩7. $(n, k, a, c) \neq (2m, 2m-2, 2m-4, 2m-2)$ for every m.
PROOF: Otherwise X is $(2m-2)$-regular on $2m$ vertices, and $X \cong CP(m)$ by Lemma (cocktail party parameters), against ⟨0⟩:4. □

⟨1⟩8. R is a nonempty, indecomposable, star-closed line system spanning W, and X is represented in R by $(\mathbf{x}_u)$.
PROOF: Lemma (the lattice roots), with ⟨0⟩:1 and ⟨0⟩:3. □

⟨1⟩9. PICK a type τ among A_ℓ ($\ell \geq 1$), D_ℓ ($\ell \geq 4$), E_6, E_7, E_8, and a bijection $f : \Phi(R) \to \Phi(\tau)$ preserving inner products.
PROOF: Theorem 3.5 with ⟨1⟩8. □

⟨1⟩10. PICK a linear isometry F from W onto $\operatorname{span} \Phi(\tau)$ that extends f.
PROOF: Lemma (transport on spans) with ⟨1⟩9, since $\Phi(R)$ spans W (⟨1⟩8). □

⟨1⟩11. The rank of τ is ρ.
PROOF: F of ⟨1⟩10 is an isometry onto $\operatorname{span} \Phi(\tau)$, and $\dim W = \rho$ by ⟨1⟩3. □

⟨1⟩12. ASSUME: τ is A_ℓ or D_ℓ.
PROVE: A contradiction.
⟨2⟩1. CASE: $\tau = \mathrm{A}_\ell$.
PROVE: $X \cong L(Y)$ for some graph Y.
PROOF: X is represented in A_ℓ by Lemma (transport of representations) with f and ⟨1⟩8; apply Theorem 4.1. □
⟨2⟩2. CASE: $\tau = \mathrm{D}_\ell$.
PROVE: $X \cong L(Y)$ for some graph Y, or $X \cong CP(b)$ for some b.
PROOF: X is represented in D_ℓ by Lemma (transport of representations) with f and ⟨1⟩8, so it is a generalized line graph by Theorem 4.2, assertion 1; it is connected and regular by ⟨0⟩:1, so apply Lemma (regular connected generalized line graphs). □
⟨2⟩3. $X \ncong L(Y)$ for every graph Y.
PROOF: Otherwise Lemma (strongly regular line graphs), with ⟨0⟩:1 and ⟨1⟩1, gives $X \cong L(K_m)$ or $L(K_{m,m})$, against ⟨0⟩:4. □
⟨2⟩4. Q.E.D.
PROOF: In both cases ⟨2⟩1, ⟨2⟩2, the conclusion contradicts ⟨2⟩3 or ⟨0⟩:4. □
⟨1⟩13. $\tau = \mathrm{E}_t$ with $t = \rho \in \{6, 7, 8\}$, and (n, k, a, c) is one of $(10, 3, 0, 1)$, $(15, 8, 4, 4)$, $(16, 10, 6, 6)$ (for $t = 6$), $(16, 6, 2, 2)$, $(21, 10, 5, 4)$, $(27, 16, 10, 8)$ (for $t = 7$) and $(28, 12, 6, 4)$ (for $t = 8$).
PROOF: By ⟨1⟩12, τ is one of E_6, E_7, E_8, of ranks 6, 7, 8 (Lemma (spans of the models)), so $t = \rho$ by ⟨1⟩11. Of the 15 tuples of ⟨1⟩6, ⟨1⟩7 removes six, and $\rho \in \{6, 7, 8\}$ removes $(9, 4, 1, 2, 5)$ and $(10, 6, 3, 4, 5)$. □
⟨1⟩14. $\det(B_t)\, n/(k+2)$ is an integer, where $\det B_t$ is 3, 2, 1 for $t = 6, 7, 8$.
⟨2⟩1. PICK $\mathbf{b}_i = F^{-1}(\mathbf{s}_i)$, $1 \le i \le t$, with the roots $\mathbf{s}_i$ of Computation C21; they lie in $\Phi(R)$ and form a basis of W with Gram matrix B_t.
PROOF: Computation C21 gives the $\mathbf{s}_i$, a basis of $\operatorname{span}\Phi(\mathrm{E}_t)$; F^{-1} is a surjective linear isometry onto W that maps $\Phi(\mathrm{E}_t)$ onto $\Phi(R)$ (⟨1⟩9 and ⟨1⟩10). □
⟨2⟩2. $\mathbf{w} \in W$, $(\mathbf{w}, \mathbf{b}_i) \in \mathbb{Z}$ for every i, and $(\mathbf{w}, \mathbf{w}) = n/(k+2)$.
PROOF: Lemma (the apex), with ⟨0⟩:1 and ⟨1⟩1; the $\mathbf{b}_i$ lie in $\Phi(R) \subseteq \Lambda$ by ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Lemma (integrality), with ⟨2⟩1 and ⟨2⟩2; the determinants are those of Computation C21. □
⟨1⟩15. (n, k, a, c) is neither $(15, 8, 4, 4)$ nor $(21, 10, 5, 4)$.
PROOF: These have $t = 6$ and $t = 7$ by ⟨1⟩13, and $3 \cdot \frac{15}{10} = \frac{9}{2}$ and $2 \cdot \frac{21}{12} = \frac{7}{2}$ are not integers, against ⟨1⟩14. □
⟨1⟩16. CASE: $(n, k, a, c) = (16, 10, 6, 6)$.
PROVE: X is isomorphic to the Clebsch graph.
PROOF: Lemma (SE2: the Clebsch graph) with ⟨0⟩:1. □
⟨1⟩17. CASE: $(n, k, a, c) = (27, 16, 10, 8)$.
PROVE: X is isomorphic to the Schläfli graph.
PROOF: Lemma (SE3: the Schläfli graph) with ⟨0⟩:1. □
⟨1⟩18. CASE: $(n, k, a, c) = (10, 3, 0, 1)$.
PROVE: X is isomorphic to the Petersen graph.
⟨2⟩1. $X \cong L(\Delta_6)^H$ for some $H \subseteq \mathcal{E}_6$.
PROOF: $t = 6$ by ⟨1⟩13, $(\mathbf{w}, \mathbf{w}) = 10/5 = 2$ by Lemma (the apex), and $n = 10 = |\mathcal{E}_6|$ by Computation C20 at the apex; apply Lemma (switching from the apex) with ⟨0⟩:1 and f of ⟨1⟩9. □
⟨2⟩2. Q.E.D.
PROOF: $L(\Delta_6)^H \cong X$ is 3-regular by ⟨2⟩1; apply Lemma (regular switchings of $T(5)$). □
⟨1⟩19. CASE: $(n, k, a, c) = (16, 6, 2, 2)$.

PROVE: X is isomorphic to the Shrikhande graph.

⟨2⟩1. $X \cong L(\Delta_7)^H$ for some $H \subseteq \mathcal{E}_7$.

PROOF: $t = 7$ by ⟨1⟩13, $(\mathbf{w}, \mathbf{w}) = 16/8 = 2$ by Lemma (the apex), and $n = 16 = |\mathcal{E}_7|$ by Computation C20 at the apex; apply Lemma (switching from the apex) with ⟨0⟩:1 and f of ⟨1⟩9. □

⟨2⟩2. Q.E.D.

PROOF: $L(\Delta_7)^H \cong X$ is 6-regular by ⟨2⟩1, so it is the Shrikhande graph or $L(K_{4,4})$ by Lemma (regular switchings of $T(6)$ with an isolated vertex); the latter is excluded by ⟨0⟩:4. □

⟨1⟩20. CASE: $(n, k, a, c) = (28, 12, 6, 4)$.

PROVE: X is isomorphic to one of the Chang graphs.

⟨2⟩1. $X \cong L(\Delta_8)^H$ for some $H \subseteq \mathcal{E}_8$.

PROOF: $t = 8$ by ⟨1⟩13, $(\mathbf{w}, \mathbf{w}) = 28/14 = 2$ by Lemma (the apex), and $n = 28 = |\mathcal{E}_8|$ by Computation C20 at the apex; apply Lemma (switching from the apex) with ⟨0⟩:1 and f of ⟨1⟩9. □

⟨2⟩2. Q.E.D.

PROOF: $L(\Delta_8)^H \cong X$ is 12-regular by ⟨2⟩1, so it is $T(8)$ or a Chang graph by Lemma (regular switchings of $T(8)$); $T(8) = L(K_8)$ is excluded by ⟨0⟩:4. □

⟨1⟩21. Q.E.D.

PROOF: By ⟨1⟩13 and ⟨1⟩15, the cases ⟨1⟩16, ⟨1⟩17, ⟨1⟩18, ⟨1⟩19 and ⟨1⟩20 are exhaustive, and each gives one of the seven graphs; it is only one of them by Proposition (the seven graphs), assertion 3. Assertion 2 is Proposition (the seven graphs), assertions 1 and 2. □

What the proof relies on. Besides Sections 2 to 4, the proof uses the computations C4 (with its transitivity on ordered stars), C9 (with its two certificate isomorphisms), C18, C20 (at the apex), C21 and C22, and the classification SE4 of `cgss/proofs/reg8/reg8.tex`. All of it is in the Lean formalization (`LineGraphs/Seidel/`, `LineGraphs/Computations/C18.lean`, `LineGraphs/Computations/C22.lean` and its data files): C18 as a checked search, C22's fourteen isomorphism certificates with the clique profiles and counts, SE4 by a case split with checked certificates, and the restated remark as `SimpleGraph.seidel_seven_classification` and `SimpleGraph.seidel_seven`.

5. Spectral characterization

We will follow Section 5 of [CGSS]. All graphs are finite and simple. For a graph X we write $V(X)$ and $E(X)$ for its vertices and edges, $A(X)$ for its adjacency matrix and $L(X)$ for its line graph. The paper's graphs G, G_1, G_2 are our X, X_1, X_2, and its auxiliary graph H in Theorem 5.8 is our Y (`cgss/notation.md`, Section 1). Valencies are k, k_1, k_2 in Lemmas 5.1 to 5.6 and in Remark 5.7; from Theorem 5.8 on, k is the block size of a design and (v, k, b, r, λ) are the design parameters, as in the paper. A generic eigenvalue is μ, since λ is a design parameter.

The spectrum $\mathrm{Spec}(M)$ of a real symmetric matrix M is the multiset of the roots of its characteristic polynomial, and $\mathrm{mult}_M(\mu)$ is the number of copies of μ in it. In a multiset $[\![\mu^m, \dots]\!]$ the exponent m counts copies, and a sum of multisets adds the counts. The spectrum of a graph is that of its adjacency matrix, two graphs are *cospectral* when their adjacency matrices have the same characteristic polynomial, and $\lambda_{\max}$ is the largest eigenvalue. Where the paper writes that two graphs have the same spectrum, we will say that they are cospectral. We cite the spectral toolkit `cgss/proofs/spectraltools/spectraltools.tex` as [ST], by the numbers of its results; its structured version `cgss/proofs/spectraltools/spectraltools-lamport.tex` has the same numbering. The toolkit is not part of this document, so we list the facts that we take from it, with its proposed Lean names, before we use them.

The paper quotes Lemmas 5.1 to 5.4 as well known (errata, Gap 8). The toolkit proves them as SP1 to SP4, and we will state them in the paper's form and cite the toolkit. Lemmas 5.5 and 5.6 and Remark 5.7 have proofs in the paper, and we will follow those proofs, with the toolkit's facts at the leaves. For Lemmas 5.1, 5.2, 5.3 and 5.5 the toolkit proposes the same Lean declarations (its Theorems 5.3, 5.4, 5.6 and 6.5), so each of these results and the corresponding toolkit result are one declaration. Lemma 5.4, Lemma 5.6 and Remark 5.7 differ in statement from the toolkit's Theorem 6.4, Theorem 6.7 and Corollary 6.8, in the forms that Theorem 5.8 needs: Lemma 5.4 does not assume an edge, and asserts $k_1 \neq k_2$; Lemma 5.6 (i) holds for every biregular bipartition of X_1 with $|P_1| \leq |Q_1|$; and Remark 5.7 assumes only that X_2 has no isolated vertices and is not bipartite. We will derive them from the toolkit's results, which we cite as inputs, and give them their own Lean names, derived from the toolkit's by the rules of `cgss/notation.md`, Section 10: `SimpleGraph.regular_or_isBiregularWith_of_lineGraph_regular`, `SimpleGraph.lineGraph_cospectral_bipartite_forall_isBiregularWith` and `SimpleGraph.lineGraph_cospectral_bipartite_spectrum_of_not_isBipartite`.

Inputs from the spectral toolkit. [repair] These are the toolkit's proofs of the facts that the paper quotes (errata, Gap 8, and `cgss/prerequisites.md`, SP1 to SP5). Matrices are real, square matrices are symmetric, and index sets are finite.

- [ST, Theorem 2.2] (`Matrix.IsHermitian.eigenvalueMultiplicity_eq_finrank`, `Matrix.IsHermitian.mem_spectrumMultiset_iff`). If M is symmetric of order n, then $\mathrm{Spec}(M)$ has n members counted with multiplicity, $\mathrm{mult}_M(\mu) = \dim\ker(M - \mu I)$, and $\mu \in \mathrm{Spec}(M)$ exactly when $M\mathbf{x} = \mu\mathbf{x}$ for some $\mathbf{x} \neq 0$.
- [ST, Lemma 2.4] (`Matrix.IsHermitian.largestEigenvalue_mem_spectrumMultiset`, `Matrix.IsHermitian.le_largestEigenvalue_of_mem`). If M is symmetric of positive order, then $\lambda_{\max}(M)$ is the largest member of $\mathrm{Spec}(M)$.
- [ST, Proposition 2.6] (`Matrix.IsHermitian.spectrumMultiset_eq_iff_charpoly_eq`, `SimpleGraph.cospectral_iff_spectrumMultiset_eq`, `SimpleGraph.Iso.cospectral`). For symmetric M, M', possibly on different index sets, $\chi_M = \chi_{M'}$ if and only if $\mathrm{Spec}(M) = \mathrm{Spec}(M')$, if and only if $\mathrm{mult}_M(\mu) = \mathrm{mult}_{M'}(\mu)$ for every real μ. Cospectral graphs have the same number of vertices, and isomorphic graphs are cospectral.
- [ST, Theorem 3.1] (`Matrix.IsHermitian.spectrumMultiset_aeval`). If M is symmetric and f is a real polynomial, then $\mathrm{Spec}(f(M))$ is the image of $\mathrm{Spec}(M)$ under f, with multiplicities. In particular $\mathrm{Spec}(cI + M)$ is $\mathrm{Spec}(M)$ shifted by c, and $\mathrm{mult}_{cI+M}(\mu + c) = \mathrm{mult}_M(\mu)$.
- [ST, Theorem 3.2] (`Matrix.IsHermitian.rank_aeval_eq_sum_multiplicity`). If M is symmetric of order n, then $\mathrm{rk}\, M = n - \mathrm{mult}_M(0)$, and for a real polynomial f, $\mathrm{rk}\, f(M)$ is the sum of $\mathrm{mult}_M(\theta)$ over the distinct eigenvalues θ of M with $f(\theta) \neq 0$.
- [ST, Corollary 3.4] (`SimpleGraph.posSemidef_two_add_adjMatrix_iff`). For a graph X, the matrix $2I + A(X)$ is positive semidefinite if and only if every member of $\mathrm{Spec}(X)$ is at least -2.
- [ST, Theorem 3.5] (`Matrix.rank_gram_eq_finrank_span`). For every matrix B, $\mathrm{rk}(B^TB) = \mathrm{rk}\, B = \mathrm{rk}(BB^T)$. The Gram matrix of a finite family of vectors in a real inner product space is positive semidefinite, and its rank is the dimension of the span of the family.
- [ST, Theorem 5.1] (`SimpleGraph.adjMatrix_cube_apply_self_eq_twice_triangles`). For a graph X with $A = A(X)$ and a vertex u, $A^3(u,u)$ is twice the number of triangles of X that contain u.
- [ST, Theorem 5.3] (SP1; `SimpleGraph.IsRegularOfDegree.largestEigenvalue_eq`, `SimpleGraph.IsRegularOfDegree.multiplicity_eq_card_components`). If X is

k-regular on a nonempty vertex set, then $\lambda_{\max}(X) = k$ and $\mathrm{mult}_X(k)$ is the number of connected components of X.

- [ST, Theorem 5.4] (SP2; `SimpleGraph.isRegular_iff_sum_sq_eigenvalues`). A graph X on $n \geq 1$ vertices is regular if and only if $\sum_{\mu \in \mathrm{Spec}(X)} \mu^2 = n\lambda_{\max}(X)$.
- [ST, Corollary 5.5] (`SimpleGraph.Cospectral.isRegularOfDegree`, `SimpleGraph.Cospectral.connected_of_regular`). If X is k-regular on a nonempty vertex set and X' is cospectral with X, then X' is k-regular, and X' is connected if X is.
- [ST, Theorem 5.6] (SP3; `SimpleGraph.IsRegularOfDegree.isBipartite_iff_neg_eigenvalue`). If X is connected and k-regular, then X is bipartite if and only if $-k \in \mathrm{Spec}(X)$, and then $\mathrm{mult}_X(-k) = 1$.
- [ST, Corollary 5.9] (`Matrix.spectrumMultiset_smul_one_add_smul_allOnes`). For real c, t and order $n > 0$, $\mathrm{Spec}(cI + tJ) = [\![(c + tn)^1, c^{n-1}]\!]$.
- [ST, Theorem 6.1] (`Matrix.eigenvalueMultiplicity_mul_comm_of_ne_zero`, `Matrix.spectrumMultiset_gram_products`). If M is $P \times Q$ and N is $Q \times P$, then $\mathrm{mult}_{MN}(\mu) = \mathrm{mult}_{NM}(\mu)$ for every $\mu \neq 0$. If B is $P \times Q$ of rank ρ, then BB^T and B^TB are positive semidefinite, have the same positive eigenvalues with multiplicities, and have $\mathrm{mult}_{BB^T}(0) = |P| - \rho$ and $\mathrm{mult}_{B^TB}(0) = |Q| - \rho$.
- [ST, Theorem 6.2] (`SimpleGraph.edgeIncMatrix_mul_transpose`, `SimpleGraph.transpose_edgeIncMatrix_mul`, `SimpleGraph.multiplicity_neg_two_lineGraph`). Let M be the $V(X) \times E(X)$ incidence matrix of a graph X and D its diagonal matrix of degrees. Then $MM^T = D + A(X)$, $M^TM = 2I + A(L(X))$, and $\mathrm{rk}\, M = |V(X)| - c$, where c is the number of bipartite connected components of X. Every eigenvalue of $L(X)$ is at least -2.
- [ST, Lemma 6.3] (`SimpleGraph.lineGraph_connected_iff_of_no_isolated`, `SimpleGraph.degree_lineGraph`). If X has at least one edge and no isolated vertices, then X is connected if and only if $L(X)$ is connected, and the edge $\{u, w\}$ has degree $\deg(u) + \deg(w) - 2$ in $L(X)$.
- [ST, Theorem 6.4] (SP4; `SimpleGraph.regular_or_semiregular_of_lineGraph_regular`). If X has at least one edge and no isolated vertices, and $L(X)$ is connected and h-regular, then X is regular, or X is bipartite with constant degree on each part.
- [ST, Theorem 6.6] (`Matrix.spectrumMultiset_fromBlocks_scalar`). Let B be $P \times Q$ of rank ρ, let $\mu_1, \ldots, \mu_\rho$ be the positive eigenvalues of BB^T with repetition, and let $s, t \in \mathbb{R}$. Then the spectrum of $C = \begin{pmatrix} sI & B \\ B^T & tI \end{pmatrix}$ consists of $\gamma_j^{\pm} = \frac{1}{2}\big(s + t \pm \sqrt{(s-t)^2 + 4\mu_j}\big)$ for $1 \leq j \leq \rho$, of s with multiplicity $|P| - \rho$ and of t with multiplicity $|Q| - \rho$.
- [ST, Theorem 6.7] (`SimpleGraph.lineGraph_cospectral_bipartite`). Let X_1 be a connected bipartite graph with an edge such that $L(X_1)$ is regular, and let X_2 be a graph without isolated vertices such that $L(X_2)$ is cospectral with $L(X_1)$. Then X_2 is connected, and precisely one of the following holds: X_2 is bipartite, and with the part sizes ordered $p \leq q$ in each graph, the part sizes, the respective degrees and the spectra of $B_1B_1^T$ and $B_2B_2^T$ agree; or X_1 is not regular, X_2 is regular and not bipartite, and $|V(X_1)| = |V(X_2)| + 1$.
- [ST, Corollary 6.8] (`SimpleGraph.lineGraph_cospectral_bipartite_spectrum`). Let X_1 be a connected graph with a bipartition (P, Q), biregular with degrees (k_1, k_2), with $k_1 > k_2 > 0$, $p = |P| \leq q = |Q|$ and biadjacency matrix B of rank ρ, and let X_2 be a

connected regular graph that is not bipartite with $L(X_2)$ cospectral with $L(X_1)$. Then k_1k_2 is a positive eigenvalue of BB^T of multiplicity 1, and if $\mu_2, \dots, \mu_\rho$ are the other positive eigenvalues of BB^T with repetition, then $\mathrm{Spec}(X_2)$ consists of $\frac{1}{2}(k_1+k_2)$, of $\pm\sqrt{\mu_j + \frac{1}{4}(k_1-k_2)^2}$ for $2 \le j \le \rho$, of $\frac{1}{2}(k_1-k_2)$ with multiplicity $p-\rho$ and of $-\frac{1}{2}(k_1-k_2)$ with multiplicity $q-\rho$.

- [ST, Theorem 7.7] (`Matrix.gram_eq_of_two_eigenvalues`). Let N be a $v \times b$ zero-one matrix with $v > 1$, constant row sum $r > 0$ and constant column sum $k > 1$, and let $\lambda > 0$ with $r(k-1) = \lambda(v-1)$. If $\mathrm{Spec}(NN^T) = [\![(rk)^1, (r-\lambda)^{v-1}]\!]$, then $NN^T = (r-\lambda)I + \lambda J$. If moreover $r > \lambda$, then $b \ge v$ (Fisher's inequality, DS3).
- [ST, Theorem 7.10] (`SimpleGraph.IsRegularOfDegree.aeval_eq_smul_allOnes_of_connected`). If X is connected and k-regular on $n > 0$ vertices, and the real polynomial f vanishes at every eigenvalue of X other than k, then $f(A(X)) = \frac{f(k)}{n} J$.

Inputs from Mathlib and from the Hall–Connor section. We will use the following declarations of Mathlib at commit `082e2d37` (4 September 2026), the version pinned in `lake-manifest.json`: `SimpleGraph.Iso.lineGraph` (an isomorphism of graphs induces one of their line graphs), `SimpleGraph.lineGraph_adj_iff_exists` (two edges are adjacent in the line graph exactly when they are distinct and share an endpoint), `SimpleGraph.sum_degrees_eq_twice_card_edges`, `SimpleGraph.degree_compl`, `SimpleGraph.adjMatrix_mul_self_apply_self` (the diagonal of $A(X)^2$ is the degree), `SimpleGraph.cycleGraph_adj'`, `SimpleGraph.Iso.completeGraph`, `SimpleGraph.completeBipartiteGraphCongr`, `Equiv.sumComm`, `Fintype.equivFin`, `Finset.card_le_univ`, `Finset.eq_of_subset_of_card_le`, `LinearEquiv.finrank_eq`, `Submodule.finrank_le`, `finrank_euclideanSpace_fin` and the structure `Matrix.IsHadamard`. Double counting is the paragraph (finite counting and order) of the Hall–Connor section. The design interface is Definition (labelled designs) of that section, with its incidence matrix N of Definition (incidence and block matrices). This is the one design interface of Section 5; the toolkit's facts concern real matrices, and we apply them to the entrywise real cast of the integer matrix N, which we also write N. The cast is taken in Lemma (design identities), assertion 3, and in Theorem 5.8, where [ST, Theorem 7.7] is applied to a real zero-one matrix. The toolkit's Section 7 proposes the same names `Design` and `Design.incMatrix` with a real incidence matrix; the shared declarations are those of the Hall–Connor section, over $\mathbb{Z}$, and we use no toolkit result stated for the toolkit's designs.

Inputs from Section 4. Theorem 5.8 rests on Theorem 4.11. We recall what we will use from Section 4, where these results are proved: Definition (cocktail party graphs), Definition (representations) and Theorem 4.11. Theorem 5.8 and Lemma (regular graphs of large valency) depend on those labels directly, and none of them depends on Section 5.

- The *cocktail party graph* $CP(m)$ (`SimpleGraph.cocktailParty`) has vertex set $\{1,\dots,m\} \times \{0,1\}$, and (i,a) is adjacent to (j,a') exactly when $i \ne j$. It is $(2m-2)$-regular on $2m$ vertices.
- A graph X is *represented in* E_8 (`SimpleGraph.IsRepresentedIn`) if there is a family $(\mathbf{x}_u)_{u \in V(X)}$ of vectors of $\Phi(\mathrm{E}_8) \subseteq \mathbb{R}^8$ (model (ii) of Definition 3.2) whose Gram matrix is $2I + A(X)$.
- Theorem 4.11 (Hoffman and Ray-Chaudhuri; `SimpleGraph.hoffman_rayChaudhuri`). If X is connected and regular and $2I + A(X)$ is positive semidefinite, then X is isomorphic to $L(Y)$ for some graph Y, or X is isomorphic to $CP(m)$ for some m, or X is represented in E_8.

Inputs for Example 5.13. The example needs one fact that this section does not prove, EX4 of `cgss/prerequisites.md`, and the existence results EX1 to EX3 for its instances. We record them as deferred inputs; none of them may depend on Section 5.

- (EX4, deferred.) If Y is a connected graph on more than three vertices that is not bipartite, and Z is a bipartite graph without isolated vertices, then $L(Y) \ncong L(Z)$. This is a case of Whitney's theorem on line graphs, which Mathlib does not have. The prerequisites prove it from four facts, which belong to Sections 3 and 4: the star closure of the standard representation of $L(Y)$ is D_n when Y is connected, not bipartite and on n vertices; that of $L(Z)$ is A_{m-1} when Z is connected, bipartite and on m vertices; isomorphic graphs have isometric representations, so the star closures are isomorphic (LN3 and LA2 there), and $\mathrm{D}_n \ncong \mathrm{A}_n$ for $n \geq 4$; and Z is connected, since $L(Z) \cong L(Y)$ is connected and Z has no isolated vertices. **[new]** The Lean proof of Example 5.13 avoids EX4, by a clique bound recorded at the step on non-isomorphism in the proof of Example 5.13; we keep EX4 for the route of the paper.
- (EX1 to EX3; computation C23.) Resolvable designs with parameters $(k(2k-1), k, 4k^2-1, 2k+1, 1)$ exist for $k = 2^d$ with $d \geq 1$ and for $k = 3$, and designs with parameters $(6, 3, 10, 5, 2)$ and $(28, 21, 36, 27, 20)$ exist. We will use these only in the remark after Example 5.13. Computation C23 gives in Lean the designs that the paper's instances $k = 2$ and $k = 4$ need; their existence for every $k = 2^d$ and for $k = 3$ is a citation, outside the scope.

5.1. Lemmas 5.1 to 5.7. Definitions and auxiliary lemmas.

Definition (bipartitions and biadjacency matrices). ASSUME: X is a graph.

A *bipartition* of X is a pair (P, Q) of disjoint sets with $P \cup Q = V(X)$ such that every edge has one end in P and one in Q; in Mathlib, `X.IsBipartiteWithPQ` (`SimpleGraph.IsBipartiteWith`) with $Q = V(X) \setminus P$. X is bipartite if it has a bipartition. The bipartition is *biregular with degrees* (k_1, k_2) if every vertex of P has degree k_1 and every vertex of Q has degree k_2. The *biadjacency matrix* of X for (P, Q) is the $P \times Q$ matrix B with $B(u, w) = A(X)(u, w)$.

Definition (incidence graphs, flag graphs, characterization).
ASSUME: D is a design with parameters (v, k, b, r, λ).
Here D is as in Definition (labelled designs), with points P and block labels $\mathcal{B}$. The *incidence graph* $I(D)$ has vertex set $P \sqcup \mathcal{B}$, and a point p is adjacent to a block B when $p \in B$; there are no other edges. The *flag graph* is $F(D) = L(I(D))$. We say that $F(D)$ is *characterized by its spectrum* if every graph cospectral with $F(D)$ is isomorphic to $F(D_2)$ for some design D_2 with the parameters (v, k, b, r, λ), on some finite types P_2 of points and $\mathcal{B}_2$ of block labels with decidable equality. In Lean, P_2 and $\mathcal{B}_2$ range over `Type`, as do the vertex types of all graphs of this section; the design D_2 of Theorem 5.8 lives on subtypes of the vertex type of a graph. **[repair]** This is the sense of the paper's proofs and of its sources (errata, Implicit hypothesis 6); in the stronger sense "determined up to isomorphism" Corollaries 5.10 and 5.11 and Remark 5.12 are false.

Lemma (design identities). ASSUME: D is a design with parameters (v, k, b, r, λ) on P and $\mathcal{B}$, with incidence matrix N.
PROVE:
1. $bk = vr$ and $r(k-1) = \lambda(v-1)$.
2. $k \leq v$, $\lambda \geq 1$ and $r \geq \lambda$; moreover $r = \lambda$ if and only if $k = v$, if and only if every block contains every point.
3. $NN^T = (r-\lambda)I + \lambda J$ and $\mathrm{Spec}(NN^T) = [\![(rk)^1, (r-\lambda)^{v-1}]\!]$.
4. If $r > \lambda$, then $b \geq v$.

PROOF SKETCH: These are DS1 and DS3 of `cgss/prerequisites.md`: double counting, and the Fisher inequality of the toolkit's Theorem 7.7.

⟨1⟩1. $bk = vr$.
PROOF: Count the pairs (p, B) with $p \in B$ by blocks and by points, with the double counting of (finite counting and order). □
⟨1⟩2. $r(k-1) = \lambda(v-1)$.
PROOF: Fix a point p and count the pairs (q, B) with $q \neq p$ and $p, q \in B$, by blocks through p and by points q, with the same double counting. □
⟨1⟩3. $k \leq v$.
PROOF: A block is a subset of P of size k; apply `Finset.card_le_univ`. □
⟨1⟩4. $\lambda \geq 1$.
⟨2⟩1. ASSUME: $\lambda = 0$.
PROVE: FALSE.
PROOF: By ⟨1⟩2, $r(k-1) = 0$, and $k > 1$ gives $r = 0$. Then ⟨1⟩1 gives $bk = 0$, against $b > 0$ and $k > 1$. □
⟨2⟩2. Q.E.D.
PROOF: λ is a natural number and ⟨2⟩1. □
⟨1⟩5. $r \geq \lambda$, with equality if and only if $k = v$.
PROOF: By ⟨1⟩2 and ⟨1⟩3, $\lambda(v-1) = r(k-1) \leq r(v-1)$, with equality exactly when $k = v$, since $r \geq 1$ by ⟨1⟩2 and ⟨1⟩4; divide by $v - 1 > 0$. □
⟨1⟩6. $k = v$ if and only if every block contains every point.
PROOF: A block of size k is a subset of P; apply `Finset.eq_of_subset_of_card_le` to it and P. □
⟨1⟩7. $NN^T = (r-\lambda)I + \lambda J$.
PROOF: $(NN^T)(p, q)$ is the number of blocks containing p and q, which is r for $p = q$ and λ otherwise, by Definition (labelled designs). □
⟨1⟩8. $\mathrm{Spec}(NN^T) = [\![(rk)^1, (r-\lambda)^{v-1}]\!]$.
PROOF: [ST, Corollary 5.9] with $c = r - \lambda$, $t = \lambda$ and $n = v$ gives the eigenvalue $r - \lambda + \lambda v$ once, and $r - \lambda + \lambda v = rk$ by ⟨1⟩2. □
⟨1⟩9. ASSUME: $r > \lambda$.
PROVE: $b \geq v$.
PROOF: [ST, Theorem 7.7] for the real zero-one $v \times b$ matrix N, whose row sums are $r > \lambda \geq 1$ and column sums $k > 1$ (Definition (incidence and block matrices)), with $v > 1$, $\lambda > 0$ by ⟨1⟩4, ⟨1⟩2, ⟨1⟩8 and ⟨1⟩. □
⟨1⟩10. Q.E.D.
PROOF: The preceding steps are the four assertions. □

Lemma (incidence and flag graphs). ASSUME: D is a design with parameters (v, k, b, r, λ) on P and $\mathcal{B}$, with incidence matrix N.
PROVE:
1. $(P, \mathcal{B})$ is a bipartition of $I(D)$, biregular with degrees (r, k), whose biadjacency matrix is N; $I(D)$ has vr edges and no isolated vertices.
2. $I(D)$ is connected, and it is regular if and only if $r = k$.
3. $F(D)$ is connected and $(r + k - 2)$-regular on vr vertices.
4. $2I + A(F(D))$ is positive semidefinite of rank $v + b - 1$.

PROOF SKETCH: We will read the degrees off the definitions, and then apply the toolkit's facts on line graphs and incidence matrices.
⟨1⟩1. Assertion 1 holds.
⟨2⟩1. $(P, \mathcal{B})$ is a bipartition with biadjacency matrix N.
PROOF: Every edge of $I(D)$ joins a point to a block, by Definition (incidence graphs, flag graphs, characterization), and $N(p, B) = 1$ exactly when $p \in B$ (Definition (incidence and block matrices)). □

⟨2⟩2. Points have degree $r \geq 1$ and blocks have degree $k \geq 2$.
PROOF: The neighbours of p are the blocks through p, and those of B are its k points; $r \geq \lambda \geq 1$ by Lemma (design identities), and $k > 1$. □
⟨2⟩3. Q.E.D.
PROOF: The number of edges is the number of pairs (p, B) with $p \in B$, which is vr by ⟨2⟩2 and double counting; there are no isolated vertices by ⟨2⟩2. □
⟨1⟩2. $I(D)$ is connected.
⟨2⟩1. Any two points are joined by a path.
PROOF: Two distinct points lie in $\lambda \geq 1$ common blocks, by Lemma (design identities); such a block is a common neighbour. □
⟨2⟩2. Q.E.D.
PROOF: P is nonempty since $v > 1$, and every block has a neighbouring point, since $k \geq 2$; combine this with ⟨2⟩1. □
⟨1⟩3. $I(D)$ is regular if and only if $r = k$.
PROOF: Both parts are nonempty ($v > 1$, $b > 0$), and the degrees are r and k by ⟨1⟩1. □
⟨1⟩4. Assertion 3 holds.
PROOF: Apply [ST, Lemma 6.3] to $I(D)$, which has an edge and no isolated vertices by ⟨1⟩1 and is connected by ⟨1⟩2: $F(D)$ is connected, and an edge $\{p, B\}$ has degree $r + k - 2$ in it. $F(D)$ has $|E(I(D))| = vr$ vertices. □
⟨1⟩5. Assertion 4 holds.
PROOF: Let M be the incidence matrix of $I(D)$. By [ST, Theorem 6.2], $M^T M = 2I + A(F(D))$ and $\operatorname{rk} M = (v + b) - 1$, since $I(D)$ is one bipartite component by ⟨1⟩1 and ⟨1⟩2. $M^T M$ is a Gram matrix, positive semidefinite with $\operatorname{rk}(M^T M) = \operatorname{rk} M$ by [ST, Theorem 3.5]. □
⟨1⟩6. Q.E.D.
PROOF: Steps ⟨1⟩1 to ⟨1⟩5. □

Lemma (regular graphs of large valency). ASSUME: X is a graph on n vertices.
PROVE: 1. If $n \geq 1$ and X is $(n-1)$-regular, then X is complete, and $X \cong K_n$.
2. If $n = 2m$ with $m \geq 1$ and X is $(2m-2)$-regular, then $X \cong CP(m)$.

PROOF SKETCH: **[repair]** The paper does not state these, but its corollaries leave them to the reader (errata, Gaps 15 and 16). We will look at the non-neighbours of each vertex.
⟨1⟩1. Every vertex has $n - 1 - \deg(u)$ non-neighbours other than itself.
PROOF: `SimpleGraph.degree_compl`. □
⟨1⟩2. Assertion 1 holds.
PROOF: By ⟨1⟩1, each vertex is adjacent to every other vertex, so X is the complete graph on $V(X)$; apply `SimpleGraph.Iso.completeGraph` to `Fintype.equivFin`. □
⟨1⟩3. Assertion 2 holds.
⟨2⟩1. Each vertex u has a unique non-neighbour $u^* \neq u$, and $u^{**} = u$.
PROOF: ⟨1⟩1 gives exactly one, and non-adjacency is symmetric. □
⟨2⟩2. PICK a bijection $e : V(X) \to \{1, \ldots, m\} \times \{0, 1\}$ such that $e(u)$ and $e(u^*)$ have the same first coordinate for every u.
PROOF: The pairs $\{u, u^*\}$ partition $V(X)$ into m pairs, by ⟨2⟩1; number the pairs with `Fintype.equivFin` and the two members of each pair by 0 and 1. □
⟨2⟩3. Q.E.D.
PROOF: For $u \neq w$, u and w are adjacent if and only if $w \neq u^*$, by ⟨2⟩1, that is, when $e(u)$ and $e(w)$ have different first coordinates by ⟨2⟩2; this is adjacency in $CP(m)$. □
⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩2 and ⟨1⟩3. □

Lemma (connected 2-regular graphs). ASSUME: X is a connected 2-regular graph on n vertices.
PROVE: $n \geq 3$ and $X \cong C_n$, the cycle `SimpleGraph.cycleGraphn`.

PROOF SKETCH: **[repair]** The paper leaves this to the reader in Corollaries 5.10 and 5.11 (errata, Gap 16). We will walk around X without turning back.
⟨1⟩1. PICK adjacent vertices u_0, u_1, and for $i \geq 1$ let u_{i+1} be the neighbour of u_i other than u_{i-1}.
PROOF: X is nonempty since it is connected, and a vertex has two neighbours; each u_i with $i \geq 1$ is adjacent to u_{i-1}, so it has exactly one other neighbour. □
⟨1⟩2. PICK the least j such that $u_j = u_i$ for some $i < j$. Then $i = 0$ and $j \geq 3$.
⟨2⟩1. Such a j exists, with $j \leq n$.
PROOF: The vertices $u_0, \ldots, u_n$ are $n+1$ vertices of a set of size n. □
⟨2⟩2. ASSUME: $u_j = u_i$ with $1 \leq i < j$.
PROVE: FALSE.
PROOF: By ⟨1⟩1, $u_{i+1} \neq u_i$ and $u_{i+2} \neq u_i$, so $j \geq i+3$; by the minimality of j, u_{i-1}, u_{i+1} and u_{j-1} are then three distinct neighbours of $u_i = u_j$, against 2-regularity. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2 give $i = 0$; and $u_1 \neq u_0$, $u_2 \neq u_0$ by ⟨1⟩1, so $j \geq 3$. □
⟨1⟩3. The set $W = \{u_0, \ldots, u_{j-1}\}$ contains all neighbours of its members.
PROOF: For $1 \leq i \leq j-1$ the two neighbours of u_i are u_{i-1} and u_{i+1}, with $u_j = u_0$; the two neighbours of u_0 are u_1 and u_{j-1}, distinct since $j \geq 3$, by ⟨1⟩2. □
⟨1⟩4. Q.E.D.
PROOF: By ⟨1⟩3 and connectedness, $W = V(X)$, so $j = n \geq 3$. By the same step, $u_i \mapsto i$ is a bijection $V(X) \to \{0, \ldots, n-1\}$ under which u_i and $u_{i'}$ are adjacent exactly when $i - i' \equiv \pm 1 \pmod{n}$; this is adjacency in C_n by `SimpleGraph.cycleGraph_adj'`. □

Lemma (isolated vertices). ASSUME: Y is a graph.
PROVE: There is a graph Y' without isolated vertices such that $L(Y') \cong L(Y)$.

PROOF SKETCH: **[repair]** Isolated vertices are invisible to the line graph (errata, Implicit hypotheses 3 and 5). We will delete them.
LET: W be the set of vertices of Y of positive degree, and Y' the subgraph of Y induced on W.
⟨1⟩1. Y' has no isolated vertices.
PROOF: A neighbour of $w \in W$ has positive degree, so it lies in W. □
⟨1⟩2. Every edge of Y has both ends in W, so $\{u, w\} \mapsto \{u, w\}$ is a bijection $E(Y') \to E(Y)$.
PROOF: The ends of an edge have positive degree; Y' is induced. □
⟨1⟩3. Q.E.D.
PROOF: The bijection of ⟨1⟩2 preserves the ends of edges, so it preserves adjacency in the line graphs by `SimpleGraph.lineGraph_adj_iff_exists`; with ⟨1⟩1 this is the claim. □

Lemmas 5.1 to 5.5.

Lemma 5.1. ASSUME: X is a k-regular graph on a nonempty vertex set.
PROVE: $\lambda_{\max}(X) = k$, and $\mathrm{mult}_X(k)$ is the number of connected components of X.

PROOF SKETCH: **[repair]** The paper quotes this as well known (errata, Gap 8); the toolkit proves it through the Laplacian $kI - A(X)$.
⟨1⟩1. Q.E.D.
PROOF: [ST, Theorem 5.3]. □

Lemma 5.2. ASSUME: X is a graph on $n \geq 1$ vertices.
PROVE: X is regular if and only if $\sum_{\mu \in \mathrm{Spec}(X)} \mu^2 = n\lambda_{\max}(X)$, where the sum counts multiplicities.

PROOF SKETCH: **[repair]** The paper quotes this as well known (errata, Gap 8); the toolkit proves it from the Rayleigh quotient of **j**.
⟨1⟩1. Q.E.D.
PROOF: [ST, Theorem 5.4]. □

Lemma 5.3. ASSUME: X is a connected k-regular graph.
PROVE: $-k \in \mathrm{Spec}(X)$ if and only if X is bipartite, and then $\mathrm{mult}_X(-k) = 1$.

PROOF SKETCH: **[repair]** The paper quotes this as well known (errata, Gap 8); the toolkit proves it through the signless Laplacian $kI + A(X)$.
⟨1⟩1. Q.E.D.
PROOF: [ST, Theorem 5.6]. □

Lemma 5.4. ASSUME: 1. X is a graph without isolated vertices.
2. $L(X)$ is connected and regular.
PROVE: X is regular, or X has a bipartition that is biregular with degrees (k_1, k_2) for some $k_1 \neq k_2$.

PROOF SKETCH: **[repair]** The paper quotes this as well known (errata, Gap 8), and it needs ⟨0⟩:1 (errata, Implicit hypothesis 3: K_2 with an isolated vertex is a counterexample). The toolkit proves it by counting, as its Theorem 6.4, which also assumes an edge and asserts $k_1 + k_2 = h + 2$ and $pk_1 = qk_2 = |E(X)|$ for the part sizes p, q. We will derive the edge and $k_1 \neq k_2$; the form without an edge is the one that Theorem 5.8 uses, so this lemma has its own Lean name.
⟨1⟩1. X has an edge.
PROOF: A connected graph is nonempty, and the vertices of $L(X)$ are the edges of X. □
⟨1⟩2. CASE: X is regular.
PROOF: This is the first alternative. □
⟨1⟩3. CASE: X is not regular.
PROOF: By [ST, Theorem 6.4], with ⟨1⟩1, ⟨0⟩:1 and ⟨0⟩:2, X has a bipartition that is biregular with some degrees (k_1, k_2), and $k_1 \neq k_2$ since X is not regular. □
⟨1⟩4. Q.E.D.
PROOF: The cases ⟨1⟩2 and ⟨1⟩3 are exhaustive. □

Lemma 5.5. ASSUME: 1. X and X' are regular graphs, each with at least one edge.
2. $L(X)$ and $L(X')$ are cospectral.
PROVE: X and X' are cospectral.

PROOF SKETCH: **[repair]** The paper states that the spectrum of a regular graph is determined by that of its line graph; this needs an edge, since all edgeless graphs have the same line graph (errata, Implicit hypothesis 3). We will follow the paper: the largest eigenvalue of the line graph gives the valency, its order gives the number of vertices, and MM^T and M^TM have the same nonzero eigenvalues.
LET: k, k' be the valencies, n, n' the numbers of vertices and m, m' the numbers of edges of X, X', and M, M' their incidence matrices.
⟨1⟩1. $m = m'$.
PROOF: The cospectral graphs $L(X)$, $L(X')$ have the same number of vertices, by [ST, Proposition 2.6]. □
⟨1⟩2. $L(X)$ is $(2k-2)$-regular and $L(X')$ is $(2k'-2)$-regular, with $k, k' \geq 1$.
PROOF: An edge gives $k, k' \geq 1$, so there are no isolated vertices; apply the degree formula of [ST, Lemma 6.3]. □
⟨1⟩3. $k = k'$.
PROOF: By Lemma 5.1 and ⟨1⟩2, $\lambda_{\max}(L(X)) = 2k-2$ and $\lambda_{\max}(L(X')) = 2k'-2$; cospectral graphs have the same largest eigenvalue by [ST, Proposition 2.6] and [ST, Lemma 2.4]. □

⟨1⟩4. $n = n'$.
PROOF: $nk = 2m$ and $n'k' = 2m'$ by `SimpleGraph.sum_degrees_eq_twice_card_edges`; use ⟨1⟩1, ⟨1⟩3 and $k \geq 1$. □
⟨1⟩5. $MM^T = kI + A(X)$, $M^TM = 2I + A(L(X))$, $M'M'^T = k'I + A(X')$ and $M'^TM' = 2I + A(L(X'))$.
PROOF: [ST, Theorem 6.2] for X and for X', with the degree matrices kI and $k'I$. □
⟨1⟩6. $\mathrm{mult}_{kI+A(X)}(\mu) = \mathrm{mult}_{kI+A(X')}(\mu)$ for every $\mu \neq 0$.
⟨2⟩1. $\mathrm{mult}_{kI+A(X)}(\mu) = \mathrm{mult}_{2I+A(L(X))}(\mu)$ and $\mathrm{mult}_{k'I+A(X')}(\mu) = \mathrm{mult}_{2I+A(L(X'))}(\mu)$.
PROOF: [ST, Theorem 6.1] for M, M^T and for M', M'^T, with ⟨1⟩5, since $\mu \neq 0$. □
⟨2⟩2. $\mathrm{mult}_{2I+A(L(X))}(\mu) = \mathrm{mult}_{2I+A(L(X'))}(\mu)$.
PROOF: Both equal the multiplicity of $\mu - 2$ in the common spectrum of $L(X)$ and $L(X')$, by [ST, Theorem 3.1] and [ST, Proposition 2.6]. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2, with $k = k'$ from ⟨1⟩3. □
⟨1⟩7. $\mathrm{mult}_{kI+A(X)}(0) = \mathrm{mult}_{kI+A(X')}(0)$.
PROOF: By [ST, Theorem 2.2], each multiplicity at 0 is the order $n = n'$ minus the sum of the multiplicities at the nonzero eigenvalues; use ⟨1⟩4 and ⟨1⟩6. □
⟨1⟩8. Q.E.D.
PROOF: By ⟨1⟩6, ⟨1⟩7 and [ST, Proposition 2.6], $\mathrm{Spec}(kI+A(X)) = \mathrm{Spec}(kI+A(X'))$. Shifting by $-k$ with [ST, Theorem 3.1] gives $\mathrm{Spec}(X) = \mathrm{Spec}(X')$, and [ST, Proposition 2.6] gives cospectrality. □

Lemma 5.6 and Remark 5.7.

We will need the paper's analysis of a connected bipartite graph with a regular line graph twice, for X_1 and for X_2 in Lemma 5.6. We state it as two lemmas, so that no step is cited from inside another proof.

Lemma (a bipartite root graph). ASSUME: 1. Z is a connected graph with at least one edge.
2. (P, Q) is a bipartition of Z, biregular with degrees (k_1, k_2), with biadjacency matrix B.
3. M is the incidence matrix of Z and $C = MM^T$.

PROVE: 1. Z has no isolated vertices, $k_1, k_2 \geq 1$, and $|P|k_1 = |Q|k_2 = |E(Z)|$.
2. $L(Z)$ is connected and $(k_1 + k_2 - 2)$-regular, and $\lambda_{\max}(L(Z)) = k_1 + k_2 - 2$.
3. In block form for $V(Z) = P \cup Q$, $C = \begin{pmatrix} k_1 I & B \\ B^T & k_2 I \end{pmatrix}$, and $BB^T\mathbf{j} = k_1k_2\mathbf{j}$.
4. $\mathrm{mult}_C(0) = 1$.
5. $|V(Z)| = \mathrm{rk}(2I + A(L(Z))) + 1$.
6. Let $\rho = \mathrm{rk}\, B$, let Λ be the multiset of positive eigenvalues of BB^T, and let $f(z) = (z - k_1)(z - k_2)$. Then Λ has ρ members, and the image of $\mathrm{Spec}(C)$ under f is $\Lambda + \Lambda + [\![0^{|V(Z)|-2\rho}]\!]$.

PROOF SKETCH: This is the first part of the paper's proof of Lemma 5.6. The kernel of C is the image of the eigenspace of C for $k_1 + k_2$ under an invertible map, and $k_1 + k_2$ is a simple eigenvalue since $L(Z)$ is connected.
⟨1⟩1. Assertion 1 holds.
⟨2⟩1. Z has no isolated vertices.
PROOF: An edge gives two vertices, so every vertex u has a vertex $w \neq u$ and a walk from u to w, by ⟨0⟩:1; its first edge is at u. □
⟨2⟩2. P and Q are nonempty and $k_1, k_2 \geq 1$.

PROOF: An edge has an end in P and one in Q by ⟨0⟩:2; the degrees of these ends are positive by ⟨2⟩1. □

⟨2⟩3. Q.E.D.

PROOF: Each edge has exactly one end in P and one in Q, so double counting the pairs (u, e) with $u \in P$ an end of e, and the pairs (w, e) with $w \in Q$ an end of e, gives $|P|k_1 = |E(Z)|$ and $|Q|k_2 = |E(Z)|$, by (finite counting and order). □

⟨1⟩2. Assertion 2 holds.

PROOF: [ST, Lemma 6.3] applies by ⟨0⟩:1 and ⟨1⟩1; an edge has one end in P and one in Q, so its degree in $L(Z)$ is $k_1 + k_2 - 2$. Then Lemma 5.1 gives $\lambda_{\max}$. □

⟨1⟩3. Assertion 3 holds.

⟨2⟩1. $C = \begin{pmatrix} k_1 I & B \\ B^T & k_2 I \end{pmatrix}$.

PROOF: By [ST, Theorem 6.2], $C = D + A(Z)$ with D the degree matrix, which is $k_1 I$ on P and $k_2 I$ on Q by ⟨0⟩:2; $A(Z)$ vanishes on $P \times P$ and $Q \times Q$ by ⟨0⟩:2, and its $P \times Q$ block is B by Definition (bipartitions and biadjacency matrices). □

⟨2⟩2. $B\mathbf{j} = k_1\mathbf{j}$ and $B^T\mathbf{j} = k_2\mathbf{j}$.

PROOF: The row sum of B at $u \in P$ is $\deg(u) = k_1$, and the column sum at $w \in Q$ is $\deg(w) = k_2$, by ⟨0⟩:2. □

⟨2⟩3. Q.E.D.

PROOF: ⟨2⟩1, and $BB^T\mathbf{j} = k_2 B\mathbf{j} = k_1 k_2 \mathbf{j}$ by ⟨2⟩2. □

⟨1⟩4. $\operatorname{mult}_C(k_1 + k_2) = 1$.

⟨2⟩1. $\operatorname{mult}_C(k_1 + k_2) = \operatorname{mult}_{2I + A(L(Z))}(k_1 + k_2)$.

PROOF: [ST, Theorem 6.1] for M and M^T, since $k_1 + k_2 \neq 0$, with $M^T M = 2I + A(L(Z))$ from [ST, Theorem 6.2]. □

⟨2⟩2. $\operatorname{mult}_{2I + A(L(Z))}(k_1 + k_2) = \operatorname{mult}_{L(Z)}(k_1 + k_2 - 2) = 1$.

PROOF: The first equality is [ST, Theorem 3.1]; the second is Lemma 5.1, since $L(Z)$ is connected and $(k_1 + k_2 - 2)$-regular by ⟨1⟩2. □

⟨2⟩3. Q.E.D.

PROOF: Steps ⟨2⟩1 and ⟨2⟩2. □

⟨1⟩5. The map $\phi(\mathbf{x}, \mathbf{y}) = (-k_2\mathbf{x}, k_1\mathbf{y})$ on $\mathbb{R}^P \oplus \mathbb{R}^Q$ is invertible and carries $\ker(C - (k_1 + k_2)I)$ onto $\ker C$.

⟨2⟩1. $C(\mathbf{x}, \mathbf{y}) = (k_1 + k_2)(\mathbf{x}, \mathbf{y})$ if and only if $B\mathbf{y} = k_2\mathbf{x}$ and $B^T\mathbf{x} = k_1\mathbf{y}$.

PROOF: Block multiplication with ⟨1⟩3: $C(\mathbf{x}, \mathbf{y}) = (k_1\mathbf{x} + B\mathbf{y}, B^T\mathbf{x} + k_2\mathbf{y})$. □

⟨2⟩2. $C\phi(\mathbf{x}, \mathbf{y}) = 0$ if and only if $B\mathbf{y} = k_2\mathbf{x}$ and $B^T\mathbf{x} = k_1\mathbf{y}$.

PROOF: Block multiplication gives $C\phi(\mathbf{x}, \mathbf{y}) = \big(k_1(B\mathbf{y} - k_2\mathbf{x}),\ k_2(k_1\mathbf{y} - B^T\mathbf{x})\big)$, and $k_1, k_2 \geq 1$ by ⟨1⟩1. □

⟨2⟩3. Q.E.D.

PROOF: ϕ is invertible since $k_1, k_2 \neq 0$; use ⟨2⟩1 and ⟨2⟩2. □

⟨1⟩6. Assertion 4 holds.

PROOF: By [ST, Theorem 2.2], $\operatorname{mult}_C(0) = \dim\ker C$ and $\operatorname{mult}_C(k_1 + k_2) = \dim\ker(C - (k_1 + k_2)I)$; these are equal by ⟨1⟩5 and `LinearEquiv.finrank_eq`, and the second is 1 by ⟨1⟩4. □

⟨1⟩7. Assertion 5 holds.

PROOF: $\operatorname{rk} C = |V(Z)| - 1$ by [ST, Theorem 3.2] and ⟨1⟩6, and $\operatorname{rk} C = \operatorname{rk}(MM^T) = \operatorname{rk}(M^T M) = \operatorname{rk}(2I + A(L(Z)))$ by [ST, Theorem 3.5] and [ST, Theorem 6.2]. □

⟨1⟩8. Assertion 6 holds.

LET: $\mu_1, \ldots, \mu_\rho$ be the members of Λ, which has ρ members by [ST, Theorem 6.1], and $\gamma_j^{\pm} = \frac{1}{2}\big(k_1 + k_2 \pm \sqrt{(k_1 - k_2)^2 + 4\mu_j}\big)$.

⟨2⟩1. $f(\gamma_j^+) = f(\gamma_j^-) = \mu_j$ for $1 \leq j \leq \rho$.

PROOF: $\gamma_j^{\pm}$ are the roots of $z^2-(k_1+k_2)z+k_1k_2-\mu_j$, by the quadratic formula, and $f(z)-\mu_j$ is this polynomial. □

⟨2⟩2. Spec(C) consists of the $\gamma_j^{\pm}$, of k_1 with multiplicity $|P|-\rho$ and of k_2 with multiplicity $|Q|-\rho$.

PROOF: [ST, Theorem 6.6] with $s=k_1$, $t=k_2$, and ⟨1⟩3. □

⟨2⟩3. Q.E.D.

PROOF: Apply f to ⟨2⟩2: by ⟨2⟩1 and $f(k_1)=f(k_2)=0$, the image is $\Lambda+\Lambda+[\![0^{|P|+|Q|-2\rho}]\!]$, and $|P|+|Q|=|V(Z)|$. □

⟨1⟩9. Q.E.D.

PROOF: Steps ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩6, ⟨1⟩7 and ⟨1⟩8. □

Lemma (two bipartite root graphs). ASSUME: 1. Z and Z' are connected graphs, each with at least one edge.

2. (P,Q) is a bipartition of Z, biregular with degrees (k_1,k_2), with biadjacency matrix B, and $|P|\le|Q|$.
3. (P',Q') is a bipartition of Z', biregular with degrees (k_1',k_2'), with biadjacency matrix B', and $|P'|\le|Q'|$.
4. $L(Z)$ and $L(Z')$ are cospectral.

PROVE: $|P|=|P'|$, $|Q|=|Q'|$, $k_1=k_1'$, $k_2=k_2'$, and Spec(BB^T) = Spec$(B'B'^T)$.

PROOF SKETCH: This is the paper's claim that the spectrum of $L(X_1)$ determines p, q, k_1, k_2 and Spec$(B_1B_1^T)$. We will recover the sizes and degrees from the number of edges, the valency and the rank, and then read Spec(BB^T) off Spec(C) through the quadratic $f(z)=(z-k_1)(z-k_2)$.

LET: $m=|E(Z)|$, $N=|V(Z)|$, $C=MM^T$ for the incidence matrix M of Z, and $m'=|E(Z')|$, $N'=|V(Z')|$, $C'=M'M'^T$ for the incidence matrix M' of Z'.

⟨1⟩1. $m=m'$.

PROOF: These are the numbers of vertices of the cospectral graphs $L(Z)$, $L(Z')$; apply [ST, Proposition 2.6]. □

⟨1⟩2. $k_1+k_2=k_1'+k_2'$.

PROOF: k_1+k_2-2 and $k_1'+k_2'-2$ are the largest eigenvalues of $L(Z)$ and $L(Z')$, by Lemma (a bipartite root graph); cospectral graphs have the same largest eigenvalue by [ST, Proposition 2.6] and [ST, Lemma 2.4]. □

⟨1⟩3. $N=N'$.

⟨2⟩1. rk$(2I+A(L(Z)))$ = rk$(2I+A(L(Z')))$.

PROOF: [ST, Theorem 3.2] with $f(z)=z+2$ expresses each rank through the common multiplicities of the two line graphs, which agree by [ST, Proposition 2.6]. □

⟨2⟩2. Q.E.D.

PROOF: Lemma (a bipartite root graph), assertion 5, for Z and Z', and ⟨2⟩1. □

⟨1⟩4. $k_1k_2N=m(k_1+k_2)$ and $k_1'k_2'N'=m'(k_1'+k_2')$.

PROOF: By Lemma (a bipartite root graph), assertion 1, for Z and for Z', $N=|P|+|Q|=m/k_1+m/k_2$ and $N'=m'/k_1'+m'/k_2'$; multiply by k_1k_2 and by $k_1'k_2'$. □

⟨1⟩5. $k_1\ge k_2$ and $k_1'\ge k_2'$.

PROOF: By Lemma (a bipartite root graph), assertion 1, for Z and for Z', $|P|k_1=|Q|k_2=m>0$ and $|P'|k_1'=|Q'|k_2'=m'>0$, with $|P|\le|Q|$ and $|P'|\le|Q'|$. □

⟨1⟩6. $k_1=k_1'$ and $k_2=k_2'$.

PROOF: By ⟨1⟩1 to ⟨1⟩4, the pairs have the same sum and the same product, so $(k_1-k_2)^2=(k_1'-k_2')^2$; both differences are nonnegative by ⟨1⟩5. □

⟨1⟩7. $|P|=|P'|$ and $|Q|=|Q'|$.

PROOF: $|P| = m/k_1$, $|Q| = m/k_2$, $|P'| = m'/k'_1$ and $|Q'| = m'/k'_2$ by Lemma (a bipartite root graph), assertion 1, for Z and Z'; use ⟨1⟩1 and ⟨1⟩6. □

⟨1⟩8. $\mathrm{Spec}(C) = \mathrm{Spec}(C')$.

⟨2⟩1. $\mathrm{mult}_C(\gamma) = \mathrm{mult}_{C'}(\gamma)$ for every $\gamma \neq 0$.

PROOF: By [ST, Theorem 6.1] and [ST, Theorem 6.2], $\mathrm{mult}_C(\gamma) = \mathrm{mult}_{2I+A(L(Z))}(\gamma)$ and $\mathrm{mult}_{C'}(\gamma) = \mathrm{mult}_{2I+A(L(Z'))}(\gamma)$; these agree by [ST, Theorem 3.1] and [ST, Proposition 2.6]. □

⟨2⟩2. Q.E.D.

PROOF: $\mathrm{mult}_C(0) = \mathrm{mult}_{C'}(0) = 1$ by Lemma (a bipartite root graph), assertion 4; with ⟨2⟩1 and [ST, Proposition 2.6]. □

LET: $f(z) = (z - k_1)(z - k_2)$, and let Λ and Λ' be the multisets of positive eigenvalues of BB^T and $B'B'^T$, and $\rho = \mathrm{rk}\, B$ and $\rho' = \mathrm{rk}\, B'$.

⟨1⟩9. The images of $\mathrm{Spec}(C)$ and $\mathrm{Spec}(C')$ under f are $\Lambda + \Lambda + \left[\!\left[0^{N-2\rho}\right]\!\right]$ and $\Lambda' + \Lambda' + \left[\!\left[0^{N-2\rho'}\right]\!\right]$.

PROOF: Lemma (a bipartite root graph), assertion 6, for Z, and for Z', whose polynomial $(z - k'_1)(z - k'_2)$ is f by ⟨1⟩6, with $N' = N$ by ⟨1⟩3. □

⟨1⟩10. $\Lambda = \Lambda'$.

PROOF: By ⟨1⟩8 the two images agree. For $\mu > 0$, the count of μ in them is $2\,\mathrm{count}_\Lambda(\mu)$ and $2\,\mathrm{count}_{\Lambda'}(\mu)$, by ⟨1⟩9; and Λ, Λ' have only positive members. □

⟨1⟩11. Q.E.D.

PROOF: By [ST, Theorem 6.1], $\mathrm{Spec}(BB^T) = \Lambda + \left[\!\left[0^{|P|-\rho}\right]\!\right]$ and $\mathrm{Spec}(B'B'^T) = \Lambda' + \left[\!\left[0^{|P'|-\rho'}\right]\!\right]$; these agree by ⟨1⟩10, which gives $\rho = \rho'$, and ⟨1⟩7. With ⟨1⟩6 and ⟨1⟩7 this is the claim. □

Lemma (biregular bipartitions). ASSUME: 1. Z is a bipartite graph without isolated vertices.
2. $L(Z)$ is connected and regular.
PROVE: Z has a bipartition (P, Q) with $|P| \leq |Q|$ that is biregular with some degrees (k_1, k_2).

PROOF SKETCH: Lemma 5.4 gives a biregular bipartition or regularity, and in a regular bipartite graph every bipartition is biregular.

⟨1⟩1. Z has a biregular bipartition.

⟨2⟩1. CASE: Z is k-regular.

PROOF: Any bipartition, which exists by ⟨0⟩:1, is biregular with degrees (k, k). □

⟨2⟩2. CASE: Z has a bipartition that is biregular with degrees (k_1, k_2), $k_1 \neq k_2$.

PROOF: This is the claim. □

⟨2⟩3. Q.E.D.

PROOF: By Lemma 5.4, with ⟨0⟩:1 and ⟨0⟩:2, the cases ⟨2⟩1 and ⟨2⟩2 are exhaustive. □

⟨1⟩2. Q.E.D.

PROOF: Exchange the two parts of the bipartition of ⟨1⟩1 if necessary; this exchanges the two degrees. □

Lemma 5.6. ASSUME: 1. X_1 is a connected bipartite graph with at least one edge, and $L(X_1)$ is h-regular.
2. X_2 is a graph without isolated vertices.
3. $L(X_1)$ and $L(X_2)$ are cospectral.
PROVE: X_2 is connected, and one of the following holds.
1. X_2 is bipartite, and for every bipartition (P_1, Q_1) of X_1 with $|P_1| \leq |Q_1|$ that is biregular with degrees (k_1, k_2) there is a bipartition (P_2, Q_2) of X_2 with $|P_2| = |P_1|$ and $|Q_2| = |Q_1|$ that is biregular with degrees (k_1, k_2), such that $\mathrm{Spec}(B_1 B_1^T) = \mathrm{Spec}(B_2 B_2^T)$ for the biadjacency matrices B_1, B_2 of X_1, X_2 for these bipartitions.

2. X_1 is not regular; X_2 is κ-regular and not bipartite, for the natural number κ with $2\kappa = h + 2$; and $|V(X_1)| = |V(X_2)| + 1$.

PROOF SKETCH: **[repair]** We need ⟨0⟩:2 (errata, Implicit hypothesis 5: $X_1 = K_{1,3}$ and X_2 a triangle with an isolated vertex have the same line graph). **[new]** We state (i) for every biregular bipartition of X_1, so that Theorem 5.8 can choose the points of a design as P_1; the toolkit's Theorem 6.7 orders the parts by size instead, and so this lemma has its own Lean name. We will take connectedness and (ii) from the toolkit's Theorem 6.7, which follows the paper: in the nonbipartite case $M_2 M_2^T$ is nonsingular by Lemma 5.3, so X_2 has $p + q - 1$ vertices. For (i) we will follow the first part of the paper's proof, which is Lemmas (a bipartite root graph) and (two bipartite root graphs).

⟨1⟩1. X_1 has no isolated vertices, and $L(X_1)$ is connected and h-regular on a nonempty vertex set.
PROOF: By ⟨0⟩:1 and [ST, Lemma 6.3]: a connected graph with an edge has no isolated vertices, since each vertex has a walk to an end of the edge that is different from it. □

⟨1⟩2. $L(X_2)$ is connected and h-regular.
PROOF: [ST, Corollary 5.5], with ⟨1⟩1 and ⟨0⟩:3. □

⟨1⟩3. X_2 has an edge and is connected, and if X_2 is not bipartite, then X_1 is not regular, X_2 is regular, and $|V(X_1)| = |V(X_2)| + 1$.
PROOF: [ST, Theorem 6.7], with ⟨0⟩:1, ⟨0⟩:2 and ⟨0⟩:3; $L(X_2)$ is nonempty by ⟨1⟩2, so X_2 has an edge. □

⟨1⟩4. CASE: X_2 is bipartite.
PROVE: (i) holds.

⟨2⟩1. ASSUME: (P_1, Q_1) is a bipartition of X_1 with $|P_1| \le |Q_1|$ that is biregular with degrees (k_1, k_2).
PROVE: There is a bipartition (P_2, Q_2) of X_2 as in (i).

⟨3⟩1. PICK a bipartition (P_2, Q_2) of X_2 with $|P_2| \le |Q_2|$ that is biregular with some degrees (k_1', k_2').
PROOF: Lemma (biregular bipartitions) for X_2, with ⟨1⟩, ⟨0⟩:2 and ⟨1⟩2. □

⟨3⟩2. Q.E.D.
PROOF: Lemma (two bipartite root graphs) with $Z = X_1$, $Z' = X_2$ applies by ⟨0⟩:1, ⟨1⟩3, ⟨2⟩, ⟨3⟩1 and ⟨0⟩:3; it gives the sizes, the degrees $k_1' = k_1$, $k_2' = k_2$ and the spectra. □

⟨2⟩2. Q.E.D.
PROOF: ⟨1⟩ and ⟨2⟩1 are (i). □

⟨1⟩5. CASE: X_2 is not bipartite.
PROVE: (ii) holds.

⟨2⟩1. X_2 is κ-regular for a natural number $\kappa \ge 1$ with $2\kappa = h + 2$.
PROOF: X_2 is regular by ⟨1⟩3 and ⟨1⟩, of some valency κ; $\kappa \ge 1$ since X_2 has an edge, by ⟨1⟩3. By the degree formula of [ST, Lemma 6.3], with ⟨0⟩:2, an edge of X_2 has degree $2\kappa - 2$ in $L(X_2)$, which is h by ⟨1⟩2. □

⟨2⟩2. Q.E.D.
PROOF: Steps ⟨1⟩3 and ⟨2⟩1, with ⟨1⟩, are (ii). □

⟨1⟩6. Q.E.D.
PROOF: Step ⟨1⟩3, and the cases ⟨1⟩4 and ⟨1⟩5, which are exhaustive. □

Remark 5.7. ASSUME: 1. X_1 is a connected graph with at least one edge, and (P, Q) is a bipartition of X_1, biregular with degrees (k_1, k_2) with $k_1 \ne k_2$, with $p = |P| \le q = |Q|$ and biadjacency matrix B of rank ρ.
2. X_2 is a graph without isolated vertices that is not bipartite, and $L(X_1)$ and $L(X_2)$ are cospectral.

PROVE: $k_1 > k_2 \geq 1$; k_1k_2 is an eigenvalue of BB^T of multiplicity 1; and if Λ' is the multiset of positive eigenvalues of BB^T with one copy of k_1k_2 removed, then

$$\mathrm{Spec}(X_2) = \left[\!\left[\tfrac{1}{2}(k_1+k_2)\right]\!\right] + \sum_{\mu\in\Lambda'} [\![\delta_\mu, -\delta_\mu]\!] + \left[\!\left[\left(\tfrac{1}{2}(k_1-k_2)\right)^{p-\rho}, \left(-\tfrac{1}{2}(k_1-k_2)\right)^{q-\rho}\right]\!\right],$$

where $\delta_\mu = \sqrt{\mu + \frac{1}{4}(k_1-k_2)^2}$.

We state only the conclusion of the remark, the spectrum of $A(X_2)$. Its intermediate assertions, that each μ gives the eigenvalues $\pm\sqrt{\mu}$ of $A(X_1)$, γ and γ' of C, and $\gamma-2$ and $\gamma'-2$ of $A(L(X_1))$, have no Lean statements of their own, and neither has the paper's comment that the multiplicities of $\pm\frac{1}{2}(k_1-k_2)$ need not be equal (the statement above gives them, as $p-\rho$ and $q-\rho$). The spectrum of a matrix of the form of C is `Matrix.spectrumMultiset_fromBlocks_scalar_of_ne`.

PROOF SKETCH: **[repair]** The paper says that each eigenvalue $\mu \neq 0$ of $B_1B_1^T$ gives two eigenvalues γ, γ' of C, the roots of $(\gamma-k_1)(\gamma-k_2)=\mu$, and two eigenvalues $\delta = \gamma - \frac{1}{2}(k_1+k_2)$ and $\delta' = \gamma' - \frac{1}{2}(k_1+k_2)$ of $A(X_2)$ (errata, Error 13, item 8, for δ'), with $\delta^2 = \mu + \frac{1}{4}(k_1-k_2)^2$. This fails for $\mu = k_1k_2$, whose roots are k_1+k_2 and 0, and 0 is not an eigenvalue of the nonsingular $M_2M_2^T$; and the multiplicities of $\pm\frac{1}{2}(k_1-k_2)$ are $p-\rho$ and $q-\rho$, whose difference, not sum, is $q-p$ (errata, Error 14). The toolkit's Corollary 6.8 is the paper's computation with these corrections, for X_2 connected, regular and not bipartite. We will derive those hypotheses from Lemma 5.6; the form that assumes only that X_2 has no isolated vertices is the one that Theorem 5.8 uses, so this remark has its own Lean name.

⟨1⟩1. $k_1 > k_2 \geq 1$.
PROOF: By Lemma (a bipartite root graph), assertion 1, $k_2 \geq 1$ and $pk_1 = qk_2$, and $p \leq q$, $k_1 \neq k_2$. □

⟨1⟩2. X_2 is connected and regular, and not bipartite.
PROOF: Lemma 5.6 applies, since X_1 is bipartite by ⟨0⟩:1 and $L(X_1)$ is (k_1+k_2-2)-regular by Lemma (a bipartite root graph), assertion 2; its alternative (i) fails since X_2 is not bipartite, so (ii) holds. □

⟨1⟩3. Q.E.D.
PROOF: [ST, Corollary 6.8], with ⟨0⟩:1, ⟨1⟩1, ⟨1⟩2 and ⟨0⟩:2; its $\mu_2, \ldots, \mu_\rho$ are the members of Λ', and its $\pm\sqrt{\mu_j + \frac{1}{4}(k_1-k_2)^2}$ are $\pm\delta_{\mu_j}$. □

5.2. **Theorem 5.8 and its consequences.**

Theorem 5.8. ASSUME: 1. D_1 is a design with parameters (v,k,b,r,λ) on P and $\mathcal{B}$, with incidence matrix N, and $b \geq v$.
2. $X_1 = F(D_1)$, and X_2 is a graph cospectral with X_1.
PROVE: One of the following holds.
1. $v+b \leq 9$, and X_2 is represented in E_8.
2. $X_2 \cong F(D_2)$ for a design D_2 with the parameters (v,k,b,r,λ).
3. There are integers $s \geq 2$ and $t \geq 1$ with $t(s+1)$ even, $s \geq 2t-1$ and $(s-1) \mid 2t(t-1)$, such that

$$v = \tfrac{1}{2}s(s+1), \quad k = st, \quad b = \tfrac{1}{2}(s+1)(s+2), \quad r = (s+2)t, \quad \lambda(s-1) = 2t(st-1),$$

and $X_2 \cong L(Y)$ for a connected $(s+1)t$-regular graph Y on $s(s+2)$ vertices with

$$\mathrm{Spec}(Y) = \left[\!\left[\left((s+1)t\right)^1, (-t)^{s+1}, (\sqrt{c})^\ell, (-\sqrt{c})^\ell\right]\!\right], \qquad c = \frac{t(s+1)(s-t)}{s-1}, \quad \ell = \tfrac{1}{2}(s-1)(s+2).$$

PROOF SKETCH: **[repair]** The hypothesis $b \geq v$ is the paper's remark after Theorem 5.8: by Lemma (design identities) it holds when $\lambda < r$, and otherwise every block contains every point

and one passes to the dual design, as Corollary 5.9 does. Without it the theorem is false (errata, Error 16): the design $(10, 10, 6, 6, 6)$ has flag graph $L(K_{6,10})$, cospectral with $L(T(6))$, and none of (i), (ii), (iii) holds. Here $v+b=16$; (ii) fails since the only design with these parameters has flag graph $L(K_{6,10})$, and $T(6)$ is not bipartite, so Whitney's theorem excludes $L(T(6)) \cong L(K_{6,10})$; and $v=10$ gives $s=4$ and $t=k/s=\frac{5}{2}$. The details are in Error 16. We will follow the paper. The spectrum makes X_2 regular and connected with $2I+A(X_2) \succeq 0$ of rank $v+b-1$, so Theorem 4.11 applies. A representation in E_8 gives (i), a cocktail party graph gives (ii), and a line graph $L(Y)$ gives (ii) or (iii) according to the two cases of Lemma 5.6.

⟨1⟩1. X_1 is connected and $(r+k-2)$-regular on vr vertices, and $2I+A(X_1)$ is positive semidefinite of rank $v+b-1$.
PROOF: Lemma (incidence and flag graphs), assertions 3 and 4. □

⟨1⟩2. X_2 is connected and $(r+k-2)$-regular on vr vertices, and $2I+A(X_2)$ is positive semidefinite of rank $v+b-1$.
⟨2⟩1. X_2 is connected and $(r+k-2)$-regular on vr vertices.
PROOF: [ST, Corollary 5.5] and [ST, Proposition 2.6], with ⟨1⟩1; the corollary is Lemma 5.2 and Lemma 5.1 for both graphs, as in the paper. □
⟨2⟩2. $2I+A(X_2)$ is positive semidefinite.
PROOF: By [ST, Corollary 3.4] and ⟨1⟩1, every eigenvalue of X_1 is at least -2; X_2 has the same eigenvalues by [ST, Proposition 2.6]; apply [ST, Corollary 3.4] again. □
⟨2⟩3. $\mathrm{rk}(2I+A(X_2)) = v+b-1$.
PROOF: [ST, Theorem 3.2] with $f(z)=z+2$ gives the rank from the multiplicities, which agree by [ST, Proposition 2.6]; use ⟨1⟩1. □
⟨2⟩4. Q.E.D.
PROOF: Steps ⟨2⟩1, ⟨2⟩2 and ⟨2⟩3. □

⟨1⟩3. $X_2 \cong L(Y)$ for some graph Y, or $X_2 \cong CP(m)$ for some m, or X_2 is represented in E_8.
PROOF: Theorem 4.11, with ⟨1⟩2. □

⟨1⟩4. CASE: X_2 is represented in E_8.
PROVE: (i) holds.
⟨2⟩1. $\mathrm{rk}(2I+A(X_2)) \le 8$.
PROOF: $2I+A(X_2)$ is the Gram matrix of a family in $\mathbb{R}^8$ (Definition 3.2), so by [ST, Theorem 3.5] its rank is the dimension of a subspace of $\mathbb{R}^8$; apply `Submodule.finrank_le` and `finrank_euclideanSpace_fin`. □
⟨2⟩2. Q.E.D.
PROOF: $v+b-1 \le 8$ by ⟨1⟩2 and ⟨2⟩1, and X_2 is represented in E_8 by ⟨1⟩. □

⟨1⟩5. CASE: $X_2 \cong CP(m)$.
PROVE: (ii) holds, with $D_2 = D_1$.
PROOF SKETCH: **[repair]** The paper does not exclude this case of Theorem 4.11 (errata, Gap 15). The errata's fill forces $v=k=b=r=2$ and hence case (i), which would also need a representation in E_8; we reach (ii) instead, since both graphs are then cocktail party graphs.
⟨2⟩1. X_2 is $(2m-2)$-regular on $2m$ vertices, and $m \ge 1$.
PROOF: $CP(m)$ is, by Definition (cocktail party graphs), and X_2 has $vr \ge 1$ vertices by ⟨1⟩2. □
⟨2⟩2. $X_1 \cong CP(m)$.
PROOF: By [ST, Corollary 5.5] and [ST, Proposition 2.6], applied to X_2 and its cospectral mate X_1, X_1 is $(2m-2)$-regular on $2m$ vertices by ⟨2⟩1; apply Lemma (regular graphs of large valency), assertion 2. □
⟨2⟩3. Q.E.D.
PROOF: $X_2 \cong CP(m) \cong X_1 = F(D_1)$ by ⟨1⟩ and ⟨2⟩2. □

⟨1⟩6. CASE: $X_2 \cong L(Y)$.

PROVE: (ii) or (iii) holds.

⟨2⟩1. PICK a graph Y' without isolated vertices with $X_2 \cong L(Y')$.
PROOF: Lemma (isolated vertices) and ⟨1⟩. □

⟨2⟩2. $L(I(D_1))$ and $L(Y')$ are cospectral.
PROOF: $L(I(D_1)) = X_1$ is cospectral with X_2 by ⟨0⟩:2, and X_2 with $L(Y')$ by ⟨2⟩1 and [ST, Proposition 2.6]; cospectrality is equality of characteristic polynomials. □

⟨2⟩3. Lemma 5.6 applies to $X_1' = I(D_1)$, with $h = r + k - 2$, and $X_2' = Y'$; in particular Y' is connected.
PROOF: $I(D_1)$ is connected and bipartite with an edge by Lemma (incidence and flag graphs), assertions 1 and 2, and $L(I(D_1)) = X_1$ is $(r + k - 2)$-regular by ⟨1⟩1; with ⟨2⟩1 and ⟨2⟩2. □

⟨2⟩4. CASE: Alternative (i) of Lemma 5.6 holds.
PROVE: (ii) holds.

⟨3⟩1. PICK a bipartition (P_2, Q_2) of Y' with $|P_2| = v$, $|Q_2| = b$, biregular with degrees (r, k), whose biadjacency matrix B_2 has $\mathrm{Spec}(B_2 B_2^T) = \mathrm{Spec}(NN^T)$.
PROOF: Apply (i) to the bipartition $(P, \mathcal{B})$ of $I(D_1)$, which is biregular with degrees (r, k) and has biadjacency matrix N by Lemma (incidence and flag graphs), and has $|P| = v \leq b = |\mathcal{B}|$ by ⟨0⟩:1. □

⟨3⟩2. $B_2 B_2^T = (r - \lambda)I + \lambda J$.
PROOF: [ST, Theorem 7.7] for the zero-one $v \times b$ matrix B_2, whose row sums are r and column sums k by ⟨3⟩1; its hypotheses $v > 1$, $k > 1$, $r > 0$, $\lambda > 0$, $r(k-1) = \lambda(v-1)$ and $\mathrm{Spec}(B_2 B_2^T) = [\![(rk)^1, (r-\lambda)^{v-1}]\!]$ hold by Definition (labelled designs), Lemma (design identities) and ⟨3⟩1. □

⟨3⟩3. The rule $D_2(w) = \{u \in P_2 : u \sim w\}$ for $w \in Q_2$ is a design with parameters (v, k, b, r, λ) on P_2 and Q_2.
PROOF: By Definition (labelled designs): $|P_2| = v$, $|Q_2| = b$; a block has k points and a point lies in r blocks by ⟨3⟩1; two points lie in $(B_2 B_2^T)(u, u') = \lambda$ common blocks by ⟨3⟩2. □

⟨3⟩4. $I(D_2) \cong Y'$.
PROOF: The map $P_2 \sqcup Q_2 \to V(Y')$ that is the inclusion on each part is a bijection, and it carries the incidences of D_2 onto the edges of Y', since every edge of Y' joins P_2 to Q_2 (Definition (bipartitions and biadjacency matrices)). □

⟨3⟩5. Q.E.D.
PROOF: $F(D_2) = L(I(D_2)) \cong L(Y') \cong X_2$ by ⟨3⟩4, `SimpleGraph.Iso.lineGraph` and ⟨2⟩1. □

⟨2⟩5. CASE: Alternative (ii) of Lemma 5.6 holds.
PROVE: (iii) holds.

⟨3⟩1. $r > k$, and Y' is connected, not bipartite and κ-regular on $v + b - 1$ vertices, with vr edges, for the natural number κ with $2\kappa = r + k$.

⟨4⟩1. $r > k$.
PROOF: $I(D_1)$ is not regular by (ii), so $r \neq k$ by Lemma (incidence and flag graphs); and $r = bk/v \geq k$ by Lemma (design identities) and ⟨0⟩:1. □

⟨4⟩2. Q.E.D.
PROOF: By (ii), Y' is not bipartite and κ-regular with $2\kappa = h + 2 = r + k$, and $|V(Y')| = (v+b)-1$; Y' is connected by ⟨2⟩3; and $|E(Y')| = |V(L(Y'))| = |V(X_2)| = vr$ by ⟨2⟩1 and ⟨1⟩2. □

⟨3⟩2. $(r + k)(v + b - 1) = 4vr$.
PROOF: `SimpleGraph.sum_degrees_eq_twice_card_edges` for Y' gives $\kappa(v+b-1) = 2vr$ by ⟨3⟩1; multiply by 2. □

⟨3⟩3. $v(r-k)^2 = k(r+k)$.
PROOF: Multiply ⟨3⟩2 by k and substitute $bk = vr$ from Lemma (design identities): $(r+k)(vk+vr-k) = 4vrk$, which rearranges to the claim (computation C11). □
⟨3⟩4. $t = \frac{1}{2}(r-k)$ is a positive integer.
PROOF: $r+k = 2\kappa$ is even by ⟨3⟩1, so $r-k = (r+k)-2k$ is even; and $r > k$. □
⟨3⟩5. $2t^2v = k(k+t)$ and $t(b-v) = k+t$; thus $t \mid k$.
PROOF: Substitute $r = k+2t$ in ⟨3⟩3 for the first; then $b-v = v(r-k)/k = 2tv/k = (k+t)/t$ by $bk = vr$ (computation C11). As $t \mid k+t$, also $t \mid k$. □
LET: $s = k/t$, a positive integer by ⟨3⟩5.
⟨3⟩6. $v = \frac{1}{2}s(s+1)$, $k = st$, $b = \frac{1}{2}(s+1)(s+2)$ and $r = (s+2)t$.
PROOF: Substitute $k = st$ in ⟨3⟩5 and in $r = k+2t$ and $bk = vr$ (computation C11). □
⟨3⟩7. $s \geq 2$.
PROOF: $\frac{1}{2}s(s+1) = v > 1$ by ⟨3⟩6. □
⟨3⟩8. $\lambda(s-1) = 2t(st-1)$ and $(s-1) \mid 2t(t-1)$.
PROOF: Substituting ⟨3⟩6 in $r(k-1) = \lambda(v-1)$ of Lemma (design identities) gives $(s+2)t(st-1) = \frac{1}{2}\lambda(s+2)(s-1)$; divide by $s+2$. Then $2t(t-1) = (\lambda-2t^2)(s-1)$ (computation C11). □
⟨3⟩9. $s \geq 2t-1$.
PROOF: **[repair]** The paper prints $s \leq 2t-1$ (errata, Error 4). By Lemma (design identities), $k \leq v$, that is $st \leq \frac{1}{2}s(s+1)$ by ⟨3⟩6; divide by $s > 0$. □
⟨3⟩10. Y' is connected and $(s+1)t$-regular on $s(s+2)$ vertices.
PROOF: By ⟨3⟩1 and ⟨3⟩6, $\kappa = \frac{1}{2}(r+k) = (s+1)t$ and $v+b-1 = (s+1)^2-1 = s(s+2)$. □
⟨3⟩11. $\mathrm{Spec}(Y') = \left[\!\left[\left((s+1)t\right)^1, (-t)^{s+1}, (\sqrt{c})^{\ell}, (-\sqrt{c})^{\ell} \right]\!\right]$.
⟨4⟩1. $\mathrm{Spec}(Y') = [\![(s+1)t]\!] + \sum_{\mu\in\Lambda'} [\![\delta_\mu, -\delta_\mu]\!] + [\![t^{v-\rho}, (-t)^{b-\rho}]\!]$, where $\rho = \mathrm{rk}\, N$, $\delta_\mu = \sqrt{\mu+t^2}$, and Λ' is the multiset of positive eigenvalues of NN^T with one copy of rk removed.
PROOF: Remark 5.7 with $X_1 = I(D_1)$ and the bipartition $(P, \mathcal{B})$, biregular with degrees (r,k), $r \neq k$, $v \leq b$ by ⟨0⟩:1, and biadjacency matrix N (Lemma (incidence and flag graphs)), and $X_2 = Y'$ by ⟨2⟩1, ⟨2⟩2 and ⟨3⟩1; here $\frac{1}{2}(r+k) = (s+1)t$ and $\frac{1}{2}(r-k) = t$. □
⟨4⟩2. $r-\lambda+t^2 = c$, $v-1 = \ell$ and $b-v = s+1$.
PROOF: Substitute ⟨3⟩6 and ⟨3⟩8 (computation C11). □
⟨4⟩3. CASE: $r > \lambda$.
⟨5⟩1. $\rho = v$ and $\Lambda' = [\![(r-\lambda)^{v-1}]\!]$.
PROOF: By Lemma (design identities), $\mathrm{Spec}(NN^T) = [\![(rk)^1, (r-\lambda)^{v-1}]\!]$ with both values positive, so $\mathrm{rk}(NN^T) = v$ by [ST, Theorem 3.2], and $\rho = \mathrm{rk}(NN^T)$ by [ST, Theorem 3.5]. □
⟨5⟩2. Q.E.D.
PROOF: By ⟨4⟩1, ⟨5⟩1 and ⟨4⟩2, the terms are $(s+1)t$, $(\pm\sqrt{c})^{v-1}$ with $v-1 = \ell$, t^0 and $(-t)^{b-v}$ with $b-v = s+1$. (Computation C11, `Computations.C11.spectrum_of_parameters`, is this substitution; it takes the spectrum of ⟨4⟩1 in this form as a hypothesis, so the rank cases and Remark 5.7 stay with this proof.) □
⟨4⟩4. CASE: $r = \lambda$.
⟨5⟩1. $\rho = 1$ and Λ' is empty.
PROOF: By Lemma (design identities), $\mathrm{Spec}(NN^T) = [\![(rk)^1, 0^{v-1}]\!]$ with $rk > 0$, so $\mathrm{rk}(NN^T) = 1$ by [ST, Theorem 3.2], and $\rho = \mathrm{rk}(NN^T)$ by [ST, Theorem 3.5]. □
⟨5⟩2. $c = t^2$, so $\sqrt{c} = t$.
PROOF: $c = r-\lambda+t^2$ by ⟨4⟩2, and $r = \lambda$. □

⟨5⟩3. Q.E.D.
PROOF: By ⟨4⟩1 and ⟨5⟩1, $\mathrm{Spec}(Y') = [[(s+1)t, t^{v-1}, (-t)^{b-1}]]$, and $v-1 = \ell$, $b-1 = (s+1)+\ell$ by ⟨4⟩2; with ⟨5⟩2 this is the claim. □
⟨4⟩5. Q.E.D.
PROOF: $r \geq \lambda$ by Lemma (design identities), so the cases ⟨4⟩3 and ⟨4⟩4 are exhaustive. □
⟨3⟩12. $t(s+1)$ is even.
LET: $A = A(Y')$, $f(z) = (z+t)(z^2-c)$ and $e = t^2(s+1)(st-1)/(s-1)$.
⟨4⟩1. $f(A) = eJ$.
PROOF: f vanishes at $-t$ and $\pm\sqrt{c}$, which by ⟨3⟩11 are the eigenvalues other than $(s+1)t$; [ST, Theorem 7.10] with ⟨3⟩10 gives $f(A) = \frac{f((s+1)t)}{s(s+2)}J$, and $f((s+1)t) = s(s+2)e$ (computation C11). □
⟨4⟩2. $A^3(u,u) = t^3(s+1)$ for every vertex u.
PROOF: The (u,u) entry of $f(A) = A^3 + tA^2 - cA - ctI$ is $A^3(u,u) + t^2(s+1) - ct$, since $A(u,u) = 0$ and $A^2(u,u) = \deg(u) = (s+1)t$ by `SimpleGraph.adjMatrix_mul_self_apply_self`; by ⟨4⟩1 it equals e, and $e + ct - t^2(s+1) = t^3(s+1)$ (computation C11). □
⟨4⟩3. Q.E.D.
PROOF: Y' has a vertex, and $A^3(u,u)$ is twice a number of triangles by [ST, Theorem 5.1]; so $t^3(s+1)$ is even by ⟨4⟩2. The difference $t^3(s+1) - t(s+1) = (t-1)t(t+1)(s+1)$ is even, as a multiple of two consecutive integers. (In computation C11, `Computations.C11.even_parameter_cube` takes the integer form of ⟨4⟩2 as a hypothesis; the passage from the real identity of ⟨4⟩1 to it stays with this proof.) □
⟨3⟩13. Q.E.D.
PROOF: Take $Y = Y'$: $X_2 \cong L(Y')$ by ⟨2⟩1; the conditions are ⟨3⟩6 to ⟨3⟩12, with $t \geq 1$ by ⟨3⟩4. □
⟨2⟩6. Q.E.D.
PROOF: By ⟨2⟩3, the cases ⟨2⟩4 and ⟨2⟩5 are the two alternatives of Lemma 5.6. □
⟨1⟩7. Q.E.D.
PROOF: By ⟨1⟩3, the cases ⟨1⟩4, ⟨1⟩5 and ⟨1⟩6 are exhaustive. □

Corollaries 5.9 to 5.11 and Remark 5.12.
The next two lemmas settle the small cases that the corollaries leave to the reader.

Lemma (a single block). PROVE: 1. If X is an $(n-1)$-regular graph on $n \geq 1$ vertices, then every graph cospectral with X is isomorphic to X.
2. If D is a design with parameters $(v, v, 1, 1, 1)$, then $F(D)$ is characterized by its spectrum.

PROOF SKETCH: **[repair]** The paper's Corollary 5.11 omits the designs with a single block (errata, Gap 16). A cospectral mate is regular of the same valency on the same number of vertices, so both graphs are complete.
⟨1⟩1. ASSUME: X is an $(n-1)$-regular graph on $n \geq 1$ vertices, and X' is cospectral with X.
PROVE: $X' \cong X$.
PROOF: X' is $(n-1)$-regular on n vertices by [ST, Corollary 5.5] and [ST, Proposition 2.6]; both graphs are isomorphic to K_n by Lemma (regular graphs of large valency), assertion 1. □
⟨1⟩2. ASSUME: D is a design with parameters $(v, v, 1, 1, 1)$.
PROVE: $F(D)$ is characterized by its spectrum.
⟨2⟩1. $F(D)$ is $(v-1)$-regular on v vertices.
PROOF: Lemma (incidence and flag graphs), assertion 3, with $r + k - 2 = v - 1$ and $vr = v$. □
⟨2⟩2. Q.E.D.

PROOF: Apply ⟨1⟩1 to $F(D)$, by ⟨2⟩1, and take $D_2 = D$ in Definition (incidence graphs, flag graphs, characterization). □

⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2. □

Lemma (the hexagon). ASSUME: D is a design with parameters $(3, 2, 3, 2, 1)$.
PROVE: $F(D)$ is characterized by its spectrum.

PROOF SKETCH: **[repair]** The paper leaves this to the reader in Corollaries 5.10 and 5.11 (errata, Gap 16): a cospectral mate of the hexagon is connected and 2-regular on six vertices.

⟨1⟩1. $F(D) \cong C_6$.
PROOF: $F(D)$ is connected and 2-regular on 6 vertices by Lemma (incidence and flag graphs), assertion 3; apply Lemma (connected 2-regular graphs). □

⟨1⟩2. ASSUME: X_2 is cospectral with $F(D)$.
PROVE: $X_2 \cong F(D)$.
PROOF: X_2 is connected and 2-regular on 6 vertices by [ST, Corollary 5.5] and [ST, Proposition 2.6], so $X_2 \cong C_6$ by Lemma (connected 2-regular graphs); use ⟨1⟩1. □

⟨1⟩3. Q.E.D.
PROOF: Step ⟨1⟩2 with $D_2 = D$ in Definition (incidence graphs, flag graphs, characterization). □

Lemma (designs with full blocks). ASSUME: $m \geq 1$ and $n \geq 2$ are integers.
PROVE: 1. The *trivial design* T on the points $\{1, \ldots, n\}$ with m block labels, each block being the whole point set, is a design with parameters (n, n, m, m, m).
2. If D is a design with parameters (n, n, m, m, m), then $F(D) \cong L(K_{m,n})$.

PROOF SKETCH: Every point lies in every block, so the incidence graph is complete bipartite.

⟨1⟩1. Assertion 1 holds.
PROOF: By Definition (labelled designs): there are $n > 1$ points and $m > 0$ block labels, every block has n points, and every point and every pair of points lies in all m blocks. □

⟨1⟩2. ASSUME: D is a design with parameters (n, n, m, m, m) on P and $\mathcal{B}$.
PROVE: $F(D) \cong L(K_{m,n})$.
⟨2⟩1. Every block of D contains every point.
PROOF: Lemma (design identities), assertion 2, since $k = v = n$. □
⟨2⟩2. $I(D) \cong K_{m,n}$.
PROOF: By ⟨2⟩1, every point is adjacent to every block in $I(D)$, so $I(D)$ is the complete bipartite graph on $P \sqcup \mathcal{B}$ (Definition (incidence graphs, flag graphs, characterization)), with $|P| = n$ and $|\mathcal{B}| = m$. We obtain an isomorphism with $K_{m,n}$ from `Equiv.sumComm`, `Fintype.equivFin` and `SimpleGraph.completeBipartiteGraphCongr`. □
⟨2⟩3. Q.E.D.
PROOF: $F(D) = L(I(D))$, and apply `SimpleGraph.Iso.lineGraph` to ⟨2⟩2. □

⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2. □

Corollary 5.9. ASSUME: 1. $m, n \geq 1$ are integers with $m + n > 9$.
2. There is no integer $t \geq 1$ with $\{m, n\} = \{2t^2 + t, 2t^2 - t\}$ for which a symmetric Hadamard matrix of order $4t^2$ with constant diagonal exists.
3. X_2 is a graph cospectral with $L(K_{m,n})$.
PROVE: $X_2 \cong L(K_{m,n})$.

PROOF SKETCH: **[repair]** The two readings of "characterized" agree here (errata, Implicit hypothesis 6), since a design with $k = v$ is unique. **[repair]** We write the exception as a condition on $\{m, n\}$ (errata, Error 17): as printed, with $m = 6$, $n = 10$ the exception does not apply,

but $L(K_{6,10})$ is cospectral with $L(T(6))$ (errata, Error 14, evidence). We will follow the paper: $L(K_{m,n})$ is the flag graph of the trivial design, and in case (iii) of Theorem 5.8 bordering a (± 1) form of $A(Y)$ gives the Hadamard matrix.

⟨1⟩1. ASSUME: $m \geq n$, and ⟨0⟩:1 to ⟨0⟩:3.
PROVE: $X_2 \cong L(K_{m,n})$.

⟨2⟩1. CASE: $n = 1$.
PROOF: $L(K_{m,1})$ is $(m-1)$-regular on m vertices by the degree formula of [ST, Lemma 6.3]; apply Lemma (a single block). □

⟨2⟩2. CASE: $n \geq 2$.
LET: T be the trivial design of Lemma (designs with full blocks) on n points with m block labels.

⟨3⟩1. T is a design with parameters (n, n, m, m, m), $b = m \geq n = v$, and $F(T) \cong L(K_{m,n})$.
PROOF: Lemma (designs with full blocks), both assertions, with $n \geq 2$ by ⟨2⟩ and $m \geq n$ by ⟨1⟩. □

⟨3⟩2. One of (i), (ii) and (iii) of Theorem 5.8 holds for $D_1 = T$ and X_2.
PROOF: X_2 is cospectral with $F(T)$ by ⟨0⟩:3, ⟨3⟩1 and [ST, Proposition 2.6]; apply Theorem 5.8. □

⟨3⟩3. CASE: (i) holds.
PROOF: Then $n + m = v + b \leq 9$, against ⟨0⟩:1. □

⟨3⟩4. CASE: (ii) holds.
PROOF: Then $X_2 \cong F(D_2)$ for a design D_2 with parameters (n, n, m, m, m), and $F(D_2) \cong L(K_{m,n})$ by Lemma (designs with full blocks), assertion 2. □

⟨3⟩5. CASE: (iii) holds.
PROVE: FALSE.
PROOF SKETCH: We will show that the exception of ⟨0⟩:2 occurs.

⟨4⟩1. $s = 2t - 1$, $n = 2t^2 - t$, $m = 2t^2 + t$, $t \geq 2$ and $c = t^2$.
PROOF: From $v = k$ in (iii), $\frac{1}{2}s(s+1) = st$, so $s + 1 = 2t$; substitute in v, b and c, and $2t - 1 = s \geq 2$ (computation C12). (Computation C12 proves the whole step, $t \geq 2$ and $c = t^2$ included, for the integers $s \geq 2$ and $t \geq 1$ of (iii), with the fractions evaluated over $\mathbb{Q}$.) □

⟨4⟩2. $A = A(Y)$ satisfies $A^2 = t^2(I + J)$ and $AJ = 2t^2 J$, on $4t^2 - 1$ vertices.
PROOF: By (iii) and ⟨4⟩1, Y is connected and $2t^2$-regular on $s(s+2) = 4t^2 - 1$ vertices with eigenvalues $2t^2$ and $\pm t$. [ST, Theorem 7.10] with $f(z) = z^2 - t^2$ gives $A^2 - t^2 I = \frac{4t^4 - t^2}{4t^2 - 1} J = t^2 J$. □

LET: $S = 2A - J$ and $H = \begin{pmatrix} -1 & -\mathbf{j}^T \\ -\mathbf{j} & S \end{pmatrix}$, of order $4t^2$.

⟨4⟩3. S is symmetric with entries ± 1 and diagonal -1, $S\mathbf{j} = \mathbf{j}$ and $S^2 = 4t^2 I - J$.
PROOF: A is a symmetric zero-one matrix with zero diagonal; by ⟨4⟩2, $S\mathbf{j} = (4t^2 - (4t^2 - 1))\mathbf{j}$ and $S^2 = 4A^2 - 2AJ - 2JA + J^2 = 4t^2 I - J$, since $J^2 = (4t^2 - 1)J$ (computation C12). □

⟨4⟩4. H is a symmetric Hadamard matrix with constant diagonal -1.
PROOF: By ⟨4⟩3, H is symmetric with entries ± 1 and diagonal -1, and block multiplication gives $H^2 = 4t^2 I$: the corner is $1 + (4t^2 - 1)$, the border is $\mathbf{j} - S\mathbf{j} = 0$, and the rest is $J + S^2$. This is `Matrix.IsHadamard` over $\mathbb{R}$. □

⟨4⟩5. Q.E.D.
PROOF: ⟨4⟩1 and ⟨4⟩4 contradict ⟨0⟩:2. □

⟨3⟩6. Q.E.D.

PROOF: By ⟨3⟩2, the cases ⟨3⟩3, ⟨3⟩4 and ⟨3⟩5 are exhaustive, and ⟨3⟩3 and ⟨3⟩5 are impossible. □

⟨2⟩3. Q.E.D.

PROOF: The cases ⟨2⟩1 and ⟨2⟩2 are exhaustive, since $n \geq 1$. □

⟨1⟩2. CASE: $m \geq n$.

PROOF: Step ⟨1⟩1. □

⟨1⟩3. CASE: $m < n$.

⟨2⟩1. $L(K_{m,n}) \cong L(K_{n,m})$.

PROOF: `Equiv.sumComm` is an isomorphism $K_{m,n} \cong K_{n,m}$ by the definition of the complete bipartite graph; apply `SimpleGraph.Iso.lineGraph`. □

⟨2⟩2. Q.E.D.

PROOF: ⟨0⟩:1 and ⟨0⟩:2 are symmetric in m and n, and X_2 is cospectral with $L(K_{n,m})$ by ⟨2⟩1 and [ST, Proposition 2.6]; apply ⟨1⟩1 to (n, m), and then ⟨2⟩1. □

⟨1⟩4. Q.E.D.

PROOF: The cases ⟨1⟩2 and ⟨1⟩3 are exhaustive. □

Corollary 5.10. ASSUME: 1. D_1 is a design with parameters (v, k, v, r, λ), that is, a symmetric design.

2. $k < v$ and $(v, k, \lambda) \neq (4, 3, 2)$.

PROVE: $F(D_1)$ is characterized by its spectrum.

PROOF SKETCH: We will follow the paper: case (iii) of Theorem 5.8 cannot occur, and case (i) leaves only $(3, 2, 1)$, the hexagon, since $(4, 3, 2)$ is excluded.

⟨1⟩1. $r = k$ and $b \geq v$.

PROOF: $vk = vr$ by Lemma (design identities) with $b = v$. □

⟨1⟩2. ASSUME: X_2 is cospectral with $F(D_1)$.

PROVE: $X_2 \cong F(D_2)$ for a design D_2 with the parameters of D_1.

⟨2⟩1. CASE: (i) of Theorem 5.8 holds.

⟨3⟩1. $(v, k, \lambda) = (3, 2, 1)$.

⟨4⟩1. $v \in \{3, 4\}$.

PROOF: $2v = v + b \leq 9$, and $v > k > 1$. □

⟨4⟩2. CASE: $v = 3$.

PROOF: $k = 2$, and $2\lambda = \lambda(v - 1) = r(k - 1) = 2$ by Lemma (design identities) and ⟨1⟩1, so $\lambda = 1$. □

⟨4⟩3. CASE: $v = 4$.

PROVE: FALSE.

⟨5⟩1. $3\lambda = k(k - 1)$ and $k \in \{2, 3\}$.

PROOF: $\lambda(v - 1) = r(k - 1)$ by Lemma (design identities), with $r = k$ by ⟨1⟩1; and $1 < k < v = 4$. □

⟨5⟩2. CASE: $k = 2$.

PROOF: Then $3\lambda = 2$ by ⟨5⟩1, which has no integer solution. □

⟨5⟩3. CASE: $k = 3$.

PROOF: Then $\lambda = 2$ by ⟨5⟩1, and $(v, k, \lambda) = (4, 3, 2)$ is excluded by ⟨0⟩:2. □

⟨5⟩4. Q.E.D.

PROOF: The cases ⟨5⟩2 and ⟨5⟩3 are exhaustive by ⟨5⟩1. □

⟨4⟩4. Q.E.D.

PROOF: The cases ⟨4⟩2 and ⟨4⟩3 are exhaustive by ⟨4⟩1. □

⟨3⟩2. Q.E.D.

PROOF: Lemma (the hexagon) with ⟨3⟩1 and $b = v = 3$, $r = k = 2$. □

⟨2⟩2. CASE: (ii) of Theorem 5.8 holds.

PROOF: This is the claim. □

⟨2⟩3. CASE: (iii) of Theorem 5.8 holds.
PROVE: FALSE.
PROOF: $r = (s+2)t > st = k$ since $t \geq 1$, against ⟨1⟩1. □

⟨2⟩4. Q.E.D.
PROOF: Theorem 5.8 applies by ⟨0⟩:1, ⟨1⟩1 and ⟨1⟩, so the cases are exhaustive. □

⟨1⟩3. Q.E.D.
PROOF: Step ⟨1⟩2 is Definition (incidence graphs, flag graphs, characterization). □

Remark. The exception is genuine. The paper's two sets of 12 vectors in E_7 give two graphs with spectrum $4^1\,2^3\,0^3\,(-2)^5$: the flag graph of the $(4,3,2)$ design, which is $L(Q_3)$, and a second graph, which computation C13 shows is not isomorphic to the first (`cgss/prerequisites.md`, HR1). A design with parameters $(4,3,4,3,2)$ is unique: each of its four blocks omits one point, each point is omitted by $b - r = 1$ block, and so the blocks are the four 3-subsets of the points. Every flag graph of such a design is therefore isomorphic to the first graph, and the second graph shows, in the sense of Definition (incidence graphs, flag graphs, characterization), that the flag graph is not characterized by its spectrum. The uniqueness of the exception (HR2) is a side claim, and we do not formalize it here. Computation C13 checks the spectra, that the two graphs are represented by the paper's vectors in E_7, and that they are not isomorphic (the first has no induced claw and the second has one); it identifies the first graph with the line graph of the incidence graph of the design on the four 3-subsets, built in `LineGraphs/Computations/C13Design.lean` rather than through Definition (incidence graphs, flag graphs, characterization). It also identifies the first graph with $L(Q_3)$, by an explicit isomorphism, where Q_3 has the eight binary triples as vertices, two of them adjacent when they differ in exactly one coordinate; we do not use this identification. The non-characterization itself is `Computations.C13.not_flagGraphCharacterized`: the second graph is cospectral with the flag graph of the design and has an induced claw, while the flag graph of every design, being a line graph, has none.

Corollary 5.11. ASSUME: D_1 is a design with parameters $(v,k,b,r,1)$.
PROVE: $F(D_1)$ is characterized by its spectrum.

PROOF SKETCH: **[repair]** The paper says that case (i) of Theorem 5.8 occurs only for $(3,2,1)$; it also occurs for a single block containing every point, $(v,v,1,1,1)$ with $2 \leq v \leq 8$, whose flag graph is complete (errata, Gap 16). We will treat that case first, and then follow the paper.

⟨1⟩1. CASE: $k = v$.
PROOF: Then $r = \lambda = 1$ by Lemma (design identities), and $b = vr/k = 1$, so D_1 has parameters $(v,v,1,1,1)$; apply Lemma (a single block). □

⟨1⟩2. CASE: $k < v$.

⟨2⟩1. $r > 1$ and $b \geq v$.
PROOF: Lemma (design identities): $r \geq \lambda = 1$ with equality only if $k = v$, and then $b \geq v$ since $r > \lambda$. □

⟨2⟩2. ASSUME: X_2 is cospectral with $F(D_1)$.
PROVE: $X_2 \cong F(D_2)$ for a design D_2 with the parameters of D_1.

⟨3⟩1. CASE: (i) of Theorem 5.8 holds.

⟨4⟩1. $(v,k,b,r) = (3,2,3,2)$.

⟨5⟩1. $v \in \{3,4\}$.
PROOF: $2v \leq v + b \leq 9$ by ⟨2⟩1, and $v > k > 1$. □

⟨5⟩2. CASE: $v = 3$.
PROOF: $k = 2$, $r = \lambda(v-1)/(k-1) = 2$ and $b = vr/k = 3$ by Lemma (design identities). □

⟨5⟩3. CASE: $v = 4$.

PROVE: FALSE.
⟨6⟩1. $r(k-1) = 3$ and $k \in \{2, 3\}$.
PROOF: $r(k-1) = \lambda(v-1)$ by Lemma (design identities), with $\lambda = 1$ and $1 < k < v = 4$. □
⟨6⟩2. CASE: $k = 2$.
PROOF: Then $r = 3$ by ⟨6⟩1, and $b = vr/k = 6$ by Lemma (design identities), so $v + b = 10 > 9$, against (i). □
⟨6⟩3. CASE: $k = 3$.
PROOF: Then $2r = 3$ by ⟨6⟩1, which has no integer solution. □
⟨6⟩4. Q.E.D.
PROOF: The cases ⟨6⟩2 and ⟨6⟩3 are exhaustive by ⟨6⟩1. □
⟨5⟩4. Q.E.D.
PROOF: The cases ⟨5⟩2 and ⟨5⟩3 are exhaustive by ⟨5⟩1. □
⟨4⟩2. Q.E.D.
PROOF: Lemma (the hexagon) with ⟨4⟩1. □
⟨3⟩2. CASE: (ii) of Theorem 5.8 holds.
PROOF: This is the claim. □
⟨3⟩3. CASE: (iii) of Theorem 5.8 holds.
PROVE: FALSE.
PROOF: Then $\lambda(s-1) = 2t(st-1) \geq 2(s-1)$, since $t \geq 1$ and $st - 1 \geq s - 1$, so $\lambda \geq 2$, as $s \geq 2$; this contradicts $\lambda = 1$. □
⟨3⟩4. Q.E.D.
PROOF: Theorem 5.8 applies by ⟨2⟩1 and ⟨1⟩, so the cases are exhaustive. □
⟨2⟩3. Q.E.D.
PROOF: Step ⟨2⟩2 is Definition (incidence graphs, flag graphs, characterization). □
⟨1⟩3. Q.E.D.
PROOF: $k \leq v$ by Lemma (design identities), so the cases ⟨1⟩1 and ⟨1⟩2 are exhaustive. □

Remark 5.12. ASSUME: D_1 is a design with parameters (v, k, b, r, λ), with $k < v$ and $2 < \lambda < 19$.
PROVE: $F(D_1)$ is characterized by its spectrum.

PROOF SKETCH: We will follow the paper. In case (iii) of Theorem 5.8, $\lambda = 2t^2 + 2t(t-1)/(s-1)$, so $t = 1$ gives $\lambda = 2$, $t \geq 3$ gives $\lambda \geq 19$, and $t = 2$ gives $s \in \{3, 5\}$; $s = 3$ is a design with $k = v$, and $s = 5$ gives the parameters $(15, 10, 21, 14, 9)$, whose complement $(15, 5, 21, 7, 2)$ does not exist. The paper cites Hall's book for this; we use the Hall–Connor section. **[new]** The paper does not mention case (i); it cannot occur for $\lambda \geq 3$.
⟨1⟩1. $r > \lambda$ and $b \geq v$.
PROOF: Lemma (design identities), since $k < v$. □
⟨1⟩2. ASSUME: X_2 is cospectral with $F(D_1)$.
PROVE: $X_2 \cong F(D_2)$ for a design D_2 with the parameters of D_1.
⟨2⟩1. CASE: (i) of Theorem 5.8 holds.
PROVE: FALSE.
⟨3⟩1. $v \in \{3, 4\}$ and $b = \lambda v(v-1)/(k(k-1))$.
PROOF: $2v \leq v + b \leq 9$ by ⟨1⟩1, and $v > k > 1$; the formula is Lemma (design identities). □
⟨3⟩2. CASE: $v = 3$.
PROOF: $k = 2$, so $b = 3\lambda \geq 9$ and $v + b \geq 12$. □
⟨3⟩3. CASE: $v = 4$.
PROOF: $k \in \{2, 3\}$, so $b \geq 12\lambda/6 = 2\lambda \geq 6$ and $v + b \geq 10$. □
⟨3⟩4. Q.E.D.

PROOF: The cases ⟨3⟩2 and ⟨3⟩3 are exhaustive by ⟨3⟩1, and both contradict $v + b \leq 9$. □

⟨2⟩2. CASE: (ii) of Theorem 5.8 holds.
PROOF: This is the claim. □

⟨2⟩3. CASE: (iii) of Theorem 5.8 holds.
PROVE: FALSE.

⟨3⟩1. $\lambda(s-1) = 2t^2(s-1) + 2t(t-1)$.
PROOF: $2t(st-1) = 2t^2(s-1) + 2t(t-1)$, and (iii) (computation C14). □

⟨3⟩2. CASE: $t = 1$.
PROOF: Then $\lambda = 2$ by ⟨3⟩1, against $\lambda > 2$. □

⟨3⟩3. CASE: $t \geq 3$.
PROOF: Then $2t(t-1) > 0$, so $\lambda > 2t^2 \geq 18$ by ⟨3⟩1, against $\lambda < 19$. □

⟨3⟩4. CASE: $t = 2$.

⟨4⟩1. $s \in \{3, 5\}$.
PROOF: By (iii), $(s-1) \mid 4$ and $s \geq 2t - 1 = 3$ (computation C14). □

⟨4⟩2. CASE: $s = 3$.
PROOF: Then $v = 6 = k$ by (iii), against $k < v$. □

⟨4⟩3. CASE: $s = 5$.
PROOF: Then $(v, k, b, r, \lambda) = (15, 10, 21, 14, 9)$ by (iii), and Corollary (the complementary parameters) of the Hall–Connor section gives FALSE. □

⟨4⟩4. Q.E.D.
PROOF: The cases ⟨4⟩2 and ⟨4⟩3 are exhaustive by ⟨4⟩1. □

⟨3⟩5. Q.E.D.
PROOF: $t \geq 1$ by (iii), so the cases ⟨3⟩2, ⟨3⟩3 and ⟨3⟩4 are exhaustive. □

⟨2⟩4. Q.E.D.
PROOF: Theorem 5.8 applies by ⟨1⟩1 and ⟨1⟩, so the cases are exhaustive. □

⟨1⟩3. Q.E.D.
PROOF: Step ⟨1⟩2 is Definition (incidence graphs, flag graphs, characterization). □

Remark. The paper adds that for $t = 2$ and $s = 3$ the parameters are those of the trivial design with $(v, k, b, r) = (6, 6, 10, 10)$, and that $L(L(K_6))$ provides an exception. The exception is genuine, and we prove it in `LineGraphs/Computations/TrivialDesignException.lean`. Let D_1 be the design $(6, 6, 10, 10, 10)$, whose ten blocks all equal its set of six points (`Computations.TrivialDesignException.design`); its incidence graph is $K_{10,6}$, so $F(D_1) \cong L(K_{10,6})$. For a graph Y with n vertices and $m \geq n$ edges, the spectrum of $L(Y)$ is the spectrum of the signless Laplacian of Y shifted by -2, together with the eigenvalue -2 of multiplicity $m-n$ (`SimpleGraph.spectrumMultiset_lineGraph_of_card_le`). Certificates for the signless Laplacian of $K_{10,6}$, of order 16, and for $A(K_6)$, of order 6, give through this identity the spectrum $14^1\, 8^5\, 4^9\, (-2)^{45}$ for both $F(D_1)$ and $L(L(K_6)) = L(T(6))$ (`Computations.TrivialDesignException.flagGraph_cospectral_double_lineGraph`). Thus $L(K_{10,6})$ and $L(T(6))$ are cospectral, the pair behind Errors 14, 16 and 17 (errata). They are not isomorphic, and $F(D_1)$ is not characterized by its spectrum in the sense of Definition (incidence graphs, flag graphs, characterization) (`Computations.TrivialDesignException.not_flagGraphCharacterized`): the flag graph of a design with replication number 10 has a clique of 10 vertices, the flags through one point, while a clique of $L(T(6))$ consists of edges of $T(6)$ through one vertex or of the three edges of a triangle, and so has at most 8 vertices, since $T(6)$ is 8-regular.

Example 5.13.

Example 5.13. ASSUME: 1. $k \geq 2$ is an integer, and D' is a design with parameters $(k(2k-1), k, 4k^2-1, 2k+1, 1)$ on P' and $\mathcal{B}'$, with incidence matrix N'.

2. $\pi : \mathcal{B}' \to \{1, \dots, 2k+1\}$ is a *resolution* of D': for each i, every point lies in exactly one block B with $\pi(B) = i$.
3. H is the graph on $\mathcal{B}'$ in which $B \neq B'$ are adjacent when B and B' are disjoint and $\pi(B) \neq \pi(B')$.
4. D_1 is a design with parameters $(v_1, k_1, b_1, r_1, \lambda_1) = (k(2k-1), (k-1)(2k-1), k(2k+1), (k-1)(2k+1), k(2k-3))$.

PROVE: 1. H is connected and κ-regular on $4k^2-1$ vertices, where $\kappa = 2k(k-1)$, and $\mathrm{Spec}(H) = [\![\kappa^1, (-(k-1))^{2k}, k^\ell, (-k)^\ell]\!]$, where $\ell = (k-1)(2k+1)$. This is the spectrum of case (iii) of Theorem 5.8 with $s = 2k-1$ and $t = k-1$.
2. $L(H)$ is cospectral with $F(D_1)$.
3. $F(D_1)$ is not characterized by its spectrum.

PROOF SKETCH: [**repair**] The paper computes the spectrum of H and leaves the rest to the reader; "not characterized" is meant in the sense of Definition (incidence graphs, flag graphs, characterization) (errata, Implicit hypothesis 6), and then it is EX4. We will split $\mathbb{R}^{\mathcal{B}'}$ into four eigenspaces of $A(H)$, compare the signless Laplacians of H and of $I(D_1)$ as in Lemma 5.5, and then apply EX4. With $s = 2k-1$, $t = k-1$ we have $(s+1)t = \kappa$, $s(s+2) = 4k^2-1$, $c = k^2$ and $\frac{1}{2}(s-1)(s+2) = \ell$ in Theorem 5.8 (iii).

LET: $c_i \in \mathbb{R}^{\mathcal{B}'}$ be the indicator vector of $\pi^{-1}(i)$ for $1 \le i \le 2k+1$, R the $\mathcal{B}' \times \mathcal{B}'$ matrix with $R(B, B') = 1$ when $B \neq B'$ and $\pi(B) = \pi(B')$ and 0 otherwise, and J the all-ones $\mathcal{B}' \times \mathcal{B}'$ matrix.

⟨1⟩1. $N'c_i = \mathbf{j}$, each $\pi^{-1}(i)$ has $2k-1$ blocks, $\sum_i c_i = \mathbf{j}$ and $R = \sum_i c_i c_i^T - I$.
PROOF: By ⟨0⟩:2, the blocks of $\pi^{-1}(i)$ partition P'; each has k points and $|P'| = k(2k-1)$. Every block lies in exactly one class. □

⟨1⟩2. $A(H) = J - I - R - (N'^T N' - kI)$.
⟨2⟩1. $(N'^T N')(B, B') = |D'(B) \cap D'(B')|$, and $(N'^T N')(B, B) = k$.
PROOF: The entry is the number of points on both blocks, and every block has k points by ⟨0⟩:1. □
⟨2⟩2. $(N'^T N')(B, B') \in \{0, 1\}$ for $B \neq B'$.
PROOF: Two distinct points lie in exactly $\lambda = 1$ common block, by ⟨0⟩:1, so distinct blocks share at most one point; apply ⟨2⟩1. □
⟨2⟩3. $(N'^T N')(B, B') = 0$ for $B \neq B'$ with $\pi(B) = \pi(B')$.
PROOF: The blocks of $\pi^{-1}(\pi(B))$ are disjoint by ⟨1⟩1; apply ⟨2⟩1. □
⟨2⟩4. Q.E.D.
PROOF: By Steps ⟨2⟩1 to ⟨2⟩3, the matrix $J - I - R - (N'^T N' - kI)$ has (B, B') entry 0 when $B = B'$ or $\pi(B) = \pi(B')$, and $1 - |D'(B) \cap D'(B')|$ otherwise, which is 1 exactly when B and B' are disjoint. Compare the entries with ⟨0⟩:3. □

⟨1⟩3. $N'N'^T = 2kI + J$, and $\mathrm{rk}\, N' = k(2k-1)$.
PROOF: Lemma (design identities), assertion 3, with $r - \lambda = 2k$; its spectrum is positive, so $\mathrm{rk}(N'N'^T) = k(2k-1)$ by [ST, Theorem 3.2], and $\mathrm{rk}\, N' = \mathrm{rk}(N'N'^T)$ by [ST, Theorem 3.5]. □

LET: $W_1 = \{\sum_i a_i c_i : \sum_i a_i = 0\}$, $W_2 = \{N'^T \mathbf{y} : \mathbf{y}^T \mathbf{j} = 0\}$ and $W_3 = \{\mathbf{x} : N'\mathbf{x} = 0,\ c_i^T \mathbf{x} = 0 \text{ for all } i\}$.

⟨1⟩4. $A(H)\mathbf{j} = \kappa\mathbf{j}$, and $A(H)$ acts on W_1, W_2, W_3 as $-(k-1)$, $-k$ and k.
⟨2⟩1. $A(H)\mathbf{j} = \kappa\mathbf{j}$.
PROOF: $J\mathbf{j} = (4k^2-1)\mathbf{j}$, $R\mathbf{j} = (2k-2)\mathbf{j}$ by ⟨1⟩1, and $N'^T N'\mathbf{j} = (2k+1)k\mathbf{j}$ from the row and column sums of N'; by ⟨1⟩2, $A(H)\mathbf{j} = (4k^2 - 2 - (2k-2) - (2k+1)k + k)\mathbf{j} = \kappa\mathbf{j}$. □
⟨2⟩2. $A(H)\mathbf{x} = -(k-1)\mathbf{x}$ for $\mathbf{x} \in W_1$.
PROOF: By ⟨1⟩1, $J\mathbf{x} = (2k-1)(\sum_i a_i)\mathbf{j} = 0$, $R\mathbf{x} = (2k-2)\mathbf{x}$ and $N'\mathbf{x} = (\sum_i a_i)\mathbf{j} = 0$; apply ⟨1⟩2. □

⟨2⟩3. $A(H)\mathbf{x} = -k\mathbf{x}$ for $\mathbf{x} \in W_2$.
PROOF: For $\mathbf{x} = N'^T\mathbf{y}$ with $\mathbf{y}^T\mathbf{j} = 0$, $c_i^T\mathbf{x} = \mathbf{j}^T\mathbf{y} = 0$ by ⟨1⟩1, so $J\mathbf{x} = 0$ and $R\mathbf{x} = -\mathbf{x}$; and $N'^T N'\mathbf{x} = N'^T(2kI + J)\mathbf{y} = 2k\mathbf{x}$ by ⟨1⟩3. Apply ⟨1⟩2. □
⟨2⟩4. $A(H)\mathbf{x} = k\mathbf{x}$ for $\mathbf{x} \in W_3$.
PROOF: $J\mathbf{x} = 0$ and $R\mathbf{x} = -\mathbf{x}$ by ⟨1⟩1, and $N'^T N'\mathbf{x} = 0$; apply ⟨1⟩2. □
⟨2⟩5. Q.E.D.
PROOF: Steps ⟨2⟩1 to ⟨2⟩4. □
⟨1⟩5. $\dim W_1 = 2k$ and $\dim W_2 = \dim W_3 = \ell$.
⟨2⟩1. $\dim W_1 = 2k$ and $\dim W_2 = \ell$.
PROOF: By ⟨1⟩1, the $2k+1$ vectors c_i have disjoint nonempty supports, so they are independent, and W_1 is a hyperplane of their span. N'^T is injective by ⟨1⟩3, so $\dim W_2 = k(2k-1) - 1 = \ell$. □
⟨2⟩2. $\dim W_3 = \ell$.
⟨3⟩1. $\dim \ker N' = (4k^2 - 1) - k(2k-1)$.
PROOF: N' has $4k^2 - 1$ columns and rank $k(2k-1)$ by ⟨1⟩3. □
⟨3⟩2. $W_1 \subseteq \ker N'$.
PROOF: $N' \sum_i a_i c_i = (\sum_i a_i)\mathbf{j}$ by ⟨1⟩1. □
⟨3⟩3. W_3 is the orthogonal complement of W_1 in $\ker N'$.
PROOF: A vector $\mathbf{x}$ is orthogonal to W_1 exactly when all $c_i^T\mathbf{x}$ are equal, so W_3 lies in the complement. Conversely, if $\mathbf{x} \in \ker N'$ and $\mathbf{x} \perp W_1$, then all $c_i^T\mathbf{x}$ are equal, and their sum is $\mathbf{j}^T\mathbf{x} = \frac{1}{k}\mathbf{j}^T N'\mathbf{x} = 0$ by ⟨1⟩1, since the column sums of N' are k; so $\mathbf{x} \in W_3$. □
⟨3⟩4. Q.E.D.
PROOF: By Steps ⟨3⟩1 to ⟨3⟩3 and ⟨2⟩1, $\dim W_3 = (4k^2 - 1) - k(2k-1) - 2k = \ell$. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩6. Assertion 1 holds.
⟨2⟩1. $\mathrm{Spec}(H) = [\![\kappa^1, (-(k-1))^{2k}, k^\ell, (-k)^\ell]\!]$.
PROOF: The values κ, $-(k-1)$, $-k$, k are distinct since $k \geq 2$, so by [ST, Theorem 2.2], ⟨1⟩4 and ⟨1⟩5, their multiplicities are at least 1, $2k$, ℓ, ℓ. These add up to $4k^2 - 1$, the order of $A(H)$, so by [ST, Theorem 2.2] they are the multiplicities and there is no other eigenvalue. □
⟨2⟩2. $\mathrm{Spec}(H)$ is the spectrum of case (iii) of Theorem 5.8 with $s = 2k-1$ and $t = k-1$.
PROOF: With these values, $(s+1)t = 2k(k-1) = \kappa$, $s+1 = 2k$, $-t = -(k-1)$, $c = t(s+1)(s-t)/(s-1) = (k-1)\cdot 2k \cdot k/(2k-2) = k^2$, so $\sqrt{c} = k$, and $\frac{1}{2}(s-1)(s+2) = (k-1)(2k+1) = \ell$. Compare with ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: H is κ-regular on $|\mathcal{B}'| = 4k^2 - 1$ vertices by ⟨1⟩4, and connected by Lemma 5.1, since $\mathrm{mult}_H(\kappa) = 1$ by ⟨2⟩1. The spectrum is given by Steps ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩7. Assertion 2 holds.
LET: M_H and M_1 be the incidence matrices of H and of $I(D_1)$.
⟨2⟩1. $\mathrm{Spec}(M_H M_H^T) = [\![(2\kappa)^1, (\kappa - k + 1)^{2k}, (\kappa + k)^\ell, (\kappa - k)^\ell]\!]$, with all members positive.
PROOF: $M_H M_H^T = \kappa I + A(H)$ by [ST, Theorem 6.2] and ⟨1⟩6; shift $\mathrm{Spec}(H)$ by κ with [ST, Theorem 3.1]. $\kappa - k = k(2k-3) > 0$ as $k \geq 2$. □
⟨2⟩2. $\mathrm{Spec}(M_1 M_1^T)$ is $\mathrm{Spec}(M_H M_H^T)$ with one copy of 0 added.
⟨3⟩1. $\mathrm{Spec}(N_1 N_1^T) = [\![(r_1 k_1)^1, (2k-1)^{v_1 - 1}]\!]$ and $\mathrm{rk}\, N_1 = v_1$, for the incidence matrix N_1 of D_1.
PROOF: Lemma (design identities), assertion 3, with $r_1 - \lambda_1 = 2k - 1 > 0$; then [ST, Theorem 3.2] and [ST, Theorem 3.5]. □

⟨3⟩2. $\mathrm{Spec}(M_1 M_1^T)$ consists of 0 and $r_1 + k_1$, of $\frac{1}{2}\big(r_1 + k_1 \pm \sqrt{(r_1 - k_1)^2 + 4(2k-1)}\big)$ with multiplicity $v_1 - 1$ each, and of k_1 with multiplicity $b_1 - v_1$.
PROOF: Lemma (a bipartite root graph), assertion 3, for $Z = I(D_1)$, which applies by Lemma (incidence and flag graphs); then [ST, Theorem 6.6] with $s = r_1$, $t = k_1$, $\rho = v_1$ and ⟨3⟩1. The roots for $r_1 k_1$ are 0 and $r_1 + k_1$. □
⟨3⟩3. Q.E.D.
PROOF: Substitute ⟨0⟩:4: $r_1 + k_1 = 2\kappa$, $r_1 - k_1 = 2(k-1)$, so the square root is $2k$ and the pair is $\kappa \pm k$; $v_1 - 1 = \ell$, $k_1 = \kappa - k + 1$ and $b_1 - v_1 = 2k$. Compare with ⟨2⟩1. □
⟨2⟩3. $|E(H)| = |E(I(D_1))| = v_1 r_1$.
PROOF: $|E(H)| = \frac{1}{2}\kappa(4k^2 - 1) = k(k-1)(2k-1)(2k+1)$ by `SimpleGraph.sum_degrees_eq_twice_card_edges` and ⟨1⟩6, and this is $v_1 r_1$, which is $|E(I(D_1))|$ by Lemma (incidence and flag graphs). □
⟨2⟩4. $\mathrm{mult}_{2I+A(L(H))}(\gamma) = \mathrm{mult}_{2I+A(F(D_1))}(\gamma)$ for every $\gamma \neq 0$.
PROOF: By [ST, Theorem 6.2], $M_H^T M_H = 2I + A(L(H))$ and $M_1^T M_1 = 2I + A(F(D_1))$; by [ST, Theorem 6.1], their multiplicities at $\gamma \neq 0$ are those of $M_H M_H^T$ and $M_1 M_1^T$, which agree by ⟨2⟩2. □
⟨2⟩5. Q.E.D.
PROOF: Both matrices have order $v_1 r_1$ by ⟨2⟩3, so their multiplicities at 0 agree by [ST, Theorem 2.2] and ⟨2⟩4. Thus $\mathrm{Spec}(2I + A(L(H))) = \mathrm{Spec}(2I + A(F(D_1)))$ by [ST, Proposition 2.6]; shift by -2 with [ST, Theorem 3.1], and apply [ST, Proposition 2.6]. □
⟨1⟩8. Assertion 3 holds.
⟨2⟩1. H is not bipartite.
PROOF: H is connected and κ-regular, and $-\kappa \notin \mathrm{Spec}(H)$ by ⟨1⟩6, since $\kappa > k$ for $k \geq 2$; apply Lemma 5.3. □
⟨2⟩2. ASSUME: D_2 is a design with the parameters of D_1, and $L(H) \cong F(D_2)$.
PROVE: FALSE.
PROOF: $F(D_2) = L(I(D_2))$, and $I(D_2)$ is bipartite without isolated vertices by Lemma (incidence and flag graphs). H is connected on $4k^2 - 1 \geq 15$ vertices by ⟨1⟩6, and not bipartite by ⟨2⟩1. This contradicts EX4 of the inputs for Example 5.13. **[new]** The Lean proof, `Design.not_nonempty_iso_flagGraph_lineGraph_of_degree`, uses a clique bound in place of EX4, so the example does not need EX4: the r_1 flags on a point of D_2 form an r_1-clique of $F(D_2)$ (`Design.exists_clique_flagGraph_card_replication`), while every clique of $L(H)$ has at most κ vertices, since H is κ-regular with $\kappa \geq 3$ (`SimpleGraph.card_clique_lineGraph_le`); and $r_1 = (k-1)(2k+1) = \kappa + k - 1 > \kappa = 2k(k-1) \geq 4$ for $k \geq 2$. □
⟨2⟩3. Q.E.D.
PROOF: $L(H)$ is cospectral with $F(D_1)$ by ⟨1⟩7, and is isomorphic to no $F(D_2)$ by ⟨2⟩2; this is the negation of Definition (incidence graphs, flag graphs, characterization) for D_1. □
⟨1⟩9. Q.E.D.
PROOF: Steps ⟨1⟩6, ⟨1⟩7 and ⟨1⟩8. □

Remark. By EX1 and EX2, the resolvable designs of the example exist for $k = 2^d$ with $d \geq 1$ and for $k = 3$, and by EX3 the designs D_1 exist for $k = 2$ and $k = 4$, with parameters $(6, 3, 10, 5, 2)$ and $(28, 21, 36, 27, 20)$; for these two, the flag graph is not characterized by its spectrum. For $k = 2$ the paper describes H as the line graph of the Petersen graph; we do not check this. Computation C15 checks that the flag graph of the design $(6, 3, 10, 5, 2)$ is cospectral with $L(L(\text{Petersen}))$ and not isomorphic to it; the two graphs have 30 and 15 cliques of size 4. In Lean the flag graph is the line graph of the incidence graph of `LineGraphs/Computations/C13Design.lean`, and the Petersen graph is the settled `SimpleGraph.petersen`. The certificate is computed on `Computations.C15.`

`petersen`, the pentagon, the pentagram and five spokes, and `Computations.C15.petersenIso` is an explicit isomorphism from it to `SimpleGraph.petersen`, through which the two statements are transported. For $k = 3$ the parameters of D_1 are $(15, 10, 21, 14, 9)$, and no such design exists, by the Hall–Connor section, as in the proof of Remark 5.12. Computation C23 (`LineGraphs/Computations/C23.lean`, with the data of `cgss/certificates/c23_lean_data.py`) builds the designs of the instances $k = 2$ and $k = 4$ and proves both instances, and the case $k = 3$ above, in Lean; the existence of the designs of EX1 for every $k = 2^d$ is a citation, outside the scope.

5.3. **The Hall–Connor nonexistence theorem.** We will follow the ordinary proof of `cgss/proofs/hallconnor/hallconnor.tex`. The argument proves the nonexistence assertion used in Remark 5.12 and Example 5.13 of [CGSS], together with its complementary parameters. We use Hall and Connor's argument [HC], including proofs of the instances they take from Connor. We have not consulted Connor's paper directly. We omit the two comparison remarks of the ordinary proof, which the argument does not use.

We follow the blueprint conventions and Lamport [L95]: ASSUME introduces hypotheses, PROVE the conclusion, LET a definition, and PICK a witness. Each numbered step has a proof. A reference $\langle 0\rangle{:}i$ means hypothesis i of the result; $\langle k\rangle$ means the assumption of the enclosing level-k step. Closing QED steps state why the preceding steps suffice. The tags **[new]** and **[repair]** mark a change of argument and a filled gap, respectively. The no-op commands `\lean` and `\uses` record the Lean declarations and the direct dependencies.

We use the notation of `cgss/notation.md`, Sections 7 and 10: `Design P B v k b r lam`, `D.incMatrix`, and `D.compl`. The Hall–Connor names not listed individually there retain the names of the ordinary proof, following the namespace and naming rules of Section 10. Points and block labels are separate finite types with decidable equality. All sets used in counts are finite sets; their cardinalities are natural numbers. Moment identities and subtractions in weighted sums are evaluated in $\mathbb{Z}$ or $\mathbb{Q}$, never by truncated natural subtraction. Matrix entries are $M(u, w)$. We write I_U for the identity, $J_{U,V}$ for the all-ones matrix, and $\mathbf{j}_U$ for the all-ones vector. Omitted index types are determined by the products. All ranks are over $\mathbb{Q}$.

5.3.1. *Definitions and Mathlib inputs.*

Definition (labelled designs).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. $v, k, b, r, \lambda \in \mathbb{N}$, with $v > 1$, $k > 1$, and $b > 0$.
A design D with these parameters consists of a function $B \mapsto D(B) \subseteq P$ such that

$$|P| = v, \quad |\mathcal{B}| = b, \quad |D(B)| = k \quad (B \in \mathcal{B}),$$
$$|\{B : p \in D(B)\}| = r \quad (p \in P),$$
$$|\{B : p, q \in D(B)\}| = \lambda \quad (p, q \in P,\ p \neq q).$$

We write $p \in B$ for $p \in D(B)$; the map D need not be injective. The predicate D.IsSymmetric means $v = b$. In Lean the parameter names are `v k b r lam`; any positivity fields of the shared interface are satisfied by the parameters here.

Definition (incidence and block matrices).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters (v, k, b, r, λ).
LET: $N \triangleq D$.incMatrix $\in \mathrm{Mat}_{P\times\mathcal{B}}(\mathbb{Z})$, where

$$N(p, B) \triangleq \begin{cases} 1, & p \in B, \\ 0, & p \notin B. \end{cases}$$

Write $N_{\mathbb{Q}}$ for its entrywise rational cast. Put $G \triangleq N_{\mathbb{Q}}^T N_{\mathbb{Q}}$ and $C \triangleq 35I_{\mathcal{B}} - 7G + 10J_{\mathcal{B}}$, both over $\mathbb{Q}$. A product containing G or C uses $N_{\mathbb{Q}}$; we abbreviate it by N in rational calculations. Products asserted integral use the original N.

Definition (the block graph and all six-cliques).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters (v, k, b, r, λ).
LET: Z be the simple graph on $\mathcal{B}$ with

$$B \sim B' \quad \Longleftrightarrow \quad B \neq B' \text{ and } |B \cap B'| = 1.$$

Here $B \cap B'$ means $D(B) \cap D(B')$. Put $\Gamma(B) \triangleq \{B' : B \sim B'\}$ and

$$\mathcal{S} \triangleq D.\text{sixCliques} \triangleq \{S \subseteq \mathcal{B} : |S| = 6 \text{ and } \forall X, Y \in S,\ X \neq Y \Rightarrow X \sim Y\}.$$

Thus $\mathcal{S}$ is the finite set of *all* six-cliques, not a choice of local cliques. Its members form a finite subtype when used as indices.

Finite counting and order. All declarations below were checked in Mathlib at commit `082e2d37` (4 September 2026), the version pinned in `lake-manifest.json`; paths are relative to `Mathlib/`. Finite double counting is interchange of indicator sums, followed by grouping fibres:

- `Finset.sum_comm`, in `Algebra/BigOperators/Group/Finset/Sigma.lean`;
- `Finset.card_eq_sum_card_fiberwise`, in `Algebra/BigOperators/Group/Finset/Basic.lean`;
- `Finset.card_sdiff_of_subset`, `Finset.card_union_add_card_inter`, `Finset.eq_of_subset_of_card_le`, `Finset.card_pos`, and `Finset.card_le_one`, `Finset.card_eq_one`, `Finset.card_eq_two`, `Finset.card_eq_three`, and `Finset.card_image_of_injective`, in `Data/Finset/Card.lean`;
- `Fintype.bijective_iff_injective_and_card` and `Fintype.equivFin`, in `Data/Fintype/EquivFin.lean`;
- `Fintype.card_sum`, in `Data/Fintype/Sum.lean`, and `Fintype.card_option`, in `Data/Fintype/Option.lean`;
- `Fintype.sum_sum_type` and `Fintype.sum_option`, in `Data/Fintype/BigOperators.lean`;
- `Finset.sum_eq_zero_iff_of_nonneg`, in `Algebra/Order/BigOperators/Group/Finset.lean`.

The last declaration says that a sum of nonnegative terms is zero exactly when every term is zero. The equal-cardinality theorem turns an injection between finite types of equal size into a bijection. An enumeration of a finite subset is an equivalence to its subtype; its inclusion in the ambient type is injective. Integer and rational identities below use ring arithmetic, nonnegative squares, and finite sum distributivity. Small numerical tables are explicit computations.
Matrix operations and reindexing. We use `Matrix.mul_apply`, `Matrix.mul_assoc`, `Matrix.transpose_mul`, `Matrix.dotProduct_mulVec`, `Matrix.mulVec_mulVec`, `Matrix.map_mul`, and `Matrix.submatrix_mul_equiv`, all in `Data/Matrix/Mul.lean`. The last identity says that, for an equivalence e of the summed index type,

$$(A \,\text{submatrix}\, f\, e)(B \,\text{submatrix}\, e\, g) = (AB) \,\text{submatrix}\, f\, g.$$

We also use `Matrix.transpose_map` and `Matrix.transpose_submatrix`, in `LinearAlgebra/Matrix/Defs.lean`, and `Matrix.map_one`, in `Data/Matrix/Diagonal.lean`. Applied to the ring homomorphism $\mathbb{Z} \to \mathbb{Q}$, these transport products, transposes, and I. The cast of J is J entrywise because 1 maps to 1; addition and scalar multiplication commute with the cast entrywise.

Ranks and determinants. The following declarations are in `LinearAlgebra/Matrix/Rank.lean`: `Matrix.rank_self_mul_transpose`, `Matrix.rank_transpose`, `Matrix.rank_of_isUnit`, `Matrix.rank_of_det_ne_zero`, `Matrix.rank_add_rank_le_card_of_mul_eq_zero`, and `Matrix.rank_submatrix_le`. The identity $\mathrm{rk}(NN^T) = \mathrm{rk}\, N$ applies over the linearly ordered field $\mathbb{Q}$; rank under transpose applies over a field. A nonzero determinant gives full rank. If $AB = 0$, the sum of their ranks is at most the size of the shared index type.
`Matrix.isUnit_iff_isUnit_det`, in `LinearAlgebra/Matrix/NonsingularInverse.lean`, turns a nonzero rational determinant into an invertible matrix.
Determinant arithmetic. We use `Matrix.det_mul`, `Matrix.det_transpose`, `Matrix.det_smul`, `Matrix.det_one`, and `Int.cast_det`, in `LinearAlgebra/Matrix/Determinant/Basic.lean`. We use the matrix determinant lemma `Matrix.det_one_add_replicateCol_mul_replicateRow`, in `LinearAlgebra/Matrix/SchurComplement.lean`, in the form $\det(I + \mathbf{u}\mathbf{v}^T) = 1 + \mathbf{v}^T\mathbf{u}$ (the shared singleton index is `Unit`). These give multiplicativity, scalar homogeneity, and compatibility with the cast $\mathbb{Z} \to \mathbb{Q}$. In particular, the determinant over $\mathbb{Q}$ of an integral matrix is the cast of its integer determinant. We do not use the analytic positive-semidefinite determinant theorem, whose scalar hypotheses do not include $\mathbb{Q}$.
Integer divisibility. We use `Prime.dvd_of_dvd_pow`, in `Algebra/Prime/Defs.lean`, `Int.prime_iff_natAbs_prime`, in `RingTheory/Int/Basic.lean`, and `Nat.prime_five`, in `Data/Nat/Prime/Defs.lean`. The last two prove that 5 is prime in $\mathbb{Z}$; the first gives $5 \mid d^2 \Rightarrow 5 \mid d$ for $d \in \mathbb{Z}$.

5.3.2. *Counts and intersections.*

Lemma (the forced parameters).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and $|P| = 15$.
2. $F : \mathcal{B} \to \mathrm{Finset}(P)$ has $|F(B)| = 5$ for every B.
3. Each pair of distinct points lies in exactly two labelled blocks of F.
PROVE: Every point is in seven labels, $|\mathcal{B}| = 21$, and

$$\exists D : \texttt{Design P B 15 5 21 7 2},\ \forall B \in \mathcal{B},\ D.\mathrm{block}(B) = F(B).$$

Here the Lean type variable `B` denotes $\mathcal{B}$.

PROOF SKETCH: We will count first through one point, then over all points.
⟨1⟩1. ASSUME: $p \in P$.
PROVE: $r_p \triangleq |\{B : p \in F(B)\}| = 7$.
⟨2⟩1. The set of pairs (q, B) with $q \neq p$ and $p, q \in F(B)$ has size $4r_p$.
PROOF: For each of the r_p labels, deleting p from its five points leaves four choices, by ⟨0⟩:2 and the fibre count in F1. □
⟨2⟩2. The same set has size 28.
PROOF: There are $15 - 1 = 14$ choices of q and two labels for each, by ⟨0⟩:1, ⟨0⟩:3 and F1. □
⟨2⟩3. Q.E.D.
PROOF: The equality $4r_p = 28$ from ⟨2⟩1 and ⟨2⟩2 gives $r_p = 7$. □
⟨1⟩2. $5|\mathcal{B}| = 105$ and $|\mathcal{B}| = 21$.
PROOF: Count (p, B) with $p \in F(B)$ by blocks and by points, using ⟨0⟩:1, ⟨0⟩:2, ⟨1⟩1, and F1: $5|\mathcal{B}| = 15 \cdot 7$. Division by five gives the second assertion. □
⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2, with ⟨0⟩:1, ⟨0⟩:2, and ⟨0⟩:3, give every field of Definition (labelled designs). Choose the structure with block function F; its block equality is reflexivity, and $15 > 1$, $5 > 1$, $21 > 0$ are numerical. □

Lemma (incidence identities).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: The integral incidence matrix has

$$NN^T = 5I_P + 2J_P, \quad N\mathbf{j}_{\mathcal{B}} = 7\mathbf{j}_P, \quad N^T\mathbf{j}_P = 5\mathbf{j}_{\mathcal{B}}.$$

Over $\mathbb{Q}$, $G(B, B') = |B \cap B'|$, $G^T = G$, and

$$G^2 = 5G + 50J_{\mathcal{B}}, \quad GJ_{\mathcal{B}} = J_{\mathcal{B}}G = 35J_{\mathcal{B}}, \quad J_{\mathcal{B}}^2 = 21J_{\mathcal{B}}.$$

PROOF SKETCH: The matrix entries count points or labels; associativity then gives the identities for G.
⟨1⟩1. $(NN^T)(p, p) = 7$ for every $p \in P$.
PROOF: By Definitions (incidence and block matrices) and (labelled designs), the entry is the sum of the squares of the seven incidence indicators equal to one; use F2. □
⟨1⟩2. ASSUME: $p, q \in P$ and $p \neq q$.
PROVE: $(NN^T)(p, q) = 2$.
PROOF: The product of the two indicators is one exactly on labels containing p, q, of which there are two by Definition (labelled designs); expand by F2. □
⟨1⟩3. $NN^T = 5I_P + 2J_P$.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2 treat the exhaustive alternatives $p = q$ and $p \neq q$. □
⟨1⟩4. $N\mathbf{j}_{\mathcal{B}} = 7\mathbf{j}_P$.
PROOF: The row sum counts the seven labels on p, by Definition (labelled designs) and F2. □
⟨1⟩5. $N^T\mathbf{j}_P = 5\mathbf{j}_{\mathcal{B}}$.
PROOF: The column sum counts the five points of a block, by Definition (labelled designs) and F2. □
⟨1⟩6. $G(B, B') = |B \cap B'|$ and $G^T = G$.
PROOF: In Definition (incidence and block matrices), the summand $N(p, B)N(p, B')$ is the intersection indicator. Its symmetry in B, B' gives symmetry of G, by F2. □
⟨1⟩7. $G^2 = 5G + 50J_{\mathcal{B}}$.
PROOF: Cast ⟨1⟩3 and ⟨1⟩5 entrywise using the product, transpose, and constant-matrix rules of F2. Then

$$\begin{aligned} G^2 &= N^T(NN^T)N && [\text{F2}] \\ &= 5G + 2N^T J_P N && [\langle 1\rangle 3] \\ &= 5G + 50J_{\mathcal{B}} && [\langle 1\rangle 5,\ 5 \cdot 5 = 25]. \end{aligned}$$

□
⟨1⟩8. $GJ_{\mathcal{B}} = 35J_{\mathcal{B}}$.
PROOF: Steps ⟨1⟩4 and ⟨1⟩5 give $G\mathbf{j}_{\mathcal{B}} = N^T(7\mathbf{j}_P) = 35\mathbf{j}_{\mathcal{B}}$; each column of J is $\mathbf{j}$, by F2. □
⟨1⟩9. $J_{\mathcal{B}}G = 35J_{\mathcal{B}}$.
PROOF: Transpose ⟨1⟩8, using the symmetry in ⟨1⟩6 and F2. □
⟨1⟩10. $J_{\mathcal{B}}^2 = 21J_{\mathcal{B}}$.
PROOF: Each entry sums $|\mathcal{B}| = 21$ ones, by Definition (labelled designs) and F2. □
⟨1⟩11. Q.E.D.
PROOF: Steps ⟨1⟩3–⟨1⟩10 give every asserted identity. □

Lemma (intersection moments).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B \in \mathcal{B}$ and $n_i \triangleq |\{B' \neq B : |B \cap B'| = i\}|$ for $0 \leq i \leq 5$.

PROVE: Over $\mathbb{Z}$, $\sum_{i=0}^{5} n_i = 20$, $\sum_{i=0}^{5} i n_i = 30$, and $\sum_{i=0}^{5} i(i-1)n_i = 20$. In particular $\sum_{B' \neq B} |B \cap B'| = 30$.

PROOF SKETCH: We will count labels, incidences, and ordered pairs of distinct points. The moment statements will be cited outside this proof.

⟨1⟩1. $\sum_{i=0}^{5} n_i = 20$.
PROOF: Intersection size is between zero and five; the fibres partition the $21 - 1$ other labels, by Definition (labelled designs) and F1. □

⟨1⟩2. $\sum_{i=0}^{5} i n_i = 30$ and $\sum_{B' \neq B} |B \cap B'| = 30$.
PROOF: Count (p, B') with $p \in B \cap B'$ and $B' \neq B$. For each of five points, six of its seven labels remain. Grouping by B' and then by i gives both sums, using Definition (labelled designs) and F1. □

⟨1⟩3. $\sum_{i=0}^{5} i(i-1)n_i = 20$.
PROOF: Count (p, q, B') with $p, q \in B \cap B'$, $p \neq q$, $B' \neq B$. Each of $5 \cdot 4$ ordered pairs in B has exactly one further label; a label with intersection size i contributes $i(i-1)$, using Definition (labelled designs) and F1. □

⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩1, ⟨1⟩2, and ⟨1⟩3 prove the four assertions. □

Lemma (block intersections). [[HC], Lemma 4.1 at $k = 5$.]
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: Distinct labels meet in one or two points. Every block has exactly ten other labels of each kind; in particular $|\Gamma(B)| = 10$.

PROOF SKETCH: A nonnegative linear combination of the moments excludes intersection sizes other than one and two.

⟨1⟩1. ASSUME: $B \in \mathcal{B}$, with n_i as in Lemma (intersection moments).
PROVE: Every other label meets B in one or two points, and exactly ten labels have each intersection size.

⟨2⟩1. $\sum_{i=0}^{5} (i-1)(i-2)n_i = 0$.
PROOF: The polynomial equals $i(i-1) - 2i + 2$; Lemma (intersection moments) gives $20 - 2 \cdot 30 + 2 \cdot 20 = 0$. □

⟨2⟩2. $n_0 = n_3 = n_4 = n_5 = 0$.
⟨3⟩1. The coefficients $(i-1)(i-2)$, in order $i = 0, 1, 2, 3, 4, 5$, are $2, 0, 0, 2, 6, 12$.
PROOF: Direct integer substitution in the polynomial of ⟨2⟩1. □
⟨3⟩2. Q.E.D.
PROOF: The n_i are nonnegative cardinalities. Apply the zero-sum criterion of F1 to ⟨2⟩1 and ⟨3⟩1; the four indicated coefficients are positive. □

⟨2⟩3. $n_1 = n_2 = 10$.
PROOF: Lemma (intersection moments) and ⟨2⟩2 give $n_1 + n_2 = 20$ and $n_1 + 2n_2 = 30$. Subtract the equations. □

⟨2⟩4. Q.E.D.
PROOF: The zero fibres of ⟨2⟩2 exclude the other intersection sizes; ⟨2⟩3 and Definition (the block graph and all six-cliques) give the multiplicities and degree. □

⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1 applies to every B; Definition (the block graph and all six-cliques) identifies its ten one-point intersections with $\Gamma(B)$. □

Lemma (common-neighbour counts). [[HC], Lemma 4.2 at $k = 5$.]

ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.

2. $B_1 \neq B_2$ are block labels, $s \triangleq |B_1 \cap B_2|$, and $x_{ij} \triangleq |\{B \notin \{B_1, B_2\} : |B_1 \cap B| = i,\ |B_2 \cap B| = j\}|$.

PROVE: The four counts are

s	x_{11}	x_{12}	x_{21}	x_{22}
1	5	4	4	6
2	4	6	6	3

.

Thus adjacent vertices have five common neighbours and distinct nonadjacent vertices have four.

PROOF SKETCH: We will expose the four filtered-sum equations, including both terms removed from the entry of G^2, before solving them.

⟨1⟩1. The four fibres partition $\mathcal{B} \setminus \{B_1, B_2\}$, and $x_{11} + x_{12} + x_{21} + x_{22} = 19$.

PROOF: Lemma (block intersections) partitions the $21-2$ remaining labels into the four disjoint fibres; use ⟨0⟩:2, F1, and Definition (labelled designs). □

⟨1⟩2. $x_{11} + x_{12} + 2x_{21} + 2x_{22} = 30 - s$.

PROOF: From the last conclusion of Lemma (intersection moments) for B_1, remove the term B_2 of size s. The four fibres have first weights $1, 1, 2, 2$, using ⟨1⟩1, Lemma (block intersections), ⟨0⟩:2, and F1. □

⟨1⟩3. $x_{11} + 2x_{12} + x_{21} + 2x_{22} = 30 - s$.

PROOF: Apply Lemma (intersection moments) to B_2 and remove the term B_1 of size s. The second weights are $1, 2, 1, 2$, using ⟨1⟩1, Lemma (block intersections), ⟨0⟩:2, and F1. □

⟨1⟩4. $x_{11} + 2x_{12} + 2x_{21} + 4x_{22} = 50 - 5s$.

⟨2⟩1. $(G^2)(B_1, B_2) = 5s + 50$.

PROOF: Substitute $G(B_1, B_2) = s$ into the identity $G^2 = 5G + 50J$ of Lemma (incidence identities). □

⟨2⟩2. The terms indexed by B_1 and B_2 in $(G^2)(B_1, B_2)$ are each $5s$.

PROOF: The products are $G(B_1, B_1)G(B_1, B_2)$ and $G(B_1, B_2)G(B_2, B_2)$; both diagonals equal five by Definition (labelled designs) and Lemma (incidence identities). □

⟨2⟩3. Q.E.D.

PROOF: Expand the matrix entry using F2, remove the two distinct terms of ⟨2⟩2 from ⟨2⟩1, then group the remaining terms by their weights ij using ⟨1⟩1, Lemma (block intersections), ⟨0⟩:2, and F1. This gives $(5s + 50) - 10s = 50 - 5s$. □

⟨1⟩5. $x_{22} = 9 - 3s$, $x_{12} = x_{21} = 2 + 2s$, and $x_{11} = 6 - s$.

PROOF: Subtract ⟨1⟩2 and ⟨1⟩3 from ⟨1⟩4, then add ⟨1⟩1, to get $x_{22} = 9 - 3s$. Subtract ⟨1⟩1 and this value from each weighted marginal, then use the total for x_{11}. □

⟨1⟩6. CASE: $s = 1$.

The count vector is $(5, 4, 4, 6)$.

PROOF: Substitute ⟨1⟩ in ⟨1⟩5. □

⟨1⟩7. CASE: $s = 2$.

The count vector is $(4, 6, 6, 3)$.

PROOF: Substitute ⟨1⟩ in ⟨1⟩5. □

⟨1⟩8. $\Gamma(B_1) \cap \Gamma(B_2)$ is exactly the x_{11} fibre; $B_1 \sim B_2$ if and only if $s = 1$.

PROOF: By Definition (the block graph and all six-cliques), membership in both neighbourhoods is intersection size one with each block; irreflexivity excludes B_1, B_2. Conversely the fibre excludes these two labels and gives both adjacencies. For the pair itself, ⟨0⟩:2 and Lemma (block intersections) leave precisely $s = 1$ for adjacency and $s = 2$ for nonadjacency. □

⟨1⟩9. Q.E.D.

PROOF: Lemma (block intersections) makes the cases ⟨1⟩6, ⟨1⟩7 exhaustive. Step ⟨1⟩8 identifies their x_{11} counts with the asserted common-neighbour counts. □

5.3.3. *Connor's matrix and its rank.*

Lemma (a scalar matrix plus a constant matrix).
ASSUME: 1. U is a finite type with decidable equality, $n = |U| \geq 1$.
2. $\alpha, \beta \in \mathbb{Q}$ and $\alpha \neq 0$.
PROVE: $\det(\alpha I_U + \beta J_U) = \alpha^{n-1}(\alpha + n\beta)$.

PROOF SKETCH: Factor out the nonzero scalar and apply the matrix determinant lemma from F4 to a rank-one product.
⟨1⟩1. $\alpha I + \beta J = \alpha\big(I + (\frac{\beta}{\alpha}\mathbf{j}_U)\mathbf{j}_U^T\big)$.
PROOF: Every entry of $\mathbf{j}_U\mathbf{j}_U^T$ is one. Distribute α and cancel $\alpha \neq 0$ from ⟨0⟩:2. □
⟨1⟩2. **[new]** $\det\big(I + (\frac{\beta}{\alpha}\mathbf{j}_U)\mathbf{j}_U^T\big) = 1 + n\frac{\beta}{\alpha}$.
PROOF: Apply `Matrix.det_one_add_replicateCol_mul_replicateRow` from F4, with singleton index `Unit`, $\mathbf{u} = \frac{\beta}{\alpha}\mathbf{j}_U$, and $\mathbf{v} = \mathbf{j}_U$. The dot product sums n copies of β/α. □
⟨1⟩3. $\det(\alpha I + \beta J) = \alpha^{n-1}(\alpha + n\beta)$.
PROOF: By `Matrix.det_smul` in F4, steps ⟨1⟩1 and ⟨1⟩2 give $\alpha^n(1+n\beta/\alpha)$. Use $n = (n-1)+1$ from ⟨0⟩:1 and cancel $\alpha \neq 0$. □
⟨1⟩4. Q.E.D.
PROOF: Step ⟨1⟩3. □

Lemma (Connor identities and entries).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: Over $\mathbb{Q}$, $C^T = C$, $C^2 = 35C$, $CN^T = 0$, and

$$C(B, B') = \begin{cases} 10, & B = B', \\ 3, & B \sim B', \\ -4, & B \neq B' \text{ and } B \not\sim B'. \end{cases}$$

PROOF SKETCH: We will substitute the identities for G into $C = 35I - 7G + 10J$ and check its three entry types separately.
⟨1⟩1. $C^T = C$.
PROOF: I, J, G are symmetric by Lemma (incidence identities); transpose the defining linear combination in Definition (incidence and block matrices). □
⟨1⟩2. ASSUME: $B = B'$.
PROVE: $C(B, B') = 10$.
PROOF: Lemma (incidence identities) and Definition (labelled designs) give $G(B, B) = 5$, so Definition (incidence and block matrices) gives $35 - 7 \cdot 5 + 10 = 10$. □
⟨1⟩3. ASSUME: $B \sim B'$.
PROVE: $C(B, B') = 3$.
PROOF: Definition (the block graph and all six-cliques) gives distinct labels and intersection size one. Definition (incidence and block matrices) and Lemma (incidence identities) give $0 - 7 \cdot 1 + 10 = 3$. □
⟨1⟩4. ASSUME: $B \neq B'$ and $B \not\sim B'$.
PROVE: $C(B, B') = -4$.
PROOF: Lemma (block intersections) and Definition (the block graph and all six-cliques) force intersection size two. Definition (incidence and block matrices) gives $0 - 7 \cdot 2 + 10 = -4$. □
⟨1⟩5. $C^2 = 35C$.

PROOF: In the following calculation I, G, J are indexed by $\mathcal{B}$:

$$\begin{aligned} C^2 &= 1225I + 49G^2 + 100J^2 - 490G + 700J - 140GJ && \text{[(incidence identities), } GJ = JG\text{]} \\ &= 1225I + 49(5G + 50J) - 490G - 2100J && \text{[(incidence identities), arithmetic]} \\ &= 1225I - 245G + 350J && \text{[ring arithmetic]} \\ &= 35C && \text{[(incidence and block matrices)].} \end{aligned}$$

□

⟨1⟩6. $CN^T = 0$.

⟨2⟩1. $N^T J_P = 5J_{\mathcal{B},P}$ and $J_{\mathcal{B}} N^T = 7J_{\mathcal{B},P}$.

PROOF: Each entry in the first product is a column sum of N, and each in the second a row sum; apply Lemma (incidence identities) and F2. □

⟨2⟩2. Q.E.D.

PROOF: By associativity from F2,

$$\begin{aligned} CN^T &= 35N^T - 7N^T(NN^T) + 10J_{\mathcal{B}}N^T && \text{[(incidence and block matrices)]} \\ &= 35N^T - 7N^T(5I_P + 2J_P) + 70J_{\mathcal{B},P} && \text{[(incidence identities), ⟨2⟩1]} \\ &= 0 && \text{[⟨2⟩1, ring arithmetic].} \end{aligned}$$

□

⟨1⟩7. Q.E.D.

PROOF: Steps ⟨1⟩2, ⟨1⟩3, ⟨1⟩4 cover every pair of indices. Steps ⟨1⟩1, ⟨1⟩5, ⟨1⟩6 give the other conclusions. □

Lemma (quadratic inequality and rank bound).

ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.

PROVE: For every $\mathbf{x} \in \mathbb{Q}^{\mathcal{B}}$, $35\mathbf{x}^T C\mathbf{x} = (C\mathbf{x})^T(C\mathbf{x}) \geq 0$, so $\mathbf{x}^T C\mathbf{x} \geq 0$; also $\operatorname{rk} C \leq 6$.

PROOF SKETCH: The equation $C^2 = 35C$ gives a sum of squares. The equation $CN^T = 0$ gives the rank bound after proving N has rank fifteen.

⟨1⟩1. For every $\mathbf{x}$, $35\mathbf{x}^T C\mathbf{x} = (C\mathbf{x})^T(C\mathbf{x}) \geq 0$.

PROOF: Lemma (Connor identities and entries) gives $35C = C^T C$. Multiply by $\mathbf{x}^T, \mathbf{x}$, using F2; the resulting value is a sum of rational squares. □

⟨1⟩2. For every $\mathbf{x}$, $\mathbf{x}^T C\mathbf{x} \geq 0$.

PROOF: Divide ⟨1⟩1 by $35 > 0$. □

⟨1⟩3. $\det(NN^T) = 5^{14} \cdot 35 \neq 0$.

PROOF: First cast Lemma (incidence identities) to $\mathbb{Q}$ by F2. Apply Lemma (a scalar matrix plus a constant matrix). Definition (labelled designs) gives $|P| = 15$; take $\alpha = 5 \neq 0$ and $\beta = 2$. □

⟨1⟩4. $\operatorname{rk} N = \operatorname{rk} N^T = 15$.

⟨2⟩1. $\operatorname{rk}(NN^T) = 15$.

PROOF: The determinant in ⟨1⟩3 is a unit in $\mathbb{Q}$. The unit determinant and full-rank criteria of F3 give rank $|P| = 15$. □

⟨2⟩2. Q.E.D.

PROOF: Apply the two transpose-rank identities of F3 over $\mathbb{Q}$ to ⟨2⟩1. □

⟨1⟩5. $\operatorname{rk} C \leq 6$.

PROOF: By $CN^T = 0$ in Lemma (Connor identities and entries) and the zero-product inequality of F3, $\operatorname{rk} C + \operatorname{rk} N^T \leq |\mathcal{B}| = 21$. Substitute ⟨1⟩4 and subtract fifteen. □

⟨1⟩6. Q.E.D.

PROOF: Steps ⟨1⟩1, ⟨1⟩2, and ⟨1⟩5. □

Lemma (principal matrices on labelled subsets).

ASSUME: 1. $P, \mathcal{B}, U$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $e : U \to \mathcal{B}$ is injective, and $C_e(u, w) \triangleq C(e(u), e(w))$.
PROVE: For all $\mathbf{x} \in \mathbb{Q}^U$, $\mathbf{x}^T C_e \mathbf{x} \geq 0$; $\operatorname{rk} C_e \leq 6$; if $|U| \geq 7$, then $\det C_e = 0$.

PROOF SKETCH: We will extend vectors by zero along the injection. Rank monotonicity also applies to the specified submatrix, so later enumerations need no unstated transport of singularity.
⟨1⟩1. For $\mathbf{x} \in \mathbb{Q}^U$, define $\widehat{\mathbf{x}}(e(u)) = \mathbf{x}(u)$ and $\widehat{\mathbf{x}}(B) = 0$ off the image. This defines $\widehat{\mathbf{x}} \in \mathbb{Q}^{\mathcal{B}}$.
PROOF: Injectivity in ⟨0⟩:2 gives at most one preimage; the two domains partition $\mathcal{B}$. □
⟨1⟩2. $\widehat{\mathbf{x}}^T C \widehat{\mathbf{x}} = \mathbf{x}^T C_e \mathbf{x} \geq 0$.
PROOF: Expand the quadratic sum using F2. Terms off the image vanish by ⟨1⟩1; change both finite indices along e using F1. Apply Lemma (quadratic inequality and rank bound) to $\widehat{\mathbf{x}}$. □
⟨1⟩3. $\operatorname{rk} C_e \leq 6$.
PROOF: Rank monotonicity for submatrices in F3 bounds this rank by $\operatorname{rk} C$; use Lemma (quadratic inequality and rank bound). □
⟨1⟩4. ASSUME: $|U| \geq 7$.
PROVE: $\det C_e = 0$.
PROOF: If the determinant were nonzero, the full-rank criterion in F3 would give $\operatorname{rk} C_e = |U| \geq 7$, contradicting ⟨1⟩3. □
⟨1⟩5. Q.E.D.
PROOF: Steps ⟨1⟩2, ⟨1⟩3, and ⟨1⟩4. □

5.3.4. *Finite certificates and local restrictions.*

Definition (the C19 matrices). The following matrices have integer entries; their rational casts are denoted by the same letters. Indices in the displays run from 1 to the indicated order, corresponding to Lean's $\mathrm{Fin}\, n$ in increasing order.

$$Q_{\text{claw}} = \begin{pmatrix} 10 & 3 & 3 & 3 \\ 3 & 10 & -4 & -4 \\ 3 & -4 & 10 & -4 \\ 3 & -4 & -4 & 10 \end{pmatrix}$$

$$Q_{\text{five}} = \begin{pmatrix} 10 & 3 & 3 & 3 & -4 \\ 3 & 10 & 3 & -4 & 3 \\ 3 & 3 & 10 & -4 & 3 \\ 3 & -4 & -4 & 10 & 3 \\ -4 & 3 & 3 & 3 & 10 \end{pmatrix}$$

$$Q_1 = \begin{pmatrix} 10 & 3 & 3 & 3 & -4 & -4 & -4 \\ 3 & 10 & 3 & -4 & 3 & 3 & 3 \\ 3 & 3 & 10 & -4 & 3 & 3 & 3 \\ 3 & -4 & -4 & 10 & -4 & -4 & -4 \\ -4 & 3 & 3 & -4 & 10 & 3 & 3 \\ -4 & 3 & 3 & -4 & 3 & 10 & 3 \\ -4 & 3 & 3 & -4 & 3 & 3 & 10 \end{pmatrix}$$

$$Q_2 = \begin{pmatrix} 10 & 3 & 3 & 3 & 3 & -4 & -4 \\ 3 & 10 & 3 & -4 & -4 & 3 & 3 \\ 3 & 3 & 10 & -4 & -4 & 3 & 3 \\ 3 & -4 & -4 & 10 & 3 & -4 & -4 \\ 3 & -4 & -4 & 3 & 10 & -4 & -4 \\ -4 & 3 & 3 & -4 & -4 & 10 & 3 \\ -4 & 3 & 3 & -4 & -4 & 3 & 10 \end{pmatrix}$$

$$K_1 = \begin{pmatrix} 20 & -12 & -12 & -6 & 8 & 8 & 8 \\ -12 & 9 & 7 & 4 & -5 & -5 & -5 \\ -12 & 7 & 9 & 4 & -5 & -5 & -5 \\ -6 & 4 & 4 & 4 & -2 & -2 & -2 \\ 8 & -5 & -5 & -2 & 5 & 3 & 3 \\ 8 & -5 & -5 & -2 & 3 & 5 & 3 \\ 8 & -5 & -5 & -2 & 3 & 3 & 5 \end{pmatrix}$$

$$K_2 = \begin{pmatrix} 17 & -11 & -11 & -7 & -7 & 6 & 6 \\ -11 & 9 & 7 & 5 & 5 & -4 & -4 \\ -11 & 7 & 9 & 5 & 5 & -4 & -4 \\ -7 & 5 & 5 & 5 & 3 & -2 & -2 \\ -7 & 5 & 5 & 3 & 5 & -2 & -2 \\ 6 & -4 & -4 & -2 & -2 & 4 & 2 \\ 6 & -4 & -4 & -2 & -2 & 2 & 4 \end{pmatrix}$$

Computation (C19: quadratic values and inverse certificates).
LET: $\mathbf{x} \triangleq (-1,1,1,1)^T$ and $\mathbf{y} \triangleq (-1,1,1,1,-1)^T$.
PROVE: $\mathbf{x}^T Q_{\text{claw}} \mathbf{x} = -2$, $\mathbf{y}^T Q_{\text{five}} \mathbf{y} = -4$, and $K_j Q_j = 14 I_7$ for each $j \in \{1,2\}$.

PROOF SKETCH: These are finite integer certificates, suitable for kernel-checked arithmetic. The script `blueprint/checks/hallconnor/check_proof.py` reads every matrix from Definition (the C19 matrices) and recomputes every entry.

⟨1⟩1. $\mathbf{x}^T Q_{\text{claw}} \mathbf{x} = -2$.
PROOF: From Definition (the C19 matrices), the diagonal sum is 40 and the upper-triangle sum is $-9-12=-21$. The quadratic sum is $40+2(-21)=-2$, by F2. □

⟨1⟩2. $\mathbf{y}^T Q_{\text{five}} \mathbf{y} = -4$.
PROOF: From Definition (the C19 matrices), the diagonal sum is 50. The upper-triangle row contributions are $-13,-4,-7,-3$, totalling -27. Thus the value is $50+2(-27)=-4$, by F2. □

⟨1⟩3. $K_1 Q_1 = 14 I_7$.
PROOF: For all $u,w \in \{1,\dots,7\}$, substitute the integer entries in Definition (the C19 matrices) into $\sum_{i=1}^{7} K_1(u,i) Q_1(i,w)$, using F2. The seven diagonal sums are 14 and all 42 other sums are zero. For example the first row starts with $200-36-36-18-32-32-32=14$ and $60-120-36+24+24+24+24=0$. □

⟨1⟩4. $K_2 Q_2 = 14 I_7$.
PROOF: For all $u,w \in \{1,\dots,7\}$, the sums $\sum_{i=1}^{7} K_2(u,i) Q_2(i,w)$ from Definition (the C19 matrices) are 14 on the diagonal and zero off it, by integer multiplication and F2. The first two sums are $170-33-33-21-21-24-24=14$ and $51-110-33+28+28+18+18=0$. □

⟨1⟩5. Q.E.D.
PROOF: Steps ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, and ⟨1⟩4. □

Lemma (the two nonsingular matrices).
ASSUME: $j \in \{1,2\}$, with Q_j, K_j as in Definition (the C19 matrices).

PROVE: $\det Q_j \neq 0$ over $\mathbb{Q}$.

PROOF SKETCH: The certified product has nonzero determinant.
⟨1⟩1. **[new]** $(\det K_j)(\det Q_j) = 14^7$.
PROOF: Cast the integral identity of Computation (C19: quadratic values and inverse certificates) to $\mathbb{Q}$ by F2, then take determinants; multiplicativity and scalar multiplication in F4 give $\det(14I_7) = 14^7$. This replaces the ordinary proof's zero-kernel argument with the determinant of the same certificate. □
⟨1⟩2. Q.E.D.
PROOF: If $\det Q_j = 0$, the left side of ⟨1⟩1 vanishes, whereas $14^7 \neq 0$ in $\mathbb{Q}$. □

Lemma (no induced claw).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: There are no four distinct blocks B, X, Y, W for which $B \sim X$, $B \sim Y$, $B \sim W$, and X, Y, W are pairwise nonadjacent.

PROOF SKETCH: The four specified labels would give a negative quadratic value of a principal matrix of C.
⟨1⟩1. ASSUME: Four blocks B, X, Y, W satisfy the forbidden pattern.
PROVE: FALSE.
⟨2⟩1. The ordered list (B, X, Y, W) defines an injection $e : \mathrm{Fin}\, 4 \to \mathcal{B}$.
PROOF: The blocks are distinct by ⟨1⟩. □
⟨2⟩2. $C_e(i, i) = 10$ for each i.
PROOF: The diagonal clause of Lemma (Connor identities and entries). □
⟨2⟩3. All entries from the centre to a leaf are 3.
PROOF: The three adjacencies in ⟨1⟩ and the adjacency clause of Lemma (Connor identities and entries). □
⟨2⟩4. All entries between distinct leaves are -4.
PROOF: Their distinctness and nonadjacency in ⟨1⟩ give the last clause of Lemma (Connor identities and entries). □
⟨2⟩5. $C_e = Q_{\mathrm{claw}}$.
PROOF: The diagonal, centre-leaf, and leaf-leaf entries are exactly those of Definition (the C19 matrices), by ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. Symmetry in Lemma (Connor identities and entries) supplies the reversed entries. □
⟨2⟩6. Q.E.D.
PROOF: Apply Lemma (principal matrices on labelled subsets) with ⟨2⟩1 to the vector of Computation (C19: quadratic values and inverse certificates). Step ⟨2⟩5 makes its value $-2 \geq 0$, a contradiction. □
⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1 negates the forbidden existence. □

Lemma (the five-block restriction). [[HC], Lemma 4.5 at $k = 5$.]
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. B_1, B_2, B_3, Y, W are five distinct blocks; B_1, B_2, B_3 form a triangle.
3. $Y \sim B_1$, $Y \not\sim B_2, B_3$, and $W \sim B_2, B_3$, $W \not\sim B_1$.
PROVE: $Y \not\sim W$.

PROOF SKETCH: An edge YW gives the second negative quadratic certificate.
⟨1⟩1. ASSUME: $Y \sim W$.
PROVE: FALSE.

⟨2⟩1. The ordered list (B_1, B_2, B_3, Y, W) defines an injection $e : \mathrm{Fin}\,5 \to \mathcal{B}$.
PROOF: Distinctness is ⟨0⟩:2. □
⟨2⟩2. Every diagonal entry of C_e is 10.
PROOF: Lemma (Connor identities and entries). □
⟨2⟩3. The three entries between B_1, B_2, B_3 are 3.
PROOF: The triangle in ⟨0⟩:2 and Lemma (Connor identities and entries). □
⟨2⟩4. The entries from Y to (B_1, B_2, B_3) are $(3, -4, -4)$.
PROOF: Apply the adjacency and nonadjacency clauses of Lemma (Connor identities and entries) to the three stipulated relations in ⟨0⟩:3; their labels are distinct by ⟨0⟩:2. □
⟨2⟩5. The entries from W to (B_1, B_2, B_3) are $(-4, 3, 3)$.
PROOF: Apply Lemma (Connor identities and entries) to the three stipulated relations of W in ⟨0⟩:3, using ⟨0⟩:2. □
⟨2⟩6. $C(Y, W) = 3$.
PROOF: By ⟨1⟩ and Lemma (Connor identities and entries). □
⟨2⟩7. $C_e = Q_{\mathrm{five}}$.
PROOF: Steps ⟨2⟩2–⟨2⟩6 give every diagonal and upper-triangle entry in Definition (the C19 matrices); symmetry from Lemma (Connor identities and entries) supplies the lower triangle. □
⟨2⟩8. Q.E.D.
PROOF: Lemma (principal matrices on labelled subsets), ⟨2⟩1, ⟨2⟩7, and Computation (C19: quadratic values and inverse certificates) give $-4 \geq 0$, a contradiction. □
⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1 proves the negation. □

5.3.5. *The local six-cliques.*

Lemma (splitting a neighbourhood).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B_1, B_2 \in \mathcal{B}$ and $B_1 \sim B_2$.
PROVE: There are $S_1, S_2 \in \mathcal{S}$ such that

$$B_1, B_2 \in S_1, \quad S_1 \cap S_2 = \{B_1\}, \quad S_1 \cup S_2 = \{B_1\} \cup \Gamma(B_1).$$

Moreover every $K \in \mathcal{S}$ containing B_1, B_2 equals S_1.

PROOF SKETCH: We will partition the other blocks by adjacency to B_1, B_2. The claw restriction gives two cliques and an exact edge count. A suitable third vertex then has a defect $j \leq 2$. The two positive defects give the certified nonsingular seven-block matrices, in conflict with the rank bound. Defect zero gives the split and uniqueness.
LET: $\mathcal{M} \triangleq \Gamma(B_1) \cap \Gamma(B_2)$, $\mathcal{U} \triangleq \Gamma(B_1) \setminus (\Gamma(B_2) \cup \{B_2\})$, $\mathcal{V} \triangleq \Gamma(B_2) \setminus (\Gamma(B_1) \cup \{B_1\})$, and $\mathcal{R} \triangleq \mathcal{B} \setminus (\{B_1, B_2\} \cup \mathcal{M} \cup \mathcal{U} \cup \mathcal{V})$.
Write $\mathcal{F}_{ij} \triangleq \{B \notin \{B_1, B_2\} : |B_1 \cap B| = i,\ |B_2 \cap B| = j\}$ for the fibres whose cardinalities are x_{ij} in Lemma (common-neighbour counts).
⟨1⟩1. $\mathcal{M} = \mathcal{F}_{11}$.
PROOF: Definition (the block graph and all six-cliques) identifies each adjacency with a one-point intersection and excludes the two endpoints by irreflexivity. Conversely the fibre gives both adjacencies. Lemma (block intersections) places these intersections among the two possible sizes for distinct labels. □
⟨1⟩2. $\mathcal{U} = \mathcal{F}_{12}$.
PROOF: Membership in $\mathcal{U}$ gives adjacency to B_1 and excludes B_2 and adjacency to B_2. Definition (the block graph and all six-cliques) gives the first intersection size one; Lemma (block

intersections) forces the second to be two. Conversely these two sizes and the exclusions in $\mathcal{F}_{12}$ give precisely the defining membership conditions of $\mathcal{U}$. □

⟨1⟩3. $\mathcal{V} = \mathcal{F}_{21}$.

PROOF: Membership in $\mathcal{V}$ gives adjacency to B_2 and excludes B_1 and adjacency to B_1. Definition (the block graph and all six-cliques) gives the second size one; Lemma (block intersections) forces the first to be two. Conversely the two sizes and endpoint exclusions in $\mathcal{F}_{21}$ give precisely the defining conditions of $\mathcal{V}$. □

⟨1⟩4. $\mathcal{R} = \mathcal{F}_{22}$.

PROOF: Outside $\{B_1, B_2\}$, Lemma (block intersections) leaves four possible pairs of sizes. Steps ⟨1⟩1–⟨1⟩3 remove the first three, leaving $(2, 2)$. Conversely this pair gives no adjacency to either endpoint by Definition (the block graph and all six-cliques), so its label lies in none of $\mathcal{M}, \mathcal{U}, \mathcal{V}$ and belongs to $\mathcal{R}$. □

⟨1⟩5. $|\mathcal{M}| = 5$, $|\mathcal{U}| = 4$, $|\mathcal{V}| = 4$, and $|\mathcal{R}| = 6$.

PROOF: Hypothesis ⟨0⟩:2 and Definition (the block graph and all six-cliques) give $s = 1$. Apply the $s = 1$ row of Lemma (common-neighbour counts) to steps ⟨1⟩1–⟨1⟩4. □

⟨1⟩6. $\mathcal{M}, \mathcal{U}, \mathcal{V}, \mathcal{R}$ are pairwise disjoint. Each is disjoint from $\{B_1, B_2\}$.

PROOF: Their four different pairs of intersection sizes in steps ⟨1⟩1–⟨1⟩4 are mutually exclusive, and each fibre excludes the endpoints. □

⟨1⟩7. $\mathcal{M} \cup \mathcal{U} \cup \mathcal{V} \cup \mathcal{R} = \mathcal{B} \setminus \{B_1, B_2\}$.

PROOF: By Lemma (block intersections), every label outside the endpoints has one of the four pairs of sizes. Use steps ⟨1⟩1–⟨1⟩4. □

⟨1⟩8. $\Gamma(B_1) = \{B_2\} \cup \mathcal{M} \cup \mathcal{U}$, with pairwise disjoint parts.

PROOF: The endpoint B_2 is a neighbour by ⟨0⟩:2. Every other neighbour has first size one, so lies in $\mathcal{F}_{11}$ or $\mathcal{F}_{12}$ by Lemma (block intersections) and Definition (the block graph and all six-cliques); both fibres consist of neighbours. Apply ⟨1⟩1, ⟨1⟩2, and ⟨1⟩6. □

⟨1⟩9. $\Gamma(B_2) = \{B_1\} \cup \mathcal{M} \cup \mathcal{V}$, with pairwise disjoint parts.

PROOF: The endpoint B_1 is a neighbour by ⟨0⟩:2. Every other neighbour has second size one, so lies in $\mathcal{F}_{11}$ or $\mathcal{F}_{21}$ by Lemma (block intersections) and Definition (the block graph and all six-cliques); both fibres consist of neighbours. Apply ⟨1⟩1, ⟨1⟩3, and ⟨1⟩6. □

⟨1⟩10. $\mathcal{U}$ is a clique.

⟨2⟩1. ASSUME: $Y, Y' \in \mathcal{U}$, $Y \neq Y'$, and $Y \not\sim Y'$.
PROVE: FALSE.

PROOF: In ⟨1⟩2, B_2, Y, Y' are distinct neighbours of B_1. Neither Y nor Y' is adjacent to B_2; ⟨2⟩ excludes their mutual edge. This is the claw forbidden by Lemma (no induced claw). □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨2⟩1 excludes every missing edge between distinct members, as required by Definition (the block graph and all six-cliques). □

⟨1⟩11. $\mathcal{V}$ is a clique.

⟨2⟩1. ASSUME: $W, W' \in \mathcal{V}$, $W \neq W'$, and $W \not\sim W'$.
PROVE: FALSE.

PROOF: Step ⟨1⟩3 makes B_1, W, W' distinct neighbours of B_2; B_1 is adjacent to neither W nor W'. With ⟨2⟩ these are the three leaves of a claw, contradicting Lemma (no induced claw). □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨2⟩1 proves the clique predicate of Definition (the block graph and all six-cliques). □

⟨1⟩12. For every $Y \in \mathcal{U}$, $|\Gamma(Y) \cap \mathcal{M}| = 2$.

⟨2⟩1. For each $Y \in \mathcal{U}$, the common neighbours of B_1, Y are the disjoint union of $\mathcal{U} \setminus \{Y\}$ and $\Gamma(Y) \cap \mathcal{M}$.

PROOF: In the first neighbourhood decomposition of ⟨1⟩8, $B_2 \not\sim Y$ by membership in $\mathcal{U}$; step ⟨1⟩10 gives every member of $\mathcal{U} \setminus \{Y\}$. These leave precisely the stated part of $\mathcal{M}$. □

⟨2⟩2. Q.E.D.

PROOF: Lemma (common-neighbour counts) gives five common neighbours of the adjacent pair B_1, Y. The first part in ⟨2⟩1 has size $4 - 1 = 3$ by ⟨1⟩5 and F1, leaving two. □

⟨1⟩13. $\sum_{X \in \mathcal{M}} |\Gamma(X) \cap \mathcal{U}| = \sum_{Y \in \mathcal{U}} |\Gamma(Y) \cap \mathcal{M}|$.

PROOF: Write each cardinality as a sum of adjacency indicators. Interchange the finite sums by `Finset.sum_comm` in F1; symmetry of adjacency from Definition (the block graph and all six-cliques) gives the other sum. □

⟨1⟩14. $\sum_{X \in \mathcal{M}} |\Gamma(X) \cap \mathcal{U}| = 8$.

PROOF: By ⟨1⟩13 the sum equals $\sum_{Y \in \mathcal{U}} 2$, using ⟨1⟩12. Step ⟨1⟩5 gives $4 \cdot 2 = 8$. □

⟨1⟩15. PICK $B_3 \in \mathcal{M}$ such that $|\Gamma(B_3) \cap \mathcal{U}| \geq 2$.

PROOF: Otherwise each summand of ⟨1⟩14 is at most one. Termwise summation gives $8 \leq \sum_{X \in \mathcal{M}} 1 = |\mathcal{M}| = 5$, using ⟨1⟩5 and F1, a contradiction. □

⟨1⟩16. LET: $j \triangleq |\mathcal{U} \setminus \Gamma(B_3)|$, and

$$h \triangleq |\Gamma(B_3) \cap (\mathcal{M} \setminus \{B_3\})|, \quad e \triangleq |\Gamma(B_3) \cap \mathcal{V}|, \quad \ell \triangleq |\Gamma(B_3) \cap \mathcal{R}|.$$

Then $0 \leq j \leq 2$ and $|\Gamma(B_3) \cap \mathcal{U}| = 4 - j$.

PROOF: The neighbours and nonneighbours of B_3 partition the four-element set $\mathcal{U}$, by ⟨1⟩5 and F1; ⟨1⟩15 supplies at least two neighbours. □

⟨1⟩17. $h = j$, $e = 4 - j$, and $\ell = j$.

⟨2⟩1. $1 + h + (4 - j) = 5$.

PROOF: Count common neighbours of B_1, B_3 using Lemma (common-neighbour counts) and ⟨1⟩8: B_2 contributes one, $\mathcal{M} \setminus \{B_3\}$ contributes h, and $\mathcal{U}$ contributes $4 - j$ by ⟨1⟩16. □

⟨2⟩2. $1 + h + e = 5$.

PROOF: Count common neighbours of B_2, B_3 using Lemma (common-neighbour counts): B_1 contributes one, the part $\mathcal{M} \setminus \{B_3\}$ contributes h, and $\mathcal{V}$ contributes e, by ⟨1⟩9 and ⟨1⟩16. □

⟨2⟩3. $2 + h + (4 - j) + e + \ell = 10$.

PROOF: Count all neighbours of B_3 in the partition ⟨1⟩7, whose parts are disjoint by ⟨1⟩6. Both B_1, B_2 are neighbours by ⟨1⟩15; the other contributions are the four counts in ⟨1⟩16. The degree is ten by Lemma (block intersections). □

⟨2⟩4. Q.E.D.

PROOF: Solve ⟨2⟩1 for h, then ⟨2⟩2 for e, then ⟨2⟩3 for ℓ, over $\mathbb{Z}$. □

⟨1⟩18. LET: $\mathcal{U}_0 \triangleq \mathcal{U} \setminus \Gamma(B_3)$ and $\mathcal{V}_1 \triangleq \mathcal{V} \cap \Gamma(B_3)$.

They are disjoint cliques of sizes $j, 4 - j$, disjoint also from $\{B_1, B_2, B_3\}$.

PROOF: Disjointness is ⟨1⟩6 and ⟨1⟩15; the sizes are ⟨1⟩16 and ⟨1⟩17. They are subsets of the cliques in ⟨1⟩10 and ⟨1⟩11. □

⟨1⟩19. No member of $\mathcal{U}_0$ is adjacent to a member of $\mathcal{V}_1$.

⟨2⟩1. ASSUME: $Y \in \mathcal{U}_0$ and $W \in \mathcal{V}_1$.

PROVE: $Y \not\sim W$.

⟨3⟩1. B_1, B_2, B_3, Y, W are distinct.

PROOF: Step ⟨1⟩18 gives disjointness, and ⟨1⟩6, ⟨1⟩15 separate B_3 from the adjacent pair B_1, B_2. □

⟨3⟩2. B_1, B_2, B_3 form a triangle.

PROOF: Use ⟨0⟩:2 and $B_3 \in \mathcal{M}$ in ⟨1⟩15. □

⟨3⟩3. $Y \sim B_1$, $Y \not\sim B_2$, and $Y \not\sim B_3$.

PROOF: Membership in $\mathcal{U}_0$ in ⟨2⟩, and the definition in ⟨1⟩18 and the fibre description in ⟨1⟩2. □

⟨3⟩4. $W \sim B_2$, $W \sim B_3$, and $W \not\sim B_1$.

PROOF: Membership in $\mathcal{V}_1$ in ⟨2⟩, and the definition in ⟨1⟩18 and the fibre description in ⟨1⟩3. □

⟨3⟩5. Q.E.D.

PROOF: Apply Lemma (the five-block restriction) to ⟨3⟩1–⟨3⟩4. □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨2⟩1 is quantified over every pair of the indicated parts. □

⟨1⟩20. On the named blocks B_1, B_2, B_3 and the members of $\mathcal{U}_0, \mathcal{V}_1$, C is symmetric with diagonal 10. Its off-diagonal entries are

	B_1	B_2	B_3	$\mathcal{U}_0$	$\mathcal{V}_1$
B_1	$*$	3	3	3	-4
B_2	3	$*$	3	-4	3
B_3	3	3	$*$	-4	3
$\mathcal{U}_0$	3	-4	-4	3	-4
$\mathcal{V}_1$	-4	3	3	-4	3

A set in this table denotes any of its members, with distinct blocks required for within-set entries.

⟨2⟩1. For every block X, $C(X, X) = 10$.

PROOF: Lemma (Connor identities and entries). □

⟨2⟩2. The three entries among B_1, B_2, B_3 are 3.

PROOF: The triangle follows from ⟨0⟩:2, ⟨1⟩1, and ⟨1⟩15; use Lemma (Connor identities and entries). □

⟨2⟩3. For $Y \in \mathcal{U}_0$, $C(B_1, Y) = 3$.

PROOF: Membership in $\mathcal{U}$ gives adjacency to B_1 by ⟨1⟩2 and ⟨1⟩18; apply Lemma (Connor identities and entries). □

⟨2⟩4. For $Y \in \mathcal{U}_0$, $C(B_2, Y) = -4$.

PROOF: Step ⟨1⟩2 excludes adjacency to B_2; distinctness in ⟨1⟩18 permits Lemma (Connor identities and entries). □

⟨2⟩5. For $Y \in \mathcal{U}_0$, $C(B_3, Y) = -4$.

PROOF: The definition and disjointness in ⟨1⟩18 give nonadjacency and distinctness; apply Lemma (Connor identities and entries). □

⟨2⟩6. For $W \in \mathcal{V}_1$, $C(B_1, W) = -4$.

PROOF: Step ⟨1⟩3 excludes adjacency to B_1; use the distinctness in ⟨1⟩18 and Lemma (Connor identities and entries). □

⟨2⟩7. For $W \in \mathcal{V}_1$, $C(B_2, W) = 3$.

PROOF: Step ⟨1⟩3 gives adjacency to B_2 for every member of the subset in ⟨1⟩18; use Lemma (Connor identities and entries). □

⟨2⟩8. For $W \in \mathcal{V}_1$, $C(B_3, W) = 3$.

PROOF: The definition of $\mathcal{V}_1$ in ⟨1⟩18 gives adjacency; use Lemma (Connor identities and entries). □

⟨2⟩9. For distinct $Y, Y' \in \mathcal{U}_0$, $C(Y, Y') = 3$.

PROOF: The clique in ⟨1⟩18 gives adjacency, so Lemma (Connor identities and entries) applies. □

⟨2⟩10. For distinct $W, W' \in \mathcal{V}_1$, $C(W, W') = 3$.

PROOF: The clique in ⟨1⟩18 gives adjacency; use Lemma (Connor identities and entries). □

⟨2⟩11. For $Y \in \mathcal{U}_0$ and $W \in \mathcal{V}_1$, $C(Y, W) = -4$.

PROOF: Step ⟨1⟩19 excludes the edge; ⟨1⟩18 gives distinctness, so Lemma (Connor identities and entries) applies. □

⟨2⟩12. Q.E.D.

PROOF: Steps ⟨2⟩1–⟨2⟩11 cover the diagonal and all ten unordered pair types. Symmetry from Lemma (Connor identities and entries) gives the reversed entries. □

⟨1⟩21. ASSUME: $j = 1$.

PROVE: FALSE.

⟨2⟩1. PICK Y, W_1, W_2, W_3 with $\mathcal{U}_0 = \{Y\}$, $\mathcal{V}_1 = \{W_1, W_2, W_3\}$, and the W_i pairwise distinct.

PROOF: By ⟨1⟩ and ⟨1⟩18, the sizes are one and three. Apply `Finset.card_eq_one` and `Finset.card_eq_three` from F1. □

⟨2⟩2. Put $a \triangleq (B_1, B_2, B_3, Y, W_1, W_2, W_3) : \mathrm{Fin}\, 7 \to \mathcal{B}$. Then a is injective.

PROOF: The first three blocks are distinct by ⟨0⟩:2, ⟨1⟩6, and ⟨1⟩15. Step ⟨1⟩18 separates them from the two disjoint parts, and ⟨2⟩1 separates the W_i. Thus the literal Lean vector `![B1, B2, B3, Y, W1, W2, W3]` has seven distinct entries. □

⟨2⟩3. $C_a = Q_1$.

PROOF: Use the fixed list in ⟨2⟩2. In zero-based indices, the named blocks occupy $0, 1, 2$, Y occupies 3, and W_1, W_2, W_3 occupy $4, 5, 6$ by ⟨2⟩1. For each of the 49 pairs, the corresponding entry of ⟨1⟩20 equals the displayed entry of Q_1 in Definition (the C19 matrices): the diagonal is 10, triangle and within-W entries are 3, the three entries from Y to B_1, B_2, B_3 are $(3, -4, -4)$, each W_i has $(-4, 3, 3)$ to those blocks, and all Y, W_i entries are -4. These cases and their reverses exhaust the entries. □

⟨2⟩4. $\det C_a = 0$.

PROOF: Apply Lemma (principal matrices on labelled subsets) to ⟨2⟩2, whose domain has size seven. □

⟨2⟩5. Q.E.D.

PROOF: Steps ⟨2⟩3 and ⟨2⟩4 contradict Lemma (the two nonsingular matrices) at $j = 1$. □

⟨1⟩22. ASSUME: $j = 2$.

PROVE: FALSE.

⟨2⟩1. PICK Y_1, Y_2, W_1, W_2 with $\mathcal{U}_0 = \{Y_1, Y_2\}$, $\mathcal{V}_1 = \{W_1, W_2\}$, $Y_1 \neq Y_2$, and $W_1 \neq W_2$.

PROOF: The two sizes in ⟨1⟩18 are both two by ⟨1⟩. Apply `Finset.card_eq_two` from F1 to each set. □

⟨2⟩2. Put $a \triangleq (B_1, B_2, B_3, Y_1, Y_2, W_1, W_2) : \mathrm{Fin}\, 7 \to \mathcal{B}$. Then a is injective.

PROOF: The initial blocks are distinct by ⟨0⟩:2, ⟨1⟩6, and ⟨1⟩15. The disjointness in ⟨1⟩18 and the two inequalities in ⟨2⟩1 separate all remaining pairs. This is the literal Lean vector `![B1, B2, B3, Y1, Y2, W1, W2]`. □

⟨2⟩3. $C_a = Q_2$.

PROOF: The list in ⟨2⟩2 puts B_1, B_2, B_3 at indices $0, 1, 2$, Y_1, Y_2 at $3, 4$, and W_1, W_2 at $5, 6$, using ⟨2⟩1. For each of the 49 pairs, ⟨1⟩20 matches Q_2 in Definition (the C19 matrices): the diagonal is 10, the triangle, Y_1, Y_2, and W_1, W_2 entries are 3, each Y_i has $(3, -4, -4)$ to the triangle, each W_i has $(-4, 3, 3)$, and the four Y_i, W_h entries are -4. The reverse entries agree by symmetry. □

⟨2⟩4. $\det C_a = 0$.

PROOF: Lemma (principal matrices on labelled subsets) applies to ⟨2⟩2 with domain size seven. □

⟨2⟩5. Q.E.D.

PROOF: Steps ⟨2⟩3 and ⟨2⟩4 contradict Lemma (the two nonsingular matrices) at $j = 2$. □

⟨1⟩23. $j = 0$; B_3 is adjacent to every member of $\mathcal{U}$ and to no member of $\mathcal{M} \setminus \{B_3\}$.

PROOF: The integer bound $0 \le j \le 2$ of ⟨1⟩16 and the exclusions ⟨1⟩21, ⟨1⟩22 force zero. The counts in ⟨1⟩16, ⟨1⟩17 are then four and zero; use F1. □

⟨1⟩24. $\mathcal{M} \setminus \{B_3\}$ is a clique of size four.

⟨2⟩1. $|\mathcal{M} \setminus \{B_3\}| = 4$.

PROOF: Use $|\mathcal{M}| = 5$, $B_3 \in \mathcal{M}$ from ⟨1⟩5, ⟨1⟩15, and deletion of a singleton in F1. □

⟨2⟩2. ASSUME: $X, Y \in \mathcal{M} \setminus \{B_3\}$ are distinct and nonadjacent.
PROVE: FALSE.

PROOF: The three distinct vertices B_3, X, Y are neighbours of B_1 by ⟨1⟩1 and ⟨1⟩15; B_3 has no edges to X, Y by ⟨1⟩23. With ⟨2⟩ they give a claw, contradicting Lemma (no induced claw). □

⟨2⟩3. Q.E.D.

PROOF: Step ⟨2⟩1 gives the size and ⟨2⟩2 gives every required edge. □

⟨1⟩25. Put $S_1 \triangleq \{B_1, B_2\} \cup (\mathcal{M} \setminus \{B_3\})$. Then $S_1 \in \mathcal{S}$ and $B_1, B_2 \in S_1$.

⟨2⟩1. $|S_1| = 6$.

PROOF: The union is disjoint by ⟨1⟩6, and ⟨1⟩24 gives $2 + 4 = 6$. □

⟨2⟩2. B_1, B_2 are adjacent.

PROOF: Hypothesis ⟨0⟩:2. □

⟨2⟩3. Every member of $\mathcal{M} \setminus \{B_3\}$ is adjacent to each of B_1, B_2.

PROOF: The definition of $\mathcal{M}$ in ⟨1⟩1. □

⟨2⟩4. Distinct members of $\mathcal{M} \setminus \{B_3\}$ are adjacent.

PROOF: Step ⟨1⟩24. □

⟨2⟩5. Q.E.D.

PROOF: The three pair types in ⟨2⟩2–⟨2⟩4 exhaust pairs in S_1. With ⟨2⟩1 this is membership in $\mathcal{S}$, by Definition (the block graph and all six-cliques); the two containments follow from the displayed union. □

⟨1⟩26. Put $S_2 \triangleq \{B_1, B_3\} \cup \mathcal{U}$. Then $S_2 \in \mathcal{S}$.

⟨2⟩1. $|S_2| = 6$.

PROOF: The two-element set is disjoint from $\mathcal{U}$ by ⟨1⟩6 and ⟨1⟩15; its union has size $2+4 = 6$ by ⟨1⟩5. □

⟨2⟩2. $B_1 \sim B_3$.

PROOF: Step ⟨1⟩15 puts $B_3 \in \mathcal{M}$. □

⟨2⟩3. Every member of $\mathcal{U}$ is adjacent to B_1.

PROOF: Step ⟨1⟩2. □

⟨2⟩4. Every member of $\mathcal{U}$ is adjacent to B_3.

PROOF: Step ⟨1⟩23. □

⟨2⟩5. Every pair of distinct members of $\mathcal{U}$ is adjacent.

PROOF: Step ⟨1⟩10. □

⟨2⟩6. Q.E.D.

PROOF: Steps ⟨2⟩2–⟨2⟩5 cover all pair types. Together with ⟨2⟩1, they are the predicate of Definition (the block graph and all six-cliques). □

⟨1⟩27. $S_1 \cap S_2 = \{B_1\}$.

PROOF: In ⟨1⟩25 and ⟨1⟩26, the parts $\{B_2\}$, $\mathcal{M} \setminus \{B_3\}$, $\{B_3\}$, and $\mathcal{U}$ are pairwise disjoint by ⟨1⟩6 and ⟨1⟩15; only B_1 is shared. □

⟨1⟩28. $S_1 \cup S_2 = \{B_1\} \cup \Gamma(B_1)$.

PROOF: The two unions in ⟨1⟩25, ⟨1⟩26 combine to $\{B_1, B_2\} \cup \mathcal{M} \cup \mathcal{U}$, since $B_3 \in \mathcal{M}$. Use the first neighbourhood decomposition of ⟨1⟩8. □

⟨1⟩29. ASSUME: $K \in \mathcal{S}$ and $B_1, B_2 \in K$.
PROVE: $K = S_1$.

⟨2⟩1. $K \setminus \{B_1, B_2\} \subseteq \mathcal{M}$ and $|K \setminus \{B_1, B_2\}| = 4$.
PROOF: Every remaining vertex is adjacent to both initial blocks by Definition (the block graph and all six-cliques). The two distinct blocks are in the six-element set K, so F1 gives size four. □
⟨2⟩2. $B_3 \notin K$.
⟨3⟩1. ASSUME: $B_3 \in K$.
PROVE: FALSE.
PROOF: By ⟨2⟩1, removing B_3 leaves three vertices in $\mathcal{M} \setminus \{B_3\}$, thus at least one, by F1. The clique K makes it adjacent to B_3, contradicting ⟨1⟩23. □
⟨3⟩2. Q.E.D.
PROOF: Step ⟨3⟩1 proves the negation. □
⟨2⟩3. $K \setminus \{B_1, B_2\} = \mathcal{M} \setminus \{B_3\}$.
PROOF: Steps ⟨2⟩1, ⟨2⟩2 give inclusion; both sets have size four by ⟨1⟩24. Apply the subset and cardinality criterion in F1. □
⟨2⟩4. Q.E.D.
PROOF: Reinsert the two members B_1, B_2 into ⟨2⟩3; the resulting union is S_1 by ⟨1⟩25. □
⟨1⟩30. Q.E.D.
PROOF: Steps ⟨1⟩25, ⟨1⟩26, ⟨1⟩27, ⟨1⟩28, and ⟨1⟩29 give the stated witnesses and uniqueness. □

Corollary (the unique six-clique through an edge).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B_1 \sim B_2$.
PROVE: There is a unique $S \in \mathcal{S}$ containing B_1, B_2. In particular, if $K, L \in \mathcal{S}$ both contain B_1, B_2, then $K = L$.

PROOF SKETCH: Choose one local witness and compare every other six-clique through the edge with that same witness.
⟨1⟩1. PICK S_1, S_2 with the conclusions of Lemma (splitting a neighbourhood) for B_1, B_2.
PROOF: Apply that lemma to ⟨0⟩:2. □
⟨1⟩2. There is a unique $S \in \mathcal{S}$ containing B_1, B_2.
PROOF: Step ⟨1⟩1 gives the witness S_1 and states that every such S equals S_1. □
⟨1⟩3. ASSUME: $K, L \in \mathcal{S}$ and $B_1, B_2 \in K \cap L$.
PROVE: $K = L$.
PROOF: The uniqueness clause of ⟨1⟩1 gives $K = S_1$ and $L = S_1$. Transitivity gives $K = L$. □
⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩2 and ⟨1⟩3. □

Corollary (exactly two six-cliques through a block).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B_1 \in \mathcal{B}$.
PROVE: There are distinct $S_1, S_2 \in \mathcal{S}$ with $\{S \in \mathcal{S} : B_1 \in S\} = \{S_1, S_2\}$ and $S_1 \cap S_2 = \{B_1\}$. In particular the filtered family has size two.

PROOF SKETCH: We will use one local split, then apply the stated adjacent-pair uniqueness lemma to each possible six-clique.
⟨1⟩1. PICK $B_2 \in \Gamma(B_1)$.
PROOF: Lemma (block intersections) gives ten neighbours; positive cardinality in F1 gives a witness. □
⟨1⟩2. PICK S_1, S_2 with all the conclusions of Lemma (splitting a neighbourhood) for B_1, B_2.

PROOF: Apply that lemma using ⟨1⟩1. □

⟨1⟩3. $S_1 \neq S_2$, and both contain B_1.

PROOF: Their intersection is $\{B_1\}$ by ⟨1⟩2. Equality would give a singleton of size six, contrary to Definition (the block graph and all six-cliques). □

⟨1⟩4. ASSUME: $K \in \mathcal{S}$ and $B_1 \in K$.

PROVE: $K = S_1$ or $K = S_2$.

⟨2⟩1. PICK $X \in K \setminus \{B_1\}$.

PROOF: This set has size five by ⟨1⟩, Definition (the block graph and all six-cliques), and F1. □

⟨2⟩2. $B_1 \sim X$ and $X \in (S_1 \setminus \{B_1\}) \cup (S_2 \setminus \{B_1\})$.

PROOF: The clique K in ⟨1⟩ and ⟨2⟩1 give adjacency. The covering conclusion of ⟨1⟩2 gives the indicated membership. □

⟨2⟩3. CASE: $X \in S_1$.

$K = S_1$.

PROOF: Both six-cliques contain the adjacent pair B_1, X by ⟨1⟩, ⟨1⟩3, and ⟨2⟩2. Apply the uniqueness conclusion of Corollary (the unique six-clique through an edge) to this pair. □

⟨2⟩4. CASE: $X \in S_2$.

$K = S_2$.

PROOF: The cliques K, S_2 contain B_1, X by ⟨1⟩, ⟨1⟩3, and ⟨2⟩2. Corollary (the unique six-clique through an edge) applied to B_1, X identifies them. □

⟨2⟩5. Q.E.D.

PROOF: Step ⟨2⟩2 makes ⟨2⟩3, ⟨2⟩4 exhaustive. □

⟨1⟩5. Q.E.D.

PROOF: Step ⟨1⟩4 gives containment of the filtered family in $\{S_1, S_2\}$; ⟨1⟩3 gives the reverse containment and size two. The intersection is part of ⟨1⟩2. □

5.3.6. *The global family.*

Lemma (seven six-cliques).

ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.

2. D is a design with parameters $(15, 5, 21, 7, 2)$.

PROVE: $|\mathcal{S}| = 7$.

PROOF SKETCH: Count incidences between block labels and six-cliques.

⟨1⟩1. There are $6|\mathcal{S}|$ pairs (B, S) with $B \in S \in \mathcal{S}$.

PROOF: Every member of $\mathcal{S}$ has size six by Definition (the block graph and all six-cliques); apply F1. □

⟨1⟩2. There are 42 such pairs.

PROOF: Each of the 21 block labels belongs to two members of $\mathcal{S}$ by Corollary (exactly two six-cliques through a block) and Definition (labelled designs); use F1. □

⟨1⟩3. Q.E.D.

PROOF: Equate ⟨1⟩1 and ⟨1⟩2, then divide $6|\mathcal{S}| = 42$ by six. □

Lemma (intersections of six-cliques).

ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.

2. $S_0, S_1 \in \mathcal{S}$ and $S_0 \neq S_1$.

PROVE: $|S_0 \cap S_1| = 1$.

PROOF SKETCH: Any two distinct cliques share at most one block. The other clique through each of the six blocks of S_0 then enumerates all six cliques different from S_0.

⟨1⟩1. For all distinct $S, T \in \mathcal{S}$, $|S \cap T| \leq 1$.
  ⟨2⟩1. ASSUME: $S, T \in \mathcal{S}$, $S \neq T$, and distinct $B, B' \in S \cap T$.
    PROVE: FALSE.
    PROOF: $B \sim B'$ by the clique predicate in Definition (the block graph and all six-cliques). Corollary (the unique six-clique through an edge) identifies S, T through this edge, contradicting $S \neq T$. □
  ⟨2⟩2. Q.E.D.
    PROOF: The cardinality-at-most-one criterion of F1 applies to ⟨2⟩1. □
⟨1⟩2. For every $B \in S_0$ there is a unique $f(B) \in \mathcal{S} \setminus \{S_0\}$ containing B.
  PROOF: Corollary (exactly two six-cliques through a block) supplies exactly two cliques through B, one of which is S_0 by ⟨0⟩:2. Deleting that one leaves a singleton by F1. □
⟨1⟩3. $f : S_0 \to \mathcal{S} \setminus \{S_0\}$ is injective.
  PROOF: If $f(B) = f(B')$ with $B \neq B'$, then S_0 and $f(B) \neq S_0$ share two blocks by ⟨1⟩2, contradicting ⟨1⟩1. □
⟨1⟩4. f is surjective.
  PROOF: Its domain has size six by Definition (the block graph and all six-cliques); its codomain has size $7 - 1 = 6$ by Lemma (seven six-cliques) and ⟨0⟩:2. The finite bijectivity criterion of F1 applies to ⟨1⟩3. □
⟨1⟩5. $S_0 \cap S_1$ is nonempty.
  PROOF: By ⟨0⟩:2 and ⟨1⟩4, choose $B \in S_0$ with $f(B) = S_1$. Step ⟨1⟩2 gives $B \in S_1$. □
⟨1⟩6. Q.E.D.
  PROOF: Combine the lower bound from ⟨1⟩5 with ⟨1⟩1, using F1. □

Lemma (points in a six-clique).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $S_0 \in \mathcal{S}$.
PROVE: For every $p \in P$, $|\{B \in S_0 : p \in B\}| = 2$.

PROOF SKETCH: Two moments of the point multiplicities give a sum of squares equal to zero.
⟨1⟩1. LET: $m(p) \triangleq |\{B \in S_0 : p \in B\}|$.
  Then $\sum_{p \in P} m(p) = 30$.
  PROOF: Count (p, B) with $B \in S_0$ and $p \in B$. There are six labels, each with five points, by Definitions (the block graph and all six-cliques), (labelled designs), and F1. □
⟨1⟩2. $\sum_{p \in P} m(p)(m(p) - 1) = 30$ in $\mathbb{Z}$.
  PROOF: Count (p, B, B') with distinct $B, B' \in S_0$ and $p \in B \cap B'$. Each of the $6 \cdot 5$ ordered pairs meets in exactly one point by Definition (the block graph and all six-cliques). Counting by p gives the sum, using F1. □
⟨1⟩3. $\sum_{p \in P} (m(p) - 2)^2 = 0$ in $\mathbb{Z}$.
  PROOF: Integer expansion gives

$$\begin{aligned} \sum_p (m(p) - 2)^2 &= \sum_p m(p)(m(p) - 1) - 3 \sum_p m(p) + 4|P| && \text{[ring arithmetic]} \\ &= 30 - 90 + 60 && \text{[⟨1⟩1, ⟨1⟩2, (labelled designs)]} \\ &= 0 && \text{[integer arithmetic].} \end{aligned}$$

  □
⟨1⟩4. For every $p \in P$, $(m(p) - 2)^2 = 0$.
  PROOF: Each integer square is nonnegative; apply the zero-sum criterion of F1 to ⟨1⟩3. □
⟨1⟩5. Q.E.D.

PROOF: An integer square vanishes only at zero, so ⟨1⟩4 gives $m(p) - 2 = 0$ and $m(p) = 2$ for every p. □

5.3.7. *The labelled extension and the matrix it supplies.*

Definition (clique incidence and extended blocks).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
LET: $T \in \mathrm{Mat}_{\mathcal{B}\times\mathcal{S}}(\mathbb{Z})$ have $T(B, S) = 1$ for $B \in S$ and 0 otherwise. Put $P^* \triangleq P \sqcup \mathcal{S}$ and $\mathcal{B}^* \triangleq \mathrm{Option}(\mathcal{B})$.
Denote their injections by $\iota : P \to P^*$ and $\eta : \mathcal{S} \to P^*$, the old labels by $\mathrm{some}(B)$, and the new label by none. Define the block function $F : \mathcal{B}^* \to \mathrm{Finset}(P^*)$ by

$$F(\mathrm{some}(B)) \triangleq \iota(B) \cup \{\eta(S) : S \in \mathcal{S},\ B \in S\},$$
$$F(\mathrm{none}) \triangleq \{\eta(S) : S \in \mathcal{S}\}.$$

Here $\iota(B)$ is the image of the point set of B. Define $N^* \triangleq D.\mathrm{hallConnorIncMatrix} \in \mathrm{Mat}_{P^*\times\mathcal{B}^*}(\mathbb{Z})$ by the four entry formulas

$$N^*(\iota(p), \mathrm{some}(B)) \triangleq N(p, B), \quad N^*(\iota(p), \mathrm{none}) \triangleq 0,$$
$$N^*(\eta(S), \mathrm{some}(B)) \triangleq T(B, S), \quad N^*(\eta(S), \mathrm{none}) \triangleq 1.$$

Lemma (parameters of the extended blocks).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: For the types and block function of Definition (clique incidence and extended blocks),

$$|P^*| = 22, \qquad |\mathcal{B}^*| = 22, \qquad \forall A \in \mathcal{B}^*,\ |F(A)| = 7,$$
$$\forall x \in P^*,\ |\{A : x \in F(A)\}| = 7,$$
$$\forall x, y \in P^*,\ x \neq y \Rightarrow |\{A : x, y \in F(A)\}| = 2.$$

PROOF SKETCH: We will verify every cardinality and each of the three unordered types of point pair. The disjoint sum distinguishes new points from old points even if their underlying representations agree.
⟨1⟩1. $|P^*| = 22$ and $|\mathcal{B}^*| = 22$.
PROOF: Definition (clique incidence and extended blocks) gives the sums $|P| + |\mathcal{S}| = 15 + 7$ and $|\mathcal{B}| + 1 = 21 + 1$, using Definition (labelled designs), Lemma (seven six-cliques), and `Fintype.card_sum` and `Fintype.card_option` in F1. □
⟨1⟩2. For every $B \in \mathcal{B}$, $|F(\mathrm{some}(B))| = 7$.
PROOF: There are five old points by Definition (labelled designs) and two new points by Corollary (exactly two six-cliques through a block). Both ι and η are injective, and their images are disjoint by Definition (clique incidence and extended blocks). Apply `Finset.card_image_of_injective` and the disjoint-union count in F1 to get $5 + 2 = 7$. □
⟨1⟩3. $|F(\mathrm{none})| = 7$.
PROOF: The map η is injective, and Lemma (seven six-cliques) gives seven members of $\mathcal{S}$; use Definition (clique incidence and extended blocks) and `Finset.card_image_of_injective` in F1. □
⟨1⟩4. Each old point $\iota(p)$ lies in seven extended blocks.
PROOF: Definition (clique incidence and extended blocks) preserves all incidences of p with old labels and puts no old point in $F(\mathrm{none})$. Its old replication number is seven by Definition (labelled designs). □
⟨1⟩5. Each new point $\eta(S)$ lies in seven extended blocks.

PROOF: Definition (clique incidence and extended blocks) puts it in exactly the old labels $B \in S$, of which there are six by Definition (the block graph and all six-cliques), and also in the distinct new label none. The count is $6 + 1 = 7$. □

⟨1⟩6. ASSUME: $p, q \in P$ and $p \neq q$.
PROVE: $\iota(p), \iota(q)$ lie together in exactly two extended blocks.
⟨2⟩1. The common old labels are precisely $\{\mathrm{some}(B) : p, q \in B\}$.
PROOF: The old-incidence clause of Definition (clique incidence and extended blocks). □
⟨2⟩2. The new block contains neither old point.
PROOF: The two injections have disjoint images in Definition (clique incidence and extended blocks). □
⟨2⟩3. Q.E.D.
PROOF: Step ⟨2⟩1 has cardinality two by Definition (labelled designs) and ⟨1⟩; ⟨2⟩2 excludes any further label. □

⟨1⟩7. ASSUME: $p \in P$ and $S \in \mathcal{S}$.
PROVE: $\iota(p), \eta(S)$ lie together in exactly two extended blocks.
⟨2⟩1. The common old labels are precisely $\{\mathrm{some}(B) : B \in S,\ p \in B\}$.
PROOF: Use both old-label incidence clauses in Definition (clique incidence and extended blocks). □
⟨2⟩2. The new block does not contain $\iota(p)$.
PROOF: Definition (clique incidence and extended blocks). □
⟨2⟩3. Q.E.D.
PROOF: Lemma (points in a six-clique) gives two labels in ⟨2⟩1; ⟨2⟩2 excludes a contribution from the new label. □

⟨1⟩8. ASSUME: $S \in \mathcal{S}$ and $p \in P$.
PROVE: $\eta(S), \iota(p)$ lie together in exactly two extended blocks.
PROOF: The set of common labels is unchanged when the two membership conditions are interchanged. Apply ⟨1⟩7 to p, S. □

⟨1⟩9. ASSUME: $S, T \in \mathcal{S}$ and $S \neq T$.
PROVE: $\eta(S), \eta(T)$ lie together in exactly two extended blocks.
⟨2⟩1. The common old labels are $\{\mathrm{some}(B) : B \in S \cap T\}$, of size one.
PROOF: Definition (clique incidence and extended blocks) gives this set, and Lemma (intersections of six-cliques) with ⟨1⟩ gives its size. □
⟨2⟩2. Both points lie in the new block.
PROOF: The clause for $F(\mathrm{none})$ in Definition (clique incidence and extended blocks). □
⟨2⟩3. Q.E.D.
PROOF: The old label of ⟨2⟩1 and the new label of ⟨2⟩2 are distinct, and together exhaust the common labels by Definition (clique incidence and extended blocks). □

⟨1⟩10. Q.E.D.
PROOF: Steps ⟨1⟩1–⟨1⟩5 verify the cardinalities, block sizes, and replication numbers. Steps ⟨1⟩6, ⟨1⟩7, ⟨1⟩8, ⟨1⟩9 exhaust ordered distinct point pairs: each point is in exactly one summand. Counting labels uses `Fintype.sum_option`, and splitting point sums uses `Fintype.sum_sum_type`, from F1. These give every asserted equality. □

Definition (the Hall–Connor extension).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
Define $D^* \triangleq D$.hallConnorExtension of type

```
Design (Sum P D.sixCliques) (Option B) 22 7 22 7 2
```

with D^*.block $\triangleq$ D.hallConnorBlocks $= F$ from Definition (clique incidence and extended blocks). The subtype of members is intended by `D.sixCliques` in the point type. The cardinality and incidence fields are exactly the equalities in Lemma (parameters of the extended blocks); the positivity fields of Definition (labelled designs) are $22 > 1$, $7 > 1$, and $22 > 0$.

Theorem (the Hall–Connor residual identity). [[HC], Theorems 2.1 and 3.2 at $k = 5$.]
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: The design $D^* = D$.hallConnorExtension is symmetric, and for every $B \in \mathcal{B}$,

$$D^*.\mathrm{block}(\mathrm{some}(B)) \setminus D^*.\mathrm{block}(\mathrm{none}) = \iota(B).$$

Thus deleting the new block and all its points recovers D label by label.

PROOF SKETCH: The parameter type gives symmetry; the two disjoint point images give the residual identity.
⟨1⟩1. D^* is symmetric.
PROOF: Its parameters in Definition (the Hall–Connor extension) have $v = b = 22$, which is Definition (labelled designs). □
⟨1⟩2. For every B, the asserted block difference is $\iota(B)$.
PROOF: Unfold the block function in Definition (the Hall–Connor extension). In Definition (clique incidence and extended blocks), every added point belongs to $F(\mathrm{none})$ and no old point does. Removing that block from $F(\mathrm{some}(B))$ leaves exactly $\iota(B)$. □
⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2. □

Lemma (integral products for the extension).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: All of the following are identities over $\mathbb{Z}$:

$$NT = 2J_{P,\mathcal{S}}, \qquad T^T T = 5I_{\mathcal{S}} + J_{\mathcal{S}}, \qquad N^* N^{*T} = 5I_{P^*} + 2J_{P^*}.$$

PROOF SKETCH: We will expose the four matrix blocks so that the main contradiction can consume the integral identity directly.
⟨1⟩1. $NT = 2J_{P,\mathcal{S}}$.
PROOF: By Definition (clique incidence and extended blocks) and F2, the (p, S) entry counts $B \in S$ with $p \in B$. Lemma (points in a six-clique) gives two. □
⟨1⟩2. $(T^T T)(S, S) = 6$ for every $S \in \mathcal{S}$.
PROOF: This entry counts the six labels in S, by Definitions (the block graph and all six-cliques), (clique incidence and extended blocks), and F2. □
⟨1⟩3. ASSUME: $S, T_0 \in \mathcal{S}$ and $S \neq T_0$.
PROVE: $(T^T T)(S, T_0) = 1$.
PROOF: The entry counts $S \cap T_0$, by Definition (clique incidence and extended blocks) and F2; Lemma (intersections of six-cliques) gives one. □
⟨1⟩4. $T^T T = 5I_{\mathcal{S}} + J_{\mathcal{S}}$.
PROOF: The diagonal and off-diagonal cases ⟨1⟩2 and ⟨1⟩3 are exhaustive and have the asserted entries 6 and 1. □
⟨1⟩5. The old-old block of $N^* N^{*T}$ is $5I_P + 2J_P$.
PROOF: Sum over old labels and the new label using Definition (clique incidence and extended blocks), F2, and `Fintype.sum_option` in F1. The result is $NN^T + 0$, equal to $5I_P + 2J_P$ by Lemma (incidence identities). □
⟨1⟩6. The old-new block of $N^* N^{*T}$ is $2J_{P,\mathcal{S}}$.

PROOF: The two label contributions are NT and zero, by Definition (clique incidence and extended blocks), F2, and `Fintype.sum_option` in F1; use ⟨1⟩1. □

⟨1⟩7. The new-old block of N^*N^{*T} is $2J_{\mathcal{S},P}$.
PROOF: It is the transpose of the old-new block, since a matrix times its transpose is symmetric by F2; apply ⟨1⟩6. □

⟨1⟩8. The new-new block of N^*N^{*T} is $5I_{\mathcal{S}} + 2J_{\mathcal{S}}$.
PROOF: Old labels contribute T^TT and the new label contributes $J_{\mathcal{S}}$, by Definition (clique incidence and extended blocks) and F2, using `Fintype.sum_option` in F1. Step ⟨1⟩4 gives their sum. □

⟨1⟩9. Q.E.D.
PROOF: Steps ⟨1⟩5–⟨1⟩8 give the four blocks of $5I_{P^*} + 2J_{P^*}$, because distinct summands never give a diagonal entry. Together with ⟨1⟩1 and ⟨1⟩4, they prove all assertions. □

Theorem (Hall–Connor, matrix form).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: There is $M \in \mathrm{Mat}_{\mathrm{Fin}\,22 \times \mathrm{Fin}\,22}(\mathbb{Z})$ with $MM^T = 5I_{22} + 2J_{22}$.

PROOF SKETCH: We will reindex both finite types in the integral identity, keeping the summed block index the same in both factors.

⟨1⟩1. $|P^*| = 22$ and $|\mathcal{B}^*| = 22$.
PROOF: Definition (clique incidence and extended blocks) gives disjoint sums of sizes $15 + 7$ and $21 + 1$, by Definition (labelled designs), Lemma (seven six-cliques), and F1. □

⟨1⟩2. PICK equivalences $a : \mathrm{Fin}\,22 \to P^*$ and $b : \mathrm{Fin}\,22 \to \mathcal{B}^*$.
PROOF: Apply the enumeration fact in F1 to ⟨1⟩1. □

⟨1⟩3. LET: $M(i, j) \triangleq N^*(a(i), b(j))$.
This is an integral matrix.
PROOF: Every entry of N^* is an integer by Definition (clique incidence and extended blocks); composition preserves its scalar type. □

⟨1⟩4. $(MM^T)(i, j) = (N^*N^{*T})(a(i), a(j))$ for all i, j.
PROOF: Both factors use the bijection b on the summed label index. Apply `Matrix.transpose_submatrix` and `Matrix.submatrix_mul_equiv` from F2 to ⟨1⟩3, using ⟨1⟩2. □

⟨1⟩5. $MM^T = 5I_{22} + 2J_{22}$.
PROOF: Lemma (integral products for the extension) and ⟨1⟩4 give entries $5\,1_{a(i)=a(j)} + 2$. The injection a in ⟨1⟩2 makes $a(i) = a(j)$ equivalent to $i = j$, so these are exactly the asserted entries. □

⟨1⟩6. Q.E.D.
PROOF: The witness ⟨1⟩3 and its equation ⟨1⟩5 prove the existential conclusion. □

5.3.8. *The determinant obstruction and the conclusion.*

Lemma (an odd power of five).
ASSUME: $m \in \mathbb{N}$ and $d \in \mathbb{Z}$.
PROVE: $d^2 \neq 49 \cdot 5^{2m+1}$. In particular $d^2 \neq 49 \cdot 5^{21}$.

PROOF SKETCH: We will induct on m, universally quantifying the integer at each stage so that the induction hypothesis applies to $d/5$.

⟨1⟩1. For every $m \in \mathbb{N}$ and every $d \in \mathbb{Z}$, $d^2 \neq 49 \cdot 5^{2m+1}$.
⟨2⟩1. ASSUME: $d \in \mathbb{Z}$ and $d^2 = 49 \cdot 5$.
PROVE: FALSE.
⟨3⟩1. PICK $e \in \mathbb{Z}$ with $d = 5e$.

PROOF: The equality in ⟨2⟩ gives $5 \mid d^2$; F5 gives $5 \mid d$. □

⟨3⟩2. $5e^2 = 49$.

PROOF: Substitute ⟨3⟩1 into ⟨2⟩ and cancel the nonzero integer factor five. □

⟨3⟩3. Q.E.D.

PROOF: Step ⟨3⟩2 would make 49 divisible by five, whereas $49 = 5 \cdot 9 + 4$. □

⟨2⟩2. ASSUME: 1. $m \in \mathbb{N}$ and $e^2 \neq 49 \cdot 5^{2m+1}$ for every $e \in \mathbb{Z}$.
2. $d \in \mathbb{Z}$ and $d^2 = 49 \cdot 5^{2m+3}$.
PROVE: FALSE.

⟨3⟩1. PICK $e \in \mathbb{Z}$ with $d = 5e$.

PROOF: The exponent $2m + 3$ is positive, so ⟨2⟩:2 gives $5 \mid d^2$. Apply F5. □

⟨3⟩2. $e^2 = 49 \cdot 5^{2m+1}$.

PROOF: Substitute ⟨3⟩1 into ⟨2⟩:2, use $5^{2m+3} = 25 \cdot 5^{2m+1}$, and cancel $25 \neq 0$. □

⟨3⟩3. Q.E.D.

PROOF: Apply ⟨2⟩:1 to the integer e in ⟨3⟩2. □

⟨2⟩3. Q.E.D.

PROOF: Natural induction using ⟨2⟩1 and ⟨2⟩2, since $2(m + 1) + 1 = 2m + 3$. □

⟨1⟩2. Q.E.D.

PROOF: Step ⟨1⟩1 proves the general assertion; its instance $m = 10$ gives exponent 21. □

Theorem (the Bruck–Ryser–Chowla instance).
ASSUME: $M \in \mathrm{Mat}_{\mathrm{Fin}\,22 \times \mathrm{Fin}\,22}(\mathbb{Z})$.
PROVE: $MM^T \neq 5I_{22} + 2J_{22}$.

PROOF SKETCH: We will take determinants over $\mathbb{Q}$, retaining the integer determinant to apply the descent at five.

⟨1⟩1. ASSUME: $MM^T = 5I_{22} + 2J_{22}$.
PROVE: FALSE.

⟨2⟩1. Put $d \triangleq \det_{\mathbb{Z}} M$. Then $d \in \mathbb{Z}$ and $\det_{\mathbb{Q}}(M_{\mathbb{Q}}) = (d : \mathbb{Q})$.

PROOF: The integer determinant is an integer by its scalar type; the cast identity is `Int.cast_det` in F4. □

⟨2⟩2. $(d : \mathbb{Q})^2 = 49 \cdot 5^{21}$.

PROOF: Cast ⟨1⟩ to $\mathbb{Q}$ using F2. Since $5 \neq 0$ in $\mathbb{Q}$, the determinant lemma applies, giving

$$\begin{aligned}(d : \mathbb{Q})^2 &= \det(M_{\mathbb{Q}} M_{\mathbb{Q}}^T) && [\langle 2\rangle 1,\ \text{F4}]\\ &= \det(5I_{22} + 2J_{22}) && [\langle 1\rangle]\\ &= 5^{21}(5 + 22 \cdot 2) && [\text{(a scalar matrix plus a constant matrix)}]\\ &= 49 \cdot 5^{21} && [\text{rational arithmetic}].\end{aligned}$$

□

⟨2⟩3. $d^2 = 49 \cdot 5^{21}$ in $\mathbb{Z}$.

PROOF: The cast $\mathbb{Z} \to \mathbb{Q}$ is injective and preserves sums and products; apply it to ⟨2⟩2. □

⟨2⟩4. Q.E.D.

PROOF: Lemma (an odd power of five) contradicts ⟨2⟩3. □

⟨1⟩2. Q.E.D.

PROOF: Step ⟨1⟩1 proves the required negation. □

Theorem (the parameter contradiction).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: FALSE.

PROOF SKETCH: The matrix supplied by the embedding is forbidden by the obstruction.

⟨1⟩1. PICK an integer matrix M of order 22 with $MM^T = 5I_{22} + 2J_{22}$.
PROOF: Apply Theorem (Hall–Connor, matrix form) to D. □
⟨1⟩2. Q.E.D.
PROOF: Apply Theorem (the Bruck–Ryser–Chowla instance) to the witness and equation in ⟨1⟩1. □

Corollary (nonexistence of a 2-$(15, 5, 2)$ design).
PROVE: The type

```
Design (Fin 15) (Fin 21) 15 5 21 7 2
```

is empty. Moreover, for every finite point type P of size 15 and every finite label type $\mathcal{B}$, both with decidable equality, there is no function $F : \mathcal{B} \to \mathrm{Finset}(P)$ whose blocks have size five and whose distinct point pairs lie in exactly two labels.

PROOF SKETCH: We will state both the bundled and unbundled forms.
⟨1⟩1. The specified design type is empty.
PROOF: Every inhabitant would give FALSE by Theorem (the parameter contradiction) with $P = \mathrm{Fin}\,15$, $\mathcal{B} = \mathrm{Fin}\,21$. □
⟨1⟩2. ASSUME: $P, \mathcal{B}, F$ have the properties in the second assertion.
PROVE: FALSE.
⟨2⟩1. PICK D : `Design P B 15 5 21 7 2` with $D.\mathrm{block}(B) = F(B)$ for every $B \in \mathcal{B}$.
PROOF: Apply Lemma (the forced parameters) to ⟨1⟩. □
⟨2⟩2. Q.E.D.
PROOF: Apply Theorem (the parameter contradiction) to ⟨2⟩1. □
⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2. □

Definition (complementary labelled blocks).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters (v, k, b, r, λ).
3. $2 \le v - k$.
Define $D.\mathrm{compl}$ with the same point and label types by

$$D.\mathrm{compl} : \texttt{Design P B v (v-k) b (b-r) (b+lam-2*r)},$$

$$D.\mathrm{compl.block}(B) \triangleq P \setminus D(B).$$

The differences are natural differences. No injectivity of the block function is assumed. The proof below constructs its fields.

PROOF SKETCH: We will give the complement constructor its full parameter type. The bound on the new block size is exactly the condition needed by Definition (labelled designs).
⟨1⟩1. Every complementary block has size $v - k$.
PROOF: Its old block is a subset of P with size k; subtract from $|P| = v$ using Definition (labelled designs) and the finite-difference formula of F1. □
⟨1⟩2. Every point belongs to $b - r$ complementary labels.
PROOF: The r old labels through it form a subset of the b labels. The other labels are exactly its complementary incidences, by Definition (labelled designs) and F1. □
⟨1⟩3. **[new]**
ASSUME: $p, q \in P$ and $p \neq q$.
PROVE: Exactly $b + \lambda - 2r$ complementary labels contain p, q.
⟨2⟩1. Let A_p, A_q be the sets of old incident labels of p, q. Then $|A_p \cup A_q| + \lambda = 2r$.
PROOF: Both sets have size r and their intersection has size λ, by ⟨1⟩ and Definition (labelled designs); use the union-intersection identity in F1. □

⟨2⟩2. If c counts labels containing neither point, then $c + 2r = b + \lambda$.
PROOF: The complement of $A_p \cup A_q$ among labels has size c, so $c + |A_p \cup A_q| = b$ by F1. Add ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Step ⟨2⟩2 gives $2r \le b + \lambda$ and $c = b + \lambda - 2r$ in $\mathbb{N}$. These are the common complementary labels by the block definition. This is the ordinary proof's inclusion-exclusion argument stated for general parameters, so the constructor has a precise type. □
⟨1⟩4. Q.E.D.
PROOF: The types keep cardinalities v, b; steps ⟨1⟩1, ⟨1⟩2, and ⟨1⟩3 give the three incidence fields. The new block size is greater than one by ⟨0⟩:3; $v > 1, b > 0$ come from the old design in Definition (labelled designs). □

Corollary (the complementary parameters).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 10, 21, 14, 9)$.
PROVE: FALSE. Thus `Design (Fin 15) (Fin 21) 15 10 21 14 9` is empty.

PROOF SKETCH: Complementing every labelled block produces the forbidden parameters.
⟨1⟩1. Put $D' \triangleq D$.compl. Then D' : `Design P B 15 5 21 7 2`.
PROOF: Apply the constructor of Definition (complementary labelled blocks) to ⟨0⟩:2, with $2 \le 15 - 10 = 5$. Its parameters reduce to $15 - 10 = 5$, $21 - 14 = 7$, and $21 + 9 - 2 \cdot 14 = 2$. These are definitionally equal numerals in Lean. □
⟨1⟩2. Q.E.D.
PROOF: Apply Theorem (the parameter contradiction) to ⟨1⟩1. Since the argument applies to arbitrary finite $P, \mathcal{B}$, it also excludes an inhabitant on Fin 15 and Fin 21. □

5.3.9. *Interfaces, provenance, and checks.* The statements above make the design-to-matrix bridge explicit: `Design.hall_connor` returns the integer matrix equation that `Matrix.bruck_ryser_chowla_twenty_two` negates. The constructor `Design.hallConnorExtension` has the separately named theorem `Design.hallConnorExtension_residual`, with point type `Sum P D.sixCliques` and label type `Option B`. Here `D.sixCliques` denotes the subtype of members when used as a type. The existential matrix theorem does not need a general residual-design API.

No mathematical repair of the ordinary proof is needed. We have made its moments, entry identities, principal-matrix bounds, and clique incidence counts separately citable, put the determinant lemma before its first use, and expanded the local and extension cases. These are changes of proof structure. The ordinary proof already proves the clique property before counting eight edges, already defines the global family as all six-cliques, and already uses rank to exclude Q_1, Q_2. Those repairs of the source are retained here. Neither the unused rational determinant-sign remark nor the unused source-local patterns are dependencies. Three steps are tagged **[new]**: using the rank-one determinant lemma, taking the determinant of each certified inverse identity, and stating the complement count for general parameters to give `Design.compl` an exact interface. They retain the same nonexistence argument.

All direct dependencies have local labels prefixed by `hc:`; there are no deferred blueprint labels. The names in `\lean` are declarations of the Lean formalization, except for the Mathlib API cited explicitly with paths. Mathlib was inspected at commit
`082e2d37e8b0463410cdb532e111cd43d5a66174`
(4 September 2026), the version pinned in `lake-manifest.json`. Its rank identities apply over $\mathbb{Q}$; the analytic theorem about nonnegative determinants over an `RCLike` field is unnecessary. The determinant obstruction uses only the even-order instance, proved here without a quadratic-form local-global theorem.

The six displayed matrices are those of the ordinary proof. The exact checks `blueprint/checks/hallconnor/check_proof.py`, with the $T(7)$ control `check_t7.py`, read them from this file and check all six matrices, the two quadratic values, and every entry of both inverse certificates; they are the checks of the ordinary proof in `cgss/proofs/hallconnor/checks/`. The control exercises all 210 ordered adjacent pairs; no existing design can exercise the hypothetical incidence structure. These checks do not establish the quantified counting and clique arguments, which are provided by the proofs above.

6. Hadamard matrices

We will follow Section 6 of [CGSS]. Throughout, $n \geq 1$ is an integer, the rows and the columns of an $n \times n$ matrix are indexed by $\{1, \dots, n\}$, and H is a Hadamard matrix of order n: an integer matrix with entries ± 1 and $HH^T = H^T H = nI$. This is Mathlib's `Matrix.IsHadamard` over $\mathbb{Z}$ (`Mathlib/LinearAlgebra/Matrix/HadamardMatrix.lean`), whose fields are `apply_mem`, `mul_conjTranspose` and `conjTranspose_mul`; over $\mathbb{Z}$ the first says that every entry is 1 or -1 (`Unitary.mem_iff_eq_one_or_eq_neg_one`). The paper writes H_n; we write H, and $H_i \in \mathbb{R}^n$ for its row i, so that $H_i(c) = H(i, c)$. As in the rest of the blueprint, $(\mathbf{x}, \mathbf{y})$ is the standard inner product of $\mathbb{R}^n$ (`EuclideanSpace ℝ (Fin n)` in Lean), $\mathbf{j}$ is the all-ones vector and J the all-ones matrix. The paper's matrix A with $PP^T = nI + xA$ is our C_H (`cgss/notation.md`, Section 7), since A is an adjacency matrix elsewhere and C_H is the matrix C of Theorem 3.7 when $d = 2$.

The paper's background paragraph asserts five things: the formula for $P^T P$, the quadratic equation for C_H, the divisibility $x \mid n$, that the lines lie in $\mathbf{j}^\perp$, and that the lines through a fixed row are mutually orthogonal. We will prove each of them. The equations $xAJ = -nJ$ and $AJ = -dJ$ are false in general (errata, Error 9), and we will not use them; the divisibility comes instead from the count of fourth rows, which the paper also states. The paragraph also gives the eigenvalues of $P^T P$ and of PP^T, and two claims about the Sylvester matrices: that they satisfy Hypothesis 6.1 for every order 2^s, and what their lines are. These are side claims, which Theorem 6.2 does not use; we prove them in `LineGraphs/SideClaims/`, and we will record them in the remark after Lemma (the matrix P) and in the last remark of this section. Norman's theorem and the uniqueness of H_{12} are side claims that the author has set aside (`cgss/prerequisites.md`, Section 10). The examples of orders 2, 4, 8, 12 and 16 belong to computation C16, which Theorem 6.2 does not use either; we will record what it proves in the first remark after the proof of Theorem 6.2.

The proof of Theorem 6.2 in the paper has two gaps. It assumes $n > 9$ (errata, Gap 10), and it asserts without proof that the lines form an indecomposable set (errata, Gap 17). We will fill both. Only the orders with $16 \mid n$ need an argument (Lemma (orders not divisible by 16)). For these we will split the lines into the components of their graph of non-orthogonality, and show that each component of rank m has exactly $\frac{1}{2}nm$ lines (Lemma (the components)). The classification of Theorem 3.5 and the line counts of Lemma 3.4 then leave two possibilities (Lemma (the possible decompositions)): a single component, of type A_{n-1} or D_{n-1}, which the paper's own argument excludes (Lemma (orthogonal lines in A_ℓ and D_ℓ)); or $n = 16$ and a component of type E_7, which a count of stars excludes (Lemma (no component of type E_7)). The last lemma is new.

The Lean names of Hypothesis 6.1 and Theorem 6.2, and of the matrices P and C_H, are the settled names of `cgss/notation.md`, Section 10, which also lists the other names of this section.

Inputs. We will use the following facts from other sections and from Mathlib.

- (LN3 of `cgss/prerequisites.md`, deferred to the opening of Section 4.) Let Σ be a finite set of vectors of squared norm 2 in a finite-dimensional real inner product space, with $\Sigma = -\Sigma$ and pairwise inner products in $\{0, \pm 1, \pm 2\}$. There is a smallest star-closed

set $\mathrm{sc}(\Sigma) \supseteq \Sigma$ of vectors of squared norm 2; it is the vector set of a line system, it spans the same subspace as Σ, and it is indecomposable if the graph on Σ in which two vectors are adjacent when their inner product is nonzero is connected. Lean: `LineSystem.starClosure`.

- (Section 3.) Theorem 3.5, with the isomorphism induced by a surjective linear isometry of spans; the ranks of the models, ℓ for A_ℓ and D_ℓ and $6, 7, 8$ for $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ (Lemma (spans of the models)); the line counts of Lemma 3.4; the coordinates of Definition 3.1; and Theorem 3.7 for E_7, with Definition (signings and the matrix convention). An isomorphism of line systems is a bijection of vector sets that preserves inner products (Section 2), so it preserves orthogonality and the sets $N_{\mathbf{r}} = \{\mathbf{u} \in \Phi : (\mathbf{u}, \mathbf{r}) = 1\}$. Lean: `LineSystem.classification`, `LineSystem.RootType.rank`, `LineSystem.numLines_E7` and the other counts, `LineSystem.signedAdj_quadratic_E7`; and, for $|N_{\mathbf{r}}| = 32$ in the model, `LineSystem.E7.card_N` (C20, `LineGraphs/Computations/C20E7.lean`).
- (Mathlib.) `Matrix.IsHadamard.four_dvd_card`: an integer Hadamard matrix of order greater than 2 has order divisible by 4. `linearIndependent_of_ne_zero_of_inner_eq_zero` (`Mathlib/Analysis/InnerProductSpace/Basic.lean`): nonzero pairwise orthogonal vectors are linearly independent. `LinearIndependent.fintype_card_le_finrank` (`Mathlib/LinearAlgebra/Dimension/Finite.lean`), `LinearIndependent.span_eq_top_of_card_eq_finrank` (`Mathlib/LinearAlgebra/FiniteDimensional/Lemmas.lean`), `finrank_span_eq_card` (`Mathlib/LinearAlgebra/Dimension/Constructions.lean`) and `finrank_euclideanSpace_fin`. `ext_inner_left` and `inner_self_eq_zero`. `Nat.pow_dvd_pow_iff`: for $k \neq 0$, $a^k \mid b^k$ if and only if $a \mid b$. `Nat.Coprime.dvd_of_dvd_mul_left`. The identities of Section 2.

6.1. The Hadamard products of pairs of rows.

Definition (Hadamard products and agreements). ASSUME: H is an $n \times n$ Hadamard matrix.

LET: $\mathcal{P}$ be the set of 2-subsets of $\{1, \dots, n\}$, so $|\mathcal{P}| = \frac{1}{2}n(n-1)$. For $\{i, j\} \in \mathcal{P}$ let $\mathbf{p}_{ij} = H_i \circ H_j \in \mathbb{R}^n$, the *Hadamard product* of the rows, with $\mathbf{p}_{ij}(c) = H(i,c)H(j,c)$. Let P be the $\mathcal{P} \times n$ matrix with rows $\mathbf{p}_{ij}$. For distinct rows i, j, k, l let $\mathrm{ag}(i,j,k,l)$ be the number of columns c with $H(i,c) = H(j,c) = H(k,c) = H(l,c)$; we say that the four rows *agree* in these places.

Thus $\mathbf{p}_{ij} = \mathbf{p}_{ji}$, and ag is symmetric in its four arguments. In Lean the index type of $\mathcal{P}$ is a proposal for the declaration `Matrix.hadamardPairs`; the pairs (i, j) with $i < j$ will do.

Lemma (the inner products of the Hadamard products). ASSUME: 1. H is a Hadamard matrix of order n.
2. $\mathbf{p}_{ij}$, $\mathcal{P}$ and ag are as in Definition (Hadamard products and agreements).

PROVE: 1. Every entry of every $\mathbf{p}_{ij}$ is ± 1, and $(\mathbf{p}_{ij}, \mathbf{p}_{ij}) = n$.
2. If $j \neq k$ and $i \notin \{j, k\}$, then $(\mathbf{p}_{ij}, \mathbf{p}_{ik}) = 0$.
3. If i, j, k, l are distinct, then $(\mathbf{p}_{ij}, \mathbf{p}_{kl}) = 8\,\mathrm{ag}(i,j,k,l) - n$.
4. $(\mathbf{p}_{ij}, \mathbf{j}) = 0$ for every $\{i, j\} \in \mathcal{P}$.

PROOF SKETCH: Each part is a sum over the columns. Part 3 is the one computation; it rests on an identity for four signs.

⟨1⟩1. For distinct rows u, w, $\sum_c H(u,c)H(w,c) = 0$.
PROOF: This is the (u, w) entry of $HH^T = nI$. □

⟨1⟩2. For all u and c, $H(u,c)^2 = 1$.

PROOF: The entries are ± 1. □

⟨1⟩3. Part 1 holds.

PROOF: $\mathbf{p}_{ij}(c)$ is a product of two signs, and $(\mathbf{p}_{ij}, \mathbf{p}_{ij}) = \sum_c H(i,c)^2 H(j,c)^2 = n$ by ⟨1⟩2. □

⟨1⟩4. Part 2 holds.

PROOF: $(\mathbf{p}_{ij}, \mathbf{p}_{ik}) = \sum_c H(i,c)^2 H(j,c) H(k,c) = \sum_c H(j,c) H(k,c)$ by ⟨1⟩2, and this is 0 by ⟨1⟩1, since $j \neq k$. □

⟨1⟩5. Part 3 holds.

⟨2⟩1. For $s_1, s_2, s_3, s_4 \in \{1, -1\}$,

$$8\,[s_1 = s_2 = s_3 = s_4] = 1 + \sum_{1 \le a < b \le 4} s_a s_b + s_1 s_2 s_3 s_4,$$

where $[\,\cdot\,]$ is 1 if the condition holds and 0 otherwise.

PROOF: A check of the 16 sign vectors (in Lean, `decide`). If all four are equal, the right side is $1 + 6 + 1$; if exactly one differs, it is $1 + (3 - 3) - 1$; if two differ from the other two, it is $1 + (2 - 4) + 1$. □

⟨2⟩2. $8\,\mathrm{ag}(i,j,k,l) = n + \sum_{\{u,w\}} \sum_c H(u,c) H(w,c) + (\mathbf{p}_{ij}, \mathbf{p}_{kl})$, where $\{u, w\}$ runs over the six 2-subsets of $\{i, j, k, l\}$.

PROOF: Sum ⟨2⟩1 over the columns c, with $s = (H(i,c), H(j,c), H(k,c), H(l,c))$; the left side sums to $8\,\mathrm{ag}(i,j,k,l)$ by Definition (Hadamard products and agreements), and $\sum_c H(i,c)H(j,c)H(k,c)H(l,c) = (\mathbf{p}_{ij}, \mathbf{p}_{kl})$. □

⟨2⟩3. Q.E.D.

PROOF: The six inner sums in ⟨2⟩2 vanish by ⟨1⟩1, since the four rows are distinct. □

⟨1⟩6. Part 4 holds.

PROOF: $(\mathbf{p}_{ij}, \mathbf{j}) = \sum_c H(i,c) H(j,c) = 0$ by ⟨1⟩1. □

⟨1⟩7. Q.E.D.

PROOF: Steps ⟨1⟩3, ⟨1⟩4, ⟨1⟩5 and ⟨1⟩6 are the four parts. □

Lemma (the matrix P). ASSUME: H is an $n \times n$ Hadamard matrix, P is as in Definition (Hadamard products and agreements), and $G = PP^T$.

PROVE: 1. $P^T P = \frac{1}{2} n^2 I - \frac{1}{2} n J$.
2. $P\mathbf{j} = 0$.
3. $G^2 = \frac{1}{2} n^2 G$.

PROOF SKETCH: Part 1 is the paper's display; we compute an entry as a sum over pairs of rows, through the square of a sum. Part 3 follows from parts 1 and 2 by associativity.

⟨1⟩1. Part 1 holds.

LET: c, c' be columns, and $u_i = H(i,c)H(i,c')$ for each row i.

⟨2⟩1. $(P^T P)(c,c') = \sum_{\{i,j\} \in \mathcal{P}} u_i u_j$.

PROOF: By Definition (Hadamard products and agreements), we have that

$$\mathbf{p}_{ij}(c)\mathbf{p}_{ij}(c') = H(i,c)H(i,c')H(j,c)H(j,c').$$ □

⟨2⟩2. $2 \sum_{\{i,j\} \in \mathcal{P}} u_i u_j = \left(\sum_i u_i\right)^2 - \sum_i u_i^2$.

PROOF: Expand the square of the sum. □

⟨2⟩3. $\sum_i u_i = (H^T H)(c,c')$, which is n if $c = c'$ and 0 otherwise, and $\sum_i u_i^2 = n$.

PROOF: $H^T H = nI$, and each u_i is ± 1. □

⟨2⟩4. Q.E.D.

PROOF: By ⟨2⟩1, ⟨2⟩2 and ⟨2⟩3, $(P^T P)(c,c')$ is $\frac{1}{2}(n^2 - n)$ if $c = c'$ and $-\frac{1}{2} n$ otherwise; these are the entries of $\frac{1}{2} n^2 I - \frac{1}{2} n J$. □

⟨1⟩2. Part 2 holds.

PROOF: The entry of $P\mathbf{j}$ at $\{i,j\}$ is $(\mathbf{p}_{ij}, \mathbf{j})$, which is 0 by Lemma (the inner products of the Hadamard products), part 4. □

⟨1⟩3. Part 3 holds.
PROOF: We have that
$$G^2 = P(P^TP)P^T = \tfrac{1}{2}n^2PP^T - \tfrac{1}{2}n(P\mathbf{j})(P\mathbf{j})^T = \tfrac{1}{2}n^2G,$$
by associativity, by ⟨1⟩1 with $J = \mathbf{j}\mathbf{j}^T$, and by ⟨1⟩2. □
⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩1, ⟨1⟩2 and ⟨1⟩3. □

Remark. The paper deduces from part 1 that P^TP has the eigenvalue 0, which is simple, and the eigenvalue $\frac{1}{2}n^2$, of multiplicity $n-1$; and that PP^T has the eigenvalues 0 and $\frac{1}{2}n^2$, with multiplicities $\frac{1}{2}(n-1)(n-2)$ and $n-1$. The second also follows from part 3, since $\frac{2}{n^2}G$ is then an idempotent symmetric matrix whose trace, by Lemma (the inner products of the Hadamard products), part 1, is $\frac{2}{n^2}\cdot n\cdot\frac{1}{2}n(n-1) = n-1$. These are side claims, and we prove them in `LineGraphs/SideClaims/HadamardSpectra.lean` for every $n \geq 1$, in the three forms of the spectral toolkit: the characteristic polynomials, the multisets of eigenvalues, and the multiplicities, which are the dimensions of the eigenspaces. For instance, `Matrix.IsHadamard.spectrumMultiset_hadamardPairs_mul_transpose` gives the spectrum of PP^T. We will not use the spectrum in the proof of Theorem 6.2.

Lemma (the lines through a fixed row). ASSUME: 1. H is an $n\times n$ Hadamard matrix, with $n \geq 2$, and $\mathbf{p}_{ij}$ is as in Definition (Hadamard products and agreements).
2. i is a row.
PROVE: 1. The $n-1$ vectors $\mathbf{p}_{ij}$ with $j \neq i$ are nonzero and pairwise orthogonal, and every $\mathbf{p}_{kl}$ lies in $\mathbf{j}^\perp$.
2. For every $\mathbf{v} \in \mathbf{j}^\perp$,
$$\mathbf{v} = \frac{1}{n}\sum_{j\neq i}(\mathbf{v},\mathbf{p}_{ij})\,\mathbf{p}_{ij}.$$
3. The vectors $\mathbf{p}_{ij}$ with $j \neq i$ span $\mathbf{j}^\perp$, which has dimension $n-1$.

PROOF SKETCH: The n vectors $\mathbf{j}$ and $\mathbf{p}_{ij}$ $(j \neq i)$ are nonzero and pairwise orthogonal in $\mathbb{R}^n$, and thus a basis. We will expand in this basis.
⟨1⟩1. Part 1 holds.
PROOF: Lemma (the inner products of the Hadamard products): part 1 gives $(\mathbf{p}_{ij},\mathbf{p}_{ij}) = n \neq 0$, part 2 orthogonality, and part 4 that $\mathbf{p}_{kl} \in \mathbf{j}^\perp$. □
⟨1⟩2. The n vectors $\mathbf{j}$ and $\mathbf{p}_{ij}$ $(j \neq i)$ are linearly independent and span $\mathbb{R}^n$.
⟨2⟩1. They are linearly independent.
PROOF: They are nonzero and pairwise orthogonal by ⟨1⟩1 and $(\mathbf{j},\mathbf{j}) = n$; apply `linearIndependent_of_ne_zero_of_inner_eq_zero`. □
⟨2⟩2. Q.E.D.
PROOF: Independence is ⟨2⟩1. There are $n = \dim\mathbb{R}^n$ of them, so `LinearIndependent.span_eq_top_of_card_eq_finrank` applied to ⟨2⟩1 gives the span. □
⟨1⟩3. ASSUME: $\mathbf{v} \in \mathbf{j}^\perp$.
PROVE: Part 2 holds for $\mathbf{v}$.
LET: $\mathbf{w} = \mathbf{v} - \frac{1}{n}\sum_{j\neq i}(\mathbf{v},\mathbf{p}_{ij})\mathbf{p}_{ij}$.
⟨2⟩1. $(\mathbf{w},\mathbf{p}_{ik}) = 0$ for each $k \neq i$.
PROOF: By bilinearity and ⟨1⟩1, only the term $j = k$ of the sum survives, and it gives $(\mathbf{w},\mathbf{p}_{ik}) = (\mathbf{v},\mathbf{p}_{ik}) - \frac{1}{n}(\mathbf{v},\mathbf{p}_{ik})\,n$. □
⟨2⟩2. $(\mathbf{w},\mathbf{j}) = 0$.
PROOF: $\mathbf{v} \in \mathbf{j}^\perp$ by ⟨1⟩, and each $\mathbf{p}_{ij} \in \mathbf{j}^\perp$ by ⟨1⟩1. □
⟨2⟩3. Q.E.D.

PROOF: By ⟨2⟩1, ⟨2⟩2 and ⟨1⟩2, $\mathbf{w}$ is orthogonal to a spanning set, so $(\mathbf{w},\mathbf{w})=0$ and $\mathbf{w}=0$ by `inner_self_eq_zero`. □

⟨1⟩4. Part 3 holds.
PROOF: Part 2 writes each $\mathbf{v}\in\mathbf{j}^{\perp}$ in the span, and the vectors lie in $\mathbf{j}^{\perp}$ by part 1. By ⟨1⟩2 they are $n-1$ linearly independent vectors, and thus the dimension is $n-1$ by `finrank_span_eq_card`. □

⟨1⟩5. Q.E.D.
PROOF: Steps ⟨1⟩1, ⟨1⟩3 and ⟨1⟩4. □

6.2. **Hypothesis 6.1.**

Hypothesis 6.1. ASSUME: H is an $n\times n$ Hadamard matrix and $x\ge 0$ is an integer.
We say that H satisfies Hypothesis 6.1 with parameter x if, for all distinct rows i,j,k,l,

$$8\,\mathrm{ag}(i,j,k,l)\in\{n,\ n-x,\ n+x\}.$$

The paper allows $x\in\mathbb{Z}$; replacing x by $-x$ does not change the condition, so we take $x\ge 0$, as in the Lean statement `H.FourRowAgreement x` with `x : ℕ`.

Definition (the matrix C_H). ASSUME: H is an $n\times n$ Hadamard matrix, $x>0$ is an integer, and P is as in Definition (Hadamard products and agreements).
LET: C_H be the $\mathcal{P}\times\mathcal{P}$ matrix with $C_H=\frac{1}{x}(PP^T-nI)$.
Thus $PP^T=nI+xC_H$, which is the paper's display with C_H for A. The diagonal of C_H is 0 by Lemma (the inner products of the Hadamard products), part 1. The Lean declaration takes x as an argument, `Matrix.hadamardSignedAdj H x`, the settled name of `cgss/notation.md`, which now takes x.

Lemma (the entries of C_H). ASSUME: 1. H is an $n\times n$ Hadamard matrix and $x>0$ is an integer.
2. H satisfies Hypothesis 6.1 with parameter x.
PROVE: Every entry of C_H lies in $\{0,1,-1\}$. For $\{i,j\}\neq\{k,l\}$ we have $C_H(\{i,j\},\{k,l\})=0$ when the two pairs meet, and $C_H(\{i,j\},\{k,l\})=0$ exactly when $8\,\mathrm{ag}(i,j,k,l)=n$ when they are disjoint.

⟨1⟩1. The diagonal entries are 0.
PROOF: Definition (the matrix C_H). □

⟨1⟩2. CASE: The pairs are distinct and meet.
PROOF: They share exactly one row, and $(\mathbf{p}_{ij},\mathbf{p}_{kl})=0$ by Lemma (the inner products of the Hadamard products), part 2. □

⟨1⟩3. CASE: The pairs are disjoint.
PROOF: $xC_H(\{i,j\},\{k,l\})=(\mathbf{p}_{ij},\mathbf{p}_{kl})=8\,\mathrm{ag}(i,j,k,l)-n$ by Lemma (the inner products of the Hadamard products), part 3, and this lies in $\{0,-x,x\}$ by ⟨0⟩:2, with 0 exactly when $8\,\mathrm{ag}(i,j,k,l)=n$; divide by $x>0$. □

⟨1⟩4. Q.E.D.
PROOF: Two entries are on the diagonal or are indexed by two distinct pairs, which meet or are disjoint. □

Lemma (the quadratic equation). ASSUME: H is an $n\times n$ Hadamard matrix, $x>0$ is an integer, and $d=n/x\in\mathbb{Q}$.
PROVE:

$$(xC_H+nI)\big(xC_H-\tfrac{1}{2}n(n-2)I\big)=0,\qquad (C_H+dI)\big(C_H-\tfrac{1}{2}d(n-2)I\big)=0.$$

PROOF SKETCH: This is Lemma (the matrix P), part 3, rewritten. It does not need Hypothesis 6.1, which only makes C_H a $(0,\pm1)$ matrix.

LET: $G = PP^T$.
⟨1⟩1. $G(G - \frac{1}{2}n^2 I) = 0$.
PROOF: Lemma (the matrix P), part 3. □
⟨1⟩2. $G = xC_H + nI$ and $G - \frac{1}{2}n^2 I = xC_H - \frac{1}{2}n(n-2)I$.
PROOF: Definition (the matrix C_H), and $n - \frac{1}{2}n^2 = -\frac{1}{2}n(n-2)$. □
⟨1⟩3. Q.E.D.
PROOF: Substituting ⟨1⟩2 in ⟨1⟩1 gives the first equation; dividing it by $x^2 > 0$ gives the second. □

Lemma (the fourth rows). ASSUME: 1. H is an $n \times n$ Hadamard matrix, with $n \geq 3$, and $x > 0$ is an integer.
2. H satisfies Hypothesis 6.1 with parameter x.
3. i, k, l are distinct rows, and m is the number of rows $j \notin \{i, k, l\}$ with $8\,\mathrm{ag}(i, j, k, l) \neq n$.
PROVE: $mx^2 = n^2$. Thus $x \mid n$, and $d = n/x$ is an integer with $d^2 = m$.

PROOF SKETCH: **[repair]** The paper derives $x \mid n$ from $xAJ = -nJ$, which is false (errata, Error 9). We will expand $\mathbf{p}_{kl}$ in the orthogonal basis of $\mathbf{j}^\perp$ attached to the row i, and compare squared norms. This is the paper's remark that d^2 is the number of fourth rows agreeing with three given rows in $\frac{1}{8}(n \pm x)$ places.
⟨1⟩1. $n^2 = \sum_{j \neq i} (\mathbf{p}_{kl}, \mathbf{p}_{ij})^2$.
PROOF: Take the inner product of the expansion of Lemma (the lines through a fixed row), part 2, for $\mathbf{v} = \mathbf{p}_{kl} \in \mathbf{j}^\perp$, with $\mathbf{p}_{kl}$; the left side is $(\mathbf{p}_{kl}, \mathbf{p}_{kl}) = n$ by Lemma (the inner products of the Hadamard products), part 1. Multiply by n. □
⟨1⟩2. The terms $j = k$ and $j = l$ vanish.
PROOF: The pairs $\{i, k\}$ and $\{k, l\}$ share only k, and $\{i, l\}$ and $\{k, l\}$ share only l; apply Lemma (the inner products of the Hadamard products), part 2. □
⟨1⟩3. For $j \notin \{i, k, l\}$, $(\mathbf{p}_{kl}, \mathbf{p}_{ij})^2$ is x^2 if $8\,\mathrm{ag}(i, j, k, l) \neq n$ and 0 otherwise.
PROOF: By Lemma (the inner products of the Hadamard products), part 3, $(\mathbf{p}_{ij}, \mathbf{p}_{kl}) = 8\,\mathrm{ag}(i, j, k, l) - n$, which is 0 or $\pm x$ by ⟨0⟩:2. □
⟨1⟩4. $mx^2 = n^2$.
PROOF: Substitute ⟨1⟩2 and ⟨1⟩3 in ⟨1⟩1, and use ⟨0⟩:3. □
⟨1⟩5. Q.E.D.
PROOF: By ⟨1⟩4, $x^2 \mid n^2$, so $x \mid n$ by `Nat.pow_dvd_pow_iff`; then $d^2 = n^2/x^2 = m$. □

Remark. **[repair]** (Errata, Error 9.) The paper's equation $xAJ = -nJ$ says that $P^T\mathbf{j} = 0$. The entry of $P^T\mathbf{j}$ at a column c is $\frac{1}{2}(\sigma(c)^2 - n)$, where $\sigma(c)$ is the sum of column c of H, by the computation of Lemma (the matrix P), part 1, with $u_i = H(i, c)$. It vanishes only if every column sum is $\pm\sqrt{n}$, which fails unless n is a square. None of our results uses it. A second route to $x \mid n$ for $n > 2$: the eigenvalue $-n/x$ of the integer matrix C_H is rational and is a root of its monic characteristic polynomial, so it is an integer by the rational root theorem ([ST, Theorem 9.6]; [ST, Theorem 10.4] proves P^TP and $x \mid n$ by this route).

6.3. **Theorem 6.2.** Theorem 6.2 is the case $x = \frac{1}{2}n$, $d = 2$: any four rows agree in $\frac{n}{16}$, $\frac{n}{8}$ or $\frac{3n}{16}$ places. We will first dispose of the orders not divisible by 16, then build the line system and study its components.

Lemma (orders not divisible by 16). ASSUME: 1. H is an $n \times n$ Hadamard matrix, with $n > 2$.
2. For all distinct rows i, j, k, l, $16\,\mathrm{ag}(i, j, k, l) \in \{n, 2n, 3n\}$.
PROVE: $16 \mid n$, and $n \geq 16$.

PROOF SKETCH: **[repair]** (Errata, Gap 10; `cgss/prerequisites.md`, HD1.) If $16 \nmid n$, every four rows agree in exactly $\frac{n}{8}$ places, and then the $\frac{1}{2}n(n-1)$ vectors $\mathbf{p}_{ij}$ are pairwise orthogonal in $\mathbb{R}^n$, which is too many.

⟨1⟩1. $4 \mid n$, so $n \geq 4$.
PROOF: `Matrix.IsHadamard.four_dvd_card` with ⟨0⟩:1. □

⟨1⟩2. ASSUME: $16 \nmid n$.
PROVE: FALSE.

⟨2⟩1. For all distinct rows i, j, k, l, $16\,\mathrm{ag}(i,j,k,l) = 2n$.

⟨3⟩1. $16\,\mathrm{ag}(i,j,k,l) \neq n$.
PROOF: Otherwise $16 \mid n$, against ⟨1⟩. □

⟨3⟩2. $16\,\mathrm{ag}(i,j,k,l) \neq 3n$.
PROOF: Otherwise $16 \mid 3n$, so $16 \mid n$ by `Nat.Coprime.dvd_of_dvd_mul_left`, since 3 and 16 are coprime, against ⟨1⟩. □

⟨3⟩3. Q.E.D.
PROOF: ⟨0⟩:2, ⟨3⟩1 and ⟨3⟩2. □

⟨2⟩2. The vectors $\mathbf{p}_\pi$, $\pi \in \mathcal{P}$, are nonzero and pairwise orthogonal.
PROOF: Nonzero by Lemma (the inner products of the Hadamard products), part 1. Two distinct pairs meet in one row or are disjoint; Lemma (the inner products of the Hadamard products), parts 2 and 3, gives inner product 0, or $8\,\mathrm{ag} - n = 0$ by ⟨2⟩1. □

⟨2⟩3. $\frac{1}{2}n(n-1) \leq n$.
PROOF: `linearIndependent_of_ne_zero_of_inner_eq_zero` makes the family of ⟨2⟩2, indexed by $\mathcal{P}$, linearly independent, and `LinearIndependent.fintype_card_le_finrank` with `finrank_euclideanSpace_fin` bounds $|\mathcal{P}|$ by n. □

⟨2⟩4. Q.E.D.
PROOF: ⟨2⟩3 gives $n \leq 3$, against ⟨1⟩1. □

⟨1⟩3. Q.E.D.
PROOF: By ⟨1⟩2, $16 \mid n$; and $n > 0$, so $n \geq 16$. □

Lemma (the line system of H). ASSUME: 1. H is an $n \times n$ Hadamard matrix, with $16 \mid n$ and $n \geq 16$.
2. For all distinct rows i, j, k, l, $16\,\mathrm{ag}(i,j,k,l) \in \{n, 2n, 3n\}$.

LET: $\mathbf{r}_\pi = \sqrt{2/n}\,\mathbf{p}_\pi$ for $\pi \in \mathcal{P}$, and $\Phi_H = \{\pm\mathbf{r}_\pi : \pi \in \mathcal{P}\}$.

PROVE:
1. H satisfies Hypothesis 6.1 with parameter $x = \frac{1}{2}n$, and then $d = 2$.
2. $(\mathbf{r}_\pi, \mathbf{r}_\pi) = 2$, and for $\pi \neq \pi'$, $(\mathbf{r}_\pi, \mathbf{r}_{\pi'}) = C_H(\pi, \pi') \in \{0, 1, -1\}$.
3. Φ_H is the vector set of a line system S_H in $\mathbb{R}^n$ with $\frac{1}{2}n(n-1)$ lines, and $\pi \mapsto \mathbf{r}_\pi$ is a signing of S_H whose matrix C of Definition (signings and the matrix convention) is C_H.
4. $(C_H + 2I)(C_H - (n-2)I) = 0$.
5. For each $\pi \in \mathcal{P}$ there are exactly $2n - 4$ pairs $\pi' \neq \pi$ with $(\mathbf{r}_\pi, \mathbf{r}_{\pi'}) \neq 0$.
6. Every entry of every $\mathbf{r}_\pi$ is $\pm\sqrt{2/n}$; every vector of Φ_H lies in $\mathbf{j}^\perp$; and for each row i, the vectors $\mathbf{r}_{ij}$, $j \neq i$, are pairwise orthogonal.

PROOF SKETCH: With $x = \frac{1}{2}n$ we rescale the Hadamard products to squared norm 2, and read off each part from the lemmas above.

⟨1⟩1. Part 1 holds.
PROOF: $x = \frac{1}{2}n$ is an integer, and $\{n, 2n, 3n\}/16 = \{n - x, n, n + x\}/8$; so ⟨0⟩:2 is Hypothesis 6.1. Then $d = n/x = 2$. □

⟨1⟩2. Part 2 holds.

PROOF: $(\mathbf{r}_\pi, \mathbf{r}_{\pi'}) = \frac{2}{n}(\mathbf{p}_\pi, \mathbf{p}_{\pi'})$. For $\pi = \pi'$ this is $\frac{2}{n} \cdot n$ by Lemma (the inner products of the Hadamard products), part 1. For $\pi \neq \pi'$ it is $\frac{2}{n} \cdot xC_H(\pi, \pi') = C_H(\pi, \pi')$ by Definition (the matrix C_H), and Lemma (the entries of C_H) with ⟨1⟩1 gives the values. □

⟨1⟩3. Part 3 holds.

⟨2⟩1. Φ_H is a line system.

PROOF: It is finite and $\Phi_H = -\Phi_H$ by definition. By ⟨1⟩2 and bilinearity, the vectors have squared norm 2 and pairwise inner products in $\{0, \pm 1, \pm 2\}$; this is the definition in Section 2. □

⟨2⟩2. For $\pi \neq \pi'$, $\mathbf{r}_{\pi'} \neq \pm \mathbf{r}_\pi$.

PROOF: By ⟨1⟩2, $(\mathbf{r}_\pi, \mathbf{r}_{\pi'}) \neq \pm 2$; apply (1). □

⟨2⟩3. $\pi \mapsto \mathbf{r}_\pi$ is a signing of S_H, and S_H has $|\mathcal{P}|$ lines.

PROOF: By ⟨2⟩2, $(\pi, \epsilon) \mapsto \epsilon \mathbf{r}_\pi$ is a bijection $\mathcal{P} \times \{1, -1\} \to \Phi_H$. □

⟨2⟩4. Q.E.D.

PROOF: By ⟨2⟩3, the matrix C of the signing has off-diagonal entries $(\mathbf{r}_\pi, \mathbf{r}_{\pi'}) = C_H(\pi, \pi')$ by ⟨1⟩2, and zero diagonal, as C_H has. □

⟨1⟩4. Part 4 holds.

PROOF: Lemma (the quadratic equation) with $d = 2$ by ⟨1⟩1: $\frac{1}{2}d(n-2) = n - 2$. □

⟨1⟩5. Part 5 holds.

⟨2⟩1. $C_H^2 = (n-4)C_H + 2(n-2)I$.

PROOF: Expand ⟨1⟩4. □

⟨2⟩2. $C_H^2(\pi, \pi) = \sum_{\pi'} C_H(\pi, \pi')^2$ is the number of π' with $C_H(\pi, \pi') \neq 0$.

PROOF: C_H is symmetric with entries in $\{0, \pm 1\}$ by Lemma (the entries of C_H). □

⟨2⟩3. Q.E.D.

PROOF: The diagonal of the right side of ⟨2⟩1 is $2(n-2)$, since $C_H(\pi, \pi) = 0$; use ⟨2⟩2, ⟨1⟩2, and that such π' differ from π. □

⟨1⟩6. Part 6 holds.

PROOF: Scale Lemma (the inner products of the Hadamard products), part 1, and Lemma (the lines through a fixed row), part 1, by $\pm\sqrt{2/n}$. □

⟨1⟩7. Q.E.D.

PROOF: Steps ⟨1⟩1 to ⟨1⟩6. □

We keep the hypotheses of Lemma (the line system of H) and its notation for the rest of the section. Let Λ_H be the graph on $\mathcal{P}$ in which $\pi \neq \pi'$ are adjacent when $(\mathbf{r}_\pi, \mathbf{r}_{\pi'}) \neq 0$. A *component* is the vertex set $\mathcal{K} \subseteq \mathcal{P}$ of a connected component of Λ_H (`SimpleGraph.ConnectedComponent`); it is nonempty. For a component $\mathcal{K}$ we write $\Phi(\mathcal{K}) = \{\pm \mathbf{r}_\pi : \pi \in \mathcal{K}\}$, $U_\mathcal{K}$ for its span, $m_\mathcal{K} = \dim U_\mathcal{K}$ for its *rank*, and $R_\mathcal{K} = \mathrm{sc}(\Phi(\mathcal{K}))$ (LN3).

Lemma (the components). ASSUME: 1. H is an $n \times n$ Hadamard matrix, with $16 \mid n$ and $n \geq 16$.
2. For all distinct rows i, j, k, l, $16\,\mathrm{ag}(i, j, k, l) \in \{n, 2n, 3n\}$.
3. $\mathcal{K}$ is a component of Λ_H, with rank $m = m_\mathcal{K}$.

PROVE: 1. For every row i, the number of rows j with $\{i, j\} \in \mathcal{K}$ is m.
2. $2|\mathcal{K}| = nm$.
3. $1 \leq m \leq n - 1$.
4. $R_\mathcal{K}$ is a nonempty indecomposable star-closed line system in $\mathbb{R}^n$, with $\Phi(\mathcal{K}) \subseteq \Phi(R_\mathcal{K})$, $|\Phi(\mathcal{K})| = 2|\mathcal{K}|$, and span $U_\mathcal{K}$.
5. There is a type t of Theorem 3.5, namely A_ℓ ($\ell \geq 1$), D_ℓ ($\ell \geq 4$), E_6, E_7 or E_8, with an isomorphism $R_\mathcal{K} \cong t$ induced by a surjective linear isometry of spans; t is unique, by Theorem 3.5, and we call it the *type* of $\mathcal{K}$. The rank of t is m, $R_\mathcal{K}$ and t have the same number L_t of lines, and $\frac{1}{2}nm \leq L_t$.

PROOF SKETCH: **[repair]** (Errata, Gap 17.) For a fixed row i, the vectors $\mathbf{r}_{ij}$ with $\{i,j\} \in \mathcal{K}$ are an orthogonal basis of $U_{\mathcal{K}}$, by the expansion of Lemma (the lines through a fixed row); so each row lies in exactly m pairs of $\mathcal{K}$. The rest is double counting and Theorem 3.5.

⟨1⟩1. Part 1 holds.

LET: i be a row, $B = \{\{i,j\} \in \mathcal{K}\}$, and U_B the span of $\{\mathbf{r}_\beta : \beta \in B\}$.

⟨2⟩1. ASSUME: $\pi \in \mathcal{K}$.
PROVE: $\mathbf{r}_\pi \in U_B$.

⟨3⟩1. If $(\mathbf{r}_\pi, \mathbf{r}_{ij}) \neq 0$ for a row $j \neq i$, then $\{i,j\} \in B$.
PROOF: Either $\{i,j\} = \pi$, or $\{i,j\} \neq \pi$ is adjacent to π in Λ_H; in both cases $\{i,j\}$ lies in the component $\mathcal{K}$ of π. □

⟨3⟩2. $\mathbf{r}_\pi = \frac{1}{2}\sum_{j \neq i} (\mathbf{r}_\pi, \mathbf{r}_{ij})\mathbf{r}_{ij}$.
PROOF: Scale the expansion of Lemma (the lines through a fixed row), part 2, for $\mathbf{v} = \mathbf{p}_\pi \in \mathbf{j}^\perp$ by $\sqrt{2/n}$. □

⟨3⟩3. Q.E.D.
PROOF: By ⟨3⟩1 each term of ⟨3⟩2 with a nonzero coefficient lies in U_B. □

⟨2⟩2. $U_B = U_{\mathcal{K}}$.
PROOF: $U_B \subseteq U_{\mathcal{K}}$ since $B \subseteq \mathcal{K}$; $U_{\mathcal{K}} \subseteq U_B$ by ⟨2⟩1, since U_B is closed under negation. □

⟨2⟩3. The vectors $\mathbf{r}_\beta$, $\beta \in B$, are linearly independent.
PROOF: They are nonzero and pairwise orthogonal by Lemma (the line system of H), part 6; apply `linearIndependent_of_ne_zero_of_inner_eq_zero`. □

⟨2⟩4. Q.E.D.
PROOF: $|B| = \dim U_B = m$ by ⟨2⟩3, `finrank_span_eq_card` and ⟨2⟩2. □

⟨1⟩2. Part 2 holds.
PROOF: Count the pairs (row i, pair $\pi \in \mathcal{K}$ with $i \in \pi$) in two ways: each π has two rows, and each row lies in m pairs of $\mathcal{K}$ by ⟨1⟩1. □

⟨1⟩3. Part 3 holds.
PROOF: $\mathcal{K}$ is nonempty, so $2|\mathcal{K}| = nm > 0$ by ⟨1⟩2; and a row i lies in at most $n-1$ pairs, so $m \leq n-1$ by ⟨1⟩1. □

⟨1⟩4. Part 4 holds.

⟨2⟩1. $\Phi(\mathcal{K})$ satisfies the hypotheses of LN3, and its graph of nonzero inner products is connected.
PROOF: Lemma (the line system of H), parts 2 and 3, give the norms and inner products and $\Phi(\mathcal{K}) = -\Phi(\mathcal{K})$. The vectors $\mathbf{r}_\pi$ and $-\mathbf{r}_\pi$ have inner product -2, and $\pm\mathbf{r}_\pi$, $\pm\mathbf{r}_{\pi'}$ are joined when π, π' are adjacent in Λ_H; $\mathcal{K}$ is connected in Λ_H. □

⟨2⟩2. $|\Phi(\mathcal{K})| = 2|\mathcal{K}|$.
PROOF: The vectors $\epsilon\mathbf{r}_\pi$ are distinct by Lemma (the line system of H), part 3. □

⟨2⟩3. Q.E.D.
PROOF: Apply LN3 to ⟨2⟩1; its span is that of $\Phi(\mathcal{K})$, which is $U_{\mathcal{K}}$; use ⟨2⟩2. □

⟨1⟩5. Part 5 holds.

⟨2⟩1. There is exactly one type t with $R_{\mathcal{K}} \cong t$, and the isomorphism is induced by a surjective linear isometry from $U_{\mathcal{K}}$ onto the span of the model of t.
PROOF: Theorem 3.5, applied to ⟨1⟩4 in $V = \mathbb{R}^n$. □

⟨2⟩2. The rank of t is m.
PROOF: A surjective linear isometry preserves dimension; the rank of the model is the dimension of its span (Lemma (spans of the models)), and $\dim U_{\mathcal{K}} = m$. □

⟨2⟩3. $R_{\mathcal{K}}$ has L_t lines, and $|\mathcal{K}| \leq L_t$.
PROOF: An isomorphism is a bijection of vector sets, so the numbers of lines agree (Lemma 3.4 gives L_t); and $2|\mathcal{K}| = |\Phi(\mathcal{K})| \leq |\Phi(R_{\mathcal{K}})| = 2L_t$ by ⟨1⟩4. □

⟨2⟩4. Q.E.D.
PROOF: Steps ⟨2⟩1, ⟨2⟩2 and ⟨2⟩3, with $|\mathcal{K}| = \frac{1}{2}nm$ by ⟨1⟩2. □
⟨1⟩6. Q.E.D.
PROOF: Steps ⟨1⟩1 to ⟨1⟩5. □

Lemma (the possible decompositions). ASSUME: 1. H is an $n \times n$ Hadamard matrix, with $16 \mid n$ and $n \geq 16$.
2. For all distinct rows i, j, k, l, $16\,\mathrm{ag}(i, j, k, l) \in \{n, 2n, 3n\}$.
PROVE: One of the following holds.
1. $\mathcal{P}$ is a single component, and its type in Lemma (the components) is A_{n-1} or D_{n-1}.
2. $n = 16$, and some component has type E_7.

PROOF SKETCH: This is the analysis in Gap 17 of the errata. Each type forces a bound on the rank through $\frac{1}{2}nm \leq L_t$, and the ranks of the components add up to $n - 1$.
⟨1⟩1. The ranks of the components add up to $n - 1$.
PROOF: The $n - 1$ pairs $\{1, j\}$ lie in the components, each in exactly one, and a component $\mathcal{K}$ contains $m_{\mathcal{K}}$ of them by Lemma (the components), part 1. □
⟨1⟩2. ASSUME: $\mathcal{K}$ is a component, with rank m and type t.
PROVE: 1. If $t = \mathrm{A}_m$, then $m = n - 1$.
2. If $t = \mathrm{D}_m$, then $m \geq \frac{1}{2}n + 1$.
3. $t \neq \mathrm{E}_6$.
4. If t is E_7 or E_8, then $n = 16$.
⟨2⟩1. Part 1 holds.
PROOF: $\frac{1}{2}nm \leq \frac{1}{2}m(m+1)$ by Lemma (the components), part 5, and Lemma 3.4; divide by $\frac{1}{2}m > 0$ to get $n - 1 \leq m$, and $m \leq n - 1$ by part 3 of that lemma. □
⟨2⟩2. Part 2 holds.
PROOF: $\frac{1}{2}nm \leq m(m-1)$ in the same way; divide by $m > 0$. □
⟨2⟩3. Part 3 holds.
PROOF: Otherwise $m = 6$ and $3n \leq 36$, so $n \leq 12$, against ⟨0⟩:1. □
⟨2⟩4. Part 4 holds.
PROOF: For E_7, $m = 7$ and $\frac{7}{2}n \leq 63$, so $n \leq 18$; for E_8, $m = 8$ and $4n \leq 120$, so $n \leq 30$. The only multiple of 16 in $[16, 30]$ is 16. □
⟨2⟩5. Q.E.D.
PROOF: Steps ⟨2⟩1 to ⟨2⟩4. □
⟨1⟩3. CASE: Some component $\mathcal{K}$ has type A_m.
PROOF: Then $m = n-1$ by ⟨1⟩2. By ⟨1⟩1 every other component has rank 0, which Lemma (the components), part 3, forbids; so $\mathcal{K} = \mathcal{P}$, and alternative 1 holds. □
⟨1⟩4. CASE: Some component $\mathcal{K}$ has type D_m.
⟨2⟩1. $\mathcal{K}$ is the only component.
⟨3⟩1. ASSUME: $\mathcal{K}' \neq \mathcal{K}$ is a component, of rank m' and type t'.
PROVE: FALSE.
⟨4⟩1. $m' \leq \frac{1}{2}n - 2$.
PROOF: $m + m' \leq n - 1$ by ⟨1⟩1, and $m \geq \frac{1}{2}n + 1$ by ⟨1⟩2. □
⟨4⟩2. t' is neither $\mathrm{A}_{m'}$ nor $\mathrm{D}_{m'}$.
PROOF: By ⟨1⟩2, these would give $m' = n - 1$ or $m' \geq \frac{1}{2}n + 1$, against ⟨4⟩1. □
⟨4⟩3. t' is not exceptional.
PROOF: By ⟨1⟩2, $t' \neq \mathrm{E}_6$, and for E_7 or E_8 we get $n = 16$ and $m' \geq 7 > 6 = \frac{1}{2}n - 2$, against ⟨4⟩1. □
⟨4⟩4. Q.E.D.
PROOF: Theorem 3.5 has no other types. □

⟨3⟩2. Q.E.D.
PROOF: Step ⟨3⟩1. □
⟨2⟩2. Q.E.D.
PROOF: By ⟨2⟩1 and ⟨1⟩1, $\mathcal{K} = \mathcal{P}$ and $m = n - 1$; alternative 1 holds. □
⟨1⟩5. CASE: No component has type A_m or D_m.
⟨2⟩1. $n = 16$, and every component has rank 7 or 8.
PROOF: $\mathcal{P}$ is nonempty, so there is a component. Each component $\mathcal{K}$ has a type t by Lemma (the components), part 5, of rank $m_{\mathcal{K}}$, and the model A_ℓ or D_ℓ has rank ℓ (Lemma (spans of the models)); so t is neither $A_{m_{\mathcal{K}}}$ nor $D_{m_{\mathcal{K}}}$ by the case hypothesis, and not E_6 by ⟨1⟩2, part 3. In the list of Theorem 3.5 this leaves E_7 and E_8, of ranks 7 and 8 (Lemma (spans of the models)); then $n = 16$ by ⟨1⟩2, part 4. □
⟨2⟩2. Some component has rank 7.
PROOF: Otherwise the ranks are all 8 and add up to 15 by ⟨1⟩1, but $8 \nmid 15$. □
⟨2⟩3. Q.E.D.
PROOF: The component of ⟨2⟩2 has type E_7, the only type of rank 7 left; alternative 2 holds. □
⟨1⟩6. Q.E.D.
PROOF: The three cases are exhaustive. □

Lemma (orthogonal lines in A_ℓ and D_ℓ). ASSUME: 1. $\ell \geq 2$ is an integer, and M is the model A_ℓ in $\mathbb{R}^{\ell+1}$ or D_ℓ in $\mathbb{R}^\ell$ of Definition 3.1.
2. $F \subseteq \Phi(M)$ has ℓ elements, and any two distinct elements of F are orthogonal.
PROVE: M is not A_ℓ, and if M is D_ℓ, then ℓ is even.

PROOF SKETCH: **[new]** The paper states this without proof, and no result of Section 3 contains it; it needs only the coordinates of Definition 3.1. Each root has a support of two coordinates, and two orthogonal roots have disjoint supports in A_ℓ, and disjoint or equal supports in D_ℓ. Theorem 3.6 is not needed.
LET: the support of a root be the set of its two nonzero coordinates.
⟨1⟩1. ASSUME: $M = A_\ell$.
PROVE: FALSE.
⟨2⟩1. Distinct $\mathbf{r}, \mathbf{s} \in F$ have disjoint supports.
PROOF: For $\mathbf{r} = \mathbf{e}_a - \mathbf{e}_b$ and $\mathbf{s} = \mathbf{e}_c - \mathbf{e}_e$, $(\mathbf{r}, \mathbf{s}) = [a = c] - [a = e] - [b = c] + [b = e]$. If the supports share exactly one coordinate, this is ± 1; if they are equal, it is ± 2; neither is 0, against ⟨0⟩:2. □
⟨2⟩2. Q.E.D.
PROOF: By ⟨2⟩1, the ℓ supports are disjoint 2-subsets of $\{1, \ldots, \ell + 1\}$, so $2\ell \leq \ell + 1$ and $\ell \leq 1$, against ⟨0⟩:1. □
⟨1⟩2. ASSUME: $M = D_\ell$.
PROVE: ℓ is even.
⟨2⟩1. Distinct $\mathbf{r}, \mathbf{s} \in F$ have disjoint or equal supports.
PROOF: If the supports share exactly one coordinate, $(\mathbf{r}, \mathbf{s})$ is the product of the two entries there, which is ± 1, against ⟨0⟩:2. □
⟨2⟩2. At most two elements of F have a given support $\{a, b\}$.
⟨3⟩1. If $\mathbf{r}, \mathbf{s} \in F$ are distinct with support $\{a, b\}$, then $\mathbf{s} = \pm(\mathbf{r}(a)\mathbf{e}_a - \mathbf{r}(b)\mathbf{e}_b)$.
PROOF: $0 = (\mathbf{r}, \mathbf{s}) = \mathbf{r}(a)\mathbf{s}(a) + \mathbf{r}(b)\mathbf{s}(b)$ with all four entries ± 1, so $\mathbf{s}(a)\mathbf{s}(b) = -\mathbf{r}(a)\mathbf{r}(b)$. □
⟨3⟩2. Q.E.D.
PROOF: If $\mathbf{r}, \mathbf{s}, \mathbf{t} \in F$ were distinct with support $\{a, b\}$, then $\mathbf{s} = \pm\mathbf{t}$ by ⟨3⟩1, so $(\mathbf{s}, \mathbf{t}) = \pm 2 \neq 0$, against ⟨0⟩:2. □
LET: k be the number of distinct supports of the elements of F.

⟨2⟩3. $2k \le \ell$ and $\ell \le 2k$.
PROOF: The k supports are pairwise disjoint 2-subsets of $\{1,\dots,\ell\}$ by ⟨2⟩1; and $|F| = \ell$ with at most two elements per support by ⟨2⟩2. □
⟨2⟩4. Q.E.D.
PROOF: By ⟨2⟩3, $\ell = 2k$. □
⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2. □

Lemma (no component of type E_7). ASSUME: 1. H is a 16×16 Hadamard matrix.
2. For all distinct rows i,j,k,l, $\mathrm{ag}(i,j,k,l) \in \{1,2,3\}$.
3. $\mathcal{K}$ is a component of Λ_H whose type in Lemma (the components) is E_7, and $R = R_{\mathcal{K}}$.
PROVE: FALSE.

PROOF SKETCH: **[new]** **[repair]** (Errata, Gap 17, the repair.) Two vectors $\mathbf{r},\mathbf{r}'$ of $\Phi(\mathcal{K})$ with $(\mathbf{r},\mathbf{r}') = -1$ span a star of R whose third vector $\mathbf{r}+\mathbf{r}'$ has zero entries, so it lies in R but not in $\Phi(\mathcal{K})$; and $\mathbf{r}$ lies in $N_{\mathbf{r}+\mathbf{r}'}$. Thus the 3136 such ordered pairs inject into the 448 pairs $(\mathbf{t},\mathbf{u})$ with $\mathbf{t} \in \Phi(R) \setminus \Phi(\mathcal{K})$ and $\mathbf{u} \in N_{\mathbf{t}}$. This is the errata's count $784 \le 7 \cdot 16$, of pairs of lines against the 16 stars through each of the 7 lines of R outside $\mathcal{K}$, multiplied by 4; we count signed vectors so that no choice of orientation, and no definition of a star as a set of lines, is needed.
⟨1⟩1. The hypotheses of Lemmas (the line system of H) and (the components) hold, with $n = 16$.
PROOF: $16 \mid 16$, and $\mathrm{ag} \in \{1,2,3\}$ is $16\,\mathrm{ag} \in \{n, 2n, 3n\}$ for $n = 16$; use ⟨0⟩:1, ⟨0⟩:2 and ⟨0⟩:3. □
LET: $\Phi = \Phi(\mathcal{K})$ and $Q = \{(\mathbf{r},\mathbf{r}') \in \Phi \times \Phi : (\mathbf{r},\mathbf{r}') = -1\}$.
⟨1⟩2. $|\Phi| = 112$ and $|\Phi(R)| = 126$, so $|\Phi(R) \setminus \Phi| = 14$.
PROOF: The rank of $\mathcal{K}$ is 7, the rank of E_7, so $|\Phi| = 2|\mathcal{K}| = 16 \cdot 7 = 112$ by Lemma (the components), parts 2, 4 and 5; R has 63 lines by that lemma and Lemma 3.4; and $\Phi \subseteq \Phi(R)$. □
⟨1⟩3. $|Q| = 112 \cdot 28 = 3136$.
⟨2⟩1. ASSUME: $\mathbf{r} \in \Phi$.
PROVE: Exactly 28 vectors $\mathbf{r}' \in \Phi$ satisfy $(\mathbf{r},\mathbf{r}') = -1$.
LET: $\mathbf{r} = \epsilon \mathbf{r}_\pi$, with $\pi \in \mathcal{K}$ and $\epsilon = \pm 1$.
⟨3⟩1. Exactly $2n - 4 = 28$ pairs $\pi' \ne \pi$ have $(\mathbf{r}_\pi, \mathbf{r}_{\pi'}) \ne 0$, and they lie in $\mathcal{K}$.
PROOF: Lemma (the line system of H), part 5; such π' are adjacent to π in Λ_H. □
⟨3⟩2. For each such π', exactly one of $\pm\mathbf{r}_{\pi'}$ has inner product -1 with $\mathbf{r}$.
PROOF: $(\mathbf{r}, \mathbf{r}_{\pi'}) = \pm 1$ by Lemma (the line system of H), part 2, and $(\mathbf{r}, -\mathbf{r}_{\pi'}) = -(\mathbf{r}, \mathbf{r}_{\pi'})$. □
⟨3⟩3. No other vector of Φ has inner product -1 with $\mathbf{r}$.
PROOF: $(\mathbf{r}, \pm\mathbf{r}_\pi) = \pm 2$, and $(\mathbf{r}, \pm\mathbf{r}_{\pi'}) = 0$ for the remaining π'. □
⟨3⟩4. Q.E.D.
PROOF: Steps ⟨3⟩1, ⟨3⟩2 and ⟨3⟩3, with the distinctness of Lemma (the line system of H), part 3. □
⟨2⟩2. Q.E.D.
PROOF: Sum ⟨2⟩1 over the 112 vectors of ⟨1⟩2. □
⟨1⟩4. ASSUME: $(\mathbf{r},\mathbf{r}') \in Q$, and $\mathbf{w} = \mathbf{r} + \mathbf{r}'$.
PROVE: $\mathbf{w} \in \Phi(R) \setminus \Phi$, and $\mathbf{r} \in N_{\mathbf{w}}(R)$.
⟨2⟩1. $\mathbf{w} \in \Phi(R)$.
PROOF: $\mathbf{r},\mathbf{r}' \in \Phi \subseteq \Phi(R)$ have inner product -1, and R is star-closed (Lemma (the components), part 4, and Definition 2.1). □
⟨2⟩2. Every entry of $\mathbf{w}$ is 0 or $\pm 2\sqrt{2/16}$, and at most four are nonzero.
PROOF: Each entry of $\mathbf{r}$ and of $\mathbf{r}'$ is $\pm\sqrt{2/16}$ by Lemma (the line system of H), part 6, so each entry of the sum is 0 or $\pm 2\sqrt{2/16}$. A nonzero entry contributes $4 \cdot \frac{2}{16} = \frac{1}{2}$ to $(\mathbf{w},\mathbf{w})$, which is $2 + 2 - 2 = 2$ by bilinearity; so there are exactly four. □
⟨2⟩3. $\mathbf{w} \notin \Phi$.

PROOF: Every vector of Φ has 16 nonzero entries by Lemma (the line system of H), part 6, and $\mathbf{w}$ has four by ⟨2⟩2. □

⟨2⟩4. $(\mathbf{r}, \mathbf{w}) = 1$.
PROOF: $(\mathbf{r}, \mathbf{r}) + (\mathbf{r}, \mathbf{r}') = 2 - 1$. □

⟨2⟩5. Q.E.D.
PROOF: Steps ⟨2⟩1, ⟨2⟩3 and ⟨2⟩4, and $\mathbf{r} \in \Phi(R)$. □

⟨1⟩5. For every $\mathbf{t} \in \Phi(R)$, $|N_{\mathbf{t}}(R)| = 32$.
⟨2⟩1. R has a signing $\sigma : I \to \Phi(R)$, $|I| = 63$, whose matrix C satisfies $(C+2I_{63})(C-16I_{63}) = 0$.
PROOF: Choose one vector on each line. $R \cong \mathrm{E}_7$ through a surjective linear isometry of spans by Lemma (the components), part 5, so R is an isometric realization of E_7; apply Theorem 3.7. □

⟨2⟩2. Each row of C has exactly 32 nonzero entries.
PROOF: From ⟨2⟩1, $C^2 = 14C + 32I_{63}$; the diagonal entry $C^2(a,a) = \sum_b C(a,b)^2$ counts the nonzero entries of row a, since the entries of C lie in $\{0, \pm 1\}$ and $C(a,a) = 0$ (Definition (signings and the matrix convention)). □

LET: $\mathbf{t} = \delta\sigma(a)$ with $a \in I$ and $\delta = \pm 1$.
⟨2⟩3. For $b \in I$, exactly one of $\pm\sigma(b)$ lies in $N_{\mathbf{t}}(R)$ if $C(a,b) \neq 0$, and neither does otherwise.
PROOF: $(\pm\sigma(b), \mathbf{t}) = \pm\delta C(a,b)$ for $b \neq a$, and $(\pm\sigma(a), \mathbf{t}) = \pm 2$. □

⟨2⟩4. Q.E.D.
PROOF: Every vector of $\Phi(R)$ is $\pm\sigma(b)$ for some $b \in I$; so $|N_{\mathbf{t}}(R)| = 32$ by ⟨2⟩3 and ⟨2⟩2. (In Lean the leaf may instead transport `LineSystem.E7.card_N` of C20 through the isomorphism, which preserves the sets $N_{\mathbf{t}}$.) □

⟨1⟩6. The map $(\mathbf{r}, \mathbf{r}') \mapsto (\mathbf{r}+\mathbf{r}', \mathbf{r})$ is an injection of Q into $T = \{(\mathbf{t}, \mathbf{u}) : \mathbf{t} \in \Phi(R) \setminus \Phi,\ \mathbf{u} \in N_{\mathbf{t}}(R)\}$.
PROOF: It maps into T by ⟨1⟩4, and $(\mathbf{r}, \mathbf{r}')$ is recovered from its image $(\mathbf{t}, \mathbf{u})$ as $(\mathbf{u}, \mathbf{t} - \mathbf{u})$. □

⟨1⟩7. $|T| = 14 \cdot 32 = 448$.
PROOF: Steps ⟨1⟩2 and ⟨1⟩5. □

⟨1⟩8. Q.E.D.
PROOF: By ⟨1⟩3, ⟨1⟩6 and ⟨1⟩7, $3136 \leq 448$, which is false. □

Remark. The same count excludes the type A_{n-1} without Lemma (orthogonal lines in A_ℓ and D_ℓ): then $\mathcal{K} = \mathcal{P}$ has $\frac{1}{2}n(n-1)$ lines, which is all of A_{n-1}, so $\Phi(R) \setminus \Phi$ is empty while Q is not. It does not exclude D_{n-1}, nor the component of type E_8 when $n = 16$ (there $|Q| = 128 \cdot 28$ and $|T| = 112 \cdot 56$). We checked the counts of E_7 (126 vectors, $|N_{\mathbf{t}}| = 32$ for each) numerically in model (ii) of Definition 3.3, in a scratch script outside the trust base.

Theorem 6.2. ASSUME: 1. H is an $n \times n$ Hadamard matrix, with $n > 2$.
2. For all distinct rows i, j, k, l, $16\,\mathrm{ag}(i,j,k,l) \in \{n, 2n, 3n\}$; that is, any four rows agree in $\frac{n}{16}$, $\frac{n}{8}$ or $\frac{3n}{16}$ places.

PROVE: FALSE. Thus there is no Hadamard matrix of order $n > 2$ for which any four rows agree in $\frac{n}{16}$, $\frac{n}{8}$ or $\frac{3n}{16}$ places.

PROOF SKETCH: We follow the paper: the $\frac{1}{2}n(n-1)$ lines lie at 60° and 90° in $\mathbf{j}^\perp$, and the $n-1$ lines through a fixed row are mutually orthogonal, which the systems A_{n-1} and D_{n-1} do not allow for odd $n-1$. **[repair]** The small orders are Lemma (orders not divisible by 16) (errata, Gap 10), and the indecomposability that the paper asserts is replaced by Lemmas (the components), (the possible decompositions) and (no component of type E_7) (errata, Gap 17). With a single component, the star closure is A_{n-1} or D_{n-1} by Theorem 3.5, as the paper says; the paper's appeal to $n > 9$ is replaced by the count of lines.

⟨1⟩1. $16 \mid n$ and $n \geq 16$.
PROOF: Lemma (orders not divisible by 16), with ⟨0⟩:1 and ⟨0⟩:2. □

⟨1⟩2. Alternative 1 or alternative 2 of Lemma (the possible decompositions) holds.
PROOF: Lemma (the possible decompositions), with ⟨1⟩1 and ⟨0⟩:2. □
⟨1⟩3. CASE: $\mathcal{P}$ is a single component, of type A_{n-1} or D_{n-1}.
LET: $R = R_{\mathcal{P}}$, $f : \Phi(R) \to \Phi(M)$ the isomorphism of Lemma (the components), part 5, onto the model M, and $F = \{f(\mathbf{r}_{1j}) : 2 \le j \le n\}$.
⟨2⟩1. F has $n-1$ elements, and any two distinct elements of F are orthogonal.
PROOF: The $\mathbf{r}_{1j}$ are distinct and pairwise orthogonal vectors of $\Phi(\mathcal{P}) \subseteq \Phi(R)$ by Lemma (the line system of H), parts 3 and 6, and Lemma (the components), part 4; f is injective and preserves inner products. □
⟨2⟩2. $n-1 \ge 2$ is odd.
PROOF: $n \ge 16$ is even by ⟨1⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Lemma (orthogonal lines in A_ℓ and D_ℓ), with $\ell = n-1$, ⟨2⟩1 and ⟨2⟩2, excludes $M = \mathrm{A}_{n-1}$, and gives $n-1$ even if $M = \mathrm{D}_{n-1}$, against ⟨2⟩2. □
⟨1⟩4. CASE: $n = 16$, and some component has type E_7.
PROOF: With $n = 16$, ⟨0⟩:2 says $\mathrm{ag}(i,j,k,l) \in \{1,2,3\}$; apply Lemma (no component of type E_7). □
⟨1⟩5. Q.E.D.
PROOF: The two cases are those of ⟨1⟩2. □

Remark. Computation C16 (`LineGraphs/Computations/C16.lean`) concerns the Sylvester matrices of orders 2, 4, 8 and 16 and the Paley matrix H_{12} of order 12, given as integer matrices in `LineGraphs/Computations/C16Data.lean`. We see that each of the five is a Hadamard matrix, by `decide +kernel`. A Boolean checker counts the agreements of every four distinct rows, and `Computations.C16.checkFourRowAgreement_sound` proves that its answer gives Hypothesis 6.1; thus H_{12} satisfies Hypothesis 6.1 with $x = 4$, and the Sylvester matrix of order n satisfies it with $x = n$ (for $n = 2$ the condition is vacuous, since there are no four distinct rows). For H_{12}, the formula of Lemma (the matrix P), part 1, is $P^TP = 72I - 6J$, a case of `Matrix.IsHadamard.transpose_hadamardPairs_mul`. The vectors $\sqrt{2/12}\,\mathbf{p}_\pi$, for the 66 pairs $\pi \in \mathcal{P}$, span 66 distinct lines in $\mathbf{j}^\perp$; this hyperplane has dimension 11 by Lemma (the lines through a fixed row) and is isometric to $\mathbb{R}^{11}$, and any two of the lines are at 90° or at $\arccos(1/3)$, both angles occurring. We note that computation C16 does not prove the uniqueness of H_{12} or Norman's theorem.

Remark. The paper says that the only known examples are the Hadamard matrices of Sylvester type, with $n = 2^s$ and $x = n$, and the unique H_{12}, with $n = 12$ and $x = 4$; and that the Sylvester case corresponds to $d = 1$ and yields $n-1$ mutually orthogonal lines, each line occurring $\frac{1}{2}n$ times. The claims about the Sylvester matrices are side claims, and we prove them for every s in `LineGraphs/SideClaims/Sylvester.lean`; the words "only known" are history, and we do not formalize them. The Sylvester matrix H of order $n = 2^s$ has its rows and columns indexed by $\{0,1\}^s$, with $H(a,b) = (-1)^{a\cdot b}$; it is the Kronecker product of s copies of the matrix H_2 with rows $(1,1)$ and $(1,-1)$ (`Matrix.sylvesterKronecker_eq_sylvester`), and it is a Hadamard matrix (`Matrix.isHadamard_sylvester`). The Hadamard product of rows a and b is row $a+b$, with addition modulo 2 (`Matrix.hadamardRowProduct_sylvester_sylvesterEquiv`); in Lean the rows are also indexed by $\{0,\dots,n-1\}$ through binary digits, and then the product of rows i and j is the row of their bitwise exclusive or (`Matrix.hadamardRowProduct_sylvester`). Thus the rows of P are the $n-1$ rows of H other than the all-ones row, each of them is the product of exactly $\frac{1}{2}n$ pairs of rows, and distinct ones are orthogonal; for the lines that the rows of P span, this is `Matrix.sylvester_lines`. By Lemma (the inner products of the Hadamard products), part 3, four distinct rows i,j,k,l then have $8\,\mathrm{ag}(i,j,k,l) = n + (\mathbf{p}_{ij}, \mathbf{p}_{kl})$, which is n

or $2n$, and H satisfies Hypothesis 6.1 with $x = n$ (`Matrix.sylvester_fourRowAgreement`); so $d = n/x = 1$. The four Sylvester matrices of computation C16 are these matrices for $s = 1, 2, 3, 4$ (`Computations.C16.h16_eq_sylvester` and the three others).

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